

Catapult HLS User View Scheduling Rules

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Abstract

This document defines a user-visible scheduling contract and modeling methodology for high-level synthesis (HLS). Its practical objective is to support a proof-backed flow in which much of the system-level functional verification and debug is performed on the pre-HLS SystemC model, for designs and implementations that satisfy the stated applicability and conformance conditions. The contract treats message-passing operations, signal I/O, and explicit synchronization through a small set of uniform rules while still allowing important implementation transformations such as loop pipelining, added FSM states, permitted memory-access reordering, direct-input late sampling, and bounded or more complex channel buffering.

The main formal safety guarantee compares observable system behavior: every covered post-HLS RTL behavior has a matching behavior in the prescribed pre-HLS reference model. This supports source-level functional verification without requiring the RTL to preserve every source-level cycle or internal state. The converse claim is deliberately narrower: showing that a particular source execution is reproduced by the RTL requires the additional alignment and witness-selection conditions described later in the document. Deadlock preservation is a separate result for internal message-passing deadlock. Its primary practical route checks one selected complete deterministic serial-issue pre-HLS execution together with finite response bounds for the relevant blocking operations. In the standard qualified-synchronous scope, a checked one-cycle source/RTL correspondence argument can be extended across all reachable implementation cycles. For functional safety properties that are prefix-closed in the Appendix-J finite-ideal sense, the completed-cycle coverage also covers finite observable prefixes within a cycle; the same cycle-by-cycle framework supplies the implementation-wide coverage used by the universal deadlock result. These guarantees remain conditional on the stated channel-capacity, fairness, progress, environment, reset, structural, and implementation-correspondence conditions.

The proofs do not by themselves certify a particular HLS invocation or RTL instance. The applicable source well-formedness, scheduling, channel realization and capacity, reset, environment, progress, and implementation-mapping conditions must be established by the tool or verification flow through documentation, generated certificates, static checks, targeted RTL/interface checks, formal verification, or equivalent evidence. Subject to those conditions, the document provides a Catapult-compatible, tool-neutral basis for industrial HLS verification and for possible scheduling-contract standardization.

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Introduction

HLS loses much of its practical advantage when design teams must return to machine-generated RTL as the primary level for functional verification and debug. RTL verification can be one of the most costly and time-consuming parts of a hardware development flow, and generated RTL may differ substantially in structure and timing from the source model. A useful HLS methodology should therefore let designers and verification engineers reason primarily at the source level while retaining a precise way to qualify the resulting implementation.

This document presents a user-visible scheduling contract intended to provide "no surprises" at process and interface boundaries. It reduces interface scheduling to uniform rules for message-passing operations, signal I/O, and explicit synchronization. A common verification intent can then be reused across the prescribed pre-HLS source reference model and the post-HLS implementation, without requiring the implementation to preserve every source-level cycle or internal state.

Subject to the conditions stated in the main text and formal appendices, the methodology accommodates sequential and combinational logic, pipelined designs, shared and external memories through the stated mapping conditions, permitted RAM-access reordering, direct-input late sampling, finite channel buffering, and localized latency-sensitive protocols. These capabilities reflect requirements encountered across a broad range of industrial HLS designs and are important for achieving competitive hardware quality of results (QOR).

The methodology builds on established verification practices such as SystemVerilog UVM, constrained-random stimulus, and functional and code coverage. The system under test is modeled and verified in SystemC, while implementation-level verification focuses on establishing that the synthesized or otherwise implemented RTL falls within the scope of the applicable theorem instance. When the selected abstract channel boundaries and back-annotated capacities faithfully represent the RTL storage or credit at those boundaries, the pre-HLS model can also be used to assess capacity and throughput effects rather than deferring those questions entirely to RTL simulation.

The formal architecture separates safety from liveness. Safety is largely compositional: design-specific evidence is supplied for individual processes, explicit channel boundaries, named atomic interactions, the environment, and reset behavior, and a generic theorem composes that evidence into a system-level observable trace-compatibility result. Liveness and deadlock preservation have a more global dependency structure and therefore require additional cutpoint-coverage, progress, fairness, drain, environment, and source-side premises.

The user-visible scheduling rules are intentionally much simpler than the formal development that supports them. The additional formal machinery is needed not because the basic rules are individually complex, but because the guarantees must remain valid under pipelining, buffering, added implementation states, memory transformations, scheduler variation, reset and environment behavior, and the composition of many interacting processes. Thus, the complexity of the formal appendices reflects the breadth and rigor of the guarantees rather than complexity that an HW architect or HLS user is expected to manage.

The most important default deadlock-avoidance rule can be stated plainly. HLS may overlap or shift message-passing operations in time, but it may not reverse the dynamic source order of message-passing calls under the default scheduling contract. It therefore cannot introduce a new deadlock merely by reversing the order of source calls. This rule is not a complete deadlock theorem: overlapped loop iterations, eager issue, finite capacities, and environment behavior can still affect deadlock, and those effects are addressed by the separate liveness machinery, whose primary practical route checks one

selected, complete serial-issue execution of the pre-HLS model with total bounded-response coverage (Appendix K.4b and K.4c).

The guarantees remain conditional. The mathematical proofs do not establish that a particular HLS tool invocation or generated RTL instance satisfies the source-side and implementation-side obligations. Those obligations must be discharged by the tool or flow through documentation, generated certificates, lint or static analysis, targeted RTL/interface checks, formal verification, or another appropriate validation method. The same framework can also be applied to hand-written, modified, or otherwise generated RTL when equivalent evidence is supplied.

Because the rules are expressed as a user-visible, tool-neutral scheduling contract rather than as the internal behavior of a particular HLS implementation, they can also serve as a starting point for standardization, for example within the Accellera Synthesis Working Group. The next section summarizes the guarantees without exposing the internal proof API. A separate Guide to the Formal Appendices appears immediately before Appendix G, Appendix F provides user-oriented modeling guidelines, and Appendix M provides a more detailed Formal Results Overview.

Formal Guarantees at a Glance

The statements below provide a user-level orientation. The formal definitions, assumptions, scope conditions, and evidence requirements appear in the appendices.

Implementation safety. Within the stated scheduling, channel, environment, reset, and qualification scope, the generated RTL does not introduce new observable functional behavior that is absent from the prescribed pre-HLS reference model. This allows most functional verification and debug to remain at the source level when those conditions are established.

Use in the reverse direction. A particular pre-HLS simulation is not automatically guaranteed to be reproduced cycle for cycle, or even as the directly corresponding execution, by the RTL. Direct comparison of a selected source execution with RTL requires the alignment and witness-selection conditions described in Appendices G and L.

Internal message-passing deadlock. The primary practical method checks one complete deterministic serial-issue execution of the pre-HLS model and verifies finite response bounds for the blocking operations relevant to the proof. For a terminating design, a clean complete execution can provide the required source-side deadlock evidence without exploring alternative scheduler interleavings, provided the tool or verification flow separately establishes that the method applies to the design and implementation. The method is deliberately conservative: it may not apply when deadlock freedom depends on favorable overlap of blocking requests, a situation most characteristic of tightly coupled rendezvous-style communication. In practice, pipelined, buffered, and credit-based communication often removes that dependence, so this conservatism is less likely to be limiting in typical modern designs. The result is specifically about internal message-passing deadlock; it does not by itself cover barrier stalls, reset-induced stalls, environment-only stalls, starvation outside the modeled internal wait structure, or protocols whose correctness depends on timing behavior outside the method's scope. Qualification may need to include internal message-passing channels as well as the design's external interfaces.

RTL coverage and qualification. A clean pre-HLS execution is only one part of the overall verification argument. A claim covering all relevant reachable RTL behavior also requires implementation-side evidence that the generated RTL follows the assumed scheduling, channel, reset, and correspondence rules, and that persistent blocking behavior cannot remain hidden indefinitely in transient internal activity. In the standard qualified-synchronous scope, the formal results allow a checked one-cycle source/RTL correspondence to be established initially and then preserved from one reachable

implementation cycle to the next. The safety result needs only the functional-correspondence part of this evidence, while the universal deadlock result requires additional blocking and progress evidence. For functional safety properties that are prefix-closed in the Appendix-J finite-ideal sense, the cycle-by-cycle safety evidence also covers finite observable prefixes within each covered cycle; it is not necessary to construct a separate source/RTL package for every such intra-cycle ideal. A flow that establishes the stronger deadlock argument therefore also supplies the cycle-by-cycle correspondence evidence used by the safety result. This must cover all reachable cycle transitions, not just one observed simulation path. Some of the more advanced deadlock-preservation proof and certificate machinery is not yet fully mechanized and qualified. Appendices G, I, and J provide the detailed safety and source/RTL correspondence results; Appendix K provides the deadlock results; and Appendix M.7 summarizes the division of verification responsibilities and current framework status.

Document Goals

This document presents HLS tool scheduling rules and a modeling methodology. The goals are:

1. To provide easy-to-understand rules to HLS users.
2. To ensure precise and consistent rules in both SystemC and C++.
3. To offer rules that are effectively compatible with how Catapult currently operates.
4. To enable the best possible *quality of results* (QOR) in Catapult synthesis.
5. To cover all known user requirements and scenarios.
6. To serve as a suitable starting point for a standardization proposal (e.g., in Accellera SWG).

To illustrate the goals of this document more specifically, consider what an engineer writing a testbench for an HLS model needs to understand about how HLS tools operate. This engineer may be using SystemVerilog UVM or may be writing a testbench in C++/SystemC.

They likely are not an expert in any specific HLS tool (and may not want to be), but their testbench should be usable across both the pre-HLS model and the post-HLS model, subject to the latency-insensitivity, observation-boundary, and witness-selection conditions stated in this document. Thus, it is crucial for the DV engineer to have a precise understanding of how the HLS tool will transform the design while still enabling the resulting RTL behavior to be checked against the prescribed source reference model. This document describes what transformations the HLS tool is allowed to perform so that the pre-HLS and post-HLS models can be effectively verified with a common verification intent, and in the deterministic or annotated RTL-consistent cases with the same directly corresponding testbench execution. The overarching philosophy of the scheduling rules is to present “no surprises” to such a DV engineer, while still giving the HLS tool ample freedom to optimize the design.

The main body of this document is intended to be sufficiently precise and complete to support the Accellera standardization effort while still being understandable to HW architects. The formal appendices are intended to have a very high degree of precision and completeness sufficient to support AI-driven Lean mechanization of the formal theorems.

Background

HLS tools generate RTL from C++ models. Broadly speaking, this conversion takes a sequential C++ model and turns it into concurrent hardware that maintains the same behavior. HLS tools identify concurrent processes within the C++/SystemC model and then independently synthesize each process.

Briefly, some of the techniques that HLS tools use to achieve good HW QOR when synthesizing each process include:

- Optimized scheduling based on the selected silicon target technology.
- Automatic HW pipeline construction according to the user's specifications.
- Automatic HW resource sharing.
- Automatic scheduling of memory accesses.

The internal behavior of each process is specified by the control and dataflow behavior of the C++ code within the process. However, the external communication that each process has with other processes and HW blocks is specified via IO operations that are coded within the model. To enable a reliable, scalable, and verifiable HLS flow that generates high quality hardware, the scheduling behavior of these IO operations needs to be precisely handled at all steps of the flow. This document specifies the rules that govern the scheduling behavior for these IO operations within HLS models.

These rules are specified with respect to an individual process, but the intent of the rules is to enable reliable and verifiable behavior of large sets of interacting processes operating as a system in real-world designs. (Appendix G provides the formal trace-compatibility guarantees between the prescribed source reference model and the post-HLS system, including the unrestricted Post → Ref direction and the restricted annotated RTL-consistent / witness-selected reverse use.)

By default, HLS tools can insert additional clock cycles (or latency) anywhere within a process -- for example, when pipelining a loop or to enable HW resource sharing. The overall approach used in this document is to make the entire design and testbench system *latency insensitive* to the maximum extent possible, while still fully enabling key HW optimizations. In addition, the overall approach enables the pre-HLS system simulation to be capacity-accurate and throughput-accurate with respect to the post-HLS RTL system.

In some cases, designs or testbenches may use protocols which are latency-sensitive. These situations can be handled by isolating the latency-sensitive portions to small, self-contained parts of the design or testbench, and then keeping the rest of the design and testbench latency insensitive. See Appendix D for more information.

In many cases the overall design will need to satisfy end-to-end latency requirements. For these designs it is still highly advantageous to use a latency-insensitive modeling approach and verify in the post-HLS model that the overall design latency requirements have been satisfied, since this is typically easy to do.

This document focuses on sequential HW processes. Combinational HW processes are supported but mostly not discussed since their synthesis is straightforward. All the examples and discussion in this document are for SystemC processes that are sensitive only to a single rising clock edge. (Appendix O discusses support for multiple clock domains.)

This document distinguishes between the following:

1. The *conceptual model* for the scheduling rules.
2. The simulation behavior of a model using the rules in C++ or SystemC.
3. The synthesis of a C++/SystemC model using the rules in an HLS tool such as Catapult.

The goal is to align each of these three cases as closely as possible, so that the user has easy-to-understand rules, while simulation and synthesis work without surprises. However, as we will see, there are practical considerations which may in certain cases cause small deviations from the conceptual model in either simulation or HLS.

A simple real-world example that illustrates the motivation for the scheduling rules is provided in Appendix A of this document.

The examples referred to in this document are available here:

<https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/tree/master>

The most up-to-date version of this document is available here:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/catapult_user_view_scheduling_rules.pdf

A summary of the scheduling rules is available here:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/brief_scheduling_rules.txt

A formal architecture guide is available here:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/formal_architecture_guide.pdf

A white paper describing the motivation for the methodology is available here:

<https://hls.academy/topics/systemc-modeling/verifying-hardware-at-the-source-why-a-user-visible-hls-scheduling-contract-matters-in-an-age-of-ai-generated-design/>

An Analogy from RTL Synthesis

To better understand the specific purpose of this document, let's consider how RTL synthesis works. Say you have a sequential block that you are modeling in Verilog RTL, and it has an output port coded like:

```
Out1 <= #some_delay new_val;
```

In Verilog simulation, if `some_delay` is less than the clock period of the block, then it will probably not affect the overall cycle-level behavior of the system during simulation. However, if `some_delay` is more than the clock period, it probably will.

During RTL synthesis, all RTL synthesis tools will ignore all delays in the input model, in this case even if `some_delay` is greater than the clock period. Some RTL synthesis tools might give a warning or error for code like the above such as "Simulation and synthesis results are likely to mismatch because the delay in

model is greater than clock period.” Some RTL synthesis tools might outright reject a model containing such delays.

One might argue that RTL synthesis tools should always match the Verilog simulation behavior of the input model. But the overall approach works well because RTL is a good and simple *conceptual model* that users and tool vendors can align around. The slight differences between the *simulation model* and the *conceptual model* used by RTL synthesis tools can be easily managed.

We’ll return to this example later in this document.

Next, let’s clarify some terminology related to signal IO. Say you have a Verilog model like:

```
forever begin
  @(posedge clk); // wait for 1st rising clock edge
  output1 <= input1 + 10;
  @(posedge clk); // wait for 2nd rising clock edge
end
```

In this example, input1 is sampled when the process wakes on the first @(posedge clk) and evaluates the RHS. For the purposes of the scheduling rules, we anchor that Read(input1,.) to the most recent synchronization point. The resulting Write(output1,.) is anchored to the next synchronization point (the cycle in which the written value is treated as stable for others to observe).

Cycle-level anchoring convention (used throughout this document). For cycle-level reasoning, each signal IO is assigned an anchor cycle relative to surrounding synchronization calls (e.g., wait() / SyncChannel / @(posedge clk)):

- Read(sig, val) is anchored to the closest preceding synchronization call, i.e. pred_S(Read).
- Write(sig, val) is anchored to the closest succeeding synchronization call, i.e. succ_S(Write).

This convention is only for cycle-level reasoning and does not depend on simulator update regions or delta-cycle ordering.

Catapult HLS Status Concerning Rules in this Document

A separate document provides a list of clarifications related to support within Catapult HLS for the rules described in this document. The items in the list are named *Cat#*, so that each item has a unique number. This document annotates certain rules with *Cat#* to refer to the items in the separate document. These *Cat#* annotations are not part of the formal proof itself.

Terms Used in this Document

Latency-Insensitive: In digital hardware design, *latency-insensitive* refers to a system that operates correctly despite variable communication delays between components. This is achieved by using mechanisms to decouple computation from communication timing. Such designs improve scalability and reliability in complex systems with unpredictable or variable latencies. ARM’s AXI4 and APB are examples of latency-insensitive protocols. ARM’s AMBA 5 CHI credit-based NOC protocol is also an example of a latency-insensitive protocol.

Process: In Verilog, a process is an *always block* and its equivalent constructs. In SystemC, a process is an instance of an SC_THREAD or SC_METHOD.

Message-passing Interface: A *message-passing interface* reliably delivers messages (or transactions) from one process to another. This document uses this term to denote the type of communication found in Kahn Process Networks. See https://en.wikipedia.org/wiki/Kahn_process_networks Message-passing read interfaces are always separate from message-passing write interfaces – there are no bidirectional message-passing interfaces. In this document, message-passing channels are assumed to be point-to-point: for each channel *c*, exactly one producer performs all Push_*c* operations and exactly one consumer performs all Pop_*c* operations.

Synchronization Interface: A *synchronization interface* synchronizes one process with another and/or with a global clock. For an example of a synchronization interface, see [https://en.wikipedia.org/wiki/Barrier_\(computer_science\)](https://en.wikipedia.org/wiki/Barrier_(computer_science))

Signal IO: In digital HW design, signals are the fundamental communication mechanism. Signals enable communication between two HW blocks/processes, but communication with signals in real HW always incurs at least some delay because communication cannot be faster than the speed of light. In HDLs and in SystemC, signal delays are modeled with the *delayed update* semantics.

blocking / non-blocking: In this document, a blocking message-passing operation is modeled as issuing a persistent request that remains asserted (and with stable payload, if applicable) until the operation commits (the message is sent or received). A non-blocking message-passing operation returns immediately with a status indicating whether the operation committed in that cycle.

One-way / two-way handshake protocols: A one-way handshake protocol only has one signal between a sender and receiver to synchronize communication. A two-way handshake protocol has two signals (one in each direction) between a sender and receiver to synchronize communication.

shall: This term indicates that a compliant tool or flow is required to follow the indicated rule.

may: This term indicates that a compliant tool or flow is allowed to follow the indicated rule but is not required to do so.

Classes of Operations Involved in Scheduling Rules

There are three classes of operations involved in the scheduling rules:

1. Calls to message-passing interfaces (which are all ac_channel methods, all SystemC Push/PushNB/Pop/PopNB calls except calls to SyncChannel)
2. Calls to synchronization interfaces (which are calls to ac_sync and Matchlib SyncChannel, also ac_wait and SystemC wait)
3. Signal IO (which are SystemC signal reads and writes, also C++ model *direct inputs*)

All the operations above are referred to as *IO operations*.

Basic Conceptual Model

The basic conceptual model encompasses processes that have no loop pipelining but may have preserved loops. If a process has a preserved loop, then the user may place a wait statement in the loop.

Wait statements explicitly placed in the model are called explicit wait statements in this document and are classified as calls to a synchronization interface. However, for a loop to be treated as an HLS-pipelined loop under the formal guarantees in this document, the loop body shall not contain any wait/ac_wait statement or other explicit synchronization call used as an in-loop wait boundary. Such synchronization boundaries must be placed outside the pipelined loop, or the loop must be treated as non-pipelined/preserved or outside the theorem instance.

The basic conceptual model rules are:

1. Synchronization interface calls within a process always remain in the source code order.
2. Signal read operations occur at the closest preceding call to a synchronization interface. (Cat1)
3. Signal write operations occur at the closest succeeding call to a synchronization interface.
4. Message-passing operations may be shifted in time, issued in the same cycle, or overlapped by simulation or synthesis, subject to the following constraints:
 - All message-passing operations before a call to a synchronization interface shall be issued before or in the same cycle in which the synchronization interface commits. (Here “the synchronization interface commits” refers to the commit of that synchronization call in the calling process’s trace. For blocking message-passing operations (Push/Pop), the process cannot complete the synchronization call until any earlier issued blocking message requests in that interval have committed.)
 - All message-passing operations after a call to a synchronization interface shall be issued after the cycle in which the synchronization interface call commits.
 - Message-passing operations on the same message-passing interface shall be issued in source order; they shall not be reversed. This same-interface order requirement applies both within a single source interval and across overlapped iterations of a pipelined loop.
 - Two message-passing operations on separate interfaces which appear in sequence in the model are source-ordered. They may be issued in the same source sequence, and the later operation may in some permitted modes issue before the earlier operation commits, including in the same clock cycle where allowed by the issue policy. They shall not be issued in the reverse sequence. (Cat2)

Here “parallel” issue means that requests may overlap or be asserted in the same cycle while the calls remain ordered by their dynamic source order. Preserving that order prevents HLS from introducing a deadlock solely by reversing source-ordered message calls. It does not by itself guarantee deadlock preservation, because loop overlap, request timing, and finite channel capacity can still affect deadlock behavior. The formal appendices address those additional cases.

Pipelined Loops

When a loop is pipelined in HLS, the body of the loop is split into pipeline stages. HLS may start the next iteration of the loop before the current iteration has completed, thus increasing throughput as compared to a non-pipelined implementation.

A loop selected for HLS pipelining shall not contain wait/ac_wait statements, SyncChannel/ac_sync calls, or any other explicit synchronization interface call inside the loop body. The pipelined-loop rules in this document therefore apply to wait-free pipelined loop bodies, with ordinary synchronization boundaries only outside the pipelined loop or around the pipelined region. If a design requires a synchronization call

inside the loop body, the flow shall either reject that loop for pipelining under this contract, require the source to be rewritten so that the synchronization boundary is outside the pipelined loop, or exclude that pipelined-loop instance from the Appendices G-K formal guarantees unless a separate specialized theorem/certificate is supplied.

To guarantee equivalent behavior between the pre-HLS and post-HLS systems, pipelined loops must be implemented with automatic flush to prevent deadlocks. See Appendix B for a formal presentation of pipelined loops.

When HLS pipelines a loop, multiple iterations of the loop are overlapped and execute at the same time. During loop pipelining, for all IO operations, HLS shall ensure that an access to a specific message-passing interface, signal, or synchronization interface shall not be reordered relative to same-interface accesses from other iterations.

During pipelining, signal reads and writes are physically embedded in specific pipeline stages, but the loop body itself contains no explicit `wait/ac_wait` or other in-body synchronization call under the contract above. For scheduling and source/RTL comparison, an ordinary signal Read or Write is therefore still associated with the surrounding real synchronization calls, not with an internal pipeline stage. Considering the entire set of signals read or written by the process, if HLS pipelining would cause the order of signal IO to differ between the pre-HLS and post-HLS models, or if any two signal IO operations that occur on the same clock edge in the pre-HLS model would not occur on the same clock edge in the post-HLS model, then the HLS tool shall detect such a situation and require the user to explicitly indicate in the model source (e.g., via a pragma) that HLS pipelining can still be used. (Cat7) In other words, if a process communicates with other processes using only latency-insensitive protocols, then HLS pipelining by default shall not break any of the protocols. Appendix G gives the formal signal-anchoring and matching rules.

Synchronization boundary for pipelined loops

When a pipelined region ends at an explicit synchronization call, work already accepted before the boundary may continue to drain. However, the implementation and the back-annotated source model must not expose capacity for work belonging to the next synchronization interval until the synchronization call completes. This prevents upstream transfers from entering the next interval prematurely. See Appendix B for the formal rule and modeling treatment.

Direct Inputs

Ordinary SystemC signal reads occur at the closest preceding synchronization interface call (e.g., `wait` statement) in both the pre-HLS and post-HLS models. (Cat1). If the HLS tool adds states to the process, or if it pipelines the process, then this implies that the HLS tool must add registers for each such read operation such that the read occurs where specified in the pre-HLS model, and the value is stored until the point where it is consumed in the post-HLS model. The area cost of such registers may be high if there are a lot of signals, and in some cases, it may be unneeded area since the value of the signals may not actually need to be stored internally to the process.

The simplest case is if such external signals are held stable after the process comes out of reset. In this case, HLS may physically sample the signal values as late as possible, with no need for register storage, while for scheduling and source/RTL comparison the Read is still treated as occurring at its prescribed

synchronization point. This case is handled with the following pragma on SystemC signals and ports (Cat6):

```
#pragma hls_direct_input
```

To ensure that there are no pre-HLS versus post-HLS simulation mismatches, the environment that drives the signal shall hold it stable within each relevant reset epoch: after all receiving processes that use this signal with this pragma have come out of a reset boundary, the signal value shall remain stable until the next reset boundary that affects those receiving processes, unless the signal is instead governed by an explicit `hls_direct_input_sync` update contract as described below. A *dynamic reset* starts a new stability epoch for the affected receiving logic and its associated direct-input signals. Thus it remains allowable to have *dynamic resets*, i.e. activation of block resets and associated resetting of direct input signals as part of the normal operation of the HW. The deterministic serial-issue deadlock route in Appendix K has additional requirements for dynamic reset epochs; see K.ResetEpoch and Appendix M.6.

A related but somewhat more complex case is where input signals to the HW block may only be changed at “agreed upon” times, typically while the portion of the HW block that relies on them is temporarily idle. For example, a block may process 2D images. At the start of each new image, it may be desirable for the TB or external environment to update the control signals for how the block will process the next image. Typically HLS designs are pipelined, and the HW pipeline for the current iteration must be fully *ramped down* before the input signals can be updated to affect the next iteration. In the pipelined-loop case, this relies on the general synchronization-boundary rule for pipelined loops: while the process is pending at the designated sync, any back-annotated message-passing capacity whose accepted data would belong to the next post-sync interval is temporarily unavailable to upstream producers until the sync commits. In this case we can use the SystemC *SyncChannel* or C++ *ac_sync* primitives to precisely synchronize the DUT with the TB/environment to enable the input signals to be updated at the correct time. The `#pragma hls_direct_input_sync` directive shown below associates the sync operation with the direct inputs that it controls. The precise synchronization scheme shown here ensures that there are no pre-HLS versus post-HLS simulation mismatches even though we are using direct inputs and also changing their values while the design is executing.

When a signal or port covered by `#pragma hls_direct_input` is also included in the scope of a later `#pragma hls_direct_input_sync` directive, the sync directive selects the effective update contract for that controlled signal set. Such signals remain direct inputs for purposes of late physical sampling, but they are not required to remain stable for the entire reset epoch; instead, they must satisfy the sync-controlled discipline described here: updates may occur only at the designated synchronization handshake, and the values must remain stable between permitted sync-controlled updates and reset boundaries.

```
// This is example 61 in Catapult Matchlib examples
sc_in<bool> SC_NAMED(clk);
sc_in<bool> SC_NAMED(rst_bar);

Connections::Out<uint32_t> SC_NAMED(out1);
Connections::In<uint32_t> sample_in[num_samples];
Connections::SyncIn SC_NAMED(sync_in);
#pragma hls_direct_input
sc_in<uint32_t> direct_inputs[num_direct_inputs];
```

```

void main() {
    out1.Reset();
    sync_in.Reset();

#pragma hls_unroll yes
    for (int i=0; i < num_samples; i++) {
        sample_in[i].Reset();
    }

    wait(); // reset state

    while (1) {
#pragma hls_direct_input_sync all
        sync_in.sync_in();

#pragma hls_pipeline_init_interval 1
#pragma hls_stall_mode flush
        for (uint32_t x=0; x < direct_inputs[0]; x++) {
            for (uint32_t y=0; y < direct_inputs[1]; y++) {
                uint32_t sum = 0;
#pragma hls_unroll yes
                for (uint32_t s=0; s < num_samples; s++) {
                    sum += sample_in[s].Pop() * direct_inputs[2 + s];
                }
                ac_int<32, false> ac_sum = sum;
                ac_int<32, false> sqrt = 0;
                ac_math::ac_sqrt(ac_sum, sqrt); // internal loop unrolled in cat .tcl
file
                if (sqrt > direct_inputs[7])
                    out1.Push(sqrt);
            }
        }
    }
};

```

From the perspective of the testbench or the environment, the updating of the signals controlled by the `hls_direct_input_sync` directive shall only occur at a precise point. The TB shall first wait for the `rdy` signal for the sync to be asserted by the DUT, and then the TB shall update all the input signals it wishes to change while simultaneously driving the `sync_vld` signal high for one cycle.

It is important to note that the only safe operation to use to synchronize the updating of direct inputs is the sync operation as shown above. Other operations such as Push/Pop or `ac_channel` operations should not be used for this. For purposes of source/RTL comparison, the logical Read(sig,val) remains associated with its synchronization point even if the implementation samples the stable physical signal later. Such additional internal sampling must not create another logical Read or change the value attributed to the original Read. Appendix G formalizes this trace treatment.

In the example above the DUT block that is being synthesized determines when to call sync, and thus it determines when the direct inputs will be updated. In some cases, it may be necessary for the environment around the DUT to determine when the direct inputs should be updated. In this case the same approach as shown above should be used; however a separate input from the environment to the DUT (either using a signal or a message-passing interface) should request that the DUT call sync as soon as feasible. This will ensure that the DUT has properly ramped down its pipeline and is ready to receive the newly updated direct inputs as per the overall synchronization scheme described above.

Additional Options for Scheduling Message-passing Interfaces

The following option may be added during HLS (Cat2):

```
STRICT_IO_SCHEDULING=relaxed
```

When this is specified, the HLS tool is allowed to reorder message-passing interface calls freely on distinct interfaces (still not moving calls across synchronization interface calls). This relaxed mode may violate the default no-reverse discipline and therefore may introduce deadlocks that are not present under the default rules. It is recommended that this relaxed option only be used when the user wants to see what order the HLS tool prefers to schedule message-passing interface calls (e.g., to achieve best QOR).

Once the user knows the preferred order, it is recommended that the user modify the pre-HLS source code to reflect the preferred order, and then return to the use of the default scheduling modes. This methodology ensures that HLS cannot introduce new deadlocks *via reversal of source order on distinct message interfaces* as described earlier. Deadlock behavior may still depend on bounded-capacity effects and on overlapped execution (e.g., loop pipelining); those cases are addressed separately by the pipelined-loop automatic-flush rules and the deadlock-preservation conditions in Appendix K. The formal equivalence and deadlock-preservation results in Appendices G–K assume STRICT_IO_SCHEDULING is not set to relaxed.

Scheduling of Array Accesses

Arrays may appear in HLS models, and they may be preserved through synthesis and mapped to RAMs. Pointers may also appear in HLS models, and pointer dereferences are resolved to array accesses during HLS.

For scheduling and verification, the physical placement of a RAM inside or outside the synthesized block does not by itself change the rules. What matters is the declared logical behavior of the mapped memory and the logical interface at which the pre-HLS and post-HLS designs are compared. The rules below apply to every array access mapped to memory. The distinction drawn after them concerns only whether the memory also carries values from one process to another; the shared rules, including the qualification requirements for transformed and for blocking memory realizations, apply in both cases.

Rules applying to all mapped memory behavior.

- For each mapped memory instance within the scheduling/verification scope, the user model or flow shall declare an *array-access mapping layer*. That declaration identifies the untransformed logical memory behavior, the logical boundary at which source and implementation behavior are compared, and how that boundary is represented using signal IO, message passing, synchronization, or a qualified logical memory interface.
- The user may authorize HLS to cache, merge, split, or reorder array accesses (for example, to improve QOR). The resulting physical RAM command/response sequence may differ below the declared logical boundary, but the transformed implementation shall preserve the declared logical behavior and the ordinary scheduling and functional requirements at that boundary. In particular, a memory mapping or transformation may not create, drop, duplicate, relabel, or illegally reorder an interaction that is supposed to be visible at the selected logical boundary. Preservation at the

declared boundary is a per-instance obligation that the qualification flow must discharge, not an assumption.

- If the selected logical boundary exposes a message channel or an anchored signal, that interface remains governed by the same rules as any other message channel or signal in this document. An otherwise visible transfer, signal Read, or signal Write cannot be hidden merely because it arose from an array or memory mapping.
- In all cases, array accesses remain constrained by explicit synchronization calls present in the process source model. All array reads or writes before a synchronization call shall commit before or in the same cycle in which that synchronization call commits, and all array reads or writes after it shall commit in a later cycle. Any cache/merge/split/reorder transformation must preserve this commit-time fence. In particular, one merged storage commit may not represent source accesses lying on opposite sides of the same synchronization boundary; such a merge is forbidden regardless of where the logical comparison boundary is placed.
- If the physical memory system can persistently backpressure or stall a process, the logical model used for the deadlock-preservation guarantee shall account for the relevant blocking behavior, including the capacity/backpressure, request, enablement, and release-support dependencies and the wait-for relationships that can keep the process blocked. Hidden physical structure may remain below the logical boundary only when the qualification flow establishes that it cannot introduce, remove, or redirect those blocking causes or the corresponding wait-for facts. This requirement applies whether or not the memory carries cross-process value flow. If transaction-changing memory behavior can persistently backpressure a process and neither such a qualification nor a suitable logical boundary above the transformation is available, that transformation is outside the deadlock-preservation guarantee for that memory instance.

Memory used for communication between processes.

1. No cross-process memory-carried value dependence

A RAM mapping by itself does not make raw physical memory transactions part of the externally compared behavior. If the memory does not carry a value from one process into another process's computation or later behavior, those raw transactions need not be exposed solely because the storage was mapped to RAM. The general mapping, synchronization, visible-interface, and blocking rules above still apply.

2. Cross-process memory-carried value dependence

If one process's memory write can affect another process's control flow, payload or value expressions, or the identity or order of its later operations, every such cross-process dependency shall remain represented in the verification model. The raw physical RAM transactions do not have to be exposed if the dependency, including the communicated value, is represented at a supported logical interface. Synchronization alone may order the accesses, but does not by itself represent the value communicated through memory. If caching, merging, splitting, or reordering changes the physical transaction stream, the standard covered route is to compare behavior at a qualified logical boundary above those transformations whose externally relevant behavior is invariant under the permitted transformation set. A separately supplied proof that directly tracks shared-memory state is an alternative. Without one of those routes, the formal source/RTL safety and associated deadlock results do not cover that shared-memory communication.

When a loop is pipelined, multiple iterations of the loop are overlapped and execute at the same time. During loop pipelining, an access to a memory interface (or array) may be moved over or in parallel with another access to the same memory if the HLS tool can prove that the reordering is conflict-free. For a mapped memory within the scheduling/verification scope, the declared mapping shall explicitly

authorize such reordering, and the reordered implementation shall still preserve the logical memory behavior, synchronization fences, visible-interface behavior, and blocking requirements stated above.

Formal treatment. The Normative Formal Core states the exact mapped-memory faithfulness (Core.MemFaithful), synchronization-order (Core.MemSyncOrder), shared-memory lowering (J.Mem), and qualified memory-profile (Core.MemProfile) conditions. Appendix J uses these conditions in the source/RTL safety proof, Appendix K gives the additional blocking and deadlock conditions, and Appendix M.7a is the definitive per-instance qualification and evidence ledger. The Core also describes the alternative direct memory-state-extension route.

Definitions: Issue vs. Commit

Informally, we can define **commit** and **issue** as follows:

Commit (message channel) — informal operational reading

For the RTL_B implementation reading at a mapped ready/valid boundary, a message commit is observed in the cycle in which an external clocked observer sees the transfer complete:

- The mapped boundary commit is observed when the observer sees both valid and ready high in that cycle.
- The committed payload is the data value seen in that same cycle. This is an implementation-facing observation rule; for the G–K theorem stack, the formal abstract commit and Commit(q)/CommitCycle(q) definitions are in Definition Core.8.

Definition: Issue (message channel) — message request assertion

Issue is the cycle in which the calling process begins requesting the transfer at the selected message-channel boundary. For a Push, its boundary manifestation is assertion of the process-owned vld request; for a Pop, it is assertion of the process-owned rdy request. A raw physical ready/valid level that instead represents channel-side storage, complementary availability, or declared coupler logic is not by itself BoundaryReq or Issue. Appendix C and Core.10/Core.IC define the precise process-owned request and the blocking back-to-back ownership rule.

For RAM reads/writes as presented above, commit denotes the cycle in which the declared mapping-layer storage semantics recognizes the access as committed. The “Scheduling of Array Accesses” constraints are therefore written in commit-time. A fixed-latency memory is one simple implementation for which the scheduler can choose an issue cycle that satisfies the synchronization fence, but fixed latency is not a general requirement; any variable-latency or transformed mapping must still preserve the same commit-time fence. The Normative Formal Core gives the exact audit condition (Core.MemSyncOrder).

Sequential processes use registered cycle semantics: a result returned by a message-passing call at a cycle boundary cannot be used to generate a dependent request in that same cycle. Such a dependent request may first issue in a later cycle. Appendix C and the Normative Formal Core (Definition Core.RegSeq) give the precise rule, including its treatment of failed non-blocking calls and independently active pipeline iterations.

Additional States Added by HLS Synthesis

By default, HLS synthesis tools may add additional states to processes (e.g. add latency to enable resource sharing), which may introduce latency differences in the interface behavior between the pre-HLS and post-HLS models. These additional states are never included in the set of *synchronization interface calls* as described above.

When the directive `IMPLICIT_FSM=true` is set on a process, the HLS synthesis tool shall ensure that the cycle-level behavior of the interfaces of the pre-HLS and post-HLS models shall be identical. With this option, the internal state machines of the pre-HLS and post-HLS models will be the same.

When the directive `IO_MODE=FIXED` is set on a process, the HLS synthesis tool shall ensure that the cycle-level behavior of the interfaces of the pre-HLS and post-HLS models shall be identical. With this option, it is still possible that the state machine internal to the process is different between the pre-HLS and post-HLS models (e.g. the post-HLS model may choose to use a pipelined multiplier where the pre-HLS model did not.)

Avoiding Pre-HLS and Post-HLS Simulation Mismatches

The scheduling rules described in this document are designed to be easy to understand, while providing good QOR via HLS and generally avoiding any mismatches between the pre-HLS and post-HLS simulation behaviors.

Non-blocking message-passing operations (PushNB/PopNB in SystemC, C++ `ac_channel nb_read/nb_write`) are a potential source of mismatches between pre-HLS and post-HLS simulations since their behavior is inherently dependent on the latency within the model, which often changes during HLS. Because of this, non-blocking message-passing interfaces should only be used when no alternative approach is possible. For example, non-blocking message-passing interfaces are required to model time-based arbitration of multiple message streams which access a shared resource. A full discussion of recommended guidelines on the use and verification of non-blocking message-passing interfaces is provided in Appendix L. Note that the scheduling rules described previously in this document fully specify how HLS tools are required to schedule such operations.

Unidirectional message-passing between two processes should not be relied on to achieve synchronization between the two processes, since in general the message latency and storage capacity between the processes may be variable. Also, depending on buffering/pipelining in the realization, the time between a producer's committed Push and the consumer's committed Pop (and the transient in-flight occupancy) may vary. Two unidirectional message-passing channels in opposite directions can be relied upon for synchronization between two processes. (An example of this is the AXI4 `ar` and `r`, and `aw` and `b`, channels). Note that such synchronization is weaker than explicit synchronization like `SyncChannel` or signal IO that uses a two-way handshake.

SystemC signal IO operations are a potential source of pre-HLS versus post-HLS simulation mismatches since timing behaviors may change between the two models. The following section provides guidance and rules to help avoid potential mismatches due to signal IO.

When signal IO operations are synchronized with a wait statement, there generally should be a proper two-way handshake associated with the wait statement so that the signal IO is latency insensitive. (Note

that this statement does not apply to the signal synchronization approaches described in the Direct Inputs section.)

For example, here's a simple two-way handshake protocol when writing the signal `out_dat`:

```
out_dat = value;
out_vld = 1;
do {
    wait();
} while (out_rdy != 1);
out_vld = 0;
```

And here's a two-way handshake example when reading signal `in_dat`:

```
in_rdy = 1;
do {
    wait();
} while (in_vld != 1);
value = in_dat;
in_rdy = 0;
```

Some signal-level protocols have different two-way handshaking approaches (e.g. ARM APB), but they are still latency insensitive.

If signal IO operations are associated with a wait statement and that wait statement does not have a proper two-way handshake, then the signal IO is likely to be latency-sensitive and may result in pre-HLS versus post-HLS simulation mismatches. In some systems a one-way signal handshake is sufficient for reliable system operation. See Appendix E for further discussion.

The scheduling rules state that signal IO operations occur at either SystemC wait statements or SyncChannel calls (`sync_in` and `sync_out`). In the remainder of this section, we will use *wait* statements to refer to both.

RULE 1: It is always best coding style to group signal write operations just before their corresponding wait statement, and signal read operations just after their corresponding wait statement. (Cat3). An example is below:

```
sc_in<int> i1;
sc_in<bool> go;
sc_out<int> o1;
void my_thread() {
    int new_val=0;
    while (1) {
        o1.write(new_val);
        do {
            wait();
        } while (!go.read());
        new_val = i1.read();
        new_val = some_function(new_val); // function has no internal IO
    }
}
```

By placing the signal IO operations as close as possible to their corresponding wait statement, the HW intent is very clear. And there is no benefit either in terms of simulation performance or HLS QOR if they are placed further away from their corresponding wait statement.

Let's look at another similar example, which now also uses a Matchlib Connections blocking Pop operation:

```

sc_in<int> i1;
sc_in<bool> go;
sc_out<int> o1;
Connections::In<int> pop1;
void my_thread() {
    int new_val=0;
    while (1) {
        o1.write(new_val);
        do {
            wait();
        } while (!go.read());
        int pop_val = pop1.Pop();
        new_val = i1.read();
        new_val = some_function(new_val + pop_val); // function has no internal
IO
    }
}

```

Under the anchoring rules, `i1.read()` remains associated with the same synchronization point regardless of the intervening Pop. However, in raw pre-HLS simulation, a blocking Pop may stall for one or more cycles, so `i1` can change before the `i1.read()` executes—creating behavior that differs from the anchored interpretation (and potentially from post-HLS scheduling). The fix is simply to place `i1.read()` immediately after its intended `wait()` anchor; this removes the mismatch risk without affecting QOR or simulation performance.

For a proof-backed flow, RULE 1 conformance is a source well-formedness obligation, not merely a recommended diagnostic. If a blocking message-passing operation separates a signal read or write operation from its corresponding synchronization interface call, then the flow shall either reject the model for purposes of applying the formal guarantees, require the source to be rewritten so the signal IO is adjacent to its intended anchor, or discharge an explicit source-level well-formedness argument showing that the signal IO has a unique unconditional real synchronization anchor on every dynamic path and cannot observe a value inconsistent with that anchor. HLS tools may implement this requirement using errors, warnings, lint checks, or other diagnostics, but an unresolved warning is not sufficient to claim the RULE 1 / RULE 2-based guarantees.

Another scenario in which RULE 1 applies is shown below:

```

sc_in<bool> go;
sc_out<int> o1;
void my_thread() {
    while (1) {
        wait(); // WAIT 1
        o1.write(some_value);
        if (go.read()) {
            some_value = some_function();
        }
    }
}

```

```

    }
    else {
        wait();           // WAIT 2
    }
    some_other_function();
    wait();               // WAIT 3
}
}

```

The signal read of `go` is clearly and uniquely associated with WAIT 1. However, the signal write of `o1` associates with WAIT 2 if `go` is false and WAIT 3 if it is not. This is a violation of RULE 1 and should be flagged as an error during HLS. The fix, as before, is to move the signal IO operation as close as possible to its intended wait statement so that the association is unconditional.

Next, let's consider *rolled* (or *preserved*) loops that perform signal IO within the loop body. Consider the following example:

```

sc_in<int> i1;
sc_out<int> o1;
void my_thread() {
    wait(); // reset state
    while (1) {
        wait(); // start of while loop
        #pragma hls_unroll no
        for (int i=0; i < 10; i++) {
            o1.write(i1.read() * i);
        }
    }
}

```

Note that the `i1.read()` operation is located inside the `for` loop, so presumably the user's intent is that it should be read as the loop iterates. *If that is not the user's intent, they simply should move the `i1.read()` operation before the loop start.*

In the post-HLS simulation, each iteration of the loop will consume at least one clock cycle, and a new value for `i1` will be read (and a new value for `o1` written) on each iteration. Again, this is the user's intent as per the code. In the pre-HLS simulation, the loop body executes in zero time, so under the final-write-at-anchor rule only the final write to `o1` supplies the anchored value. The solution to avoid this mismatch is to manually place a `wait()` statement within the `for` loop body so that the signal IO synchronization is explicit in the pre-HLS simulation.

RULE 2: If you have signal IO operations within *rolled* (or *preserved*) loops, manually place a `wait` statement within the body of the loop to avoid pre-HLS versus post-HLS simulation mismatches, and while doing so also follow **RULE 1**.

Scope of RULE 2 for pipelining. RULE 2 applies only to rolled or preserved loops that are not selected for pipelining under this contract. A loop body selected for pipelining shall not contain `wait/ac_wait` statements, `SyncChannel/ac_sync` calls, or any other explicit synchronization interface call. Therefore, if ordinary per-iteration signal IO inside a loop requires a per-iteration wait anchor, that loop is outside the pipelined-loop contract unless the source is rewritten: move the signal IO outside the pipelined loop, model the interaction through message-passing, use `hls_direct_input / hls_direct_input_sync` under their stated contracts, or forgo pipelining for that loop. Without an in-body synchronization anchor, repeated per-iteration signal reads or writes inside one surrounding source interval do not acquire

separate logical synchronization points merely because HLS later pipelines the loop; they remain associated with the surrounding synchronization calls. Appendix G formalizes the corresponding signal-event treatment.

For a proof-backed flow, a rolled loop with signal IO operations but no wait statement in the loop body shall be treated as a source well-formedness failure unless an explicit source-level argument shows that each observable signal Read/Write still has a unique real synchronization anchor on every dynamic path. HLS tools may implement this requirement using errors, warnings, lint checks, or other diagnostics. However, if a diagnostic is emitted as a warning rather than an error, the warning must be resolved, waived with an explicit well-formedness justification, or treated as excluding that model from the formal guarantees. (Cat4)

If a pre-HLS model has established RULE 1 and RULE 2 conformance, then all signal IO in the post-HLS model shall occur only at clock cycles that correspond to explicit wait statements or explicit synchronization statements. For direct-input signals/ports (e.g., `#pragma hls_direct_input` and `#pragma hls_direct_input_sync`), this requirement refers to the logical anchor cycle of the Read/Write: the Read or Write is still treated as occurring at its synchronization point even if HLS physically samples a stable signal later. Such later physical samples are internal implementation activity; they do not create additional observable Read/Write events. User designs that do not adhere to both RULE 1 and RULE 2 are ill-formed and are outside the formal guarantees unless the nonconformance has been eliminated or explicitly discharged by a valid source-level well-formedness argument.

Returning to the Analogy from RTL Synthesis

At the beginning of this document, we presented the example of a Verilog sequential block with an output coded like:

```
Out1 <= #some_delay new_val;
```

Recall that in Verilog simulation, if `some_delay` is less than the clock period of the block, then it will probably not affect the overall cycle-level behavior of the system during simulation. However, if `some_delay` is more than the clock period, it probably will.

During RTL synthesis, all RTL synthesis tools will ignore all delays in the input model, in this case even if `some_delay` is greater than the clock period. Some RTL synthesis tools might give a warning for the code above like “Simulation and synthesis results are likely to mismatch because delay in model is greater than clock period.”

HLS tools or associated lint/qualification flows that adhere closely to the conceptual model presented in this document should provide a concrete mechanism for detecting or discharging violations of RULE 1 and RULE 2 as described in the section above. This is analogous to the error or warning that the RTL synthesis tool would provide in the example directly above.

However, the formal guarantees in this document do not depend on any particular diagnostic mechanism. If an HLS synthesis tool does not report a RULE 1 / RULE 2 issue, the design is not thereby made well-formed. The proof-backed flow must still establish that the source model satisfies the RULE 1 / RULE 2 well-formedness obligation before applying the trace-compatibility guarantees.

Summary

At the beginning of this document, we said that the intent was to present “no surprises” to a DV engineer who wants a common verification methodology for the pre-HLS and post-HLS models. The key aspects of the document which support this are:

- Three groups of IO operations are defined (message-passing, signal IO, and synchronization calls) and each is treated uniformly. These IO operations are easy for verification engineers to understand because they are already using them in their testbenches.
- The document specifically avoids complex constructs such as *protocol regions* used in some HLS tools.
- The document preserves the ability of the pre-HLS SystemC model to be *throughput accurate* by using a library such as Matchlib.
- Synchronization calls can affect the scheduling (anchoring) of signal IO operations, and synchronization calls can affect the scheduling of message-passing calls. In the conceptual anchoring semantics of this document, message-passing calls do not affect the anchor cycle of signal IO operations (and signal IO does not affect the legality/order of committed message transfers) except through explicit synchronization. Practically, in raw pre-HLS SystemC simulation a blocking message operation may introduce additional cycles of waiting, so signal IO statements that are lexically separated from their intended anchor can observe different values; RULE 1 addresses this by recommending placement of signal IO adjacent to its anchor.
- HLS cannot by default reverse the order of message-passing calls, so it cannot introduce new deadlocks *via call-order reversal*. For pipelined loops (overlapped iterations) and other transformations that change request/commit timing under bounded capacities, deadlock preservation depends on the additional automatic-flush, blocking, and progress conditions described later in the document, not on the no-reverse rule alone.
- HLS pipelining is largely a don't care from the perspective of the verification engineer when the design and testbench are insensitive to latency changes, pipelines flush automatically, and any mapped memory transformations preserve the declared logical memory behavior and the synchronization and blocking rules. When the design or testbench is latency-sensitive, or when memory transactions are rearranged, the affected behavior must instead be handled through the explicit latency-sensitive or mapped-memory contracts described in this document. Appendix G gives the formal source/RTL trace-compatibility guarantee and its scope rather than an unrestricted bidirectional equivalence theorem.
- For sequential processes, signal IO operations in the post-HLS model are logically associated with synchronization points (e.g., wait statements) that are explicit in the pre-HLS source. For direct-input signals/ports, the implementation may physically sample a stable signal later, but that later sample is internal and does not move or duplicate the logical Read/Write. For combinational processes, observable signal behavior is instead taken only after the combinational logic has reached its stable fixed point for the cycle (R2C). Appendix G gives the precise formal treatment.

The principal formal safety and deadlock-preservation results, their scope limitations, and the evidence required to apply them are summarized in Appendix M. Appendix F provides user-oriented modeling guidelines.

Appendix A – Factory Analogy

The scheduling rules described in this document apply to pre-HLS and post-HLS HW models. Although the rules may seem abstract and perhaps even arbitrary, they are shaped by the need to model systems in the real world that are required to have predictable behavior.

To understand the motivation behind the rules, it might be helpful to draw a simple real-world analogy and its correspondence to the rules described earlier.

Consider a factory that produces various types of wooden furniture:

The factory consists of people (processes) stationed at workbenches with various tools.

Each person is given written instructions about the specific tasks they are to perform (C++ code within a process).

People are instructed to send or receive objects (messages) to or from other people in the factory.

Sending or receiving objects may be blocking or non-blocking from the perspective of a person.

There is a clock with a second hand on the factory wall that everyone can see. (HW clock).

People have colored flags they can raise or lower to communicate with other people (signals).

If someone raises or lowers a flag, this is only seen by others the next time the clock second hand is at the top of the clock. (propagation delay of signals between sequential processes)

A person can choose to pipeline the tasks that they were assigned by hiring subordinates (pipeline stages) and having each one do a subtask. In general, this will improve the throughput for the tasks that the person was assigned.

It is possible for two people to explicitly synchronize their work by communicating via a synchronization protocol such as a barrier (synchronization).

It is possible to restart the work of some or all the people by raising a reset flag (HW resets).

Assume:

1. That the time that people take to complete their various tasks is in general variable, and similarly that the time that objects (messages) take to pass between people is in general variable.
2. That each person in general wants to complete their tasks as quickly and efficiently as possible.
3. That the factory needs to be able to make multiple types of furniture at the same time. For example, it might make chairs that need to be different colors or have different styles.

To reliably produce output, each person in the factory will need to adhere to rules like those described in this document.

Here's a sketch of a specific way the above example relates to the scheduling rules:

Person 1 sends chair seats and chair backs to Person 2. This is done with blocking operations, and there is no storage capacity between the people when the objects are sent. (This means that a Push operation cannot complete until the corresponding Pop operation is performed.) Assume that the fastest an object can be sent between people is 1 minute.

Timing convention for the examples below. The timelines show a Push at the time the calling process executes the Push() call in its source-process context—i.e., for the clocked SystemC analogy, during the process execution immediately following the preceding clock edge. This is **not necessarily the formal Issue cycle** defined elsewhere in this document; formal Issue refers to the cycle in which the corresponding request becomes visible at the selected message-channel boundary (for example, as vld for a Push). A displayed Pop instead denotes the time at which the blocking Pop() completes and returns its value to the calling process. In these zero-capacity examples, the corresponding blocking Push completes when that Pop completes.

The written instructions that person 1 is given are:

```
while (1) {
    // internal processing code for seats and backs ...
    seats.Push(seat_object);
    backs.Push(back_object);
}
```

The written instructions that person 2 is given are:

```
while (1) {
    seat_object = seats.Pop();
    back_object = backs.Pop();
    // internal processing code for seats and backs ...
}
```

If both person 1 and person 2 choose to perform their tasks sequentially as written, the resulting sequence of Push calls and Pop completions is:

	Person 1	Person 2
Minute 1:	seats.Push(seat1);	
Minute 2:	backs.Push(back1);	seat1 = seats.Pop();
Minute 3:	seats.Push(seat2);	back1 = backs.Pop();
Minute 4:	backs.Push(back2);	seat2 = seats.Pop();
Minute 5:		back2 = backs.Pop();

If both person 1 and person 2 choose to perform their IO operations in parallel, the resulting sequence of Push calls and Pop completions is:

	Person 1	Person 2
Minute 1:	seats.Push(seat1); backs.Push(back1);	
Minute 2:	seats.Push(seat2); backs.Push(back2);	seat1 = seats.Pop(); back1 = backs.Pop();
Minute 3:		seat2 = seats.Pop(); back2 = backs.Pop();

If person 1 chooses to perform their IO operations in parallel, and person 2 chooses to perform their IO operations sequentially as written, the resulting sequence of Push calls and Pop completions is:

	Person 1	Person 2
Minute 1:	seats.Push(seat1); backs.Push(back1);	
Minute 2:		seat1 = seats.Pop();
Minute 3:	seats.Push(seat2); backs.Push(back2);	back1 = backs.Pop();
Minute 4:		seat2 = seats.Pop();
Minute 5:		back2 = backs.Pop();

If person 1 chooses to perform their IO operations sequentially as written, but person 2 chooses to perform their IO operations sequentially in the reverse order as it was written, the resulting sequence of Push calls and Pop completions is:

	Person 1	Person 2
Minute 1:	seats.Push(seat1);	
Minute 2:	backs.Push(back1);	back1 = backs.Pop();
Minute 3:		seat1 = seats.Pop();

In this case the person 1 seats.Push(seat1) operation at minute 1 will not complete at the start of minute 2. This means that the person 1 backs.Push(back1) operation will never start, and thus the Person 2 back1 backs.Pop() operation will never complete. So, the system will be in deadlock.

This example directly corresponds to the ordering rules related to message-passing operations in the *basic conceptual model* within this document, and in particular the specific rule that disallows reversing the order of message-passing operations.

Appendix B – Formal Presentation of Pipelined Loops

Modeling convention for pipelined-loop overlap.

Pipelined-loop body no-wait condition. The formal pipelined-loop model in this appendix applies only to HLS-pipelined loop bodies that contain no wait/ac_wait statement, SyncChannel/ac_sync call, or other explicit synchronization interface call inside the loop body. All ordinary synchronization calls selected into S must be outside the pipelined loop body or at explicit boundaries around the pipelined region. This is a well-formedness condition for applying the Appendices G-K theorem stack to a pipelined-loop instance; an in-loop synchronization call requires source rewriting, non-pipelined/preserved-loop treatment, or a separate proof/certificate outside this appendix.

The post-HLS pipelined-loop implementation may overlap iterations and may internally accept/stage up to the finite bound $B(c_in)$ of input values for younger overlapped iterations before older iterations' outputs are produced. This internal accept/stage behavior is part of the back-annotated channel realization counted in $B(c_in)$ between the selected producer and consumer boundary points. Here $B(c_in)$ is an effective abstract-boundary storage/credit bound, not a claim that the RTL contains a single physical FIFO named c_in . In the ordinary Sys_B case, FIFO availability completely determines the complementary ready/valid signal at that boundary. In Sys_B^elab, the same finite storage/credit is represented by explicit bounded channels and any additional mutual-stall behavior is represented by declared stateless combinational ready/valid couplers; those couplers may be located at an internal elaborated boundary or at the process-facing outer boundary selected by E. The proof obligation is that committed transfers remain E4-legal and that the settled ready/valid behavior used by

ChannelEnabled/ChannelDisabled is exactly the behavior declared by E. Internal movements that do not cross an explicit abstract boundary remain ϵ -steps. Σ records only the committed transfers at the selected observable boundary. In particular, the consumer-side boundary $\text{Pop}_{\{c_in\}}$ is the loop-body $\text{Pop}_{\{c_in\}}$ operation from the pre-HLS source when that Pop is included in the chosen Σ / Σ_DUT projection. The externally relevant effect of pipelining is therefore captured by the declared bounded storage together with the explicit $\text{Sys_B}^{\text{elab}}$ handshake network, without introducing duplicate outer Σ -visible Pops. Once the pipeline reaches a quiescent blocked state under the automatic-flush discipline below, it does not initiate new blocking request assertions for younger overlapped iterations.

B-faithfulness obligation

The use of $B(c_in)$ in this modeling convention assumes that the selected abstract boundary and the reported capacity faithfully summarize the corresponding RTL storage/credit realization: all state that can accept or hold messages is either counted in $B(c_in)$ or represented as a separate explicit $\text{Sys_B}^{\text{elab}}$ finite-capacity channel, and movements inside a represented storage element do not create, delete, duplicate, reorder, or relabel the selected boundary transfers. In $\text{Sys_B}^{\text{elab}}$, B-faithfulness additionally requires the declared stateless ready/valid coupler network to match the RTL boundary handshake behavior after same-cycle settling. A coupler adds no capacity and is not a new process; it may gate a complementary ready/valid signal only through logic explicitly declared in E. This is a tool/RTL-flow compliance obligation, not a fact derived by Appendix B.

Multi-input Pop points and coupler policies.

When a pipelined loop consumes from multiple input channels (e.g., $c1, c2$), back-annotation still assigns finite storage/credit bounds and cycle-boundary occupancies to the explicit storage elements that represent those paths. In the multi-input / coupler cases of Appendix Q the formal source-side proof target is $\text{Sys_B}^{\text{elab}}$ of Definition Core.1 and Appendix J, Remark 2: a finite network of explicit bounded/rendezvous channels plus stateless combinational ready/valid couplers.

ChannelEnabled/ChannelDisabled is evaluated from the settled handshake at the selected boundary. A conjunctive/join dependency is therefore not modeled as extra state or as an unexplained predicate; it is the ordinary consequence of the declared combinational ready/valid equations of E, which may include a coupler at an internal boundary or at the process-facing outer boundary. Appendix K contracts those internal handshake dependencies onto the original modeled-process WFG as specified by Core.WFG and Appendix Q. For the detailed elaboration patterns and obligations, see Appendix Q.

Practical note (outer Σ_DUT projection and proof-internal Σ)

The loop-body $\text{Pop}_{\{c_in\}}$ in the pre-HLS source (i.e., the consumer-side $\text{Pop}_{\{c_in\}}$ operation, conceptually after the in-flight capacity $B(c_in)$) may be exposed in the outer DUT/testbench-visible alphabet Σ_DUT if desired, but is typically omitted since it is latency-insensitive and may be difficult to expose directly in RTL. The liveness/deadlock reasoning still accounts for this blocking point without requiring it to be Σ_DUT -visible: Appendix K evaluates the corresponding pending frontier item using the settled ChannelEnabled/ChannelDisabled value. In ordinary Sys_B this reduces to the familiar E4 occupancy/rendezvous test. In $\text{Sys_B}^{\text{elab}}$, the same value is obtained by settling the finite explicit FIFO/rendezvous/coupler network declared by E, and the resulting blocking dependency is projected onto modeled processes by Core.WFG. Any proof-internal channel used to account for storage remains explicit even if its committed events are projected away from Σ_DUT by Π_elab .

Drain-only blocked-frontier discipline and automatic flush.

Potential deadlocks are avoided by having the pipeline automatically flush.

In this document, “automatic flush” does not mean a global pipeline clear or a mandatory drain-to-empty phase. It is the pipelined or otherwise overlapped implementation of the more general drain-only blocked-frontier discipline used by Appendix K. Once the minimal pending blocking frontier is fully channel-disabled after ε -settling, the process is blocked at that frontier; no new later source work may issue past that frontier; already asserted blocking requests remain asserted until commit; and already-issued downstream work may continue to drain only insofar as it does not create new WFG-relevant blocking causes. For ordinary non-overlapped control, the same Appendix K discipline may instead be discharged structurally, as described below.

Terminology (pending blocking frontier / $\text{Front_P}(\sigma)$)

At a settled boundary-evaluation cutpoint, consider the set of pending source-level blocking message-operation frontier items $x \in \Omega_{\text{msg},P}$ in the current interval between adjacent synchronization calls, where each such item has an owning dynamic message-call instance $q = \text{MsgOf_P}(x)$ whose $\text{Issue}(q)$ has occurred and whose $\text{Commit}(q)$ has not yet occurred. By the Issue/Commit linkage (Appendix C) together with BoundaryReq ’s definition at the Σ -visible boundary (as given in “Normative Formal Core for Appendices G–K”), each such pending blocking frontier item x has its endpoint-owned boundary request asserted through q (Push: vld, Pop: rdy) until q commits; purely combinational channel-side structure (including couplers) may affect only the complementary mirror signal (Push: rdy, Pop: vld) and thus whether x commits in that cycle after ε -settling. Define $\text{Front_P}(\sigma)$ to be the set of such pending blocking message-operation frontier items that are minimal with respect to the source-induced order $\leq_P$ within that interval. By Lemma $\text{Core.OrdFrontSingleton}$, an ordinary non-elaborated source interval has at most one minimal pending blocking message-operation frontier item. If $\text{Front_P}(\sigma)$ has multiple message-operation members, that multiplicity must come from explicit $\text{Sys_B}^{\text{elab}}$ proof-internal structure such as staged/bundle/join/epoch-gate abstract channels; it is not inferred merely from the fact that ordinary source calls use distinct interfaces. We refer to $\text{Front_P}(\sigma)$ as the pending blocking frontier (or “minimal pending blocking frontier”). This $\text{Front_P}(\sigma)$ notion is distinct from NextFront_P (Definition Core.6 ; reconstructed from KObs by Lemma J.KObs-Proc and Corollary J.1), which denotes the next source-level frontier-item set over Ω_P and may include synchronization-call items, signal-anchored action items, and message-operation items. A message-operation item in NextFront_P may be pending before it contributes any committed Σ -event.

A pipelined loop supplies the drain-only blocked-frontier discipline by automatically flushing if, whenever the pending blocking frontier $\text{Front_P}(\sigma)$ is non-empty and every frontier item $x \in \text{Front_P}(\sigma)$ cannot make progress because it is not channel-enabled (evaluated after combinational ready/valid/backpressure has ε -settled for the cycle), then:

1. **Block point.** The process is blocked on message passing at that pending blocking frontier $\text{Front_P}(\sigma)$, as in the unpipelined source.
2. **No new later work.** While blocked at that frontier (i.e., while $\text{Front_P}(\sigma)$ remains non-empty and all of its members are channel-disabled), the implementation does not issue any new blocking Pop/Push message-operation frontier item $y \in \Omega_{\text{msg},P}$ that is later in source order than some $x \in \text{Front_P}(\sigma)$, including any younger overlapped iteration, and therefore does not begin asserting $\text{BoundaryReq}(y, \cdot)$ for such later work. Here issue is read through the owning message instance $q_y = \text{MsgOf_P}(y)$.
3. **No withdrawal.** While blocked, the implementation does not withdraw any already-asserted blocking Pop/Push request (no “try-and-withdraw”); outstanding blocking requests remain asserted until their owning message instances commit.

4. **Clarification (Frontier-minimality vs. other asserted requests).** The Wait-For Graph in Appendix K is intentionally defined from the minimal pending blocking frontier $\text{Front_P}(\sigma)$, i.e., the current blocking causes. Accordingly, an already-asserted blocking request whose frontier item is not in $\text{Front_P}(\sigma)$ (for example, a request from a younger overlapped iteration that was issued before the blocked condition, or a downstream request draining work already past the frontier) may persist as protocol state under the “no withdrawal” rule, but it is not treated as a wait-for source while an earlier pending blocking frontier item remains frontier-minimal. Such a request contributes WFG blocking only if/when its frontier item later becomes a member of $\text{Front_P}(\sigma)$.
5. **Drain-only downstream progress.** Work that has already passed the blocked frontier point(s) may continue to drain, and any already-asserted resulting output-Push requests may commit when channel-enabled, provided no work advances past the blocked frontier point(s). (Formally: “no work advances past” is meant in the $\leq_P$ / issue sense—while $\text{Front_P}(\sigma)$ is fully disabled, P does not issue any new blocking message-operation frontier item y , with owning instance $q_y = \text{MsgOf_P}(y)$, such that, for some $x \in \text{Front_P}(\sigma)$, $x \leq_P y$ and $x \neq y$; downstream drain may only complete/commit operations whose BoundaryReq was already asserted prior to reaching this blocked condition.) (Equivalently: this is the “ALL disabled $\Rightarrow$ stall” policy— P stalls only when none of the frontier items in $\text{Front_P}(\sigma)$ are channel-enabled; downstream work may drain while $\text{Front_P}(\sigma)$ is fully disabled.)
6. **Finite explicit boundary storage/credit.** Every selected abstract boundary that can hold or credit already accepted work has a finite declared capacity $B(c)$, or is represented by an explicit finite $\text{Sys_B}^{\text{elab}}$ boundary chain whose component capacities are finite and B -faithful. Together with “no new later work,” this makes the set of drainable work finite and non-replenishing.
7. **Synchronization-boundary withdrawal of future-epoch acceptance.**
Let s be an explicit synchronization interface call that separates the current source interval from later work. While the process is pending at s , the implementation shall not present producer-facing availability for any back-annotated capacity intended only for $\text{suff_S}(s)$; equivalently, any asserted upstream producer request targeting such capacity shall remain channel-disabled until s commits. This does not prevent drain of work already accepted before reaching s . **Formal elaboration requirement.** For purposes of Appendices J–K, this withdrawal shall be represented in Sys_ref by the elaborated source model $\text{Sys_B}^{\text{elab}}$, not by adding a hidden extra enablement predicate to an otherwise E4-enabled outer channel. $\text{Sys_B}^{\text{elab}}$ shall introduce explicit sync-boundary / epoch-gate abstract channels, or equivalent explicit staged realization structure, so that capacity usable only by $\text{suff_S}(s)$ is separated from capacity available to the current interval. While s is pending, the gate/boundary for future-epoch acceptance is channel-disabled in the ordinary E4 sense at that explicit abstract boundary. After s commits, the corresponding post-sync capacity may again become producer-facing available according to the ordinary ChannelEnabled rules for the elaborated channels.

General drain-only blocked-frontier obligation used by Appendix K

Appendix K applies the drain-only blocked-frontier discipline to every source interval in which Policy L can leave source-later same-interval blocking work issued while an earlier minimal blocking frontier remains uncommitted. This includes pipelined-loop overlap and other explicitly modeled eager/overlapped issue admitted by the selected Sys_ref transition system; it is not a license to treat ordinary source-sequenced operations as incomparable, and it is not limited to physically pipelined control. Under the generalized drain profile, “work already accepted before the block” includes explicit proof-internal $\text{Sys_B}^{\text{elab}}$ staged/bundle/join state represented in the qualified Q.2 action universe; the finite admitted domain is not limited to residual source offers owned by the later candidate D. Appendix

B remains implementation-side evidence for Core.16/K.DrainEvidence and does not by itself establish the reconstructed finite-path opportunity theorem J.InFlightPathSound.

For pipelined or otherwise overlapped control, Appendix B automatic flush supplies the implementation-side evidence for the drain-only discipline; it does not by itself prove path-indexed Core.Drain for an arbitrary Sys_ref witness. On the standard normal J.GWR-Comp route, Lemma K.DrainReach-Comp performs the checked implementation-to-source lift to the exact constructed τ_{ref} , using the bound LocalGWR/I.A0/E3 construction-action and issue-prefix evidence described there. For ordinary non-overlapped control, the source-side obligation may instead be discharged directly by proving that, after the process reaches a channel-disabled blocked point, its control cannot launch any new source-later dynamic operation until the blocker commits. In both cases, Appendix C supplies the non-withdrawal discipline, and E4/B-faithfulness plus explicit Sys_B^elab accounting supply finite boundary storage or credit. The resulting drain set may shrink through already-issued commits but cannot be replenished across the blocked frontier.

Example (II=1, latency=4, automatic flush): a loop that performs one blocking Pop_{c_in} and one blocking Push_{c_out} per iteration in the pre-HLS source may, after pipelining in the post-HLS RTL, internally accept/stage up to four input values into pipeline staging (pipeline overlap) before the first output Push_{c_out} commits, subject to the back-annotated bound B(c_in) (E4). Values may then reside for several cycles in internal pipeline storage; such internal staging/hand-off is ϵ -state within the back-annotated realization and does not create any new auxiliary or duplicate Σ -visible Push_c/Pop_c events beyond the boundary commits at the Σ -visible endpoints (it may change only the timing of those boundary commits under overlapped execution). The drain-only blocking behavior is as defined above under “automatic flush”; in particular, if the blocked operation is in a late pipeline stage (e.g., the final Push), there may be little or no downstream work to drain.

Appendix C – Issue/Commit Terminology and Linkage

Scope / shared definitions

The definitions below are with respect to the Σ -visible endpoints of the chosen source reference model Sys_ref (Definition Core.1 and Appendix J, Remark 2)—i.e., the ready/valid interfaces at which Σ records message-transfer events in the raw rendezvous case Sys, the ordinary bounded-FIFO case Sys_B, or the elaborated back-annotated case Sys_B^elab. All shared notions about (i) Σ vs. ϵ steps, (ii) BoundaryReq / ChannelEnabled / ChannelDisabled, (iii) non-blocking polls as ϵ on failure, and (iv) the B(c) / occ_c(t) (cycle-boundary, non-fall-through) interpretation of E4 are as defined in “Normative Formal Core for Appendices G–K” and are to be interpreted over the explicit abstract channels of Sys_ref. Appendix C uses Definition Core.8 for the formal Commit(q)/CommitCycle(q) meaning and Definition Core.9 for the formal Issue(q) meaning. It defines ReturnedAtCycle and records the Issue/Commit attribution and linkage conditions used elsewhere in the document.

Commit (message channel) — informal operational reading

Informally, on RTL_B at a mapped ready/valid boundary, a message commit is observed when an external clocked observer sees both valid and ready high in the same cycle, with the committed payload equal to the mapped data value seen in that cycle. This is the RTL-facing realization of the abstract boundary commit; for the G–K theorem stack, Definition Core.8 is authoritative and defines the abstract E4 commit transition, Commit(q) as the source operation’s process-facing commit occurrence, and CommitCycle(q) as its cycle index.

Issue (message channel) — operational restatement of Core.9

Issue(q) is the cycle index of the operational event in which the calling process begins requesting the transfer corresponding to the source-level message operation instance q at the chosen explicit abstract boundary endpoint. Scheduler/witness evidence may record that cycle for rules such as R4, E3, and E5, but the evidence records the operational Issue event rather than selecting an independent timestamp.

Definition: ReturnedAtCycle (message result availability). $\text{ReturnedAtCycle}(q,t)$ means that the process-visible result of message-call instance q becomes available at the cycle- t boundary. For a successful blocking or non-blocking call, this is the transfer-commit cycle. For a failed PushNB/PopNB call, no message commits, but the success/failure status is nevertheless returned at the cycle- t boundary and is therefore a cycle- t result. ReturnedAtCycle is an operational timestamp used for sequential-result availability; it is not a KObs field.

Same-cycle rendezvous clarification. For a rendezvous channel $B(c)=0$, if a matched Push_c operation instance and Pop_c operation instance first assert their endpoint-owned requests in the same cycle t , and the ready/valid rendezvous is selected/observed in that cycle, then both operation instances have $\text{Issue} = t$ and both committed transfer events occur in cycle t . There is no implicit extra cycle between same-cycle matching endpoint requests and the corresponding rendezvous commit. If one endpoint request was already asserted from an earlier blocking operation, then its Issue may be earlier than t , but the matched Push_c/Pop_c commit is still a same-cycle commit in cycle t .

Issue/Commit linkage (well-formedness conditions)

The list below is intentionally broader than Definition Core.IC: the request-lifecycle, attribution, serialization, and synchronization-boundary clauses restate Core.IC(1)–(5), while “Sequential-result availability” is the separate Core.RegSeq result-availability rule included here for operational completeness. Fix an endpoint direction at an explicit abstract boundary, and let $\text{req}(t)$ denote the endpoint-owned boundary request signal in cycle t (valid for Push, ready for Pop), i.e., the process-owned request level captured by BoundaryReq at that boundary (per “Normative Formal Core for Appendices G–K”). A physical ready/valid level that represents complementary channel/storage availability or declared coupler logic rather than the process-owned request is not $\text{req}(t)$ for this linkage rule. For each source-level message operation instance q on that endpoint direction:

- Boundary-request soundness (blocking requests are non-withdrawable). $\text{req}(\text{Issue}(q)) = 1$. If q is a blocking Push/Pop, then within its reset epoch, once issued it remains the unique outstanding request on that endpoint direction until it commits: $\text{req}(t)=1$ in every cycle t from $\text{Issue}(q)$ through $\text{CommitCycle}(q)$ (inclusive), with stable payload while asserted for Push. At every cycle-aligned state, a process-owned blocking req has exactly one live issued, not-yet-committed/reset owner; after that ownership ends, a continuing high physical level is not such a req unless the back-to-back rule below transfers ownership. If an affecting RESET occurs first, RW2 ends that dynamic instance rather than withdrawing it within the epoch. (This is the “no deassert-before-commit” discipline used throughout the document.)
- Stable attribution for a single outstanding source-level request. If $\text{req}(t)=1$ and no transfer commits on that endpoint direction in cycle t , then any continued assertion $\text{req}(t+1)=1$ is attributed to the same outstanding source-level message-operation instance q only so long as the requesting source-level operation remains that same outstanding instance. For blocking Push/Pop, this is the mandatory non-withdrawable case above, so attribution cannot change before commit. For non-blocking PushNB/PopNB, continuous assertion of the endpoint request signal, by itself, does not identify later executed non-blocking call instances as the same q ; a new executed source-level non-blocking call creates a new dynamic instance $q' \in Q_{\text{exec}}, P$ for the owning process P , even if req does not deassert between cycles.

- Back-to-back request attribution under continuously asserted req. Suppose blocking q_0 commits in cycle t and $\text{req}(t+1)=1$ on the same endpoint direction. Let q_1 be the next same-direction blocking source-level message operation in the same source interval between adjacent ordinary synchronization calls selected into S , if such an operation exists. If q_1 is that next operation, the continuing process-owned request is owned by q_1 beginning in cycle $t+1$, and $\text{Issue}(q_1)=t+1$ even though req did not deassert. The process-owned blocking request may not remain asserted but ownerless for one or more later cycles and then be attributed to q_1 . Genuine scheduler/HLS delay before q_1 issues remains permitted by $\text{Core.SelectLatitude}$, but during such delay q_1 's process-owned BoundaryReq has not yet begun; a high physical ready/valid value caused by channel-side storage, complementary availability, or declared coupler logic is interpreted by the fixed selected boundary/channel/elaboration mapping and is not an ownerless req. Non-blocking calls retain the separate dynamic-instance treatment stated below.
- **Same-endpoint blocking issue serialization.** Let q_0 and q_1 be distinct blocking source-level message operations on the same scalar explicit abstract boundary and endpoint direction governed by the Core single-transfer-per-cycle endpoint constraint in one reset epoch, with $q_0 <_{\text{src}} q_1$. If both issue, then $\text{Issue}(q_0) < \text{Issue}(q_1)$. A scalar endpoint has one operation-attributed process-owned request direction at a time; two distinct blocking dynamic instances may not acquire that same request direction in the same issue cycle. This is the strict same-endpoint fact used by mechanization; R4/E3 retain their general non-strict form because they also cover distinct interfaces and non-blocking cases.
- Sequential-result availability (no same-cycle dependent continuation). Let $\text{ReturnedAtCycle}(q_0, t)$ hold in a sequential process P . If a dynamically succeeding operation q_1 is reachable only after q_0 returns, or if q_1 's control choice, endpoint, address, payload, or other required operand depends on the value or success/failure status returned by q_0 , then $\text{Issue}(q_1) > t$. This rule applies to successful commits and failed non-blocking returns, across distinct channels and endpoint directions as well as on the same endpoint. Same-cycle issue or commit remains permitted only when q_1 was independently reached and all required inputs, payload values, and control predecessors were available before the cycle- t boundary under Core.RegSeq ; overlap between independently active pipeline iterations is not prohibited, but an unresolved NB result in an earlier iteration places any dependent later-iteration action in a strictly later Horizon.
- Sync-boundary discipline. If $q_0 \in \text{pref_S}(s)$ and $q_1 \in \text{suff_S}(s)$ for an ordinary synchronization call s selected into S , then $\text{Issue}(q_1)$ is strictly after s 's commit in the same trace (i.e., $\text{Issue}(q_1) > \text{clk_pre}(s)$ in τ_{pre} and $\text{Issue}(q_1) > \text{clk_post}(s)$ in τ_{post}). If q_0 is a blocking Push/Pop, q_0 must process-facing commit no later than s ; no issued-but-uncommitted blocking operation in $\text{pref_S}(s)$ may survive the commit of s into the succeeding source interval. Same-cycle completion is permitted: q_0 and s may both commit in cycle t when q_0 's process-facing commit is part of the selected cycle- t boundary actions and is accounted for before the interval advance caused by s . Any boundary request assertion that persists while the process is still pending at s is attributed to the outstanding operation(s) in $\text{pref_S}(s)$ and does not constitute issuance of any operation in $\text{suff_S}(s)$. Under the pipelined-loop body no-wait condition, there are no in-body wait events that create additional sync-boundary cuts inside an HLS-pipelined loop.

For non-blocking message passing (PushNB/PopNB), a failed poll produces no Σ -event and is modeled as an ϵ -step, while a successful poll contributes the corresponding committed Push_c(v) / Pop_c(v) event; see “Non-blocking message-passing” in “Normative Formal Core for Appendices G–K” and Appendix I.2. Each executed non-blocking call is a distinct dynamic source-level message-call instance $q \in Q_{\text{exec}, P}$ for the owning process P . Thus, if repeated PopNB/PushNB polls occur in later loop iterations or at later source-level call instances, they are distinct q 's even if the endpoint request remains continuously

asserted and no intervening commit occurs. Stable attribution therefore applies to a single outstanding source-level request instance; it does not collapse distinct non-blocking dynamic call instances across iterations into one q . Successful non-blocking calls that commit induce message-operation frontier items $x \in \Omega_msg, P$ when their committed transfer is relevant to Ω_P , with owning message instances $q = \text{MsgOf_P}(x)$ in Q_P . Failed non-blocking polls remain only in Q_exec, P and contribute no Ω_P or Ω_msg, P item.

Appendix D – Modeling Latency-Sensitive Protocols

In some cases, designs or testbenches may use protocols or local control policies which are latency-sensitive. These situations are handled by isolating the latency-sensitive portions into small, self-contained parts, and then keeping the rest of the design and testbench latency insensitive. A latency-sensitive portion is any local block of logic whose functional or externally visible behavior depends on cycle-level timing that is not represented by the ordinary latency-insensitive Σ interface. This includes ordinary signal-level protocols with fixed cycle timing, global interrupt/request ordering logic, non-blocking arbitration whose winner depends on request arrival timing, and non-blocking retry/timeout logic whose later behavior depends on the number or cadence of failed PushNB/PopNB polls before success.

For such logic, the intended methodology is not to treat the hidden timing dependence as part of the ordinary latency-insensitive source-core proof. Instead, the latency-sensitive portion should be isolated behind an explicit boundary. On one side of the boundary, the transactor or local protocol block may be modeled cycle-accurately, snooped, or otherwise verified with its detailed timing behavior. On the other side of the boundary, the rest of the system should see a latency-insensitive interface, such as message-passing transactions, synchronization events, or a properly handshaken signal protocol.

Consider a case where a signal-level protocol is latency-sensitive. The protocol will have specific timing behavior that it must meet. To handle this, we create a transactor that has two sides: the first side interacts with the signals and handles the detailed timing requirements, and the second side sends and receives messages with the rest of the system. The message-passing side is latency insensitive. The same pattern applies to cadence-sensitive non-blocking polling logic: if retry counters, timeout counters, or arbitration decisions depend on the number or cadence of failed non-blocking polls, that polling loop is part of the latency-sensitive local protocol block. Its detailed failed-poll cadence should be contained inside the isolated block or aligned by the Appendix L snooping mechanism; it should not leak as an unmodeled hidden timing dependence into the ordinary Appendix I/J latency-insensitive proof. To properly model the detailed timing behavior, the latency-sensitive transactor or local protocol block is modeled at the cycle-accurate level in SystemC, or is otherwise covered by an explicit snooping/alignment argument. There is an example of such a transactor model in the following document and example:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/examples/53_transactor_modeling/transactor_modeling.pdf

For purposes of the formal appendices, once the latency-sensitive portion has been isolated, the ordinary Appendix I/J proof is applied at the chosen observable boundary of that isolated block. The failed-poll cadence internal to the isolated block remains local timing behavior; the rest of the system observes only the latency-insensitive or explicitly modeled boundary behavior.

Appendix E – One-Way Handshake Protocols

In some systems a one-way signal handshake is sufficient for reliable system operation. When using a one-way handshake, the implicit assumption is that the “missing” handshake isn’t required since it is always true.

For example, in time-domain signal processing hardware designs, signal processing HW blocks may input new samples at a fixed rate. The HW blocks are always ready to receive new samples on each clock, but they need to know if the samples are valid. These types of designs can be modeled with the one-way handshake dat/vld protocol demonstrated in this example:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/tree/master/matchlib_examples/examples/32_dat_vld

Appendix F - Modeling Guidelines and Design Constraints

The following are simple modeling guidelines that align with the modeling methodology described in this document.

- 1) Prefer to use `Connections::In/Out + SyncChannel` over signal IO and `wait()`.
- 2) Prefer to use `Pop()/Push()` over `PopNB()/PushNB()`.
- 3) When pipelining a loop, prefer to use `hls_stall_mode flush`.

When using signal IO:

- 4) If modeling a cycle-accurate process, use `disable_spawn` and follow the example `53_transactor_modeling` style and do not use `Push/Pop` in the process.
- 5) If modeling a *direct input*, use `hls_direct_input` and possibly `hls_direct_input_sync`, and only change the signal at allowed times.
- 6) If combining signal IO with `Connections::In/Out` in the same process, use proper signal IO handshake that does not rely on `In/Out` ports for process synchronization. Place signal IO operations very close to their `wait()` statements.
- 7) Unless you are modeling a cycle-accurate process, you should expect that latency will change between the pre-HLS and post-HLS models.

Design Constraints for Applying the Formal Guarantees

This section consolidates the design choices that most directly determine whether the formal guarantees in this document can be applied. It is written for hardware architects and HLS users rather than for the tool or proof implementer. A design can synthesize and function correctly while still lying outside the stated theorem scope if one of these behaviors is left implicit.

This section is informative. Each constraint restates a normative requirement located in the referenced sections; the referenced text is authoritative.

The architect is responsible for choosing interfaces, reset behavior, and communication structures that admit the required correspondence. The HLS tool and verification flow are responsible for constructing and checking the detailed process, channel, attribution, coverage, and other certificates. The architect does not need to produce those certificates, but the design must expose enough explicit structure for the flow to do so.

The first list applies to the implementation-safety and observable-correspondence result. The second list contains additional constraints for the internal message-passing deadlock-preservation result. A design outside the simple route is not necessarily prohibited; it generally requires explicit modeling, witness alignment, or a separate proof route.

Design constraints for the safety and correspondence guarantee

These constraints support the Post $\rightarrow$ Ref guarantee that the covered RTL behavior introduces no new observable functional behavior absent from the prescribed source reference model.

1. **Give every signal read and write an unambiguous synchronization anchor.** For a sequential process, place signal reads immediately after the synchronization call that defines their observation point and place signal writes immediately before the synchronization call at which they become visible. Branches must not allow one write to be associated with different later waits on different paths. Signal I/O inside a rolled or preserved loop needs an explicit per-iteration wait when each iteration is intended to create a distinct observation. A loop selected for HLS pipelining cannot use an in-body wait under this contract; move the signal interaction outside the loop, use a supported message-passing or direct-input mechanism, or do not pipeline that loop.
2. **Keep direct inputs stable, or update them through the supported synchronization scheme.** A direct input may be physically sampled later than its logical source-level read only when its value remains stable for the applicable reset epoch. If software or the testbench must change control inputs while the design is running, use the documented `hls_direct_input_sync` handshake and change the values only at the designated synchronization point. An ordinary Push or Pop is not a substitute for that update contract.
3. **Make timing-sensitive environment behavior explicit.** The ordinary safety result assumes that the same external behavior is selected from the same prior observable history, not from hidden differences in absolute cycle count. Testbench branches, watchdogs, stimulus changes, or pass/fail decisions that depend on raw latency must be isolated as latency-sensitive logic, made observable through an explicit protocol, or checked at RTL rather than assumed to transfer automatically from an arbitrary pre-HLS run.

4. **Treat non-blocking polling as an architectural proof-route decision.** The simple route applies when a failed PopNB or PushNB does not change the next observable action, request, payload, endpoint choice, or control path. Typical arbitration code, retry counters, timeouts, backoff, and "work on B if A is unavailable" behavior do change what happens next. Such designs require an approved witness-alignment mechanism, commonly the Appendix L snooping route, or an isolated and explicitly modeled latency-sensitive protocol block. Prefer blocking interfaces when the architecture does not genuinely require poll-dependent control.
5. **Do not encode an unbounded wait as an invisible busy-poll loop in the ordinary proof fragment.** A loop that repeatedly executes failed non-blocking polls continues to take local internal steps and is not a stable waiting state. Use a blocking operation, an explicit stable-wait or handshake construct, or isolate and model the polling loop as timing-sensitive logic. A finite, statically bounded polling sequence can be handled when its outcomes and later effects satisfy the applicable witness and dependency conditions.
6. **Expose every J.Mem cross-process shared-memory dependency through a supported proof boundary.** Synchronization alone can order two processes' memory accesses, but the present safety proof does not carry an independent global shared-memory state. Keep the existing J.Mem trigger: whenever one process's memory write can affect another process's control flow, payload/value expressions, or the identity or order of its later operations, every such dependency must be represented through covered proof-visible operations and the flow must establish that no other unlowered cross-process memory-carried path satisfying that trigger exists. Raw physical RAM transactions may remain outside Σ only under Core.MemFaithful. If caching, merging, splitting, or reordering changes the transaction stream of a memory carrying such cross-process value flow, use a qualified Core.MemProfile whose logical boundary is transformation-invariant, use the separately supplied memory-state extension, or treat the design as outside Theorem J.1. The qualified-library route is intended for reusable patterns such as the shared-memory classes discussed in Appendix H.1.
7. **Model the channel behavior that the implementation actually provides.** The default buffered-channel abstraction is a bounded FIFO whose storage legality is evaluated from beginning-of-cycle occupancy and which does not permit first-word fall-through at the selected abstract boundary. Internal RTL bypass/fall-through may be hidden inside the realization only when the selected abstraction boundary absorbs that behavior so that the resulting explicit positive-capacity channel still satisfies the Core.12 non-fall-through E4 semantics. If an empty-channel value pushed in cycle t remains observable as a Pop at that same selected positive-capacity boundary in cycle t , the ordinary $B(c) > 0$ E4 model does not cover that boundary; use a separately qualified bypass/rendezvous elaboration template or choose a different abstract boundary that restores the stated semantics. Debug drains, hidden credit paths, side entrances, or other structure that changes enablement or transfer ordering must likewise be represented explicitly. Back-annotated capacity alone is not sufficient when hidden structure changes enablement or transfer ordering. The same boundary-discipline applies to mapped memories: enabling cache/merge/split/reorder beneath a memory boundary used by the proof requires Core.MemFaithful; when a reusable transformation-invariant boundary is needed, use a qualified Core.MemProfile.

8. **Use an elaborated model for coupled multi-input pipelines and other joined structures.** When acceptance on one input depends on another input, a bundle, join, shared stage, coupler, or epoch gate, do not interpret the outer capacity of each input as independent slack. The verification flow must expose the relevant staged, rendezvous, bundle, or join structure in the elaborated source reference model. A declared stateless coupler may also sit directly at a process-facing outer boundary; in that case $B(c)/occ_c$ still describe storage legality while the settled ready/valid handshake determines actual transfer eligibility. The outer interface remains the user-visible boundary, while proof-relevant storage, handshake, and dependency facts may be carried by the selected Sys_B^{elab} structure.
9. **Give every dynamic reset a channel-consistent disposition.** Reset both endpoints of a live channel together with the channel state, or define an explicit flush, abort, cancellation, or end-of-epoch protocol that determines what happens to outstanding requests and in-flight data. A reset of only one endpoint cannot silently discard, retain, or reattribute a transaction across the reset boundary. The protocol must be represented in both the source model and RTL using ordinary observable actions or another explicitly verified reset-state correspondence.

Additional constraints for the internal message-passing deadlock guarantee

The Appendix K result excludes the defined reachable closed internal wait-cycle cutpoints. It therefore needs the safety constraints above plus the following structural and operational conditions. These are additional requirements for the deadlock claim; they should not be read as extra premises for every use of the Appendix J safety theorem.

1. **Give each analyzed logical channel one producer and one consumer.** A shared bus, crossbar, multi-client queue, or resource with multiple requesters should be represented by an explicit arbiter process and point-to-point channels. This makes the complementary endpoint and wait-for ownership unambiguous. A custom shared-channel abstraction requires its own ownership and arbitration proof rather than being treated as an ordinary point-to-point channel.
2. **Represent multi-lane, burst, and multi-transfer resources explicitly.** The standard scalar-channel model permits at most one Push and one Pop commit per channel direction in one cycle. A resource that transfers several elements, lanes, or beats at once must expose that behavior in its abstract interface and capacity model. It should not be encoded as several hidden transfers on a channel that the proof treats as scalar.
3. **Keep blocking requests asserted and Push payloads stable until completion.** A blocking Push or Pop is a persistent request, not a speculative try that may be withdrawn when back-pressure lasts longer than expected. Request ownership must remain with the same dynamic operation until commit or an explicit reset. A separate nonterminal cancellation/withdrawal action belongs to a different modeled protocol and is not part of the ordinary blocking wait-for interpretation used by the G–K theorem stack.
4. **Ensure that a blocked pipelined process can only drain previously accepted work.** Once a process has reached a disabled blocking frontier, later blocking work from that process must not continue to enter the system behind the blocked operation. Previously accepted pipeline work may finish and release finite downstream state, but the pipeline must not replenish that work indefinitely or expose next-interval capacity before the synchronization boundary completes. This is the practical meaning of the automatic-flush and drain-only discipline used by the deadlock proof.

5. **Do not make deadlock freedom depend on favorable overlap or same-cycle issue.** The primary practical check deliberately uses a conservative serial-issue source execution. If the design avoids deadlock only because two source-ordered blocking operations overlap, issue in the same cycle, or are arbitrated in a particularly favorable order, a clean serial execution cannot justify scheduler-robust deadlock freedom. Rewrite or buffer the protocol, add explicit synchronization, or use a different scheduler-specific proof.
6. **Check deadlock against the actual capacities and explicit boundary structure.** The relevant source model is the back-annotated bounded or elaborated model, not an idealized unbounded network and not necessarily the raw rendezvous model. A capacity-induced deadlock is real for the selected capacity configuration even if it disappears at a larger depth. Conversely, a rendezvous-only run can be a useful early screen but may report a deadlock that the intended buffered or elaborated design does not have.

Practical interpretation of the Policy-S conservatism. The serial check is most restrictive when liveness fundamentally depends on a source-later blocking request becoming outstanding before a source-earlier blocking operation commits. A canonical minimal shape is two processes connected by two oppositely directed rendezvous channels ($B(c)=0$) with crossed source-order dependencies that require the relevant requests to overlap. This is different from pipelining itself being problematic. If the implementation contains pipeline stages, skid/elastic buffers, FIFOs, interconnect queues, or credits that can accept, hold, or backpressure messages, the selected Sys_B/Sys_B^elab model shall reflect that communication slack through $B(c)$ or explicit elaborated boundaries under B-faithfulness. The Policy-S check is then performed on that buffered/elaborated communication architecture, where the added slack can break the tight rendezvous dependency that made favorable overlap essential. Because modern designs commonly pipeline both computation and communication and frequently contain buffering, queues, or credit-based interconnect, this makes the conservative Policy-S restriction substantially less likely in practice to exclude a design than the raw rendezvous example might suggest. Do not infer deadlock freedom merely from $B(c)>0$: finite storage can fill. Where needed, prove the relevant cycle-breaking occupancy, token, or credit invariant directly (for example through Theorem K.Str-3) or use another valid A5-L discharge route.

7. **Account explicitly for reset during the serial comparison.** For a candidate blocked component, the primary serial-reduction route requires either a comparison continuation with no affecting reset or an already-triggered, non-waived response monitor whose exact pending operation is cleared by the matched reset. A reset that occurs before such a monitor exists, or whose scope cannot be matched to the pending operation, is outside that route and needs another deadlock argument.
8. **Keep other kinds of stalls separate from the internal message-passing claim.** Barrier participation errors, a process waiting for a peer that has terminated without an explicit end-of-stream protocol, environment-only stalls, and timed/reactive cases with non-ordinary release structure are not automatically proved safe by the internal wait-cycle theorem. In particular, a single frontier item behind a joined/coupled boundary can require the same explicit release-structure treatment as a multi-frontier. Model EOF, done, abort, and barrier protocols explicitly, and use the supported environment/release fragment or a separate proof when external timing can release a candidate wait cycle. For the primary Policy-S serial-reduction route, a timed/reactive or otherwise nondefault environment is supported only when every relevant candidate has a single internal frontier with the ordinary/single-release-support structure required by K.EnvMF; a joined/coupled candidate with a co-frontier, multiple frontier items, or multiple release alternatives requires another A5-L discharge.

9. Treat mapped-memory backpressure as explicit K structure unless a qualified boundary absorbs it.

A mapped memory that can create persistent process-facing backpressure cannot be omitted from Appendix-K reasoning merely because its raw RAM transactions are outside the outer Σ_{DUT} trace. The selected K abstraction must either show that the hidden realization is WFG-inert in the sense required by Core.MemFaithful/ $K\epsilon$, or expose a transformation-invariant, K-faithful logical boundary whose point-to-point/arbitration structure represents the relevant capacity, backpressure, release support, and wait dependencies. If no such qualified boundary exists, transaction-changing memory transformations are not covered by Appendix K for that instance.

Formal references. The detailed rules are in RULE 1 and RULE 2, Direct Inputs, Scheduling of Array Accesses, Appendices B, D, I, J, K, L, and Q, and the shared channel, Core.MemFaithful/Core.MemProfile memory-boundary rules, reset, and registered-interface definitions in the Normative Formal Core. Appendix M.7a remains the definitive per-instance evidence checklist for the tool and verification flow.

Guide to the Formal Appendices

This guide is informative. It explains how the formal appendices fit together and suggests reading paths; it does not introduce new assumptions, definitions, or proof obligations. The appendix letters and theorem identifiers remain authoritative.

How to interpret the qualification detail. The formal appendices deliberately make explicit many assumptions and implementation facts that are often left implicit in hardware verification, whether the RTL is synthesized from HLS source or written manually. The underlying engineering conditions were largely already there: a claimed model-to-implementation result already depends on facts about scheduling, interfaces, buffering, reset, environment behavior, progress, and implementation correspondence, whether or not those dependencies have been individually named. The formalization makes those dependencies explicit and auditable. It should not be read as requiring every design user to understand or manually discharge every formal condition; many conditions are expected to be established automatically, satisfied by simple design restrictions, or discharged once through reusable tool, library, or implementation-flow qualification. The methodology may add explicit evidence and accounting where a theorem-backed guarantee requires it, but that should not be mistaken for creating all of the underlying engineering conditions.

Recommended reading paths

HLS users and hardware architects. Read the main body through Appendix F for the scheduling contract and modeling guidance, then read Appendix M for the formal conclusions and their practical limits. Consult Appendix L when the design uses non-blocking operations or cadence-sensitive behavior. If you need the Policy-S deadlock-preservation route, read Appendix M.6 followed by Appendix K.4b, K.4c, and K.4f. These sections explain the selected serial-mode run, the bounded-response check, and the standard applicability fragment. The design-user activity is one complete instrumented Policy-S execution; the flow separately establishes simulator determinism, applicability, response coverage, and the implementation-side QSync/correspondence evidence from which RTL cutpoint coverage is derived or otherwise discharged. Tool, checker, and formal-proof implementers should additionally read Appendix K.4e and Appendix K-P.

Safety reviewers. Read the Normative Formal Core for Appendices G–K, then Appendix G, Appendix I, and Appendix J. Appendix J.2b is optional for a one-prefix Theorem J.1 safety review or an independently discharged J.TraceCertCov domain, but it is used by the standard QSync implementation-wide safety route: J.QSyncTraceInd/J.TraceCertCov-QSyncCycle plus J.CycleIdealCover/J.1-Safe-QSync extend safety to all reachable finite observable prefixes. Its richer J.QSyncCertInd route additionally supplies the K-facing package-totality result used by the universal Appendix K conclusion. Appendix R provides a compact derived companion, and Appendix T highlights important examples of the scheduling-specific obligations that remain beyond generic transition-system and simulation theory.

Liveness and deadlock reviewers. Begin with the Normative Formal Core and Appendix G, then read Appendix J for the source/RTL cutpoint connection and Appendix J.2b for the standard QSync canonical-cycle package-totality route, Appendix K for the preservation theorem, Appendix K-P for the serial-reduction proof companion, Appendix L for witness selection and environment-sensitive cases, and Appendix N for fairness and progress assumptions. Appendix Q is required when the selected source reference model uses elaborated staged, bundle, join, coupler, or epoch-gate structure.

Mechanization work. Use the Normative Formal Core as the semantic skeleton and Appendix R as its derived compact map, Appendix S for the Lean proof strategy and mechanization decomposition/import anchors, and Appendix T as a selective explanation of why some obligations require scheduling-specific machinery beyond standard libraries. The Normative Formal Core and Appendices G–K are the

normative sources for the definitions and theorem interfaces; Appendix M.7a supplies the definitive consolidated side-condition discharge record; Appendix R is derived.

Appendix map

Normative Formal Core for Appendices G–K. Defines the shared reference model, operational issue/admissibility semantics, observation and frontier objects, channel/request predicates, drain discipline, KObs record, capacity conventions, reset treatment, and terminology used by the G–K theorem stack. Appendix M.7a supplies the consolidated side-condition discharge record. Appendix M.7b gives the standard QSync logical certificate-profile packaging of that record.

Appendix G - Trace Compatibility Rules. Defines the observable trace-compatibility framework, the E1-E5 rules, the selected source-reference/RTL observation boundary, and the asymmetric Post $\rightarrow$ Ref and restricted Ref $\rightarrow$ Post uses.

Appendix H - Possible Criticisms. Collects anticipated objections, limitations, and responses. It is useful for understanding why the formal contract has its present scope and why certain stronger-looking claims are not made.

Appendix I - Per-Process Replay and Preparation Proof. Establishes the process-local replay and finite preparation machinery used by the compositional safety construction, including the treatment of deferred non-blocking decisions and registered phase behavior.

Appendix J - System-Level Trace Compatibility Proof. Composes local process, channel, atomic-interaction, environment, reset, footprint, request-snapshot, dependency, agreement, and coverage evidence into a global source witness and the system-level Post $\rightarrow$ Ref trace-compatibility result. Section J.2b additionally gives the QSync-specialized base-and-step routes used to derive cycle-aligned safety package coverage and total K-facing package coverage for Appendix K. The cycle-aligned route discharges J.TraceCertCov on J.CycleReach_post; Lemma J.CycleIdealCover and Corollary J.1-Safe-QSync then extend ideal-prefix-closed safety to all reachable finite outer observable prefixes without asserting a separate J.TraceCertCov package for each non-cycle-aligned prefix. General safety-side J.TraceCertCov remains the interface when an application needs direct per-prefix correspondence packages or uses a broader non-QSync route.

Appendix K - Preservation of Internal Wait-Cycle Freedom. States the certified-cutpoint deadlock-preservation result, the exact admissibility and drain bridge, the A5-L premise, and K.CertCov for a universal reachable-cutpoint conclusion. In the standard QSync normal-compositional scope, J.QSyncCertInd/J.CertExist-QSync plus K.CertCov-QSync provide the preferred theorem-derived K.CertCov route; alternate independently checked coverage routes remain permitted.

Appendix K-P - Proofs Companion for the Serial-Reduction Route. Develops the principal Policy-S/A5-S route for discharging A5-L, including deterministic serial-source dominance and bounded-response evidence, while recording the route's environment/frontier applicability limits.

Appendix L - Snooping Introduction, Implementation and Concerns. Explains witness selection, non-blocking and latency-sensitive cases, and the conditions under which a source execution can be aligned with the RTL for restricted reverse use.

Appendix M - Formal Results Overview. Provides a readable but technically precise summary of the models, principal safety theorem, restricted reverse direction, liveness theorem, coverage requirements, and implementation-qualification obligations.

Appendix N - Scheduler and Environment Fairness Assumptions. Restates the scheduler, channel-progress, and quiescence assumptions in operational ready/valid form and clarifies that they are theorem-applicability assumptions, not automatic guarantees, whose discharge may be owned by

source/scheduler semantics, HLS-generated or hand-written RTL arbitration and back-pressure, channel-library behavior, or external masters.

Appendices O and P. Appendix O gives an informative sketch for multiple clock domains. Appendix P records alternative formal frameworks and their relationship to the chosen architecture.

Appendix Q - Multiple Input Pipelines Back-Annotation. Defines the local elaboration structures and projection/accounting rules used when ordinary bounded-FIFO abstraction is insufficient for staged, bundled, joined, coupled, or sync-gated implementations.

Appendix R - Compact Derived Formal Core and Mechanization View. Provides a downstream review and mechanization map of the Core $\rightarrow I \rightarrow J \rightarrow K$ dependency chain; it introduces no normative definition or theorem interface used by earlier sections.

Appendices S and T. Appendix S records the Lean proof strategy. Appendix T identifies the scheduling-specific proof obligations that cannot be obtained merely by importing generic labeled-transition-system, simulation, or temporal-logic libraries.

Appendix U - Revision Record. Records material changes to the formal interfaces and explanatory text so that reviewers can trace the evolution of the document.

Normative Formal Core for Appendices G–K

Scope of comparison. Appendix G first presents the raw rendezvous source model Sys and the R/E scheduling rules. For the Appendices I–K proof stack, and for the Sys_ref-specific uses in Appendix G, the formal comparison is between RTL_B and Sys_ref (Definition Core.1 and Appendix J, Remark 2), i.e., the same source processes interpreted under the channel semantics of E4 with capacities $B(c)$ matching the RTL. The E4 channel semantics are case-split: $B(c)=0$ is interpreted as a capacity-zero rendezvous channel, and $B(c)>0$ is interpreted as a cycle-boundary, non-fall-through bounded FIFO. The rendezvous case is therefore recovered as the $B(c)=0$ case of the channel model, not by applying the $B(c)>0$ FIFO-availability inequalities with $B(c)=0$.

Authority of the Core. The shared definitions in this Core are normative for the G–K theorem stack. Later appendices may refine these objects and prove properties about them, but they do not redefine them. Material that merely explains the completed J/K architecture or records certificate-discharge choices is placed downstream in Appendix R or Appendix M rather than inside this definitional layer.

Symbol / Rule	Definition — concise but complete
BoundaryReq(x, σ) (shorthand BoundaryReq(q, σ) only in ordinary unique-boundary case)	For any dynamic source-level message-operation frontier item $x \in \Omega_{\text{msg},P}$ on an explicit abstract channel $c = \text{BoundaryOf_P}(x)$ of Sys_ref, with owning source-level message instance $q = \text{MsgOf_P}(x)$ and $\text{kind}(q) \in \{\text{Push_c}, \text{Pop_c}\}$, BoundaryReq(x, σ) means that q’s own endpoint request for item x is asserted at the selected BoundaryEvalPoint σ : producer valid for Push, consumer ready for Pop, with stable Push payload as required. BoundaryReq is operation- and boundary-attributive and is not an already-committed Σ event. In the ordinary unelaborated case, BoundaryReq(q, σ) is shorthand for the unique x-form. In Sys_B^elab, the frontier-item form is authoritative. Stateless elaboration logic may compute the complementary mirror signal at that boundary (Push: ready; Pop: valid), including through a coupler located at the process-facing outer boundary, but it does not withdraw or reattribute the owning source operation’s BoundaryReq. When cycle-index notation is needed,

Symbol / Rule	Definition — concise but complete
	BoundaryReq(x,t) is shorthand for BoundaryReq(x, σ_t), where σ_t is the selected BoundaryEvalPoint for cycle t.
ChannelEnabled(x, σ) (shorthand ChannelEnabled(q, σ) only in ordinary unique-boundary case)	ChannelEnabled(x, σ) means BoundaryReq(x, σ) holds and, after same-cycle combinational settling, the complementary handshake signal at BoundaryOf_P(x) is asserted: ready for a Push endpoint and valid for a Pop endpoint. For a primitive uncoupled bounded FIFO, that complementary signal is equivalent to the E4 occupancy test (Push: $\text{occ_c}(\sigma) < B(c)$; Pop: $\text{occ_c}(\sigma) > 0$). For a primitive rendezvous channel, it is equivalent to the matching complementary endpoint request in the same settled cycle. In Sys_B ^{elab} , E may place stateless deterministic coupler logic between the primitive storage/request roots and the evaluated boundary, so the complementary signal is the value produced by that declared network; E4 still forbids a Push when the relevant storage is full and a Pop when it is empty, but nonfull/nonempty status alone need not be sufficient for enablement at a coupling-sensitive boundary. If the owning operation represented by x commits, it contributes the corresponding committed Σ event at that boundary. In the ordinary unique-boundary case ChannelEnabled(q, σ) is shorthand; in Sys_B ^{elab} the x-form is authoritative.
ChannelDisabled(x, σ) (shorthand ChannelDisabled(q, σ) only in ordinary unique-boundary case)	ChannelDisabled(x, σ) means BoundaryReq(x, σ) holds but the settled complementary handshake signal required by ChannelEnabled is not asserted. At an ordinary primitive FIFO/rendezvous boundary this is exactly the familiar full/empty/missing-peer condition. At a coupling-sensitive Sys_B ^{elab} boundary it may instead be caused by the declared stateless ready/valid network—for example a join withholding ready/valid because another input is absent or downstream storage has no space. Such a cause must be explicit in E and reconstructible at the settled cutpoint; it is not hidden state. In the ordinary unique-boundary case ChannelDisabled(q, σ) is shorthand; in Sys_B ^{elab} the x-form is authoritative.
Σ	Set of observable actions for the equivalence check (committed IO events) at the chosen observable boundary (often the DUT IO boundary for functional equivalence, but optionally expanded—e.g., for Appendix K deadlock reasoning): • channel commit events Push_c(v) and Pop_c(v) on channels designated as observable, labeled with the transferred message value (or abstract message identity) v • synchronization events wait(), Sync, START_P, FINISH_P • single-cycle signal actions Write(sig, val) and Read(sig, val) on signals designated as observable, labeled with the value written/read. Internal-only channels/signals may be excluded from Σ (treated as ϵ -microstate) when they are not part of the chosen observable boundary. For mapped memory activity, that existing selection freedom is admissible only under Core.MemFaithful relative to the declared untransformed logical mapping-layer semantics. A mapped message channel or anchored signal that is designated observable at the selected boundary remains a Σ event governed by the ordinary E4 or R2/E2/Core.17 rules and cannot be selectively hidden. If

Symbol / Rule	Definition — concise but complete
	an excluded channel/signal is later brought into the observable boundary (e.g., by snooping for debug or analysis), it becomes Σ-visible and must match under $\approx$. Deadlock-analysis requirement (Appendix K): for Appendix K, Σ is instantiated to include every message-passing channel whose endpoint transfers at the chosen Sys_ref boundary can contribute a dependency edge in the Appendix K Wait-For Graph, either because the channel is on the declared observable boundary or because it is snooped. The WFG itself is defined over pending blocking message-operation frontier items $x \in \Omega_msg, P$ for the relevant owning process P, with owning dynamic message-call instance $q = \text{MsgOf_P}(x)$, not over already-committed Σ-event labels and not over failed non-blocking polls in Q_exec, P. Thus, when Appendix K refers to a blocked Push or Pop at a boundary, the relevant frontier object is x, and its owning pending source-level operation instance is q with $\text{kind}(q) \in \{\text{Push_c}, \text{Pop_c}\}$; the edge is determined by $\text{BoundaryReq}(x, \sigma)$ together with $\text{ChannelEnabled}(x, \sigma)/\text{ChannelDisabled}(x, \sigma)$ at the selected Core.KCut σ (and hence at a settled BoundaryEvalPoint). Internal implementation micro-interfaces that are not declared as explicit Sys_B^elab storage boundaries remain ϵ/snoop detail. Appendix K evaluates the selected frontier boundary using its settled $\text{ChannelEnabled}/\text{ChannelDisabled}$ value. In ordinary Sys_B this reduces to the primitive E4/occupancy or rendezvous test; in Sys_B^elab the value is reconstructed by the E-declared finite stateless ready/valid network and the corresponding WaitOwners process projection. A stateless coupler need not be a separate Σ channel, residual offer, or modeled process. Non-blocking message-passing (PushNB/PopNB): A non-blocking call that returns “not completed” produces no Σ-event and is modeled as an internal ϵ-step. When a non-blocking call succeeds (returns “completed”), that success is represented in Σ as the corresponding committed Push_c(v) or Pop_c(v) event on that channel, labeled with the same transferred value/message v (i.e., Σ records transfers and their payloads, not failed polls). A non-blocking call is considered issued in the cycle in which it executes (its Issue cycle). A call that returns “completed” commits in that same cycle and therefore yields a committed transfer $m(q) \in M$; a call that returns “not completed” produces no committed transfer event ($m(q)$ undefined / absent) and may be treated as an ϵ-step with respect to Σ-visible message-transfer events. Each executed non-blocking call is still a distinct dynamic source-level message-call instance $q \in Q_exec, P$ for the owning process P. Since failed non-blocking q's have no Σ-label, their cross-trace matching is witness-relative: it is supplied by the chosen μ_Q / WC witness, by NB-CF when failed polls cannot affect the next Σ-visible control flow or frontier, or by Appendix L's snooping / witness-selection mechanism. Continuous assertion of an endpoint request does not by itself identify multiple executed non-blocking calls as one q, and the Σ trace alone is not required to determine how failed non-blocking q's are matched. A successful non-blocking call may also be a member of Q_Q, P when it

Symbol / Rule	Definition — concise but complete
	contributes a committed Push _c (v) / Pop _c (v) event; a failed non-blocking poll remains in $Q_{exec,P} \setminus \Omega_P$.
START_P	First observable action emitted by process P after reset. Marks the moment P begins executing user code. A new START _P is emitted each time P exits reset.
FINISH_P	Last observable action of process P in the current reset epoch. If P executes an unbounded loop within that reset epoch, FINISH _P never occurs for that epoch, and the epoch-local trace is treated as infinite until a later reset boundary, if any. If a later reset affecting P occurs and P exits reset again, a new dynamic START _P begins a fresh reset epoch; any later termination is represented by a distinct dynamic FINISH _P occurrence.
ε-step	An <i>internal</i> , unobservable RTL state transition (pipeline advance, FSM micro-state, etc.).
B(c)	Back-annotated effective capacity of channel c at the chosen explicit abstract Sys_{ref} boundary used by E4 and Appendix K. In the ordinary non-elaborated case this is the chosen Σ-observable Sys_B boundary; in elaborated cases it is the corresponding explicit abstract channel boundary of Sys_B^{elab}, which may be proof-internal and later projected away from the outer Σ_{DUT} trace. Finite, $B(c) \geq 0$. It reflects the total bounded “composite buffer” storage/credit in the RTL realization represented by that explicit abstract channel boundary, including FIFO storage, skid/elastic buffering, HLS-inserted pipeline staging that can hold in-flight messages, and any explicit staged/bundle/join channel capacity introduced by Sys_B^{elab}. • $B(c)=0 \Rightarrow$ rendezvous (capacity-zero) channel. • $B(c)>0 \Rightarrow$ bounded FIFO (of that effective capacity). B-faithfulness. For the formal results to apply to a concrete RTL_B instance, each back-annotated B(c) and, when used, the surrounding Sys_B^{elab} handshake network must be faithful to the chosen boundary. This means: • Boundary completeness: every RTL storage element, credit, skid/elastic buffer, pipeline stage, or elaborated staged/bundle/join element that can accept or hold messages across the represented boundary is either included in B(c) or represented as a separate explicit finite-capacity Sys_B^{elab} channel. • E4 legality plus declared handshake behavior: no committed Push/Pop may violate the corresponding E4 capacity/order rule. In ordinary Sys_B, E4 availability also completely determines the complementary ready/valid value. In Sys_B^{elab}, an additional restriction on that complementary value is permitted only when it is produced by the finite stateless coupler network declared in E; that logic may be at an internal or process-facing outer boundary and must be included in same-cycle settling/reconstruction. • ε-opacity of internal movement: movement of an already accepted message within one represented storage realization is internal ε-state unless E makes it a separate explicit proof boundary; it does not create, delete, duplicate, reorder, or relabel the selected outer transfers. • Evidence obligation: B-faithfulness is a tool/RTL-flow compliance obligation.

Symbol / Rule	Definition — concise but complete
	It may be discharged by a trusted back-annotation algorithm, generated certificate, structural RTL check, interface monitor, formal equivalence/checking flow, or other validation method appropriate to the implementation flow. It is not derived by E4, Proposition J.0, or Theorem J.1. Point-to-point source/process ownership assumption. Each source-level logical message channel has exactly one producer process that owns its Push operations and exactly one consumer process that owns its Pop operations. Shared source resources must be remodeled through explicit arbitration and point-to-point process-facing channels. In $\text{Sys_B}^{\text{elab}}$, an explicit internal channel endpoint may instead be driven by a stateless coupler; that coupler is not a process and need not be invented as one. Appendix Q therefore declares endpoint roles plus a process-level wait-dependency projection for WFG construction. For an ordinary primitive channel that projection is the singleton complementary process owner; for a coupling-sensitive boundary it may contain multiple modeled process owners whose process-facing behavior can release the settled blocking condition. Single-transfer-per-cycle endpoint constraint (scalar channel interface): For every message-passing channel c, in any clock cycle t, at most one Push_c may commit and at most one Pop_c may commit. Equivalently, the set of commits on a fixed channel in a single cycle contains ≤ 1 Push and ≤ 1 Pop (it may contain one of each in the same cycle). For rendezvous channels ($B(c)=0$), a committed transfer occurs only as a matching same-cycle $\text{Push}_c(v)$ and $\text{Pop}_c(v)$ pair. For buffered channels ($B(c)>0$), a same-cycle Push_c and Pop_c do not imply fall-through; the channel semantics are cycle-boundary, non-fall-through as stated in E4. Reset-empty FIFO convention: after each reset boundary at which the compared execution begins or restarts, every buffered message-passing channel with $B(c)>0$ has logical FIFO occupancy 0 at the chosen Σ boundary. Equivalently, the reset state contains no preloaded in-flight messages in the abstract channel realization used by Sys_ref / RTL_B. If a design requires preloaded channel contents, those contents must be modeled explicitly as post-reset producer actions or by a separate initialized-state extension outside the present empty-FIFO formal model. Modeling note (multi-lane / burst transfers): If an implementation can transfer $k>1$ messages per cycle on a logical resource, model it as k parallel point-to-point channels (or as a single widened message transferred by one $\text{Push}_c/\text{Pop}_c$ per cycle) so that the Σ-level event model continues to satisfy the scalar constraint above.
Core.MemFaithful	Mapped-memory boundary faithfulness. Core.MemFaithful is an admissibility condition on the existing Σ-selection freedom; it is not a new hiding mechanism. Fix (i) the untransformed logical array-access mapping-layer semantics for the selected memory instance as the reference, (ii) the selected abstract memory boundary and its channel/signal alphabet, and (iii) the declared set of permitted cache/merge/split/reorder transformations. The transformed implementation may keep lower-level mapped-memory activity

Symbol / Rule	Definition — concise but complete
	outside Σ only when it induces at the selected boundary the same observable behavior required from the reference by Core.17/E1–E5: it does not create, delete, duplicate, or relabel selected-boundary events, preserves the per-endpoint occurrence order and every required ordering/atomicity constraint, and preserves the values carried by those events. A mapped message channel or anchored signal that is selected at that boundary remains governed by E4 or R2/E2 respectively. For an Appendix-K application, hidden memory realization shall additionally be K-faithful: it shall not introduce, remove, or redirect a process-facing blocking cause, frontier/request attribution, ChannelEnabled/ChannelDisabled fact, release-support fact, or WFG dependency except through structure represented at the selected Sys_ref/Sys_B^elab boundary; any proof-hidden ϵ activity on the relevant normalization segments satisfies $K\epsilon$. If a hidden physical memory resource can persistently backpressure a process, the selected K boundary must lie above the transaction-changing realization and faithfully represent that capacity/backpressure and release dependency; otherwise the transaction-changing transformation is not covered by Appendix K for that instance. Core.MemFaithful may be discharged by a trusted mapping/back-annotation algorithm, generated certificate, structural RTL check, interface monitor, formal refinement/equivalence check, or other qualified mapping-layer validation. It is not derived by E4, Core.17, Proposition J.0, or Theorem J.1.
occ_c(t)	Abstract FIFO occupancy at the chosen explicit abstract Sys_ref channel boundary: number of items committed by Push_c but not yet committed by Pop_c at that boundary, measured from the most recent reset boundary under the reset-empty FIFO convention. In the ordinary non-elaborated case this is the chosen Sys_B channel boundary; in elaborated cases Sys_ref = Sys_B^elab, this is the occupancy of the relevant explicit abstract staged/bundle/join channel boundary. When a clause explicitly identifies c as an outer Sys_B/Chan_outer refinement-target channel, as in Core.ElabProj and Q.1(a)–(c), occ_c, B(c), and the associated E4 legality denote the corresponding outer-channel semantics rather than the explicit elaborated-channel B_E(d)/occ_d state. Equivalently (E4), $\text{occ}_c(t) = \text{push}(\pi_t) - \text{pop}(\pi_t)$, where π_t is the Σ-visible boundary-commit prefix for that explicit abstract channel strictly before cycle t (i.e., excluding any Push_c/Pop_c commits that occur in cycle t), so occ_c(t) is the abstract occupancy at the beginning of cycle t. The subtraction $\text{push}(\pi_t) - \text{pop}(\pi_t)$ is the mathematical signed balance. E4 proves that a legal channel history places this balance in $[0, B(c)]$; non-negativity is therefore a semantic legality property, not the result of truncating natural-number subtraction. For pure-FIFO channels with $B(c) > 0$, the FIFO availability predicates (not-full/not-empty) are evaluated against this begin-of-cycle occupancy; hence a Pop_c cannot commit in a cycle that begins empty solely because a Push_c also commits in that same cycle (no fall-through), and likewise a Push_c cannot commit in a cycle that begins full solely because a Pop_c also commits in that same cycle.

Symbol / Rule	Definition — concise but complete
	Clarification (no “second Push/Pop”): movements of an item within the RTL channel realization (e.g., FIFO→staging/skid buffering, staging/skid buffering→pipeline-stage registers, pipeline-stage→pipeline-stage handoff) are internal ε-steps; they do not constitute additional Σ-visible committed Push_c/Pop_c events beyond the boundary events counted by $\text{push}(\pi_t)/\text{pop}(\pi_t)$. Derived cutpoint shorthand (used in the Core and Appendix K): if σ is a BoundaryEvalPoint occurring in cycle t_σ, write $\text{occ}_c(\sigma) \triangleq \text{occ}_c(t_\sigma)$. Likewise write $\text{ChannelEnabled}(q, \sigma) \triangleq \text{ChannelEnabled}(q, t_\sigma)$ and $\text{ChannelDisabled}(q, \sigma) \triangleq \text{ChannelDisabled}(q, t_\sigma)$ when enablement is evaluated at that settled boundary point. Every source SettlePoint and every Core.KCut used by Appendix K is a BoundaryEvalPoint. This is only notational shorthand for cutpoint reasoning; the primary semantic definition of occupancy remains the cycle-boundary quantity $\text{occ}_c(t)$. Note: $\text{occ}_c(t)$ is an abstract analysis quantity derived from Σ-visible boundary commits; in both Sys_B and RTL_B, processes do not directly read $\text{occ}_c(t)$ and instead proceed solely based on the per-channel ready/valid outcome of their own Push/Pop attempts.
P_push(c)	Primitive-channel finite-progress invariant (producer): this is a persistent-request premise, not a ChannelEnabled premise. In this paragraph, “the producer’s request remains asserted continuously” means that the producer presents a persistent, non-withdrawable Push_c endpoint request from the starting cycle through the matching discharge event, with no deassert-before-commit and with stable payload while asserted. • Buffered case ($B(c) > 0$): If the producer’s Push_c request remains asserted continuously from some cycle t_0 onward while the channel is full ($\text{occ}_c = B(c)$) —with stable payload while asserted—and the complementary endpoint continuously requests the matching Pop_c (persistent request asserted; Pop_c boundary request remains true), then some cycle $t' \geq t_0$ commits a complementary Pop_c (i.e., the full condition is eventually discharged). • Rendezvous case ($B(c) = 0$): If the producer’s Push_c request remains asserted continuously from some cycle t_0 onward (with stable payload while asserted) and the complementary endpoint continuously requests the matching Pop_c (persistent request asserted; Pop_c boundary request remains true), then some cycle $t' \geq t_0$ commits the rendezvous transfer—i.e., a matching Push_c(v) and Pop_c(v) commit in the same cycle. In Sys_B^{elab} this clause is used when the blocked cause is the primitive full/rendezvous condition of the represented storage element; coupler-mediated disabledness is handled by the Q.1 compositional handshake/progress obligations.
P_pop(c)	Primitive-channel finite-progress invariant (consumer): this is a persistent-request premise, not a ChannelEnabled premise. In this paragraph, “the consumer’s request remains

Symbol / Rule	Definition — concise but complete
	asserted continuously” means that the consumer presents a persistent, non-withdrawable Pop_c endpoint request from the starting cycle through the matching discharge event, with no deassert-before-commit. In Sys_B^{elab} this clause is used when the blocked cause is the primitive empty/rendezvous condition of the represented storage element; coupler-mediated disabledness is handled by the Q.1 compositional handshake/progress obligations. • Buffered case ($B(c) > 0$): If the consumer’s Pop_c request remains asserted continuously from some cycle t_0 onward while the channel is empty ($occ_c = 0$)—and the complementary endpoint continuously requests the matching Push_c (persistent request asserted with stable payload; Push_c boundary request remains true)—then some cycle $t' \geq t_0$ commits a complementary Push_c (i.e., the empty condition is eventually discharged). • Rendezvous case ($B(c) = 0$): If the consumer’s Pop_c request remains asserted continuously from some cycle t_0 onward and the complementary endpoint continuously requests the matching Push_c (persistent request asserted with stable payload; Push_c boundary request remains true), then some cycle $t' \geq t_0$ commits the rendezvous transfer—i.e., a matching Push_c(v) and Pop_c(v) commit in the same cycle.
Trace-compatibility rules (E1–E5)	E1 Synchronization-order preservation: For each process P, the sequence of synchronization events executed by P (wait(), SyncChannel, START_P, FINISH_P) appears in the same source order in the pre-HLS and post-HLS traces. E2 Signal visibility: For each sequential process P, each signal Read in P occurs at the closest preceding synchronization event in P, and each signal Write in P occurs at the closest succeeding synchronization event in P, in both the pre-HLS and post-HLS traces (per R2/E2). For each combinational process or explicitly modeled combinational network component C_comb, observable combinational signal Read/Write labels appear only inside emitted R2C fixed-point signal-observation units for the selected R2C observation component, as defined by R2C. Each emitted R2C unit is evaluated at one ϵ-quiescent combinational fixed point, and all Read/Write labels inside that unit are atomic with respect to that fixed point. Emitted R2C units are matched by component identity and local emitted-occurrence ordinal, not by global observable-cycle bucket number. R2C stutter cycles whose selected observable slice is unchanged do not create separate Σ-visible units unless an explicitly modeled observer/sampler requires a fresh emitted unit. Interpretation (logical timestamping). For sequential processes, E2 defines the logical timestamps of Read/Write Σ-events at their anchors. An implementation may physically sample a Read later than $\text{pred}_S(r)$, or physically apply the canonical Write earlier than $\text{succ}_S(w)$, provided the anchored value is preserved; if several source Writes share that anchor, R2/Core.17 select the last executed Write value. For combinational processes, the logical observation point is the emitted R2C fixed-point signal-observation

Symbol / Rule	Definition — concise but complete
	unit, not a global observable-cycle bucket number; internal delta-cycle propagation before the fixed point and R2C stutter cycles with unchanged selected observable slices are ϵ/stutter and are not separately Σ-visible. Direct-input pragmas (e.g., <code>#pragma hls_direct_input</code> and <code>#pragma hls_direct_input_sync</code>) do not change E2 and are not exceptions to E2. For sequential processes, they impose stability/timing contracts that allow later physical sampling, or controlled updates at a synchronization boundary, while the logical Read/Write Σ-events used for $\approx$ checking remain timestamped at the E2 synchronization anchor points and retain the anchored value labels. For combinational processes, ordinary direct-input pragmas do not introduce a separate anchoring rule; any observable combinational signal Read/Write label remains part of an emitted R2C fixed-point signal-observation unit under the R2C unit-generation and matching conventions. Thus the pragma affects only implementation sampling freedom under its stated contract, not the logical Σ observation used for compatibility. E3 (Safe message issue ordering). (Post-HLS preservation of R4). For any two executed message-call instances $q_1, q_2 \in Q_{\text{exec}, P}$ of the same process P: (i) Same-interface order is always preserved: if q_1 and q_2 access the same message-passing interface/channel and $q_1 <_{\text{src}} q_2$, then $\text{issue_post}(q_1) \leq \text{issue_post}(q_2)$. (ii) Distinct-interface no-reverse: if q_1 and q_2 access distinct message-passing interfaces and appear in sequence in the source ($q_1 <_{\text{src}} q_2$), then $\text{issue_post}(q_1) \leq \text{issue_post}(q_2)$ (i.e., they shall not be issued in the reverse sequence). (iii) Prefix-closed issue obligation: at every cycle-aligned cutpoint σ of the post-HLS execution / witness, if $q_1 <_{\text{src}} q_2$ and q_2 has issued by σ, then q_1 has also issued by σ and $\text{Issue}(q_1) \leq \text{Issue}(q_2)$. This is an implementation-side obligation on issued sets, not merely an inequality over pairs whose timestamps are already defined. (Note: any pipelined-loop internal staging/dequeue discussed in “Pipelined Loops” is ϵ-microstate within the back-annotated channel realization and is not a Σ-visible boundary issue event associated with any $q \in Q_{\text{exec}, P}$ nor a committed-transfer event in M; therefore it does not constitute a counterexample to (ii) or (iii).) E4 FIFO legality: for each channel c, the post-HLS committed ($\text{Push}_c(v)$, $\text{Pop}_c(v)$) history is a legal bounded-capacity execution consistent with Sys_B for that channel (capacity $B(c)$), and reduces to rendezvous consistency when $B(c)=0$. In particular, on each channel, the committed $\text{Pop}_c(v)$ events return exactly the FIFO-ordered sequence of values v previously committed by $\text{Push}_c(v)$, with no drops, duplications, or reordering. E5 (Messages cannot cross their immediate synchronization boundary). For every synchronization call s (in the source order used by R3 / pref_S / suff_S): $\forall q \in \text{pref}_S(s) : \text{issue_post}(q) \leq \text{clk_post}(s)$

Symbol / Rule	Definition — concise but complete
	$\forall q \in \text{suff_S}(s) : \text{issue_post}(q) > \text{clk_post}(s)$ For every blocking $q \in \text{pref_S}(s)$: $m(q)$ is defined and $\text{clk_post}(m(q)) \leq \text{clk_post}(s)$.

Frontier-item predicate convention. BoundaryReq, ChannelEnabled, and ChannelDisabled are authoritative in their frontier-item form. For a message-operation frontier item $x \in \Omega_{\text{msg},P}$, let $q = \text{MsgOf_P}(x)$ and $c = \text{BoundaryOf_P}(x)$. BoundaryReq(x, σ), ChannelEnabled(x, σ), and ChannelDisabled(x, σ) are evaluated at that specific explicit abstract boundary and are undefined for non-message frontier items unless the statement explicitly restricts $x \in \Omega_{\text{msg},P}$. In the ordinary unelaborated case, where q has a unique frontier item and a unique evaluated boundary, the q -forms BoundaryReq(q, σ), ChannelEnabled(q, σ), and ChannelDisabled(q, σ) may be used as shorthands for the corresponding x -forms. In Sys_B^{elab}, one source-level call q may own several frontier items at different explicit boundaries, so the x -form is authoritative.

Declared combinational coupling; no hidden state or predicate. In ordinary Sys_B, the complementary ready/valid value at a channel boundary is determined entirely by the primitive E4 buffered/rendezvous semantics. In Sys_B^{elab}, a cross-channel or join dependency may additionally affect that complementary ready/valid value only through the finite stateless combinational network explicitly declared by E. Such logic may sit at an internal elaborated boundary or at a process-facing outer boundary. It does not alter the owning source operation's persistent BoundaryReq, add storage, or create a new process. After L2 settling, its effect is visible directly in ChannelEnabled/ChannelDisabled and must be reconstructible from E, the explicit channel state, and the attributed outstanding requests. A grant, gate, coupler, or other condition not represented in that declared network is a hidden enablement predicate and is outside the model.

Prescribed reference model

Definition Core.1 (Prescribed source reference model Sys_ref).

For a fixed design, fixed external stimulus, and selected elaboration package, define the prescribed source reference model Sys_ref as follows: • Sys_ref = Sys_B in the ordinary abstract back-annotated presentation, where $B(c)=0$ is a primitive rendezvous channel and $B(c)>0$ is a primitive cycle-boundary non-fall-through bounded FIFO. • Sys_ref = Sys_B^{elab} in the elaborated/back-annotated case, where the source model is refined by a finite network of explicit finite-capacity/rendezvous channels together with stateless deterministic ready/valid/data couplers needed for staging, joins, bundles, shared-capacity realization, or sync/epoch gating. A coupler may be placed at an internal elaborated boundary or at a process-facing outer boundary; it introduces no storage and is not a modeled process. All source-side semantic notions in Appendices G–K are interpreted with respect to this prescribed source reference model. Appendix J, Remark 2 explains the model ladder and practical back-annotation interpretation but does not redefine Sys_ref.

Definition Core.ElabProj (Elaboration projection/accounting map Π_{elab}).

When Sys_ref = Sys_B^{elab}, fix the coordinated projection/accounting and handshake interface supplied by the selected elaboration package E: the channel map Π_{elab} , outer trace map $\pi_{\Sigma,E}$, finite token projection $\Pi_{\text{token},E}$, signed state-accounting map A_E , and the declared handshake/dependency description. This interface has five roles:

(1) Trace projection. $\pi_{\Sigma,E}$ preserves the labels and order of the selected outer Σ_{DUT} -visible boundary events and hides or projects away introduced staged/bundle/join events that are not part of that outer

observation boundary.

(2) Occupancy projection. For every selected outer abstract channel c , Π_{elab} identifies the explicit storage-channel fiber contributing to c , and A_E computes the corresponding signed outer accounting value from the elaborated state. At an applicable ϵ -quiescent cycle-boundary state, Appendix Q requires that state-computed value to agree with the E4 outer-history occupancy.

(3) Conservation/accounting. Internal ϵ -steps and transfers projected away from the outer view preserve the projected outer history and A_E accounting. When an elaboration token contributes outer effects, $\Pi_{\text{token},E}$ may identify a finite set of outer channels; $\pi_{\Sigma,E}$ supplies the corresponding one-or-more outer Push/Pop effects and A_E supplies their accounting. Each projected outer Push_c/Pop_c commit updates the outer accounting exactly as required by E4.

(4) Settled-handshake visibility. E declares the finite stateless combinational ready/valid/data network connecting process-owned requests and explicit storage/rendezvous boundaries. At every BoundaryEvalPoint, the declared network determines the complementary ready/valid value used by ChannelEnabled/ChannelDisabled; no additional hidden gate or coupler state exists.

(5) WFG dependency projection. E supplies the process-level dependency projection used by Core.WFG: combinational couplers and local elaboration nodes are contracted away, while every process-facing condition that can release a disabled frontier remains represented as a dependency on its modeled process owner. The projection is required to be sound and deadlock-reflecting as stated in Appendix Q. Appendix Q supplies the permitted elaboration constructions and the detailed trace/token/storage accounting, settling, handshake-reconstruction, process-dependency, and locality obligations that instantiate this Core interface.

Remark Core.1 (Interpretation at explicit abstract channels).

Whenever $\text{Sys_ref} = \text{Sys_B}^{\text{elab}}$, $B(\cdot)$, $\text{occ_}\cdot(t)$, and E4 are interpreted on the explicit storage/rendezvous channels declared by E , while BoundaryReq and ChannelEnabled/ChannelDisabled are evaluated at the selected explicit handshake boundary after the complete declared ready/valid network has settled. For a primitive uncoupled boundary, the settled handshake reduces exactly to the ordinary E4 full/empty or rendezvous test. For a coupling-sensitive boundary, E4 remains the storage/transfer-legality rule, but nonfull/nonempty status alone need not assert the complementary ready/valid signal; that signal is computed by the stateless coupler network. Outer observations are recovered through $\Pi_{\text{elab}}/\pi_{\Sigma,E}$. A coupler at a process-facing outer boundary is therefore permitted when it is part of E and satisfies the same reconstruction and projection obligations.

Remark Core.1a (Operational Sys_ref transition system and issue policies).

Sys_ref is the selected source-reference transition system defined operationally in this Core. Appendix G constrains that system through R1–R5 and E1–E5. Its states contain process-local control/variable state, reached/issued/committed dynamic message-call instances, explicit abstract channel state for every Sys_B or $\text{Sys_B}^{\text{elab}}$ boundary, operation-attributive pending endpoint requests, synchronization/signal-anchor state, and ϵ -microstate. Its steps are pure source-evaluation ϵ -steps, message issue steps, E4 channel-commit steps, synchronization/signal-anchor steps, and admissible ϵ -settling/drain steps. Core.Phase is part of the legality of this transition system: after the selected L1 window closes, L1-only process evaluation and issue steps are phase-disabled until the next phase permits them. This phase restriction may be represented by an explicit phase tag or by a phase-indexed transition relation; it is not inferred merely from which enabled step a scheduler happened to select. Core.SelectLatitude below makes the complementary selection rule explicit: neither L1 issue selection nor L3 boundary-action selection is required to be maximal in a finite source prefix. The phase-advance markers that close L1 and select L2, advance L2 to L3, advance L3 to L4, and begin the next cycle are meta-level control transitions of the cycle semantics: they are neither ϵ -steps nor Σ -events and do not

themselves change the proof-visible state or history; they only select which phase-indexed operational subrelation is active. The optional L3Obs observation subphase defined by Core.KObsSettle is a phase index within L3 after the selected L3 action set is irrevocably closed; it introduces no additional phase-advance marker and no additional scheduling opportunity. State changes associated with L3 boundary actions and L4 returned-result/registered updates occur through the operational actions already named by Core.Phase, not through the phase-advance marker itself. Policy L and Policy S are two issue policies over this same transition system. Policy L permits source-later blocking requests to be issued before source-earlier same-interval blockers process-facing commit, subject to source-order prefix closure and all R/E/E4/E5 obligations. Policy S adds the conservative serial-blocking restriction: a later source-order blocking Push/Pop may issue only after every earlier source-order blocking Push/Pop in that interval has process-facing committed in a strictly earlier cycle. In Sys_B^{elab}, process-facing commit is the commit that releases the owning source-level operation at its selected boundary, not an arbitrary internal staged/bundle/join transfer. The authoritative definitions are Definition Core.LLegal for Policy-L step legality and fixed-instance binding, Definition Core.SelectLatitude for finite-prefix non-maximal source selection, Definition Core.Phase for the L0–L4 cycle phases and pre-settle/settled-frontier boundary, and Definition Core.LAdm for finite-prefix admissibility and complete admissible executions; this remark introduces no alternate legality, phase, or admissibility relation.

Observable alphabet and executions

Definition Core.2 (Observable alphabet Σ).

Let Σ be the observable event alphabet consisting of:

- (1) committed message-transfer events Push_c(v) and Pop_c(v),
- (2) synchronization events selected into S, and
- (3) signal events Read(sig, v) and Write(sig, v), interpreted as anchored signal events for sequential processes under R2 and as labels carried by emitted ϵ -fixed-point observation units for combinational processes under R2C; those R2C units are identified by component identity and local emitted-occurrence ordinal, not by global cycle bucket.

Under the pipelined-loop body no-wait condition, an HLS-pipelined loop body contributes no additional in-body wait/synchronization events to S or Σ .

Observable-unit carrier convention. Core.17 Σ Match and downstream references to the selected observable-unit projection range over the selected Σ -events together with any additional proof-internal observable units explicitly included by the theorem instance, notably RESET(S_p, S_c). Such proof-internal units need not be members of Σ and may be projected away from the outer Σ_{DUT} view. Their matching and ordering rules are stated explicitly where introduced; including them in the observable-unit carrier does not convert them into ordinary Σ commits.

Definition Core.3 (ϵ -steps and compact trace notation).

Let ϵ denote internal micro-steps that are not observable at the proof-internal alphabet Σ , including internal staging/hand-off within the concrete realization of a single explicit abstract channel, late direct-input sampling between anchors, and failed non-blocking polls. In elaborated cases Sys_{ref} = Sys_B^{elab}, transfers across introduced staged/bundle/join channels are not ϵ merely because Π_{elab} hides them from the outer DUT/testbench-visible alphabet Σ_{DUT} ; if such a channel is an explicit abstract boundary used by Appendix K, its committed Push/Pop transfers are Σ -events for the proof-internal comparison.

A finite execution prefix τ is understood in the same sense as in Appendix J: it carries the inherited cycle partition / per-cycle observable buckets of the execution, together with any ϵ -steps between successive Σ -visible cutpoints. For compact notation one may write a linearization over $(\Sigma \cup \{\epsilon\})$, but only modulo

permutation of Σ -events that occur in the same cycle and are not additionally ordered by the document's explicit Σ -visible constraints. Accordingly, $\tau|\Sigma$ denotes the observable projection together with that inherited cycle partition; same-cycle observable events are not assigned any extra order merely by this compact notation.

Definition Core.4 (ϵ -quiescent cutpoint).

For any proof model M in this formal stack with a declared Σ/ϵ transition relation (including Sys_ref , Sys_K , RTL_B , and explicit lowered/elaborated variants), write $\text{Quiescent_M}(s)$ when no ϵ -step of M 's full proof transition relation is enabled from state s at its current cycle/phase. For any finite prefix τ of M , $\text{Norm_M}(\tau, v, s)$ means that v is a finite ϵ -only extension of τ ending at s , s satisfies Quiescent_M , and v preserves the inherited Σ -visible history and cycle-bucket structure of τ . A selected maximal finite normalization may still be written $\bar{\tau}$. Core.4 quiescence excludes enabled ϵ -work only; it does not exclude enabled Σ -actions and is therefore weaker than a B5/global fixed point. For $\text{Sys_ref}/\text{Sys_K}$, Core.Phase is part of the full transition legality used by Quiescent_M , and the meta-level phase-advance markers of Remark Core.1a are not ϵ -steps. RTL_B has its own cycle/ ϵ relation and is not assigned the source-side L0–L4 phases. Appendix J/K comparison cutpoints use the model-specific Quiescent predicate through Core.KCut below.

Definition Core.Settle (L2 settled boundary-evaluation point).

For a Policy-L or Policy-S cycle t , after the selected L1 process-owned pre-settle window closes, let $\rightarrow_{\epsilon, L2}$ denote the complete phase-L2 ϵ subrelation: process-owned L1 source-evaluation and message-issue steps are disabled; process-owned request assertions and Push payloads that were fixed at the L1 endpoint remain fixed; and the cycle- t exogenous/environment-driven values treated as inputs to the selected G–K system remain fixed throughout L2. Every ϵ -step that remains phase-legal at L2 is a combinational, explicit-boundary-evaluation, or other cycle-local L2 settling step in $\rightarrow_{\epsilon, L2}$.

L2 frame (normative). Every $\rightarrow_{\epsilon, L2}$ step preserves the cycle index; registered process/FSM/pipeline state; registered explicit-channel storage, occupancy, and credit state; and the selected cycle- t exogenous/environment-driven input values. It emits no Σ -visible commit and performs no L4 or other next-cycle registered update. It may modify only cycle-local ϵ /settling microstate and derived combinational boundary values such as complementary ready/valid, guard, join/coupler, and $\text{ChannelEnabled}/\text{ChannelDisabled}$ evaluation results. A same-cycle reactive component whose value is intentionally combinational dependent on DUT outputs is not a changing exogenous input under this clause: it must be included inside the same qualified settling network or isolated behind an explicitly verified boundary.

$\text{SettlePoint_I}(t, \sigma)$ holds when σ is the first fixed point of $\rightarrow_{\epsilon, L2}$ reached from that cycle's selected L1 endpoint. This definition is conditional on such a fixed point being reached; it does not by itself prove termination or existence. Define $\text{BoundaryEvalPoint_I}(\sigma)$ to mean a state at which the explicit boundary request and $\text{ChannelEnabled}/\text{ChannelDisabled}$ network has reached the applicable combinational fixed point; every SettlePoint is a BoundaryEvalPoint . SettlePoint is the L2 activation/evaluation notion used by Core.16 before L3 boundary actions; Lemma Core.SettleIsKCUT below records its relationship to Core.4 quiescence for the phase-bearing source models. The standard existence/uniqueness result is supplied by Definition Core.QSync and Theorem Core.QSync-Ref below.

Definition Core.KObsSettle (post-L3 observation-only K-facing settling).

For a phase-bearing source proof model $M \in \{\text{Sys_ref}, \text{Sys_K}\}$ and its explicit lowered/elaborated variants, after the selected L3 boundary-action set for cycle t has been closed and every selected A3/A4

decision/commit in that set has completed, but before the meta-level advance to L4, designate the distinguished observation-only subphase of L3 as L3Obs and let $\rightarrow_{\epsilon, \text{KObs}}$ denote the phase-L3Obs ϵ subrelation used to form a K-facing source observation. Its roots are the actual post-L3 architectural state: the cycle index and selected cycle-t environment inputs; process/control and result state fixed for the remainder of the cycle; explicit channel/storage state after the selected L3 commits; surviving operation-attributive endpoint requests and payloads; and the current reset epoch. During L3Obs, all ordinary L1 actions, additional L3 actions/decisions, and every other source ϵ transition outside $\rightarrow_{\epsilon, \text{KObs}}$ are phase-disabled; the only phase-legal ϵ work is the observation-network evaluation described here. The relation re-evaluates only cycle-local derived boundary/observation values—such as ready/valid, guards, joins/couplers, ChannelEnabled/ChannelDisabled, and release-support values—using the same finite deterministic stateless boundary network qualified by Core.QSync/Q.1, but with those post-L3 roots fixed. It emits no Σ -visible or proof-internal observable unit; initiates, resolves, or retires no witness-significant non-blocking attempt/decision or deferred hole; makes no witness-significant arbitration, signal, failed-NB, or other ϵ choice retained by Core.18a; creates, removes, or reattributes no blocking request; changes no reset epoch; and performs no L4 or other next-cycle registered update.

The selected L3 action set is irrevocably closed before $\rightarrow_{\epsilon, \text{KObs}}$ begins. A value recomputed by this relation is observation-only: it may not enable, select, cancel, or reorder an additional same-cycle L3 action; it may not cause J.BufSameCycle to re-evaluate an intervening MirrorAvail value; and it may not create fall-through behavior forbidden by E4. Thus Core.KObsSettle is not a second scheduling settle and does not alter the L2 snapshot from which the cycle-t L3 action set was selected. $\text{KObsEvalPoint_I}(t, \sigma)$ holds when σ is the first fixed point of $\rightarrow_{\epsilon, \text{KObs}}$ from the exported post-boundary J.GWR state. Every KObsEvalPoint is a BoundaryEvalPoint . Define $\text{Core.KObsNormRef_I}(\tau_ref^G, v_ref, \sigma_ref)$ to mean $\text{NormRef_I}(\tau_ref^G, v_ref, \sigma_ref)$ together with the requirement that v_ref consists only of $\rightarrow_{\epsilon, \text{KObs}}$ steps, begins at the post-L3 endpoint of τ_ref^G in L3Obs after L3 selection is closed, and ends at $\text{KObsEvalPoint_I}(t, \sigma_ref)$ while L3Obs remains the active phase index. Write $\tau_ref^K := \tau_ref^G \cdot v_ref$ for the resulting exact K-facing source prefix. Core.QSync-Ref continues to provide the L2 settle used by J.RegPhaseNF/J.UnitExt to select L3 actions; Core.KObsNormRef is the distinct post-boundary observation normalization used when Corollary J.1 needs a later K-facing source cutpoint. Define $\text{KObsSettleQualified_M}(I)$ to hold when, for every reachable post-L3 construction endpoint used by the K-facing source route: (KSQ1) the same declared finite stateless boundary-node equations, ownership roots, and dependency topology qualified for the instance remain valid when re-instantiated from the actual post-L3 architectural/request roots; (KSQ2) $\rightarrow_{\epsilon, \text{KObs}}$ is complete for every ϵ transition phase-legal in L3Obs and every non-observation source ϵ /action transition is phase-disabled there; (KSQ3) the resulting observation network is deterministic and finite acyclic, so its fixed point exists and is order independent; and (KSQ4) the closed-L3/no-feedback rule above is enforced. This is a K-facing qualification obligation, not part of the safety-only Core.QSync premise.

Definition Core.KCut (K-facing normalized cutpoint domain).

Fix a theorem instance I and the explicit abstract boundaries/elaboration applicable to the model being evaluated. For any proof model M used by Appendix K, $\text{KCut_M}, I(s)$ holds when s is reachable, Quiescent_M(s) holds, and every explicit boundary used for BoundaryReq/ChannelEnabled/ChannelDisabled has reached its selected combinational fixed point. Write $\text{KCutReach_M}(I)$ for the reachable domain. For the prescribed source reference model use the aliases $\text{KCut_ref}, I$ and $\text{KCutReach_ref}(I)$; for RTL_B use $\text{KCut_post}, I$ and $\text{KCutReach_post}(I)$. In Appendices J–K, “K-relevant cutpoint” means a member of the appropriate KCutReach_M domain. For a reachable RTL_B prefix τ_post , $\text{NormPost_I}(\tau_post, v_post, p_post)$ abbreviates $\text{Norm_RTL_B}(\tau_post, v_post, p_post) \wedge \text{KCut_post}, I(p_post)$; the analogous generic prescribed-source normalization is NormRef_I . For a

phase-bearing source model, a KCut may be the L2 SettlePoint used to select L3 actions or a later post-L3 KObsEvalPoint formed by Core.KObsSettle; the latter is an observation-only fixed point and does not reopen L3 selection. By Lemma Core.SettleIsKCUT, every source-side Sys_ref/Sys_K L2 SettlePoint is itself a K-facing cutpoint. In the Corollary-J.1/K-facing path from a J.GWR post-boundary endpoint, the specialized source normalization is Core.KObsNormRef rather than an implicit identification of that endpoint with the earlier L2 SettlePoint. RTL_B has no L0–L4 phases; its KCut existence remains expressed through Core.KCutClosure_post, which Theorem Core.QSync-Post derives from the same qualified synchronous-model discipline in the standard scope.

Lemma Core.SettleIsKCUT (source L2 settle implies K-facing cutpoint).

For each phase-bearing source proof model $M \in \{\text{Sys_ref}, \text{Sys_K}\}$ and its explicit lowered/elaborated variants, $\text{SettlePoint_I}(t, \sigma)$ implies $\text{KCUT_M}(I, \sigma)$. Proof. The SettlePoint is reachable from the selected L1 endpoint. At L2, all L1-only ϵ /evaluation/issue steps are phase-disabled; by Remark Core.1a the phase-advance marker is not an ϵ -step; and by Definition Core.Settle, $\rightarrow_{\epsilon, L2}$ contains every ϵ -step that remains phase-legal at L2 and has reached its fixed point. Hence Quiescent_M(σ). Definition Core.Settle also gives BoundaryEvalPoint_I(σ), so the remaining Core.KCut condition holds. $\square$

Definition Core.CycleState (model-indexed cycle-entry theorem state).

For a proof model M and theorem instance I , Core.CycleState_M, I is the full cycle-entry theorem state: registered/internal machine state, time-aware environment state, global cycle index, and every cycle-valued auxiliary or monitor component declared by I . For RTL_B, the K-origin used by Core.QSync and Appendix J is not a second unrelated state type: it is a same-cycle registered/input/request frame associated with such a Core.CycleState after the cycle's registered/FSM/pipeline state and process-owned registered request outputs have been established. Definition Core.RTLKOriginFrame below types that bridge explicitly.

Definition Core.RTLKOriginFrame (typed RTL cycle-state/K-origin bridge).

Fix theorem instance I . For $s_t \in \text{Core.CycleState_RTL_B}, I$ with cycle index t , $\text{RTLKOriginFrame_I}(s_t, o_t, e_t)$ holds when o_t is the reachable RTL K-origin associated with that same registered cycle state and e_t is the complete theorem-instance reset-epoch valuation at that origin. The epoch valuation is a finite vector over every process/channel/reset scope whose epoch is modeled by I ; a design with one global reset epoch is the one-component special case. Require: (OF1) o_t has cycle index t and is reached from the registered cycle-entry state represented by s_t without any cycle-advancing registered update; (OF2) o_t carries the same registered/FSM/pipeline state, time-aware environment state and selected cycle- t exogenous inputs, explicit-channel registered storage/occupancy/credit, and cycle-valued auxiliary/monitor state as s_t , while exposing the process-owned registered request/payload drives and other fixed frame values from which K-facing same-cycle settling begins; (OF3) e_t is exactly the reset-epoch valuation after all preceding real cycles and before any RESET unit attributed to cycle t ; and (OF4) no selected cycle- t complete observable/proof unit or next-cycle registered update has occurred between the state represented by s_t and o_t . If the concrete RTL semantic state already contains all request/frame values, o_t may be definitionally the same state as s_t . This is a typing bridge only: it does not assert a successor cycle from an arbitrary state, introduce a new hardware transition, or impose uniqueness by itself. Definition Core.QSync/QS2a below supplies the standard-scope uniqueness/recoverability condition.

Definition Core.RTLDesignatedOriginFrame (designated standard-QSync RTL K-origin).

Fix theorem instance I . $\text{RTLDesignatedOriginFrame_I}(s_t, o_t, e_t)$ holds when $\text{RTLKOriginFrame_I}(s_t, o_t, e_t)$ holds and o_t is the single theorem-instance-designated state at the declared cycle- t registered/input/request frame boundary from which the complete RTL K-facing

Core.QSync settling relation begins. The designation is part of the fixed RTL operational/QSync boundary mapping, not merely the fact that OF1–OF4 happen to hold: a same-cycle intermediate state may be compatible with the broad RTLKOriginFrame relation without being designated. For the fixed registered cycle state s_t and reset-epoch valuation e_t , at most one o_t is designated. In the standard QSync scope the concrete RTL/QSync qualification shall establish existence of this designated origin for every reachable cycle frame used by the theorem and shall establish the recoverability/uniqueness facts of QS2/QS2a. This definition does not assert that a successor cycle exists from an arbitrary designated origin; it only selects the canonical pre-settle origin view used by Core.QSync, Core.RTLCycleStep, and the carrier chain below.

Hardware interpretation of the qualified synchronous model (informative). At the architectural level, the intended pre-HLS and post-HLS comparison models are ordinary single-clock, cycle-level synchronous hardware systems. At the beginning of cycle t , registered process, pipeline, and explicit-channel state is fixed. Sequential processes determine their cycle- t request, payload, and other process-owned outputs from that registered state and from operands fixed for the cycle; they do not provide a same-cycle combinational through-path from a value newly returned or sampled at an interface in cycle t to a dependent process-owned output in that same cycle. Ordinary combinational derivation of outputs from registered state is permitted. The complete system-level same-cycle network—including R2C combinational processes, ready/valid and enablement logic, and stateless elaboration logic—then settles before the boundary action to one deterministic valuation. Pure combinational feedback is outside the standard theorem scope; feedback that affects later behavior passes through explicit registered or finite-capacity state. Boundary actions are selected from the settled valuation, after which the next registered state is formed.

Relation to the proof vocabulary. Definition Core.QSync formalizes the qualified same-cycle network rather than introducing an additional hardware mechanism. On the phase-bearing source side, Core.Phase presents the scheduling cycle as L1 reach/issue, L2 settle, L3 boundary commit/decision, and L4 registered update; RTL_B need not expose those proof phases. SettlePoint names the pre-L3 scheduling snapshot. BoundaryEvalPoint is the broader fixed-point notion used by the K-facing predicates: it includes both the L2 SettlePoint and a post-L3 KObsEvalPoint produced by Core.KObsSettle. Core.KCut names a reachable ϵ -quiescent BoundaryEvalPoint when it is used by the J–K proof stack. Core.KObsSettle is only an observation view of the already-selected post-L3 state and creates no second L3 scheduling opportunity. On RTL_B, Core.RTLKOriginFrame types the relation between the full registered cycle state and the same-cycle K-origin view, while Core.RTLCycleStep below types one complete physical clock transition between consecutive such origins; neither introduces a new designer-visible state or transition. These are proof views/cutpoints of the cycle-level machine, not additional designer-visible hardware states or extra physical timing phases. HLS-added silent cycles remain real clock cycles and are hidden from the selected observable behavior except where the contract declares timing significant. WC/latency-sensitive choices, Policy-L issue latitude, arbitration, and other explicitly modeled choices remain available, so this interpretation does not assert that Sys_ref has one unique execution.

Definition Core.QSync (qualified synchronous same-cycle settling model).

Fix a theorem instance I and a proof model M used by the G–K stack. For a phase-bearing source model, the model-specific same-cycle settling relation is $\rightarrow_{\epsilon, L2}$ from Definition Core.Settle. For RTL_B, it is the complete K-facing same-cycle ϵ /combinational settling relation beginning at a reachable designated K-origin of Definition Core.RTLDesignatedOriginFrame after the current cycle’s registered/FSM/pipeline state and process-owned registered request outputs have been established and ending before any cycle-advancing registered update. QualifiedSynchronousModel_M(I) holds when all of the following are true.

(QS1) Single-clock scope. There is one selected global clock/cycle relation shared by every process and explicit abstract channel inside the G–K theorem instance. The cycle indices used by K-origins, boundary evaluation, commits, and registered updates are all defined relative to that clock, and no unmodeled cross-clock-domain boundary lies inside the selected instance. The Appendix M.7a ClockScope entry records the concrete evidence for this condition.

(QS2) Registered-phase/frame separation. The same-cycle settling relation satisfies the L2/K-origin frame: the cycle index, registered process/FSM/pipeline state, registered explicit-channel storage/occupancy/credit state, and selected cycle-t exogenous inputs are invariant during settling; no Σ -visible commit and no next-cycle registered update occurs in the settling relation. Sequential process request/payload drives covered by Core.RegSeq remain fixed after their selected pre-settle value is established. R2C combinational-process outputs, and stateless elaboration logic such as couplers, are instead part of the settling network. (QS2a) RTL KCut origin recoverability. For RTL_B in the standard K-facing scope, every reachable state p_post satisfying Core.KCut_post, I belongs to one cycle-t QS2 registered/input frame and has a designated triple (s_t, o_post, e_t) satisfying Core.RTLDesignatedOriginFrame_I(s_t, o_post, e_t) for that same frame after the cycle's registered/FSM/pipeline state and process-owned registered request outputs are established and before K-facing same-cycle settling. The reachable suffix from o_post to p_post before any later cycle-advancing registered update consists only of the qualified same-cycle settling relation. The designated Core.CycleState/K-origin/epoch frame is unique for that KCut under the selected single-clock operational semantics, and the same concrete boundary mapping supplies one designated origin for each reachable registered cycle frame used by a carrier node. This clause is vacuous for the phase-bearing source models, whose L1/L2 phase boundary already names the source K-origin explicitly.

(QS3) Settling-network realization and deterministic local evaluation. The complete same-cycle settling relation is realized by the qualified settling network: every state component that may change during settling is represented by a cycle-local settling node or by a derived value of such nodes; each settling step changes only such represented values by evaluating one or more node functions from their predecessor values; every phase-legal same-cycle settling step is accounted for by this representation; and no additional settling transition exists outside it. A state is a fixed point of the settling relation iff all represented node equations are satisfied. Every combinational or other cycle-local settling node has a deterministic value for fixed predecessor values. Any state that can affect a later cycle is represented at an explicit registered/process/channel state boundary rather than being hidden inside the settling relation.

(QS4) Finite acyclic settling dependency. After cutting the network at those explicit registered/storage boundaries, the dependency graph for the complete same-cycle settling network is finite and acyclic. Pure combinational feedback is outside the standard theorem scope; required feedback shall pass through explicit state. An equivalent independently checked structural theorem establishing the same finite acyclic dependency property may be used.

(QS5) Explicit-boundary completeness. Every value used by BoundaryReq, ChannelEnabled/ChannelDisabled, rendezvous pairing, join/coupler/guard/epoch-gate evaluation, or Appendix K WFG construction at a selected boundary is produced by that qualified settling network from the fixed cycle-t state/inputs. No hidden enablement or storage predicate may bypass E4/Core.ElabProj/B-faithfulness. For Sys_B^elab(E), Appendix Q supplies the elaboration-specific obligations; for RTL_B, the selected boundary mapping and B-faithfulness evidence supply the

implementation side.

QualifiedSynchronousModel is a same-cycle structural property. It does not assert one unique Sys_ref execution: latency-sensitive/WC choices, Policy-L issue latitude, arbitration/environment choices across cycles, and other explicitly modeled choices may still admit multiple source executions. Nor does it imply B1; B1 remains the separate observable RTL trace-determinism assumption.

Theorem Core.QSync-Ref (qualified source/elaboration settling gives a unique source KCut).

For $M \in \{\text{Sys_ref}, \text{Sys_K}\}$ and its explicit lowered/elaborated variants, assume QualifiedSynchronousModel_M(I). Then every reachable source K-origin has exactly one L2 fixed-point endpoint σ_{set} in that cycle. That endpoint satisfies SettlePoint_I(t, σ_{set}); by Lemma Core.SettleIsKCut it lies in KCutReach_M(I). Hence KCutClosure_M(I) holds, and a NormRef that begins at that L1 K-origin may select this unique settled endpoint canonically. This theorem does not identify a later J.GWR post-L3 endpoint with σ_{set} and does not normalize such a post-boundary endpoint; the K-facing observation normalization from a J.GWR endpoint is Core.KObsNormRef/Core.KObsSettle defined above.

Proof sketch. QS2 freezes registered state and cycle-t inputs and excludes $\Sigma/L4$ updates during settling. QS3 realizes the complete settling relation as evaluation of the represented cycle-local nodes, makes each node functional, and identifies relation fixed points with satisfaction of all node equations. QS4 gives a finite acyclic dependency order, so evaluation terminates and produces one order-independent valuation. QS5 makes that valuation complete for every K-facing boundary predicate. Therefore the complete $\rightarrow_{\epsilon, L2}$ relation reaches one fixed point; Lemma Core.SettleIsKCut supplies the KCut conclusion. In the elaborated case, Appendix Q supplies or cites the settling facts for the elaboration-owned template subnetwork. Those template-side facts are not by themselves the whole-source QS2–QS5 discharge: the concrete theorem instance must also establish the applicable design-owned source-settling facts and a checked template/design settling-composition condition, including QS4 for the complete combined same-cycle settling network. Definition Core.Settle alone is not used as an existence axiom.

Lemma Core.KObsSettle-Ref (qualified post-L3 observation settling).

Assume QualifiedSynchronousModel_M(I) and KObsSettleQualified_M(I) for a phase-bearing source model M, and let s_{post} be a reachable exported post-boundary state after the selected L3 action set is closed and before L4. Hold the post-L3 architectural/request/environment roots named by Definition Core.KObsSettle fixed and re-evaluate the declared observation network. The ordinary Core.QSync/Q.1 qualification supplies the typed node equations used by the instance, while KSQ1–KSQ3 establish that those equations/topology are valid and complete under the actual post-L3 roots and give a finite deterministic acyclic evaluation order. Therefore $\rightarrow_{\epsilon, \text{KObs}}$ reaches one finite order-independent fixed point σ_K before L4. By KSQ2 every other source ϵ transition is phase-disabled in this interval, so that fixed point is Quiescent_M; it is reachable, its explicit boundary network is fixed, and hence $\sigma_K \in \text{KCutReach}_M(I)$. Thus for any reachable $\tau^{\wedge}G$ ending at s_{post} , the finite observation path v_K satisfies Core.KObsNormRef_I($\tau^{\wedge}G, v_K, \sigma_K$). KSQ4 makes the conclusion observation-only: the lemma does not reopen the closed L3 action set, permit a second same-cycle transfer, or alter J.BufSameCycle/E4 frozen-snapshot semantics. $\square$

Theorem Core.QSync-Post (qualified RTL settling gives a unique post KCut and KCut closure).

Assume QualifiedSynchronousModel_{RTL_B}(I). Then every reachable designated RTL K-origin o_t represented by Core.RTLDesignatedOriginFrame_I(s_t, o_t, e_t) has exactly one same-cycle settled endpoint ρ_{post} before the next cycle-advancing registered update. There is a finite ϵ /combinational

normalization from that origin to p_post , $p_post \in KCutReach_post(I)$, and therefore $Core.KCutClosure_post(I)$ holds. The theorem also supplies a canonical K-facing NormPost normalization/endpoint for that origin. With $K\epsilon$, Lemma $Core.KCutObserve$ then gives the existing persistent-blocker observability consequence.

This canonicalization is limited to K-facing same-cycle RTL settling. It does not eliminate Appendix I/J $ReplayNormalizationConvention$ or the selected source replay-normalization/congruence machinery, which may include witness-fixed replay-changing ϵ behavior and preparation steps unrelated to combinational settling. In particular, $Core.QSync_Post$ does not prove that a later clock cycle exists, that a temporarily idle RTL state advances to the next registered state, or that a cross-cycle RTL execution satisfies B2/B3. Lemma $Core.RTLCycleCarrier$ below is separate: it supplies only the carrier/completion fact for a legal RTL real-cycle transition that a reachable prefix has already entered; it is not a progress theorem from an arbitrary K-origin. The finite-prefix Appendix J correspondence and Appendix K RTL cutpoint-closure results do not require a generic RTL fair-extension premise; any later theorem that needs complete RTL continuation must state its own continuation/progress assumptions.

Lemma $Core.QSync_KCutCanonical$ (every reachable post $KCut$ is the canonical endpoint of its K-origin).

Assume $QualifiedSynchronousModel_RTL_B(I)$. Let $p_post \in KCutReach_post(I)$ in the standard K-facing theorem scope. By QS2a, p_post determines a unique cycle-t $Core.RTLDesignatedOriginFrame_I(s_t, o_post, e_t)$, including its designated reachable K-origin o_post and reset-epoch valuation, and the reachable suffix from o_post to p_post before any later registered update consists only of the qualified same-cycle settling relation. Then p_post is exactly the unique settled endpoint selected for o_post by Theorem $Core.QSync_Post$. Consequently, in the standard $QSync$ scope, $KCutReach_post(I)$ is the image of the reachable designated RTL K-origin domain under the canonical $QSync_Post$ endpoint map; every member of $KCutReach_post(I)$ therefore admits the canonical finite NormPost witness supplied by $Core.QSync_Post$. Proof. QS2a supplies the origin/frame recovery that is not implied by $Core.KCut$ quiescence alone. Because p_post is a $KCut$ reached by settling from that designated origin, it is ϵ /combinationally quiescent at the selected explicit boundaries in the fixed QS2 frame. QS3 identifies such a fixed point with satisfaction of the complete settling-network equations, and QS4 makes that fixed point unique. $Core.QSync_Post$ produces the unique fixed point for o_post , so p_post must be that endpoint. The argument is same-cycle only and adds no cross-cycle progress or fairness premise. $\square$

Definition $Core.CycleResult$, $CompleteCycle$, and $SilentCycle$.

For a proof model M and theorem instance I , $CycleResult_M, I(s_t, r)$ is a typed legal result $r = \langle nextState, committedUnits, resetUnits \rangle$. The $nextState$ field is a $Core.CycleState_M, I$ at cycle $t+1$, so the global clock advances exactly once; its environment component is related to the cycle-t environment by $EnvStep_I$ using the applicable canonical Σ -causal history, and its auxiliary/monitor components advance according to their definitions. In particular, Appendix K's $K.Resp$ age is part of the cycle-valued auxiliary state and advances in $nextState$ on each selected Policy-S cycle while its monitor remains pending. $committedUnits$ is the finite collection of selected complete observable/proof units produced in cycle t at the boundary of proof model M , including that model's Σ commits and START/FINISH and emitted R2C fixed-point units when those are selected units of the theorem instance. This inventory is model-relative and precedes any later outer-history projection: for $Sys_ref/Sys_K = Sys_B^{elab}(E)$, complete commits at explicit proof-internal abstract channel boundaries remain in $committedUnits$ even when Π_elab/Π_DUT later erases them from Σ_DUT . Projection changes only the retained outer history, not the typed per-cycle unit inventory or its finiteness. $resetUnits$ is the finite collection of dynamic proof-internal

RESET(S_p, S_c) occurrences in that cycle; each occurrence carries its reset scope and corresponds exactly to the state-changing reset transition governed by RW1-RW5. For a phase-bearing source model, a legal result is witnessed by a finite L1 preparation/issue segment, finite L2 settling, the selected legal L3 boundary actions/decisions and reset boundary units, and the L4 registered/result update. For RTL_B it is witnessed by the model-specific real clock transition through its K-facing same-cycle settling/boundary behavior and next registered update; RTL_B is not assigned source L0-L4 phases. CompleteCycle_M, I(s_t, s_{t+1}) holds exactly when there exists r with CycleResult_M, I(s_t, r) and $r.nextState = s_{t+1}$. Thus CompleteCycle remains the next-state projection used by existing theorem statements; CycleResult is the primary typed object carrying the full cycle evidence. SilentCycle_M, I(s_t, s_{t+1}) holds when such a result exists with no selected Σ event in $r.committedUnits$. A SilentCycle is a real cycle, not an ϵ -step and not an identity transition: registered/internal machine state, environment state, and cycle-valued monitor state may change. A reset-only cycle may be SilentCycle with respect to the selected outer Σ alphabet when RESET is proof-internal/projected away; that does not make the cycle event-free, because the corresponding occurrence remains in $r.resetUnits$ and in the proof-internal history. RESET is therefore not an alternative cycle outcome: a partial reset changes only its named scope while the unaffected state still proceeds to the same result's nextState under RW5. When the fixed semantics permits more than one reset occurrence in one cycle, resetUnits retains each dynamic occurrence and only the order required by RW1-RW5 and $\prec_J$ is imposed; independent disjoint resets are not artificially serialized. Lemma J.RS continues to consume the emitted RESET unit itself, not the enclosing cycle. Persistent blocking requests follow Core.IC: once issued, an outstanding blocking dynamic instance remains asserted and attributed until its process-facing commit or an affecting RESET governed by RW2; a DeclaredTerminal state may instead end the execution while the request remains pending. No other nonterminal transition in the G-K theorem model may deassert, retire, reattribute, cancel, or replace that outstanding blocking instance. Protocols with independent nonterminal cancellation, withdrawal, or retry require a separately defined operational extension and are outside this theorem stack. An activated Core.16 no-new-later-work restriction remains in force while its fully disabled frontier persists.

Definition Core.RTLCycleStep (typed one-real-cycle K-origin step).

Fix theorem instance I. RTLCycleStep_I($\kappa_t, s_t, o_t, e_t, r_t, \kappa_{t+1}, s_{t+1}, o_{t+1}, e_{t+1}$) holds when the following one-cycle data are bound to one legal RTL_B execution carrier. (RS1) κ_t and κ_{t+1} are finite legal RTL_B prefixes on that carrier, κ_t ends exactly at o_t , κ_{t+1} ends exactly at o_{t+1} , and $\kappa_t \leq \kappa_{t+1}$. (RS2) RTLDesignatedOriginFrame_I(s_t, o_t, e_t) and RTLDesignatedOriginFrame_I($s_{t+1}, o_{t+1}, e_{t+1}$) hold. (RS3) s_t is not DeclaredTerminal_RTL_B, I, CycleResult_RTL_B, I(s_t, r_t) holds, $r_t.nextState = s_{t+1}$, s_t/o_t have cycle index t , and s_{t+1}/o_{t+1} have cycle index $t+1$. Thus the step advances exactly one real clock cycle even when no selected outer unit is produced. (RS4) The complete Core.17 proof-unit delta between κ_t and κ_{t+1} is exactly the cycle- t content recorded by r_t : every ordinary complete unit appears exactly once in $r_t.committedUnits$ and every dynamic scoped RESET appears exactly once in $r_t.resetUnits$; no complete cycle- t unit is omitted, duplicated, or attributed to another cycle. Permitted same-cycle settling/non-unit microsteps may occur between the endpoints but add no complete unit. (RS5) e_{t+1} is the reset-epoch valuation obtained from e_t by applying exactly $r_t.resetUnits$ under RW1-RW5: every affected scope enters its prescribed fresh epoch and every RW5-framed unaffected scope retains its prior epoch value; when multiple reset occurrences are legal, only their normative RW1-RW5/reset-boundary order is imposed. (RS6) κ_t and κ_{t+1} are consecutive designated K-origin prefixes for this real cycle: there is no distinct designated RTL K-origin strictly between them on the carrier. If s_{t+1} is DeclaredTerminal, o_{t+1} is still the final successor designated K-origin produced by this step; Definition Core.DeclaredTerminal forbids any later ordinary RTLCycleStep from that state. The canonical same-

cycle $KCut\ p_t$ is intentionally not stored in this record: under $QualifiedSynchronousModel_RTL_B(I)$, Theorem $Core.QSync-Post$ derives it uniquely from o_t . This record is a one-cycle typing object, not a cross-cycle totality, fairness, or carrier-chain premise.

Lemma $Core.RTLCycleCarrier$ (RTL finite prefixes inherit complete real-cycle carriers).

Assume $QualifiedSynchronousModel_RTL_B(I)$. In the standard RTL_B semantics, fix a finite legal RTL_B prefix κ_t ending at a reachable designated cycle- t K-origin o_t together with $RTLDesignatedOriginFrame_I(s_t, o_t, e_t)$. Let τ_{post} be a finite extension of κ_t on the same execution carrier that contains at least one complete selected observable/proof unit attributed to real cycle t ; in particular, the hypothesis holds when τ_{post} contains a retained outer unit of that cycle. Then there exist $r_t, s_{t+1}, o_{t+1}, e_{t+1}$, and a finite successor prefix κ_{t+1} such that $RTLCycleStep_I(\kappa_t, s_t, o_t, e_t, r_t, \kappa_{t+1}, s_{t+1}, o_{t+1}, e_{t+1})$ holds and $\tau_{post} \leq \kappa_{t+1}$. A DeclaredTerminal s_{t+1} may supply the final successor origin without any later step. Proof. Definition $Core.3$ gives the inherited real-cycle partition of the legal RTL execution prefix. The complete cycle- t unit already present in τ_{post} belongs to the enclosing model-specific real clock transition represented by Definition $Core.CycleResult$. Completing that already-entered transition supplies $r_t.nextState = s_{t+1}$; the fixed RTL/QSync boundary mapping establishes the unique successor designated registered/input/request origin o_{t+1} without another clock advance. Definition $Core.RTLDesignatedOriginFrame$ records the source and successor frame/epoch valuations; exact CycleResult unit accounting and RW1–RW5 give RS4–RS5; and the one-clock CycleResult together with uniqueness of the designated boundary in each frame gives RS6. No successor existence is asserted from a designated K-origin whose cycle has not already been entered. $\square$

Definition $Core.KOriginCarrier$ (finite designated RTL K-origin carrier).

Fix a standard-QSync theorem instance I . A finite $KOriginCarrier_I$ object C of length n consists of nodes $N_i = (\kappa_i, s_i, o_i, e_i)$ for $0 \leq i \leq n$ and CycleResult witnesses r_i for $0 \leq i < n$, all bound to one legal RTL_B execution carrier, satisfying: (KC0) N_0 is the initial/reset-established node on this execution carrier: κ_0 is that carrier's exact finite prefix from the selected RTL initial state through its first designated K-origin, s_0 is the corresponding initial/reset-established $Core.CycleState$, $RTLDesignatedOriginFrame_I(s_0, o_0, e_0)$ holds, and e_0 is exactly the post-initialization/reset epoch valuation prescribed by I (so different reachability witnesses may not choose a different logical initial epoch merely because their internal pre-origin ϵ representation differs); (KC1) every κ_i is a reachable finite prefix ending exactly at its designated o_i and $RTLDesignatedOriginFrame_I(s_i, o_i, e_i)$ holds; (KC2) for each $i < n$, $RTLCycleStep_I(\kappa_i, s_i, o_i, e_i, r_i, \kappa_{i+1}, s_{i+1}, o_{i+1}, e_{i+1})$ holds; (KC3) the successor fields of step i are the actual node $i+1$ fields, so the prefix, full state, designated origin, and reset-epoch valuation are shared exactly rather than existentially reselected, and $\kappa_0 \leq \kappa_1 \leq \dots \leq \kappa_n$; and (KC4) if $DeclaredTerminal_RTL_B, I(s_i)$ holds then $i = n$. The final node need not be terminal and the definition imposes no obligation to start a later real cycle from it. The canonical K-facing settled state p_i is deliberately derived, not stored: when needed, $Core.QSync-Post$ maps o_i to its unique same-cycle NormPost/KCut endpoint. Thus $KOriginCarrier$ is only the finite historical cycle/origin decomposition already present in a reachable RTL execution, not a fair-extension or cross-cycle progress assumption.

Lemma $Core.KOriginCarrier-Extend$ (append one typed real cycle).

Assume $QualifiedSynchronousModel_RTL_B(I)$. Let C be a $KOriginCarrier_I$ object whose final node is $N_n = (\kappa_n, s_n, o_n, e_n)$, assume s_n is not DeclaredTerminal $_RTL_B, I$, and let a reachable $RTLCycleStep_I(\kappa_n, s_n, o_n, e_n, r_n, \kappa_{n+1}, s_{n+1}, o_{n+1}, e_{n+1})$ be supplied on the same execution carrier. Then appending that step and successor node yields a $KOriginCarrier_I$ object of length $n+1$. Proof. KC0 is unchanged; RS1–RS6 give KC1–KC3 for the appended edge/node; the explicit

nonterminal premise preserves KC4 at the old final node, and the new final node is permitted to be terminal or nonterminal. $\square$

Lemma Core.KOriginCarrier-AtOrigin (every reachable designated origin has a finite carrier).

Assume QualifiedSynchronousModel_RTL_B(I). Let κ be a finite reachable RTL_B prefix ending exactly at a designated K-origin o with $\text{RTLDesignatedOriginFrame}_I(s,o,e)$. Then there exists a finite KOriginCarrier_I object whose final node is exactly (κ,s,o,e) . Proof. Definition Core.3 supplies the inherited finite real-cycle partition of the reachability witness from the selected initial/reset state to κ . The first declared registered/input/request boundary gives the KC0 node and its prescribed e_0 . Every completed real cycle before κ is already present in that finite reachability witness, so Definition Core.CycleResult supplies its typed result, exact ordinary/reset unit content, and nextState; the fixed designated-origin mapping supplies the unique successor node in the same reachability witness. For each completed cycle, the one-clock Core.CycleResult together with uniqueness of the designated frame boundary for its source and successor cycle excludes any intervening designated K-origin, so RS6 holds. Package each completed cycle as Core.RTLCycleStep. RS4 accounts for every complete unit exactly once and RS5 threads the reset-epoch vector without re-selection. Finite iteration over those completed real cycles yields C. No successor is postulated beyond κ . $\square$

Lemma Core.KOriginCarrier-Decomp (arbitrary finite RTL prefixes have a carrier-relative cycle position).

Assume QualifiedSynchronousModel_RTL_B(I), and let τ be any finite reachable RTL_B prefix in the standard QSync scope with its selected finite reachability carrier. Let $N_0=(\kappa_0,s_0,o_0,e_0)$ be that carrier's first designated origin node with the KC0 initial/reset-established state and prescribed e_0 valuation. Exactly one of the following positional cases applies. (KD0) τ is a strict prefix of κ_0 ; τ therefore ends before the first designated K-origin on this reachability carrier. (KD1) There exist a finite KOriginCarrier_I object C with this same base and final node $N_t=(\kappa_t,s_t,o_t,e_t)$, with $\kappa_t \leq \tau$, such that no designated K-origin prefix κ' on the same reachability carrier satisfies $\kappa_t < \kappa' \leq \tau$. Thus κ_t is the greatest designated K-origin prefix at or before τ and is a direct projection of C, not a separately assumed state. If τ itself ends at a designated K-origin, then $\tau=\kappa_t$. Proof. The reachability witness is finite and inherits the real-cycle partition of Core.3. Under the assumed QualifiedSynchronousModel_RTL_B(I), the designated-origin existence obligation of Definition Core.RTLDesignatedOriginFrame supplies the first designated frame boundary and hence the KC0 node $N_0=(\kappa_0,s_0,o_0,e_0)$ with the prescribed initial/reset-established state and e_0 valuation. Before that first designated frame boundary KD0 holds. Otherwise the finite set of designated origin boundaries already passed by τ is nonempty and linearly ordered by exact prefix inclusion/cycle index; choose its last member. Lemma Core.KOriginCarrier-AtOrigin supplies C ending at that member, and maximality gives the no-intermediate-origin clause. $\square$

Corollary Core.KOriginCarrier-KCut (reachable post KCut has a carrier/origin representation).

Assume QualifiedSynchronousModel_RTL_B(I). Let $p_post \in \text{KCUTReach_post}(I)$, and take any finite reachability witness prefix ending at p_post . Lemma Core.QSync-KCutCanonical/QS2a recovers its unique designated cycle-t origin triple (s_t,o_t,e_t) and the exact subprefix κ_t of that witness ending at o_t . Lemma Core.KOriginCarrier-AtOrigin then supplies a finite KOriginCarrier_I object whose final node is (κ_t,s_t,o_t,e_t) , while Core.QSync-Post supplies the canonical v_post/p_post normalization from o_t . Thus a reachable KCut has a carrier-indexed representation; a bare state value alone is not the constructive input used by Appendix J certificate extraction. $\square$

Definition Core.DeclaredTerminal (modeled terminal-state predicate)

DeclaredTerminal_M,I(s) remains a predicate on the full Core.CycleState, not a field or alternative constructor of CycleResult. It means that the selected machine/test contract explicitly declares the execution terminated, aborted, or at end-of-test at s and that this disposition is absorbing for the ordinary cycle semantics: no ordinary CycleResult_M,I(s,r), CompleteCycle_M,I(s,s'), or other ordinary cycle successor from s is legal. A finite execution may therefore end at s. Only a disposition modeled as part of the selected theorem instance's machine/test semantics counts as DeclaredTerminal. An external verification-harness watchdog that merely stops host simulation or reports a timeout, without being a modeled transition or termination disposition of M, is not DeclaredTerminal and does not truncate Core.CycleResult or Core.CompleteCycle. A modeled cycle-count, watchdog, abort, or end-of-test rule that is part of the selected machine/test semantics does count. Mere absence of a currently enabled Σ -action or ϵ -step is not DeclaredTerminal when the time-aware environment or registered machine can evolve on a later cycle. If a theorem extends a terminal execution indefinitely under the separate B6 idle-tick convention, that extension additionally requires Core.TerminalIdle and is not an ordinary CycleResult/CompleteCycle continuation from s.

Definition Core.TerminalIdle (permanent terminal-idle eligibility)

TerminalIdle_M,I(s,H,t,e) holds exactly when s is the stronger B6-eligible terminal-idle state: s is a genuine B5/global fixed point and every future machine/environment evolution permitted by the fixed theorem instance preserves the no-future-enabling condition, so no registered/internal update and no environment step can change non-clock state or re-enable work. Core.EnvTerminal below supplies the environment component of this obligation; the machine-side permanent-idle fact must also be established. A stable DeclaredTerminal state is one sufficient machine-side route, but B6 may also apply to a nonterminating service model proved permanently idle after stimulus exhaustion.

Definition Core.EnvStep and Core.EnvTerminal (time-aware environment evolution).

Let EnvStep_I(H,t,e,e') be the fixed theorem-instance environment transition from environment state e at cycle t and canonical Σ -causal history H to the next-cycle environment state e'. Every legal CycleResult uses this relation for the environment component of nextState. The requirement that a compatible next-cycle environment evolution exist at every reachable nonterminal source cycle-entry state is folded into Core.CycleClosure below rather than exposed as a separate environment-totality premise. The cycle/environment state is part of the relation: identical Σ history at two different cycles need not imply identical future availability. Core.EnvTerminal_I(H,t,e) is the environment component of Core.TerminalIdle: every future environment evolution allowed by the fixed instance preserves the no-future-enabling condition required for an indefinite B6 idle suffix. StandardEnv stimulus exhaustion is one sufficient environment-side discharge route. A timed environment may satisfy Core.CycleClosure's environment-definedness clause while delivering a new input later and therefore fail EnvTerminal at the earlier idle cycle.

Definition Core.CycleClosure (source full-cycle qualification record).

For a selected source theorem instance I, Core.CycleClosure_I is the normative full-cycle qualification record used whenever a theorem constructs a complete/fair source continuation, including Lemma K.2b.E. It packages the following obligations. (C1) Full-state cycle typing: every legal CycleResult has a nextState with cycle index advanced exactly once, environment state related by EnvStep_I, every declared cycle-valued auxiliary/monitor component advanced according to its semantics, and committedUnits/resetUnits consistent with the selected source transition semantics. Each resetUnits occurrence corresponds exactly to one scoped state-changing RESET transition and satisfies RW1-RW5; no reset is represented as an alternative to the enclosing cycle result. (C2) Nonterminal source-cycle

existence, finite phase execution, and partial-cycle extension: every reachable nonterminal source cycle-entry state admits at least one legal CycleResult / CompleteCycle successor; its L1 part reaches the selected finite, not necessarily maximal, preparation/issue endpoint under Core.ReachPrep/Core.Phase and its L2 part reaches the required settled endpoint before L3/L4. The selected L1 issue segment may be empty when no operational rule requires an L1 action. In addition, every finite phase-legal partial source-cycle segment selected from the operational transition relation—respecting Core.Phase/Core.16, the fixed reset contract, and an EnvStep-compatible environment choice—admits a legal completion to a CycleResult whose phase projection contains that selected segment in the chosen order. This extension clause is what permits a later proof to complete a finite phase-legal issue/boundary selection to a typed complete cycle; bare successor existence is insufficient. Completion to a CycleResult does not impose maximal L3 selection: any otherwise legal nonselection remains governed by Core.SelectLatitude. It does not itself construct a fair selection or prove that a proposed issue/L3 combination is jointly phase-legal. When Lemma K.2b.E constructs a complete fair continuation, the existing Core.IssueFair and B2/B3 discharges are therefore supplied in the constructive Core.FairPick form below and combined with C2 by Lemma Core.FairCycleDovetail. The standard discharge combines finite source-control/FPD evidence for L1 with Core.QSync-Ref for L2 and an operational cycle-completion theorem, or an allowed equivalent source-cycle theorem. (C3) Environment definedness: at every reachable nonterminal source cycle-entry state, the fixed time-aware environment supplies at least one EnvStep compatible with the fixed stimulus/history and with some legal CycleResult. The former standalone source-cycle-closure and environment-totality obligations are exactly the source-cycle-existence/finiteness and environment-definedness clauses of this record and are no longer independent premises. Core.CycleClosure is an existence/typing qualification, not a fairness premise: Core.IssueFair and B2/B3/B5 remain separate where a complete/fair continuation is required. No generic RTL cross-cycle totality/fair-extension premise is part of Core.CycleClosure: finite-prefix Appendix J uses Reach_post, while the universal Appendix K RTL cutpoint result uses Core.QSync-Post/Core.KCutClosure_post. Any infinite B6 suffix is governed separately by Core.TerminalIdle and B6.

Lemma Core.CycleChoiceExt (phase-legal source-cycle selection extends to a CycleResult).

Assume Core.CycleClosure_I. Let s be a reachable nonterminal source cycle-entry state and let α be a finite phase-legal partial source-cycle segment from s whose steps are operationally legal in sequence, satisfy Core.Phase/Core.16, respect the selected reset contract, and are compatible with a selected EnvStep_I environment evolution. Then clause C2 supplies a legal CycleResult r from s whose phase projection extends α and completes the remaining L2/L3/L4 work required by the source semantics. Consequently, a later proof may extend any already-constructed finite phase-legal selection, but it may invoke dependent choice only after obtaining a typed CycleResult for each selected cycle. This lemma neither constructs the fair selection nor proves that independently proposed L1 and L3 choices compose legally; Core.IssueFair and B2/B3 remain separate obligations. Definition Core.FairPick and Lemma Core.FairCycleDovetail below provide the constructive bridge used by Appendix K. $\square$

Definition Core.FairPick (constructive discharge of source issue/boundary fairness).

Fix a phase-bearing source theorem instance I and issue policy Π . Core.FairPick_I, Π is the constructive witness form used when a theorem must build, rather than merely assume, a complete execution satisfying Core.IssueFair and B2/B3. It does not add a new behavioral fairness property to Core.LAdm or Exec_L^adm; instead it packages prefix-local selection functions and the starvation/progress evidence needed by the particular constructed execution. The witness contains the following. (FP1) Issue selection. At each reachable nonterminal cycle-entry prefix, IssuePick returns a finite phase-legal L1 segment α_{issue} (possibly empty). Any fairness-selected issue in α_{issue} is a currently IssueEnabled_ Π operation; α_{issue} may also contain permitted preparation or other operationally required L1 work. The

whole segment is operationally legal in sequence, respects Core.Phase/Core.16 and source-order prefix closure, uses the fixed witness/reset/scheduling semantics, and is compatible with at least one EnvStep_I environment evolution for that cycle, so that Core.CycleChoiceExt is applicable. (FP2) Boundary selection. After an actual finite L2 settling prefix of that same cycle has reached its SettlePoint, BoundaryPick returns a finite phase-legal L3 segment γ_bdy (possibly empty). Any fairness-selected boundary action in γ_bdy is a currently B2-enabled Push/Pop/rendezvous/Sync action. The combined prefix through L2 followed by γ_bdy is operationally legal in sequence, respects E4, Core.SyncFence, the fixed arbitration/reset semantics and endpoint cardinality, and is compatible with at least one EnvStep_I evolution under the fixed environment semantics. Any same-cycle predecessor commits required by Core.SyncFence are included in γ_bdy before the selected Sync. (FP3) Local finiteness and stable obligation identity. At every finite operational prefix, the currently live issue and boundary obligations presented to the selectors are finite and have stable dynamic identities until issue/commit, RESET, or DeclaredTerminal. In the standard hardware scope this follows from the finite process and explicit-boundary sets together with finite generation in every finite source prefix and the endpoint-cardinality rules; an equivalent checked locally-finite enumeration is allowed. (FP4) Starvation freedom and fixed-choice discipline. Along any nonterminal suffix within one applicable reset epoch, repeated per-cycle IssuePick selections eventually select every operation that remains IssueEnabled_I at every applicable L1 opportunity, and repeated BoundaryPick selections eventually select every B2 action whose enabling predicate remains continuously true at the applicable settled boundary opportunities. Where the selected Policy-L operational semantics leave issue-selection latitude proof-selectable, an oldest-enabled/round-robin queue over the finite live obligations, with consecutive-enable age reset when an obligation ceases to be enabled or is discharged, is a sufficient construction. Instance-fixed choices—including any fixed scheduler component, arbitration/grant choice, environment evolution, witness outcome, reset behavior, or other choice bound by Core.LLegal/I—are not proof-selectable: IssuePick/BoundaryPick must respect those actual choices and the discharge must prove their required starvation-freedom. BoundaryPick may not replace a fixed unfair arbiter with an invented fair one. (FP4a) Certified finite-segment progress. When a consuming theorem needs finite preparation/materialization progress in the execution it is constructing, its FairPick discharge may additionally bind a prefix-local segment-designation rule SegReq before dependent choice begins. SegReq is a fixed deterministic transition rule on the current finite prefix together with its proof-construction active-segment state and theorem-instance artifacts already bound by the application; it is not a choice made later by a downstream proof. When no top-level obligation is active it returns either no obligation or one certified top-level finite operational segment obligation. Once a top-level obligation is active, re-evaluation retains that same obligation until completion or case rejection; only lower-ranked nested obligations required by its fixed certificate may be pushed, and a newly eligible unrelated obligation cannot replace it mid-segment. Segment obligations may be nested only through a separately supplied well-founded rank; the currently active obligation is the top of that stack, and completion of a nested lower-ranked obligation resumes the suspended suffix of its caller. Each segment certificate fixes a finite sequence of whole phase-bearing operational units and the required step-local legality. A partial realization may suspend only between declared atomic operational units; a rendezvous pair or any other declared atomic unit is indivisible. For an active segment, the consuming proof must establish continuous admittedness: at every applicable prefix the next whole unit is operationally and phase legal, respects Core.16 and every additional invariant/freeze condition supplied by that proof, is compatible with the fixed scheduler/arbitration/environment/reset choices, and its realization leaves the remaining suffix or resumed caller admitted. The unit is realized by the phase mechanism that owns it: selectable L1 work is included in IssuePick, a certified L2 unit must be forced by or contained in the actual finite settle/CycleChoiceExt path supplied by the fixed semantics, selectable L3 work is included in BoundaryPick, and any other mandatory typed phase work is carried by the applicable CycleChoiceExt

completion. No selector is deemed to select a unit outside its phase. While a SegReq-designated obligation remains continuously admitted, the combined FairPick/CycleChoiceExt construction eventually realizes its complete finite stack. The construction may interleave other work only when that work is compatible with the active-segment admittedness proof; because the stack is finite and well-founded, this requirement does not waive the FP4 starvation obligations. If continuous admittedness cannot be established, FP4a gives no progress conclusion and the consuming theorem must reject that proof case or discharge it by another route. FP4a is a construction obligation only; it does not strengthen Core.IssueFair for executions that are merely assumed rather than built. (FP5) B3 exposure. For every B3 P_push/P_pop or rendezvous antecedent used by the theorem, the corresponding discharge action is exposed to BoundaryPick as a continuously B2-enabled action under the selected explicit-boundary semantics, or the witness supplies the equivalent elaborated-boundary progress segment and proves it is selected under FP4. Thus the same constructive witness discharges the B3 progress cases used by the continuation. For theorem applications that only assume an already supplied complete fair execution, Core.FairPick need not be materialized; it is required only at proof sites, such as Lemma K.2b.E, that construct the fair execution from a finite prefix.

Lemma Core.FairCycleDovetail (typed fair per-cycle construction).

Assume Core.CycleClosure_I and Core.FairPick_I, Π . Let π end at a reachable nonterminal source cycle-entry state s . One fair typed successor cycle is constructed in two C2 applications. First evaluate any FP4a SegReq rule bound by this FairPick witness at the current prefix and retain its active certified stack, if any. Take the finite L1 segment α_issue returned by IssuePick, including the next active-segment unit when that unit belongs to selectable L1 work. By FP1 and FP4a admittedness it is operationally and phase legal and is compatible with some EnvStep_I evolution, so Lemma Core.CycleChoiceExt yields a provisional CycleResult r_0 extending α_issue . Let β be the finite phase prefix of r_0 from s through the end of its actual L2 settling segment and its SettlePoint; β is finite and phase legal because it is a prefix of a legal CycleResult. If the active segment's next unit belongs to L2, its certificate must prove that the fixed operational settling/CycleChoiceExt path contains or forces that whole unit; FP4a does not invent an L2 selector. Evaluate BoundaryPick only at that actual settled state. Let γ_bdy be its returned finite L3 segment, including the next active-segment unit when that unit belongs to selectable L3 work. By FP2 and FP4a admittedness, $\beta \cdot \gamma_bdy$ is a finite operationally legal, phase-legal partial cycle and supplies compatibility with an EnvStep_I evolution under the same fixed environment semantics. That environment evolution need not be the provisional r_0 next-state choice: it is the one compatible with the final selected $\beta \cdot \gamma_bdy$ history. Apply Lemma Core.CycleChoiceExt a second time to $\beta \cdot \gamma_bdy$ to obtain the final CycleResult r for this cycle, with any remaining mandatory typed active-segment unit included only when its certified owning phase is part of that completion. The second application is the step that proves the selected L1 fairness/segment work, the actual L2 settle path, and the selected L3 fairness/segment work are jointly realizable in one typed cycle; no boundary choice is guessed before settling and no declared atomic unit is split. A lower-ranked nested segment is discharged before its caller resumes. The FP4a admittedness proof is part of the invariant supplied by the consuming construction, so each typed successor preserves either the admitted remaining suffix/stack or completion of that obligation. Now suppose a prefix invariant Inv is preserved by these selected/completed cycles and Inv implies that every constructed cycle-entry state remains nonterminal and that Core.FairPick remains applicable. Repeating the construction and applying dependent choice only to the resulting typed CycleResult successors yields an infinite CompleteCycle continuation. FP4 proves Core.IssueFair_ Π and B2 on that continuation: a continuously enabled obligation cannot be skipped forever by the certified selectors. FP4a additionally proves completion of every continuously admitted obligation actually emitted by the fixed SegReq rule. FP5 gives the B3 progress obligations. Any

separately assumed B5 quiescence-closure property continues to apply to the constructed operational continuation; it is not manufactured by CycleChoiceExt or by FairPick. □

Remark Core.Settle-a (δ -settling, silent τ -cycles, and B6 idle ticks).

The following names separate three phenomena that are easy to conflate but are different objects in the formal model.

- δ -settling is the cycle-local same-cycle settling relation. On the phase-bearing source side, a δ_{ref} step is exactly a step of $\rightarrow_{\epsilon, L2}$; on RTL_B in the standard qualified synchronous scope, δ_{post} denotes the same-cycle settling relation of Definition Core.QSync. By the L2/QSync frame, δ -steps preserve the cycle index and registered state. The converse is not intended: other ϵ -steps, including L1 source-evaluation activity, may also preserve the cycle index but are not δ -settling steps. δ is therefore a settling classification, not a renaming of all ϵ .
- A silent τ -cycle is the explanatory name for a Core.SilentCycle: a real CompleteCycle in which no selected Σ event commits. Hidden registered/internal progress may occur, but need not; a completely idle-looking yet nonterminal cycle is still a SilentCycle if the clock/environment advances. Silent cycles carry the CompleteCycle state updates described above, including time-aware environment evolution and any active cycle-valued monitor aging. They are neither ϵ -steps nor identity stutter.
- A B6 idle tick is the special real clock tick used only when Core.TerminalIdle holds: a genuine B5/global fixed point together with machine-side permanent-idle evidence and Core.EnvTerminal. It performs no ϵ -prefix or ϵ -suffix work and leaves all non-clock state unchanged. It is treated as ϵ -stutter only for E1–E5 prefix/trace matching. Core.CycleClosure's environment-definedness clause or Core.EnvTerminal alone does not justify this branch.

The τ name remains explanatory cycle-level terminology and should not be confused with execution-prefix variables such as $\tau_{\text{ref}}/\tau_{\text{post}}$; a mechanization may use the constructor name SilentCycle. No theorem may replace the model-specific ϵ relation by δ or by SilentCycle without the stated typing.

Remark Core.Det (determinism taxonomy and discharge status).

The document uses several distinct determinism notions; they shall not be conflated. (1) Same-cycle settling determinism is the model-class restriction captured by R2C/Core.QSync: fixed registered state, fixed cycle- t inputs, and fixed explicit issue/request drives yield one settled boundary valuation. (2) Conditional synchronous-step determinism is the corresponding functional-cycle view after the currently legal witness/issue/environment choices for that cycle have been fixed; it does not remove Policy-L scheduling latitude or WC-sensitive source nondeterminism. (3) B1 is observable RTL trace determinism under fixed stimulus and initial state; internal ϵ behavior may still differ. (4) I.DR/I.DR' are derived replay theorems, not assumptions to be asserted in a discharge record. (5) Definition K.CV is the Appendix K causal/value-determinism premise. (6) SDet fixes the selected Policy-S simulator/scheduler/arbitration semantics and excludes a second incompatible committed history for that selected instance. (7) Normalization selection/congruence records which proof-relevant normalization is used and why the replay quotient is preserved; Core.QSync canonicalizes only the K-facing same-cycle settling part, not the Appendix I/J replay normalization.

For any item that is a theorem premise, Appendix M.7a's normative evidence axis applies directly: the record states whether the premise is inapplicable to the selected theorem scope, satisfied by a simple syntactic/design restriction, discharged by tool/RTL evidence, or discharged by a source/RTL witness

construction. Derived theorems such as I.DR are not separate premises and therefore require no independent M.7a discharge entry.

Definition Core.KCutClosure (K-facing cutpoint existence / non-vacuity).

For a proof model M used by Appendix K, a K-origin is a reachable per-cycle state from which the K-facing boundary network is to be settled without first advancing to the next cycle: for Sys_ref/Sys_K it is the selected L1 endpoint immediately before $\rightarrow_{\epsilon}$, L2 settling; for RTL_B it is the o_t component of a reachable Core.RTLDesignatedOriginFrame_I(s_t, o_t, e_t), after the current cycle's registered/FSM/pipeline state and process-owned request outputs are established and before K-facing same-cycle settling. KCutClosure_M(I) holds when every reachable K-origin admits a finite ϵ -only/same-cycle settling normalization, within that same cycle and before any later cycle-advancing registered update, to some $\sigma_K \in \text{KCutReach}_M(I)$. For the RTL use $\text{KCutClosure_post}(I) := \text{KCutClosure_RTL_B}(I)$.

In the standard K-facing/cutpoint theorem scope, Core.KCutClosure is a derived interface rather than an independent modeling freedom. Theorem Core.QSync-Ref derives source-side KCutClosure_M from QualifiedSynchronousModel_M(I), including existence of the source SettlePoint; Theorem Core.QSync-Post derives Core.KCutClosure_post from QualifiedSynchronousModel_RTL_B(I). Downstream Appendix K statements continue to consume Core.KCutClosure_post so that the liveness proof is factored from the particular structural settling proof. A future non-QSync semantic extension could reuse that downstream interface only by separately proving the same closure property and explicitly extending the theorem scope; it is not part of the standard user-facing G–K model defined here.

Lemma Core.KCutObserve (persistent blocker observability).

Assume KCutClosure_M(I) and $K\epsilon$ for the selected model. Let s_0 be a K-origin and let σ_K be a KCut supplied by the closure witness. If a K-relevant boundary request/frontier/enablement configuration is present at s_0 after the owning requests are fixed and persists through the selected ϵ -normalization, then the same attributed K-facing configuration is present at σ_K ; WFG-relevant facts cannot be changed solely by the intervening ϵ segment. Therefore a persistent K-relevant blocker cannot be excluded from the Appendix K observation domain merely because unrelated ϵ -work would otherwise continue indefinitely. A configuration that commits, withdraws under an allowed reset disposition, or otherwise ceases to persist before the KCut is transient and is not asserted to satisfy Dead_K. Proof. KCutClosure supplies the finite endpoint; $K\epsilon$ supplies WFG-inertness of the intervening ϵ segment; BoundaryEvalPoint/Core.KCut supply settled enablement at the endpoint. $\square$

Definition Core.SettleKBridge (L2 trigger to K-facing cutpoint).

Fix a Core.LAdm-admissible source prefix or continuation. Core.SettleKBridge_I holds when every Core.16 activation at an L2 SettlePoint σ_{set} is related to every later K-facing cutpoint σ_K used to evaluate the persisting blocked condition as follows: while the same fully disabled frontier persists, the only intervening changes are phase-legal WFG-inert ϵ -normalization and Core.16-permitted already-admitted in-flight action occurrences; no new source-later work is launched/issued; blocking requests are nonwithdrawable; $K\epsilon$ makes each applicable ϵ -only segment WFG-inert; and every intervening complete proof-model action unit that changes explicit channel/request/history state is represented explicitly in H/E4/E before Front, ChannelEnabled/Disabled, ReleaseOwnerSets, and WFG are re-evaluated at σ_K . The action unit may be an ordinary already-issued boundary commit, an E-declared proof-internal staged/bundle/join transfer, or—when the selected Stage-2 profile is active—a typed SyncFire/EpochGateAdvance occurrence; it is not required to be owned by a later candidate D. Persistence is a hypothesis of each represented bridge instance, not a conclusion: the bridge does not prove that the blocked frontier persists. For mechanization, a finite activation-to-KCut subpath may be

decomposed into maximal WFG-inert ϵ -only segments separated by complete in-flight action occurrences and proved by induction over that alternation, applying $K\epsilon$ to each ϵ -only segment, $Q.2/E4/Core.7d-Core.7e$ exact action effects at each applicable occurrence, and the applicable $Core.16/K.DrainEvidence$ clauses throughout. If the frontier/root family changes, commits, or otherwise ceases to persist, that particular bridge instance terminates; candidate handling then belongs to $K.InFlightRelocate/redispach$ rather than an assertion that the old D survives. When the final K -facing source cutpoint is selected from a $J.GWR$ post-L3 construction endpoint, the terminal ϵ -only portion is the $Core.KObsSettle/Core.KObsNormRef$ observation segment and is additionally qualified by $K.NormStable$; it does not reopen L3 selection. By Lemma $Core.SettleIsKCut$, σ_set is itself a source $KCut$. The bridge remains needed because actual reached in-flight actions can occur before a later candidate is evaluated; generalized finite future-path soundness beyond the exact reached prefix is supplied separately by $K.InFlightQual/J.InFlightPathSound$. $K.DrainReach$ shall discharge this bridge for the exact source K -facing prefix used by the certificate.

Source order and source-level frontier items

Definition Core.5 (Per-process execution-call universe, frontier-item universe, observable projection, and source order).

For each process P , let Q_exec, P denote its execution-level set of dynamic source-level message-call instances for the particular execution / witness under the fixed external stimulus being analyzed. Q_exec, P includes blocking Push/Pop calls, successful non-blocking PushNB/PopNB calls, and failed non-blocking PushNB/PopNB polls.

Let Ω_P denote P 's local dynamic source-level frontier-item universe for that same execution / witness. Thus Ω_P is stimulus-dependent: it contains exactly the local synchronization-call items, signal-action items, and message-operation frontier items that belong to the realized execution being compared, and excludes items on untaken control-flow branches, unentered cases, and unexecuted loop iterations. For sequential processes, signal-action items are positioned by the R2 anchoring rule; for combinational processes, signal-action items are positioned in the observable cycle bucket determined by the R2C ϵ -fixed-point rule. Message-operation frontier items include reached blocking Push/Pop instances, whether pending or committed, and successful non-blocking PushNB/PopNB instances that contribute committed $Push_c(v)$ / $Pop_c(v)$ events. Failed non-blocking polls are ϵ -steps: they are members of Q_exec, P when executed, but they are not members of Ω_P merely because they executed.

Because Q_exec, P is a universe of dynamic message-call instances while Ω_P is a universe of source-level frontier items, their message-operation overlap is not a literal set intersection. Let $MsgOf_P(x)$ be the partial projection from a frontier item $x \in \Omega_P$ to its owning dynamic message-call instance, defined exactly for message-operation frontier items and undefined for pure synchronization or signal-anchor items. Define the message-frontier slice

$$\Omega_msg, P := \{ x \in \Omega_P \mid \exists q \in Q_exec, P : MsgOf_P(x) = q \}.$$

For any message-operation frontier item $x \in \Omega_msg, P$, let $BoundaryOf_P(x)$ denote the explicit abstract channel boundary at which x is evaluated for $BoundaryReq$, $ChannelEnabled$, $ChannelDisabled$, $Front_P$, and Appendix K WFG reasoning. In the ordinary unelaborated case, each source-level message-call instance q has a unique corresponding frontier item x and a unique evaluated boundary. In Sys_B^{elab} , $MsgOf_P$ need not be injective: one owning source-level message-call instance q may project to a chain or bundle of proof-internal frontier items x at several explicit abstract boundaries. The elaboration

package must then provide $\text{BoundaryOf_P}(x)$, $\text{MsgOf_P}(x)$, and the partial order $\leq_P$ for each introduced item.

Define the corresponding message-instance slice induced by Ω_P as

$$Q_{\Omega,P} := \{ q \in Q_{\text{exec},P} \mid \exists x \in \Omega_{\text{msg},P} : \text{MsgOf_P}(x) = q \}.$$

Let Σ_P denote the corresponding observable event/label projection: synchronization and signal items contribute their Σ labels when observed, and a message-operation frontier item $x \in \Omega_{\text{msg},P}$ with $\text{MsgOf_P}(x)=q$ contributes the committed Σ -event $m(q)$ only if q commits. Pending blocking message-operation frontier items may therefore be in Ω_P and $\Omega_{\text{msg},P}$, and their owning message-call instances may be in $Q_{\Omega,P}$, without yet contributing any committed Σ -event. Let $\leq_P$ denote the source order on Ω_P , i.e. the order generated by the document's scheduling rules for synchronization calls, signal actions as positioned by R2 or R2C, message-operation frontier items, and any explicit partial-order structure declared by the selected Sys_ref elaboration. The execution-level order constraints E3/E5 use the corresponding source-induced order on $Q_{\text{exec},P}$.

Definition Core.6 (Done_P , Remain_P , and NextFront_P relative to the fixed witness).

Fix the particular execution / witness under analysis, hence the associated $Q_{\text{exec},P}$, Ω_P , $\Omega_{\text{msg},P}$, $Q_{\Omega,P}$, Σ_P , and $\leq_P$ from Definition Core.5. For a state σ occurring as a cutpoint of that compared execution, let

$$\text{Done_P}(\sigma) \subseteq \Omega_P$$

be the set of local frontier items already realized by σ . For synchronization and signal items, “realized” means that the corresponding Σ -event has occurred. For a message-operation frontier item $x \in \Omega_{\text{msg},P}$ with $\text{MsgOf_P}(x)=q$, “realized” means that q has committed and its committed Σ -event $m(q)$ has occurred. A pending issued-but-uncommitted blocking message-operation frontier item x is therefore not in $\text{Done_P}(\sigma)$, even though $\text{BoundaryReq}(x,\sigma)$, $\text{ChannelEnabled}(x,\sigma)$, and $\text{ChannelDisabled}(x,\sigma)$ may be defined at σ , read operation-attributively through $q = \text{MsgOf_P}(x)$. A failed non-blocking poll $q \in Q_{\text{exec},P}$ that induces no frontier item in $\Omega_{\text{msg},P}$ is not considered by Done_P , Remain_P , or NextFront_P , because it is an ϵ -step with no frontier item and no committed Σ -event.

Let

$$\text{Remain_P}(\sigma) := \Omega_P \setminus \text{Done_P}(\sigma).$$

Then define the next source-level frontier-item set by

$$\text{NextFront_P}(\sigma) := \min_{\leq P}(\text{Remain_P}(\sigma)).$$

This is the shared next-frontier notion used by Theorem J.1 and reconstructed from KObs by Lemma J.KObs-Proc, interpreted relative to the fixed witness universe Ω_P . It may contain synchronization-call items, signal-anchored action items, and message-operation items. It is a frontier of source-level items, not a set of already-committed Σ -event labels.

Definition Core.7 (Message-operation slice of the next frontier).

For the same fixed compared execution / witness, define

$$\text{MsgNextFront_P}(\sigma) := \{ x \in \text{NextFront_P}(\sigma) \mid x \in \Omega_{\text{msg},P} \text{ and, for } q = \text{MsgOf_P}(x), q \text{ is defined and } \text{kind}(q) \in \{\text{Push_c}, \text{Pop_c}\} \text{ for some channel } c \}.$$

This is only the message-operation frontier-item slice of the candidate next frontier; it is not yet the pending blocking frontier of Appendix B / K.

Definition Core.7a (Blocking message-operation slice of the next frontier).

For the same fixed compared execution / witness, define

$\text{BlockingMsgNextFront_P}(\sigma) := \{ x \in \text{NextFront_P}(\sigma) \mid x \in \Omega_{\text{msg},P} \text{ and, for } q = \text{MsgOf_P}(x), q \text{ is defined and } q \text{ is a blocking Push or Pop instance} \}$.

This is the blocking-message candidate-frontier slice used by the Appendix K Front/NextFront bridge and WFG classification. It may contain a reached blocking operation before that operation has issued; BoundaryReq and Pending determine whether such an item is currently an outstanding request and therefore a member of Front_P. Successful non-blocking transfer items may belong to MsgNextFront_P but never to BlockingMsgNextFront_P.

Definition Core.7b (Synchronization-call slice and typed synchronization carrier).

For the same fixed compared execution / witness, let $S_{\text{exec},P}$ denote P's execution-level set of dynamic source-level synchronization endpoint-call instances selected into the synchronization set S. Let $\text{SyncOf_P}(x)$ be the partial projection from a frontier item $x \in \Omega_P$ to its owning dynamic synchronization endpoint-call instance $s \in S_{\text{exec},P}$, defined exactly for synchronization-call items and undefined for message-operation and signal-anchor items. Define $\Omega_{\text{sync},P} := \{ x \in \Omega_P \mid \exists s \in S_{\text{exec},P} : \text{SyncOf_P}(x)=s \}$. MsgOf_P remains defined exactly on $\Omega_{\text{msg},P}$ and is not widened by this definition. For $s \in S_{\text{exec},P}$, let $\text{SyncTxn_I}(s)$ be the theorem-instance dynamic synchronization-transaction key. The key records the synchronization interface/site, dynamic occurrence/ordinal, reset epoch, and transaction kind. For a paired two-party SyncChannel/ac_sync transaction, the matching endpoint calls have the same SyncTxn key and commit as the existing atomic two-endpoint synchronization unit; for a unary clock/wait anchor or another declared synchronization form, the key records that distinct kind rather than inventing a message peer. Let $\text{SyncParticipants_I}(\kappa)$ be the finite set of modeled process participants associated with transaction key κ ; clock/environment participants, when present, are typed separately and are not modeled-process owners merely because they can release the synchronization. For $x \in \Omega_{\text{sync},P}$ with $s=\text{SyncOf_P}(x)$, $\text{SyncReached_P}(x,\sigma)$ holds when the operational source/replay control state of P has reached that dynamic synchronization call in its current reset epoch. $\text{SyncPending_P}(x,\sigma)$ holds when $\text{SyncReached_P}(x,\sigma)$ holds and the synchronization endpoint instance has not committed/returned and has not been ended by an affecting reset/terminal disposition. $\text{SyncReady_I}(x,\sigma)$ is the current-state synchronization-completion readiness predicate supplied by the declared synchronization semantics: all required current transaction participants/conditions needed for the synchronization action are present at σ . Define $\text{SyncBlocked_I}(x,\sigma) := \text{SyncPending_P}(x,\sigma) \wedge \neg \text{SyncReady_I}(x,\sigma)$. SyncReady is a state predicate only; it does not assert that the synchronization action is selected in the current phase or that a future CycleResult exists. Neither SyncPending nor SyncReady is a message BoundaryReq/ChannelEnabled predicate. BoundaryReq, ChannelEnabled, ChannelDisabled, MsgOf, $\Omega_{\text{msg},P}$, Pending_P, Front_P, and the literal message-passing WFG remain message-typed exactly as before.

Definition Core.7c (Synchronization release support and carrier well-formedness).

For a pending source synchronization carrier $x \in \Omega_{\text{sync},P}$, define $\text{SyncReleaseOwnerSets_I}(x,\sigma)$ only for the internal modeled-process portion of a blocked synchronization cause. When $\text{SyncBlocked_I}(x,\sigma)$ holds, $\text{SyncReleaseOwnerSets_I}(x,\sigma)$ is a finite antichain of minimal non-empty sets of modeled processes whose process-facing progress is jointly sufficient, under the declared synchronization semantics and fixed transaction pairing, to remove that particular synchronization obstruction. Members of one support set are conjunctive requirements; distinct support sets are alternative release supports. The family is exact: no declared support is a spurious internal rescue alternative, and every genuine internal modeled-process release alternative contains a declared minimal support. For the ordinary two-party barrier case, if P is pending and the matching peer Q has not yet supplied the

required participation, the internal family is $\{\{Q\}\}$; once the complete transaction is SyncReady, no further modeled-process release support is required for readiness and the blocked-support family is empty. If only a clock/environment event can release the pending synchronization, the internal process family is empty and the synchronization is not reclassified as an internal process dependency.

Core.SyncCarrierWF_I holds when: (SC1) every synchronization-call frontier item has a unique SyncOf endpoint identity and reset-epoch-consistent SyncTxn key; (SC2) the fixed transaction-pairing/participant schema agrees with the paired synchronization semantics already required by E1/J.1, including matching endpoint identity for SyncChannel/ac_sync; (SC3) SyncReached/SyncPending/SyncReady are determined from the current replay/control states of the declared transaction participants, committed synchronization history, fixed synchronization semantics, and reset/terminal state, with no hidden timing or Issue-like timestamp; if current readiness depends on environment state not otherwise determined by those process/history facts, the selected instance supplies EnvReplayAdequate or an equivalent typed reconstruction sufficient for that readiness predicate; (SC4) SyncReleaseOwnerSets is finite and satisfies the soundness/completeness/minimality rule above; and (SC5) affecting reset never matches or carries one dynamic synchronization instance across reset epochs as though it were the same transaction. Write SyncCarrier_P(x,σ) for the typed record $\langle P, x, \text{SyncOf_P}(x), \text{SyncTxn_I}(\text{SyncOf_P}(x)), \text{epoch}, \text{SyncParticipants}, \text{SyncPending}, \text{SyncReady}, \text{SyncReleaseOwnerSets} \rangle$ when x is a reached synchronization-call item. This carrier is Stage-2 lowering/correspondence infrastructure only. It does not add synchronization vertices or edges to the literal message-passing WFG/Dead_K, does not weaken the A8 barrier-well-formedness scope, and its action semantics are supplied separately by Definitions Core.7d–Core.7e.

Definition Core.7d (Source synchronization in-flight completion unit SyncFire).

Fix a theorem instance I and a currently represented dynamic synchronization transaction key κ . SyncFire_I(κ) is one complete source synchronization boundary-action unit, admitted to the in-flight universe only when its required transaction is already present in the current typed synchronization carrier. Concretely: (SF1) every modeled endpoint whose participation is required for this occurrence is represented by a reached, pending Core.7b carrier with SyncTxn key κ , and SyncReady_I holds for the transaction under the complete declared participant/environment state; a peer that has not yet reached the synchronization is therefore a release owner, not hidden in-flight work; (SF2) Core.SyncFence holds for every committing endpoint owner, so every applicable pref_S blocking message operation has already process-facing committed or is represented by a distinct earlier same-cycle in-flight action in the reconstructed path before SyncFire; (SF3) the fixed reset/terminal/environment semantics permit the transaction and no affecting reset rekeys or carries it across epochs; and (SF4) selecting SyncFire requires no new source ReachPrep, launch, message Issue, or source-later operation to make the transaction ready. For a paired SyncChannel/ac_sync occurrence, SyncFire is the existing indivisible two-endpoint synchronization transaction; for another declared synchronization kind it is the corresponding already-defined atomic synchronization unit. Any E2/R2 signal actions required to share that synchronization anchor are included in the same declared atomic observable unit rather than split into separately orderable in-flight actions.

The exact SyncFire successor commits that synchronization transaction once, appends the corresponding atomic synchronization/anchored-signal unit to the H-retained history, marks exactly the participating Ω_sync items realized, advances each participating process across that synchronization boundary to its immediate post-sync source-interval/control point, and retires the corresponding pending synchronization carriers. It does not execute post-sync source evaluation, ReachPrep, or message Issue and does not assert any new suff_S BoundaryReq as part of the action. Existing message

residual offers and explicit message-channel contents are framed except for effects already produced by distinct earlier action units; any synchronization-dependent stateless handshake/epoch-gate roots are then deterministically re-settled. Thus completion of an already-pending synchronization is permitted in-flight progress, but launching work from the newly entered source interval is not. SyncFire is synchronization-typed: it is not MsgOf, BoundaryReq, ChannelEnabled, or a message WFG edge.

Definition Core.7e (Lowering-only epoch-gate in-flight unit EpochGateAdvance).

Fix a lowered model $\text{Sys_K} = \Lambda_W(\text{Sys_ref})$, an E-declared lowering-created epoch gate g bound by the $K.\text{Low}$ certificate to synchronization transaction κ , and a current explicit gate token/request/credit state represented in the selected proof model. $\text{EpochGateAdvance}_{\{I,E\}}(g,\kappa)$ is the complete E-typed gate action unit that advances or retires that already-present lowering state when its declared E4/handshake precondition is satisfied. It is state-based admitted from the explicit gate carrier; it is not admitted merely because κ exists historically. Its successor changes exactly the declared gate channel/token/credit/frontier state, applies the E projection/accounting effect, frames source synchronization history and source process-control state, launches/issues no source-later work, and deterministically re-settles the affected handshake network.

EpochGateAdvance is a lowering-only action. Under the $K.\text{Low}$ execution projection it is erased/stutters: it emits no source synchronization event and is not itself SyncFire. This distinction is mandatory because Λ_W may insert real-cycle gate skew after the source synchronization transaction has already projected as committed. A half-completed enabled gate may therefore be an in-flight escape on Sys_K even when its Sys_ref projection is already past the synchronization. The exact cross-model ordering/insertion relation between SyncFire and zero or more lowering-only gate steps is not asserted by this local definition; it is established, for a qualified lowering, by $K.\text{Low}.\text{InFlightPathCorr}$.

Clarification Core.5a (ordinary source order versus elaborated partial order).

For ordinary user-written sequential source code, dynamic message-call instances that appear in sequence in one process are ordered by dynamic program order, including distinct-interface calls, repeated loop iterations, and unrolled instances whose selected source expansion is ordered. Same-cycle issue, same-cycle commit, or issue before an earlier operation commits is a timing/issue-policy property, not incomparability. A message-operation multi-frontier exists only when an explicit $\text{Sys_B}^{\text{elab}}$ elaboration package introduces incomparable proof-internal frontier items and supplies the corresponding projection, boundary, order, and attribution facts. Ordinary distinct-interface source calls are not made incomparable merely because they access different channels.

Lemma Core.OrdFrontSingleton (ordinary-source frontier uniqueness).

Fix a process P , a particular execution/witness, and an ordinary non-elaborated source interval. Assume every message-operation frontier item in that interval is induced by a unique ordinary dynamic source-call instance and no $\text{Sys_B}^{\text{elab}}$ partial-order structure is introduced. Then the restriction of $\leq_P$ to the message-operation frontier items in that interval is total. Consequently, for every cutpoint σ in the interval, $|\text{Front_P}(\sigma)| \leq 1$.

Proof. By Clarification Core.5a, all ordinary dynamic source-call instances encountered by P in the fixed execution are ordered by dynamic program order, including distinct-interface calls, repeated loop iterations, and ordered unrolled instances. In the ordinary unelaborated case, each such call induces one frontier item and $\leq_P$ is the corresponding reflexive source-induced order. By Definitions Core.14 and Core.15, every member of $\text{Front_P}(\sigma)$ belongs to $\text{Pending_P}(\sigma)$ and hence is a blocking message-operation frontier item in the same interval. Hence any two members of $\text{Front_P}(\sigma)$ are comparable.

Two distinct elements therefore cannot both be $\leq_P$ -minimal. Thus $\text{Front}_P(\sigma)$ contains at most one member. $\square$

Operational interpretation of Sys_ref and source issue policies

The R-rules stated in Appendix G are constraints on a selected source-reference transition system, not merely constraints on a finished list of labels. For a fixed design, fixed stimulus, and selected reference-model choice, Sys_ref has the following operational reading.

A Sys_ref state consists of: (1) one local source state per process, including control location, local variables, call-stack/loop state, the current synchronization interval, and the reached/issued/committed status of dynamic message-call instances; (2) one state for each explicit abstract channel of the selected reference model, including $B(c)$, $\text{occ_c}(t)$, FIFO order when $B(c)>0$, rendezvous endpoint requests when $B(c)=0$, and any explicit $\text{Sys_B}^{\text{elab}}$ staged/bundle/join/epoch-gate channel state; (3) signal and synchronization state at the selected observable anchors; and (4) ϵ -microstate for internal staging, direct-input sampling between anchors, and other proof-hidden activity. In $\text{Sys_B}^{\text{elab}}$, introduced abstract channels are explicit proof-internal boundaries: their commits may be Σ -events for Appendices J-K even when projected away from the outer DUT/testbench-visible alphabet Σ_{DUT} by Π_{elab} .

The transition relation contains the following kinds of steps. Pure source-evaluation ϵ -steps advance ordinary C++/SystemC computation and allocate dynamic operation instances in dynamic source order. For a sequential process, ordinary call-return state and the L1 preparation/reach state are distinguished by Definition Core.ReachPrep below: a later operation may become reached before an earlier blocking operation has process-facing committed only through a legal dependency-independent preparation segment, and this does not make the operations unordered. A message-issue step for q may occur only after q has been reached, the endpoint request can be attributed to q , the R3 synchronization-boundary constraints are respected, and the issued set is prefix-closed under $\leq_{\text{src}}$. Issuing a blocking Push or Pop asserts the endpoint-owned persistent request for q and records $\text{Issue}(q)$; issuing PushNB/PopNB asserts its one-cycle attributed attempt. A channel-commit step for a bounded FIFO boundary $B(c)>0$ may occur when $\text{BoundaryReq}(q, \sigma)$ and $\text{ChannelEnabled}(q, \sigma)$ hold at the selected abstract boundary; it emits the corresponding Push_c(v) or Pop_c(v) event, updates the E4 channel state, and releases the owning call at the process-facing boundary. For a rendezvous boundary $B(c)=0$, commit is one atomic two-endpoint transfer step with matching producer and consumer endpoint requests in the same settled cycle and one shared payload value. A failed non-blocking boundary decision emits no Σ event but produces a ReturnedAtCycle status and retires its one-cycle attempt. A synchronization boundary action for call s may be selected only when Core.SyncFence is satisfied by the already-completed and same-cycle selected process-facing message commits; the interval advance caused by s occurs only after those predecessor commits have been accounted for. Synchronization and signal-anchor steps then emit wait/Sync and anchored Read/Write events according to R1/R2/R2C. ϵ -settling and drain steps update only proof-hidden or explicitly elaborated internal state and must respect $K\epsilon$ / B-faithfulness when used by Appendices J–K. For a sequential process, Definition Core.RegSeq below governs which process-owned evaluation, request, and payload actions may share one Horizon.

Definition Core.ReachPrep (overlap-aware source reach/preparation semantics).

Fix a sequential process P , its current source interval, and a cycle t . Sys_ref retains the logical source/replay control position together with reached/issued/committed status for dynamic calls. During the L1 pre-settle phase, an operational preparation cursor may advance through a finite source-control segment and mark a later dynamic operation q as reached without claiming that an earlier blocking operation q_0 has committed or returned. Such preparation is legal only when: (1) q remains in the current synchronization interval and all dynamic source-order relations are preserved; (2) the reachability, control choice, endpoint, address, payload, and required operands of every prepared action

are independent of any unresolved earlier returned data/status and satisfy Core.RegSeq and the applicable field-sensitive NoDependentUse/footprint restrictions; (3) no synchronization boundary is crossed; (4) no already-activated Core.16 blocked-frontier discipline forbids launching that source-later work; and (5) the selected source/tool witness or certificate validates the preparation segment. Reaching q does not itself assert a request. Issue remains a separate transition and remains prefix-closed under $<_{src}$.

This permission applies both to overlapped/pipelined iterations and to ordinary straight-line source sequence $q_0; q_1$ when q_1 is dependency-independent of the unresolved q_0 result and the selected Sys_ref theorem instance admits eager preparation. It does not make q_0 and q_1 incomparable, and it does not assert that a raw serial SystemC execution has returned from q_0 early; it is the source-reference scheduling permission used to represent allowed RTL-matched eager issue. Examples K.CE and K.CE-2, together with the Appendix K A5-L blocking-rendezvous example, use this straight-line case. Coverage direction for Post $\rightarrow$ Ref: for every RTL_B prefix included in a Post $\rightarrow$ Ref coverage claim, the selected Sys_ref ReachPrep semantics shall admit every dependency-independent preparation/reach segment required by the accepted J.LocalGWR or J.GWR construction for that prefix. The instance may be more permissive than the observed RTL scheduling, but it shall not exclude preparation needed to realize a covered RTL witness. Once the L2 SettlePoint exposes a nonempty fully disabled pending frontier, Core.16 prohibits every new source-later launch while that frontier persists. Every L1 preparation segment used by a theorem instance must be finite. Lemma I.3/FPD supplies that fact for its certified finite construction segments; Core.CycleClosure's source-cycle-existence/finiteness clause supplies the generic per-cycle finiteness obligation when a complete/fair source continuation is required.

Definition Core.RegSeq (registered sequential-interface semantics).

The Core classifies each process for the Appendices I–K proof stack as either combinational under R2C or sequential under Core.RegSeq. For a sequential process P , every process-driven endpoint request and Push-data output assigned to cycle/Horizon t is a function of registered process or pipeline state and operands fixed before the L2 settle point of cycle t . Combinational derivation from that registered state is permitted, but the request and payload must remain stable for the selected boundary action. Direct inputs count as fixed operands only when their applicable E2, hls_direct_input, or hls_direct_input_sync contract makes their value stable at the relevant anchor. If ReturnedAtCycle(q, t) holds, any P -owned evaluation, control selection, endpoint/address selection, payload computation, or message issue that depends on q 's returned data or success/failure status has Horizon strictly greater than t . A same-Horizon source-later action is permitted only when its reachability, control, operands, endpoint, address, and payload are independent of every unresolved cycle- t NB outcome and when its writes do not change the ReqSnapshot of that unresolved decision except through an explicitly ordered dependency. This is a normative theorem-scope restriction, not a consequence of the word “clocked” alone. Tool-side Cat# or equivalent conformance evidence and source-side simulation/lint/witness evidence shall discharge it. Intentional zero-time outcome-dependent NB cascades or retry loops shall be identified by checked source/tool analysis and then rejected or explicitly excluded from the theorem instance, isolated as latency-sensitive logic, remodeled as one explicit atomic multi-way decision, or aligned by Appendix L; silent scope exclusion is not permitted.

Definition Core.PolicyL (liberal, witness-relative issue).

Policy L is the most permissive source-side issue policy used by the safety witness and by the K.2b dominance/extraction bridge. It may issue a reached source-level message operation whenever the transition-system preconditions above, R3/R4/R5, E4, E5, Core.RegSeq, the selected non-blocking outcome witness, and the source-order prefix-closure requirement are satisfied. A source-later

operation may have become reached through Core.ReachPrep even while a source-earlier same-interval blocker remains uncommitted. Policy L may therefore leave the source-later blocking request issued in that state. This is how the reference model represents loop overlap, ordinary dependency-independent eager preparation, RTL-matched eager issue, and other allowed same-cycle/overlapped request timing. Policy L is not a claim that every raw pre-HLS simulator mode can generate every such witness state. Policy L does not override returned-result availability: it cannot issue an operation in cycle t when its Core.ReachPrep reachability, control selection, endpoint or address selection, payload, or any required operand depends on data/status first returned at the cycle- t boundary. Core.LLegal fixes Policy-L step legality, Core.Phase fixes the cycle-settle boundary and the trigger for Core.16, and Core.LAdm below combines those pieces with the activated Core.Drain qualification used by liveness admissibility.

Definition Core.LLegal (Policy-L step legality and fixed-instance binding).

Fix a selected liveness instance I consisting of Sys_ref, the initial/reset state, capacities and elaboration/lowering, the fixed stimulus and reactive environment, witness-selected non-blocking and other ϵ outcomes, arbitration choices, and the selected synchronization, signal-anchor, and ϵ -normalization conventions. A finite or infinite path p is Policy-L-legal for I iff it begins in the selected initial/reset state, every transition is a transition of the operational Sys_ref relation above permitted by Policy L, and every fixed choice in I is respected. R1–R5, E4/E5, dynamic-operation identity, request attribution and persistence, endpoint cardinality, reset-epoch typing, and sequential-result availability are part of this step legality. Core.SelectLatitude is constitutive of this source legality: a legal finite source cycle or prefix need not select a maximal set of otherwise legal L1 issue or L3 boundary actions.

Definition Core.SelectLatitude (finite-prefix source selection latitude).

Fix a phase-bearing source theorem instance I and issue policy $\Pi \in \{\text{Policy L}, \text{Policy S}\}$. The operational Sys_ref relation has no maximal-progress requirement at either (SL1) L1 issue selection or (SL3) L3 boundary-action selection. Subject to Core.Phase, Π , Core.16, Core.SyncFence, R1–R5/E4/E5, reset/atomicity/endpoint-cardinality rules, and the fixed witness, environment, and arbitration choices, an otherwise legal reached L1 message-issue transition or currently legal L3 boundary action may remain unselected at a given finite-prefix opportunity. Such nonselection is ordinary legal source-cycle behavior under the selected issue policy; it does not disable the action, withdraw or reattribute an already-issued request, change a stable Push payload, or license any step prohibited by Core.16 or another legality rule. Conversely, absence of an issue or boundary commit does not by itself establish that the action was disabled or that Core.16 had activated. When the fixed reset/environment semantics admit the cycle and no mandatory atomic or proof-internal unit requires otherwise, L1 and/or L3 selection may be empty and the source may advance through an ordinary legal CycleResult/CompleteCycle; if no selected Σ event commits, that result is a Core.SilentCycle with the ordinary full-state next-cycle updates of Definition Core.CycleResult. Core.IssueFair and B2 are separate complete-execution requirements that prevent indefinite deferral under their respective continuous-enablement antecedents; neither is a premise merely for this finite-prefix legality. The same latitude applies under Policy S: Policy S narrows which issue transitions are legal; a later theorem or qualified simulator may separately fix deterministic selection choices without changing this constitutive legality.

Definition Core.Phase (Policy-L cycle phases and settled-frontier trigger).

Each cycle t of a Policy-L-legal path is interpreted using the following normative phase boundary. (L0/L1) Cycle entry and pre-settle issue window. Starting from the cycle-entry state after all prior-cycle boundary updates, sequential processes may take a finite selected sequence of pure source-evaluation ϵ -steps and message-issue steps for operations that were already reached at cycle entry or become reached during this phase through Core.ReachPrep and evaluation independent of every cycle- t

boundary result. Their control choice, endpoint, address, payload, status inputs, direct-input values, and other required operands must be available under `Core.RegSeq` independently of any cycle- t commit or failed/non-failed NB decision outcome. These issue steps remain subject to R3–R5, source-order prefix closure, stable payload, and the deferred-hole restrictions of Lemma I.3. Thus a source-later outcome-independent operation, including an ordinary straight-line successor admitted by `Core.ReachPrep`, may issue during this window while a source-earlier blocker or unresolved NB attempt remains uncommitted/undecided. A value or status first returned at the cycle- t boundary cannot be used in this window. The selected L1 issue sequence is not required to be maximal: under `Core.SelectLatitude` an otherwise legal issue may remain unselected at this L1 opportunity, subject to `Core.16` and every other step-legality condition. Any theorem that constructs a complete/fair source continuation additionally requires the selected L1 preparation/issue segment to terminate finitely as part of `Core.CycleClosure`.

(L2) settle point. After the selected process-owned pre-settle steps for cycle t , the L1 window closes. Process-owned request assertions and Push payloads are then held fixed while the explicit channel network and every other phase-legal L2 ϵ -step settle under $\rightarrow_{\epsilon, L2}$ to the first `SettlePoint_I(t,  $\sigma_t^{\text{set}}$ )` of Definition `Core.Settle`. The phrase “the frontier settles” and every Appendix K test of whether a frontier is fully channel-disabled at the activation boundary refer to this L2 `SettlePoint`. By Lemma `Core.SettlesKCut`, the settled source state is also `Core.4`-quiescent and lies in the applicable source `KCut` domain; its special role here is the `Core.16` activation boundary, not necessarily the later drain-closed `Dead_K` evaluation point.

(L3) Freeze and boundary action. `Front_P` and `ChannelEnabled/ChannelDisabled` are evaluated at σ_t^{set} . If `Front_P` is non-empty and every frontier item is disabled there, Definition `Core.16` activates immediately at σ_t^{set} : no new source-later operation may issue after that snapshot in cycle t or in a later cycle while the same fully disabled frontier persists, although already-admitted work may progress only through `Core.16`-permitted in-flight actions, including qualified proof-internal staging/join movement already represented in explicit state. If the frontier is not fully disabled, the selected E4/rendezvous, synchronization, signal-anchor, arbitration, and endpoint-cardinality rules determine the boundary actions that commit in cycle t . The selected L3 boundary-action set is likewise not required to be maximal: `Core.SelectLatitude` permits finite deferral of an otherwise legal boundary action without treating that action as disabled. Any synchronization action s in that selected L3 boundary-action set is additionally gated by `Core.SyncFence`: every blocking $q \in \text{pref}_S(s)$ must have process-facing committed earlier or be selected to process-facing commit in the same cycle before s advances P into the next source interval. After the selected L3 action set has been closed and its selected actions have completed, a proof that requires a K -facing observation of the resulting post-boundary state may enter the distinguished `L3Obs` subphase of `Core.KObsSettle` and take its observation-only $\rightarrow_{\epsilon, KObs}$ segment before the existing meta-level advance to L4. `L3Obs` is a subphase of L3 for phase-indexed transition legality, not a peer scheduling phase: it cannot feed recomputed handshake values back into L3 action selection and therefore cannot change the cycle- t scheduling decision.

(L4) Next-cycle result availability. Boundary commits and failed non-blocking decisions produce their process-visible `ReturnedAtCycle` results and update registered process/channel state for the next cycle. Process-owned evaluation or issue whose reachability, control, endpoint, address, payload, or required operand depends on a value or success/failure status returned in cycle t may begin only in a strictly later cycle. Every deferred NB hole created for Horizon t is resolved or retired before the exported post-boundary construction state for that Horizon.

The L2/L3 freeze is not retroactive. An issue that was legal in the L1 pre-settle window is not invalidated merely because the subsequent settled snapshot exposes an earlier fully disabled frontier. Conversely, once that settled snapshot exists, an operation not already issued before it cannot use the same-cycle pre-settle window to bypass the activated `Core.16` block.

Dependency convention. Core.Phase defines the settled-frontier trigger and phase ordering without assuming the continuation discipline. Core.16 separately defines the drain-only continuation obligations activated by that trigger. Core.LAdm, stated after Core.16, combines Policy-L legality, Core.Phase, nonwithdrawal, $K\epsilon$, and the activated Core.Drain qualification; there is therefore no circular definitional dependency between admissibility and the drain discipline.

Definition Core.PolicyS (conservative serial-blocking issue).

Policy S uses the same Sys_ref state space, channels, values, witness choices, and transition rules as Policy L, but adds one process-side restriction: within a source interval between adjacent ordinary synchronization calls selected into S, a later source-order blocking Push/Pop operation may not issue until every earlier source-order blocking Push/Pop operation in that same interval has committed in a strictly earlier clock cycle. In Sys_B^elab, the relevant commit for this restriction is the process-facing commit that completes, accepts, or releases the owning source-level blocking operation at its selected boundary; internal staged/bundle/join/epoch-gate transfers do not by themselves satisfy the serial condition unless the elaboration package identifies that transfer as the process-facing commit for the source operation. Policy S serializes only process-side issue of source-level blocking calls. It does not serialize independent peer processes, E4 channel-side drainage, FIFO movement, or ϵ -steps that are otherwise admissible. Core.SelectLatitude applies under Policy S as well: the serial rule restricts which issue transitions are legal but does not require an earlier legal issue to be selected at its first opportunity. Core.IssueFair remains the complete-execution eventual-issue condition and B2 remains the corresponding boundary-selection condition; a later deterministic Policy-S instantiation may separately fix the concrete choices without changing this legality.

Remark Core.PolicyS-Sim (Relation to practical pre-HLS simulation modes).

A raw serial source simulation mode implements the Policy-S discipline for ordinary unelaborated blocking calls. A Matchlib-style eager or skidbuffered source simulation mode may execute source calls in dynamic program order while issuing requests without consuming a clock cycle until data or space is unavailable; that mode is Policy-L-compatible only to the extent that its request attribution, channel states, and non-blocking outcomes satisfy the transition-system conditions above. Neither mode treats ordinary distinct-interface source calls as unordered. The difference between serial and eager simulation modes is a difference in issue timing and buffering, not a difference in $<_{src}$.

Engineering interpretation (informative). Policy S is intentionally conservative about request overlap, not about pipelining as such. Its potential false-negative case is concentrated in protocols whose liveness requires source-later blocking requests to be exposed before earlier blocking calls complete; the simplest example is a crossed two-process/two-rendezvous-channel pattern. In a concrete buffered or elaborated instance, any HLS-created storage or credit that affects acceptance/backpressure is represented by B(c) or explicit Sys_B^elab structure under B-faithfulness. If that stored slack breaks the wait dependency, the selected instance is no longer the rendezvous-only case and Policy-S applicability is assessed on the actual buffered/elaborated architecture. Thus, although the rendezvous counterexample is important for stating the theorem correctly, the corresponding Policy-S conservatism is expected to be much less significant in practice for modern pipelined systems with buffered or credit-based communication. This observation does not strengthen Corollary K.2c and does not make positive finite capacity sufficient for A5-L; a finite buffer may fill, and a required cycle-breaking credit/occupancy property remains a separate proof obligation or alternate A5-L route.

Commit, issue, and boundary requests

Definition Core.8 (Commit).

A boundary-commit occurrence at an explicit abstract boundary endpoint is the corresponding legal E4 channel-commit transition of the selected proof model: it produces the applicable committed $\text{Push}_c(v)$ or $\text{Pop}_c(v)$ observable unit and the channel-state update prescribed by the selected rendezvous or bounded-FIFO semantics. For RTL_B , the selected explicit-boundary mapping and B-faithfulness/conformance evidence relate that abstract commit to the implementation clock cycle in which the mapped valid and ready conditions are simultaneously satisfied; the committed payload is the mapped data value observed at that boundary in that cycle. For a source-level message operation q , $\text{Commit}(q)$ denotes q 's process-facing commit occurrence. In the ordinary unelaborated case, this is the boundary-commit occurrence at q 's unique selected explicit abstract boundary. In $\text{Sys}_B^{\text{elab}}$, it is the commit/accept/release occurrence designated by the elaboration package as completing or releasing q ; proof-internal boundary commits that do not release q are not $\text{Commit}(q)$. Let $\text{CommitCycle}(q)$ denote the cycle index of $\text{Commit}(q)$. When a rule compares q 's process-facing commit with a clock cycle, it compares $\text{CommitCycle}(q)$. Per-boundary E4 commit occurrences at explicit abstract channels retain the cycle index of their own boundary-commit occurrence and are not mapped to $\text{CommitCycle}(q)$ unless that boundary commit is the designated process-facing commit for q . Process-facing commit identifies completion/release of the source operation; it does not imply that returned data or status is available for dependent use in that same cycle, which remains governed by Core.Phase L4 and Core.RegSeq .

Definition Core.9 (Issue).

For a source-level message operation instance q at a chosen explicit abstract boundary endpoint, $\text{Issue}(q)$ is the first cycle in which the calling process begins requesting the transfer corresponding to q at that endpoint. $\text{Issue}(q)$ denotes the cycle index of that operational issue event. Scheduler/witness records may carry this value for rules such as R4, E3, and E5, but they do not define a different event or an independently selectable timestamp.

Definition Core.IssueFair (weak fairness of source issue).

Fix an issue policy $\Pi \in \{\text{Policy L}, \text{Policy S}\}$ and a complete phase-bearing source execution p . $\text{IssueEnabled}_\Pi(q, t)$ holds at an applicable L1 opportunity when q has been reached, has not yet issued or been retired/reset, and its message-issue transition is legal under the operational Sys_ref relation, Core.Phase , Π , R3–R5/E4/E5, Core.RegSeq , source-order prefix closure, Core.16 , and the fixed witness/environment/reset/arbitration choices. $\text{IssueFair}_\Pi(p)$ holds iff, whenever from some cycle t_0 onward within one reset epoch q remains IssueEnabled_Π at every applicable L1 opportunity until it issues or the execution reaches an affecting RESET for q or an explicit DeclaredTerminal disposition, q is issued after finitely many such opportunities. This is weak fairness: an operation whose issue legality merely recurs intermittently does not satisfy the antecedent. An operation forbidden by Core.16 is not IssueEnabled_Π and imposes no IssueFair obligation. IssueFair governs assertion/issue of a reached source operation; B2/B3 separately govern selection/commit of already-issued boundary actions. A constructive complete-source continuation theorem may use a stronger recurrent-opportunity dovetailing scheduler, but the required property remains Core.IssueFair itself. Appendix K consumes Core.IssueFair as a premise of complete admissible continuations.

Definition Core.10 (Endpoint-owned boundary request).

Fix an endpoint direction at an explicit abstract boundary. For any message-operation frontier item $x \in \Omega_{\text{msg}, P}$ with owning source-level message instance $q = \text{MsgOf}_P(x)$ and evaluated boundary $c = \text{BoundaryOf}_P(x)$, let $\text{req}_x(t)$ denote the endpoint-owned boundary request signal for that endpoint in cycle t , namely valid for a Push endpoint and ready for a Pop endpoint. The predicate $\text{BoundaryReq}(x, \sigma)$

records that this process-owned request for frontier item x is asserted at state σ . $\text{BoundaryReq}(x, \sigma)$ is operation- and boundary-specific: it is true only when the asserted endpoint-owned request at σ is the request of the specific frontier item x , its owning source-level operation q , and its evaluated boundary $\text{BoundaryOf_P}(x)$. On RTL_B , identifying that process-owned request is an implementation-side boundary-mapping/conformance fact. For each admitted boundary/direction, the selected mapping is fixed for the theorem instance and shall exhibit the implementation request condition and owner decode; it is bidirectionally complete (BoundaryReq holds exactly when that fixed mapping identifies the applicable x/q as the process-owned request) and may not vary by cycle or dynamic instance to satisfy E3/E5. A raw physical valid/ready level that instead represents complementary channel/storage availability or declared coupler logic is not BoundaryReq merely because it is high. In particular, if $\text{Issue}(q)$ has not yet occurred at σ , then $\text{BoundaryReq}(x, \sigma)$ is false. In the ordinary unelaborated case, where q has a unique frontier item and a unique evaluated boundary, $\text{BoundaryReq}(q, \sigma)$ may be used as shorthand for $\text{BoundaryReq}(x, \sigma)$. In $\text{Sys_B}^{\text{elab}}$, the x -form is authoritative. Elsewhere, “operation-attributive” means operation-specific in this Core.10 sense, not witness-selectable attribution.

Definition Core.IC (Issue/Commit linkage invariant).

Fix an endpoint direction at an explicit abstract boundary. For source-level message operation instances on that endpoint direction, the following all hold:

(1) Boundary-request soundness / non-withdrawal.

If q is blocking, then once issued its endpoint-owned request remains asserted, with stable payload while asserted for Push, until q process-facing commits or an affecting RESET governed by RW2 clears that dynamic instance. Ordinary back-pressure, scheduler delay, a silent cycle, or any other nonterminal transition cannot deassert, retire, reattribute, cancel, or replace it. These are the only nonterminal mechanisms in the G–K theorem model by which an outstanding blocking dynamic instance may cease to be outstanding: process-facing commit or an affecting RW2 RESET. At every cycle-aligned state, an asserted process-owned blocking $\text{BoundaryReq}(x, \sigma)$ has exactly one live owner $q = \text{MsgOf_P}(x)$: q has issued and has not yet process-facing committed or been cleared by an affecting RW2 RESET. After that ownership ends, any continuing high physical level is not BoundaryReq unless clause (3) transfers ownership. A DeclaredTerminal disposition may instead end the execution with q still pending. Protocols that require independent nonterminal cancellation, withdrawal, retry, or reattribution of a blocking request are outside the G–K theorem model unless an explicit operational extension and its correspondence/progress obligations are separately defined and proved.

(2) Stable attribution while the same source-level request remains outstanding.

If a blocking request remains asserted across a cycle in which no transfer commits on that endpoint direction, the continued assertion remains attributed to the same outstanding q until its process-facing commit or an affecting RW2 RESET. For non-blocking PushNB/PopNB, continuous assertion of the endpoint request signal does not collapse later executed non-blocking call instances into the same q ; each later executed source-level poll is a distinct $q' \in Q_{\text{exec}, P}$ unless the witness explicitly identifies the same still-outstanding source-level request instance.

(3) Back-to-back request attribution.

If blocking q_0 commits in cycle t and the same process-owned endpoint request remains asserted in cycle $t+1$, let q_1 be the next same-direction blocking source-level message operation in the same source interval, if one exists. If q_1 is that next operation, the continuing request is owned by q_1 beginning in cycle $t+1$ and $\text{Issue}(q_1) = t+1$. The process-owned blocking request may not remain asserted but ownerless for one or more later cycles and then be attributed to q_1 . $\text{Core.SelectLatitude}$ still permits genuine scheduler/HLS delay before q_1 issues, but then q_1 's process-owned BoundaryReq has not yet begun; a high physical ready/valid value that is not q_1 's process-owned BoundaryReq is interpreted through the fixed selected boundary/channel/elaboration mapping rather than through this clause.

Non-blocking calls retain the separate clause (2) treatment.

(4) Same-endpoint blocking issue serialization.

For distinct blocking q_0 and q_1 on the same scalar explicit abstract boundary and endpoint direction governed by the Core single-transfer-per-cycle endpoint constraint within one reset epoch, if $q_0 <_{\text{src}} q_1$ and both issue, then $\text{Issue}(q_0) < \text{Issue}(q_1)$. Equal issue cycles are not legal for two such scalar blocking instances.

(5) Sync-boundary discipline.

If q_0 and q_1 are same-direction source-level message operation instances and $q_0 \in \text{pref_S}(s)$ and $q_1 \in \text{suff_S}(s)$ for a synchronization call s , then $\text{Issue}(q_1)$ occurs strictly after the commit of s . Any request assertion that persists while the process is still pending at s is attributed to the outstanding operation(s) in $\text{pref_S}(s)$, not to issuance of any operation in $\text{suff_S}(s)$. Synchronization-completion legality itself is governed by Core.SyncFence below.

Definition Core.SyncFence (synchronization-completion legality).

Let s be a dynamic synchronization-call instance of process P selected into S , and let $\text{BPref_S}(s)$ be the blocking Push/Pop instances in $\text{pref_S}(s)$. A boundary action that commits s is legal only if every $q \in \text{BPref_S}(s)$ has process-facing committed no later than that same cycle. Equivalently, after s commits and P advances into the succeeding source interval, no $q \in \text{BPref_S}(s)$ may remain issued and uncommitted. Same-cycle completion is permitted: q and s may both commit in cycle t provided q 's process-facing commit is included in the selected L3 boundary-action set for t and is accounted for before the interval advance caused by s . This “accounted for before” convention is an ordering of the source operational update and of the proof/certificate attribution of the same-cycle boundary batch; on RTL_B it does not require an observable physical sub-cycle ordering between two events that commit on the same active clock edge. In $\text{Sys_B}^{\text{elab}}$, “process-facing commit” means the commit/accept/release event designated by the elaboration package as completing the owning source-level operation q ; an internal staged, bundle, join, guard, or epoch-gate transfer that does not release q does not satisfy this fence. Appendix G rules R3 and E5 state the corresponding source-side and implementation-side trace constraints explicitly.

Status and conformance split. Core.IC and Core.SyncFence are constitutive legality/invariant clauses of the selected Sys_ref operational transition system: source executions outside these rules are not executions of the theorem's prescribed source model. They are not global proof axioms to be admitted independently of that semantics. On RTL_B, the corresponding request lifecycle, process-owned BoundaryReq identification, process-facing completion, and synchronization-fence behavior is not obtained by definition. The selected boundary mapping/B-faithfulness and request-lifecycle/attribution evidence establish the applicable Core.IC request facts; E3/E5 then check the resulting Issue/Commit ordering at their stated scope. The named Appendix M.7a Core.IC/Core.SyncFence audit entries record the required implementation-side conformance evidence.

Remark Core.10a (Non-blocking polls).

For non-blocking message passing, a failed poll produces no Σ event and is modeled as an ϵ -step, while a successful poll contributes the corresponding committed $\text{Push_c}(v)$ or $\text{Pop_c}(v)$ event. Each executed PushNB/PopNB call is nevertheless a distinct dynamic source-level instance $q \in Q_{\text{exec},P}$ for its owning process P ; thus repeated failed polls in later loop iterations are distinct q 's even if the endpoint request remains continuously asserted. The ϵ -modeling concerns observability in Σ , not the dynamic-instance identity of the underlying source-level calls.

Channel enablement

Definition Core.11 (ChannelEnabled / ChannelDisabled).

Let x be a message-operation frontier item on channel $c = \text{BoundaryOf_P}(x)$, and write $q = \text{MsgOf_P}(x)$ with $\text{kind}(q) \in \{\text{Push_c}, \text{Pop_c}\}$. At a $\text{BoundaryEvalPoint_I}(\sigma)$, let $\text{MirrorAvail_E}(x, \sigma)$ denote the settled complementary handshake value at that boundary: ready as seen by a Push endpoint, or valid as seen by a Pop endpoint. Define $\text{ChannelEnabled}(x, \sigma) \Leftrightarrow \text{BoundaryReq}(x, \sigma) \wedge \text{MirrorAvail_E}(x, \sigma)$, and $\text{ChannelDisabled}(x, \sigma) \Leftrightarrow \text{BoundaryReq}(x, \sigma) \wedge \neg \text{MirrorAvail_E}(x, \sigma)$. For an ordinary primitive Sys_B boundary, MirrorAvail_E is exactly the E4 full/empty or rendezvous complement described in Core.12/Core.13. For $\text{Sys_B}^{\text{elab}}$, MirrorAvail_E is the deterministic value produced by the finite stateless ready/valid network declared in E from the fixed roots of the applicable boundary evaluation: the pre-L3 roots for a $\text{Core.Settle L2 SettlePoint}$, or the actual post-L3 roots for a $\text{Core.KObsSettle KObsEvalPoint}$. A coupler may contribute to this value at an internal or process-facing outer boundary but does not own q , does not alter BoundaryReq , and introduces no storage. In the ordinary unique-boundary case, q -forms may be used as shorthand. Every L2 SettlePoint and every Core.KCut used by Appendix K is a BoundaryEvalPoint ; a post-L3 KObsEvalPoint is observation-only and cannot create another L3 commit opportunity.

Definition Core.12 (Buffered channels).

If $B(c) > 0$, E4 imposes the primitive storage legality condition at c : a Push may commit only when $\text{occ_c}(\sigma) < B(c)$, and a Pop may commit only when $\text{occ_c}(\sigma) > 0$, using the begin-of-cycle non-fall-through occupancy. At an ordinary uncoupled Sys_B boundary these conditions also completely determine MirrorAvail_E , so:

- Push: $\text{ChannelEnabled}(x, \sigma) \Leftrightarrow \text{BoundaryReq}(x, \sigma) \wedge \text{occ_c}(\sigma) < B(c)$;
- Pop: $\text{ChannelEnabled}(x, \sigma) \Leftrightarrow \text{BoundaryReq}(x, \sigma) \wedge \text{occ_c}(\sigma) > 0$.

Hence the familiar disabled-Push $\Rightarrow$ full and disabled-Pop $\Rightarrow$ empty characterizations are valid for ordinary primitive buffered boundaries. At a coupling-sensitive $\text{Sys_B}^{\text{elab}}$ boundary, the same occupancy tests remain necessary E4 legality conditions, but they need not be sufficient for MirrorAvail_E : a stateless declared coupler may withhold the complementary ready/valid value because another input, downstream space, epoch gate, or other explicitly modeled handshake condition is not satisfied. Such withholding is legal only when it is part of E 's settled combinational network and process-dependency projection; it is not additional channel state and it is not a hidden predicate.

Definition Core.13 (Rendezvous channels).

If $B(c) = 0$, a committed transfer remains an atomic same-cycle rendezvous $\text{Push_c}(v)/\text{Pop_c}(v)$ pair. At an ordinary uncoupled rendezvous boundary, $\text{MirrorAvail_E}(x, \sigma)$ is exactly the complementary endpoint request, so ChannelEnabled is BoundaryReq for x conjoined with the matching complementary Push/Pop request. At a coupling-sensitive $\text{Sys_B}^{\text{elab}}$ rendezvous boundary, both source-attributed endpoint requests may be present while the declared stateless coupler network still withholds the complementary mirror value because another explicit join/downstream condition is unsatisfied; the rendezvous is then ChannelDisabled at that evaluated boundary. Conversely, a coupler cannot synthesize a committed rendezvous without the required source-attributed requests and E4/rendezvous legality. Thus the ordinary missing-complementary-request characterization is used only for primitive uncoupled rendezvous boundaries; elaborated rendezvous disabledness is characterized by the settled network declared in E .

Pending blocking frontier

Definition Core.14 (Pending blocking operations).

Fix a $\text{BoundaryEvalPoint}_I(\sigma)$ for process P within the current interval $I_P(\sigma)$ between the adjacent synchronization calls that bracket σ in P 's current execution. Define

$\text{Pending}_P(\sigma) := \{ x \mid x \in \Omega_{\text{msg},P} \text{ is a blocking source-level message-operation frontier item in that same interval } I_P(\sigma), \text{ and for } q = \text{MsgOf}_P(x), q \text{ is defined, Issue}(q) \text{ has occurred, and Commit}(q) \text{ has not yet occurred in } \sigma \}.$

By Appendix C's Issue/Commit linkage together with the definition of BoundaryReq , each member x of $\text{Pending}_P(\sigma)$ has its endpoint-owned request asserted at the cutpoint, with ownership read through its dynamic message-call instance $q = \text{MsgOf}_P(x)$ under Core.10/Core.IC. By Core.SyncFence, an issued blocking operation from the preceding interval cannot survive the synchronization commit that advances P into $I_P(\sigma)$; therefore the interval restriction cannot discard an outstanding blocker. This definition is valid both at an L2 SettlePoint used by Core.16 and at a Core.KCut comparison cutpoint; WFG and Dead_K use only the latter.

Definition Core.15 (Pending blocking frontier $\text{Front}_P(\sigma)$).

Define

$\text{Front}_P(\sigma) := \min_{\leq P}(\text{Pending}_P(\sigma))$

If several $\leq_P$ -minimal pending blocking message-operation frontier items are simultaneously outstanding, they form a frontier. Lemma Core.OrdFrontSingleton proves that this set has at most one member for an ordinary non-elaborated source interval. This is the Appendix B / K notion of the pending blocking frontier (or minimal pending blocking frontier). It is distinct from $\text{NextFront}_P(\sigma)$, which is the broader source-level frontier-item set over Ω_P .

Wait-for graph

Definition Core.WFG (Wait-for graph).

At a K -facing cutpoint $\sigma \in \text{KCUTReach_ref}(I)$, or at the corresponding certified RTL K -facing view, define $\text{WFG}(\sigma)$ over the original modeled processes. For every channel-disabled frontier item $x \in \text{Front}_P(\sigma)$, the selected elaboration E supplies a finite release-support family $\text{ReleaseOwnerSets}_E(x, \sigma)$. Each member S of this family is a finite non-empty set of modeled processes whose process-facing progress is jointly sufficient, under the qualified E template, to remove that particular settled blocking condition; members of one S are conjunctive requirements, while different S values are alternative release supports. The family is maintained as an antichain of minimal supports. Define the coarse owner set $\text{WaitOwners}_E(x, \sigma) := \bigcup \text{ReleaseOwnerSets}_E(x, \sigma)$, and add $P \rightarrow Q$ to WFG for every internal modeled process $Q \in \text{WaitOwners}_E(x, \sigma)$. Thus WFG deliberately retains the conservative ordinary directed-graph union used by structural analyses, while Definition K.Dead below uses the unflattened release family for exact closure. At an ordinary primitive point-to-point boundary, $\text{ReleaseOwnerSets}_E(x, \sigma) = \{\{Q\}\}$ for the unique complementary endpoint owner Q , so $\text{WaitOwners}_E(x, \sigma) = \{Q\}$ and the former definition is recovered. Couplers, FIFO stages, and other elaboration-local nodes are never WFG process vertices merely because they appear in the ready/valid network.

Here "internal" is a property of the projected process dependency: the members of $\text{ReleaseOwnerSets}_E$ are modeled processes of $\text{Sys_ref} / \text{RTL_B}$ rather than the external environment/testbench. A low-level elaborated boundary may have a stateless coupler on one side; that does not make the boundary external and does not require inventing a coupler process. Appendix Q requires the release-support family to be correct as a family, not merely edge-by-edge: no declared support may be a spurious rescue alternative, and every genuine internal modeled-process release alternative must contain a declared minimal support. External/terminal alternatives remain outside

ordinary WFG/Dead_K and are handled by the existing environment-waiver/diagnostic machinery. Purely local combinational paths cannot manufacture a process self-dependency merely by starting and ending inside P's elaboration; a $P \rightarrow P$ dependency is retained only when a genuine release support includes process behavior owned by P. Buffered and rendezvous storage elements may both occur in the dependency cone. Core.16 itself still uses the ALL-disabled frontier test at the L2 SettlePoint and reaches the K-facing WFG domain only through Core.SettleKBridge/K.DrainReach.

Clarification Core.WFGa (WFG edge versus blocked process).

A directed WFG edge records a channel-disabled dependency of one frontier item; it does not by itself imply that the owning process P is blocked on message passing. If $\text{Front_P}(\sigma)$ contains another explicitly incomparable channel-enabled member, P has an outgoing WFG edge but is not blocked under Appendix K's ALL-disabled test. The internal message-passing-deadlock predicate therefore quantifies only over nonempty sets D in which every $P \in D$ is blocked, so every edge used to close a deadlocked component originates from a process with no channel-enabled frontier item.

Clarification Core.WFGb (Multi-frontier outgoing edges).

When $\text{Front_P}(\sigma)$ has multiple members, P may have multiple outgoing coarse WFG edges from one or more disabled frontier items. In an ordinary non-elaborated interval, Lemma Core.OrdFrontSingleton and point-to-point ownership reduce the active frontier to at most one item with the singleton release family $\{\{Q\}\}$. In $\text{Sys_B}^{\text{elab}}$, either one coupling-sensitive frontier item or several explicit frontier items may have non-singleton/multiple ReleaseOwnerSets_E supports; WaitOwners_E is only their per-item owner union for graph display and structural over-approximation. The ALL-disabled test remains across frontier items, while conjunction/disjunction inside one item is retained by its ReleaseOwnerSets_E family. Ordinary source-sequenced calls do not become incomparable, and local coupler/FIFO nodes do not become processes, merely because the realization contains several internal ready/valid stages.

Drain-only blocked-frontier discipline and automatic flush

Definition Core.16 (Drain-only blocked-frontier discipline).

A source interval satisfies the drain-only blocked-frontier discipline if, at the L2 SettlePoint_I(t, σ_t^{set}) of Definitions Core.Phase/Core.Settle, $\text{Front_P}(\sigma_t^{\text{set}})$ is non-empty and every frontier item $x \in \text{Front_P}(\sigma_t^{\text{set}})$ is channel-disabled. The discipline activates at that first settled snapshot. It does not retroactively invalidate source-later operations legally issued during the preceding L1 pre-settle window, but it forbids any new such issue after σ_t^{set} in that cycle and while the same fully disabled frontier persists. Set $\sigma := \sigma_t^{\text{set}}$ at activation. This activation state is a BoundaryEvalPoint and, by Lemma Core.SettlesKCut, is itself a source Core.KCut in the applicable phase-bearing source proof model; it is not necessarily the later K-facing cutpoint at which Dead_K is evaluated, and that connection is supplied by Core.SettleKBridge/K.DrainReach. Once activated, the following all hold:

- (1) Block point. P is blocked on message passing at $\text{Front_P}(\sigma)$.
- (2) No new later work. While $\text{Front_P}(\sigma)$ remains non-empty and fully disabled, no new later blocking message-operation frontier item $y \in \Omega_{\text{msg},P}$ is issued, where "later" means that, for some $x \in \text{Front_P}(\sigma)$, $x \leq_P y$ and $x \neq y$, with issue understood through $q_y = \text{MsgOf_P}(y)$. More generally, no new source-later dynamic operation is launched past the blocked point. "Launched" is intentionally broader than Issue(q): it includes a source-evaluation or message-issue transition that advances new source-later dynamic work past the blocked point. The no-new-Issue and broader no-new-launch clauses remain distinct.
- (3) No withdrawal. Already-asserted blocking requests are not withdrawn before the owning message instances commit.
- (4) Frontier-minimality. An already-asserted blocking request whose frontier item is not in $\text{Front_P}(\sigma)$ is

not treated as a wait-for source unless and until that frontier item later becomes frontier-minimal.

(5) Drain-only already-admitted progress. Work admitted before Core.16 activation may continue to advance through complete proof-model action units, including E-declared proof-internal staged/bundle/join transfers and ordinary already-issued boundary commits, provided those actions do not launch or issue new source-later work past the blocked frontier. Under a Stage-2 synchronization-qualified profile, an already-reached/pending Core.SyncCarrier whose transaction is currently SyncReady may also complete through Core.7d SyncFire, and a currently represented lowering-created epoch-gate token/request/credit may advance through Core.7e EpochGateAdvance. SyncFire advances only across the synchronization boundary to the immediate post-sync control/interval entry and does not include post-sync ReachPrep/Issue. “Already admitted” is a current-state property: the work is represented by a current attributed outstanding request, explicit reconstructed channel/token/storage/gate state, or a current typed synchronization carrier; no historical Issue timestamp is required. A peer that still must execute source control to reach a synchronization is not admitted drain work and remains a SyncReleaseOwnerSets release owner. An in-flight action need not be owned by a process in a later deadlock candidate D; actions outside D remain relevant when they can change D’s reconstructed blocked-frontier roots.

(6) Finite non-replenishing admitted domain. Every selected explicit abstract boundary that can hold or credit already-admitted work has finite B-faithful storage/credit, directly or through a finite Sys_B^elab chain, and the selected elaboration declares a finite action-unit inventory over that state. Under a Stage-2 synchronization profile, the admitted population additionally contains only the finite current set of reached/pending SyncTxn keys, counted once per atomic transaction rather than once per endpoint, plus the finite explicit lowering-created gate state bound to those keys. Together with clause (2), the currently admitted population is finite and receives no new source work. The applicable drain qualification additionally supplies a well-founded stage/rank or equivalent finite multiset measure strictly decreased by every selected in-flight progress action, including SyncFire/EpochGateAdvance when present. Finite capacity alone is not a termination proof, and the measure is activation-local: no old activation certificate or rank is reused after source progress, reset, or candidate termination creates a fresh activation.

(7) Sync-boundary capacity withdrawal — elaborated-boundary interpretation.

While a process is pending at an explicit synchronization call s , producer-facing availability for any back-annotated capacity belonging only to $\text{suff}_S(s)$ is withheld until s commits; work already accepted before s may still drain. For the Sys_ref proof model, this withdrawal is not an additional hidden enablement predicate on an otherwise ordinary E4 channel. It is represented only by choosing $\text{Sys_ref} = \text{Sys_B}^{\text{elab}}$ with explicit sync-boundary / epoch-gate abstract channels, or equivalent explicit staged realization structure, so that the refusal appears as ordinary ChannelDisabled behavior at an explicit abstract boundary.

For a pipelined or otherwise overlapped implementation, Appendix B’s automatic-flush rules are the implementation-side construction that supplies Definition Core.16. For ordinary non-overlapped control, the same definition may be discharged by a structural proof that no new source-later dynamic operation can be launched after the blocked point is reached. Thus the K.2b dominance/extraction premise ranges over every liberal interval that can contain eager issue, not only loop pipelines. In particular, eager issue permitted by Policy L must occur in Core.Phase’s L1 pre-settle window; the no-new-launch obligation begins at the L2 settled snapshot, not at the earlier issue of the frontier request.

Definition Core.Drain (activated drain-discipline qualification).

For a fixed liveness instance I and a Policy-L path or prefix p , Core.Drain_I(p) holds when every Core.Phase L2 SettlePoint at which Definition Core.16 activates obeys all seven Core.16 clauses for as

long as the same fully disabled frontier persists. The qualification is bound to the fixed instance, witness, environment, arbitration, reset, elaboration, boundary attribution, and the certified already-admitted in-flight domain. Its finite-drain clause is interpreted over the selected B-faithful explicit capacities/credits together with the finite action inventory and well-founded progress measure of Core.16(6); it is not restricted to D-owned downstream source offers. For Appendix K applications, the same prefix/continuation must additionally satisfy Core.SettleKBridge_I so that an L2 activation is never silently treated as the later global K-facing comparison state. Core.Drain is the continuation-level no-replenishment/activation qualification consumed by Core.LAdm. It does not itself assert that an arbitrary future cycle or selected action opportunity exists; the separate Core.InFlightCycleExt/J.InFlightPathSound interface supplies that continuation-existence justification when generalized DrainClosedNow uses reconstructed finite paths. Later theorem layers may require design- or cutpoint-specific evidence that discharges these qualifications, but that evidence does not redefine them.

Definition Core.InFlightStep (already-admitted proof-model progress action).

Fix a source proof model M and elaboration package E . $\text{Core.InFlightStep}_{\{M,E\}}(s,u,s')$ means that u is one complete E -typed proof-model action unit, or one E/J -declared indivisible atomic group, that advances work already represented in the current state s and produces the exact successor s' under the selected operational semantics. Stage-1 action kinds include an ordinary already-issued Push/Pop commit and a proof-internal staged/bundle/join transfer. Under a Stage-2 synchronization-qualified schema, the relation may additionally contain Core.7d SyncFire for an already-pending ready synchronization carrier and, on Sys_K, Core.7e EpochGateAdvance for explicit lowering-created gate state. It may cross an explicit abstract channel and therefore need not be ϵ . It is not a source launch/Issue transition and does not introduce new source-later work; SyncFire may advance process control across the synchronization boundary only to the immediate post-sync point and includes no post-sync ReachPrep/Issue. It respects Core.Phase/Core.16, Core.SyncFence where applicable, the fixed reset/environment/arbitration/witness choices, $E4$ and any declared atomicity, and the E -declared projection/accounting semantics. The relation has no candidate- D argument and no ownership/relevance filter: D later scopes only which blocked-frontier roots are observed. A reconstructed in-flight edge names one distinguished already-admitted action unit and its exact state effect. Its operational realization may require phase advancement or crossing one or more real cycle boundaries before that action is selected; those opportunity/embedding steps are justified by Core.InFlightCycleExt/J.InFlightPathSound and are not silently treated as additional source Issues or as part of Q.InFlightActionEffectSound.

State-based admission rule. An action is eligible for the in-flight universe only because its required work is currently represented in the reconstructed state: an attributed outstanding request, explicit abstract channel/token/storage state, or—under the Stage-2 synchronization profile—a reached/pending Core.SyncCarrier whose transaction is already ready without additional source reach/launch. Historical Issue timestamps are neither required nor reconstructed. A peer process that still must execute source control to reach a synchronization is represented through SyncReleaseOwnerSets and is not converted into an in-flight action. Likewise, an epoch-gate action is admitted only from currently represented lowering-created gate state. A template that needs hidden state to decide action identity, enablement, or effect shall expose that state in the qualified proof model or prove it irrelevant.

Definition Core.InFlightDrainWF (finite admitted population and well-founded progress).

For a Core.16 activation and selected in-flight action schema, Core.InFlightDrainWF holds when the currently admitted action-bearing population is finite and a declared well-founded rank/multiset measure μ is defined on its reconstructed state such that every Core.InFlightStep selected as drain

progress strictly decreases μ . Clause Core.16(2) supplies non-replenishment by forbidding new source launch/Issue into the activation-local domain. Capacity bounds contribute finiteness of the current population but are not by themselves the decrease argument. Under a Stage-2 synchronization profile the admitted population and measure also account for current synchronization transaction keys and lowering-created gate state without double-counting one multi-endpoint SyncTxn per participant. A sufficient simple local ranking for an epoch lowering is pending-ready synchronization $>$ residual epoch-gate realization $>$ retired, provided the qualified action/effect theorem proves that SyncFire and EpochGateAdvance strictly move the same admitted obligation downward and never create a fresh source operation. The measure and admission certificate are activation-local and end when the applicable fully-disabled frontier/activation ends, an affecting reset occurs, or later source progress creates a fresh activation.

Definition Core.InFlightCycleExt (restricted in-flight continuation-existence qualification).

Fix M, I, E and a qualification profile Φ over reachable K -facing/cycle-entry states and the no-new-launch in-flight segments exposed by the selected action schema. $\text{Core.InFlightCycleExt}_{\{M, I, E, \Phi\}}$ is the restricted C2/C3-style existence record used to justify reconstructed paths. Its load-bearing requirement is compositional closure: after every finite operational realization prefix admitted by Φ , either a modeled reset/terminal/activation-termination branch ends that profile instance or the reached successor remains in Φ so the next reconstructed in-flight edge can be realized by the same rule. For every required cross-cycle edge, Φ also supplies: (ICE1) existence of the needed legal CycleResult/CompleteCycle and compatible EnvStep; (ICE2) a finite phase-legal path through any required L1 preparation (possibly empty under Core.SelectLatitude), the actual L2 SettlePoint, and the selected in-flight L3 action; (ICE3) compatibility with Core.16, E4, fixed arbitration/witness/reset semantics, and no new source launch/Issue; and (ICE4) a resulting K -facing successor whose reconstructed state is the declared action successor. This is an existence/embedding qualification, not a fairness premise.

Discharge relation to Core.CycleClosure. Full Core.CycleClosure plus the applicable phase/action legality facts is a sufficient discharge route for Core.InFlightCycleExt, via Core.CycleChoiceExt, but the restricted interface quantifies only over the profile Φ rather than all reachable nonterminal cycle-entry states and all partial cycles. StandardEnv may simplify the environment-stutter component but does not by itself prove CycleResult existence. B2/B3 likewise cannot bootstrap this existence: their eventuality conclusions quantify over an execution whose existence must already be justified. In the profile-G1 no-lowering serial route, $\text{Sys_K} = \text{Sys_ref}$ and $K.\text{SerialRoutePremises}$ already binds the relevant Core.CycleClosure; alternative A5-L routes shall supply Core.InFlightCycleExt, full Core.CycleClosure, or an equivalent checked path-existence theorem when generalized drain is used.

Definition Core.SyncInFlightOpportunityQual (synchronization/epoch opportunity specialization).

For a model M , instance I , schema E , and in-flight profile Φ that exposes Core.7d SyncFire and/or Core.7e EpochGateAdvance edges, $\text{Core.SyncInFlightOpportunityQual}_{\{M, I, E, \Phi\}}$ holds when the generic Core.InFlightCycleExt obligation is discharged for those constructors with the following specialization.

(SIO1) If a reconstructed SyncFire edge is exposed, the operational transaction is already pending/ready and Core.SyncFence-valid; the embedding reaches an applicable L3 synchronization opportunity without performing any source-later ReachPrep/Issue, selects the exact atomic synchronization unit, and stops at a K -facing successor whose reconstructed state is the declared SyncFire successor. (SIO2) If a reconstructed EpochGateAdvance edge is exposed on Sys_K , the embedding reaches the exact E4/handshake gate opportunity, selects that complete lowering action, and reaches its declared successor without treating the gate step as a source synchronization event. (SIO3) Every required intervening real cycle has a compatible EnvStep/reset disposition and preserves the fixed

transaction/gate identity until the selected action or an explicit profile-termination branch; B2/B3 do not manufacture this continuation. (SIO4) After each realized action, either reset/terminal/activation termination ends the profile instance or the successor remains in Φ , so a later finite-path proof may compose another edge. Full model-indexed Core.CycleClosure plus the local action legality facts is a sufficient discharge; a narrower route-specific theorem is also permitted. If a profile cannot prove opportunity for a locally plausible synchronization/gate action, it shall omit that edge conservatively rather than expose a fictitious escape.

Remark Core.16a (ALL disabled $\Rightarrow$ stall).

The document’s intended policy is exactly the “ALL disabled $\Rightarrow$ stall” interpretation: the process stalls only when none of the frontier items in $\text{Front_P}(\sigma)$ is channel-enabled. Downstream drain is compatible with that stall provided no new later work is issued past the blocked frontier.

Definition Core.LAdm (admissible Policy-L prefixes and complete executions).

For Appendix K, a finite Policy-L prefix is admissible under Core.LAdm iff it is Policy-L-legal, follows L0–L4, respects nonwithdrawal and $K\epsilon$, satisfies Core.Drain from every activation point reached in the prefix, and satisfies $\text{Core.SettleKBridge}$ for every K-facing cutpoint selected to evaluate a persisting blocked frontier. For a finite prefix, “follows L0–L4” means that every transition actually present in the prefix obeys the phase ordering up to that prefix endpoint; it does not require the prefix to execute later subphases or the L4 update after its selected endpoint. In particular, when an Appendix-K prefix ends at σ_{ref} after a recorded Core.KObsNormRef suffix, that suffix is phase-legal in the observation-only L3Obs subphase of Core.KObsSettle and the admissible prefix may end there before L4. A $\text{Core.SelectLatitude}$ -permitted nonselection, and any resulting legal $\text{CompleteCycle/SilentCycle}$, is ordinary Policy-L-legal behavior; finite nonselection is not itself a fairness failure or evidence that Core.16 has activated. Write $\text{Pref_L}^{\text{adm}}(I)$ for these finite prefixes. A complete Policy-L-admissible execution follows $\text{Core.CompleteCycle}$ for every real cycle and is either infinite or ends at an explicitly DeclaredTerminal state; it additionally satisfies the selected B2/B3/B5 progress/quiescence obligations and Core.IssueFair for Policy L. A state is not finite-maximal merely because no Σ -action or ϵ -step is enabled at the present cycle: if it is nonterminal, $\text{Core.CycleClosure/CompleteCycle}$ continues it, possibly through a SilentCycle . If a complete execution is extended indefinitely by the B6 convention, the terminal state must also satisfy Core.TerminalIdle (including its Core.EnvTerminal component). Write $\text{Exec_L}^{\text{adm}}(I)$ for this set. A5-L’s reachability clause quantifies over the source K-facing cutpoints $\text{KCutReach_ref}(I)$ reached by members of $\text{Pref_L}^{\text{adm}}(I)$, not over L2 SettlePoints merely because they are combinationally settled. K.2b.Q’s continuation, stabilization, and limit arguments additionally require the existence of a member of $\text{Exec_L}^{\text{adm}}(I)$ extending the finite counterexample prefix. Lemma K.2b.E constructs such an extension from its stated premises, using Core.FairPick and Lemma $\text{Core.FairCycleDovetail}$ to turn the existing $\text{Core.IssueFair/B2/B3}$ discharges into typed per-cycle choices; the extension is not a theorem-application certificate field or an additional witness-package assumption. Fairness is therefore not decided by a finite prefix, but a finite prefix that has already violated the phase, freeze, or settle-to-K-cutpoint bridge rules is not admissible.

Normative-use convention. Elsewhere in this document, “Policy-L execution” means a Policy-L-legal path unless the context says otherwise. “Admissible Policy-L execution/prefix” in A5-L, K.1A, K.2b, K.Ser, the K.CE examples, and Appendix K-P means Core.LAdm -admissible. Core.Drain is therefore not silently part of the basic Policy-L issue rule; it is an explicit qualification of liveness admissibility. Policy S uses the same L0–L4 cycle phases, with its stricter serial-blocking issue condition substituted for the Policy-L issue permission in L1.

Observable compatibility and cutpoint correspondence

Definition Core.17 (System-level observable compatibility).

For a witness prefix τ_{ref} of Sys_ref and a post-HLS prefix τ_{post} of RTL_B , define $\Sigma\text{Match}(\tau_{\text{ref}}, \tau_{\text{post}})$ to mean that the selected Σ -visible events/observable units match under the canonical occurrence convention of this theorem stack. For each message-passing channel c , the projected sequences of committed Push_c events and of committed Pop_c events have the same length; the k -th Push_c on one side matches the k -th Push_c on the other, and likewise for Pop_c , with the same endpoint and payload/value label. For each observable signal sig of a sequential process P , the canonicalized sequences of $\text{Read}(\text{sig}, \cdot)$ and $\text{Write}(\text{sig}, \cdot)$ occurrences have the same length and the k -th occurrences match with the same process/signal/direction and value label; within each source anchor interval, canonicalization records at most one Σ -visible Read and one Σ -visible Write for each (P, sig) at the E2 anchor; if several source Writes share it, the last executed Write supplies its value and earlier writes are ϵ -internal. For each R2C combinational process or explicitly modeled combinational network component C_{comb} , the emitted fixed-point observation-unit sequences have the same length and the k -th units match by component identity, included process/signal/direction labels, and fixed-point values, including any cycle-locked participation requirement. For proof-internal $\text{RESET}(S_p, S_c)$ units included in the selected theorem comparison, project each trace to the RESET occurrences of each reset scope (S_p, S_c) ; the k -th RESET occurrence for a scope on one side matches the k -th RESET occurrence for the same scope on the other, so matched RESET units have identical S_p and S_c . Independent disjoint RESET units may commute except where the reset well-formedness rules or another selected observable-order constraint imposes an order. Synchronization events match by their per-process occurrence index in source order. Independent matched Σ events may commute or occupy different global cycle buckets except where E1–E5 or the selected observable-unit semantics impose an ordering or same-cycle constraint. An unmatched selected Σ event/unit on either side makes ΣMatch false. Define $\tau_{\text{ref}} \approx \tau_{\text{post}} :\Leftrightarrow \Sigma\text{Match}(\tau_{\text{ref}}, \tau_{\text{post}}) \wedge E1 \wedge E2 \wedge E3 \wedge E4 \wedge E5$

This is an ordered compatibility predicate, not a mathematical equivalence relation. The left argument supplies the Sys_ref witness prefix, and the right argument supplies the RTL_B post-HLS prefix. ΣMatch and E1–E5 are normative conjuncts of the Core definition. Appendix G expands the canonical Σ -event matching convention and the E1–E5 rules but does not redefine this relation or add a separate matching conjunct. In particular, the R2C matching statement in E2 below is the Appendix-G expansion of the R2C clause already contained in ΣMatch , not a second R2C matching requirement.

Definition Core.18 (Cross-model KObs correspondence).

For an exact selected source/RTL package with common typed record $\langle E, w, H, O \rangle$, cross-model KObs correspondence is equality of the source K-observation record with a post-side K-observation record of that same Core.18c type. On the source side, $\text{KObs_E}, w(\sigma_{\text{ref}}) = \langle E, w, H, O \rangle$. The Core fixes only the type and equality notion of this correspondence; it does not independently define how an RTL_B cutpoint yields such a record. Appendix J supplies that post-side representation as $\text{KObsPost_J}(\rho_{\text{post}}; E, w, H, O)$ and, for the selected package of Corollary J.1, proves by $J.\text{KObsConf}$ that $\text{KObsPost_J}(\rho_{\text{post}}; E, w, H, O) = \text{KObs_E}, w(\sigma_{\text{ref}})$. This definition introduces no additional state clauses. Process, channel, frontier, request, enablement, WFG, and snapshot-drain facts are obtained by applying Definitions Core.18d–Core.18e to the common record. Appendix K separately defines and factors its deadlock, waiver, environment-terminal, and diagnostic predicates from those reconstructed inputs.

K-observation quotient and residual-offer layer

Status of this subsection. Definitions Core.18a–Core.18e provide the source-side KObs record and its pure Core reconstruction functions. Appendix J proves the source-side reconstruction lemmas $J.\text{KObs-}$

Proc, J.KObs-Chan, J.KObs-Offers, and J.KObs-WFG, constructs the corresponding J-facing RTL record, and uses J.OfferReach/J.OfferConf/J.KObsConf to prove equal KObs at matched cutpoints. Appendix K then defines and proves the deadlock-, waiver-, and diagnostic-predicate factoring results over that record. This subsection does not characterize the image of arbitrary well-typed records beyond the explicit J.OfferReach predicate, and it does not alter the operational Issue, Policy-L/Policy-S, or DrainDiscipline semantics.

Definition Core.18a (Enriched replay history H).

Fix a theorem instance I consisting of the design, fixed stimulus, declared environment model, reset policy, chosen observation boundaries, and KObs-facing waiver set W. For the Policy-S A5-L discharge, I may additionally reference a response-monitor schema, finite-bound function, reset/end-of-test disposition, and response-waiver policy; those are operational certificate data and are not fields of KObs. I also records an environment-reconstruction mode for optional downstream diagnostics: either StandardEnv, meaning the normative B1-avail pull-based, consumption-ordered input model with default always-ready outputs, or EnvReplayAdequate, meaning a certificate that supplies a typed reconstructed environment state EnvStateRec_I(H,w) and proves that permanent next-interaction availability is determined by H, w, and that reconstructed state. The exact environment-terminal and diagnostic predicates are defined in Appendix K. Fix the full elaboration package E and witness w. For a reachable source K-facing cutpoint $\sigma \in \text{KCutReach_ref}(I)$, let $H_{E,w}(\sigma)$ be the finite canonical annotated replay history through σ . It contains the observable units and annotations needed by Theorem J.1 and I.DR/I.DR': committed Push_c(v)/Pop_c(v) events with dynamic identities and per-channel ordinals; synchronization and wait units; anchored Read/Write units; START_P, FINISH_P, and RESET(S_p,S_c) units where applicable; and the WC/witness-selected non-blocking, arbitration, signal, and ϵ -choice outcomes required to replay the selected execution. H retains explicitly atomic unit grouping, per-process replay order, per-channel FIFO order, reset-epoch structure, and the required-prefix constraints used by $\prec_J$. It quotients away incidental model-specific clock-bucket placement and ordering among independent units not required by those constraints. Raw source or RTL cycle buckets may be retained as certificate metadata, but they are not fields compared by KObs equality. $H_{E,w}(\sigma)$ contains no historical Issue(q) timestamps merely because requests were asserted before commit.

Definition Core.18b (Attributed residual offer and residual-offer set).

At a reachable source K-facing cutpoint $\sigma \in \text{KCutReach_ref}(I)$, an attributed residual offer is a tuple $o = (P, x, q, c, \text{dir}, e)$ satisfying all of the following: P owns x; $x \in \Omega_{\text{msg}, P}$; $q = \text{MsgOf_P}(x)$; $c = \text{BoundaryOf_P}(x)$; q is a blocking Push or Pop instance that has issued but not committed in reset epoch e; dir identifies the endpoint-owned producer-valid or consumer-ready direction; and BoundaryReq(x, σ) is asserted and attributed to that same x, q, c, direction, and epoch. Define the derived operation-kind projection $\text{kind}(o) := \text{kind}(q)$, with the typing invariant $\text{dir} = \text{producer-valid}$ iff $\text{kind}(q) = \text{Push}$ and $\text{dir} = \text{consumer-ready}$ iff $\text{kind}(q) = \text{Pop}$. Let $\text{ResOffers}_{E,w}(\sigma)$ be exactly the set of such tuples. $\text{ResOffers}_{E,w}(\sigma)$ is well typed and issue-prefix consistent under the existing E3/Core.IC semantics. On reachable source cutpoints, Core.SyncFence additionally ensures that no residual blocking offer owned by P belongs to a source interval earlier than P's current interval: such an offer would have had to process-facing commit before the intervening synchronization could commit. In particular: (a) its item and boundary fields are supplied by E; (b) if a source-later residual offer exists while an earlier blocking item in the applicable source prefix is also issued, uncommitted, and still requesting at an evaluated boundary, that earlier offer is also represented; (c) reset clears all offers in the ending epoch under RW2; and (d) for each explicit abstract boundary and endpoint direction there is at most one attributed outstanding source-level offer. A concrete token or transfer already accounted for as explicit channel

occupancy is not additionally represented as an independent source-level residual offer unless E declares a distinct source-attributed frontier item at that boundary.

Definition Core.18c (K-observation KObs).

For the fixed instance I, elaboration E, witness w, and reachable source K-facing cutpoint $\sigma \in$

KCutReach_ref(I), define the K-observation record by

$\text{KObs}_{E,w}(\sigma) := \langle E, w, H_{E,w}(\sigma), \text{ResOffers}_{E,w}(\sigma) \rangle$.

The fixed stimulus, environment model and reconstruction mode, reset policy, observation-boundary selection, and KObs-facing waiver set W are ambient parameters of the theorem instance I rather than duplicated in every record. K.Resp configuration and monitor ages are likewise ambient or CF-S operational data, not KObs fields. Equality of KObs records means typed equality of E and w and equality of the canonical replay-history quotient and residual-offer components. It does not require equality of the source and RTL clock-bucket decompositions permitted to differ by Theorem J.1. It is not equality of arbitrary process, scheduler, channel-realization, environment, response-monitor, or RTL microstate.

Definition Core.18c-1 (Observational waiver closure).

For fixed I,E,w, define $\text{KObsClosed}_{I,E,w}(W)$ to mean: for all $\sigma_1, \sigma_2 \in \text{KCutReach_ref}(I)$ of the same fixed instance, $\text{KObs}_{E,w}(\sigma_1) = \text{KObs}_{E,w}(\sigma_2)$ implies $(\sigma_1 \in W \text{ iff } \sigma_2 \in W)$. This is an observational-closure premise for any waiver-parameterized cutpoint predicate later claimed to be determined by KObs; it is not a claim that every well-typed KObs record is reachable. Environment-terminal adequacy and the optional diagnostic reconstructions are defined in Appendix K, where the relevant terminal predicates are introduced.

Definition Core.18d (Process and channel reconstruction from KObs).

For $K = \langle E, w, H, O \rangle$, define $\text{ProcRec}_P(K)$ by deterministic replay of P's projection of H under w, as before. Pending/source-offer inventory remains separate from ProcRec. Define $\text{ChanRec}_c(K)$ for every explicit storage/rendezvous channel c declared by E from its reset/initial state and committed Push/Pop history in H: for $B(c) > 0$ this yields occupancy, empty/full status, and logical contents/head information; for $B(c) = 0$ occupancy is zero. For $\text{Sys}_B^{\text{elab}}$, define $\text{HandshakeRec}_E(K)$ to be the unique settled valuation of every declared boundary ready/valid/data and derived enablement wire obtained by evaluating E's finite acyclic stateless coupler network from ChanRec, the process/replay roots that E permits, and the attributed outstanding requests O. This is a pure reconstruction; coupler state is not added to KObs because the couplers have no state. Projected outer occupancy remains obtained through Π_{elab} , while every explicit storage channel used by Appendix K retains its own ChanRec state.

Definition Core.18d-1 (Synchronization-carrier reconstruction from KObs).

Assume Core.SyncCarrierWF_I. For $K = \langle E, w, H, O \rangle$ and process P, define $\text{SyncCarrierRec}_P(K)$ by evaluating the Core.7b/Core.7c synchronization typing on $\text{ProcRec}_P(K)$ together with the ProcRec slices of every modeled participant in the selected SyncTxn, the committed synchronization/reset information in H, the fixed theorem-instance synchronization pairing/participant schema, and—only when SC3 requires it—the typed environment state supplied by EnvReplayAdequate or an equivalent reconstruction. Its projections SyncReachedRec, SyncPendingRec, SyncReadyRec, SyncBlockedRec, SyncTxnRec, SyncParticipantsRec, and SyncReleaseOwnerSetsRec are pure functions of the same data. No synchronization request is inserted into O, and no message BoundaryReq is synthesized. Under the Stage-2B action schema, the same reconstructed carrier defines the local SyncFireRec_I(k,r) applicability/effect function of Definition Core.7d; on a lowered Sys_K reconstruction, explicit gate channel/token state together with E defines EpochGateAdvanceRec_{I,E}(g,k,r) of Definition Core.7e. These are local reconstructed action/effect functions only: future opportunity is supplied by

Core.SyncInFlightOpportunityQual/Core.InFlightCycleExt, and cross-model path correspondence is supplied later by K.Low.InFlightPathCorr.

Definition Core.FlightStateRec and Core.InFlightStepRec (KObs-pure in-flight reconstruction).

For a fixed model M and qualified elaboration/action schema E , $\text{Core.FlightStateRec}_{\{M,E\}}(K)$ is the derived exploration state rooted at $K = \langle E, w, H, O \rangle$. It contains only the current abstract facts required by the declared in-flight actions: the relevant ProcRec slices, explicit ChanRec contents/occupancies/credits, HandshakeRec values, current attributed outstanding requests O , reset/epoch facts reconstructible from H , and fixed model/ E constants; when the Stage-2B synchronization action schema is selected it also consumes the separately typed SyncCarrierRec facts of Core.18d-1 without changing the KObs record. It is a derived state for finite exploration, not a new KObs record and not a new operational cutpoint. It contains no historical Issue timestamp, Core.16 activation certificate, future-opportunity fact, or scheduler-history field.

$\text{Core.InFlightStepRec}_{\{M,E\}}(r, u, r')$ is the pure D-independent reconstructed action relation obtained by evaluating the qualified Q in-flight action/effect schema on r . State-based admission, action identity, enablement, atomic grouping, exact successor channel/request/history effects, projection/accounting effects, and deterministic re-settling are functions of r and fixed E . Operational activation/DR evidence and continuation existence are deliberately outside this relation and are supplied by K.InFlightQual/J.InFlightPathSound. Equal KObs records under the same M/E therefore reconstruct equal FlightStateRec values and equal declared in-flight transition relations.

Definition Core.18e (K-relevant derived functions).

For $K = \langle E, w, H, O \rangle$ define the following typed functions. $\text{DoneRec}_P(K)$, $\text{RemainRec}_P(K)$, and $\text{NextFrontRec}_P(K)$ are computed from $\Omega_P/\leq_P$ supplied by E and the realized-item history reconstructed from H . $\text{PendingRec}_P(K)$ is the current attributed blocking-offer set for P represented by O , with Core.SyncFence providing the current-interval property on reachable records.

$\text{BoundaryReqRec}(x, K)$ is the attributed own-side request for x . $\text{FrontRec}_P(K) := \min_{\leq P}(\text{PendingRec}_P(K))$. $\text{HandshakeRec}_E(K)$ supplies $\text{MirrorAvailRec}_E(x, K)$, so $\text{ChannelEnabledRec}(x, K) := \text{BoundaryReqRec}(x, K) \wedge \text{MirrorAvailRec}_E(x, K)$, with $\text{ChannelDisabledRec}$ defined by negation of the mirror value under BoundaryReq. $\text{ReleaseOwnerSetsRec}_E(x, K)$ is the template-qualified minimal release-support family obtained from the same reconstructed settled dependency semantics; $\text{WaitOwnersRec}_E(x, K) := \bigcup \text{ReleaseOwnerSetsRec}_E(x, K)$, and $\text{WFGRec}(K)$ applies Definition Core.WFG to FrontRec, ChannelDisabledRec, and that coarse union. When Core.SyncCarrierWF_I is selected, Core.18d-1 additionally supplies SyncCarrierRec/SyncPendingRec/SyncReadyRec/SyncBlockedRec/SyncReleaseOwnerSetsRec as pure KObs-derived synchronization facts; they are not message PendingRec/BoundaryReqRec facts and are not inputs to literal FrontRec/WFGRec in the current theorem scope.

Core.FlightStateRec/Core.InFlightStepRec provide the additional pure current-state transition reconstruction used by Appendix K's generalized DrainClosedNowRec. The legacy selector $\text{DrainCandidatesRec}_D(K)$ —enabled residual non-frontier source offers owned by D —is retained only as an ordinary/diagnostic fast path and is no longer the general definition of drain closure. Appendix K defines WaitRoots_D, InFlightEscape, DrainClosedNowRec, DeadKRec, and the corresponding qualification requirements. At this stage these functions are interpreted on records extracted from reachable source cutpoints; no image characterization is asserted.

Historical issue-time quotient. The record deliberately retains whether an operation survives as a current attributed offer, but not the earlier cycle in which that request first became asserted. Therefore two reachable source cutpoints with the same E , w , H , and O have the same KObs even if their operational executions used different legal historical issue times. This is only an observation equivalence

at the selected cutpoint; it does not assert that the two executions have the same admissible continuations or satisfy the same transition-level DrainDiscipline for reasons independent of the common cutpoint. Generalized in-flight reconstruction preserves this quotient: its declared action/effect relation is current-state/KObs-pure. The separate J.InFlightPathSound/Core.InFlightCycleExt evidence justifies operational continuations and is not inferred from KObs equality.

Shared-memory and reset side conditions

Array/synchronization commit-order conformance (Core.MemSyncOrder). Core.MemSyncOrder holds when the selected source/HLS/mapping-layer schedules satisfy the commit-time fence stated in “Scheduling of Array Accesses”: for every dynamic array or memory access a and every synchronization call s selected into S that is ordered with a in the process’s dynamic source execution, every mapped storage commit representing a occurs no later than the corresponding commit of s when a is source-before s , and occurs strictly after s when a is source-after s . Any allowed cache/merge/split/reorder transformation shall carry mapping-layer evidence that preserves those synchronization fences. In particular, no one transformed storage commit may jointly represent source accesses lying on opposite sides of the same synchronization boundary; such a merge cannot satisfy both fence inequalities and is forbidden. This is a local scheduling-conformance obligation whether the storage is process-internal or externally mapped. It does not add memory read/write events to Σ and does not extend the Theorem J.1 replay state. Accordingly, Core.MemSyncOrder is a separate scheduling-contract/tool-conformance obligation recorded in Appendix M.7a; it is not a premise consumed by Appendix I replay or Theorem J.1. For an application of J.1 with cross-process shared-memory value flow, J.Mem is separately required; when transaction-changing transformations are hidden below a selected memory boundary, Core.MemFaithful is additionally required.

Shared-memory lowering condition (J.Mem). The proof-internal observable alphabet Σ of Appendices I–K contains committed channel transfers, synchronization events, and signal Read/Write actions; it does not contain mapped memory read/write events, and the Theorem J.1 invariant carries per-process replay state and per-channel commit histories but no global shared-memory state. Accordingly, the formal results of Appendices I–K apply to cross-process shared-memory communication only under the following lowering condition, named J.Mem: every shared-memory dependency through which one process’s memory write can affect another process’s control flow, payload/value expressions, or the identity or order of its later operations shall be lowered by the array-access mapping layer onto operations already covered by the proof alphabet — explicit point-to-point message-passing channels governed by E4 (including a qualified logical memory boundary), and/or Σ -visible anchored signal IO governed by E2 with the producing Writes observable — so that every memory-carried value covered by the trigger is represented by matched, value-labeled proof-visible behavior at that logical boundary. Excluding raw physical mapped-memory transactions from Σ does not waive J.Mem. The discharge is dependency-level rather than per physical RAM transaction, but it is complete: the flow shall establish that every dependency satisfying the trigger is represented and that no other unlabeled cross-process memory-carried path satisfying that trigger remains. Synchronization alone (E1) fixes access order but does not, within the present invariant, establish that the two models contain the same memory value. If a transaction-changing transformation would alter the command/response or signal event stream otherwise used for J.Mem, Theorem J.1 covers that memory only through a transformation-invariant Core.MemProfile boundary satisfying Core.MemFaithful, or through the separately supplied memory-state extension. Designs lacking either route are outside Theorem J.1, Corollary J.1, and Appendix K as proved here. K.CV’s “declared memory serialization” clause remains a liveness-side premise and does not substitute for J.Mem in the safety proof.

Qualified memory mapping profile (Core.MemProfile). A reusable qualified memory mapping profile is the standard route for a memory whose physical realization uses transaction-changing transformations

while the logical memory interface remains in the J/K proof scope. A profile shall declare: (MP1) the untransformed logical mapping-layer reference semantics, selected logical boundary, boundary alphabet, and permitted transformation set; (MP2) a once-per-profile Core.MemFaithful proof for that boundary and transformation set; (MP3) a J.Mem lowering at that logical boundary, including the completeness argument that no qualifying cross-process memory dependency remains unrepresented; (MP4) Core.MemSyncOrder preservation for all permitted transformations; (MP5) the Appendix-K disposition — either the hidden realization is WFG-inert at the selected K abstraction, or the profile exposes a transformation-invariant, K-faithful logical boundary above the physical transaction changes that faithfully represents capacity/backpressure, request attribution, release support, and WFG dependencies; (MP6) A6-compatible ownership for shared/multi-client resources, using explicit arbitration and point-to-point process-facing channels rather than treating a raw multi-client port as an ordinary point-to-point channel; when that exposed logical boundary is staged, coupled, joined, or otherwise elaborated as Sys_B^elab, the applicable Appendix Q.1 qualified-template obligations also apply; and (MP7) RW1–RW5/reset behavior for every exposed logical channel, including an explicit flush/abort or other cross-epoch disposition when reset scope would otherwise split a live channel. The profile identity/version and its qualification evidence are bound to the concrete theorem instance through K.CertHdr. If MP5 cannot be satisfied for a persistently backpressuring memory, transaction-changing transformations are forbidden for the covered J/K instance (or that memory is outside the theorem scope). Concrete reusable library instances, such as the ping_pong_mem/native_ram_fifo patterns discussed in Appendix H.1, remain qualification work until their profile obligations are discharged; the Core.MemProfile interface itself is normative.

Informative note (direct memory-state extension, not developed here). The alternative to J.Mem/Core.MemProfile lowering — adding mapped memory Read(addr, val)/Write(addr, val) events to the proof-internal Σ , extending the required-order relation $\prec_J$ with the mapping layer's declared serialization constraints, and adding a new global memory-state agreement clause to the Theorem J.1 invariant together with a memory-state component in the enriched replay/KObs reconstruction and the corresponding J.OfferReach/J.OfferConf interface — is compatible with the present proof architecture but is not developed in this document. A flow that requires direct coverage of non-lowered shared memory shall supply that extension and re-verify Lemmas I.DR/I.DR', J.HCanon, and J.KObs-Proc–J.KObs-WFG and the corresponding Appendix K deadlock-factorization result, together with Corollary J.1, with the added clause.

Reset-boundary observable unit (RESET) and reset well-formedness (RW1–RW5). When dynamic reset is used, each dynamic reset occurrence is represented in the proof-internal event model as an explicitly atomic observable unit RESET(S_p, S_c): a proof-internal boundary marker whose scope names the set S_p of processes and the set S_c of explicit abstract channels affected by that reset occurrence. RESET units may be projected away from the outer Σ_{DUT} alphabet, like other proof-internal units. The unit is ordered strictly after every observable unit of the ending epoch owned by the affected scope and strictly before every observable unit of the new epoch of that scope, and the first observable action of each $P \in S_p$ in the new epoch is its fresh dynamic START_P. The state-changing reset transition itself — which occurs before the new epoch's START_P — is exactly what the RESET unit represents. In Core.CycleResult that dynamic occurrence is carried in resetUnits of the enclosing real cycle rather than replacing the cycle with a reset outcome; it therefore retains its own correspondence obligation (Lemma J.RS in Appendix J) rather than being folded into START_P matching. The following reset well-formedness conditions are implementation-side obligations for designs using dynamic reset:

RW1 (channel-consistent scope). For every explicit abstract channel c , interpret its endpoint-process set through the selected source/elaboration mapping: it consists of the modeled source-level process owner(s) associated with c by E 's boundary attribution/projection; elaboration-local stateless couplers are not endpoint processes for reset-scope purposes. Either all such modeled endpoint processes are in

S_p and $c \in S_c$ — in which case c re-initializes to reset-empty occupancy under the Core reset-empty FIFO convention at the boundary and any in-flight (accepted but not yet delivered) contents are discarded consistently on both sides in both models — or none of those modeled endpoint processes are in S_p and $c \notin S_c$, in which case c 's state and history are unaffected. A reset whose scope intersects but does not contain all modeled endpoint processes of a live channel is permitted only if the design models the cross-epoch disposition explicitly (for example, an explicit flush/abort protocol on that channel); that protocol's events are ordinary Σ events, and the channel is treated per this clause after the protocol completes.

RW2 (request disposition). Every outstanding endpoint-owned boundary request of a process in S_p is cleared at the RESET unit; its owning dynamic instance does not commit, is not re-attributed across the boundary, and any post-reset re-execution is a fresh dynamic instance in the new epoch's universe. The Appendix C non-withdrawal discipline is scoped per reset epoch.

RW3 (corresponding reset transitions). At the RESET unit, RTL_B's post-reset state at the chosen abstract boundary corresponds to Sys_ref's prescribed epoch-initial state: control at epoch-initial locations for every $P \in S_p$, source-visible values at their reset values, reset-empty occupancy on every $c \in S_c$, no stale synchronization state, no unaccounted pending endpoint request, no hidden accepted message absent from the source reference model, and direct-input signals obeying their per-epoch stability or sync-controlled contracts (main text, Direct Inputs).

RW4 (atomic boundary). The reset state transition is not interleaved with same-cycle observable actions of the affected scope. If the implementation asserts reset in a cycle in which affected-scope observable actions also commit, those actions belong to the ending epoch and precede the matching RESET occurrence in Core.CycleResult.resetUnits in the required order; the two reset epochs of the affected scope share no observable cycle bucket ambiguity at the boundary. The enclosing CycleResult may still carry unaffected-scope progress because RW5 frames state outside the reset scope.

RW5 (locality). The RESET unit changes no state of processes outside S_p and no history or occupancy of channels outside S_c . Cross-scope effects, if any, must be expressed by ordinary modeled events.

Additional Semantic Assumptions (B-rules)

The following B rules are process assumptions used within the formal proofs. These B rules are in addition to the R rules presented in Appendix G.

ID	Basic-process property (informal statement)	Why it is needed / how it is used
B1 Deterministic Trace Property	For any fixed test stimulus and initial state, the sequence of observable actions emitted by the post-HLS design is unique, including the channel identity and payload/value of each committed Push_c(v)/Pop_c(v) and the value of each Write(sig,val)/Read(sig,val). Internal ϵ -steps may differ between runs, but the externally visible Σ -trace (with labels) cannot. Clarification: B1 is assumed of RTL_B (post-HLS) only. The source-	Ensures the per-process and system-level trace-compatibility proofs (App. I & J) can match one post-HLS trace to one pre-HLS trace without branching on scheduler nondeterminism.

ID	Basic-process property (informal statement)	Why it is needed / how it is used
	level models Sys/Sys_B may admit multiple ϵ-/latency-different executions with the same Σ-projection. When a single concrete witness execution of Sys_B is required for simulation/debug, Appendix L's snooping wrapper may be used to select one admissible witness whose latency-sensitive events align with the unique RTL Σ-trace. Derived per-process consequence. For every process P, the projection of the unique RTL_B Σ-trace onto the observable units owned by or attributed to P is also unique. This follows from the system-level B1 property; it does not assert that P is deterministic when considered in isolation under arbitrary peer, channel, arbitration, or environment behavior.	
B2 Weak Fairness of the Scheduler (WF)	If an action's enabling predicate remains continuously true from cycle t onward, the scheduler/arbiter must eventually select that action, and the selected action then commits in the selected cycle, within a finite but unspecified number of cycles. Applies to Push, Pop, and Sync operations. For message-passing operations, "enabled" means ChannelEnabled at the selected settled boundary. In ordinary Sys_B this reduces to the primitive E4 full/empty/rendezvous test. In Sys_B^{elab} it is the settled ready/valid result of the declared finite FIFO/rendezvous/coupler network, subject to E4 transfer legality. Enablement is eligibility for selection, not a same-cycle commit. Thus an enabled Push/Pop may be delayed by fair arbitration, but it may not be delayed forever while it remains continuously enabled.	Required for all progress arguments, especially the liveness lemmas in Appendix K and the channel-progress corollaries.

ID	Basic-process property (informal statement)	Why it is needed / how it is used
B3 Channel Progress Invariants	B3 is the family of channel-progress obligations used by complete/fair liveness arguments. For an ordinary primitive boundary, B3 consists of the familiar $P_push(c)$ and $P_pop(c)$ obligations below: a full buffered Push is discharged by a complementary Pop, an empty buffered Pop by a complementary Push, and a persistent rendezvous pair by the rendezvous commit. In those ordinary cases the discharge follows from B2 plus primitive E4 once the complementary request remains continuously asserted. In $Sys_B^{\wedge}elab$, do not classify every ChannelDisabled boundary as full, empty, or missing its direct peer. A coupling-sensitive boundary may be disabled because the declared stateless ready/valid network withholds the complementary signal. The elaboration package must therefore supply the compositional progress/enablement facts of Q.1: the coupler network itself has no state or autonomous progress, every persistent blocking cause is exposed through explicit finite channel state or modeled process-facing request roots, and when those causes are discharged the next settled valuation reflects the resulting ready/valid change. B2 then governs any transfer that remains continuously ChannelEnabled. The ordinary P_push/P_pop obligations are still checked for the primitive storage/rendezvous elements that supply those discharge steps; they are not silently applied to a coupler-disabled boundary as though nonfull/nonempty status alone implied enablement. This distinction does not add B2/B3 to finite-prefix Theorem J.1. B3 remains a liveness assumption/certificate interface	Required by complete/fair progress arguments, principally Appendix K, to rule out permanent back-pressure loops in which both endpoints continuously request a transfer until the matching discharge event but the matching discharge never occurs; such progress is logically independent of the cycle-by-cycle R-rules. It is not a premise of finite-prefix Theorem J.1.

ID	Basic-process property (informal statement)	Why it is needed / how it is used
	used only where a complete/fair continuation is required. Persistent-request and payload-stability scopes are unchanged: requests remain asserted through the first matching discharge/commit or until an affecting RESET, as specified below. • $P_push(c)$, primitive channel case: if a blocking Push request persists while the represented primitive buffered storage is full, and the complementary consumer request persists, a complementary Pop eventually commits and frees space; for a primitive rendezvous channel, the rendezvous eventually commits. • $P_pop(c)$, primitive channel case: if a blocking Pop request persists while the represented primitive buffered storage is empty, and the complementary producer request persists, a complementary Push eventually commits and provides data; for a primitive rendezvous channel, the rendezvous eventually commits. After the ordinary discharge, or after a declared elaborated blocking cause clears, the actual settled ChannelEnabled predicate—not occupancy alone—controls the subsequent B2 obligation.	
B4 Finite Stutter Bound	Every process P has a finite constant depth D_P such that whenever P is not externally stalled (i.e., P is advancing internal micro-state toward its next observable Σ-action, or its next Σ-action is locally enabled), P can execute at most D_P consecutive ϵ-steps before either (i) executing a Σ-action, or (ii) reaching a stable waiting state in which P has no further ϵ-step available until some	This guarantees the ϵ-stutter closure needed to construct witness mappings in Appendix I and to bound internal micro-latency (pipeline/FSM bookkeeping) by $\leq D_P$. Unbounded waiting due to back-pressure or missing peer stimulus is governed by WF/B2 and the progress obligations B3, not by B4.

ID	Basic-process property (informal statement)	Why it is needed / how it is used
	external/peer/environment condition changes the enabling predicates.	
B5 System Quiescence Closure	B5 carries a finite system-level cycle bound C_{sys} for the selected bucket semantics. A standard discharge defines per-process cycle bounds C_P and takes $C_{sys} = \max_P C_P$. Each C_P may be supplied independently or derived from the B4 epsilon-step bound D_P by a proved charging lemma showing that every relevant silent cycle in which P remains internally advancing consumes at least one P-owned epsilon-step; under that lemma $C_P \leq D_P$. If, at some cycle t_0, no observable actions are enabled in any process, and no external or environmental change from t_0 through the closure interval changes the relevant enabling predicates, then within at most C_{sys} further cycles the system reaches an epsilon-fixed point. Thereafter, while no external or environmental change re-enables observable or internal work, the system executes no additional epsilon-steps.	Needed to finish the starvation-vs-deadlock analysis in Appendix K: ensures an all-disabled state cannot hide behind an infinite tail of unobservable activity.
B6 Time-Divergence / Idle-Stutter Convention	After <code>Core.TerminalIdle</code> is established — a genuine B5/global fixed point together with proof that neither future registered/internal machine evolution nor any permitted future environment evolution can re-enable work — the global clock continues to advance. <code>Core.EnvTerminal</code> supplies the environment component of this obligation. We model each subsequent B6 idle clock tick as a stutter step that produces no Σ-event, performs no ϵ-prefix or ϵ-suffix work, and leaves all non-clock state unchanged. A present-instant observation that no action is enabled is not sufficient to establish this antecedent.	When a theorem explicitly represents a genuinely terminal-idle execution by an infinite B6 suffix, those idle ticks are treated as ϵ-stutter for the applicable E1–E5 prefix/trace matching. Thus B6 gives a time-divergent infinite extension only for a reachable state already proved to satisfy <code>Core.TerminalIdle</code>. <code>Core.CycleClosure</code>'s environment-definedness clause or <code>Core.EnvTerminal</code> alone is insufficient: a total timed/reactive environment may legally deliver new stimulus at a later cycle. B6 is not an arbitrary deadlock-cutpoint or temporary-idle extension principle. Ordinary nonterminal cycles in which no Σ event commits are represented by <code>Core.SilentCycle/CompleteCycle</code> and may advance registered state, environment state,

ID	Basic-process property (informal statement)	Why it is needed / how it is used
		and cycle-valued monitors. A partial internal message-passing deadlock cutpoint in Appendix K need not be a global fixed point because outside-D activity, environment activity, or already-issued non-frontier drain work may still be possible. Such cutpoints are handled by the separate Appendix K complete-cycle/fair drain continuation; B6 is used only for a genuine terminal-idle suffix.

Appendix G – Trace Compatibility Rules

This document informally states that a primary goal is to present “no surprises” to a DV engineer using the same testbench to verify the pre-HLS and post-HLS models.

Somewhat more formally, we can show how the scheduling rules within this document establish a set of trace-compatibility rules between the pre-HLS and post-HLS models that provide strong guarantees about their behaviors.

The *basic conceptual model* establishes the base behavior for a single process:

- 1) Synchronization interface calls always remain in source code order.
- 2) Signal reads occur at closest preceding synchronization interface call.
- 3) Signal writes occur at closest succeeding synchronization interface call.
- 4) All message-passing operations before a call to a synchronization interface shall be issued before or in the same cycle in which the synchronization interface commits.
(For blocking message-passing operations (Push/Pop), this implies the corresponding commits occur no later than the synchronization call’s commit cycle.)
- 5) All message-passing operations after a call to a synchronization interface shall be issued after the cycle in which the synchronization interface call commits.
- 6) Two message-passing operations on separate interfaces which appear in sequence in the model are source-ordered. They may issue with timing overlap or in the same clock cycle where the selected issue policy allows, but they shall not be issued in the reverse sequence. Operations on the same message-passing interface are issued in source order (they are never reversed).
- 7) Synchronization calls can affect the scheduling of signal IO operations, and synchronization calls can affect the scheduling of message-passing calls, but message-passing calls cannot affect the scheduling of signal IO operations and vice-versa.
- 8) The STRICT_IO_SCHEDULING=relaxed option is not used.
- 9) We distinguish the rendezvous source model Sys (B(c)=0 for all channels) from the buffered source interpretation Sys_B, which uses bounded-FIFO channel capacities B(c) matching the post-HLS RTL (Appendices I–K).

Conceptually, systems are composed of many such processes which follow the basic conceptual model, and which communicate using only synchronization interface calls, latency-insensitive signal-level protocols, and committed message-passing transfers. If a design uses non-blocking message-passing

calls (PushNB/PopNB), then unsuccessful polls are treated as internal ϵ -steps (no Σ -event), and each successful completion is represented in Σ as the corresponding committed Push_c/Pop_c transfer (Σ records transfers, not failed polls). If the resulting latency-sensitive behavior is externally visible at the DUT boundary, the snooping wrapper of Appendix L may be used to align the selected admissible pre-HLS execution with the RTL. Here we call this system of processes the *conceptual system*.

Practically, the modeling approach described above is too restrictive for real-world systems with optimized HW, so the scheduling rules in this document allow for key HW optimizations. But this is done in a carefully controlled manner, such that the HW optimizations do not break equivalent behavior with the *conceptual system*.

Here is a summary of the HW optimizations that the scheduling rules allow, and a brief description of how we maintain equivalent behavior with the *conceptual system*:

1. Pipelined loops: In the post-HLS RTL, the pipeline may overlap iterations to improve throughput. See Appendix B for a formal presentation of pipelined loops.
2. Pipelined loops: HLS by default will not break any latency-insensitive signal IO protocols when pipelining loops.
3. Direct inputs: Signals that do not change after a process comes out of reset can be read as late as possible using *hls_direct_input*, saving register area.
4. Direct inputs: When *hls_direct_input_sync* is used, signals read by a process are updated at the precise point at which the HW pipeline is guaranteed to be ramped down, so the signal values do not need to be saved within pipeline registers, saving register area.
5. Memory accesses: HLS can reorder memory accesses within a process only if it can prove that the reordering is conflict-free.
6. Shared memories: Memories shared between processes are modeled as simple C arrays but always include explicit synchronization between processes in both the pre-HLS and post-HLS models. For the formal results of Appendices I–K, the value-carrying accesses must additionally satisfy the J.Mem lowering condition (Normative Formal Core).
7. Non-blocking message-passing interfaces: They are modeled at the level of committed transfers (successful completion), with unsuccessful polls treated as ϵ -steps and thus not part of Σ . If latency differences in these committed-transfer events are externally visible at the DUT boundary and must be aligned for verification, equivalent behavior between pre-HLS and post-HLS systems can be maintained by snooping the post-HLS system and using this to delay pre-HLS message arrival (Appendix L).
8. Latency-sensitive global signal IO: If latency differences are externally visible to the DUT, equivalent behavior between pre-HLS and post-HLS systems can be maintained by snooping the post-HLS system and using this to delay pre-HLS signals.

9. Latency-sensitive local signal IO: Isolate local latency-sensitive protocols to small, dedicated transactors, and use latency-insensitive signal IO or message-passing to communicate with the rest of the system.
10. One-way signal handshake protocols: Only use in cases where backpressure is not possible and embed assertions into the model to check that this is always true.
11. Combinational processes: If the pre-HLS model contains combinational processes, their observable signal Read/Write behavior is interpreted at each cycle's ϵ -quiescent combinational fixed point. Under the well-formedness requirement that the combinational network is deterministic and reaches a unique fixed point, the RTL/tool-flow conformance obligation is to preserve the fixed-point signal relation, so combinational signal IO can be included in Σ without using synchronization anchors.

Taken as a whole, these rules support a verification methodology in which full-system verification is performed primarily on the pre-HLS system model, and post-HLS RTL verification can be reduced to targeted checks that the formal side conditions are met. The precise guarantee is the execution-level guarantee of Appendix J: for every reachable RTL_B prefix whose local certificate package satisfies Definition J.LocalGWR and whose selected source model satisfies the source-side Core.QSync premise required by J.GWR-Comp, Theorem J.GWR-Comp constructs a globally witness-realizable package and a matching Sys_ref behavior (Post $\rightarrow$ Ref). A directly validated J.GWR package carrying equivalent per-Horizon settling evidence is an alternate route. The reverse direction applies only to annotated RTL-consistent source-side prefixes or to source-side executions selected by the Appendix L snooping / witness-selection mechanism. Thus a source-side pass justifies the corresponding RTL_B behavior only within that annotated RTL-consistent / witness-selected subset. For liveness/deadlock guarantees, the results additionally assume the Appendix K/N progress, fairness, drain, and A5 premises; the cycle-by-cycle R/E rules and the safety construction alone do not imply those liveness properties.

Formalization of the Trace Compatibility Rules in Appendix G

The following notation is used in this section.

Symbol	Meaning
P	A process (thread or method) in the design
Σ	The alphabet of observable IO actions
τ_P	A (finite or infinite) per-cycle trace of observable actions executed by process P: for each global clock cycle t , $\tau_P[t]$ is the (possibly empty) set of Σ -actions committed by P at t . Actions committed in the same cycle are treated as simultaneous (unordered); only cross-cycle order (and the explicit E1–E5 constraints) is semantically significant.
S	The subset of Σ that are synchronization calls (explicit wait, SyncChannel, START_P/FINISH_P indicating process start/finish)
R	The subset of Σ that are signal reads
W	The subset of Σ that are signal writes

Symbol	Meaning
M	The subset of Σ that are committed message-passing transfers (i.e., committed Push_c / Pop_c events on message-passing channels, including successful non-blocking transfers). Unsuccessful non-blocking polls contribute no Σ -event and are modeled as ϵ -steps, and therefore are not members of M .
Q_exec	The execution-level set of dynamic source-level message-call instances executed by a process at the Σ -visible endpoints (blocking Push/Pop, and non-blocking PushNB/PopNB calls). Elements of Q_exec may or may not commit. In particular, failed non-blocking polls are elements of Q_exec , have no committed Σ -event, and are modeled as ϵ -steps. The frontier-item universe Ω_P , defined below, is separate from Q_exec .
Ω_P	The local dynamic frontier-item universe for process P in the selected execution / witness. It contains synchronization-call items, signal-action items, and message-operation frontier items; it is distinct from Q_exec, P , the universe of dynamic source-level message-call instances.
Ω_msg, P	The message-operation slice of Ω_P : frontier items x for which $MsgOf_P(x)$ is defined.
$MsgOf_P(x)$	Partial projection from a message-operation frontier item $x \in \Omega_msg, P$ to its owning dynamic source-level message-call instance $q \in Q_exec, P$. In Sys_B^{elab} , $MsgOf_P$ need not be injective: one source-level q may own several elaborated frontier items.
$BoundaryOf_P(x)$	The explicit abstract channel boundary at which frontier item x is evaluated for BoundaryReq, ChannelEnabled, ChannelDisabled, Front_P, and Appendix K WFG reasoning. In ordinary unelaborated cases this boundary is unique for the owning source-level operation; in Sys_B^{elab} it is declared by the elaboration package.

A *system trace* is the tuple

$\tau = (\tau_P)_P$, one component per process.

For any action $a \in \Sigma$, let $clk_pre(a)$ / $clk_post(a)$ be the clock cycle at which a is committed.

For any issued message-call instance $q \in Q_exec$, $issue_pre(q)$ / $issue_post(q)$ denotes the auxiliary Issue timestamp for q in the pre-HLS / post-HLS trace, respectively, as defined in Appendix C and required to satisfy the associated Issue/Commit linkage (well-formedness) conditions with respect to the Σ -visible boundary request signal on q 's endpoint direction.

If q commits, it contributes a unique committed transfer $m(q) \in M$ with commit cycle $clk_pre(m(q))$ / $clk_post(m(q))$. If q does not commit (e.g., an unsuccessful non-blocking call), then $m(q)$ is undefined / absent and no Σ -event is added to M .

(Q_exec-matching / witness convention.) Because unsuccessful non-blocking polls are ϵ -steps and produce no Σ -event, Q_exec is not in general reconstructible from the Σ -projection alone. Whenever E3, E5, or any later proof compares executed message-call instances across Sys_ref and RTL_B , the comparison is made relative to the chosen execution witness. That witness supplies a partial matching μ_Q between dynamic source-level message-call instances in the compared traces, preserving process identity, endpoint/interface identity, dynamic source-order position, and source-level call-instance identity where applicable. If q commits, then $\mu_Q(q)$ commits the corresponding matched transfer $m(\mu_Q(q))$ with the same Σ -label/value under the event-matching convention. If q is an unsuccessful

non-blocking poll, then q has no $m(q)$ and no Σ -label; its presence and its match are supplied only by the witness-compatibility condition WC, by the sufficient NB-CF condition when applicable, or by the Appendix L snooping / witness-selection construction. Thus E3/E5 quantify over Q_exec only relative to this chosen witness; they do not require failed non-blocking polls to be inferable from Σ alone.

(No deassert-before-commit assumption.) For blocking message requests $q \in Q_exec$, once q is issued, its Σ -visible boundary request is not withdrawable within that reset epoch: it remains asserted in every cycle from $issue_pre(q)$ (resp. $issue_post(q)$) through q 's commit cycle if it commits, and for Push the payload remains stable while the request is outstanding. If an affecting RESET occurs first, RW2 ends the dynamic instance at the epoch boundary; no other nonterminal transition may withdraw or reattribute it.

Notation convention (Issue/Commit functions). The function-like forms clk_pre/clk_post , $issue_pre/issue_post$, and $m(q)$ abbreviate the unique values supplied when the corresponding Appendix C existence/uniqueness and Issue/Commit-linkage conditions hold. In particular, $m(q)$ remains partial: a rule that requires a committed transfer must also require or entail that $m(q)$ is defined. This notation does not replace the underlying relational Issue/Commit semantics.

Dynamic source-order convention

The order relation used below is over dynamic source-level operation/frontier items in the compared execution / witness, not over all syntactic call sites in the source text.

For a process P , let Q_exec,P denote the execution-level set of dynamic source-level message-call instances of P at the chosen abstract boundary. Q_exec,P includes blocking Push/Pop calls, successful non-blocking PushNB/PopNB calls, and failed non-blocking PushNB/PopNB polls.

Separately, let Ω_P denote the witness-specific dynamic universe of source-level frontier items that are actually reached under the fixed stimulus / selected witness and that are relevant to observable/frontier/deadlock reasoning. Items in Ω_P include synchronization-call instances, anchored signal Read/Write action instances, and message-operation frontier items at the chosen abstract boundaries. A message-operation frontier item is either (i) a blocking Push/Pop item that has been reached, whether pending or committed, or (ii) a successful non-blocking PushNB/PopNB item that contributes a committed $Push_c(v)$ / $Pop_c(v)$ event. A failed non-blocking poll q is an element of Q_exec,P , has no committed Σ -event $m(q)$, and is not an element of Ω_P merely because it executed.

Because Q_exec,P is a universe of dynamic message-call instances while Ω_P is a universe of source-level frontier items, their message-operation overlap is not a literal set intersection. Let $MsgOf_P(x)$ be the partial projection from a frontier item $x \in \Omega_P$ to its owning dynamic message-call instance, defined exactly for message-operation frontier items and undefined for pure synchronization or signal-anchor items. Define the message-frontier slice

$$\Omega_msg,P := \{ x \in \Omega_P \mid \exists q \in Q_exec,P : MsgOf_P(x) = q \}.$$

Define the corresponding message-instance slice induced by Ω_P as

$$Q_Q,P := \{ q \in Q_exec,P \mid \exists x \in \Omega_msg,P : MsgOf_P(x) = q \}.$$

Let Σ_P denote the corresponding observable labels/events contributed by items of Ω_P when they are

Σ -visible. For a message-call instance $q \in Q_{\text{exec},P}$, the committed Σ -event $m(q) \in M$ exists only if q commits; if q is a failed non-blocking poll, $m(q)$ is absent and q contributes no Σ -event. For a message-operation frontier item $x \in \Omega_{\text{msg},P}$ with $\text{MsgOf}_P(x)=q$, item x contributes the committed Σ -event $m(q)$ only if q commits. A pending blocking message-operation frontier item x may therefore be in Ω_P and $\Omega_{\text{msg},P}$, and its owning message instance q may be in $Q_{\Omega,P}$, even though $m(q)$ is not yet present.

Operations on untaken branches and unexecuted loop iterations are not members of Ω_P , $\Omega_{\text{msg},P}$, $Q_{\Omega,P}$, or $Q_{\text{exec},P}$. Distinct loop iterations, unrolled instances, and repeated calls are represented by distinct dynamic instances.

Default source order for ordinary message calls. For dynamic items x and y of process P , write $x <_{\text{src}} y$, or equivalently $x <_P y$ when the process must be explicit, to mean that P 's source control flow and the R-rules force x to precede y in that dynamic execution / witness. For ordinary user-written source-level message-call instances in a sequential process, this is the dynamic program order along the executed control path. Thus, if two executed message-call instances $q_1, q_2 \in Q_{\text{exec},P}$ appear in sequence in the same process, exactly one of $q_1 <_{\text{src}} q_2$ or $q_2 <_{\text{src}} q_1$ holds according to that dynamic program order, including when q_1 and q_2 use distinct message-passing interfaces, occur in the same synchronization interval, arise from repeated loop iterations, or arise from unrolled instances whose source-level expansion is ordered by the selected source elaboration. Same-cycle issue, same-cycle commit, or issue of a later operation before an earlier operation commits is a timing/issue-policy freedom; it does not make ordinary source message-call instances incomparable. The no-reverse rule R4/E3 is therefore stated over this ordered dynamic source relation.

Limited sources of incomparability. The relation $\leq_P$ used for frontier reasoning is the reflexive closure of the restriction of this source-induced order to Ω_P , extended only by explicitly modeled partial-order structure. Ordinary distinct-interface source calls are not incomparable merely because they are on different interfaces. A message-operation multi-frontier may arise only from explicit $\text{Sys}_B^{\text{elab}}$ proof-internal structure such as staged/bundle/join/epoch-gate abstract channels whose elaboration package declares the relevant frontier items, evaluated boundaries, partial order, and projection/attribution facts. Such elaborated frontier items are part of the selected Sys_{ref} proof model; they are not an implicit weakening of source order among the ordinary user-written message calls.

For sequential-process signal IO, $<_{\text{src}} / <_P$ is interpreted through the R2 anchor semantics rather than through the raw lexical statement location of the signal read or write. Thus a signal Read is ordered, for formal frontier and trace-compatibility purposes, at its pred_S anchor, and a signal Write is ordered, for formal frontier and trace-compatibility purposes, at its succ_S anchor. A lexically earlier signal Write whose succ_S anchor is a later synchronization call does not, merely by lexical placement, precede or block intervening message-operation instances in the same source interval; conversely, a lexically later signal Read whose pred_S anchor is an earlier synchronization call does not, merely by lexical placement, wait behind intervening message-operation instances. Signal IO may constrain message IO only through the explicit synchronization anchors and the stated R/E rules, not by raw lexical adjacency inside an interval.

Conceptual Model for a Single Process

For every process P the pre-HLS trace τ_P must satisfy the following *partial-order constraints* over these dynamic operation instances:

R1 (Program order of S).

$\forall s1, s2 \in S : s1 <_{\text{src}} s2 \Rightarrow \text{clk_pre}(s1) < \text{clk_pre}(s2)$

R2 (Location of R/W for sequential processes).

For a sequential process P, ordinary signal Read/Write events are anchored to synchronization events as follows:

$\forall r \in R_P : \text{clk_pre}(r) = \text{clk_pre}(\text{pred_S}(r))$

$\forall w \in W_P : \text{clk_pre}(w) = \text{clk_pre}(\text{succ_S}(w))$

where $\text{pred_S}(r)$ (resp. $\text{succ_S}(w)$) is the closest dynamic synchronization-call instance that precedes (resp. follows) the signal Read/Write on the dynamic source path of the current execution / witness, after applying the anchoring convention above. This anchoring determines the signal event's formal position for $\leq_P$ / $<_P$, Done_P, Remain_P, and NextFront_P reasoning. In particular, a sequential-process Write w is formally placed at $\text{succ_S}(w)$, regardless of intervening message operations; when several Writes by P to the same sig share that anchor, the last executed source Write supplies the canonical value and earlier writes are overwritten. Likewise, a signal Read anchored to $\text{pred_S}(r)$ is not delayed by intervening message operations merely because the read statement appears lexically after them.

Every observable signal Read/Write of a sequential process must have a real bounding synchronization anchor on the dynamic path on which it occurs. A process-start synchronization event START_P may serve as the initial predecessor anchor when the modeling convention explicitly includes START_P in S. A terminating process event FINISH_P may serve as the terminal successor anchor when the process actually terminates and FINISH_P occurs. If no real predecessor anchor exists for a Read, or no real successor anchor exists for a Write on the dynamic path being modeled, the sequential-process signal IO is ill-formed for this formal model or lies outside the finite observable prefix being compared. No observable Read/Write event is assigned clock time ∞ .

R2C (Combinational signal IO at the ϵ -fixed point).

A combinational process P has no process-owned synchronization events and no message-passing events in this formal clause; it may, however, contribute observable signal Read(sig,val) and Write(sig,val) events to Σ through explicitly selected R2C observation components. These events are not anchored by $\text{pred_S}/\text{succ_S}$. In every clock cycle t, the global state first reaches the cycle's ϵ -quiescent combinational fixed point after all sequential clock-boundary updates and all internal combinational propagation for that cycle have settled. Let μ_t denote this unique fixed-point valuation of the signal network.

R2C cycle-entry valuation. The fixed point μ_t is reached from the cycle-t entry valuation established after the applicable sequential clock-boundary updates, with the registered/process-owned request and signal drives and the external/direct-input values selected for cycle t fixed. Internal combinational propagation then settles from that valuation without advancing the architectural cycle.

R2C unit-generation convention

Fix an explicitly modeled combinational process or combinational network component C_comb and its selected observable slice $\text{Obs}(C_comb)$, consisting of the process/signal/direction labels whose fixed-point values are included in that component's R2C observation. Define $\text{ComponentParticipatesInCycle}(C_comb, t)$ to mean that C_comb is active in the selected reset epoch and $\text{Obs}(C_comb)$ is defined at the cycle-t ϵ -quiescent fixed point μ_t ; when cycle-locked observation is

used, the selected observation/elaboration package shall define this participation predicate explicitly. The fixed-point valuation μ_t exists every cycle, but for an ordinary non-cycle-locked component a Σ -visible R2C fixed-point signal-observation unit is emitted only at canonical observation occurrences: (i) the first cycle t for which $\text{ComponentParticipatesInCycle}(C_{\text{comb}}, t)$; (ii) any later participating cycle in which the vector of fixed-point values on $\text{Obs}(C_{\text{comb}})$ differs from the vector in the most recent emitted R2C unit; or (iii) any participating cycle in which an explicitly modeled observer/sampler requires a fresh observation unit. Participating cycles in which the observable fixed-point slice is identical to the most recent emitted unit and no explicit observer/sampler requires a fresh unit are R2C stutter cycles for C_{comb} and emit no new Σ -visible unit. Cycles in which the component does not participate likewise emit no C_{comb} unit. Delta-cycle re-evaluation, intermediate combinational values, redundant re-drive before the fixed point, and ordinary R2C stutter cycles are $\epsilon/\text{stutter}$ for this component and are not separately Σ -visible.

R2C cycle-locked completeness.

If C_{comb} is declared cycle-locked, then for every cycle t satisfying $\text{ComponentParticipatesInCycle}(C_{\text{comb}}, t)$, exactly one R2C fixed-point observation unit for C_{comb} shall be emitted from μ_t , even when $\text{Obs}(C_{\text{comb}})$ is unchanged and no separate observer/sampler trigger occurs. Conversely, no cycle-locked unit is emitted for a cycle in which $\text{ComponentParticipatesInCycle}(C_{\text{comb}}, t)$ is false. This is a positive completeness obligation as well as an emission guard: proving only that every emitted unit occurs in a permitted cycle is insufficient if a required participating-cycle unit is omitted.

If a combinational process P reads sig in an emitted R2C unit of C_{comb} , the corresponding label is $\text{Read}(\text{sig}, \mu_t(\text{sig}))$ for the cycle t at which that unit is emitted. If P drives sig in an emitted R2C unit of C_{comb} , the corresponding label is $\text{Write}(\text{sig}, \mu_t(\text{sig}))$, where $\mu_t(\text{sig})$ is the final resolved fixed-point value of sig at the chosen observation boundary. All Read/Write labels inside one emitted R2C unit are interpreted at the same ϵ -quiescent fixed point and may not be split.

Combinational-process signal IO is well formed only when the relevant combinational network is deterministic and reaches a unique ϵ -fixed point in each cycle being compared, and when each observable R2C component has a well-defined selected observable slice $\text{Obs}(C_{\text{comb}})$. For a multi-input combinational component whose inputs may be updated by independent latency-insensitive sources, the component is Σ -observable under the stutter-canonicalized R2C rule only if the chosen observation boundary provides a single explicit anchoring domain for the slice, or if an additional component-specific well-formedness proof shows that all admissible interleavings induce the same emitted R2C unit sequence. Otherwise the component's internal combinational propagation must remain ϵ -hidden, or the design must expose the relevant behavior through explicit synchronization, message passing, a cycle-accurate transactor, or a cycle-locked R2C observation boundary. Pure combinational oscillations, ambiguous multiple-driver resolution, hidden stateful behavior, or intentional combinational-loop behavior outside a unique fixed-point interpretation are outside this R2C model unless represented by explicit state or synchronization elsewhere in the formal model.

R2C determinism clarification. Here “deterministic” means that fixed cycle-entry registered state/drives and fixed cycle- t inputs determine one ϵ -quiescent fixed-point valuation for the selected observation network. It does not require the composed internal combinational microstep relation or evaluation order itself to be functional. Value-preserving re-evaluation, intermediate values, and redundant re-drive may be $\epsilon/\text{stutter}$ provided they do not change the unique settled result.

R3 (Isolation around S).

Let s be a dynamic synchronization-call instance of process P . Let $\text{pref}_S(s)$ be the set of executed dynamic message-call instances $q \in Q_{\text{exec}, P}$ in the immediately preceding source interval of P whose

source-induced order places q before or at the boundary controlled by s . Let $\text{suff_S}(s)$ be the set of executed dynamic message-call instances $q \in Q_{\text{exec},P}$ in the immediately following source interval of P whose source-induced order places q after s . Then

$\forall q \in \text{pref_S}(s) : \text{issue_pre}(q) \leq \text{clk_pre}(s)$ and $\forall q \in \text{suff_S}(s) : \text{issue_pre}(q) > \text{clk_pre}(s)$.

$\forall q \in \text{pref_S}(s)$ with q a blocking Push/Pop : $m(q)$ is defined and $\text{clk_pre}(m(q)) \leq \text{clk_pre}(s)$.

The third clause is the synchronization-completion fence of Core.SyncFence ; it permits a blocking transfer and s to commit in the same cycle. In $\text{Sys_B}^{\text{elab}}$, $m(q)$ here is the designated process-facing commit that completes/releases q , not an arbitrary internal elaboration transfer. Failed non-blocking polls in $Q_{\text{exec},P}$ satisfy the issue-side synchronization-boundary rule, but they contribute no Σ -event and do not become Ω_P frontier items merely because they executed.

R4 (Safe message issue ordering).

For every pair $q_1, q_2 \in Q_{\text{exec},P}$ within the same process and within the compared dynamic execution / witness,

$q_1 <_{\text{src}} q_2 \Rightarrow \text{issue_pre}(q_1) \leq \text{issue_pre}(q_2)$.

(For same-channel operations, this states that dynamic calls on a single interface are issued in source-induced program order. For distinct channels, it forbids issuing dynamic message-call instances in the reverse of their source-induced order. This rule does not impose ordering between syntactic operations on untaken branches or between dynamic instances that are not ordered by the source-induced partial order of the selected witness. Failed non-blocking polls are included in this issue-order universe when they are executed, but remain ϵ -steps with no Σ -event.) Strict scalar same-endpoint case: when q_1 and q_2 are distinct blocking instances on the same scalar explicit abstract boundary and endpoint direction governed by the Core single-transfer-per-cycle endpoint constraint in one reset epoch, $\text{Core.IC}(4)$ strengthens this to $\text{issue_pre}(q_1) < \text{issue_pre}(q_2)$.

Prefix-closed reading of issue order. R4 and E3 are to be read not only as inequalities between already-defined issue timestamps, but also as closure conditions on issue itself. Within a compared dynamic execution / witness, if $q_1 <_{\text{src}} q_2$ and q_2 has issued by cutpoint σ , then q_1 has also issued by σ , and $\text{Issue}(q_1) \leq \text{Issue}(q_2)$ on the same side of the comparison (Issue_pre for R4/source-side prefixes, Issue_post for E3/implementation-side prefixes). Equivalently, in each source interval, the set of issued message-operation instances is downward closed under the source-induced order used by R4/E3. This rule is what excludes the case where a source-later message operation has issued while a required source-earlier message operation in the same interval remains merely reached but unissued.

R5 (Channel capacity semantics; Sys vs. Sys_B).

Appendix G defines the raw rendezvous source model Sys , in which every channel has effective capacity 0. Concretely: for every channel c and any clock cycle t , the number of unmatched committed Push_c actions is zero and the number of unmatched committed Pop_c actions is zero; equivalently, no Push_c shall commit unless a matching Pop_c also commits at t , and no Pop_c shall commit unless a matching Push_c also commits at t .

In Appendices I–K we additionally use the back-annotated source interpretation Sys_B : the same source processes as Sys , but interpreted under the case-split channel semantics of E4 with capacities $B(c)$ matching RTL_B . Thus:

- $B(c)=0$ is interpreted as the rendezvous case of E4: matching $\text{Push_c}(v)$ and $\text{Pop_c}(v)$ endpoint events commit in the same cycle, and no unmatched committed endpoint event may remain after any cycle boundary.
- $B(c)>0$ is interpreted as the cycle-boundary, non-fall-through bounded-FIFO case of E4, using the

occupancy predicate $\text{occ_c}(t)$ and the FIFO availability predicates $\text{occ_c}(t) < B(c)$ for Push_c and $\text{occ_c}(t) > 0$ for Pop_c .

The source conceptual semantics for a process is thus a labeled partial order $(\Sigma, \leq_P)$ generated by R1–R4 plus the applicable E4 channel semantics: raw rendezvous for Sys, or the case-split back-annotated channel semantics of Sys_B / Sys_ref.

Operational semantics location

The selected Sys_ref transition system, registered sequential-interface semantics, Policy L and Policy S, and the admissible Policy-L execution domain are defined normatively in the preceding “Normative Formal Core for Appendices G–K.” Appendix G uses that core while stating the R1–R5 and E1–E5 trace-compatibility rules; it introduces no second operational semantics.

Trace Compatibility Relation

Let τ_{pre} and τ_{post} be the Σ -traces (finite or infinite) of the same process before and after HLS, where each Σ -event is labeled exactly as in “Normative Formal Core for Appendices G–K” (e.g., $\text{Push_c}(v)$, $\text{Pop_c}(v)$, $\text{Read}(\text{sig}, \text{val})$, $\text{Write}(\text{sig}, \text{val})$, and synchronization events).

Notation note. We write $\tau_{\text{pre}} \approx \tau_{\text{post}}$ as a named, ordered trace-compatibility predicate. The symbol $\approx$ is local to this document and is not used here to assert a mathematical equivalence relation; in particular, reflexivity, symmetry, and transitivity are not being claimed. Its arguments are ordered: the left trace supplies the pre-HLS/source reference witness, and the right trace supplies the post-HLS implementation trace constrained by E1–E5. In later appendices the same convention applies to $\tau_{\text{ref}} \approx \tau_{\text{post}}$, with τ_{ref} as the prescribed Sys_ref witness.

Convention (S): all references to S, $\text{pred_S}/\text{succ_S}$, $\text{pref_S}/\text{suff_S}$, and $<_{\text{src}}$ in E1–E5 use the dynamic source-induced order defined in R1–R4 for the compared execution / witness. Thus τ_{pre} denotes the Σ -trace of the pre-HLS source over the dynamic operation instances reached under the fixed stimulus / selected witness.

Σ -event matching convention (expansion of Definition Core.17 ΣMatch). We use a canonical per-endpoint occurrence matching to make “the same Σ -events” precise. For each channel c, match committed Push_c and Pop_c events by their ordinal occurrence on that channel (e.g., the k-th committed Push on c in τ_{pre} matches the k-th committed Push on c in τ_{post} ; likewise for Pop_c). For each such matched pair, the endpoint (c) and the payload/value label must agree. For the ordinal index k to be well-defined under the per-cycle “set/bucket” trace representation used in these appendices, we assume that for any fixed channel endpoint c, at most one committed Push_c and at most one committed Pop_c may occur in any single clock cycle.

For each observable signal sig of a sequential process, match $\text{Read}(\text{sig}, \cdot)$ and $\text{Write}(\text{sig}, \cdot)$ by their ordinal occurrence on that signal within the process trace, again requiring the value labels to agree.

For sequential processes, within each anchor interval between consecutive synchronization events in S, record at most one Σ -visible $\text{Read}(\text{sig}, \text{val})$ and one Σ -visible $\text{Write}(\text{sig}, \text{val})$ for each (process, sig), at the E2 anchor. If several source Writes share that succeeding anchor, the last executed source Write supplies val and earlier same-interval Writes are overwritten. Additional physical sampling/re-driving that does

not alter the retained anchor value is ϵ -internal. If an intermediate intra-interval value must be observable, use an explicit synchronized interface rather than ordinary sequential-process signal IO.

For combinational processes, the canonical observable unit is the emitted R2C fixed-point signal-observation unit for the relevant combinational process/network component C_comb : the set of $Read(sig, val)$ and $Write(sig, val)$ labels for the selected observable slice $Obs(C_comb)$, recorded at one ϵ -quiescent combinational fixed point under the R2C unit-generation convention. Intermediate delta-cycle reads/writes, repeated re-evaluations before the fixed point, and stutter cycles in which $Obs(C_comb)$ is unchanged and no explicit observer/sampler requires a fresh unit are ϵ /stutter and are not separately Σ -visible.

R2C unit matching/index convention

For each combinational process or explicitly modeled combinational network component C_comb , project each trace to the subsequence of emitted R2C fixed-point signal-observation units for C_comb . The k -th emitted such unit in Sys_ref is matched with the k -th emitted such unit in RTL_B . This local emitted-occurrence ordinal, together with the component identity C_comb , is the correspondence rule for R2C fixed-point units; it is not the global observable-cycle bucket number of the whole-system trace. Thus repeated identical fixed-point observations across stutter cycles do not create additional units, while repeated equal-valued units created by explicit observer/sampler occurrences are distinguished by their local emitted-occurrence ordinal for C_comb .

Matched pre-HLS and post-HLS fixed-point units must agree on the included process/signal/direction labels and fixed-point values. The same-cycle agreement required here is local to the matched emitted R2C fixed-point unit: all $Read/Write$ labels inside that unit are interpreted at one ϵ -quiescent fixed point and may not be split. This local atomicity requirement is one of the explicit same-cycle semantic constraints and does not require unrelated Sys_ref and RTL_B events to share the same global bucket decomposition. If a component is cycle-locked, its local emitted-occurrence sequence is the ordered sequence of cycles t satisfying $ComponentParticipatesInCycle(C_comb, t)$, with one required unit per such cycle. The pre-HLS and post-HLS traces must therefore agree both on the participation-indexed unit values and on the absence of omitted required units. Cycle-locked components fall outside the bucket-stutter freedom claimed for ordinary latency-insensitive R2C components.

R2C cross-model/cycle-lock clarification. Sys_ref and RTL_B are not assumed to share internal combinational state, a common combinational microstep model, or the same global cycle-bucket placement; each side settles its own cycle-entry valuation. For a cycle-locked component, both sides must satisfy the per-side participation/completeness rule above, and the k -th matched units are compared within those respective component-local participation-indexed emitted sequences; equality of absolute Sys_ref and RTL_B cycle numbers is not required. If RTL_B does not expose an explicit combinational-network model from which participation can be derived, the implementation/certificate evidence for the selected annotated prefix shall justify that its emitted sequence obeys the declared cycle-lock/participation rule. The R2C matching statement in E2 is the Appendix-G expansion of this same R2C clause of Core.17 $\Sigma Match$, not an additional independent matching conjunct.

For proof-internal $RESET(S_p, S_c)$ units included in the selected theorem comparison, match by reset scope and local occurrence within that scope's reset-epoch history: the k -th $RESET$ for a given (S_p, S_c) scope in τ_pre matches the k -th $RESET$ for the same scope in τ_post . Matched $RESET$ units therefore have the same scope. Independent disjoint $RESET$ occurrences may commute unless an explicit reset-boundary or other observable-order constraint orders them.

For synchronization events, match by their per-process occurrence index in the source order (R1; preserved on the post-HLS side by E1 below). Independent Σ -events on different endpoints (including distinct message-passing interfaces) may commute and/or be grouped differently within a clock cycle as permitted by the R-rules and E3. Here “independent” is meant in the Σ -equivalence sense: the pair is not additionally ordered by any explicit Σ -visible ordering/timestamp constraint in E1–E5 (notably E3/E5); it is not a statement about (in)comparability under the Appendix G causal-dependency relation ($\rightarrow$). Accordingly, τ_{pre} and τ_{post} are not required to be identical as a single total order, nor to agree on same-cycle “grouping” of distinct-interface Push/Pop operations, beyond the explicit ordering/timestamp constraints in E1–E5 below.

Consistently with Definition Core.17, we say $\tau_{\text{pre}} \approx \tau_{\text{post}}$ iff $\Sigma\text{Match}(\tau_{\text{pre}}, \tau_{\text{post}})$ holds under the convention above and all the following hold:

E1 (Per-process synchronization-order preservation).

For every process P ,

$$\forall s_1, s_2 \in S_P : s_1 <_P s_2 \Rightarrow \text{clk_post}(s_1) < \text{clk_post}(s_2).$$

Equivalently, synchronization events are matched and ordered by their per-process source occurrence index. E1 does not preserve incidental global clock ordering between synchronization events of different processes unless that ordering is induced by an explicit inter-process synchronization relation already represented in Σ .

E2 (Signal-visibility preservation).

Partition signal Read/Write events by process kind:

- R_{seq} and W_{seq} are the ordinary signal Read/Write events of sequential processes.
- R_{comb} and W_{comb} are the signal Read/Write events of combinational processes, interpreted by R2C.

For sequential-process signal IO:

$$\forall r \in R_{\text{seq}} : \text{clk_post}(r) = \text{clk_post}(\text{pred}_S(r))$$

$$\forall w \in W_{\text{seq}} : \text{clk_post}(w) = \text{clk_post}(\text{succ}_S(w))$$

Equivalently, ordinary signal Reads/Writes of sequential processes are logically timestamped at their synchronization anchors, as in R2. The same anchoring convention is used on the pre-HLS side when forming the matched trace.

For combinational-process signal IO, $\text{pred}_S/\text{succ}_S$ anchoring does not apply. Instead, each R2C fixed-point signal-observation unit is matched by the component-local occurrence ordinal defined above: for each combinational process or explicitly modeled combinational network component C_{comb} , the k -th R2C fixed-point unit of C_{comb} in Sys_ref matches the k -th R2C fixed-point unit of C_{comb} in RTL_B . The matched units must contain the same corresponding combinational Read/Write events and fixed-point values at their respective ϵ -quiescent combinational fixed points, as defined by R2C. Any delta-cycle re-evaluation, intermediate combinational value, or redundant re-drive before the fixed point is an ϵ -step and is not separately Σ -visible. This condition does not require the matched R2C units to occur in the same global observable-cycle bucket number, nor to be grouped with the same unrelated events, in the two executions.

E3 (Safe message issue ordering). (Post-HLS preservation of R4).

For every pair of executed message-call instances $q1, q2 \in Q_{\text{exec}}, P$ of the same process P :

(i) Same-interface source order: if $q1$ and $q2$ access the same message-passing interface and $q1 <_{\text{src}} q2$, then $\text{issue_post}(q1) \leq \text{issue_post}(q2)$. This preserves source order at cycle granularity; any stricter serialization required by the concrete interface is supplied by the channel/interface semantics, not by E3 itself.

(ii) Distinct-interface no-reverse: if $q1$ and $q2$ access distinct message-passing interfaces and appear in sequence in the source ($q1 <_{\text{src}} q2$), then $\text{issue_post}(q1) \leq \text{issue_post}(q2)$ (i.e., they shall not be issued in the reverse sequence).

(iii) Prefix-closed issue obligation: for every cycle-aligned cutpoint σ of the post-HLS execution / witness, if $q1 <_{\text{src}} q2$ and $q2$ has issued by σ , then $q1$ has also issued by σ and $\text{Issue}(q1) \leq \text{Issue}(q2)$. This applies across same-interface and distinct-interface source order. It is an implementation-side obligation on issued sets, not merely an inequality over pairs whose timestamps are already defined.

(Note: HLS internal pipeline staging, dequeue, and hand-off inside the back-annotated process/channel realization are not, by themselves, Σ -visible boundary issue events in Q_{exec}, P nor committed-transfer events in M ; therefore such internal movement does not constitute a counterexample to (ii) or (iii).)

Strict scalar same-endpoint case: when $q1$ and $q2$ are distinct blocking instances on the same scalar explicit abstract boundary and endpoint direction governed by the Core single-transfer-per-cycle endpoint constraint in one reset epoch, Core.IC(4) strengthens clause (i) to $\text{issue_post}(q1) < \text{issue_post}(q2)$. For this ordinary scalar blocking special case, once the selected boundary mapping has qualified the process-owned BoundaryReq, Core.IC persistence/back-to-back ownership, scalar endpoint serialization, and the dynamic same-endpoint source sequence derive the same-interface Issue order; the implementation audit therefore cites the corresponding Core.IC/BoundaryReq evidence rather than selecting Issue annotations to satisfy clause (i). This does not weaken clause (ii) or the cross-interface content of clause (iii). Core.17/ Σ Match separately checks the matched committed per-channel occurrence/value observations.

E4 (Per-channel channel semantics: rendezvous and bounded FIFO).

For every observable channel c with capacity $B(c)$, the projection of τ_{post} to events on c is legal for that channel's capacity model. The channel model splits into two cases.

- Rendezvous case, $B(c)=0$. Channel c is a capacity-zero rendezvous channel. A committed transfer on c consists of a matching same-cycle pair $\text{Push}_c(v)$ and $\text{Pop}_c(v)$ with the same payload value v . No unmatched committed Push_c or Pop_c may exist after any cycle boundary. Equivalently, the k -th committed $\text{Push}_c(v)$ and the k -th committed $\text{Pop}_c(v)$ occur in the same clock cycle and carry the same value. A Push_c endpoint can commit in cycle t only when the matching Pop_c endpoint boundary request is also asserted in cycle t , and symmetrically a Pop_c endpoint can commit in cycle t only when the matching Push_c endpoint boundary request is also asserted in cycle t .

- Buffered case, $B(c)>0$. Channel c is a bounded FIFO with cycle-boundary, non-fall-through semantics. The committed $\text{Pop}_c(v)$ events return exactly the FIFO-ordered sequence of values v previously committed by $\text{Push}_c(v)$, with no drops, duplications, or reordering. For every cycle-aligned prefix π_t of the c -projection—i.e., π_t contains exactly the events on c committed in cycles $< t$ —letting $\text{push}(\pi_t)$ and $\text{pop}(\pi_t)$ be the number of Push_c and Pop_c events in π_t , we have

$$0 \leq \text{push}(\pi_t) - \text{pop}(\pi_t) \leq B(c).$$

Equivalently, the abstract occupancy after cycle $t-1$ is $\text{occ}_c(t) = \text{push}(\pi_t) - \text{pop}(\pi_t)$. E4 storage legality for commits in cycle t is evaluated against $\text{occ}_c(t)$ at the start of cycle t : Pop_c may commit only if $\text{occ}_c(t) > 0$, Push_c may commit only if $\text{occ}_c(t) < B(c)$, and if both Push_c and Pop_c commit in the

same cycle then necessarily $0 < \text{occ}_c(t) < B(c)$. At an ordinary uncoupled primitive buffered boundary, these occupancy conditions together with the attributed own-side request also determine the complementary availability used by Core.12. At a coupling-sensitive $\text{Sys_B}^{\text{elab}}$ boundary they are necessary storage-legality conditions only; actual ChannelEnabled additionally requires the settled MirrorAvail_E value produced by the declared elaboration network.

Thus the inequality-based FIFO availability predicates are used only for $B(c) > 0$. The $B(c) = 0$ case is not interpreted by substituting $B(c) = 0$ into the bounded-FIFO inequalities; it is interpreted by the rendezvous rule above.

Note: $\text{occ}_c(t)$ is the logical occupancy of channel c at the chosen Σ boundary, induced solely by the Σ -visible boundary-commit history ($\text{Push}_c/\text{Pop}_c$) and the annotated capacity $B(c)$ in the buffered case $B(c) > 0$. For $B(c) = 0$, the corresponding channel fact is the same-cycle rendezvous request/commit predicate rather than a nonzero FIFO occupancy predicate.

E5 (Messages cannot cross their immediate synchronization boundary).

For every synchronization call s (in the source order used by $R3 / \text{pref}_S / \text{suff}_S$ over Q_{exec}, P):

$\forall q \in \text{pref}_S(s) : \text{issue_post}(q) \leq \text{clk_post}(s)$

$\forall q \in \text{suff}_S(s) : \text{issue_post}(q) > \text{clk_post}(s)$

$\forall q \in \text{pref}_S(s)$ with q a blocking Push/Pop : $m(q)$ is defined and $\text{clk_post}(m(q)) \leq \text{clk_post}(s)$.

The third clause is the implementation-side synchronization-completion fence: no earlier blocking request may survive s into the succeeding source interval, although q and s may commit in the same cycle. In $\text{Sys_B}^{\text{elab}}$, $m(q)$ denotes the designated process-facing commit that completes/releases q . For a same-cycle q/s completion, the Core.SyncFence statement that q is “accounted for before” the interval advance is an abstract trace/certificate accounting convention used to preserve operation and interval attribution; E5 does not require a separately observable physical sub-cycle ordering between q and s on RTL_B . Under the Issue/Commit linkage above, any boundary request assertion that persists while the process is still pending at s reflects only already-issued outstanding operation(s) in $\text{pref}_S(s)$; no operation $q \in \text{suff}_S(s)$ may have $\text{Issue}(q) \leq \text{clk_post}(s)$, and any back-to-back request handoff on that endpoint direction, if present, is attributed in the first cycle strictly after $\text{clk_post}(s)$. For non-blocking calls, a later executed source-level PushNB/PopNB in $\text{suff}_S(s)$ is a distinct q and cannot be identified with an earlier $\text{pref}_S(s)$ call solely because the endpoint request signal remained asserted.

See Appendix I – Per-Process Replay and Preparation Proof for the local result used by the system theorem:

For any process P participating in a selected finite RTL_B prefix, the fixed stimulus, WC witness, dynamic-operation matching, and P -local observable-unit history determine the corresponding immediate source replay state. Appendix I proves that, from that state, a finite P -owned ϵ preparation path reaches the exact dynamic source operation for the next selected unit—either as next-frontier or as an already issued persistent request—while preserving E3/E5 and committing no unmatched observable unit. This is a local readiness theorem, not an independent channel-commit or global-trace existence theorem. Appendix J combines these local paths with channel and atomic-interaction certificates to construct the complete Sys_ref witness. The reverse direction remains restricted to $\text{RTLConsistentRefPrefix} / \text{Appendix L}$ witness-selected executions.

Allowed Transformations and Why They Preserve $\approx$

Transformation permitted by the rules	Formal justification
Loop pipelining (overlaps iterations)	Introduces internal pipeline stages (modeled as ϵ -steps) and overlaps iterations. R1–R5 (and hence E1–E5) are applied to the explicit S set. Loop pipelining may commute independent Σ -actions across iterations (e.g., across distinct channels)—where “independent” here means “not additionally ordered by any explicit Σ -visible constraint in E1–E5 between the pair (notably E3/E5),” rather than “incomparable under Appendix G’s causal-dependency relation ($\rightarrow$)”—but it must still preserve: (i) per-channel FIFO legality (E4), (ii) the required issue discipline (E3) for source-ordered pairs, and (iii) the sync-boundary constraints (E1/E5) with respect to the (explicit + implicit) S-actions.
Overlap of internal dequeue/staging vs. later pipeline work (ϵ)	Permitted because the internal staging/acceptance activity is ϵ -internal within the back-annotated channel realization (as defined in “Appendix B - Pipelined Loops”) and does not constitute a Σ -visible boundary message-issue reorder. Σ -visible boundary message issues must still satisfy E3, and per-channel committed-transfer legality is ensured by E4; automatic flush / Appendix K addresses deadlock preservation for overlap timing.
#pragma hls_direct_input (late sampling of static signals)	This pragma imposes a designer/environment stability contract: the signal’s value must remain stable after reset (or after its last permitted update) while the receiving process may consume it. The logical signal Read actions r remain anchored exactly as in E2 ($\text{clk_post}(r) = \text{clk_post}(\text{pred_S}(r))$). HLS may implement this by sampling the signal later (instead of inserting internal storage), but because the signal is stable the sampled value equals the value at $\text{pred_S}(r)$; therefore E2 and $\approx$ are preserved.
#pragma hls_direct_input_sync (controlled updates)	This pragma imposes an explicit timing contract on the environment: the controlled direct-input signals may change only on the cycle of the designated synchronization call s^* (i.e., during the sync handshake). With that contract, the logical anchoring required by E2 is preserved: for the receiving process, the relevant Reads are anchored at the closest preceding synchronization call $\text{pred_S}(r) = s^*$, and any environment update Write w_update is aligned with synchronization ($\text{succ_S}(w_update) = s^*$). HLS may still implement late sampling internally, but the value observed is the same as the value at the E2 anchor point, so $\approx$ is preserved without modifying E2.

Transformation permitted by the rules	Formal justification
	Instrumentation note (logical signal timestamping). In the Σ-traces used for $\approx$ checking, sequential-process signal Read/Write events are timestamped at their E2 anchor points (e.g., <code>clk_post(pred_S(r))</code> for Reads), not at any later physical RTL sample cycle that HLS may introduce. Any internal late-sampling micro-steps are treated as ϵ-steps. Combinational-process signal Read/Write labels are handled by R2C rather than by <code>pred_S/succ_S</code> anchoring or by global observable-cycle bucket number. A monitor records a combinational fixed-point value only after the simulator/RTL has reached an ϵ-quiescent combinational fixed point, and that value contributes to Σ only when the relevant R2C observation component emits an R2C fixed-point signal-observation unit under the R2C unit-generation convention. Intermediate delta-cycle reads/writes, redundant re-evaluations, transient combinational values before the fixed point, and R2C stutter cycles whose selected observable slice is unchanged are ϵ/stutter and are not separately Σ-visible. Therefore, equivalence checking must instrument/record the appropriate logical Read/Write event: the anchored event for sequential-process signal IO, or the emitted R2C fixed-point signal-observation unit for combinational-process signal IO. A wrapper/transactor may be used when needed to expose only those logical events rather than naive physical sample, delta-cycle, or per-cycle stutter events.
Memory-access reordering proven conflict-free	Let <code>addr(a)</code> be the symbolic address of access <code>a</code>. If the tool proves <code>addr(a₁) $\neq$ addr(a₂)</code> for the overlapped window, then reordering the underlying memory operations may be treated as observationally silent only when that activity lies below a selected boundary satisfying <code>Core.MemFaithful</code> relative to the untransformed logical mapping semantics. If the reordered mapped message/signal events themselves are selected into Σ, the ordinary <code>Core.17/E1–E5</code> ordering and matching obligations apply; the table entry does not authorize changing that selected event sequence.
Implicit FSM states / added latency	Extra states introduce <i>silent</i> ϵ-steps between observable actions; the partial order on Σ is unchanged, so E1–E5 are unaffected.
Shared-memory arrays with explicit synchronization	When a memory is accessed by multiple processes, the designer explicitly coordinates access with synchronization operations modeled as S-actions, and <code>J.Mem</code> applies to every shared-memory dependency through which one process's write can affect another process's control flow, payload/value expressions, or later operation identity/order. Raw physical memory transactions may remain below the

Transformation permitted by the rules	Formal justification
	proof boundary only under Core.MemFaithful. If caching, merging, splitting, or reordering changes the physical transaction stream, a covered J.1 route uses a transformation-invariant Core.MemProfile logical boundary that discharges J.Mem and Core.MemSyncOrder; otherwise the separately supplied memory-state extension (not developed here) is required. E4 applies to the logical message-passing channels actually selected at that boundary; it is not weakened to match variable-cardinality raw memory traffic. For Appendix K, the same profile must either establish WFG-inert hidden realization or expose a K-faithful logical boundary above the transformation. A persistently backpressuring memory for which no such boundary exists cannot combine transaction-changing transformations with the Appendix-K theorem scope.
Non-blocking message-passing (PushNB/PopNB) verified by post-HLS snooping	If the latency differences are externally visible to the DUT, a verification wrapper delays the pre-HLS side so that the committed transfer events (successful completions) occur in the same order/timing as observed on the post-HLS channel. This wrapper is itself purely synchronising (S), so it cannot violate R1–R5. Once the wrapper is assumed, τ_{pre} and τ_{post} coincide on the committed message-transfer history recorded in Σ (i.e., the Push_c/Pop_c commit events), so E3–E5 are preserved. Unsuccessful non-blocking polls remain ϵ -steps and are not required to match. This is not a transformation that inherently preserves $\approx$, but rather a verification technique to enforce equivalence for inherently latency-sensitive operations.
Latency-sensitive <i>global</i> signal IO matched by snooping	If the latency differences are externally visible to the DUT, the same “mirror-and-delay” wrapper used for NB channels is applied to any globally-visible signal whose timing matters. Because the wrapper inserts only S-actions, the partial-order constraints are unchanged. This is not a transformation that inherently preserves $\approx$, but rather a verification technique to enforce equivalence for inherently latency-sensitive operations.
Latency-sensitive <i>local</i> signal IO isolated in cycle-accurate transactors	Local timing-exact protocols are confined to dedicated transactor processes that expose only latency-insensitive M or S operations to the rest of the design. Since the transactor boundary is now the observable interface, R1–R5 apply <i>outside</i> the timing-sensitive region, so system-level $\approx$ still holds.
One-way signal-handshake protocols with run-time assertions	For single-direction vld or rdy signals, an assertion proves that back-pressure can <i>never</i> occur. That assertion establishes a refinement in which the missing handshake is

Transformation permitted by the rules	Formal justification
	equivalent to a permanent logical '1', so the protocol is observationally identical to the corresponding two-way handshake with the missing side tied true. E2 then applies only to the logical timestamping of the signal Read/Write events, not to the protocol-refinement step itself.
Combinational processes	Combinational processes have no process-owned S actions and no message-passing M actions in this formal clause, but they may have observable signal Read/Write events in Σ through explicitly selected R2C observation components. Their signal IO is interpreted at ϵ-quiescent combinational fixed points as specified by R2C: intermediate delta-cycle propagation is ϵ, and only emitted fixed-point Read/Write labels in the selected observable slice are Σ-visible. Repeated cycles in which the selected fixed-point slice is unchanged are R2C stutter for that component unless an explicitly modeled observer/sampler requires a fresh emitted unit. Under the well-formedness requirement that the combinational network is deterministic, reaches a unique fixed point in each compared cycle, and has a well-defined R2C unit-generation rule for the selected component, the RTL/tool-flow conformance obligation is to preserve the emitted fixed-point signal-observation unit sequence at the chosen observation boundary. Thus the required correspondence is not equality of simulator delta-cycle behavior, nor equality of global cycle-bucket numbers, nor an automatic consequence of ordinary combinational synthesis; it is equality of matched emitted R2C fixed-point units by component identity and local emitted-occurrence ordinal. Designs with oscillation, ambiguous multiple-driver resolution, hidden stateful behavior, intentional combinational-loop behavior outside a unique fixed-point interpretation, or multi-input observation slices whose interleavings can change the emitted unit sequence are outside R2C unless represented by explicit state, synchronization, a cycle-accurate transactor, a cycle-locked R2C observation boundary, or another formally modeled boundary.

For the purposes of the formal analysis, non-blocking message-passing is modeled at the level of committed transfers: a successful PushNB/PopNB contributes the same committed Push_c/Pop_c event to Σ as a blocking Push/Pop, while unsuccessful polls are ϵ -steps.

If a design contains latency-sensitive global signals or non-blocking transfers whose timing or ordering is externally visible at the DUT boundary and must be reproduced in the selected comparison, an Appendix L snooping/witness-selection wrapper may be used. When a witness satisfying WC and L.Dir exists, the wrapper selects an admissible pre-HLS execution whose relevant Σ -events align with the observed RTL execution. The wrapper does not enlarge the set of admissible pre-HLS behaviors; it selects one

admissible execution. If the wrapper cannot realize such a witness, the compared run is outside the theorem scope and must be handled according to the Appendix L triage rules.

Appendix L defines the witness-existence conditions, monitoring, and triage required to establish this alignment for a concrete compared run.

Proof Structure

The proofs proceed in three stages:

First, Appendix I establishes process-local deterministic replay and finite preparation. For each process participation in a selected RTL_B observable unit, it identifies the exact dynamic source operation, proves the relevant issue/order and witness facts, and constructs a finite process-owned ε path that reaches the operation or its persistent issued request without committing an unmatched unit. These facts are packaged by J.PCert. Appendix I deliberately does not assert that the complementary endpoint is present, that a channel transfer is globally enabled, or that the local paths already form one Sys_ref execution.

Second, Appendix J combines J.PCert with J.CCert, J.ACert, environment/reset certificates, the normative field-sensitive footprint vocabulary, deferred-hole records, Dep_J, and LocalAgree. J.CanonicalRank is derived from the graph. J.Frame/J.LocalLift/J.PrepareCommutate and J.RegPhaseNF prove that outcome-independent actions can be moved into L1, requests freeze before the L2 snapshot, and A3/A4 decisions and units execute in L3 modulo J.HCanon. Horizon-batched J.UnitExt and J.GWR-Comp construct J.GWR and J.RegPhaseReach; Proposition J.0 proves observable compatibility. This safety construction does not by itself prove K.DrainReach, A5-L, or RTL deadlock freedom.

Finally, Appendix K addresses a separate liveness-preservation obligation for internal message-passing deadlock. Appendix K does not discharge the Appendix I/J weak-fairness, finite-depth, channel-progress, realization-correctness, K.Drain, K ε , J.KObsConf, K.RLAdmReach, QualifiedSynchronousModel/Core.KCutClosure_post, K.CertCov, or A6–A11 side conditions. Nor does it derive SDet, A5-L, A5-S, K.RespCov, K.RespClean, K.EnvMF, K.InterruptionGate/K.ResetEpoch/K.TerminalDisposition applicability, or any alternate A5-L discharge from the local R-rules. The core Theorem K.1A consumes A5-L and one K.CertifiedCutpoint package directly; Corollary K.1A-Total additionally consumes Core.KCutClosure_post and K.CertCov; in the standard scope Core.QSync-Post derives the closure premise from the qualified synchronous structural evidence for the universal conclusion. The primary user-facing route fixes the actual Policy-S pre-HLS simulator and its selected execution through SDet, satisfies K.EnvMF and the shared K.InterruptionGate applicability, instantiated first as K.ResetEpoch and then, only on K.ResetEpoch.Continue, as K.TerminalDisposition, supplies maximality/fairness/completeness and complete bounded-response evidence through A5-S/CF-S, uses Lemma K.2b and Corollary K.2c to discharge A5-L, and uses Core.KCutClosure_post plus K.CertCov to justify the universal reachable K-facing RTL conclusion. Structural, invariant, exploration, direct Policy-L, and tool-certificate routes remain alternatives only for instances in which A5-L is actually true; they cannot rescue a design containing an admissible serial deadlock prefix.

Thus the architecture is non-circular. Appendix I supplies only local replay/preparation facts; J.GWR-Comp proves, rather than assumes, their global coexistence and complete-unit extension; Proposition J.0 consumes an already constructed trace pair; and Appendix K separately preserves internal message-

passing deadlock freedom under A5-L and its recorded discharge route. The liveness theorem remains conditional on the named fairness, progress, drain, and serial-dominance premises and does not retroactively justify the safety construction.

Causal Dependency Between Processes

To prove system-level trace compatibility, we must distinguish between incidental timing and functionally significant ordering.

Formal Definition of Causal Dependency

To formalize the notion of functionally significant dependency and resolve ambiguity, we define a causal-dependency relation, denoted by $\rightarrow$, on the set of observable IO actions (Σ) across all processes in the system. The relation is causal/information-flow rather than purely temporal: events related by $\rightarrow$ may commit in the same clock cycle, and the direction of $\rightarrow$ indicates which event may be depended on by the other.

Important: $\rightarrow$ is a causal-dependency graph, not by itself the source order used for the Appendix J induction. In particular, because the methodology permits distinct-interface message operations to be issued with timing overlap or in the same cycle while still preserving dynamic source order, and because rendezvous transfers are same-cycle atomic Push/Pop transactions, the unquotiented dependency graph $\rightarrow$ need not be acyclic in every legal execution. A cycle in $\rightarrow$ can represent a legal same-cycle mutually enabling dependency among atomic transactions; it is not, by itself, an observable mismatch. When Appendix J needs a well-founded induction order, it uses the separate required-order relation $<_J$ over observable units. That relation is generated only by the explicit ordering, timestamp, FIFO, atomicity, and synchronization constraints that must be preserved by the trace-compatibility predicate, not by the full causal-dependency graph $\rightarrow$ and not by raw per-process source order on distinct-interface message commits.

Well-formedness use of causal-dependency. The well-formedness rule below uses $\rightarrow$ as a design-intent and source-admissibility relation: if correctness depends on the relative timing, value, availability, or cross-process observation of two events, that dependence must be represented by an explicit modeled causal mechanism such as message passing, synchronization, signal handshakes/anchors, declared memory serialization, or an Appendix L snooped/witness-selected boundary. This use of $\rightarrow$ does not add proof-preserved observable ordering beyond $<_J$, FIFO legality, rendezvous/synchronization atomicity, signal-anchor constraints, synchronization-boundary constraints, and declared memory-serialization constraints.

The causal-dependency relation $\rightarrow$ is the smallest directed relation satisfying the conditions below:

1. **Intra-Process Order:** If actions a and b occur within the same process P and a precedes b in the program's execution trace, then $a \rightarrow b$. This directly reflects the sequential nature of the code within a single thread. **Clarification (non-temporal).** Here “precedes” refers to the order of IO action *instances* in the process's trace/program order (i.e., the order induced by the process's execution), **not** the relative order of their **commit cycles**; distinct-interface message actions may be **issued** in order while **committing** in either order depending on backpressure.
2. **Inter-Process Communication:** The relation is established by the direct exchange of information or synchronization between processes.

- **Message-passing (committed transfer):** If a is the commit of a send on channel c in process $P1$ (a committed $Push_c$, including a successful $PushNB$), and b is the commit of the corresponding receive on c in process $P2$ (a committed Pop_c , including a successful $PopNB$), for the same logical message instance, then $a \rightarrow b$. Here “corresponding receive” is defined by the channel’s FIFO semantics: b is the Pop_c commit that consumes the element produced by a (matching is by FIFO position / transfer instance, not by payload-value equality; duplicate payload values do not introduce ambiguity). Equivalently, if we enumerate committed Pushes on c as $Push_c^1, Push_c^2, \dots$ in commit order and committed Pops as $Pop_c^1, Pop_c^2, \dots$ in commit order, then $Push_c^k \rightarrow Pop_c^k$. For buffered channels with $B(c) > 0$, E4 also imposes the required FIFO legality used by Appendix J’s required-order relation $\leq_J$. For rendezvous channels with $B(c) = 0$, the matching $Push_c$ and Pop_c commits occur in the same clock cycle and are treated in Appendix J as one explicitly atomic observable unit. The directed dependency $a \rightarrow b$ records unidirectional information flow — the Pop observes the value supplied by the Push — but it does not make the two rendezvous endpoints separately orderable for the Appendix J induction, and it does not imply an earlier commit cycle. No reverse edge $b \rightarrow a$ is implied unless there is an explicit reverse-direction communication, such as another channel or a signal handshake. Unsuccessful non-blocking polls are ϵ -steps and do not participate in $\rightarrow$.
- **Signal Handshake:** If a is a signal write action $w(sig)$ in $P1$ and b is a signal read action $r(sig)$ in $P2$ that reads the value written by a as part of an explicit handshake protocol, then $a \rightarrow b$. A complete two-way handshake (e.g., vld/rdy) creates a causal chain, such as $w(vld)@P1 \rightarrow r(vld)@P2 \rightarrow w(rdy)@P2 \rightarrow r(rdy)@P1$.
- **Explicit Synchronization (two-party SyncChannel / ac_sync):** Let $sout$ be the commit of a SyncChannel $sync_out$ (or equivalent ac_sync call) in process $P1$, and let sin be the commit of the matching $sync_in$ in process $P2$ for the same synchronization instance. Then this synchronization acts as a two-process barrier:
 - Every observable action in $P1$ that occurs before $sout$ (in $P1$ ’s trace/program order) satisfies $a \rightarrow b$ for every observable action b in $P2$ that occurs after sin .
 - Symmetrically, every observable action in $P2$ that occurs before sin satisfies $a \rightarrow b$ for every observable action b in $P1$ that occurs after $sout$.
 - (The two sync commits $sout$ and sin are treated as simultaneous; we do not impose $sout \rightarrow sin$ or $sin \rightarrow sout$.)

3. **Transitivity:** The relation is transitive. If $a \rightarrow b$ and $b \rightarrow c$, then $a \rightarrow c$.

Using this relation, we now provide a formal definition for causal dependency between synchronization events, which are the fundamental ordering anchors in the scheduling model.

Definition: Causal Dependency

A synchronization event $s2$ in process $P2$ is **causally dependent** on a synchronization event $s1$ in process $P1$ if and only if $s1 \rightarrow s2$.

If neither $s1 \rightarrow s2$ nor $s2 \rightarrow s1$ holds true, the events $s1$ and $s2$ are considered **causally independent** (or concurrent). Any change in the observed temporal ordering of causally independent events between the pre-HLS and post-HLS simulations is considered an incidental, functionally insignificant variation in latency. A **well-formed design**, under this methodology, **shall not rely** on a specific ordering of causally independent events for functional correctness.

To be clear, the shall-not-rely requirement is a design rule. To adhere to this design rule, designs must express functionally significant cross-process dependencies through explicit modeled mechanisms such as message passing, SyncChannel/ac_sync operations, signal handshakes/anchors, declared memory-serialization constraints in the array-access mapping layer, or Appendix L snooping/witness-selection boundaries. Designs that violate this design rule are outside the formal guarantees. This rule concerns design well-formedness; it does not convert every causal-dependency edge into a strict proof-order edge.

System-Level Theorem (B1-Specialized Appendix G Statement)

Appendix G gives the high-level system theorem in the same standing-assumption regime used by Appendices J–K: B1 is assumed for RTL_B under the fixed stimulus, so the practical “unique RTL_B trace” reading is available throughout the main theorem stack. The authoritative finite-prefix trace-set theorem, including the exact reachable-prefix quantification and the conditional use of B4/B5 when ϵ -normalization or uniqueness up to finite ϵ -stutter is invoked, is Theorem J.1. Therefore, if this Appendix G statement and Theorem J.1 are read together, Theorem J.1 controls; Appendix G is the high-level summary of the same B1-standing finite-prefix theorem regime.

Fix an initial state and a fixed external stimulus/environment behavior for both the prescribed source reference model Sys_ref and RTL_B, with Sys_ref as defined in Definition Core.1 and Appendix J, Remark 2. (The raw rendezvous model Sys is the special case obtained when Sys_ref = Sys_B and all effective capacities are zero.)

Interpret “fixed stimulus” per Appendix I.2, “Fixed-stimulus interpretation (reactive environments).” Let Reach_post / Reach_ref denote the corresponding finite reachability sets with no fair-extension qualifier (Definition J.Reach, Appendix J).

Assume the Appendix I / Appendix L witness machinery and WC are available, together with the local certificate package of Definition J.LocalGWR: J.PCert for every process, J.CCert for every explicit abstract channel, J.ACert for every explicitly atomic unit, the applicable environment/reset certificates, and J.LocalAgree. Also assume the capacities/occupancies/enableness semantics of Sys_ref, finite capacities and B-faithfulness at every explicit abstract boundary, the standing RTL_B determinism assumption B1 under the fixed stimulus, and—on the normal compositional route—QualifiedSynchronousModel_Sys_ref(I), so Core.QSync-Ref supplies the finite L2 settled snapshot used by J.RegPhaseNF/J.UnitExt. A directly validated J.GWR package may instead carry equivalent per-Horizon settling evidence. No B2/B3 or generic fair-extension premise is used for either finite-prefix direction; use B4/B5 only for replay ϵ -normalization or uniqueness up to finite ϵ -stutter not discharged by the QSync same-cycle settle. Complete/fair execution obligations belong to later theorems that explicitly require them.

Then the B1-specialized system-level result is:

(Post $\rightarrow$ Ref existence) For every $\tau_{\text{post}} \in \text{Reach_post}$ equipped with a local certificate package satisfying Definition J.LocalGWR and the standard source-side QualifiedSynchronousModel_Sys_ref(I) premise of J.GWR-Comp, Theorem J.GWR-Comp constructs W with GlobalWitnessRealizable(τ_{post}, W), and there exists $\tau_{\text{ref}} \in \text{Reach_ref}$ such that $\tau_{\text{ref}, \text{system}} \approx_{\text{sys}} \tau_{\text{post}, \text{system}}$ under E1–E5. A directly validated J.GWR package may instead supply the equivalent checked per-Horizon settling evidence.

(Ref $\rightarrow$ Post selected-pair compatibility, annotated RTL-consistent prefixes only) For every $\tau_{\text{ref}} \in \text{Reach_ref}$ equipped with a selected RTL_B prefix $\tau_{\text{post}}^* \in \text{Reach_post}$ and a compatible witness/annotation package satisfying the annotated RTL-consistency requirement of Theorem J.1 (RTLConsistentRefPrefix), there exists $\tau_{\text{post}} \in \text{Reach_post}$ such that $\tau_{\text{ref}, \text{system}} \approx \tau_{\text{post}, \text{system}}$.

Moreover, because B1 holds in this Appendix G specialization, the matching Σ -label sequence is unique (up to insertion/removal of finite ϵ -steps permitted by B4/B5). That uniqueness is only projection uniqueness; it does not by itself determine μ_Q , failed-poll instances, issue annotations, request attribution, E4 boundary histories, or the corresponding source-side witness prefix/state in Sys_ref.

Proof sketch. Appendix I proves the process-local replay and finite-preparation facts packaged by J.PCert. Theorem J.GWR-Comp combines those facts with J.CCcert, J.ACcert, environment/reset evidence, and the operational frame/commutation lemmas to construct one reachable Sys_ref witness and J.GWR package. Proposition J.0 then lifts the chosen compatible trace pair to system-level $\approx_{\text{sys}}$, and Theorem J.1 supplies the finite reachable-prefix trace-set statement. Thus the main result is explicitly about sets of finite reachable execution prefixes under fixed stimulus, not about fair-extension-qualified prefixes. The Ref $\rightarrow$ Post direction is not a symmetric arbitrary-source construction; it applies only to annotated RTL-consistent source prefixes, and under this Appendix G specialization B1 supplies only the unique RTL_B Σ -trace against which the projection component of that consistency is judged. The compatible annotation package remains a separate witness obligation and is not inferred from bare Σ -label matching.

Practical Implication (“No Surprises” Guarantee)

If a verification environment exercises only the alphabet Σ (signals, message ports, synchronization calls) and does not test internal latency, then—under the assumptions of Theorem J.1 in Appendix J—the prescribed source reference model Sys_ref and the post-HLS RTL agree at the chosen observable boundary in the following qualified finite-prefix sense:

- for every $\tau_{\text{post}} \in \text{Reach_post}$ whose local certificate package satisfies Definition J.LocalGWR and whose selected Sys_ref satisfies the source-side Core.QSync premise of J.GWR-Comp, Theorem J.GWR-Comp constructs a J.GWR package and a matching $\tau_{\text{ref}} \in \text{Reach_ref}$; a directly validated J.GWR package carrying equivalent per-Horizon settling evidence may be used instead; and
- in the reverse direction, only those source-side $\tau_{\text{ref}} \in \text{Reach_ref}$ that are annotated RTL-consistent in the sense of Theorem J.1—i.e., equipped with a selected RTL_B prefix $\tau_{\text{post}}^* \in \text{Reach_post}$ and a compatible witness/annotation package satisfying RTLConsistentRefPrefix, not merely a bare Σ -projection match—are guaranteed to have a matching reachable τ_{post} .

Direction-of-proof clarification for source-side safety signoff. Although users often describe this use case as transferring a pre-HLS result “from Ref to Post,” the deductive direction ordinarily used for safety is the Post $\rightarrow$ Ref clause. If every reachable RTL_B prefix has a matching Sys_ref prefix with the same canonical Σ_{DUT} observation, then RTL_B introduces no Σ_{DUT} -observable safety behavior absent

from Sys_ref. A safety property proved for every covered Sys_ref prefix therefore transfers without the restricted Ref $\rightarrow$ Post clause. When Definition J.RefProjCover holds, one complete pre-HLS execution covers those source-reference observations and can support the same transfer. The restricted Ref $\rightarrow$ Post clause instead concerns realization of a selected source behavior by RTL_B and remains annotation-dependent.

Latency-sensitive testbench caveat

A testbench that branches on raw cycle counts, uses timeout watchdogs, checks performance/latency bounds, or otherwise changes its stimulus or pass/fail behavior based on timing not represented in Σ is not covered by the same-stimulus trace-compatibility guarantee unless that timing dependence is itself modeled through explicit Σ -visible causal mechanisms such as synchronization events, handshakes, or message-passing transactions. Such checks may still be useful, but they should be treated as latency-sensitive testbench logic, isolated as described in Appendix D, or performed as post-HLS / RTL-level performance checks rather than as properties transferred automatically from an arbitrary pre-HLS simulation run.

Accordingly, the precise “no surprises” reading distinguishes two source-side uses. For safety, no Σ -observable behavior can appear in RTL_B under the fixed stimulus without a matching behavior in Sys_ref; therefore universal Sys_ref safety, or one complete source run satisfying Definition J.RefProjCover, transfers through Post $\rightarrow$ Ref as stated by Corollary J.1-Safe. Separately, a claim that a particular source-side behavior is realized by RTL_B uses the restricted Ref $\rightarrow$ Post clause and requires RTLConsistentRefPrefix or the applicable Appendix L snooping / witness-selection construction.

Thus, for a Σ -only testbench under the same fixed stimulus, any mismatch found on RTL_B corresponds to a matching mismatch on Sys_ref. A source-side safety pass justifies RTL_B through Corollary J.1-Safe under universal source verification or J.RefProjCover, provided J.TraceCertCov supplies the applicable Post $\rightarrow$ Ref construction evidence, including source-side Core.QSync on the normal compositional route or equivalent checked per-Horizon settling evidence on the direct J.GWR route; realization of a selected source trace still uses the restricted reverse direction. In the standard qualified-synchronous normal-compositional route, J.QSyncTraceInd supplies the cycle-aligned J.TraceCertCov evidence and Lemma J.CycleIdealCover / Corollary J.1-Safe-QSync extend safety predicates satisfying Definition J.IdealPrefixClosed to finite observable prefixes within those cycles, without requiring an independent correspondence package for each such intra-cycle ideal. Here Sys_ref means Sys_B in the ordinary bounded-FIFO case and Sys_B^elab in the elaborated back-annotated cases of Appendix B / Appendix Q. See Appendix J / Theorem J.1, Lemma J.CycleIdealCover, and Corollaries J.1-Safe/J.1-Safe-QSync for the full execution-level statements.

Appendix G states the trace-compatibility rules and a B1-specialized high-level implication. Appendix I supplies process-local deterministic replay and finite preparation certificates; Appendix J proves that these local process facts, together with local channel and atomic-interaction certificates, construct a single globally realizable source witness, and then proves the system theorem for the prescribed source reference model. Taken together, these appendices establish the stated execution-level correspondence at the chosen observable interfaces under fixed stimulus and the named local/tool obligations. The unrestricted guarantee is Post $\rightarrow$ Ref; the reverse direction ranges only over annotated RTL-consistent source-side behaviors covered by Theorem J.1.

Appendix H – Possible Criticisms

This section highlights several potential challenges in the verification approach of Appendix G.

1. Designer-Managed Shared-Memory Synchronization

Appendix G requires that any memory shared between processes use explicit synchronization primitives inserted by the designer. The HLS tool does not perform static verification of those primitives. In practice, this risk can be mitigated by encapsulating shared memories within parameterized library classes (for example, examples 12_ping_pong_mem and 15_native_ram_fifo) that are checked for correct memory access coordination in both the pre-HLS and post-HLS models. Such classes are natural first candidates for reusable Core.MemProfile qualification: the profile places a transformation-invariant logical boundary above any permitted cache/merge/split/reorder behavior, discharges Core.MemFaithful, J.Mem, Core.MemSyncOrder, the applicable A6 ownership/arbitration structure, reset treatment, and the Appendix-K disposition once per library pattern, and binds that qualified profile to each concrete theorem instance through K.CertHdr. Until such a profile has been qualified, the class must present value-carrying traffic at boundaries satisfying the existing J.Mem route without transaction-changing mismatch, use the separately supplied memory-state extension, or carry a correctness argument outside the Appendix I–K theorems.

2. Latency-Sensitive Signal Alignment (“Snooping”)

See Appendix L for detailed discussion on snooping.

3. By Default, Matchlib Connections Model a Skidbuffer inside In<> Ports

(Note that this overall document and specifically Appendix G are not intended to be strictly tied to Matchlib. Instead, this document is intended to be applicable to any pre-HLS model written in SystemC using signals, message-passing channels, and synchronization calls.)

Matchlib Connections supports throughput accurate modeling in the pre-HLS simulation. Currently the throughput accurate modeling mechanism relies on the Pre() and the Post() methods using a 1-place buffer to store transactions that are in flight on Connections::In<> ports. This means that by default In<> ports function like a skidbuffer. Connections::Out<> ports do not introduce any additional storage capacity into the pre-HLS simulation.

By default, Catapult adds skidbuffers on input ports into the post-HLS design, so the formal trace-compatibility relationship described in this document still holds. In other words, the default pre-HLS and post-HLS designs both still model rendezvous semantics, since both explicitly include a structural instantiation of a skidbuffer on input ports. In effect, we define the rendezvous boundary *before* the skidbuffer.

It is possible to remove some or all the skidbuffers during Catapult HLS. Because the abstract rendezvous boundary is defined before the skidbuffer, this choice is orthogonal to the trace-compatibility rules (R1–R5/E1–E5) as long as skidbuffer-internal behavior is not included in Σ . However, to keep the pre-HLS simulation’s port micro-architecture aligned with the RTL configuration, the following compile flag must then be used in this case:

```
-DFORCE_AUTO_PORT=Connections::SYN_PORT
```

This will remove all the skidbuffers from the pre-HLS model. In this case, if some of the skidbuffers are still inserted during HLS, the capacity effects of those specific skidbuffers should be added into Sys_B so that the formal trace-compatibility relationship still holds.

In this case the recommended methodology is to use both approaches:

- Use the default throughput accurate mode in Matchlib for most analysis and debug, and for pre-HLS performance verification.
- To keep the pre-HLS simulation's *port micro-architecture* strictly aligned with the RTL configuration, then use -DFORCE_AUTO_PORT=Connections::SYN_PORT, and rerun final verification tests.

Appendix I – Per-Process Replay and Preparation Proof

I.1 Overview

This appendix establishes the process-local replay and preparation results used by Appendix J. Under a fixed initial state, fixed external stimulus/environment behavior, and WC-compatible witness, a selected RTL_B observable-unit prefix determines the corresponding source-local replay state for each process and a finite process-owned ε preparation path to the exact dynamic operation participating in the next selected unit. Appendix I does not by itself commit a channel transfer, pair rendezvous or synchronization endpoints, or construct a complete global Sys_ref execution; those composition steps are proved in Appendix J by Lemmas J.Frame–J.UnitExt and Theorem J.GWR-Comp.

Downstream citation discipline. Later appendices may cite Appendix I for Implementation assumption I.A0; Definitions I.ReplayObs and I.CutpointOfferObs; Lemmas I.1, I.ReplayObs-Congruence, I.DR, I.DR', and I.3; and the process-local certificate obligations later packaged as J.PCert. Appendix I does not justify an unrestricted per-process or system-level existence theorem in isolation. A complete matching Sys_ref prefix is obtained only after Appendix J combines the process-local paths with the common explicit-channel state, endpoint co-presence, atomic grouping, environment/reset evidence, current request-issue evidence, and interface agreement. The converse direction likewise remains restricted to RTLConsistentRefPrefix / Appendix L witness-selected executions.

I.2 Formal Preconditions

- **Observable alphabet** Σ is as defined in *Normative Formal Core for Appendices G–K*: it consists of the event schemas {Push_c, Pop_c, Sync, wait, START_P, FINISH_P, Write, Read} instantiated only for channels/signals designated observable at the chosen proof boundary. For Appendices I–K, Σ is the proof-internal observable alphabet of the prescribed Sys_ref / RTL_B comparison. Thus all message-passing channels that can participate in Appendix K deadlock reasoning are designated observable at that proof boundary and hence included in Σ . In elaborated cases Sys_ref = Sys_B^elab, this proof-internal Σ may include introduced staged/bundle/join channels even when the projection Π_{elab} hides those internal events from the outer DUT/testbench-visible alphabet Σ_{DUT} . Such events are therefore Σ -visible for the proof-internal comparison, but may be projected away from the outer observation boundary.

- **Non-blocking message-passing interpretation.** In both RTL_B and Sys_ref , a non-blocking call ($PushNB$ / $PopNB$) is treated as a poll. A successful poll contributes the same observable event as the corresponding committed $Push_c$ / Pop_c on that channel. An unsuccessful non-blocking call (returns “false” / “not completed”) contributes no Σ -event and is modeled as an ϵ -step. A non-blocking poll may still assert a Σ -visible endpoint request in the cycle of that call. However, each executed non-blocking call is a distinct dynamic source-level instance $q \in Q_exec, P$ for its owning process P . Accordingly, continuous assertion of an endpoint request with no intervening commit does not by itself collapse later executed non-blocking call instances into an earlier q . Only a poll that succeeds contributes the corresponding committed transfer $m(q) \in M$ for that call instance. If a request assertion persists without a new executed source-level call instance being entered, its attribution remains with that same currently outstanding q ; but repeated non-blocking polls in later loop iterations or later source-level call instances create additional $q \in Q_exec, P$ even if req does not deassert.
- **Silent step** Any internal RTL microstate transition is written ϵ . In particular, unsuccessful non-blocking polls, arbitration bookkeeping, and pipeline advances are ϵ -steps.
- **Finite-pipeline-depth premise (FPD)** There exists a finite constant $pipe_depth(P) = D_P < \infty$ such that once P begins internal micro-progress toward its next observable Σ -action (or that action is locally enabled), P executes at most D_P consecutive ϵ -steps before committing a Σ -action or reaching a stable wait with no further ϵ -step available until external/peer conditions change. In particular, a process cannot perform an unbounded number of ϵ -only failed non-blocking polls while making no progress toward any Σ -action; designs that can spin indefinitely without reaching either a Σ -action or a stable wait state violate FPD/B4 and are outside the model.
- **Weak-fairness premise (WF)** If a micro-operation remains continuously enabled, the scheduler eventually selects it for progress. For an already-issued blocking message request whose $BoundaryReq$ is asserted and whose $ChannelEnabled$ predicate remains true, this means the enabled transfer is eventually allowed to commit; it does not mean that the same source-level operation is “issued” again. This is an assumption on the execution/scheduling environment (see Appendix N), not an automatic guarantee of “clock-synchronous RTL.” In practice it requires that any arbiters/back-pressure logic that can affect whether an enabled operation is selected or allowed to commit must be fair in the sense of B2/B3.
- **Finite-progress invariants (B3).** For ordinary primitive storage/rendezvous boundaries, $P_push(c)$ and $P_pop(c)$ retain their existing discharge meaning: full is discharged by a Pop , empty by a $Push$, and a persistent primitive rendezvous pair by the atomic rendezvous. In $Sys_B^{\wedge}elab$, those primitive obligations are applied to the explicit storage/rendezvous elements that actually hold the blocking capacity condition. A coupling-sensitive boundary may instead be $ChannelDisabled$ because the declared stateless coupler network withholds its complementary ready/valid value even though the local storage is nonfull/nonempty. Definition Q.1(j)–(k) handshake/release reconstruction exposes the actual process-facing/storage causes, and when the serial route consumes a coupling-sensitive boundary Q.1(l) supplies the additional handshake-support/CouplerPersistence obligation; when those causes change, the next settled valuation changes accordingly, and B2 governs any transfer that remains continuously $ChannelEnabled$. B3 is still a liveness assumption/certificate interface, not a consequence of the cycle-by-cycle R-rules and not a premise of finite-prefix Theorem J.1.
- **Channel enablement semantics.** Let $q \in Q_exec, P$ be a dynamic source-level message operation on boundary c and let $BoundaryReq(q, \sigma)$ hold at a settled $BoundaryEvalPoint$. $ChannelEnabled$ is the actual settled ready/valid eligibility of that operation, not merely a capacity test. Write $MirrorAvail_E(q, \sigma)$ for the complementary signal: ready for $Push$, valid for Pop . Then

$\text{ChannelEnabled}(q, \sigma) \Leftrightarrow \text{BoundaryReq}(q, \sigma) \wedge \text{MirrorAvail_E}(q, \sigma).$

Ordinary primitive boundary. If $\text{Sys_ref} = \text{Sys_B}$ and $B(c) > 0$, MirrorAvail_E is exactly not-full for Push and not-empty for Pop, giving the familiar E4 formulas. If $B(c) = 0$, MirrorAvail_E is exactly the matching complementary endpoint request, and an enabled pair may commit atomically when selected.

Elaborated boundary. If $\text{Sys_ref} = \text{Sys_B}^{\text{elab}}$, E4 still governs storage legality and committed FIFO/rendezvous behavior, but E's declared stateless coupler network may further determine the complementary ready/valid value from other explicit channel/request conditions. A nonfull Pop/Push boundary is therefore not automatically ChannelEnabled if the coupler withholds the mirror signal. The coupler may be at an internal elaborated boundary or at a process-facing outer boundary; it has no stored state and does not withdraw the owning source BoundaryReq . $\text{Core.QSync/Core.Settle}$ provide the unique settled valuation used for the test.

Commit occurs only when the settled ChannelEnabled action (or enabled rendezvous/bundle action) is selected. B2 prevents starvation only while that actual settled enablement remains continuously true. No unmodeled grant, arbitration, or backpressure predicate may alter this result: all such same-cycle logic must be part of E or the selected scheduler/arbitrator semantics, and every storage/credit effect must be represented by $B(c)$ or an explicit finite-capacity channel.

- **No ill-formed signal IO.** Every observable signal Read/Write has a unique real bounding synchronization operation on each dynamic path on which it occurs; otherwise, the design is ill-formed. START_P may be used as an initial anchor only when it is included as a real synchronization event in S, and FINISH_P may be used as a terminal anchor only when the process actually terminates. The formal model does not use a virtual synchronization event at ∞ . (From RULE 1 and RULE 2.)
- **Fixed-stimulus interpretation (reactive environments).** If the “same stimulus” includes a reactive testbench and/or peer processes, it is interpreted as the same global Σ -causal behavior (same externally supplied inputs, and the same peer/environment responses to the same prior Σ -visible history), not as a function of P-local history alone. In Appendix I this interpretation is used only as a witness-replay condition: when a next RTL_B event e is already fixed in the chosen post trace prefix, any peer/environment enabling needed for e is part of that same fixed global Σ -causal behavior and is therefore replayed consistently for the matching Sys_ref construction. This is not a standalone assumption that arbitrary Sys_ref continuations are globally live.
- **WC (Witness-compatible non-blocking outcomes).** Because failed polls are modeled as ϵ (unobservable), all correspondence results in Appendices I–K are stated for witness-compatible comparisons: any ϵ -modeled non-blocking poll outcome (and any other ϵ -modeled internal choice that can affect the next Σ -visible control flow / next Σ -visible blocking frontier) is fixed/selected to be consistent with the compared RTL_B execution prefix τ_{post} (e.g., via the Appendix L witness-selection/snooping wrapper). WC also supplies the μ_Q matching for ϵ -modeled failed non-blocking call instances when E3/E5 or the proof construction refers to $q \in Q_{\text{exec}, P}$; those failed q 's are not inferred from Σ alone.
- **Remark (NB-CF as a sufficient condition).** If a design satisfies the following non-blocking control-flow property—failed PopNB/PushNB polls do not influence the next Σ -visible control flow/frontier—then the witness is trivial and WC holds automatically. Equivalently: between two consecutive Σ -events of a process, inserting/removing any finite sequence of failed

PopNB/PushNB polls may change only internal ϵ -behavior/timing, not which Σ -visible action (Push/Pop/synchronization anchor) is next in program order.

I.3 Auxiliary Lemmas

Implementation assumption I.A0 (RTL message-issue discipline).

The post-HLS RTL_B execution satisfies the full E3 obligation as an implementation-side scheduling obligation: (i) per-interface issue order is preserved for all message operations; (ii) for distinct interfaces, the no-reverse rule holds; and (iii) issue is prefix-closed under source order, so at every compared cutpoint a source-later issued message operation implies that every required source-earlier message operation has also issued, with Issue timestamps ordered accordingly. This condition is not derived inside Appendix I; it is the RTL/tool-compliance obligation corresponding to the scheduling rules stated in the main text and Appendix G. For hand-written RTL or other non-HLS implementations, I.A0 is the corresponding proof/qualification obligation on that implementation.

Lemma I.1 (Bounded ϵ -chain to Σ -action or stable wait).

Assume the finite-pipeline-depth / finite-internal-stutter premise FPD/B4. From any reachable well-formed cycle-boundary state of P in the Appendix I construction, if P is not already stably waiting on an external/peer enabling condition, then within at most D_P consecutive P-owned epsilon-steps of internal progress, P either:

- (i) commits its next observable Sigma-action, or
- (ii) reaches a stable waiting state in which no further P-local epsilon-step is available until some external, peer, channel, or stimulus condition changes the relevant enabling predicates.

Equivalently, P cannot produce an infinite P-local epsilon-only execution segment from a reachable well-formed state while remaining internally enabled. Any unbounded delay that remains after the stable-wait point is therefore attributable to external/channel/stimulus enabling and is governed by WF/B2 and B3, not by local epsilon-stutter. When a clock-cycle bound is required, the proof uses the separate B5 bound C_{sys} , or a proved bucket-semantic charging lemma deriving a per-process cycle bound C_P from D_P ; Lemma I.1 does not identify one epsilon-step with one clock cycle.

Proof. Direct from FPD/B4 for the epsilon-step conclusion, together with the separately stated B5 or bucket-semantic bridge only when a cycle-count conclusion is used. ■

Lemma I.2 (Eventual space / data). Assume P is blocked on either:

- Push_c with $occ_c = B(c)$, P's Push_c request remains asserted with stable payload for the outstanding transfer, and the complementary endpoint continuously requests the matching Pop_c until the matching discharge event (Pop_c endpoint request asserted; Pop_c boundary request remains true); or
 - Pop_c with $occ_c = 0$, P's Pop_c request remains asserted for the outstanding transfer, and the complementary endpoint continuously requests the matching Push_c until the matching discharge event (Push_c endpoint request asserted with stable payload; Push_c boundary request remains true).
- Then a complementary Pop_c (respectively Push_c) commits within finite time.

Proof. (By B3 / Appendix N.) In either case, P is stalled at a blocking channel call while its persistent request for that outstanding transfer remains asserted until commit, with stable payload if it is a Push. By the additional premise, the complementary endpoint also continuously requests the matching operation until the corresponding discharge event, with its boundary request remaining true. Therefore the interval-scoped premises of P_push/P_pop (Appendix N) hold, and the matching complementary discharge must eventually occur: for Push_c at full, some complementary Pop_c eventually commits, freeing space; for Pop_c at empty, some complementary Push_c eventually commits, providing data. The premise does not require either endpoint request or Push payload to remain asserted after that

matching transfer has committed.

■

Definition I.ReplayObs (replay-derived Σ -relevant source-state slice).

Fix a process P , a fixed external stimulus, and a fixed witness w satisfying WC for the chosen observable boundary. Define $\text{ReplayObs}_{\{\Sigma, P, w\}}(\sigma)$ to be the logical P -local replay checkpoint represented at operational state σ : the tuple of source-level facts determined by the complete P -local observable-unit replay history under that stimulus and witness and needed to determine later P -local Σ -visible behavior or replay-derived Appendix I–K predicates. ReplayObs is a projection of σ , not the complete raw operational source state at σ . Its control-position field is the logical replay control position immediately after the last matched observable unit (or at the selected initial/reset checkpoint), and its value fields contain only replay-relevant logical source variables and latched values reconstructed from committed history, the fixed stimulus, and w . Optional outcome-independent L1 preparation may advance the raw operational cursor, calculate prepared temporaries, or assert source-later requests without changing this logical checkpoint, but only when the field-sensitive footprint, alias-closure, and NoDependentUse obligations certify that the segment is replay-inert.

The tuple includes: the logical replay control position and reset epoch; replay-relevant source variables and latched values used to evaluate future control predicates Pred_x , payload/value expressions, anchored Write values, and anchored Read labels; witness-selected epsilon outcomes that can affect later Σ -visible control flow; $Q_{\text{exec}, P} / Q_{\Omega, P}$ dynamic-operation attribution facts determined by the replay prefix; completed/realized-item status; candidate NextFront_P ; anchored signal values and returned Pop/NB results reconstructed from the committed history and witness; and P 's normal-completion status. Raw preparation progress—including the operational cursor, prepared local temporaries, current issue attribution, and current request assertions—is not part of ReplayObs . In particular, Pending_P , Front_P , and BoundaryReq are not ReplayObs fields. Define $\sigma \sim^{\text{replay}}_{\{\Sigma, P, w\}} \sigma'$ iff $\text{ReplayObs}_{\{\Sigma, P, w\}}(\sigma) = \text{ReplayObs}_{\{\Sigma, P, w\}}(\sigma')$. The replay-derived source-state slice of σ is its equivalence class $[\sigma]^{\text{replay}}_{\{\Sigma, P, w\}}$.

Replay-abstraction and selected-normalization convention. Not every epsilon-step is replay-inert. A witness-fixed epsilon transition, such as a failed non-blocking poll that writes a returned status or another replay-relevant field, may change ReplayObs . WC fixes the selected outcome; it does not make that transition neutral. Deterministic normalized replay therefore uses paired congruence: from equal logical replay checkpoints, corresponding transitions under the same witness produce equal successor checkpoints. A selected finite epsilon-normalization may contain both (a) certified replay-inert preparation segments, which leave each logical checkpoint unchanged, and (b) witness-fixed replay-changing transitions, which must map equal checkpoints to equal successors. The selected normalization convention is proof-relevant and must satisfy this congruence requirement; the document does not assert that every legal normalization has the same raw operational endpoint or that each normalization leaves one execution's ReplayObs unchanged.

Definition I.CutpointOfferObs (current blocking-offer slice).

For a reachable source state σ at the selected abstract boundaries, define $\text{CutpointOfferObs}_{\{P, E\}}(\sigma)$ to be the P -owned current blocking-offer facts: the uncommitted blocking message-operation frontier items whose endpoint requests are asserted, their Pending_P and Front_P membership, BoundaryReq values, owning dynamic message-call instances, evaluated explicit boundaries, endpoint directions, reset epochs, and the stable Push payload obligation where applicable. CutpointOfferObs is indexed by the actual operational issue schedule and raw L1 preparation progress and is not determined by the committed observable-unit history alone: two legal source executions can share the same logical ReplayObs checkpoint while differing in how far outcome-independent preparation has advanced and whether a source-later blocking operation has already issued. Appendix J constructs these request

assertions through its phase-normal preparation/issue actions, represents the normalized cutpoint slice by the residual-offer set O , proves source reachability through $J.OfferReach$, and audits the corresponding RTL request inventory through $J.OfferConf$. The complete K -facing cutpoint observation therefore combines replay history H with current residual offers O ; deterministic replay never manufactures O from H .

This quotient intentionally ignores purely internal diagnostic state and epsilon-only micro-timing state that cannot affect any future Σ -visible label, replay-derived frontier predicate, payload/value expression, or Appendix I–K predicate. $ChannelEnabled$ / $ChannelDisabled$ facts are not independently added to $ReplayObs$. They are obtained by combining the logical $ReplayObs$ checkpoint with the operational $I.CutpointOfferObs$ slice—including $BoundaryReq$, $Pending_P$, and $Front_P$ —and with the corresponding channel-state facts in the chosen Sys_ref boundary model, such as $B(c)$, $occ_c(\sigma)$, rendezvous complementary endpoint requests, and any explicit Sys_B^{elab} staged/bundle/join boundary state. For cadence-sensitive non-blocking polling sites, such as retry/timeout counters whose externally visible behavior depends on the number or cadence of failed polls, this definition is applied only after the site has been handled as an isolated or snoop-aligned latency-sensitive local protocol block under Appendix D / Appendix L. The ordinary Appendix I source-core proof does not derive retry/timeout behavior from an unconstrained hidden failed-poll cadence.

Lemma I.ReplayObs-Congruence (one-step closure of the replay-derived slice).

Fix P , the external stimulus, the chosen observable boundary, and a witness w satisfying WC . Let σ_1 and σ_2 be source states of P such that $ReplayObs_{\Sigma,P,w}(\sigma_1) = ReplayObs_{\Sigma,P,w}(\sigma_2)$. Suppose that, from σ_1 and σ_2 , the same next P -local observable unit u is produced under the same witness w and the same fixed-stimulus interpretation, where u may be a committed Push/Pop value, successful non-blocking transfer, synchronization/wait unit, anchored Read/Write unit, $START_P/finish_P$ unit, R2C fixed-point observation unit, or another explicitly atomic Σ -visible boundary unit. Let $\hat{\sigma}_1$ and $\hat{\sigma}_2$ be the corresponding canonical immediate post-unit states. Then their $ReplayObs$ checkpoints are equal. If σ'_1 and σ'_2 are instead the epsilon-quiescent endpoints selected from $\hat{\sigma}_1$ and $\hat{\sigma}_2$ by the fixed proof-relevant normalization convention, require that convention to preserve replay equivalence as specified above: replay-inert preparation segments leave the logical checkpoint unchanged, while corresponding witness-fixed replay-changing epsilon transitions map equal checkpoints to equal successors. Under that selected convention:

$$ReplayObs_{\Sigma,P,w}(\sigma'_1) = ReplayObs_{\Sigma,P,w}(\sigma'_2).$$

Proof. By case analysis on the next source construct and on the kind of observable unit u .

For deterministic local computation, equality of the logical replay control position and of all replay-relevant variables/latched values feeding future predicates, payloads, anchored writes/reads, guards, and replay-derived proof predicates implies equality of the updated $ReplayObs$ fields. Raw L1 cursor movement, prepared temporaries, and request assertions are outside $ReplayObs$ and may differ only under the certified replay/preparation noninterference obligations.

For anchored sequential signal reads/writes and emitted R2C fixed-point units, the relevant value and component-local observation identity are fixed by the common stimulus, witness, anchor/fixed-point semantics, and common unit u . Therefore all future-relevant latched values, anchored-label facts, and R2C replay facts agree after the unit.

For committed blocking Push/Pop and synchronization units, the common unit u , together with equality of the logical replay control position, relevant payload variables, replay-prefix dynamic-operation attribution, and completed-item history, determines the same replay-derived control/data update and candidate $NextFront_P$. Whether a not-yet-committed blocking operation was already issued before the unit is not a $ReplayObs$ fact and is not concluded equal here; that current assertion state belongs to $CutpointOfferObs$ and is established by explicit operational issue evidence in Appendix J.

For successful non-blocking PushNB/PopNB, the successful transfer is included in u as the corresponding committed Push_c(v) / Pop_c(v) unit. The common unit and equal current ReplayObs determine the same control, returned-data, completed-item, dynamic-attribution, and candidate-next-front updates. For failed non-blocking PushNB/PopNB, the failed poll is an epsilon-step. If NB-CF holds, failed polls cannot affect the next Σ -visible control flow, dynamic-operation selection, guard values, payload values, or candidate NextFront, so any difference is outside ReplayObs. If NB-CF does not hold but WC is supplied by Appendix L, the selected failed-poll outcome may update returned-status, witness, or other replay fields. WC fixes that outcome; it does not make the step replay-neutral. The normalized proof therefore pairs the corresponding failed-poll transitions and uses equal prior checkpoints plus the same witness-selected outcome to obtain equal successor ReplayObs. Cadence-sensitive sites remain subject to the Appendix D / Appendix L isolation or snoop-alignment requirement.

For any other epsilon-only implementation choice, WC requires that every outcome capable of affecting later Σ -visible behavior or replay-derived Appendix I–K predicates is selected by w and is causally consistent with the matched prefix. A certified replay-inert preparation step preserves each logical checkpoint by framing. A witness-fixed replay-changing step need not be neutral, but equal prior ReplayObs checkpoints and the same selected outcome must produce equal successor checkpoints. Current request assertions remain outside this conclusion.

Therefore equality of ReplayObs is preserved by the canonical immediate same-unit replay step and, for normalized cutpoints, by the two-sided congruence of the exact selected normalization paths. This is not a claim that each selected normalization preserves one side’s ReplayObs individually, nor a claim of global normalization confluence. No equality of CutpointOfferObs is asserted. ■

Lemma I.DR (Deterministic replay / replay-derived source-state determinacy under fixed stimulus + witness). Fix a process P , a fixed external stimulus, a fixed witness w satisfying WC, and the proof-relevant epsilon-normalization convention selected for the theorem instance. Consider two executions of the same P source code that start from the same initial source-level state and are run with that same witness. If their chosen P -local observable-unit replay projections are the same, then for every k , $\text{ReplayObs}_{\{\Sigma, P, w\}}$ at the selected epsilon-quiescent cutpoint after the first k P -local observable units is identical in the two executions. The selected normalization convention must satisfy the paired replay-congruence requirement of Definition I.ReplayObs and Lemma I.ReplayObs-Congruence; arbitrary maximal epsilon-extensions are not quantified here. “ P -local observable-unit replay projection” means the sequence obtained from the document’s event-matching / Appendix J observable-unit convention: explicitly atomic same-cycle transactions are grouped as single units; independent same-cycle bucket members are not given an intrinsic order by the bucket itself; and any ordering used for deterministic replay is the selected witness/replay order, together with w and μ_Q for epsilon-modeled failed non-blocking $q \in Q_{\text{exec}, P}$ when those instances are needed by E3/E5. Thus Lemma I.DR is neither a determinacy theorem from a bare unordered Σ -bucket projection nor a theorem that reconstructs current CutpointOfferObs/Pending_P/Front_P/BoundaryReq from committed history.

Proof. At $k = 0$ the executions start from the same initial source-level state, so their logical $\text{ReplayObs}_{\{\Sigma, P, w\}}$ checkpoints agree. For the induction step, assume the checkpoints agree after k P -local observable units. The two executions have the same $(k+1)$ -st P -local observable unit and are run under the same fixed stimulus and witness. Lemma I.ReplayObs-Congruence first gives equal canonical immediate post-unit checkpoints and then applies the paired congruence of the two exact selected normalization paths. Therefore the selected normalized ReplayObs checkpoint agrees after every finite P -local observable-unit prefix. ■

Lemma I.DR’ (Immediate post-unit deterministic replay under fixed stimulus + witness). Fix a process P , a fixed external stimulus, and a fixed witness w satisfying WC for the chosen observable boundary. Consider two executions of the same P source code that start from the same initial source-level state

and are run with that same witness. Suppose that their chosen P-local observable-unit replay projections through the first k units are the same. Stop each execution at the canonical immediate post-unit semantic state: after the transition, or explicitly atomic transition group, that realizes the k th P-local observable unit, and before any optional following L1 preparation or epsilon-normalization. Then the two executions agree on the P-owned logical ReplayObs checkpoint at that point, including:

- the logical replay control position, not the raw post-preparation operational cursor;
- all replay-relevant source variables and latched values that can influence future Pred_x predicates, payload/value expressions, or anchored Read/Write values, excluding prepared temporaries that are certified outside the replay projection;
- the dynamic-operation attribution and completed/realized-item facts determined by the replay prefix;
- the candidate NextFront_P , returned-result, anchored-signal, reset-epoch, and normal-completion facts determined by that prefix.

This lemma does not assert that the immediate post-unit state is epsilon-quiescent or drain-closed, and it does not identify that state with a later epsilon-normalized cutpoint. Lemma I.DR remains the corresponding theorem for endpoints selected by the fixed paired-congruent normalization convention. Neither lemma determines raw L1 preparation progress or the current asserted blocking-request inventory: Pending_P , Front_P , and BoundaryReq belong to $\text{I.CutpointOfferObs}$ and, at the K -facing normalized cutpoint, are represented by O and justified by $J.\text{OfferReach}/J.\text{OfferConf}$.

Proof. By induction on k . The base case follows from equality of the initial source-level state. For the induction step, equality of the prior immediate ReplayObs state, together with the same fixed stimulus, witness-selected ϵ outcomes, dynamic-instance attribution, and next observable unit, determines the same finite source transition segment through the transition or explicitly atomic transition group that realizes the next unit. Therefore the two canonical immediate post-unit states agree on every listed replay-derived fact. The proof makes no claim about issue-schedule-dependent CutpointOfferObs , and no maximal ϵ -closure is invoked. ■

I.4 Local Preparation Construction for Finite Observable-Unit Prefixes (Post $\rightarrow$ Ref support)

Fix a process P , a finite selected RTL_B prefix τ_{post} , and a finite rank-independent required-prefix candidate H as defined by $J.\text{RequiredPrefixCandidate}$ below. Let $\text{Replay}_P(H, w)$ be the deterministic immediate post-unit source replay state obtained from the P-local projection of H under the fixed stimulus, WC witness w , and μ_Q dynamic-instance matching. Let γ be a not-yet-matched observable unit in which P participates, and let $q_P(\gamma)$ denote P 's matched dynamic source operation, if γ has a process-owned operation. This local construction is indexed without reference to Dep_J , $J.\text{DepAcyclic}$, or $J.\text{CanonicalRank}$; Appendix J later instantiates it at the prefixes selected by the validated construction order.

The process-local input to the construction consists only of P 's source code and state, P-owned endpoint request state, the fixed witness and dynamic identities, and read-only summaries of the incident explicit boundaries named by γ . It does not include a premise that γ is globally enabled or that a complete Sys_{ref} extension exists.

Definition I.DeferredNBHole (carried non-blocking decision obligation).

A deferred hole for dynamic call q at boundary c is the proof-construction record $\text{NBHole}(q, c, \kappa, t, \text{dst}, e, \chi)$, where κ is the snapshot key, t is the selected Horizon, dst identifies the returned status/data destinations, e is the reset epoch, and χ records the WC-selected expected outcome together with $\text{NoDependentUse}(q)$. $\text{NoDependentUse}(q)$ is a locally checkable assertion that no action executed while the hole remains unresolved reads $\text{NBResult}(q)$, uses q 's outcome to select reachability/control/endpoint/address/payload, or otherwise depends on the returned value/status.

The dependence test is semantic and transitively closed: it includes aliases and pointer-derived accesses,

loads from result-containing locations, derived dataflow expressions, and control, address, endpoint-selection, and payload dependence. The local certificate or checker shall justify this closure using the normative field-sensitive location/footprint model rather than a purely syntactic scan for $\text{NBResult}(q)$. The record also retains the attributed one-cycle attempt and every location needed to reconstruct $\text{ReqSnapshot_c}(q)$. It is ghost construction state, not KObs state. All deferred holes for Horizon t must be resolved or retired before the exported post-boundary construction state of t .

Lemma I.3 (Finite process-local preparation script with carried shared-decision holes). Assume I.A0, WC, Core.RegSeq for sequential P , the source well-formedness conditions, and equality between P 's current source state and $\text{Replay_P}(H, w)$. Assume the P -local certificate identifies $q_P(y)$, the finite dynamic source-control segment leading toward it, and every locally generated Dep_J predecessor required by that segment. Then there exists a finite preparation script $\text{Prep_P}(H, y)$ consisting of P -owned ϵ /request actions plus zero or more carried I.DeferredNBHole records. Each action has the field-sensitive footprint and Horizon metadata required by J.ConstructionAction, respects E3/E5, synchronization boundaries, same-interface order, issue prefix closure, persistence, stable payload, and commits no unmatched Σ unit. When a witness-significant PushNB/PopNB decision is reached, the script asserts/records the one-cycle attempt, creates $\text{NBHole}(q, c, k, t, \text{dst}, e, \chi)$, and does not locally choose success or failure. The script may nevertheless continue within Horizon t through source-later actions certified by NoDependentUse(q) and by disjoint snapshot/result footprints; this includes a later pipeline iteration when it is outcome-independent. It stops at the first action that depends on q 's result, would change $\text{ReqSnapshot_c}(q)$ without an ordered dependency, crosses a synchronization boundary, or leaves the L1 pre-settle phase. Every outcome-dependent continuation is assigned a Horizon strictly greater than t . A successful deferred decision remains a complete observable unit; a failed deferred decision is an ϵ boundary result. Filling all holes and completing the selected Horizon yields the applicable next-frontier/issued-pending state for y , with no unresolved hole exported at the post-boundary construction state.

Proof. Induct over the finite source-control segment certified for $q_P(y)$, using Core.RegSeq and the typed location footprints. Deterministic local statements preserve the replay slice by Lemma I.ReplayObs-Congruence; anchored and direct-input values are fixed by their E2/stability contracts. Blocking requests and NB attempts may be asserted in the certified L1/source-order prefix and remain attributed with stable payload. At a witness-significant NB call, emit the enriched deferred-hole record instead of forcing a global stop. Continue only through actions whose reachability and operands are independent of the unresolved result and whose writes are disjoint from the hole's snapshot/result locations; record NoDependentUse(q) for each such step. The first dependent use receives a strictly later Horizon by ReturnedAtCycle/Core.RegSeq. Appendix J later reconstructs the common settled snapshot and applies J.NBDecision. Success is a complete unit; failure writes the returned status and retires the attempt without channel commit. FPD/B4 makes each L1 segment and the complete per-Horizon script finite. Field-sensitive framing ensures that carried holes and independent continuation coexist without changing one another's semantics. ■

Corollary I.4 (J.PCert construction). Applying Lemma I.3 to every process participation and every applicable finite J.RequiredPrefixCandidate, and packaging its outputs with the dynamic-operation identities supplied by its P -local certificate hypotheses, yields the process-local records required by Definition J.PCert: deterministic replay state, those exact dynamic-operation identities, finite per-Horizon preparation scripts, issued/pending attribution, stable-payload facts, enriched deferred NB holes with result destinations and NoDependentUse evidence, field-sensitive action footprints, ReturnedAtCycle/Horizon facts, and locally generated dependency facts. This corollary remains local; J.CCert and J.NBDecision must evaluate each shared-state poll from the common settled boundary snapshot, J.RegPhaseNF must normalize the complete Horizon, and J.ACert/J.UnitEnable must establish the complete observable units.

I.5 Countable Local Replay Chains and Scope

For a process whose selected RTL_B projection contains countably many observable units, repeated application of Lemma I.3 and Lemma I.DR' constructs a countable family of process-local replay/preparation scripts. Each finite record is indexed by a finite rank-independent $J.RequiredPrefixCandidate$. $J.GWR-Comp$ later instantiates that family only at the candidates that occur as prefixes of the theorem-derived validated construction order. The construction does not independently evaluate shared-boundary non-blocking decisions, select channel commits, or choose an infinite global scheduler.

If P eventually emits no further observable unit, its local replay record simply stabilizes after its last participation; this does not justify appending a global $B6$ idle suffix. Appendix I constructs only the countable family of finite local replay/preparation records and makes no existence claim for an infinite global Sys_ref execution. If a later theorem requires a complete/fair source execution, that theorem must supply or construct it under its own continuation/progress premises; Appendix K, for example, uses Definition $Core.LAdm$, $Exec_L^{adm}$, and Lemma K.2b.E. $B6$ may supply global idle stutter only when such a theorem explicitly uses an infinite terminal-idle suffix and the complete system satisfies $Core.TerminalIdle$, including both the genuine $B5/global-fixed-point$ condition and its machine/environment no-future-enabling proof.

Thus Appendix I proves local replay determinacy and finite local preparation. Appendix J is solely responsible for gluing the local paths into one reachable global execution, validating shared channel and atomic interaction state, and establishing the $Post \rightarrow Ref$ system witness. ■

Appendix J – System-Level Trace Compatibility Proof

Synchronization-pair well-formedness for $\approx$.

For $SyncChannel/ac_sync$ operations with paired endpoints, the compared Sys_ref and RTL_B executions use the same dynamic synchronization-transaction pairing. That is, the two endpoint events of a matched $sync_out/sync_in$ transaction are matched by dynamic sync-instance identity, commit as one two-party barrier transaction, and are treated as simultaneous events in the same observable cycle bucket. This synchronization transaction induces the same cross-process before/after ordering as described in the causal-dependency definition.

Equivalently, $E1$ preserves the per-process source order of synchronization endpoint events, but $E1$ does not weaken or replace the paired-barrier semantics of $SyncChannel/ac_sync$. The system-level trace-compatibility predicate $\tau_{ref,system} \approx \tau_{post,system}$ is defined only over synchronization-well-formed traces in this sense: if one endpoint of a dynamic $SyncChannel/ac_sync$ transaction appears in the trace, then the matching endpoint appears in the same observable synchronization transaction, with the same dynamic pairing, in both Sys_ref and RTL_B .

Theorem J.1 (Execution-Level / Trace-Set Compositional Compatibility)

Fix an initial state and a fixed external stimulus/environment behavior (e.g., the same testbench-driven inputs) for both the prescribed source reference model Sys_ref and RTL_B , with Sys_ref as defined in Remark 2. The raw rendezvous model Sys is the special case obtained when $Sys_ref = Sys_B$ and all effective capacities are zero.

Clarification (reactive environments). Interpret “fixed stimulus” per Appendix I.2, “Fixed-stimulus interpretation (reactive environments).”

Definition J.Reach (finite reachability sets Reach_post / Reach_ref). Let Reach_post denote the set of all finite execution prefixes of RTL_B reachable under the fixed stimulus, with no fair-extension qualifier; define Reach_ref analogously for Sys_ref . Both finite-prefix directions of Theorem J.1 and the cutpoint correspondence of Corollary J.1 are stated over Reach_post / Reach_ref . Their finite-ideal/selected-pair arguments never use the existence of a fair infinite extension, so a genuinely bad reachable prefix (for example, one ending at a deadlock cutpoint) remains in scope regardless of whether it can be embedded in a fair infinite execution. Appendix J therefore introduces no generic fair-extension-qualified execution domain and no generic reachable-prefix-to-fair-execution theorem. Any later theorem that requires a complete/fair execution states its own execution domain and continuation premises; Appendix K uses $\text{Exec_L}^{\text{adm}}$ and Lemma K.2b.E for the liberal source continuation and K.CompleteS for the selected Policy-S run.

Cycle-bucket and required-prefix convention for this theorem. A system execution may be represented as a sequence of observable cycle buckets $\beta_1 \beta_2 \dots$, where β_i is the finite set of Σ -events committed in one observable clock cycle of that particular execution. Same-cycle events inside a bucket are unordered except for the explicit constraints imposed by E1–E5, the channel semantics, signal-anchor semantics, and explicit synchronization constructs.

The bucket decomposition records the clock timing of one execution; it is not, by itself, part of the cross-model compatibility obligation. The relation $\tau_{\text{ref},\text{system}} \approx \tau_{\text{post},\text{system}}$ does not require Sys_ref and RTL_B to use the same observable cycle-bucket decomposition. Events that are not ordered by the explicit required constraints may be placed in different cycles or grouped into different buckets in the two executions, provided the matched executions preserve all ordering, timestamp, FIFO-legality, and atomicity constraints required by E1–E5 and by the explicitly modeled channel, signal, and synchronization semantics.

The prefix order used by Theorem J.1 is required-prefix order over matched observable units, not RTL_B bucket-prefix order and not the full Appendix G causal-dependency graph $\rightarrow$. An observable unit is either:

- a singleton Σ -event; or
- an explicitly atomic same-cycle transaction that the semantics requires not to be split, including a rendezvous $\text{Push_c}(v)/\text{Pop_c}(v)$ transfer, the two endpoints of a paired $\text{SyncChannel}/\text{ac_sync}$ transaction, an R2C combinational fixed-point signal-observation unit, a proof-internal RESET boundary unit when dynamic reset is used (Normative Formal Core, reset well-formedness RW1–RW5), and any other same-cycle transaction explicitly identified by the signal/channel semantics.

E2 signal-anchor grouping convention: a synchronization commit together with the sequential-process signal Read/Write events that E2/R2 anchor to it (and therefore require to share its commit cycle) may — and, for mechanization, should — be grouped as one explicitly atomic same-cycle unit. This grouping replaces the same-cycle equality constraint between separately enumerated units by unit-internal atomicity and is consistent with the $\text{J.RegPhaseNF}/\text{J.UnitExt}$ treatment of a synchronization commit and its anchored signal actions as one same-cycle observable unit. When anchored signal events are instead enumerated as singleton units, their E2 timestamp equality with the anchor is enforced by the Sys_ref realization semantics (R2) and the within-cycle listing convention, not by the strict order $<_J$, which never orders same-cycle-aligned events.

Let $<_J$ be the strict required-order relation over observable units generated only by the explicit constraints that must be preserved by $\tau_{\text{ref},\text{system}} \approx \tau_{\text{post},\text{system}}$:

- E1 synchronization source-order constraints within each process;
- E2 signal-anchor timestamp constraints, interpreted as required alignment with the relevant

- synchronization or R2C fixed-point unit rather than as raw ordering among all source actions;
- E4 channel/FIFO legality for explicit abstract channels, including same-channel FIFO order for $B(c)>0$ and atomic rendezvous grouping for $B(c)=0$;
 - E5 immediate synchronization-boundary constraints;
 - paired SyncChannel/ac_sync barrier semantics; and
 - any other explicitly stated signal/channel/synchronization atomicity constraint.

Per-process source order contributes to $\prec_J$ only where it is reflected in one of these explicit required constraints, such as synchronization order, same-endpoint/same-channel order, signal anchoring, or immediate synchronization-boundary legality. E3 contributes issue-order and request-attribution obligations, but it does not by itself create a strict predecessor edge between distinct-interface committed observable units merely because their source calls are lexically ordered. The Appendix G causal-dependency relation $\rightarrow$ is not a generator of $\prec_J$.

A theorem prefix is a finite ideal of observable units under $\prec_J$, i.e. a finite down-closed set H such that $\gamma \in H$ and $\gamma' \prec_J \gamma$ imply $\gamma' \in H$. The proof below proceeds by induction along the finite ideal chain induced by the selected J .ValidatedConstructionOrder. On the normal compositional route, that order is derived from Definition J.DepJ, Lemma J.DepRank, Lemma J.DepAcyclic, and J.CanonicalRank. On the optional direct route, it is supplied with an independently checked well-founded construction order satisfying the same required-predecessor and ideal-prefix obligations. Acyclicity of $\prec_J$ on the fixed execution prefix. For the chosen legal RTL_B execution prefix, assign each observable unit $\gamma \in U(\tau_{\text{post}})$ its RTL_B commit cycle $\text{clk}(\gamma)$. If γ is an explicitly atomic same-cycle unit, such as a rendezvous Push_c(v)/Pop_c(v) transfer or a paired SyncChannel/ac_sync transaction, $\text{clk}(\gamma)$ is the common commit cycle of the endpoints in that atomic unit.

Every strict generator edge $\gamma_1 \prec_J \gamma_2$ implies a strict cycle inequality $\text{clk}(\gamma_1) < \text{clk}(\gamma_2)$ in that same legal execution prefix:

- E1 synchronization source-order constraints impose strict ordering between the relevant synchronization commits.
- E4 buffered-channel FIFO/order/capacity constraints impose strict commit-cycle ordering through reset-empty FIFO order, bounded occupancy, and non-fall-through cycle-boundary availability. In particular, for $B(c)>0$, a Pop_c can consume only a value already present at the relevant cycle boundary, and a Push_c that depends on space becoming available after a full condition is enabled only in a later cycle after the required Pop_c effect has occurred.
- E5 immediate synchronization-boundary constraints impose strict separation between a synchronization commit and later suff_S work whose issue/commit is after that synchronization boundary.
- Paired synchronization/barrier before/after constraints impose the corresponding strict before/after cycle relation between distinct observable units when such a relation is explicitly part of the synchronization semantics.
- E2 signal-anchor constraints contribute timestamp equality/alignment to the relevant synchronization or R2C fixed-point unit, or same-cycle atomicity when explicitly specified; they do not by themselves introduce an independent strict edge between distinct observable units.
- Rendezvous Push/Pop endpoints, paired same-cycle synchronization endpoints, and other explicitly atomic same-cycle transactions are represented as single observable units, not as strict internal edges.
- Non-strict issue simultaneity permitted by E3 does not create a strict $\prec_J$ edge.

Therefore any cycle $\gamma_0 <_J \gamma_1 <_J \dots <_J \gamma_0$ over $U(\tau_{\text{post}})$ would imply $\text{clk}(\gamma_0) < \text{clk}(\gamma_1) < \dots < \text{clk}(\gamma_0)$, an impossibility. Hence $<_J$ is acyclic on $U(\tau_{\text{post}})$, which is the only acyclicity needed for the finite-ideal induction. For the proof of Theorem J.1, Definition J.DepJ and Lemmas J.DepSound–J.DepAcyclic derive J.CanonicalRank from the chosen RTL_B witness’s concrete issue/decision/commit annotations and the local operational footprints. The induced observable-unit enumeration refines commit-cycle order and $<_J$. Its prefixes are ideals under $<_J$, but stable tie ordering between commuting actions has no semantic significance and must not be read as adding an observable order among independent events. There is no semantic cutpoint after only one endpoint of an explicitly atomic multi-endpoint unit has been taken; such units are split neither in Sys_ref nor in RTL_B.

Scope note (intentional). The finite-prefix Post $\rightarrow$ Ref clause of Theorem J.1 and the cutpoint correspondence of Corollary J.1 are stated over Reach_post / Reach_ref and do not require an infinite fair extension. Their finite-ideal construction uses the process-local preparation records and the local channel/atomic certificates of Definition J.LocalGWR, combined by Lemma J.UnitExt and Theorem J.GWR-Comp with the Sys_ref operational transition semantics. On the normal compositional route, J.RegPhaseNF/J.UnitExt also require a finite source L2 settle for every constructed Horizon; in the standard scope that same-cycle existence obligation is supplied by QualifiedSynchronousModel_Sys_ref(I) and Theorem Core.QSync-Ref. A directly supplied J.GWR package may instead carry independently checked finite L2-settled segments for its Horizons and need not separately assert global source-side Core.QSync solely for Theorem J.1. Weak fairness B2, channel progress B3, and the time-divergence / idle-stutter convention B6 are not premises of either finite-prefix direction of Theorem J.1. Any finite L1 issue or L3 boundary-action deferral used by the Post $\rightarrow$ Ref construction is ordinary Core.SelectLatitude source legality, not an invocation of Core.IssueFair or B2/B3. They remain available only to later theorems that explicitly construct or validate complete/fair runs, such as Core.LAdm/K.2b.E and K.CompleteS as applicable. B4/B5 remain separate normalization premises for replay/quiescence uses not discharged by the same-cycle Core.QSync theorem.

Assume the Appendix I / Appendix L local replay machinery is available—namely I.A0, WC for ε -modeled choices, deterministic replay, and the finite process-local preparation theorem—with NB-CF as a sufficient WC route where applicable. Also assume the capacities/occupancies/enablement semantics of Sys_ref, finite capacities and B-faithful E4 behavior on every explicit abstract channel, the complete local certificate package of Definition J.LocalGWR, and the standing RTL_B determinism assumption B1 under the fixed stimulus. For the normal compositional Post $\rightarrow$ Ref route, additionally assume QualifiedSynchronousModel_Sys_ref(I); Theorem Core.QSync-Ref then supplies the finite L2 SettlePoint required by J.RegPhaseNF and J.UnitExt for every reachable constructed Horizon. The direct semantic route may instead supply a J.GWR package whose checked reachable phase-normal chain already contains the required finite L2-settled segment for each Horizon; such a package need not separately discharge global source-side Core.QSync merely to apply finite-prefix Theorem J.1. Core.QSync-Post is not a premise of plain finite-prefix J.1 unless a post-side KCut/NormPost interface is additionally invoked. No B2/B3 premise is used by the finite-prefix clause. Use B4/B5 additionally when replay ε -normalization or uniqueness up to finite ε -stutter outside the QSync same-cycle settle is invoked. When a later theorem explicitly asserts a complete/fair execution, separately require that theorem’s applicable B2/B3 and continuation/closure premises. Additionally assume the shared-memory lowering condition J.Mem for every cross-process shared-memory dependency in the compared design, and, when dynamic reset is used, the reset well-formedness conditions RW1–RW5 (all in Normative Formal Core). Then:

Compatibility-relation typing convention. The symbol $\approx$ is an ordered cross-model compatibility predicate, not a mathematical equivalence relation. It is intentionally directed from a Sys_ref

execution/projection to an RTL_B execution/projection, and no symmetry or transitivity property is implied. Its carrier is not a bare Σ -label trace alone.

For each process P , the per-process carrier is an annotated observable projection record. Such a record contains the P -local bucketed Σ projection, the relevant observable-unit projection, the dynamic source/witness universe Ω_P and Q_exec, P used for that comparison, the Issue/Commit annotation for dynamic message-call instances, the selected witness w , and the μ_Q matching supplied by WC / Appendix L when ε -modeled failed non-blocking calls or other witness-selected choices are relevant. Forgetting these annotations yields the ordinary P -local observable Σ projection, but E3, E5, BoundaryReq attribution, and the witness construction may depend on the annotations and are not reconstructed from the Σ projection alone.

Accordingly, write

$$\approx_P \subseteq \text{AnnTraces}_{\{\Sigma, P, w\}}(\text{Sys_ref}) \times \text{AnnTraces}_{\{\Sigma, P, w\}}(\text{RTL_B})$$

for the per-process annotated observable-compatibility predicate induced by the P -local R/E obligations: synchronization ordering, signal anchoring or R2C fixed-point agreement, message issue-order constraints, synchronization-boundary issue constraints, and the P -local witness/request-attribution facts needed for those obligations. Here $\text{AnnTraces}_{\{\Sigma, P, w\}}(\cdot)$ denotes the annotated projection records just described, not merely $\text{Traces}_{\{\Sigma, P\}}(\cdot)$.

Similarly, write

$$\approx_{\text{sys}} \subseteq \text{AnnTraces}_{\{\Sigma, w\}}(\text{Sys_ref}) \times \text{AnnTraces}_{\{\Sigma, w\}}(\text{RTL_B})$$

for system-level annotated observable compatibility. A system-level annotated record includes the per-process annotated projections plus the explicit abstract-channel histories and boundary-state facts needed to check E4 at the chosen Sys_ref / RTL_B boundaries.

For system executions $\tau_{\text{ref}, \text{system}}$ and $\tau_{\text{post}, \text{system}}$, the assertion $\tau_{\text{ref}, \text{system}} \approx_{\text{sys}} \tau_{\text{post}, \text{system}}$ is therefore an abbreviation: there exist compatible annotation/witness records for the two executions such that every process projection satisfies the corresponding $\tau_{\text{ref}, P} \approx_P \tau_{\text{post}, P}$ obligation and every explicit abstract channel in the chosen Sys_ref / RTL_B comparison satisfies the applicable E4 channel semantics at the agreed abstract boundary. When the annotations are clear from context, the same $\tau_{\text{ref}} \approx \tau_{\text{post}}$ notation may be used for readability; it should not be read as typing $\approx$ over bare Σ -traces alone.

Definition J.RTLConsistentRefPrefix (annotated RTL-consistent source prefix). A source-side prefix $\tau_{\text{ref}} \in \text{Reach_ref}$ is RTL-consistent for the $\text{Ref} \rightarrow \text{Post}$ clause only when it is equipped with a selected RTL_B prefix $\tau_{\text{post}}^* \in \text{Reach_post}$ and a witness/annotation package W such that: (1) τ_{post}^* is admitted by RTL_B under the same fixed stimulus; (2) the complete selected observable-unit projection of τ_{ref} matches the complete selected observable-unit projection of τ_{post}^* under the canonical occurrence matching of Definition Core.17; in particular, neither selected projection may contain an additional unmatched observable unit beyond the other; (3) W provides compatible annotated projection records for τ_{ref} and τ_{post}^* , including the dynamic operation universes Ω_P and Q_exec, P , μ_Q matching, the operational Issue/Commit and E3/E5 annotations needed by the execution-level construction, request-attribution facts (including failed non-blocking poll witnesses where relevant), and the applicable per-channel E4 histories and abstract boundary facts; and (4) whenever Corollary J.1, Appendix K, or any cutpoint transfer uses the selected pair, write the source prefix used by that cutpoint package as $\tau_{\text{ref}}^G := \tau_{\text{ref}}$, record v_{ref} and σ_{ref} with $J.\text{OfferReach_I, E, w}(H, O; \tau_{\text{ref}}^G, v_{\text{ref}}, \sigma_{\text{ref}})$, and write $\tau_{\text{ref}}^K := \tau_{\text{ref}}^G \cdot v_{\text{ref}}$ for the exact source prefix ending at σ_{ref} ; the selected RTL cutpoint satisfies $J.\text{OfferConf}$, and the composite $J.\text{KObsConf}$ relation holds for that same exact H/O package. Label-projection matching alone is therefore not sufficient to establish $\text{Ref} \rightarrow \text{Post}$; the compatible annotations and, when needed, the KObs cutpoint package are part of the $\text{RTLConsistentRefPrefix}$ witness.

Then:

(Post $\rightarrow$ Ref existence, compositional route) For every $\tau_{\text{post}} \in \text{Reach_post}$ together with a local certificate package L satisfying $\text{LocalGWRPackage}(\tau_{\text{post}}, L)$ (Definition J.LocalGWR), there exist a package W and a prefix $\tau_{\text{ref}} \in \text{Reach_ref}$ such that $\text{GlobalWitnessRealizable}(\tau_{\text{post}}, W)$ holds and the annotated system projections satisfy $\tau_{\text{ref}}, \text{system} \approx_{\text{sys}} \tau_{\text{post}}, \text{system}$ under E1–E5.

(Post $\rightarrow$ Ref existence, direct semantic route) The same core conclusion holds when the application directly supplies a validated package W satisfying $\text{GlobalWitnessRealizable}(\tau_{\text{post}}, W)$. The compositional route is the normal theorem path; the direct route permits translation validation, symbolic replay, or another independently checked global certificate to target the stable J.GWR interface.

(Ref $\rightarrow$ Post selected-pair compatibility, annotated RTL-consistent prefixes only) For every $\tau_{\text{ref}} \in \text{Reach_ref}$ for which there exist $\tau_{\text{post}}^* \in \text{Reach_post}$ and a witness/annotation package W satisfying $\text{RTLConsistentRefPrefix}(\tau_{\text{ref}}, \tau_{\text{post}}^*, W)$ under the same fixed stimulus, Definition J.RTLConsistentRefPrefix supplies the already-selected annotated source/RTL pair and its compatible per-process projection records. The theorem-wide finite-capacity and B-faithful abstract-boundary assumptions, together with the per-channel E4 histories and abstract-boundary facts carried by W , supply the corresponding per-channel well-formedness premises of Proposition J.0. Proposition J.0 therefore applies to this selected pair. Hence there exists $\tau_{\text{post}} \in \text{Reach_post}$ with the required compatibility, witnessed by $\tau_{\text{post}} := \tau_{\text{post}}^*$, and the annotated system projections satisfy $\tau_{\text{ref}}, \text{system} \approx_{\text{sys}} \tau_{\text{post}}, \text{system}$. In Appendix L snooping/witness-selection applications, the selected τ_{post}^* and W are the RTL_B reachable prefix and witness package selected by the snooping wrapper/witness.

Moreover, because B1 is a standing assumption in this theorem instance (deterministic Σ -projected RTL_B trace under fixed stimulus), the matching observable Σ projection of a selected RTL_B prefix is unique (up to insertion/removal of finite ε -steps permitted by B4/B5). This uniqueness is only a uniqueness statement about the observable projection; it does not by itself determine μ_Q , failed-poll instances in Q_{exec} , issue annotations, request-attribution facts, the selected package W , or the full stateful Sys_ref witness. The corresponding execution prefix witness τ_{ref} need not be unique as a stateful prefix, since Sys_ref may admit multiple ε -different realizations with the same Σ projection. Sys_ref may also admit executions whose Σ projection does not match RTL_B's Σ trace when latency-sensitive / non-blocking effects are Σ -visible; such executions are outside the scope of (Ref $\rightarrow$ Post) unless constrained by RTLConsistentRefPrefix / the snooping/witness-selection technique (Appendix L).

Definition J.OuterCanonHist (outer canonical history projection).

Fix a theorem instance I , elaboration E , and selected DUT-boundary observation alphabet Σ_{DUT} . For any finite reachable source or RTL prefix $\tau \in \text{Reach_ref}(I) \cup \text{Reach_post}(I)$, let $\text{Units}_I(\tau)$ be the complete selected observable/proof units already carried by that prefix under the Core.17 canonical occurrence convention and the theorem-instance observation/elaboration boundary. Define $\Pi_{\text{DUT}, E}$ unitwise by composing the applicable elaboration projection $\Pi_{\text{elab}} / \pi_{\Sigma, E}$ with restriction to Σ_{DUT} . Each explicitly atomic unit or contracted atomic group is either mapped as a whole to its declared complete outer unit/group or erased as a whole; $\Pi_{\text{DUT}, E}$ may not expose only one endpoint, participant, signal member, transfer leg, or other proper fragment. Internal staged/bundle/join/epoch-gate units with no selected outer image are erased. A proof-internal RESET(S_p, S_c), START_P, FINISH_P, or other proof-only unit is likewise erased when Σ_{DUT} does not select it; if RESET is selected at the outer boundary, its scope is preserved exactly. Let $\prec_{\text{req}}$ be the strict required-predecessor relation on $\text{Units}_I(\tau)$. On

ordinary selected observable units, the generators of $\prec_{\text{req}}$ are exactly the generators of $\prec_J$ stated for Theorem J.1/Core.17/E1–E5. If $\text{Units}_I(\tau)$ also contains a selected proof-internal unit that lies outside the ordinary $\prec_J$ carrier, $\prec_{\text{req}}$ additionally contains only that unit's explicitly stated normative predecessor/successor edges, such as the applicable RESET/reset-epoch constraints, together with their transitive consequences when composed with ordinary $\prec_J$ edges. Full per-process replay order or lexical source order retained by Core.18a is replay metadata and is not by itself a $\prec_{\text{req}}$ generator; in particular, E3 does not add a strict predecessor edge between distinct-interface committed units merely because their source calls are ordered. Thus the restriction of $\prec_{\text{req}}$ to the ordinary J observable-unit carrier is exactly $\prec_J$, not a stronger source-order relation. Define the induced outer order $\prec_{\Sigma_DUT}$ on distinct retained outer units a, b by $a \prec_{\Sigma_DUT} b$ whenever representatives u, v with $\Pi_DUT, E(u)=a$ and $\Pi_DUT, E(v)=b$ are connected by a nonempty $\prec_{\text{req}}$ path $u \prec_{\text{req}}^+ v$; erased intermediate units are therefore contracted rather than deleting a dependency. $\text{CanonHist}_{\Sigma_DUT}(\tau)$ is the finite quotient consisting of the retained Π_DUT, E images of $\text{Units}_I(\tau)$, their canonical labels/values/occurrence identities, and this induced strict order, modulo arbitrary ordering and raw clock-bucket placement among units unordered by $\prec_{\Sigma_DUT}$. An incomplete atomic unit contributes nothing until complete. This definition is intrinsic to τ and does not require a matching J.GWR, J.TraceCertCov, or J.HCanon package. When an enriched $\text{CanonHist}_{E,w}(\tau)$ record is available, its retained complete observable/proof units, the Core.18a required-prefix constraints used by $\prec_J$, and the retained proof-unit/reset-epoch fields that determine any explicitly selected proof-unit edges reconstruct exactly this $\prec_{\text{req}}$ relation and therefore project to exactly this outer quotient; Lemma J.OuterCanon-HCanon records the resulting cross-model consequence. This is the authoritative meaning of $\text{CanonHist}_{\Sigma_DUT}$ used by J.RefProjCover, J.IdealPrefixClosed, J.1-Safe, and J.2b.

Lemma J.OuterCanon-HCanon (enriched canonical agreement descends to the outer history).

For fixed I, E, w, Σ_DUT , whenever Appendix J supplies enriched histories for annotated prefixes τ_1 and τ_2 and $\text{CanonHist}_{E,w}(\tau_1) = \text{CanonHist}_{E,w}(\tau_2)$ in the J.HCanon/Core.18a quotient, the intrinsic outer histories of Definition J.OuterCanonHist satisfy $\text{CanonHist}_{\Sigma_DUT}(\tau_1) = \text{CanonHist}_{\Sigma_DUT}(\tau_2)$. Proof. The enriched quotient contains the same complete selected observable/proof-unit identities, atomic grouping, reset scopes, labels, the required-prefix constraints used by $\prec_J$, and the retained proof-unit/reset-epoch structure needed to reconstruct any explicitly selected proof-internal predecessor edges. Core.18a also retains per-process replay order, but that replay-order field is not automatically a $\prec_{\text{req}}$ generator unless the corresponding relation is independently one of the explicit required-predecessor constraints. Therefore both enriched records reconstruct the same $\prec_{\text{req}}$ relation of Definition J.OuterCanonHist. Applying the deterministic Π_DUT, E whole-unit projection and contracting erased intermediate required-order vertices produces the same retained outer units and induced $\prec_{\Sigma_DUT}$ relation on both sides. $\square$ A predicate Φ stated extensionally on the quotient-valued $\text{CanonHist}_{\Sigma_DUT}$ is consequently J.HCanon-invariant automatically; a checker that represents this quotient by a chosen linearization or other noncanonical encoding shall prove that its evaluation of Φ depends only on the quotient defined here.

Lemma J.RequiredPredCycleOrder (ordinary required predecessors precede completion in both execution models).

Let $M \in \{\text{Sys_ref}, \text{RTL_B}\}$, let τ be a finite legal execution prefix of M , and let u, v be distinct complete ordinary observable units of τ with a strict generator edge $u \prec_J v$. Then $\text{clk}_M(u) < \text{clk}_M(v)$. Proof. For RTL_B this is exactly the generator-by-generator strict-cycle argument in the acyclicity part of Theorem J.1. For Sys_ref the same cases follow from the normative phase/cycle semantics rather than from an imported RTL argument: E1/synchronization and E5 interval-boundary edges require the predecessor commit before the successor interval/action; E4 buffered FIFO/non-fall-through and scalar endpoint-

cardinality rules force a required distinct transfer predecessor into an earlier cycle; paired barrier/synchronization before/after constraints give the same strict separation; E2 alignment and rendezvous/paired same-cycle semantics are represented as timestamp alignment or one atomic unit rather than a strict edge between distinct units; and E3 lexical/cross-interface order does not create such an edge. Any additional ordinary strict generator admitted by the theorem instance must carry its separately stated strict-before completion rule; a same-cycle alignment/atomicity relation is not silently converted into a strict edge. Core.Phase and Core.CycleResult provide the source commit-cycle indices and prevent a later required successor from completing before its strict predecessor. Therefore the strict inequality holds on both legal carriers. $\square$

Lemma J.RequiredPredPrefix (legal execution prefixes are closed under required predecessors).

Let $\tau \leq \tau'$ be finite prefixes on the same legal source or RTL execution carrier, and let u and v be complete selected observable/proof units of τ' with v already present in τ . If $u \prec_{\text{req}}^+ v$ under Definition J.OuterCanonHist, then u is already present in τ . Proof. It is enough to check one generator edge and compose transitively. For an ordinary $\prec_J$ generator on either RTL_B or Sys_ref, Lemma J.RequiredPredCycleOrder gives $\text{clk}(u) < \text{clk}(v)$; hence an execution prefix that already contains the complete successor v also contains u . For a selected proof-internal RESET edge outside the ordinary $\prec_J$ carrier, RW1–RW5 give the epoch-boundary order directly: affected-scope ending-epoch units precede RESET, RESET precedes new-epoch units, RW4 prevents an affected-scope boundary interleaving, and RW5 does not introduce an order on framed unaffected scope. Any other proof-internal generator is handled by its separately stated normative predecessor/successor rule. Chaining these one-edge facts along the nonempty $\prec_{\text{req}}$ path yields the result. $\square$

Lemma J.OuterCanon-Ideal (outer projection preserves required-prefix ideals).

Let $\tau \leq \tau'$ be reachable finite prefixes on the same legal source or RTL execution carrier. Then $\text{CanonHist}_{\Sigma_DUT}(\tau)$ is a finite ideal of $\text{CanonHist}_{\Sigma_DUT}(\tau')$ under $\prec_{\Sigma_DUT}$. More generally, if H_1 is a finite required-prefix ideal of a larger complete-unit history H_2 under the same theorem-instance required order, then their Π_DUT, E outer projections have the same ideal relation. Proof. Take a retained outer unit b already present in the smaller prefix/projection and any retained outer predecessor a of b in the larger outer history. By Definition J.OuterCanonHist, $a \prec_{\Sigma_DUT} b$ is witnessed by a nonempty $\prec_{\text{req}}$ path between retained representatives, possibly through units erased by Π_DUT, E . Definition J.OuterCanonHist builds $\prec_{\text{req}}$ only from the theorem's actual required-predecessor generators—ordinary $\prec_J$ plus any explicitly normative selected proof-unit edges—and not from full replay/lexical source order. In the execution-prefix case, Lemma J.RequiredPredPrefix implies that presence of the representative of b in τ entails presence of every required predecessor on that path, including the retained representative of a . In the abstract H_1/H_2 case, the same fact is exactly the assumed required-prefix ideal property. Hence a is already in the smaller outer history. Whole-unit projection prevents a partial atomic predecessor from being manufactured by erasure. $\square$ If the extension from τ to τ' adds no retained outer unit—only ϵ /non-unit activity and/or complete units erased by Π_DUT, E —the two outer histories are equal. When enriched J.HCanon histories are available, this intrinsic prefix/ideal fact is equivalently the projection-of-enriched-ideals rule.

Definition J.RefProjCover (selected-reference-run observable coverage)

Fix a theorem instance I , a DUT-boundary observable projection Σ_DUT , and a selected complete source-reference execution ρ_ref^0 . Define $\text{RefProjCover}_{I, \Sigma_DUT}(\rho_ref^0)$ to hold iff, for every $\tau_ref \in \text{Reach_ref}$ of I , the finite outer canonical history $\text{CanonHist}_{\Sigma_DUT}(\tau_ref)$ of Definition J.OuterCanonHist is a finite ideal (a down-closed canonical prefix) of $\text{CanonHist}_{\Sigma_DUT}(\rho_ref^0)$ under the induced outer required-order relation $\prec_{\Sigma_DUT}$. Lemma J.OuterCanon-HCanon makes this statement independent of

the particular enriched $J.HCanon$ representative, and Lemma $J.OuterCanon-Ideal$ supplies the projection-of-ideals rule. The definition covers observable ideals/down-closed prefixes; it does not require identical cycle buckets or a fixed linear order among independent observable units.

Definition J.IdealPrefixClosed (ideal-prefix closure for safety predicates)

Fix theorem instance I and outer observation projection Σ_DUT . For a predicate Φ on the quotient-valued finite canonical Σ_DUT histories of Definition $J.OuterCanonHist$, $J.IdealPrefixClosed_I, \Sigma_DUT(\Phi)$ holds iff, for all finite outer canonical histories H and H' in the selected theorem instance, whenever H' is a finite ideal (down-closed canonical prefix) of H under $<_ \Sigma_DUT$, $\Phi(H)$ implies $\Phi(H')$. In the safety results below, ‘prefix-closed’ means this ideal-prefix notion, not merely closure under deleting a suffix from one chosen linearization of otherwise independent events. By Lemma $J.OuterCanon-HCanon$, any extensional predicate on this quotient is automatically insensitive to the enriched $J.HCanon$ representative.

Practical scope. Typical qualifying properties include assertion-style “no bad observable event has occurred” invariants, per-channel FIFO/order/value-legality invariants, and protocol-legality predicates explicitly formulated to be downward closed over the retained outer required-order history. A cross-interface transaction checker need not satisfy $J.IdealPrefixClosed$ when it requires an explanatory event that the induced outer relation $<_ \Sigma_DUT$ permits an ideal to omit—for example, an output on one channel checked against an otherwise independent input on another. Conversely, source/replay order that is not an explicit $<_req$ generator does not by itself force that explanatory event into every outer ideal. Conversely, projecting away a proof-internal RESET does not erase the applicable epoch boundary: contraction through the erased RESET preserves the required predecessor relation between retained affected-scope units on opposite sides of that reset. RW5-framed unaffected units and independent disjoint reset scopes are not ordered merely because such a RESET occurred. Such a property is not transferred to arbitrary intra-cycle ideals by $J.CycleIdealCover$ alone; use Definition $J.TraceCertCov$ on the relevant arbitrary-prefix domain, or refine the selected observation/order so that the needed dependency is retained.

Definition J.TraceCertCov (Post $\rightarrow$ Ref certificate coverage)

Fix a theorem instance I and a coverage domain $D \subseteq Reach_post$. $TraceCertCov_I(D)$ holds iff every $\tau_post \in D$ is equipped with one of the following route-complete Post $\rightarrow$ Ref packages: (a) a $J.LocalGWR$ package accepted by $J.GWR-Comp$, together with $QualifiedSynchronousModel_Sys_ref(I)$ so that $Core.QSync-Ref$ supplies the finite L2 settle required by $J.RegPhaseNF/J.UnitExt$, and the remaining applicable Post $\rightarrow$ Ref premises of Theorem $J.1$; or (b) a directly validated $J.GWR$ package whose checked reachable phase-normal chain supplies the required finite L2-settled segment for every represented Horizon, together with the remaining applicable Post $\rightarrow$ Ref premises of Theorem $J.1$. All evidence shall be bound to the same theorem instance, stimulus, observation boundary, and selected RTL_B prefix. The definition names only safety-side certificate/package coverage; it does not strengthen Theorem $J.1$ and does not imply $K.CertCov$.

Corollary J.1-Safe (source-side safety transfer through Post $\rightarrow$ Ref)

Assume $J.TraceCertCov_I(D)$ for a selected coverage domain $D \subseteq Reach_post$ containing every reachable RTL_B prefix relevant to the claimed property. $J.TraceCertCov$ includes the source-settling evidence appropriate to the selected Post $\rightarrow$ Ref construction route: source-side $Core.QSync$ for the normal compositional route, or equivalent checked per-Horizon settling evidence for a direct $J.GWR$ route. Let Φ be an extensional predicate on the quotient-valued finite canonical Σ_DUT histories of Definition $J.OuterCanonHist$ satisfying $J.IdealPrefixClosed_I, \Sigma_DUT(\Phi)$. Lemma $J.OuterCanon-HCanon$ supplies the former ‘ $J.HCanon$ -invariance’ condition automatically. Then:

- (1) If every $\tau_{\text{ref}} \in \text{Reach_ref}$ satisfies Φ , then every $\tau_{\text{post}} \in D$ satisfies Φ .
- (2) If one selected complete pre-HLS execution ρ_{ref}^0 satisfies Φ at every finite canonical ideal (down-closed prefix) and $\text{RefProjCover_I}, \Sigma_{\text{DUT}}(\rho_{\text{ref}}^0)$ holds, then every $\tau_{\text{post}} \in D$ satisfies Φ .

Proof. For any $\tau_{\text{post}} \in D$, J.TraceCertCov supplies the applicable $\text{Post} \rightarrow \text{Ref}$ package, and Theorem J.1 constructs a reachable matching Sys_ref prefix with equal enriched canonical history under J.HCanon . Lemma $\text{J.OuterCanon-HCanon}$ therefore gives the same $\text{CanonHist_}\Sigma_{\text{DUT}}$ observation on the source and RTL prefixes. Clause (1) follows from the universal Sys_ref premise. For clause (2), J.RefProjCover places that matching outer observation among the finite canonical ideals/down-closed prefixes covered by ρ_{ref}^0 , and $\text{J.IdealPrefixClosed}$ supplies the RTL_B conclusion. $\square$

Common-case discharge note. For blocking deterministic dataflow, or for designs in which NB-CF makes failed non-blocking polls irrelevant to later Σ_{DUT} behavior, J.RefProjCover may be discharged from fixed stimulus, deterministic/value-consistent process behavior, deterministic or witness-fixed arbitration and environment choices, and completeness of the selected source execution. Lemma K.2b.P1 supplies the payload and dynamic-operation-identity determinacy ingredient in its blocking-only scope, but it is not by itself the full observable-prefix coverage theorem. Static extraction of B(c) and any elaboration package define Sys_ref ; B-faithfulness and J.TraceCertCov remain separate obligations.

Scope. Corollary J.1-Safe transfers safety predicates only. It does not transfer raw cycle latency or performance properties, prove eventual RTL realization of every source event, or establish unrestricted equality of the two behavior sets.

Proof strategy. Theorem J.1 is not obtained from pairwise composition alone. Appendix I proves the local replay/preparation scripts with carried deferred NB holes, packaged as J.PCert . Definition J.LocalGWR adds per-channel, atomic-unit, environment/reset, field-sensitive footprint, request-snapshot, generated-dependency, and interface-agreement certificates. Lemmas $\text{J.DepSound/J.HorizonOrderSafe/J.DepComplete/J.DepRank/J.DepAcyclic}$ derive a canonical action order rather than accepting one as a premise. Lemma J.RegPhaseNF then normalizes each Horizon into the single $\text{Core.Phase L1/L2/L3}$ phase vocabulary, preserving the J.HCanon history quotient. Horizon-batched J.UnitExt and Theorem J.GWR-Comp construct one coherent source execution and the theorem-derived $\text{J.ValidatedConstructionOrder/J.GWR}$ package. J.GWR-Direct instead supplies an equivalent directly validated order, phase-normal chain, and cutpoint invariants. Proposition J.0 remains only the conditional lifting step for an already selected compatible trace pair.

Definition J.ConstructionAction and operational footprints

Fix a finite RTL_B prefix τ_{post} and a candidate local package L . $\text{Act_L}(\tau_{\text{post}})$ is the finite set of primitive source-construction actions named by the process scripts, channel records, deferred-hole records, and complete observable units. An action is one of: (A1) a process-owned ϵ evaluation/control transition, including outcome-independent continuation past a carried NB hole; (A2) assertion or retention of an endpoint-owned blocking request, or assertion of a one-cycle attributed NB attempt; (A3) a shared-boundary NB decision/return that reads the settled ReqSnapshot , writes $\text{NBResult}(q)$, retires $\text{NBPending}(q)$ /the attempt, and on success is contracted with the corresponding transfer commit; (A4) any other complete explicit-channel or observable atomic-unit commit; or (A5) an environment, reset, or explicit $\text{Sys_B}^{\text{elab}}$ transition. A3 does not write the ordinary control/operand locations used by certified same-Horizon outcome-independent actions; dependent continuation is an A1 action at a strictly later Horizon. Every action has a typed identity, owner set, reset epoch, Horizon, associated unit, semantic stage, local ordinal, and finite ReadSet/WriteSet over a normative location vocabulary. That vocabulary distinguishes at least $\text{Ctrl}(P)$, $\text{Local}(P,x)/\text{Operand}(P,e)$, $\text{NBResult}(q)$, $\text{NBPending}(q)$, endpoint request and Push-payload locations, each explicit channel/elaboration location, environment locations, and reset locations. Certificates may realize these locations as components, fields, or lenses, but shall

map them to this common vocabulary and prove the corresponding disjointness/frame laws. The vocabulary is intentionally finer than a monolithic “process-local state” component. Custom Sys_B^{elab} components require local footprint/commutation and well-founded ordinal certificates.

Definition J.ReqSnapshot and J.NBDecision

For every witness-significant NB dynamic call q at explicit boundary c , let $\kappa(q)$ be its unique snapshot key and let $\text{NBPending}(q)$ be the carried I.DeferredNBHole record. $\text{ReqSnapshot}_c(q)$ is the common source-boundary state at the L2 settled point of q ’s Horizon, after every L1 request/attempt write and every other Dep_J predecessor of the decision has executed, but before q ’s A3 success/failure return and before any channel-state update caused by that decision. The snapshot contains the reset epoch; E4 channel state; every attributed persistent request and one-cycle attempt present at c , including q ’s own attempt with stable Push payload; and every explicit gate/join/arbitration/environment field affecting c . $\text{J.NBDecision}(q, \text{ReqSnapshot}_c(q))$ is the unique result prescribed by the source call semantics, E4 storage/rendezvous legality, and the settled boundary handshake reconstructed from that same snapshot. For $B(c) > 0$ at an ordinary uncoupled primitive boundary, PopNB succeeds exactly when nonempty and PushNB exactly when nonfull. At a coupling-sensitive $B(c) > 0$ boundary those occupancy tests remain necessary, but success additionally requires the q -side settled MirrorAvail_E value. For $B(c) = 0$, success requires the matching complementary attributed request/attempt with value agreement and, when the boundary is coupling-sensitive, the corresponding settled MirrorAvail_E condition. Failure is an ϵ A3 return with $\text{ReturnedAtCycle}(q, t)$ and no channel commit; success is the contracted A3/committed-transfer unit. WC selects the implementation outcome and J.CCcert proves equality. Process-local preparation may create $\text{NBPending}(q)$ and may continue through certified outcome-independent actions, but it may not execute A3 or read the settled snapshot locally.

Lemma J.BufSameCycle (same-cycle buffered Push/Pop sequentialization)

Let c have $B(c) > 0$, let the pre-cycle logical FIFO contents be Q with $n = |Q|$, and suppose one $\text{Push}_c(v)$ and one $\text{Pop}_c(w)$ both commit in the same RTL cycle under the non-fall-through E4 semantics. Then $0 < n < B(c)$ and $w = \text{head}(Q)$. For an ordinary uncoupled primitive buffered boundary, the common L2 snapshot also establishes both endpoint enablements, and the two FIFO state updates may be proof-sequentialized in either order: $\text{Push}_c(v); \text{Pop}_c(w)$ and $\text{Pop}_c(w); \text{Push}_c(v)$ terminate with logical contents $\text{tail}(Q)$ followed by v , occupancy n , the same producer and consumer ordinals, and the same unordered same-cycle observable bucket. For a coupling-sensitive boundary, occupancy alone is not an enablement proof: $\text{J.CCcert}/\text{ACert}$ shall either provide a same-snapshot commutation certificate showing that the two already-selected L3 actions may be sequentialized without re-evaluating the intervening MirrorAvail_E value, or represent the pair as one declared atomic boundary-action unit. If $n = 0$, E4 legality forbids the Pop from being created by the same-cycle Push; if $n = B(c)$, E4 legality forbids the Push from being created by the same-cycle Pop. For $B(c) = 0$ the endpoints remain one indivisible rendezvous action and this lemma does not apply. Proof: E4 gives the interior-occupancy necessity; the ordinary case uses the FIFO update equations, while the coupling-sensitive case uses the stated same-snapshot commutation/atomicity certificate. ■

Definition J.DepJ (generated construction-dependency relation)

After contracting every explicitly atomic action, let a $\prec_D^{\text{sem}}$ b denote a direct semantic-prerequisite edge. The semantic relation is generated only by (D1) same-process dynamic control, data, source-issue, synchronization-boundary, and ReturnedAtCycle dependencies; (D2) request/attempt-before-settle/snapshot-before-A3-decision edges for every location read by $\text{ReqSnapshot}_c(q)$, plus A3-decision-before-continuation edges only for actions whose reachability or operands depend on $\text{NBResult}(q)$; (D3) per-channel FIFO/history, occupancy/head, and non-commuting same-boundary

update predecessors, with $J.\text{BufSameCycle}$ as the named buffered exception; and (D4) reset, environment, and explicit elaboration-component predecessors. Let $a \prec_D b$ denote the full construction-dependency relation obtained by adding (D5) concrete earlier-Horizon proof-stratification edges from the selected RTL annotation. A D5 edge that is not already implied by $(\prec_D^{\text{sem}})^+$ is admissible only when the local footprint/frame certificate proves that the two actions commute modulo $J.\text{HCanon}$, so choosing the earlier-Horizon action first preserves transition legality, complete units, and the post-settle semantic state. Thus D5 selects a canonical construction representative; it is not by itself a universal Sys_ref happens-before fact. An outcome-independent same-Horizon action reached past an unresolved q acquires no A3-to-action edge: its $I.\text{DeferredNBHole}$ NoDependentUse fact and field-sensitive footprints must instead show that it is independent of $\text{NBResult}(q)$ and of q 's settled snapshot, so ordinary framing/commutation applies. An edge may not be inserted merely for replay convenience. Mere source order between distinct-interface operations creates the applicable issue-order obligations, but not a decision/commit-to-later-request edge when Core.RegSeq and the local certificate establish independent co-issue.

Lemma J.DepSound (semantic generated edges are operational prerequisites)

If $a \prec_D^{\text{sem}} b$, then no legal Sys_ref construction segment realizing both actions can execute b before a unless the actions are contracted into one explicit atomic unit. Proof: D1 follows from source control/data/issue rules and $\text{ReturnedAtCycle}/\text{Core.RegSeq}$; D2 from the settled-snapshot read and true NB-result dependence; D3 from E4/FIFO state dependence except $J.\text{BufSameCycle}$; and D4 from the named reset/environment/elaboration rule. Soundness of every declared footprint against the normative location semantics, including frame and disjointness laws, is part of the local-certificate obligation. $\square$

Lemma J.HorizonOrderSafe (D5 proof stratification is construction-safe).

Let $a \prec_D b$ be a D5 edge with no path $a (\prec_D^{\text{sem}})^+ b$. The LocalGWR package must then supply the D5 commutation certificate required by Definition J.DepJ: a and b may be exchanged without changing transition legality, the complete observable units, the terminal post-settle semantic state, or the $J.\text{HCanon}$ quotient. Hence the canonical construction may choose a before b even though a is not an operational prerequisite of b in every legal Sys_ref execution. If the pair does not commute, the actual overlap must instead be represented by D1–D4 and is covered by J.DepSound. $\square$

Lemma J.DepComplete (all non-commuting construction actions are ordered)

Let a and b be distinct construction actions that are not endpoints of one contracted atomic action. If neither $a \prec_D^+ b$ nor $b \prec_D^+ a$, then either (i) their field-sensitive ReadSet/WriteSet interference tests are disjoint and the transitions commute by $J.\text{Frame}$, or (ii) they are the legal same-cycle buffered Push/Pop dual update of $J.\text{BufSameCycle}$. Every pair that cannot be swapped while preserving transition legality, the same complete observable units, the same post-settle semantic state, and the same $J.\text{HCanon}$ history quotient is therefore ordered in at least one direction. In particular, narrowed $\text{A3}(q)$ commutes with a same-Horizon owner action carried past q only when $\text{NoDependentUse}(q)$ and the location footprints show disjointness from $\text{NBResult}(q)$, $\text{NBPending}(q)$, q 's attempt, and every location in $\text{ReqSnapshot_c}(q)$. The finite A1–A5 case analysis invokes D1–D4 for every actual overlap; D5 may additionally order already-commuting cross-Horizon pairs under Lemma J.HorizonOrderSafe, and custom components use their local certificate. ■

Definition J.FoundationRank (rank independent of graph acyclicity)

For each construction action a , define $\text{FR}(a)$ lexicographically as $\langle \text{EpochIndex}(a), \text{Horizon}(a), \text{SemanticStage}(a), \text{LocalOrdinal}(a), \text{ComponentTie}(a) \rangle$. Horizon is the selected RTL cycle/decision horizon or enclosing unit cycle. If an action consumes a value or status with $\text{ReturnedAtCycle}(q, t)$, Core.RegSeq

requires $\text{Horizon}(a) > t$. Same-Horizon actions carried past q must instead satisfy $\text{NoDependentUse}(q)$. SemanticStage is the fixed order: reset/environment prerequisites; local evaluation plus request/attempt writes (Core.Phase L1); completed settled-snapshot formation (L2); shared A3 decision or contracted/other boundary commit (L3); and a reserved continuation-bookkeeping stage usable only for within-Horizon sinks whose outgoing same-Horizon edges never return to an earlier stage. Outcome-dependent process continuation after an NB result is not assigned to that reserved stage; it is an ordinary evaluation/request action at a later Horizon. LocalOrdinal is the action-kind-specific script, request/snapshot, transfer, environment, reset, or component ordinal. ComponentTie is used only when all prior coordinates agree and a local certificate proves commutation or a component-local order. FR is computed before Dep_J and does not use Height or J.CanonicalRank .

Lemma J.DepRank (every generated dependency advances the foundation rank)

For every well-formed LocalGWRPackage and generated direct edge $a \prec_D b$, $\text{FR}(a) < \text{FR}(b)$. D5 strictly increases Horizon; reset edges increase EpochIndex . D1 either advances the within-L1 local ordinal/stage, or—when b depends on a boundary result $\text{ReturnedAtCycle}(q, t)$ —strictly increases Horizon under Core.RegSeq . Independently reached same-Horizon actions carried past an unresolved q have no A3-to-action edge; their true control/data/issue predecessors advance the L1 ordinal or stage. D2 advances request/attempt $\rightarrow$ settled snapshot $\rightarrow$ A3 decision/contracted commit. A decision-to-dependent-continuation edge strictly increases Horizon; any retained within-Horizon continuation-bookkeeping target is a certified sink and therefore cannot generate a later stage regression. D3 advances transfer/history ordinal or a non-commuting substage, with J.BufSameCycle removing the legal dual-update ordering; D4 advances the certified component ordinal/substage. Thus the prior failed-NB decision(t) $\rightarrow$ dependent request(t) case is impossible by construction, and the decision(t) / independent request(t) pair is unordered and commuting. ■

Lemma J.DepAcyclic (well-foundedness of the generated dependency graph)

For every well-formed LocalGWRPackage , the graph Dep_J is acyclic after atomic contraction. Proof. Suppose a directed cycle $a_0 \prec_D a_1 \prec_D \dots \prec_D a_0$ existed. Repeated application of Lemma J.DepRank would give $\text{FR}(a_0) < \text{FR}(a_1) < \dots < \text{FR}(a_0)$, contradicting irreflexivity of the well-founded lexicographic order on the finite rank components. This argument uses one global semantic rank that every D1–D5 edge advances; it does not infer acyclicity from a union of separately acyclic process, channel, request, environment, reset, or component orders. Height and J.CanonicalRank are defined only after this lemma and play no role in proving it. ■

Definition J.CanonicalRank (derived construction order)

Because $\text{Act_L}(\tau_{\text{post}})$ is finite and Dep_J is acyclic by the independent FR argument, define $\text{Height}(a) = 0$ for actions with no incoming $\prec_D$ edge and $\text{Height}(a) = 1 + \max\{\text{Height}(b) \mid b \prec_D a\}$ otherwise. Define $\text{CanonicalRank}(a) := \langle \text{FR}(a), \text{Height}(a), \text{KindKey}(a), \text{StableId}(a) \rangle$ lexicographically. FR preserves the concrete epoch/horizon and semantic prerequisite structure; Height is a theorem-derived topological depth used only after acyclicity has been established; and $\text{KindKey}/\text{StableId}$ only totalize otherwise commuting actions and create no observable $\prec_J$ edge. The rank of an observable unit is the rank of its contracted commit action. J.CanonicalRank is reconstructed from checked action identities, local operational coordinates, and Dep_J ; no phase or replay rank is supplied by the implementation flow.

Definition J.RequiredPrefixCandidate (rank-independent local-certificate index)

For a finite RTL_B prefix τ_{post} , $\text{RequiredPrefixCandidate}(\tau_{\text{post}}, H)$ means that H is a finite set of complete observable units of τ_{post} satisfying only order-neutral prefix conditions: (R1) H is down-closed under the execution-specific required-order relation $\prec_J$; (R2) for every process P , the P -local projection of H is a prefix of the selected P -local observable-unit projection, with no dynamic operation

identity used twice and with reset epochs respected; (R3) every explicitly atomic unit is either wholly present or wholly absent; and (R4) each explicit-channel projection is a reset-epoch-consistent prefix of the selected rendezvous or FIFO history. The definition does not mention Dep_J , $J.\text{DepAcyclic}$, $J.\text{CanonicalRank}$, a proposed global replay order, or reachability of a common Sys_ref state. It is only the finite index domain on which Appendix I can compute process-local replay states and preparation scripts. Theorem J.GWR-Comp later selects from this domain the candidates that arise along its derived construction chain.

Definition J.PCert (process-local replay and preparation certificate)

For a process P and finite RTL_B prefix τ_{post} , $\text{PCert}_P(\tau_{\text{post}}, L_P)$ holds when L_P supplies, for every relevant $\text{RequiredPrefixCandidate}$ and process participation: (P1) WC/μ_Q identity, reset epoch, and matched dynamic operation; (P2) $\text{Replay}_P(H, w)$ with future- Σ -relevant values, control, completion, and operation ordinals; (P3) the finite per-Horizon Lemma I.3 script, every enriched $I.\text{DeferredNBHole}$, its $\text{NBResult}/\text{NBPending}/\text{result-destination}$ locations, and NoDependentUse evidence for all same-Horizon actions carried past it; (P4) $\text{Issue}/E3/E5$, prefix closure, persistence, stable payload, Core.RegSeq , direct-input stability, and synchronization facts; (P5) exact dynamic NB attempt identity, desired WC outcome, and snapshot key, with no local A3 decision; (P6) complete A1/A2/A3 inventory, normative field-sensitive footprints, Horizon/stage/local ordinal, and every D1/D2 generator; and (P7) proof that local actions modify only their declared locations and export no unmatched commit or unresolved same-Horizon hole at a cutpoint. PCert contains no $J.\text{CanonicalRank}$ and does not assume the global order or phase-normal chain.

Definition J.CCert (explicit-channel history, request, and enablement certificate)

For every explicit abstract channel c of E , $\text{CCert}_c(\tau_{\text{post}}, L_c)$ supplies: (C1) boundary, $B(c)$, ownership, epochs, and B -faithfulness/ ϵ -opacity; (C2) complete prefix-indexed $E4$ history, ordinals, payloads, occupancy/head, and reset-empty restarts; (C3) producer/consumer agreement on transfer identity, value, ordinal, direction, and epoch; (C4) for every Horizon and witness-significant q , the complete L2 settled $\text{ReqSnapshot}_c(q)$, including all absent/present requests/attempts, stable payloads, and explicit gate/join/environment fields; (C5) proof that $\text{WC}(q) = J.\text{NBDecision}(q, \text{ReqSnapshot}_c(q))$; (C6) exact local channel/A3 state updates with no hidden availability; (C7) $J.\text{BufSameCycle}$ evidence for every legal same-cycle buffered pair; (C8) every D2/D3 edge and field-sensitive channel/snapshot/result footprint; and (C9) proof that every one-cycle NB attempt is retired and every deferred-hole record for the Horizon is resolved before the exported post-boundary construction state.

Definition J.ACert, J.EnvCert, and J.ResetCert

ACert_y identifies the complete participant set, dynamic identities, values, reset epoch, and atomic transition rule for every rendezvous transfer, paired $\text{SyncChannel}/\text{ac_sync}$ transaction, $E2$ anchor/signal group, $R2C$ fixed-point unit, RESET unit, and other explicitly atomic unit. It proves that every participant record names the same contracted action and supplies the endpoint requests, values, guards, and component state from which the generic Sys_ref rule can derive enablement; it does not take complete-unit enablement itself as an unexplained certificate field. An endpoint cannot be certified independently of its counterpart. EnvCert proves, for every selected external interaction, that the source and RTL records use the same response of the fixed reactive environment to the same preceding Σ -causal history and supplies the corresponding footprints, horizon/stage/ordinal fields, and dependencies. ResetCert supplies RW1 – RW5 for each reset occurrence and scope, including request clearing, reset-empty channel state, source/ RTL reset-state correspondence, atomic reset splicing, framing outside the reset scope, and the D4 dependency and $J.\text{DepRank}$ coordinate facts.

Definition J.LocalAgree and J.LocalGWR

LocalAgree(L) is the finite conjunction of equality, uniqueness, ownership, and coverage checks on duplicated local-record fields: shared action/unit identity; endpoint pairing; boundary identity; payload/value; transfer ordinal; request/attempt attribution; deferred-hole identity and result destination; snapshot key/contents; NoDependentUse facts; reset epoch; witness/elaboration identity; Horizon and Core.LAdm phase; semantic stage/local ordinal; the normative field-sensitive ReadSet/WriteSet locations; and every generated dependency edge. Every process, explicit boundary, construction action, atomic unit, deferred hole, and witness-significant ϵ choice in τ_{post} must be covered, and each endpoint request/attempt has at most one dynamic owner. LocalAgree does not assert a global execution, acyclicity, phase-normality, or extension.

Definition J.LocalGWR. LocalGWRPackage(τ_{post}, L) consists of PCert_P for every process, CCert_c for every explicit boundary, ACert for each atomic unit, applicable EnvCert/ResetCert, the complete A1–A5 action/deferred-hole inventory, the normative field-sensitive footprints and footprint-soundness proofs, Core.RegSeq/direct-input conformance facts, locally checkable epoch/Horizon/stage/ordinal metadata, every local fact from which Dep_J is generated, custom-component certificates, and LocalAgree, all bound to one instance/stimulus/E/w/observation boundary/RTL prefix. J.DepRank, J.DepAcyclic, J.CanonicalRank, J.RegPhaseNF, J.RegPhaseReach, and the global source chain are theorem-derived, not supplied as unexplained local premises.

Definition J.LocalPrefixAgree and J.LocalGWRExtCert (cross-prefix local-certificate extension)

Fix two finite reachable RTL_B prefixes $\tau_{\text{post}}^0 \leq \tau_{\text{post}}^1$ on the same legal execution carrier and LocalGWR packages L_0, L_1 for the same theorem instance, stimulus, elaboration E, witness w, observation boundary, environment, reset interpretation, and fixed implementation choices.

LocalPrefixAgree_I($\tau_{\text{post}}^0, L_0; \tau_{\text{post}}^1, L_1$) holds when the newer package restricts exactly to the older package on every object already present at τ_{post}^0 : dynamic operation/unit identities, WC/ μ_Q matches, Issue/ReturnedAtCycle/Commit annotations, channel histories and ordinals, request/attempt attribution, snapshots and deferred-hole identities/results, environment/reset identities, atomic groups, ConstructionAction identities, field-sensitive ReadSet/WriteSet footprints, Horizon/stage/local ordinals, and all LocalAgree equalities are unchanged. In particular, the old action carrier is literally the restriction of the new action carrier, Dep_J¹ restricted to old actions equals Dep_J⁰, and no newly introduced action is a required predecessor of an old action. Hence the old construction-action set is Dep_J-down-closed and its complete-unit projection remains a $\prec_J$ required-prefix ideal. Every old FR coordinate and stable key is unchanged; because no new predecessor enters the old action set, old Height values are unchanged as well, so J.CanonicalRank and the theorem-derived ValidatedConstructionOrder restrict to the old order rather than being recomputed with a different old prefix. New dependencies that cross the boundary run only from already-present prerequisites into genuinely new actions and are completely represented by the new package; atomic groups are never split across the restriction.

LocalGWRExtCert_I($\tau_{\text{post}}^0, L_0; \tau_{\text{post}}^1, L_1$) means LocalGWRPackage($\tau_{\text{post}}^0, L_0$), LocalGWRPackage($\tau_{\text{post}}^1, L_1$), and LocalPrefixAgree, together with the following extension-specific facts. (LE1) Closed-Horizon suffix: every newly added construction action/unit belongs to a Horizon strictly after the final nonempty Horizon already exported by the old J.GWR prefix, or to a certified unit-empty bridge cycle after that old endpoint; no new action is inserted into an old Horizon whose L3 action set/post-boundary endpoint has already been closed. (LE2) Bridge-choice stability: every evidence-determined source/environment/reset bridge cycle lying wholly before the old endpoint is the same fixed choice in both packages; the larger package may append later bridge cycles but may not retcon the earlier source chain. (LE3) Cross-prefix join coherence: every new action that reads state written or constrained by the old construction has the corresponding Dep_J/footprint/snapshot/atomic prerequisite, and every shared channel/environment/reset object begins the new suffix from the exact

old exported state. (LE4) Exact reset/epoch and request continuity: old reset epochs, outstanding request identities, payloads, and no-unresolved-hole facts are preserved at the join, with any newly occurring RESET represented only in the new suffix under RW1–RW5. These are checked extension facts, not consequences of two unrelated successful LocalGWR packages. For the standard QSync carrier-edge use, RS1–RS5 and exact cycle-aligned package generation are the normal source of the $\tau_{\text{post}}^0 \leq \tau_{\text{post}}^1$, unit-accounting, epoch, and closed-Horizon evidence.

Definition J.ConstructionOrderProvenance and J.ValidatedConstructionOrder

For fixed τ_{post} and W , a ValidatedConstructionOrder contains a complete unit enumeration $\gamma_1 \dots \gamma_m$, the finite predecessor relation $\prec_W$, and the partition of actions/units into nondecreasing Horizon buckets. It proves: (V1) well-foundedness; (V2) refinement of $\prec_J$, reset epochs, atomic grouping, and every strict Horizon constraint; (V3) exact action/unit/deferred-hole coverage; (V4) coverage of every preparation, request, settled snapshot, A3 decision, channel, environment, reset, and atomic prerequisite; (V5) every flattened enumeration prefix is a finite required-prefix ideal; and (V6) within each Horizon, the order admits the Core.LAdm phase normal form of J.RegPhaseNF, with all L1 requests fixed before L2 settle, all boundary decisions/commits in L3, and no deferred hole surviving the bucket. The theorem-derived form uses LocalGWR, Dep_J, J.CanonicalRank, and J.RegPhaseNF. The direct form independently validates V1–V6. Stable keys order only actions/units that commute modulo J.HCanon.

Definition J.GWR (GlobalWitnessRealizable(τ_{post}, W))

A selected package W is globally witness-realizable when it contains one coherent annotation package—WC/ μ_Q , Issue/ReturnedAtCycle/Commit attribution, E4 histories, atomic groups, environment/reset identities, dynamic operations, deferred-hole/snapshot identities, and construction actions—and there exist: (G1) one J.ValidatedConstructionOrder with finite Horizon buckets and a flattened complete-unit enumeration; (G2) required-prefix ideals for the flattened prefixes and matched cycle-entry/post-boundary bucket prefixes; and (G3) reachable Sys_ref states such that each nonempty Horizon bucket extends by one finite legal phase-normal segment: L1 process-owned A1/A2 evaluation and request/attempt assertion, then L2 freeze/ ϵ -settle to the common snapshots, then L3 A3/A4 boundary decisions and complete units in commit-action rank order, with J.BufSameCycle used for legal buffered dual updates. Where the selected source witness, environment, reset-state contract, or other cycle-sensitive construction evidence requires a later source cycle-entry state before the next nonempty bucket can be realized, the chain may first contain a finite evidence-determined sequence of invariant-preserving unit-empty CompleteCycle transitions permitted by Core.SelectLatitude; a no- Σ -commit member of that sequence is a Core.SilentCycle. These bridge cycles carry no unmatched selected J observable/proof-internal unit and are not Horizon buckets or rank elements: Height, FR, J.DepRank/J.DepAcyclic, and J.CanonicalRank remain defined only over the construction actions in Act_L(τ_{post}). The bridge length is not inferred merely from the numerical gap between RTL Horizon indices unless the selected cycle-sensitive evidence itself requires that alignment. No new L1 request occurs after the bucket's L2 settle; every action depending on a ReturnedAtCycle result is placed in a later Horizon; and every deferred hole is resolved or retired before the exported post-boundary state. The segment has the same complete units and terminal post-settle state and the same CanonHist/J.HCanon quotient as the selected construction order, even when same-Horizon independent actions are permuted. The chain preserves replay, channel, witness, attribution, reset, phase, and snapshot invariants and is compatible with τ_{post} under E1–E5. The direct route supplies the same bucketed interface. The terminal state of the selected J.GWR prefix is the exported post-boundary construction endpoint; J.GWR does not identify that state with a Core.KCut merely by construction. A K-facing Corollary J.1 application may subsequently append the distinct Core.KObsNormRef observation-

only normalization, producing τ_{ref}^K without changing the matched J.GWR observable prefix. Historical operational data remains evidence rather than KObs fields.

J.GWR is a stable semantic interface. The normal route derives its ValidatedConstructionOrder and the complete witness chain by Theorem J.GWR-Comp from Definition J.LocalGWR. A flow may instead establish the same interface directly through the direct-validated construction-order provenance and a checked reachable chain. No theorem below infers J.GWR from bare Σ -label equality, and the direct route may not refer to J.CanonicalRank unless it also supplies the LocalGWR package and theorem-derived provenance that define that rank. When the same normal-compositional flow must preserve an already selected exact source witness across a larger cycle-aligned RTL prefix, Definition J.LocalGWRExtCert and Theorem J.GWR-CompExt below are the authoritative incremental interface; two independent applications of J.GWR-Comp do not by themselves imply that their selected source prefixes are nested.

Lemma J.Frame (local transition framing)

Let Sys_ref state be viewed through the normative typed location vocabulary of J.ConstructionAction. A transition segment whose certified WriteSet is F changes only locations in F. Every location disjoint from F is equal before and after, and every guard/expression whose ReadSet is disjoint from F has the same value. This includes separate framing of Ctrl/Local/Operand, NBResult, NBPending, endpoint request/payload, channel/elaboration, environment, and reset locations. Each certificate must prove soundness of its declared locations against the actual Sys_ref transition semantics; the representation may use components, fields, or lenses. ■

Lemma J.LocalLift (lifting a certified process-local preparation path)

If the P-owned locations of a reachable global state equal the start state of a PCert_P segment, every shared location read by that segment equals its certified summary, and the field-sensitive footprint conditions hold, then the segment lifts to a finite global ϵ /request path. The lifted path has the same P-local result and endpoint writes, commits no unmatched Σ unit, and frames every unrelated location. At an NB call it creates/carries I.DeferredNBHole and performs no A3 decision; unlike the previous stop-at-hole formulation, it may continue through certified NoDependentUse same-Horizon actions. Proof: induction over the segment using J.Frame. ■

Lemma J.PrepareCommutate (commutation of local preparation paths)

Consider the executable L1 preparation actions and other construction actions in one Horizon. Field-sensitive disjoint ReadSet/WriteSet actions commute by J.Frame. Separately owned request writes may accumulate before the common settle when neither reads a snapshot whose fields remain mutable. A narrowed A3(q) never moves before an action whose write appears in ReqSnapshot_c(q), but it commutes with a carried same-Horizon owner action when NoDependentUse(q) and the footprints establish disjointness from NBResult(q), NBPending(q), q's attempt, and all snapshot locations. J.DepComplete orders every other non-commuting pair; J.BufSameCycle handles the legal buffered L3 dual update. Thus topological executions can be permuted into the normal form stated next, preserving terminal state and the J.HCanon quotient. No fairness premise is used. ■

Lemma J.RegPhaseNF (registered sequential-interface phase normal form)

Assume Core.RegSeq, a well-formed LocalGWRPackage, and that the frozen L2 state of the selected Horizon admits a finite $\rightarrow \epsilon, L2$ path to a SettlePoint. In the standard compositional scope, QualifiedSynchronousModel_Sys_ref(l) and Theorem Core.QSync-Ref discharge this existence premise uniformly for every reachable Horizon; a directly validated J.GWR route may instead carry the equivalent checked per-Horizon settling evidence. For every finite construction fragment whose actions have one selected Horizon t, there is a permutation respecting Dep_J and atomic contraction with the following

order: (1) all required A1 local evaluation/control and A2 request/attempt assertions permitted by Core.Phase L1, including outcome-independent actions carried past deferred NB holes; (2) one L2 request freeze and ϵ -settle producing every ReqSnapshot for t ; and (3) the A3/A4 boundary decisions and complete units of L3, ordered by commit-action rank except for explicitly atomic units and J.BufSameCycle sequentializations. Actions depending on a value/status ReturnedAtCycle in t occur only at a later Horizon. The normalized fragment contains the same complete observable units, reaches the same post-boundary semantic state, and yields the same canonical enriched history under J.HCanon; literal ordering of independent same-Horizon labels need not be identical. Every deferred hole for t is resolved or retired at the end. Proof: repeatedly commute adjacent unordered actions using J.Frame/J.PrepareCommute, use NoDependentUse and field-sensitive footprints for decision/independent-continuation swaps, use J.BufSameCycle within L3, and use J.DepRank to exclude any required edge from a later phase back to an earlier phase. Termination follows from finiteness. ■

Lemma J.NBDecision (globally valid non-blocking decision step)

Suppose the matched Horizon has completed its J.RegPhaseNF L1 actions and L2 settle, every Dep_J predecessor of NBPending(q) has executed, and J.CCert reconstructs ReqSnapshot_c(q)= S . Then q has exactly one legal A3 result J.NBDecision(q, S). On failure, A3 writes the status to NBResult(q), retires NBPending(q)/the one-cycle attempt, produces ReturnedAtCycle(q, t), and changes no channel history. On success, A3 is contracted with the corresponding committed transfer and writes the returned result. A3 does not perform ordinary outcome-independent control advance: such same-Horizon actions have already occurred in L1 under NoDependentUse, while every outcome-dependent continuation is an A1 action at a later Horizon. The result agrees with WC by C5 and no process-local script may choose or overwrite it. ■

Lemma J.UnitEnable (local certificates imply complete-unit enablement)

After J.RegPhaseNF has executed all required L1 actions and formed the common L2 snapshots for a Horizon, the endpoint requests/attempts, stable payloads, channel state, participant state, and deferred-hole records satisfy the premises of each selected L3 complete unit. J.NBDecision supplies failed or successful A3 results from those snapshots; failure is an ϵ result and success is the contracted unit. Buffered storage legality follows from the common pre-update occupancy/head and non-fall-through E4, while actual enablement is taken from the same settled snapshot through MirrorAvail_E/ChannelEnabled; J.BufSameCycle handles legal dual updates under its ordinary or certified coupling-sensitive case. Rendezvous units require the attributed endpoint requests/value agreement plus any declared coupling-sensitive settled handshake condition. Synchronization, E2 anchor, R2C, environment, and RESET units use their named certificates. ACert/CCert supply constituent facts while this lemma derives complete-unit enablement. ■

Lemma J.UnitExt (one Horizon-batched locally certified extension)

Assume the matched cycle-entry invariant for the next nonempty Horizon bucket B_t of the J.ValidatedConstructionOrder and LocalGWRPackage. Before executing B_t , if the selected source witness, environment, reset-state contract, or other cycle-sensitive construction evidence requires a later source cycle-entry state, use Core.SelectLatitude and the ordinary Sys_ref cycle semantics to take the finite evidence-determined sequence of invariant-preserving unit-empty CompleteCycle transitions needed to reach that entry. Any bridge transition with no selected Σ commit is a Core.SilentCycle; a bridge used here carries no unmatched selected J observable/proof-internal unit. The bridge creates no synthetic Horizon bucket or rank element, and its length is not inferred from the raw numerical difference between RTL Horizon indices unless the instance's cycle-sensitive evidence requires that alignment. Because no selected unit commits, the required-prefix ideal is unchanged; typed cycle

evolution together with the existing local/channel/environment/reset and LocalAgree evidence preserves or re-establishes the process/replay, channel, request/witness, reset, environment, phase, and canonical-history components of the matched cycle-entry invariant. Then execute the finite Dep_J-minimal L1 A1/A2 predecessor set by J.LocalLift/J.PrepareCommute, carrying deferred holes through NoDependentUse work. Apply J.RegPhaseNF to obtain the canonical L1/L2/L3 arrangement, freeze requests, settle the explicit network, reconstruct all snapshots, and execute the bucket's failed A3 ϵ returns and complete A3/A4 units in commit-action rank order, using J.BufSameCycle where applicable. J.UnitEnable supplies each complete transition. No L1 action is executed after the L2 settle; dependent continuation is deferred to a later Horizon; all t-holes are empty at the exported post-boundary state. The resulting finite extension preserves replay, channel/atomic, attribution, reset, and canonical-history invariants and advances from the prior bucket ideal to the ideal containing all units in B_t. The bridge and bucket extension use no B2/B3 fairness.

Theorem J.GWR-Comp (local certificates construct global witness realizability)

Fix a finite reachable RTL_B prefix τ_{post} , the theorem instance, stimulus, elaboration E, observation boundaries, witness w, and reset interpretation. Assume Core.RegSeq, LocalGWRPackage(τ_{post} , L), the Sys_ref decomposition used by J.Frame–J.UnitExt, I.A0/WC, the source/RW premises, B-faithful E4 boundaries, and QualifiedSynchronousModel_Sys_ref(I). By Theorem Core.QSync-Ref, every reachable constructed Horizon therefore has the finite L2 SettlePoint required by J.RegPhaseNF/J.UnitExt. Then Dep_J satisfies J.DepSound/J.HorizonOrderSafe/J.DepComplete; FR/J.DepRank prove J.DepAcyclic; J.CanonicalRank and the Horizon partition define the theorem-derived J.ValidatedConstructionOrder; J.RegPhaseNF proves its V6 phase property; and Horizon-batched J.UnitExt constructs W and a reachable τ_{ref} satisfying GlobalWitnessRealizable(τ_{post} , W).

Proof. Generate the field-sensitive actions, deferred-hole records, Dep_J, and FR metadata.

J.DepSound/J.HorizonOrderSafe/J.DepComplete establish prerequisite coverage and commutation; J.DepRank/J.DepAcyclic derive the order. Partition the canonical order by Horizon. For each bucket, J.RegPhaseNF permutes unordered actions into the Core.Phase L1/L2/L3 order modulo J.HCanon, and J.UnitExt performs the finite phase-normal extension. J.CCerts/ACerts reconstruct common snapshots and updates; J.NBDecision derives shared outcomes; J.BufSameCycle handles legal buffered dual updates; J.Frame and LocalAgree preserve unrelated locations and coherence; ResetCert/J.RS handle resets. Outer induction over Horizons, with inner commit-action order within L3, yields a reachable chain covering every unit and no exported deferred holes. Assemble W from the records, order, phase-normal chain, snapshots, and invariants; G1–G3 and E1–E5 follow. ■

Non-circularity. LocalGWR supplies field-sensitive actions, deferred-hole/NoDependentUse records, snapshots, local dependencies, and Core.RegSeq evidence, but no global order, phase-normal execution, or extension. J.DepComplete/J.DepRank/J.DepAcyclic derive the order; J.RegPhaseNF derives the per-Horizon phase arrangement; J.NBDecision derives shared outcomes; and J.UnitExt/J.GWR-Comp prove the reachable bucketed chain. The normal form is computed from graph-independent Horizon/stage/footprint facts and therefore does not assume the order it proves replayable.

Definition J.GWRPrefixExt (exact cross-prefix GWR restriction/extension)

For $\tau_{\text{post}}^0 \leq \tau_{\text{post}}^1$ and selected J.GWR results ($W_0, \tau_{\text{ref}}^0\{G, 0\}$) and ($W_1, \tau_{\text{ref}}^1\{G, 1\}$), write GWRPrefixExt_I($W_0, \tau_{\text{ref}}^0\{G, 0\}; W_1, \tau_{\text{ref}}^1\{G, 1\}$) when: (GX1) $\tau_{\text{ref}}^0\{G, 0\} \leq \tau_{\text{ref}}^1\{G, 1\}$ as exact Sys_ref execution prefixes, with the state ending $\tau_{\text{ref}}^0\{G, 0\}$ literally the join state from which the new suffix proceeds; (GX2) W_1 restricted to the old RTL/action/unit domain is W_0 , including annotations, histories, snapshots, deferred-hole dispositions, atomic groups, environment/reset choices, and source chain; (GX3) the new ValidatedConstructionOrder and Horizon chain restrict to the old one, so no old complete unit, action, bridge cycle, or canonical-history fact is dropped, duplicated, relabeled, or

reordered; and (GX4) the suffix after $\tau_{\text{ref}}^{\{G,0\}}$ contains only the newly required later-Horizon phase-normal construction and any later evidence-determined unit-empty bridge cycles. GWRPrefixExt is stronger than equality of J.HCanon histories: it records exact source-prefix nesting and exact witness-package restriction. It concerns the J.GWR construction prefix τ_{ref}^G only; it says nothing about a previously selected K-facing v_{ref} or τ_{ref}^K suffix.

Theorem J.GWR-CompExt (local-certificate extension preserves the exact J.GWR source prefix)

Fix $\tau_{\text{post}}^0 \leq \tau_{\text{post}}^1$ on one legal RTL_B carrier and assume the standing normal-compositional premises of Theorem J.GWR-Comp for the larger prefix. Let L_0, L_1 satisfy J.LocalGWRExtCert_I($\tau_{\text{post}}^0, L_0; \tau_{\text{post}}^1, L_1$), and let a prior application of J.GWR-Comp to L_0 have selected W_0 and the exact reachable source construction prefix $\tau_{\text{ref}}^{\{G,0\}}$. Then the J.GWR-Comp construction for L_1 can be chosen to produce W_1 and $\tau_{\text{ref}}^{\{G,1\}}$ such that GlobalWitnessRealizable($\tau_{\text{post}}^1, W_1$) and GWRPrefixExt_I($W_0, \tau_{\text{ref}}^{\{G,0\}}; W_1, \tau_{\text{ref}}^{\{G,1\}}$) hold. In particular $\tau_{\text{ref}}^{\{G,0\}} \leq \tau_{\text{ref}}^{\{G,1\}}$; the theorem extends the old construction rather than replacing it with an unrelated matching source witness. Proof. LocalPrefixAgree makes the old ConstructionAction carrier an exact Dep_J-closed restriction of the new one and excludes new predecessors of old actions. Therefore old FR and Height values, stable keys, J.CanonicalRank order, Horizon partition, snapshots, atomic contractions, and LocalAgree facts are unchanged on the old domain; Lemma J.DepRank remains the same edge-order proof on that restriction. LE1 prevents the larger package from reopening an already closed old L3 Horizon; LE2 fixes every bridge/source/environment/reset choice already consumed before the old endpoint. Reuse the previously constructed J.GWR phase-normal chain verbatim through $\tau_{\text{ref}}^{\{G,0\}}$. Starting at its exact exported state, LE3–LE4 provide the channel/request/environment/reset join invariant for the genuinely new suffix. Apply J.RegPhaseNF/J.UnitExt only to the later new Horizons, inserting only the later evidence-determined unit-empty bridge cycles allowed by LE1–LE2. The ordinary J.GWR-Comp preservation argument then proves the replay, channel, attribution, reset, no-hole, E1–E5, and J.HCanon invariants for the appended suffix. The assembled W_1 restricts to W_0 and the exact source chain has $\tau_{\text{ref}}^{\{G,0\}}$ as a prefix, giving GX1–GX4. $\square$ The closed-Horizon premise is essential: without it, an arbitrary larger RTL prefix could require inserting a newly discovered same-Horizon unit behind an already exported post-L3 source endpoint, and exact source-prefix extension would not follow.

Proposition J.0 (Pairwise Composition Lemma)

For each process P , let $A_{\text{ref},P}$ and $A_{\text{post},P}$ be the annotated P -local projection records induced by $\tau_{\text{ref},\text{system}}$ and $\tau_{\text{post},\text{system}}$ under the chosen witness/annotation package. Their forgotten observable components are the ordinary projections $\tau_{\text{ref},P}$ and $\tau_{\text{post},P}$, but the records also carry the issue annotations, dynamic message-call universe, μ_Q matching, and request-attribution facts needed by E3/E5 and WC. Assume that for every process P , these annotated projections satisfy the per-process compatibility predicate $A_{\text{ref},P} \approx_P A_{\text{post},P}$. In readable prose below, this may be abbreviated as $\tau_{\text{ref},P} \approx_P \tau_{\text{post},P}$ when the annotation package is fixed by context.

Thus Proposition J.0 is only a conditional composition result for a chosen compatible trace pair. It neither constructs a source execution nor proves J.GWR. Theorem J.GWR-Comp performs the local-to-global execution construction; Theorem J.1 then instantiates Proposition J.0 with that constructed pair. The restricted reverse clause continues to use the selected pair carried by RTLConsistentRefPrefix. Assume every explicit abstract channel c in the chosen Sys_{ref} / RTL_B comparison has finite capacity $B(c) \geq 0$. Also assume, as a per-channel Sys_{ref} / RTL_B well-formedness condition, that both chosen traces satisfy the applicable E4 channel semantics at the chosen abstract boundaries. Thus, for $B(c)=0$

channels, committed Push_c / Pop_c events obey the same-cycle rendezvous semantics; for $B(c) > 0$ channels, the committed Push_c / Pop_c history obeys the reset-empty, cycle-boundary, non-fall-through bounded-FIFO semantics with capacity $B(c)$, including FIFO order, no underflow, no overflow, no duplication, and no loss. Proposition J.0 does not derive these per-channel E4 facts; it assumes them as part of the definition of the compared well-formed Sys_ref / RTL_B traces.

No weak-fairness or channel-progress premise is needed for this conditional pairwise lifting lemma: Proposition J.0 does not construct a matching trace, but only lifts already-compatible per-process trace projections and already well-formed per-channel communication histories to the aggregate system trace. The finite selected-unit witness-construction premises are used later in Theorem J.1 when proving the finite execution-prefix existence result; weak fairness and channel progress are used only when a corresponding infinite fair execution is asserted. Then the aggregate traces are compatible: $\tau_{\text{ref}, \text{system}} \approx_{\text{sys}} \tau_{\text{post}, \text{system}}$ under rules E1–E5 (given the R rules and B rules).

Notation: The order relation $<_{\text{src}}$ used below is the dynamic source-induced order from Appendix G / R1–R4: it ranges over dynamic operation instances in the compared execution / witness, not over all syntactic call sites in the source text. Equivalently, when the process must be explicit, write $<_P$. Untaken branches and unexecuted loop iterations are not members of the witness-specific dynamic universe.

Proof

We prove each compatibility property. The per-channel E4 facts are not derived inside this proposition; they are part of the per-channel Sys_ref / RTL_B well-formedness hypothesis for the chosen traces.

E1 (Synchronization-order preservation):

Fix any process P . By the proposition hypothesis, $\tau_{\text{ref}, P} \approx_P \tau_{\text{post}, P}$, so the synchronization events of P occur in the same source order in both traces. Since this holds for every P , E1 holds for the system traces $\tau_{\text{ref}, \text{system}}$ and $\tau_{\text{post}, \text{system}}$.

E2 (Signal-visibility preservation):

Fix any process P . If P is a sequential process, then by the proposition hypothesis (using the E2 anchoring relation embodied in $\tau_{\text{ref}, P} \approx_P \tau_{\text{post}, P}$), every signal Read in P is anchored to P 's closest preceding synchronization event, and every signal Write in P is anchored to P 's closest succeeding synchronization event, in both $\tau_{\text{ref}, P}$ and $\tau_{\text{post}, P}$.

If P is a combinational process, then R2C rather than $\text{pred}_S/\text{succ}_S$ anchoring applies: P 's observable signal Read/Write events are represented by the R2C fixed-point signal-observation unit for the relevant combinational process/network component, matched by that component's local R2C occurrence ordinal as specified in the R2C unit matching/index convention. By the combinational well-formedness condition and the compatibility hypothesis for the fixed stimulus / selected witness, the pre-HLS and post-HLS combinational networks reach corresponding unique ϵ -fixed-point valuations for the matched R2C occurrence unit, so the matched Read/Write labels agree within that fixed-point unit. This local R2C-unit agreement is an explicit atomicity constraint; it does not require Sys_ref and RTL_B to use the same global observable cycle-bucket decomposition for unrelated independent events.

Since the appropriate sequential or combinational signal-visibility condition holds for every process P , E2 holds for $\tau_{\text{ref}, \text{system}}$ and $\tau_{\text{post}, \text{system}}$.

E3 (Safe message issue ordering):

Fix any process P . E3 is the post-side issue-order conjunct of the per-process compatibility relation $\approx_P$, not an additional ordering property derived from the Sys_ref trace. Thus what must be checked here is that P 's post-HLS execution satisfies E3 as stated in Appendix G: for any $q_1, q_2 \in Q_{\text{exec}, P}$ with $q_1 <_{\text{src}} q_2$, including ordinary distinct-interface message calls that appear in dynamic source sequence, $\text{issue_post}(q_1) \leq \text{issue_post}(q_2)$, and the issued set is prefix-closed under that source order. Same-cycle issue is permitted where allowed, and a later operation may be issued before an earlier operation commits under Policy L, but a source-later operation may not be issued before the source-earlier

operation has issued. This post-side property is part of the local per-process hypothesis $\tau_{\text{ref},P} \approx_P \tau_{\text{post},P}$; in the execution-level use of Proposition J.0 / Theorem J.1, the needed post-side witness/order fact is supplied by Appendix I / Implementation assumption I.A0. Since the E3 post-side conjunct holds for every P , the aggregate system pair satisfies E3.

E4 (Per-channel channel semantics):

E4 holds by the per-channel Sys_ref / RTL_B well-formedness hypothesis of Proposition J.0. That hypothesis states that, for every explicit abstract channel c in the chosen comparison, the Sys_ref and RTL_B traces already satisfy the applicable E4 channel semantics at the chosen abstract boundary: same-cycle rendezvous semantics when $B(c)=0$, and reset-empty, cycle-boundary, non-fall-through bounded-FIFO semantics when $B(c)>0$. Thus Proposition J.0 uses E4 as an assumed channel-realization condition for the chosen traces; it does not attempt to derive E4 from the per-process compatibility relation. Accordingly, Proposition J.0 should be read as a composition theorem under already-established process and channel conformance hypotheses. It does not prove that the implementation realizes the chosen abstract channel boundaries, nor that the back-annotated capacities $B(c)$ satisfy B-faithfulness as defined in the Normative Formal Core. Those facts are separate RTL/tool-flow compliance obligations required before the theorem is applied to a concrete generated design.

E5 (Messages cannot cross syncs):

For every synchronization call s (in the source order used by R3 / pref_S / suff_S over Q_{exec},P):

$\forall q \in \text{pref_S}(s) : \text{issue_post}(q) \leq \text{clk_post}(s)$

$\forall q \in \text{suff_S}(s) : \text{issue_post}(q) > \text{clk_post}(s)$

$\forall q \in \text{pref_S}(s)$ with q a blocking Push/Pop : $m(q)$ is defined and $\text{clk_post}(m(q)) \leq \text{clk_post}(s)$.

The third clause is the E5/Core.SyncFence completion fence. Same-cycle completion is allowed; on the post side, the “accounted for before interval advance” convention is proof/certificate attribution for the same-cycle batch, not a requirement for observable physical sub-cycle ordering in RTL. Failed non-blocking polls in Q_{exec},P are constrained at the issue level by E5 but contribute no committed Σ -event when they fail.

This property is guaranteed per-process by the proposition hypothesis on the chosen compatible pair (with Appendix I supplying the witnesses in the execution-level Post $\rightarrow$ Ref use case) and requires no inter-process reasoning, so it lifts directly to the system level.

Conclusion:

All five compatibility properties hold at the system level for this chosen compatible trace pair. The composition is valid because:

- intra-process properties are preserved by the per-process compatibility hypothesis;
- inter-process communication respects the applicable E4 channel semantics by the per-channel Sys_ref / RTL_B well-formedness hypothesis; and
- no progress or fairness argument is needed inside Proposition J.0, because the lemma composes already-given traces.

The existence of the source trace used in the normal Post $\rightarrow$ Ref application is supplied by Theorem J.GWR-Comp; a J.GWR-Direct package supplies the alternate entry route. Therefore $\tau_{\text{ref},\text{system}} \approx_{\text{sys}} \tau_{\text{post},\text{system}}$. $\square$

Remark 1. When Sys_ref = Sys_B and all $B(c)=0$, Sys_ref coincides with the rendezvous interpretation of Sys, so Proposition J.0 and Theorem J.1 both specialize to the rendezvous case. In the elaborated cases, the source-side proof target is Sys_ref = Sys_B^elab.

Remark 2 (Reference-model ladder: raw Sys, abstract Sys_B, and elaborated Sys_B^elab).

Definition Core.1 is the normative definition of Sys_ref. This remark explains the model ladder and back-annotation intent. Sys is the raw rendezvous interpretation of the user-written processes. Sys_B is the

ordinary bounded-FIFO interpretation with primitive E4 boundaries and finite B(c). $\text{Sys_B}^{\text{elab}}$ is used when the RTL capacity/handshake structure cannot be represented faithfully by isolated primitive boundaries: it refines the same source processes with explicit finite-capacity/rendezvous channels plus stateless deterministic ready/valid/data couplers for staging, joins, bundles, shared-capacity realization, and sync/epoch gates. A coupler may be located at an internal elaborated boundary or at a process-facing outer boundary. It contributes no stored state and is not a modeled process.

The formal target of Appendices J–K is the prescribed Sys_ref selected by the theorem instance. For $\text{Sys_B}^{\text{elab}}$, the outer Σ_{DUT} projection and E4 occupancy/accounting remain fixed by $\Pi_{\text{elab}}/\pi_{\Sigma,E}$, while the actual ChannelEnabled/ChannelDisabled value at a coupling-sensitive boundary is the settled ready/valid result of E’s declared combinational network. Thus elaboration does not add an arbitrary hidden guard; it makes the simple hardware handshake structure part of the source reference model. Appendix Q gives the well-formedness, settling/reconstruction, WFG dependency-projection, and serial-route template obligations for that structure.

Operational note. In all three cases, Sys_ref is a transition system with explicit process states, explicit abstract channel states, operation-attributive pending requests, and ϵ -microstate as defined in Appendix G’s operational interpretation of Sys_ref . Sys_B and $\text{Sys_B}^{\text{elab}}$ differ in the channel/elaboration component of that transition system, not in the ordinary dynamic source order of the user-written process code. Source-level message-call instances that appear in sequence in a process remain ordered by $<_{\text{src}}$; elaboration may introduce additional explicit proof-internal frontier items and abstract channels, but it does not make ordinary source-sequenced calls incomparable.

Note that in all cases the elaborated structure (staged/bundle/join channels, combinational couplers, and the projection map Π_{elab}) introduced for a particular pipelined realization is local to that process’s side of the abstract channels at its boundaries. The complementary endpoint processes see only the abstract Σ -visible channels at the boundary and are unaffected by the elaboration choices.

For Lean-style formalization, the phrase “chosen Sys_ref ” should be read as an explicit theorem parameter: either Sys_B itself, or $\text{Sys_B}^{\text{elab}}(E)$ for a well-formed elaboration package E as defined in Appendix Q. Likewise, references to the Appendix I/L witness machinery assume that the actual WC premise has been discharged. The corresponding audit entry records whether WC was proved through NB-CF, an Appendix L snooping / witness-selector construction satisfying Lemma L.2, another direct witness construction, or complete mixed per-site coverage. NB-CF and Appendix L snooping are provenance routes for proving WC, not independent theorem premises alongside WC.

Side-condition audit before theorem application. Applying Theorem J.1, Corollary J.1, or Appendix K to a concrete design requires the side-condition discharge record in Appendix M.7a. The normal $\text{Post} \rightarrow \text{Ref}$ route shall identify Sys_ref , discharge WC, and provide Definition J.LocalGWR: complete J.PCert, J.CCert, J.ACert, environment/reset, construction-action inventory, typed footprint, request-snapshot, locally generated Dep_J , LocalAgree, and coverage evidence, together with per-channel B-faithfulness. J.CanonicalRank is derived by the theorem. A flow may instead record J.GWR-Direct together with its direct-provenance J.ValidatedConstructionOrder and independently checked V1–V6 obligations. When Corollary J.1 or Appendix K is used, the application must additionally discharge J.OfferReach, J.OfferConf, and J.KObsConf for the same theorem-produced or directly supplied W, source prefix, RTL prefix, H, O, and cutpoints. Liveness uses the additional K.Drain, K_{ϵ} , K.CV, A5-L, selected A5-L discharge route, progress, NB-Align, and L.Dir obligations as applicable. Ordinary synthesis does not automatically discharge any of these named conditions.

Remark 3 (Practical rendezvous liveness screening under determinism).

When B1 holds and the design’s control flow does not branch on empty/full observations of non-blocking interfaces (i.e., it depends only on the ordered history of observable Push/Pop/synchronization actions, not on “probe” outcomes (NB-CF, Appendix I.2)), the rendezvous specialization $B(c)=0$ is often an effective practical screen for design-level deadlocks: it exercises the most constrained communication discipline and can expose true cyclic data-dependency deadlocks early. Because rendezvous is strictly more constrained than buffered/elaborated operation, it can also yield false positives: a deadlock observed under $B(c)=0$ may disappear once finite buffering or explicit elaborated staged/bundle/join structure is present. However, bounded buffering can still introduce capacity-induced deadlocks (FIFO-depth artifacts) when FIFO depths are underestimated; these deadlocks are real for that modeled capacity configuration but may disappear once capacities are increased to the intended design point. Increasing buffer capacity (or applying dynamic buffer-growth strategies) is a standard way to distinguish depth-limited deadlocks from true cyclic dependency deadlocks. Accordingly, the formal results of Appendices J–K deliberately target the prescribed source reference model Sys_ref of Remark 2, with the actual back-annotated capacities and explicit abstract boundaries required by the chosen realization, rather than relying on rendezvous alone.

Remark 4 (When raw Sys may be used instead of the prescribed source reference model for safety-only checks).

If verification is concerned only with safety properties over the chosen DUT-boundary projection Σ (denote it Σ_{DUT}) (and not with deadlock/liveness), the prescribed source reference model Sys_ref of Remark 2 remains the default reference model. This is because Σ_{DUT} includes committed channel-transfer events, and bounded buffering / explicit staged-bundle-join structure generally changes the set/timing of those committed Push_c/Pop_c events relative to the raw rendezvous case.

Practical note (early bring-up): Before accurate capacities $B(c)$ are available (or before back-annotation/elaboration can be used), running the rendezvous specialization Sys against the intended testbench is still useful to validate functional intent and catch gross protocol/ordering bugs early; however, it remains a screening check, not a proof about the buffered/elaborated RTL-correspondent model.

Important (soundness): When any channel or explicit abstract staged/bundle/join boundary that can influence Σ_{DUT} -observable behavior has positive effective capacity or nontrivial enablement structure, the raw rendezvous model Sys is often a restriction of Sys_ref / RTL_B at Σ_{DUT} (it can admit fewer Σ_{DUT} behaviors). Therefore, using Sys in place of Sys_ref is generally suitable only as a debug/screening check: a failure can be useful diagnostically, but a pass does not by itself prove Σ_{DUT} -safety of the buffered/elaborated implementation.

Sys may be used as a proof-equivalent substitute for Sys_ref only when Sys and Sys_ref are Σ_{DUT} -equivalent for the chosen Σ_{DUT} (i.e., their Σ_{DUT} -projected trace sets coincide under the intended stimulus). A simple sufficient condition is that every Σ_{DUT} -visible explicit abstract channel/boundary in Sys_ref has effective capacity zero and that buffering/elaboration elsewhere is Σ_{DUT} -opaque (it cannot enable/disable/reorder Σ_{DUT} -events or change when Σ_{DUT} -events occur). In all other cases—i.e., if any Σ_{DUT} -visible explicit abstract boundary has positive effective capacity, or if buffered/elaborated internal channels can affect when Σ_{DUT} -events occur—use the prescribed Sys_ref of Remark 2.

Formal soundness condition. Let $\text{Traces}_{\Sigma_{\text{DUT}}}(\text{M})$ denote the set of Σ_{DUT} -projected traces of model M under the intended fixed stimulus. Sys may replace Sys_ref for proving a universal Σ_{DUT} -safety property ϕ only if $\text{Traces}_{\Sigma_{\text{DUT}}}(\text{Sys}) = \text{Traces}_{\Sigma_{\text{DUT}}}(\text{Sys_ref})$; under that condition, $\text{Sys} \models \phi$ iff $\text{Sys_ref} \models \phi$ (and otherwise Sys is only a diagnostic/screening check as stated above).

Lemma J.RS (MatchedResetEpochSplice).

Assume RW1–RW5 and the fixed-stimulus interpretation. Let H be a matched Theorem J.1 prefix satisfying $\text{Inv}(H)$ whose next unit in the selected validated construction-order enumeration is a RESET unit $\gamma_R = \text{RESET}(S_p, S_c)$, matched between RTL_B and Sys_ref with the same scope. Then Sys_ref extends from $\sigma_{\text{ref}}[H]$ by its corresponding reset transition, and $\text{Inv}(H \cup \{\gamma_R\})$ holds: (I1) holds because γ_R is matched with the same scope and the only required orders at the boundary are the epoch-boundary orders above; (I2) holds because, for every $P \in S_p$, the P -local replay state after γ_R is the deterministic epoch-initial state prescribed by RW3 — identical in both models by the reset-correspondence obligation — while for $P \notin S_p$ the replay state is unchanged by RW5; RW2 guarantees that no request-attribution fact survives the boundary for S_p , so Lemma I.DR' applies epoch-locally with γ_R delimiting the P -local projection; and (I3) holds because channels in S_c restart with reset-empty histories per RW1 and the Core reset-empty FIFO convention — the $\text{occ}_c(t)$ definition is already measured from the most recent reset boundary — while channels outside S_c retain their histories by RW5. ■ Subsequent induction steps within the new epoch proceed exactly as in the initial epoch; START_P for $P \in S_p$ is matched as the new epoch's first ordinary unit. Open dispositions that are not covered by RW1–RW5 — for example, an un-modeled partial reset of one endpoint of a live channel, retention of in-flight messages across the boundary, or cross-epoch request re-attribution — place the design outside Theorem J.1, Corollary J.1, and Appendix K until modeled explicitly. Under the Core.CycleResult interface, γ_R is the emitted resetUnits occurrence corresponding to that scoped reset transition; the induction index remains γ_R itself, not the enclosing cycle result. Thus the CycleResult refactor does not change the reset splice case or its epoch-local replay/channel-history obligations.

Expanded proof of Theorem J.GWR-Comp and Theorem J.1

Post $\rightarrow$ Ref, compositional route. Fix $\tau_{\text{post}} \in \text{Reach_post}$ and $\text{LocalGWRPackage}(\tau_{\text{post}}, L)$. Generate the complete field-sensitive action and deferred-hole inventory, Dep_J , FR, and $J.\text{CanonicalRank}$. Partition the complete units by their selected Horizon, retaining explicit atomic groups. Within each Horizon, stable proof keys order only otherwise commuting actions; $J.\text{RegPhaseNF}$ later removes dependence on those incidental choices modulo $J.\text{HCanon}$.

Let $B_1 \dots B_n$ be the nonempty Horizon buckets and H_i the finite required-prefix ideal containing all units through B_i ; within B_i retain the commit-action suborder only for the inner L3 pass. Construct by outer induction reachable cycle-entry and post-boundary states. The invariant is: (I1) exactly H_i has committed with matching units/values/identities and required order; (I2) process replay and every pending/deferred-hole record agree with the local certificates at cycle entry, while no deferred hole survives a post-boundary state; (I3) explicit boundary/channel/atomic state equals the CCert/ACert reconstruction; (I4) requests, witness choices, reset epochs, direct-input anchors, and environment history agree; and (I5) the constructed prefix through H_i satisfies $J.\text{RegPhaseReach}$ and has the same CanonHist quotient as the selected RTL prefix through those units.

Base case. Use the matched initial state, reset-empty channel state, empty request inventory, fixed environment history, and LocalAgree. For later dynamic resets, ResetCert and Lemma J.RS re-establish the same invariant per reset epoch while framing components outside the reset scope.

Gap bridge. The nonempty bucket list does not introduce empty $J.\text{CanonicalRank}$ buckets: Height and FR are ranks of construction actions, and an empty RTL Horizon contributes no action in $\text{Act}_L(\tau_{\text{post}})$. Before the next nonempty bucket, if the selected source witness, environment, reset-state contract, or other cycle-sensitive construction evidence requires a later source cycle-entry state, apply Core.SelectLatitude through the ordinary Sys_ref cycle semantics to obtain the finite evidence-determined sequence of unit-empty invariant-preserving CompleteCycle transitions required to reach that entry. Any no- Σ -commit bridge transition is a Core.SilentCycle. Each bridge leaves (I1) unchanged

because no selected unit commits; the typed full-state transition plus the existing local/channel/environment/reset and LocalAgree evidence preserves or re-establishes (I2)–(I4); and (I5) retains Policy-L/Core.Phase legality and the same CanonHist quotient because no selected J unit is added. The bridge length is not equated to the raw difference between consecutive RTL Horizon indices unless the selected cycle-sensitive evidence requires such alignment. Thus the construction advances through any required source-only latency without changing Dep_J, FR, J.DepRank/J.DepAcyclic, or J.CanonicalRank and without invoking B2/B3 fairness.

Induction step. Let B_t be the next Horizon bucket. Execute all finite L1 A1/A2 predecessors, including actions carried past unresolved NB calls under NoDependentUse. Field-sensitive framing and J.DepComplete permit the required commutations; true dependencies remain ordered. J.RegPhaseNF then freezes the complete request set, performs the L2 settle, and orders all A3/A4 decisions/commits in L3. A failed NB writes its returned status and retires its hole; a successful NB is one of the bucket's complete units. No decision-dependent continuation remains in t, and no new request is issued after settle.

Every witness-significant failed or successful NB call is therefore evaluated from one common settled incident-boundary snapshot. The construction cannot choose failure with a matching rendezvous request present, success in an empty/full state that forbids it, or an outcome for the wrong dynamic call. WC identifies the selected result; J.CCert/J.NBDecision prove it unique. Outcome-independent same-Horizon work is already in L1; outcome-dependent work begins only in a later bucket.

J.UnitEnable proves every complete unit in the bucket enabled at the appropriate L3 point. The buffered case uses common pre-update occupancy/head and J.BufSameCycle; rendezvous uses simultaneous attributed requests/attempts and payload identity; synchronization/E2/R2C/environment/reset use their named certificates. Executing the inner commit-action order reaches the post-boundary state for H_iUB_t. The phase-normal segment has the same complete units, same post-boundary semantic state, and same J.HCanon quotient as any permitted same-Horizon topological ordering.

Lemma I.DR' updates participant replay state; J.CCert/ACert update exact channel/atomic state; J.Frame preserves nonparticipants; LocalAgree preserves coherence; and the all-holes-resolved invariant establishes the clean post-boundary construction state. Thus (I1)–(I5) hold for the next bucket. Finiteness follows from the local certificates/FPD, not fairness.

After the final Horizon bucket, the chain covers every unit of τ_{post} . Assemble W from the local certificates, generated graph, derived bucketed order, phase-normal segments, snapshots, and invariants. Definition J.GWR and J.RegPhaseReach hold. The terminal source prefix and τ_{post} have compatible annotated projections and well-formed explicit-channel histories, so Proposition J.0 proves Theorem J.1's compositional Post $\rightarrow$ Ref clause.

Post $\rightarrow$ Ref, direct route. If a validated W satisfying Definition J.GWR is supplied directly, take the reachable terminal source prefix from its chain and apply Proposition J.0 to the annotated pair. This route has the same semantic conclusion but bypasses the local-certificate derivation.

Normalization and cutpoints. Neither route appends a maximal ε -normalization suffix after each matched unit. Corollary J.1 forms the RTL K-facing endpoint through the recorded NormPost/Core.QSync-Post normalization. On the source side it retains the J.GWR construction prefix $\tau_{\text{ref}}^{\wedge}G$ and separately forms the K-facing endpoint through the recorded Core.KObsNormRef suffix v_{ref} , with K.NormStable supplying the required request/replay/NB/reset invariance. Generic B4/B5 normalization arguments may discharge their independently stated bounded-progress uses, but they do not by themselves establish the post-L3 L3Obs placement or observation-only restrictions of Core.KObsNormRef. The theorem-produced W remains the authoritative source of the operational identities and history consumed by these obligations.

Ref $\rightarrow$ Post selected-pair clause. Let $\tau_{\text{ref}} \in \text{Reach}_{\text{ref}}$ and choose $\tau_{\text{post}}^* \in \text{Reach}_{\text{post}}$ and W with $\text{RTLConsistentRefPrefix}(\tau_{\text{ref}}, \tau_{\text{post}}^*, W)$. By definition, the selected pair already carries compatible

annotated projections, Issue/Commit and request-attribution evidence, dynamic identities, WC/ μ_Q data, and E4 histories. Proposition J.0 therefore gives $\tau_{\text{ref}, \text{system}} \approx_{\text{sys}} \tau_{\text{post}^*, \text{system}}$; take $\tau_{\text{post}} = \tau_{\text{post}^*}$. B1 supplies uniqueness only of the selected RTL Σ projection, not of the annotation package. This is not a construction from an arbitrary source run. $\square$

Lemma J.HCanon (Canonical replay-history quotient agreement)

Fix the same theorem instance I, elaboration E, witness w, reset interpretation, dynamic operation identities, and theorem-produced or directly validated J.GWR annotation package. Let τ_{post} be a finite RTL_B prefix and τ_{ref} the Theorem J.1 source witness prefix selected for τ_{post} . If τ_{ref} and τ_{post} satisfy the annotated observable-unit compatibility relation of Theorem J.1 under that fixed package, then their canonical enriched replay-history quotients are equal:

$\text{CanonHist}_{E,w}(\tau_{\text{ref}}) = \text{CanonHist}_{E,w}(\tau_{\text{post}})$.

Proof. The compatibility relation matches the same atomic observable units with the same labels, values, dynamic identities, and reset scopes. The fixed witness w and μ_Q /WC package identify the same witness-selected failed non-blocking, arbitration, signal, and ϵ -choice outcomes used by replay. Theorem J.1 preserves each process-local required order, each per-channel committed-transfer ordinal/FIFO order, every explicit atomic rendezvous or paired-synchronization grouping, every RESET marker, and every predecessor required by $<_J$. CanonHist retains exactly those fields. It discards only incidental source-versus-RTL clock-bucket placement and ordering among units that are independent under the retained constraints. Therefore both annotated prefixes map to the same typed quotient H. This lemma asserts equality of the canonical replay quotient, not equality of concrete cycle buckets or historical Issue timestamps. $\blacksquare$

Definition J.RegPhaseReach (phase-correct reachability of the selected source witness)

For fixed I, E, w and a selected J.GWR source prefix τ_{ref}^G ending at s_{ref} , $\text{RegPhaseReach}_{I,E,w}(\tau_{\text{ref}}^G, s_{\text{ref}})$ means that τ_{ref}^G is Policy-L-legal under Core.RegSeq and admits the Horizon-bucket decomposition of J.RegPhaseNF: every cycle's process-owned A1/A2 actions occur in L1, one L2 settle fixes the requests/snapshots, all A3/A4 decisions/commits occur in L3, ReturnedAtCycle-dependent work occurs only in later Horizons, and no deferred NB hole remains at s_{ref} . A J.GWR-Direct package must supply this checked property for its selected construction prefix. RegPhaseReach establishes the L0–L4 ordering/legality component for the J.GWR-constructed prefix; it does not assert that s_{ref} is a Core.KCut and does not establish nonwithdrawal, $K\epsilon$, K.NormStable, or the continuation-level Core.Drain/K.Drain discipline.

Corollary J.RegPhaseReach-Comp (compositional derivation of phase-correct reachability)

Under the hypotheses of Theorem J.GWR-Comp, every constructed source prefix τ_{ref}^G ending at s_{ref} satisfies $\text{RegPhaseReach}_{I,E,w}(\tau_{\text{ref}}^G, s_{\text{ref}})$. Proof. Lemma I.3 carries each enriched deferred NB hole through exactly the same-Horizon work certified by NoDependentUse; J.RegPhaseNF normalizes each Horizon modulo J.HCanon into L1 process actions, one L2 settled snapshot, and L3 decisions/commits, using field-sensitive framing and J.BufSameCycle; and Horizon-batched J.UnitExt preserves that normal form while constructing the reachable prefix. ReturnedAtCycle-dependent work is assigned only to later Horizons, and the exported-construction-endpoint invariant removes every unresolved deferred hole at s_{ref} . No K-facing normalization is appended by this corollary. $\blacksquare$

Definition K.NormStable (K-facing stability of the source observation normalization).

Fix I, E, w, a J.GWR source prefix τ_{ref}^G ending at s_{ref} , and a recorded Core.KObsNormRef_I($\tau_{\text{ref}}^G, v_{\text{ref}}, \sigma_{\text{ref}}$); write $\tau_{\text{ref}}^K := \tau_{\text{ref}}^G \cdot v_{\text{ref}}$.

$\text{KNormStable}_{I,E,w}(\tau_{\text{ref}}^G, v_{\text{ref}}, \sigma_{\text{ref}})$ holds when the checker validates the following primitive facts for the exact v_{ref} segment: (NS1) it is the post-L3 observation-only Core.KObsSettle path recorded by

Core.KObsNormRef, with the L3 action set already closed and no L4 update; (NS2) after the WFG-relevant request/attribution roots are fixed, $K\epsilon_{\{I, \text{Sys_ref}\}}$ applies throughout the segment, and no source-level issue/attribution step creates, removes, withdraws, or reassigns a pending blocking request; (NS3) no witness-significant PushNB/PopNB attempt or A3-style success/failure decision is initiated, resolved, retired, or otherwise changed, and no deferred NB hole is created or discharged; (NS4) no witness-significant arbitration, signal, failed-NB, or other ϵ -choice record retained by Core.18a is created, removed, or changed; (NS5) no selected Σ -visible or proof-internal observable unit occurs and the reset epoch does not change; and (NS6) the no-unresolved-hole invariant that holds at s_ref is preserved to σ_ref . These are the primitive checker obligations. Canonical-history equality is a derived consequence below rather than an independently trusted certificate field.

Lemma K.NormStable-H (observation normalization preserves the enriched canonical history).

If $K\text{NormStable}_{I,E,w}(\tau_ref^G, v_ref, \sigma_ref)$ and $\tau_ref^K := \tau_ref^G \cdot v_ref$, then $\text{CanonHist}_{E,w}(\tau_ref^K) = \text{CanonHist}_{E,w}(\tau_ref^G)$. Proof. NS5 excludes new selected observable/proof-internal units and reset changes; NS3 excludes any new or resolved witness-significant non-blocking outcome/deferred-hole event; and NS4 excludes every remaining arbitration, signal, or ϵ -choice field retained by Core.18a. NS2 preserves the request/frontier attribution layer but is not used as a substitute for the Core.18a field checks. Therefore the finite enriched replay-history quotient is unchanged across v_ref . $\square$

Definition J.OfferReach (reachable source realization of a KObs record)

Fix I, E, w, H, O , and the particular Sys_ref J.GWR witness prefix τ_ref^G selected for the current Theorem J.1 / Corollary J.1 application. $\text{OfferReach}_{I,E,w}(H,O;\tau_ref^G, v_ref, \sigma_ref)$ holds when τ_ref^G is a finite reachable Sys_ref prefix under the same stimulus; the package records $\text{Core.KObsNormRef}_I(\tau_ref^G, v_ref, \sigma_ref)$; $\tau_ref^K := \tau_ref^G \cdot v_ref$ is therefore a finite reachable source prefix ending at $\sigma_ref \in \text{KCutReach_ref}(I)$; $H = H_{E,w}(\sigma_ref)$; $O = \text{ResOffers}_{E,w}(\sigma_ref)$; the Issue/ReturnedAtCycle/Commit, E3/E5, Core.IC, reset, elaboration, and explicit-boundary rules hold for that exact K-facing prefix; and no proof-internal deferred NB hole is exported at σ_ref . OfferReach remains an image/realizability predicate, not a Core.LAdm qualification. $K.\text{NormStable}$ is a separate stability obligation connecting the J.GWR history/phase endpoint to this K-facing normalization. OfferReach is an image/realizability predicate. It is not implied by type-correctness, source-prefix closure, or endpoint cardinality of an arbitrary tuple $\langle E, w, H, O \rangle$. In Corollary J.1 it is discharged for the K-facing observation-normalized extension τ_ref^K of the exact τ_ref^G selected by Theorem J.1, not for an unrelated existential source state. A source execution log, mechanized reachability proof, or tool/witness certificate may discharge it. The safety-only Theorem J.1/J.QSyncTraceState path continues to use τ_ref^G and does not require $\tau_ref^K, \sigma_ref, \text{OfferReach}$, or $K.\text{NormStable}$. Historical Issue and ReturnedAtCycle data may be used internally to prove reachability, phase legality, and the current residual-offer set, but they are not KObs fields and are not compared as cross-model cycle timelines. $J.\text{RegPhaseReach}$ records the theorem-derived/direct L0–L4 phase property at the J.GWR construction endpoint s_ref ; $\text{Core.KObsNormRef}/K.\text{NormStable}$ qualify the observation-only extension to the K-facing endpoint σ_ref ; OfferReach records the realized H/O cutpoint there; and $K.\text{DrainReach}$ records the remaining Appendix K continuation discipline on the exact K-facing prefix τ_ref^K . $K.\text{RLAdmReach}$ composes those three distinct layers into the $\text{Pref_L}^{\text{adm}}(I)$ qualification.

Definition K.DrainReach and K.RLAdmReach (composed Core.LAdm qualification)

Fix $I, E, w, H, O, \tau_ref^G, v_ref, \sigma_ref$ satisfying $J.\text{OfferReach}_{I,E,w}(H,O;\tau_ref^G, v_ref, \sigma_ref)$, let s_ref be the endpoint of τ_ref^G , and define $\tau_ref^K := \tau_ref^G \cdot v_ref$. $\text{DrainReach}_{I,E,w}(H,O;\tau_ref^K, \sigma_ref)$ holds

when the exact K-facing prefix and instance binding satisfy the Sys_ref request-nonwithdrawal obligation and $K\varepsilon_{\{I, \text{Sys_ref}\}}$; discharge Core.Drain on Sys_ref from every L2 fully-disabled-frontier SettlePoint reached by τ_ref^K ; and discharge Core.SettleKBridge for every represented activation-to-KCut path ending at σ_ref , including no-new-later-work, the finite non-replenishing already-admitted action domain/Core.InFlightDrainWF, exact reached in-flight action accounting, sync-boundary, attribution, and ε -inertness obligations. RLAdmReach_I,E,w(H,O; $\tau_ref^G, v_ref, \sigma_ref$) is the three-part composition of (RLA1) J.RegPhaseReach_I,E,w(τ_ref^G, s_ref), (RLA2) KNormStable_I,E,w($\tau_ref^G, v_ref, \sigma_ref$), and (RLA3) DrainReach_I,E,w(H,O; τ_ref^K, σ_ref), all under the same fixed I/E/w/stimulus/environment/arbitration/reset choices. The K.NormStable bridge transports the phase/no-hole/request/history invariants across the observation-only suffix without adding a scheduling action; therefore the exact prefix that reaches the K-facing cutpoint, τ_ref^K , satisfies every finite-prefix clause of Core.LAdm and RLAdmReach proves $\tau_ref^K \in \text{Pref_L}^{\text{adm}}(I)$. Neither equal KObs nor OfferReach alone implies K.NormStable or DrainReach. A K-facing CF-0 package shall carry/refer to all three components and the exact v_ref. On the standard normal-compositional route, Lemma K.DrainReach-Comp below combines the independently instantiated Sys_ref K.DrainEvidence/DR1–DR4 fields, including Core.IC/nonwithdrawal, $K\varepsilon_{\{I, \text{Sys_ref}\}}$, Q.2 action inventory, Core.InFlightDrainWF, and finite-drain/channel evidence with the theorem-derived J.RegPhaseReach on τ_ref^G , K.NormStable on v_ref, and the same-package attribution/history/Core.QSync/Core.KObsSettle facts; absent those conditions, direct cutpoint-specific validation remains required.

Lemma K.DrainReach-Comp (normal-route compositional discharge of K.DrainReach)

Fix the standard normal-compositional instance I and the exact source package ($\tau_ref^G, v_ref, \sigma_ref, E, w, H, O$) selected by J.GWR-Comp/Corollary J.1 and satisfying J.OfferReach; write $s_ref := \text{last}(\tau_ref^G)$ and $\tau_ref^K := \tau_ref^G \cdot v_ref$. Require J.RegPhaseReach_I,E,w(τ_ref^G, s_ref) and KNormStable_I,E,w($\tau_ref^G, v_ref, \sigma_ref$); on the normal route Corollary J.RegPhaseReach-Comp supplies the former and the Core.KObsSettle-Ref/checker package supplies the latter. Assume that, for this fixed instance and witness binding: (DR1) the Sys_ref K.DrainEvidence activation field covers every Core.16 activation a reached on the exact constructed prefix τ_ref^G and records one route for each activation, with different activations permitted to use different routes. (DR1a, RTL-backed activation) For a pipelined or otherwise overlapped activation, Appendix-B automatic-flush conformance supplies a bound RTL_B fact for the corresponding region/cycle and the same dynamic fully disabled blocking frontier/request attribution: once that frontier has settled, no new source-later request/operation is launched past it and only already accepted finite work may drain. J.LocalGWR/J.GWR-Comp transfers that restriction to activation a on τ_ref^G : I.AO/E3 and the PCert/ConstructionAction identity make the constructed L1 A2 request assertions/retentions the matched source operations of the bound RTL issue/attribution package and preserve issue-prefix closure; any L1 A1 preparation that would itself count as the broader Core.16(2) source-later launch is covered by the same K.DrainEvidence construction-action check. If those existing J/I fields do not establish that the RTL flush fact and a name the same dynamic blocked frontier, the application supplies a named activation-correspondence theorem/certificate. (DR1b, direct-source activation) For an ordinary non-overlapped activation, the record may instead contain the direct source structural proof of the same applicable Core.16 blocked-point/frontier, no-new-launch, frontier-minimality/non-frontier-exclusion, and drain-only clauses. K.NormStable closes the only segment not constructed by J.GWR-Comp: NS1–NS6 prove that v_ref is post-L3 observation-only and initiates no source operation, request/attribution change, NB decision, replay-significant choice, reset, or L4 update, so it creates no new L2 Core.16 activation and extends the DR1 no-new-launch result from τ_ref^G across v_ref to τ_ref^K . Thus every L2 fully-disabled-frontier activation reached by τ_ref^K is covered rather than being filtered out after construction. (DR2) Core.IC/Appendix C supplies Sys_ref request nonwithdrawal. (DR3) $K\varepsilon_{\{I, \text{Sys_ref}\}}$ holds on the

applicable intervening normalization segments, including the request-frozen v_ref segment through $K.NormStable$. (DR4) E4/B-faithfulness, the selected Sys_ref Q.2 finite action inventory, finite explicit storage/credit, and $Core.InFlightDrainWF$ establish a finite activation-local already-admitted population and a well-founded rank/multiset that strictly decreases on each selected in-flight drain action; $Core.16(2)$ supplies non-replenishment. When the selected source-side G2 schema includes a pending/ready synchronization carrier, $SyncFire$ is part of that Sys_ref action inventory and its rank/opportunity obligations are discharged through the Stage-2 synchronization qualification; lowering-only $EpochGateAdvance$ remains a Sys_K action and is not part of this Sys_ref $DrainReach$ proof. Any applicable same-interval synchronization-boundary withdrawal still satisfies $Core.16(7)$. DR1, DR2, and DR4 together are the completed per-activation $Core.16$ discharge fields of $K.DrainEvidence$ for this exact prefix; DR3 supplies the separately required model-indexed ϵ -inertness used by $K.DrainReach/Core.SettleKBridge$. No separate DR5 hypothesis is required: $J.GWR-Comp/I.A0$ supplies dynamic-operation and boundary attribution through $\tau_ref^A G$; $J.HCanon$ fixes H on $\tau_ref^A G$; Lemma $K.NormStable-H$ transports that same H across v_ref to $\tau_ref^A K$; and $J.OfferReach$ binds $H=H_E, w(\sigma_ref)$ and $O=ResOffers_E, w(\sigma_ref)$. By $K.NormStable$ NS1 and NS5, v_ref is observation-only and contains no selected Σ -visible or proof-internal observable unit; therefore every intervening selected in-flight action occurrence on the exact constructed prefix is represented by the corresponding $H/E4/E$ proof-model unit before v_ref ; NS1/NS5 ensure v_ref itself contains no such action occurrence. Then $DrainReach_I, E, w(H, O; \tau_ref^A K, \sigma_ref)$ holds, including $Core.SettleKBridge$ for every persisting activation through the represented finite drain to the exact selected K -facing cutpoint σ_ref . Proof. $J.RegPhaseReach$ supplies Policy-L legality and the phase-normal constructed prefix. The activation-indexed DR1a/DR1b field supplies the path-indexed $Core.16$ blocked-point/frontier, no-new-launch, frontier-minimality, and drain-only discipline on $\tau_ref^A G$; $K.NormStable$ extends the no-new-launch/no-hole/request/history invariants over the exact observation suffix without creating a new activation; DR2 and DR4 complete the $Core.16$ record, and DR3 supplies model-indexed $K\epsilon$. $Core.KObsSettle$ makes the final boundary valuation observational only and cannot create another L3 selection. Together with the finite alternating WFG-inert ϵ -segment/in-flight-action decomposition of $Core.SettleKBridge$, these facts establish every conjunct of $DrainReach$ on $\tau_ref^A K$. $\square$ This is a compositional lifting theorem only: it adds no scheduling capability or weaker liveness route, and it does not infer $DrainReach$ from $KObs$, $OfferReach$, or $K.NormStable$ alone. When this lemma is used after Theorem $J.GWR-CompExt$, the uniform instance-level $K.DrainEvidence/DR1-DR4$ proof schema is instantiated afresh on the enlarged $\tau_ref^A G$ and newly selected v_ref/σ_ref package. There is no $DrainReach$ or $K.DrainEvidence$ prefix-extension rule: an old fact on an earlier prefix is not transported merely by prefix inclusion, because new activations, commits, reset facts, and drain obligations may have appeared in the enlarged construction.

Definition J.RelevantReq (KObs-relevant RTL endpoint request)

At an RTL K -facing cutpoint $p_post \in KCutReach_post(I)$, an asserted request level r is $KObs$ -relevant for elaboration E when it is an own-side source-attributed boundary request selected by E and its presence or absence can affect Pending/BoundaryReq/Front or the settled handshake/WFG reconstruction. Relevance is determined from the selected boundary and operational attribution package, not from whether the flow happened to supply an attribution; an asserted relevant own-side request with missing attribution causes $J.OfferConf$ to fail. Stateless coupler-derived complementary mirror values are not independent residual offers and need not be added to O : they are reconstructed from E , the explicit channel state, and the attributed requests by $HandshakeRec_E$. $K.Resp$ timers remain outside $KObs$.

The dynamic operation/frontier/epoch attribution used below is supplied by $J.GWR-Comp$ or $J.GWR-Direct$ under Appendix C and I.A0; it is not inferred from wire level alone. A post-cycle request proved to belong only to an operation that committed in the selected cycle, with no next dynamic owner, is not a

residual offer. For a continuing assertion across a back-to-back handoff, the annotation identifies the next owning dynamic operation.

Obligation J.OfferConf (bidirectional RTL residual-offer conformance)

For an RTL K-facing cutpoint $\rho_{\text{post}} \in \text{KCutReach_post}(I)$, fixed E and w, and candidate residual-offer set O, $\text{OfferConf_I}, E, w(\rho_{\text{post}}, O)$ holds only if the following conditions are satisfied for every KObs-relevant own-side source-attributed request selected by E. A stateless coupler's derived complementary ready/valid output is checked through the elaboration/handshake conformance of J.KObsConf and is not required to appear as a separate offer. All attribution uses the theorem-produced or directly supplied operational package, Appendix C, and I.A0; no attribution is inferred solely from an asserted wire level.

(1) Offer-to-request completeness. For every $o = (P, x, q, c, \text{dir}, e) \in O$, the corresponding source-attributed own-side RTL request at c/direction dir is asserted at ρ_{post} , is KObs-relevant, and is attributed by the selected operational package to the same frontier item x and source operation q in reset epoch e, with operation kind derived from q. If E propagates that request through stateless coupler logic to another evaluated boundary, the propagated handshake value is derived by E and is not a new owning operation.

(2) Request-to-offer completeness and uniqueness. Every KObs-relevant source-attributed own-side RTL request asserted at the selected RTL K-facing cutpoint is attributed by the selected operational package, Appendix C, and I.A0 to exactly one member of O, unless the package proves that the level belongs only to an operation that committed in the selected cycle and no next operation owns the continuing assertion. A back-to-back handoff identifies the next dynamic owner. No relevant own-side request may be hidden, left unattributed, or omitted. Stateless complementary ready/valid values computed by E are deliberately excluded from this one-offer-per-request inventory; their completeness is instead the boundary-network completeness/reconstruction obligation of E/J.KObsConf.

(3) Snapshot issue-order consistency. The issued-summary represented by the committed/replay history H, the WC witness information, and the current residual offers O is prefix-closed under the applicable dynamic source order and is consistent with E3, E5, reset epochs, and the selected elaboration. This is a cutpoint set invariant; OfferConf does not compare historical first-issue timestamps across models.

(4) Boundary and reset typing. Every x, q, c, direction, kind, owner, and epoch in O is supplied by E and agrees with the live RTL boundary attribution. Reset clears ending-epoch offers as required by RW2. Internal movement already represented as E4 occupancy is not also exposed as a second source-level offer.

(5) Payload scope. O need not contain pending Push payload values. Stable request payload is still required by Appendix C/Core.IC, and committed payload equality is checked by the value-labeled replay history. A flow that needs mid-flight data-wire correspondence may add it as a separate implementation obligation.

Audit-critical status of J.OfferConf(2). J.OfferConf is the authoritative RTL abstraction boundary for current source-attributed own-side blocking requests. Omitting such a request can remove a frontier member or a root of the settled handshake dependency network and invalidates J.OfferConf/J.KObsConf. Conversely, a stateless coupler output must not be fabricated as a separate source offer merely to make the graph work: its ready/valid value and process dependency are reconstructed from E and checked by the elaboration/handshake conformance. B-faithfulness/E4, canonical-history agreement, and the declared combinational-network check remain separate independent obligations.

Obligation J.KObsConf (composite KObs cutpoint conformance)

For every RTL K-facing cutpoint $p_post \in \text{KCutReach_post}(I)$ used by Corollary J.1 or Appendix K, the flow shall select one exact package $(\tau_ref^G, v_ref, \sigma_ref, p_post, I, E, w, H, O)$, with $\tau_ref^K := \tau_ref^G \cdot v_ref$, such that: (a) source and RTL use the same fixed instance, elaboration, witness, stimulus, reset interpretation, and observation boundaries; (b) $J.HCanon$ yields H as the common canonical replay-history quotient of τ_ref^G and the matched RTL prefix; (c) Core.KObsNormRef and $K.NormStable$ hold for $(\tau_ref^G, v_ref, \sigma_ref)$, and Lemma $K.NormStable-H$ transports that same H to τ_ref^K/σ_ref without changing any Core.18a -retained field; (d) $J.OfferReach$ realizes O at σ_ref on that exact K-facing extension; (e) $J.OfferConf$ identifies O bidirectionally with the complete $KObs$ -relevant source-attributed own-side RTL request inventory; and (f) the selected RTL boundary mapping satisfies $E4/B$ -faithfulness, reset correspondence, ϵ -opacity, and the E -declared stateless handshake/release-support conformance, so evaluating the same qualified template from the reconstructed channel/request roots gives the same settled mirror values, ReleaseOwnerSets family, WaitOwners union, and coarse WFG. When the generalized drain profile is used, the package also binds the exact Q.2 in-flight action/effect schema identity/version whose current-state carrier is reconstructed from that same $E/H/O$ record; this binding does not assert future RTL/source path equivalence, which is supplied separately on the source side by $J.InFlightPathSound$. $J.KObsConf$ is tied to the particular τ_ref^G witness selected by Theorem J.1 and its recorded K-facing observation normalization; an unrelated source cutpoint or unrealizable request record does not discharge it.

Define the J-facing RTL observation certified by that package as $\text{KObsPost}_J(p_post; E, w, H, O) := \langle E, w, H, O \rangle$. $J.HCanon$ plus Lemma $K.NormStable-H$ identify H with the canonical enriched history through τ_ref^K , and $J.OfferReach$ plus Definition Core.18c give $\text{KObs}_E, w(\sigma_ref) = \langle E, w, H, O \rangle$. Hence $J.KObsConf$ entails $\text{KObsPost}_J(p_post; E, w, H, O) = \text{KObs}_E, w(\sigma_ref)$. This typed equality is the sole cutpoint-level correspondence used by Appendix K. It is not equality of arbitrary RTL and source microstates, and it creates no independent clause-based state relation.

J.2a KObs Reconstruction Theory

The following lemmas are source-side reconstruction results over reachable Core.18c $KObs$ records. They are placed here because Corollary J.1 consumes them after $J.OfferReach$ and $J.OfferConf$ establish one common typed record across the selected source and RTL cutpoints.

Lemma J.KObs-Proc (Process replay reconstruction).

Let $\sigma \in \text{KCutReach_ref}(I)$ under I, E, w and let $K = \text{KObs}_E, w(\sigma)$. For every process P , $\text{ProcRec}_P(K)$ equals the $I.\text{ReplayObs}$ source slice of σ used by Lemmas $I.DR/I.DR'$ and the Appendix J finite-ideal construction. Consequently $\text{DoneRec}_P(K)$, $\text{RemainRec}_P(K)$, $\text{NextFrontRec}_P(K)$, all replay-derived next-item selection predicates, all values needed to determine future committed Σ labels, and P 's normal-completion status agree with their operational definitions at σ . ProcRec does not reconstruct Pending_P , Front_P , or BoundaryReq ; those current offer facts are supplied by O and proved separately by Lemma J.KObs-Offers. Proof. Project H to P , include the WC /witness choices and $RESET$ epoch markers, and apply deterministic replay $I.DR/I.DR'$; the replayed process state together with Definition Core.6 determines completed/remaining items and candidate NextFront . $\square$

Lemma J.KObs-SyncCarrier (Synchronization-carrier reconstruction).

Let $\sigma \in \text{KCutReach_ref}(I)$, $K = \text{KObs}_E, w(\sigma)$, and assume $\text{Core.SyncCarrierWF}_I$. For every process P and synchronization-call frontier item $x \in \Omega_sync, P$, $\text{SyncCarrierRec}_P(K)$ agrees with the corresponding operational Core.7b/Core.7c synchronization carrier at σ : endpoint-call identity, dynamic SyncTxn key, reset epoch, participant set, reached/pending/ready/blocked state, and the exact minimal internal $\text{SyncReleaseOwnerSets}$ family are equal. Proof. Apply Lemma J.KObs-Proc to P and to every modeled

process in $\text{SyncParticipants_I}(\text{SyncTxn_I}(\text{SyncOf_P}(x)))$; the resulting ProcRec slices equal their operational replay/control slices at σ , including the current synchronization intervals and replay-derived dynamic item occurrences. Core.18a retains every committed synchronization unit and RESET marker, while the fixed Core.SyncCarrierWF transaction schema determines the pending dynamic transaction key/participant matching from those replay occurrences and the reset epoch without introducing a historical Issue timestamp. SC3 reconstructs current reached/pending/readiness from the complete declared participant state and, when explicitly required, the typed EnvReplayAdequate environment reconstruction; SC4 supplies exact minimal internal release supports. No residual-offer fact from O and no message BoundaryReq is used. $\square$ Scope. This is a state/carrier theorem. Stage 2B separately reconstructs the local SyncFire action/effect relation; neither theorem by itself supplies a future CycleResult or path opportunity.

Lemma J.KObs-SyncInFlight (source synchronization action/effect reconstruction).

Let σ, K be as in Lemma J.KObs-SyncCarrier and assume the Stage-2B Q.2 synchronization action schema. For every currently reconstructed transaction key κ , the local reconstructed predicate/effect $\text{SyncFireRec_I}(\kappa, \text{Core.FlightStateRec_}\{\text{Sys_ref}, E\}(K))$ agrees with Definition Core.7d at σ : it is exposed only when the transaction is already pending and SyncReady , all participant identities and Core.SyncFence preconditions agree, and no additional source ReachPrep/Issue is needed to make it applicable. If the operational SyncFire action occurs, the reconstructed successor marks exactly the same synchronization items realized, appends the same atomic synchronization/E2-anchored unit to H , advances the same participant replay slices to the immediate post-sync interval entry, retires the same carriers, frames message O /channel state as specified, and obtains the same deterministic re-settled roots. Proof. Combine J.KObs-SyncCarrier with J.KObs-Proc, the existing synchronization-pair/atomic-unit semantics, Core.SyncFence, and Q.SyncInFlightActionEffectSound. $\square$ This remains local action/effect reconstruction; Core.SyncInFlightOpportunityQual/J.InFlightPathSound owns same-/cross-cycle realization.

Lemma J.KObs-Chan (Abstract channel reconstruction).

Let σ and K be as in Lemma J.KObs-Proc. For every explicit storage/rendezvous channel c declared by E , $\text{ChanRec_c}(K)$ equals the E4 abstract channel state of σ . For $B(c) > 0$, occupancy and logical contents are uniquely obtained from reset/initial contents and the committed Push/Pop sequence and per-channel ordinals in H ; for $B(c) = 0$ occupancy is identically zero. In $\text{Sys_B}^{\text{elab}}$, the same statement holds at each explicit storage boundary and Π_{elab} supplies the declared outer accounting. This lemma reconstructs stored channel state only; the current settled ready/valid values of coupling-sensitive boundaries are reconstructed separately by HandshakeRec_E from E , ChanRec , ProcRec roots where permitted, and O . Proof. Induction over the committed channel-history projection of H using E4, B-faithfulness, RESET/RW1–RW3, and Π_{elab} conservation. $\square$

Lemma J.KObs-Offers (Residual-offer, BoundaryReq, and frontier reconstruction).

Let σ and $K = \langle E, w, H, O \rangle$ be as above. For every process P and blocking message-operation frontier item x , $x \in \text{Pending_P}(\sigma)$ iff $x \in \text{PendingRec_P}(K)$, and $\text{BoundaryReq}(x, \sigma)$ iff $\text{BoundaryReqRec}(x, K)$; hence $\text{Front_P}(\sigma) = \text{FrontRec_P}(K)$. O is the exact set of current source-attributed own-side blocking requests selected by E . For an ordinary primitive rendezvous channel, the producer/consumer requests used by Core.13 are therefore exactly the corresponding direction projections of O . In $\text{Sys_B}^{\text{elab}}$, O supplies the source-attributed request roots while any additional complementary ready/valid value produced by stateless coupling is reconstructed through HandshakeRec_E rather than represented as a new offer. Proof. Core.SyncFence excludes cross-interval survivors; Definitions Core.14–Core.15 and the one-offer-per-own-side invariant then give Pending/Front . $\square$

Lemma J.KObs-WFG (Enablement and wait-for-graph reconstruction).

For every residual frontier item x at σ , $\text{HandshakeRec}_E(K)$ reconstructs the same settled MirrorAvail_E value as the operational source model; therefore $\text{ChannelEnabledRec}(x,K)=\text{ChannelEnabled}(x,\sigma)$ and likewise for ChannelDisabled . The same template-qualified reconstruction yields $\text{ReleaseOwnerSetsRec}_E(x,K)=\text{ReleaseOwnerSets}_E(x,\sigma)$; taking unions gives $\text{WaitOwnersRec}_E(x,K)=\text{WaitOwners}_E(x,\sigma)$, and consequently $\text{WFGRec}(K)=\text{WFG}(\sigma)$. Proof. Use J.KObs-Chan for explicit stored channel state, J.KObs-Offers for attributed source-request roots/frontier membership, and ProcRec for any E-declared replay roots. Apply Q.1(j) to establish that these reconstructed channel/request/replay roots, together with E, exhaust the inputs to the complete settled handshake and release-family reconstruction; then use the finite deterministic acyclic settling property of Core.QSync/Q.1 to evaluate the declared ready/valid network and obtain the same $\text{MirrorAvail}/\text{ChannelEnabled}$ values. Apply Q.1(k)'s family-level support soundness/completeness theorem to the resulting disabled dependency semantics to obtain the correct ReleaseOwnerSets family and its WaitOwners union. No separate coupler state or coupler offer is required because the couplers are stateless and their complete equations/release-support schema are part of E. $\square$

Lemma J.KObs-InFlight (current in-flight carrier and transition reconstruction).

Let $\sigma \in \text{KCutReach_ref}(I)$, $K=\text{KObs}_E,w(\sigma)$, and assume the selected qualified Sys_ref Q.2 schema, either G1 or the source side of G2. $\text{Core.FlightStateRec}_{\{\text{Sys_ref},E\}}(K)$ equals the corresponding current abstract operational state needed by every declared Sys_ref action: ProcRec, explicit channel/token/storage state, settled handshake values, current attributed requests, and reconstructible reset/epoch facts agree; when the Stage-2 synchronization schema is selected, J.KObs-SyncCarrier/J.KObs-SyncInFlight additionally supply the typed SyncCarrier/SyncFire current-state facts. Sys_ref contains no lowering-only EpochGateAdvance state. Consequently the instantiated source action identities, state-based carriers, local enablement equations, and declared successor-effect functions are pure functions of K and E. This lemma does not assert that every locally applicable reconstructed action can be realized from the current phase or across a future cycle.

Definition J.InFlightPathSound (finite reconstructed-path operational realization).

Fix a source proof model $M \in \{\text{Sys_ref}, \text{Sys_K}\}$, a qualification profile Φ_M , and a reachable K-facing realization σ_0 together with an exact current-state reconstruction witness r_0 for the selected M/E schema. For $M=\text{Sys_ref}$ the normal witness is $r_0=\text{Core.FlightStateRec}_{\{\text{Sys_ref},E\}}(\text{KObs}_E,w(\sigma_0))$ from J.KObs-InFlight; for $M=\text{Sys_K}$ the lowering/Q certificate supplies the corresponding exact reconstructed current state used by K.Low.ModalQual. $\text{J.InFlightPathSound}_{\{I,M,E,\Phi_M\}}$ holds when every finite nonempty reconstructed path $r_0 \xrightarrow{u_0} r_1 \xrightarrow{u_1} \dots \xrightarrow{u_{n-1}} r_n$ admitted by $\text{Core.InFlightStepRec}_{\{M,E\}}$ has a finite legal M-operational extension that realizes those action units in order, launches/issues no new source-later work, and reaches K-facing successor observations/reconstructed checkpoints whose $\text{Core.FlightStateRec}$ values are exactly $r_1, \dots, r_n$. The load-bearing closure clause is prefix-compositional: after each realized proper prefix, either an explicit reset/terminal/activation-termination disposition ends that profile instance or the reached successor remains in Φ_M so the induction can realize the next edge. Single-edge realizability without this closure does not discharge the theorem. This definition is for source proof models only; it is not an RTL_B path theorem.

Proof/discharge structure. Use the Q.InFlightActionEffectSound theorem for the selected M/E schema, including Q.SyncInFlightActionEffectSound when applicable, and $\text{Core.InFlightCycleExt}_{\{M,I,E,\Phi_M\}}$ for any same-cycle or cross-cycle opportunity/embedding required by the selected edge; induct on path length, reapplying profile closure after each operational successor. Both local effect fidelity and path/opportunity soundness are dangerous-direction obligations: a fictitious action/effect or an

unrealizable reconstructed edge can manufacture InFlightEscape and hide a real deadlock. By contrast, operational incompleteness of the declared relation is conservative for deadlock absence but loses precision/applicability and shall be recorded as such.

Route note. On the primary serial route, Core.CycleClosure already bound by K.SerialRoutePremises is a Sys_K theorem and may discharge Core.InFlightCycleExt/J.InFlightPathSound for $M = \text{Sys_K}$ once the Q.2 action legality and profile-closure proof are supplied. In G1, $\text{Sys_K} = \text{Sys_ref}$, so the same evidence also serves the source theorem. In G2, it does not automatically establish the separate $M = \text{Sys_ref}$ path theorem required by K.Low.ModalQual; that theorem needs its own Core.InFlightCycleExt/CycleClosure discharge or a named continuation-preservation result. Structural, invariant, exploration, direct Policy-L, and tool-certificate A5-L routes likewise bind the path theorem for the proof model on which their generalized Dead_K claim is interpreted. B2/B3 cannot serve as the constructor because their eventuality is interpreted along an execution whose existence is the goal.

Three-level semantic separation. The current KObs record is level 1; Core.FlightStateRec/InFlightStepRec is a finite reconstructed transition system at level 2; J.InFlightPathSound is the operational realization theorem at level 3. Continuation existence is required only at level 3 to justify that the reconstructed modal relation contains no fictitious paths. It is not a permanence/liveness field stored in KObs and does not change Dead_K into a claim that the observed blocked state persists forever.

Corollary J.1 (K-observation Correspondence at the End of a Matched Observable Prefix)

Purpose.

This corollary is the bridge between execution-level trace compatibility and the K-facing cutpoint observation used directly by Appendix K. Its primary and authoritative cutpoint conclusion is equality of the compact KObs record, not equality of arbitrary source and RTL state. K-relevant frontier, request, channel, enablement, WFG, drain, and deadlock facts are derived from that equality by the named reconstruction functions; no clause expansion is required by the Appendix K transfer layer. Corollary J.1 establishes a reachable K-facing extension of the selected J.GWR witness but does not by itself establish membership of that extension in $\text{Pref_L}^{\text{adm}}(I)$. An Appendix K application supplies the separate K.RLAdmReach proof for the exact $\tau_{\text{ref}}^{\text{K}}$ prefix.

Statement. Assume the hypotheses of Theorem J.1 for Sys_ref versus RTL_B under the fixed stimulus, including B1 for the RTL_B Σ projection. Let $\tau_{\text{post}} \in \text{Reach_post}$ carry the J.GWR package constructed by J.GWR-Comp or supplied by J.GWR-Direct. Select and record a finite RTL normalization v_{post} with $\text{NormPost_I}(\tau_{\text{post}}, v_{\text{post}}, \rho_{\text{post}})$, so $\rho_{\text{post}} \in \text{KCutReach_post}(I)$. Let $\tau_{\text{ref}}^{\text{G}}$ be the matching Theorem J.1 source witness prefix, let $s_{\text{ref}} := \text{last}(\tau_{\text{ref}}^{\text{G}})$, and select the recorded post-boundary source observation normalization v_{ref} with $\text{Core.KObsNormRef_I}(\tau_{\text{ref}}^{\text{G}}, v_{\text{ref}}, \sigma_{\text{ref}})$; write $\tau_{\text{ref}}^{\text{K}} := \tau_{\text{ref}}^{\text{G}} \cdot v_{\text{ref}}$, so $\tau_{\text{ref}}^{\text{K}}$ ends at $\sigma_{\text{ref}} \in \text{KCutReach_ref}(I)$. On the standard normal-compositional K-facing route the separately checked $\text{Core.KObsSettleQualified_Sys_ref}(I)$ together with $\text{Core.KObsSettle-Ref}$ supplies the finite observation fixed point once the J.GWR post-L3 roots are fixed; an alternate route must validate the same Core.KObsNormRef interface directly. Fix the same I, E, w, H, O throughout, assume K.NormStable for v_{ref} , and assume J.KObsConf for this exact package, namely: canonical-history agreement on $\tau_{\text{ref}}^{\text{G}}$ from Lemma J.HCanon, the derived K.NormStable-H transport to $\tau_{\text{ref}}^{\text{K}}$, J.OfferReach_I, E, w(H, O; $\tau_{\text{ref}}^{\text{G}}, v_{\text{ref}}, \sigma_{\text{ref}}$), J.OfferConf_I, E, w(ρ_{post}, O), B-faithful/E4 interpretation at E's explicit boundaries, reset correspondence, and ϵ -opacity. Then:

- (1) $\tau_{\text{ref}}^{\text{K}}, \text{system} \approx \tau_{\text{post}}, \text{system}$ under E1–E5; and
- (2) $\text{KObsPost_J}(\rho_{\text{post}}; E, w, H, O) = \text{KObs_E, w}(\sigma_{\text{ref}})$.

Derived K-facing consequence. Let K denote the common record in conclusion (2). Every typed reconstruction function in Definitions Core.18d–Core.18e has the same value on the selected post and source views. In particular, ProcRec, ChanRec, NextFrontRec, PendingRec, BoundaryReqRec, FrontRec,

ChannelEnabledRec/ChannelDisabledRec, ReleaseOwnerSetsRec, WaitOwnersRec, WFGRec, and DrainClosedNowRec agree. These are functional consequences of conclusion (2), not additional implementation-side state-equality assumptions. Corollary J.1 makes no Dead_K, Dead_K^W, Dead_KS^W, or K.RespFail conclusion; Appendix K applies its own factoring results to the common record when those predicates are needed.

Proof of Corollary J.1

Step 0 (Choose the K-facing normalized cutpoints). Use the recorded NormPost_I witness to obtain $\rho_{\text{post}} \in \text{KCutReach_post}(I)$. On the source side, retain the exact J.GWR prefix $\tau_{\text{ref}}^{\text{G}}$ ending at its post-L3 construction endpoint s_{ref} and use the recorded Core.KObsNormRef witness v_{ref} to form $\tau_{\text{ref}}^{\text{K}} := \tau_{\text{ref}}^{\text{G}} \cdot v_{\text{ref}}$ ending at $\sigma_{\text{ref}} \in \text{KCutReach_ref}(I)$. The source v_{ref} segment lies entirely in the post-L3 observation-only Core.KObsSettle interval before L4; it is not the earlier L2 SettlePoint, does not cross the L4 registered update, and cannot feed its fixed-point result back into same-cycle L3 selection. Core.KObsSettleQualified_Sys_ref(I) plus Core.KObsSettle-Ref supplies existence/uniqueness on the standard K-facing QSync route; alternate routes must establish the same specialized source interface directly. Generic B4/B5 normalization arguments do not by themselves establish the post-L3 phase placement or observation-only restrictions required by Core.KObsNormRef.

Step 1 (Obtain the matched replay history). Apply Theorem J.1 to the selected RTL prefix and its theorem-produced or directly supplied J.GWR package to obtain the matching Sys_ref construction prefix $\tau_{\text{ref}}^{\text{G}}$ under the same stimulus. Lemma J.HCanon gives

$$\text{CanonHist_E}, w(\tau_{\text{ref}}^{\text{G}}) = \text{CanonHist_E}, w(\tau_{\text{post}}) = H.$$

K.NormStable supplies only primitive path-inertness facts; Lemma K.NormStable-H derives $\text{CanonHist_E}, w(\tau_{\text{ref}}^{\text{K}}) = \text{CanonHist_E}, w(\tau_{\text{ref}}^{\text{G}}) = H$. Thus conclusion (1) remains a statement about the J.GWR prefix $\tau_{\text{ref}}^{\text{G}}$, while the replay-history field used by the K-facing source cutpoint σ_{ref} is the same H for a separately proved reason. The retained fields and the erased incidental ordering are exactly those stated in Lemma J.HCanon/Core.18a.

Step 2 (Establish the common residual offers). J.OfferReach identifies O exactly with $\text{ResOffers_E}, w(\sigma_{\text{ref}})$ at the selected reachable source K-facing cutpoint on $\tau_{\text{ref}}^{\text{K}}$. The Core.KObsSettle/K.NormStable path creates, removes, or reattributes no blocking request and is $\text{K}\epsilon$ -inert after the request roots are fixed; J.OfferConf identifies that same O bidirectionally and uniquely with the relevant asserted endpoint-owned RTL requests at ρ_{post} . The reset-epoch, source-prefix-closure, boundary-cardinality, and no-hidden-request conditions ensure that O is a well-typed residual-offer component for both sides.

Step 3 (Conclude equal KObs). Both records have the same elaboration E, witness w, replay history H, and residual-offer set O. Therefore $\text{KObsPost_J}(\rho_{\text{post}}; E, w, H, O) = \langle E, w, H, O \rangle = \text{KObs_E}, w(\sigma_{\text{ref}})$, proving conclusion (2). B-faithfulness and E4 ensure that the post-side interpretation of the committed history at each explicit boundary is the same abstract channel interpretation used by ChanRec; source-side post-boundary normalization opacity is supplied by Core.KObsNormRef/K.NormStable rather than by treating all ϵ steps as replay-inert.

Step 4 (Derive the K-facing facts). Lemmas J.KObs-Proc–J.KObs-WFG reconstruct the process slice, explicit abstract channel state, NextFront, residual endpoint requests, Pending/BoundaryReq attribution, enablement, Front, WFG, and DrainClosedNow from the common record K. These conclusions follow by functional extensionality of the named reconstruction functions; no broad state correspondence is assumed. Deadlock-, waiver-, environment-terminal-, and diagnostic-predicate factoring is deliberately deferred to Appendix K, which consumes this equal-KObs conclusion. The Policy-S K.Resp discharge is separately checked over $\rho_{\text{S}}^{\text{det}}$ and is not a KObs cross-model predicate. $\square$

J.2b QSync-Specialized Certificate-Existence, Cycle Coverage, and Ideal Closure

The preceding results prove one matched RTL/source cutpoint package. The standard QSync scope permits two related instance-wide consequences from one-cycle induction. On the K-facing side, Core.QSync-Post supplies one canonical RTL KCut for each reachable designated RTL K-origin. On the source construction side, Core.QSync-Ref supplies the L2 settled snapshots required by J.RegPhaseNF/J.UnitExt; after each selected J.GWR post-L3 construction endpoint, Core.KObsSettle-Ref supplies the separate observation-only source K-facing normalization used by J.OfferReach/J.KObsConf. The richer correspondence data can therefore be carried from one canonical RTL cycle cutpoint to the next without identifying the J.GWR endpoint with the earlier source L2 SettlePoint. On the safety side, the same per-cycle construction has a smaller exact-prefix projection over RTL prefixes ending at reachable K-origins. Definition J.LocalGWRExtCert and Theorem J.GWR-CompExt make the source construction incremental on those carrier edges, and Definition J.QSyncTraceStep below packages exactly the safety-producing part of each edge. This subsection formalizes both routes. The 6' K-facing endpoint separation does not insert K-facing cutpoint or residual-offer fields into Definition J.QSyncTraceState, J.QSyncTraceStep/J.QSyncTraceInd, Theorem J.TraceCertCov-QSyncCycle, Lemma J.CycleIdealCover, or the M.7b safety-profile composition: those safety interfaces retain only the cycle-aligned RTL prefix, the J.GWR construction prefix τ_ref^G , the local/extension construction evidence, E,w, and J.HCanon history H. The subsection does not make Core.QSync itself a source/RTL simulation theorem and does not assume J.TraceCertCov or K.CertCov. The cycle-aligned safety theorem below discharges J.TraceCertCov only on Definition J.CycleReach_post; it does not manufacture route-complete J.TraceCertCov packages for arbitrary non-cycle-aligned finite prefixes. Instead, Lemma J.CycleIdealCover proves that every reachable finite outer observable history is a finite ideal of a reachable cycle-aligned history, and Corollary J.1-Safe-QSync uses J.IdealPrefixClosed to transfer safety to those ideals. During the qualified same-cycle settling relation, Core.QSync/QS2 and the L2 frame emit no Σ -visible commit. R2C intermediate delta-cycle evaluations are ε /stutter, and an emitted R2C fixed-point observation unit is atomic and may not be split.

Definition J.CycleReach_post (cycle-aligned RTL safety domain).

For a fixed theorem instance I in the standard QSync scope satisfying QualifiedSynchronousModel_RTL_B(I), define CycleReach_post(I) := { $\tau_post \in \text{Reach_post}(I)$ | there exist s_t, o_t, e_t such that τ_post ends exactly at o_t and Core.RTLDesignatedOriginFrame_I(s_t, o_t, e_t) }. The K-origin is therefore the designated cycle-aligned proof point associated with one full Core.CycleState and reset-epoch valuation after the current registered/FSM/pipeline frame and process-owned registered request outputs have been established and before the qualified same-cycle settling relation. By Lemma Core.KOriginCarrier-AtOrigin, every member has a finite carrier from the explicit initial node (κ_0, s_0, o_0, e_0) to that exact prefix; the carrier is theorem-derived historical typing, not a new progress premise. A reachable cycle-entry state is not excluded from supplying the final K-origin merely because it also satisfies DeclaredTerminal: when the Core.QSync registered/input/request frame is established, that node is the final ordinary cycle-aligned origin, and Definition Core.DeclaredTerminal forbids any later ordinary Core.RTLCycleStep; no additional post-terminal clock transition is required. Thus a member of CycleReach_post contains the selected observable history of the preceding completed real cycles and is the exact τ_post carrier used by the safety projection below. The definition does not identify arbitrary intra-cycle required-prefix ideals with cycle-aligned prefixes.

Definition J.QSyncTraceState (cycle-aligned safety correspondence state).

Fix the standard normal-compositional theorem instance I and its standing/static Theorem J.1 premises that do not themselves quantify a LocalGWR package over every reachable prefix. A tuple $T = (\tau_post; \tau_ref^G, E, w, H)$ satisfies QSyncTraceState_I(T) when all of the following hold. (TS1) $\tau_post \in$

CycleReach_post(I). (TS2) The exact τ_{post} carries the route-(a)-complete Post $\rightarrow$ Ref evidence of Definition J.TraceCertCov: a J.LocalGWR package accepted by J.GWR-Comp, QualifiedSynchronousModel_Sys_ref(I)/Core.QSync-Ref supplies the finite L2 settling required by J.RegPhaseNF/J.UnitExt, all remaining applicable Post $\rightarrow$ Ref premises of Theorem J.1 hold for the fixed instance, and J.GWR-Comp selects $\tau_{\text{ref}}^{\wedge}G \in \text{Reach_ref}$. (TS3) For that same selected package and witness, the theorem-derived J.HCanon relation holds and $H = \text{CanonHist_E},w(\tau_{\text{post}}) = \text{CanonHist_E},w(\tau_{\text{ref}}^{\wedge}G)$. Corollary J.RegPhaseReach-Comp may re-derive J.RegPhaseReach from the retained J.GWR-Comp evidence whenever a downstream use needs it; J.RegPhaseReach is therefore not an additional safety-side premise or required stored field. QSyncTraceState contains no J.OfferReach, J.OfferConf, J.KObsConf, K.DrainReach, or K.RLAdmReach requirement.

Definition J.QSyncTraceStep (safety-only carrier-edge refinement certificate).

Let $T_t = (\kappa_t; \tau_{\text{ref}}^{\wedge}\{G, t\}, E, w, H_t)$ and $T_{t+1} = (\kappa_{t+1}; \tau_{\text{ref}}^{\wedge}\{G, t+1\}, E, w, H_{t+1})$ be candidate safety tuples, and let R_t be a typed Core.RTLCycleStep_I($\kappa_t, s_t, o_t, e_t, r_t, \kappa_{t+1}, s_{t+1}, o_{t+1}, e_{t+1}$) appendable to the represented Core.KOriginCarrier. A record S_t satisfies QSyncTraceStep_I($T_t, R_t, T_{t+1}; S_t$) when it contains only the following safety-side evidence. (ST1) Exact edge binding: T_t is a QSyncTraceState for κ_t , R_t is the exact carrier edge, and the target post prefix is exactly κ_{t+1} ; RS4/RS5 provide the exact new complete-unit/reset delta and successor epoch. (ST2) Local package extension: S_t binds the old LocalGWR package L_t retained by the TS2 route-(a) evidence and a successor package L_{t+1} , together with J.LocalGWRExtCert_I($\kappa_t, L_t; \kappa_{t+1}, L_{t+1}$); the successor package is not an unrelated per-prefix certificate. (ST3) Exact source construction extension: Theorem J.GWR-CompExt is applied to the same old selected $W_t/\tau_{\text{ref}}^{\wedge}\{G, t\}$ and produces $W_{t+1}/\tau_{\text{ref}}^{\wedge}\{G, t+1\}$ with J.GWRPrefixExt, hence $\tau_{\text{ref}}^{\wedge}\{G, t\} \leq \tau_{\text{ref}}^{\wedge}\{G, t+1\}$. The extension target is $\tau_{\text{ref}}^{\wedge}G$, never an old $\tau_{\text{ref}}^{\wedge}K$. (ST4) Safety construction legality: source-side Core.QSync-Ref supplies every newly required L2 settle inside the appended J.GWR construction; newly occurring RESET units are spliced by J.RS/RW1–RW5 using e_{t+1} ; I.A0/E3/request-attribution, LocalAgree, channel/atomic, environment, and no-unresolved-hole facts are updated exactly for the new suffix. (ST5) History update: J.HCanon for the extended package establishes $H_{t+1} = \text{CanonHist_E},w(\kappa_{t+1}) = \text{CanonHist_E},w(\tau_{\text{ref}}^{\wedge}\{G, t+1\})$ and restricts to H_t on the old complete-unit ideal. (ST6) Safety-only type discipline: the record has no fields for $v_{\text{post}}, p_{\text{post}}, v_{\text{ref}}, \sigma_{\text{ref}}, O$, Core.KObsNormRef, K.NormStable, J.OfferReach/J.OfferConf/J.KObsConf, K.DrainReach, K.RLAdmReach, or any K-facing waiver/deadlock fact, and none of those facts may be cited to discharge ST1–ST5. Thus a QSyncTraceStep is independently checkable from the carrier edge, local J/I construction evidence, source QSync settling, reset handling, and J.HCanon alone.

Lemma J.QSyncTraceStep-Sound (a safety step produces the successor safety state).

If QSyncTraceState_I(T_t) and QSyncTraceStep_I($T_t, R_t, T_{t+1}; S_t$) hold, then QSyncTraceState_I(T_{t+1}) holds. Proof. ST1 gives $\kappa_{t+1} \in \text{CycleReach_post}$. ST2–ST4 give the exact route-(a) LocalGWR/J.GWR-Comp evidence for κ_{t+1} , with J.GWR-CompExt selecting the exact extended $\tau_{\text{ref}}^{\wedge}\{G, t+1\}$ and Core.QSync-Ref supplying its internal L2 settling; hence TS2 holds. ST5 gives TS3. ST6 guarantees that this derivation has no K-facing premise. $\square$

Definition J.QSyncTraceInd (base-and-step cycle-aligned safety-package induction).

QSyncTraceInd_I holds when the following proof obligations are discharged over the fixed theorem instance. (TIO) Carrier base. For every exact initial/reset-established carrier node (κ_0, s_0, o_0, e_0) admitted by KC0 for the fixed instance, Core.RTLDesignatedOriginFrame_I(s_0, o_0, e_0) holds, e_0 is the post-initialization/reset epoch valuation prescribed by I, $\kappa_0 \in \text{CycleReach_post}(I)$, and there exists a QSyncTraceState_I tuple T_0 for that exact prefix. This allows harmless internal pre-origin

representation differences without permitting a different logical base state or epoch. (TI1) Safety-step preservation. Assume a finite $\text{Core.KOriginCarrier_I}$ object whose final node is $(\kappa_t, s_t, o_t, e_t)$ and a $\text{QSyncTraceState_I}(T_t)$ whose exact post prefix is κ_t . For every reachable Core.RTLCycleStep R_t appendable to that carrier, the flow constructs a record S_t and target tuple T_{t+1} satisfying $J.QSyncTraceStep_I(T_t, R_t, T_{t+1}; S_t)$. Lemma $J.QSyncTraceStep\text{-Sound}$ then supplies $\text{QSyncTraceState_I}(T_{t+1})$ for the exact successor κ_{t+1} . In particular, TI1 must provide the cross-prefix $J.LocalGWRExtCert$ and use Theorem $J.GWR\text{-CompExt}$ to extend the already selected $\tau_ref^G\{G, t\}$; a fresh unrelated $J.GWR$ witness is not a valid incremental discharge. TI1 contains no $J.OfferReach$, $J.OfferConf$, $J.KObsConf$, Core.KObsNormRef , $K.NormStable$, $K.DrainReach$, or $K.RLAdmReach$ evidence by the ST6 type discipline. Corollary $J.RegPhaseReach\text{-Comp}$ remains theorem-derived from the retained/extended $J.GWR\text{-Comp}$ evidence rather than an extra TI1 input. (TI2) Exhaustive carrier-position/successor coverage. TI1 is proved for every finite $\text{Core.KOriginCarrier}$ reachable from the TI0 base and every reachable typed RTLCycleStep appendable to its final node, not merely the nodes/edges observed in one simulation. Equivalently, every nonfinal position of every resulting carrier is covered by the same safety-step construction. A deterministic implementation theorem may collapse that successor set only if, for each reachable carrier-final node, the reachable appendable Core.RTLCycleStep set is a singleton or all of its members agree on every field consumed by TI1; B1 alone does not license dropping unobserved internal successors. TI0/TI1 may be supplied by an inductive simulation/refinement invariant or a verified certificate-extraction procedure.

Definition $J.QSyncCertState$ (canonical-cutpoint correspondence state).

Fix the standard normal-compositional theorem instance I , including the fixed source/RTL artifacts, Sys_ref elaboration E , capacities, stimulus, environment/reset interpretation, witness w , and the standing/static Theorem $J.1$ premises that do not themselves quantify a LocalGWR package over every reachable prefix (including B1, E4/B-faithfulness, witness/replay machinery, and the qualified source/RTL structural semantics). No $J.TraceCertCov$ or other prefix-package-totality premise is assumed. A tuple $C = (\tau_post, v_post, \rho_post; \tau_ref^G, v_ref, \sigma_ref, E, w, H, O)$ satisfies $\text{QSyncCertState_I}(C)$ when, writing $\text{TraceOf}(C) := (\tau_post; \tau_ref^G, E, w, H)$, $s_ref := \text{last}(\tau_ref^G)$, and $\tau_ref^K := \tau_ref^G \cdot v_ref$, all of the following hold. (QC0-Safe) $\text{QSyncTraceState_I}(\text{TraceOf}(C))$ holds. This is a stored/provable safety projection and is the only part of C consumed by the safety induction. (QC1) $\tau_post \in \text{Reach_post}$ ends exactly at a reachable designated RTL K -origin o_t with a recorded $\text{Core.RTLDesignatedOriginFrame_I}(s_t, o_t, e_t)$ witness for the represented cycle, and v_post is the canonical finite Core.QSync-Post normalization to $\rho_post \in \text{KCutReach_post}(I)$. (QC2) The per-prefix $J.LocalGWR/J.GWR\text{-Comp}$ evidence used by the K -facing package is exactly the same route-(a) evidence retained by QC0-Safe; it selects the same τ_ref^G and may not independently reselect a different source witness. Core.QSync-Ref supplies every L2 settle required inside that construction, but does not identify s_ref with the K -facing source endpoint. (QC3) The H component is exactly the $J.HCanon$ history already retained by QC0-Safe; Core.KObsNormRef and $K.NormStable$ then hold for $(\tau_ref^G, v_ref, \sigma_ref)$, Lemma $K.NormStable\text{-H}$ carries that same H to τ_ref^K , and $J.OfferReach/J.OfferConf/J.KObsConf$ hold for the exact common K -facing tuple $(\rho_post, \tau_ref^G, v_ref, \sigma_ref, E, w, H, O)$. (QC4) Corollary $J.RegPhaseReach\text{-Comp}$ supplies $J.RegPhaseReach_I, E, w(\tau_ref^G, s_ref)$; this is theorem-derived from the QC0/QC2 $J.GWR\text{-Comp}$ machinery rather than an independent input obligation. (QC5) $K.DrainReach$ holds for the exact K -facing prefix τ_ref^K and endpoint σ_ref , including $\text{Core.SettleKBridge}$ for every newly or previously activated persisting blocked frontier, so the three-part $K.RLAdmReach$ and $\tau_ref^K \in \text{Pref_L}^{\text{adm}}(I)$ follow. QSyncCertState is deliberately a strict extension of its QC0-Safe projection: the K -facing fields may depend on the safety payload, but QC0-Safe may not depend on $v_post, \rho_post, v_ref, \sigma_ref, O, KObs$, normalization-stability, or drain fields. It contains no $J.TraceCertCov$ assumption, no $K.CertCov$ assumption, and no successor $K.CertifiedCutpoint$ premise. On the standard

normal-compositional route, QC3 may cite the separately checked $\text{Core.KObsSettleQualified_Sys_ref}(I)$, $\text{Core.KObsSettle-Ref}$, and the K.NormStable primitive facts, and QC5 may cite Lemma K.DrainReach-Comp once the independent DR1–DR4 evidence is bound to this exact package. An alternate/direct route may validate the corresponding K-facing fields independently while preserving QC0-Safe as the safety projection.

Definition J.QSyncCertInd (base-and-step certificate induction discharge).

QSyncCertInd_I holds when the following proof obligations are discharged over the fixed theorem instance. (CI0) Carrier base. For every exact initial/reset-established carrier node $(\kappa_0, s_0, o_0, e_0)$ admitted by KC0 for the fixed instance, $\text{Core.RTLDesignatedOriginFrame_I}(s_0, o_0, e_0)$ holds, e_0 equals the theorem-instance post-initialization/reset epoch valuation, and Core.QSync-Post supplies its canonical endpoint p_post^0 . There exists a QSyncCertState_I tuple C_0 whose QC1 origin prefix/frame is exactly that base node, whose post endpoint is p_post^0 , and whose QC0-Safe projection $\text{TraceOf}(C_0) = T_0$ is the TI0 QSyncTraceState for the same base. (CI1) Carrier-step preservation is structurally split into two stages. Safety stage: for every reachable Core.RTLCycleStep R_t appendable to a carrier ending at the QC1 prefix of C_t , take $T_t := \text{TraceOf}(C_t)$ and construct S_t, T_{t+1} with $\text{J.QSyncTraceStep_I}(T_t, R_t, T_{t+1}; S_t)$. This stage binds the old/new LocalGWR packages, J.LocalGWRExtCert , and J.GWR-CompExt result, so the exact successor construction prefix $\tau_ref^{\{G, t+1\}}$ extends $\tau_ref^{\{G, t\}}$; Lemma $\text{J.QSyncTraceStep-Sound}$ establishes $\text{QSyncTraceState_I}(T_{t+1})$. By ST6 this stage is complete without using any K-facing field of C_t or the successor package. K-facing stage: only after T_{t+1} is fixed, Core.QSync-Post supplies the canonical RTL endpoint from designated o_{t+1} ; then $\text{Core.KObsSettleQualified}/\text{Core.KObsSettle-Ref}$ and the checked K.NormStable facts select a fresh v_ref and σ_ref from the newly extended $\tau_ref^{\{G, t+1\}}$; $\text{J.OfferReach}/\text{J.OfferConf}/\text{J.KObsConf}$ bind O there; Corollary $\text{J.RegPhaseReach-Comp}$ supplies the phase-correct $\tau_ref^{\{G, t+1\}}$; and the independently qualified uniform instance-level DR1–DR4 proof schema/evidence is instantiated afresh for this exact enlarged construction. Lemma K.DrainReach-Comp re-derives successor $\text{K.DrainReach}/\text{Core.SettleKBridge}$ on the fresh $\tau_ref^K := \tau_ref^{\{G, t+1\}} \cdot v_ref$. DrainReach is not incrementally extended from the old τ_ref^K , and neither the old v_ref nor old τ_ref^K is an extension target. RESET handling needed for the safety construction is already in S_t through $\text{J.RS}/\text{RW1}–\text{RW5}$ and $\text{RS5}/\text{KC3}$; any K-facing reset/normalization consequence is established separately in this second stage. The resulting C_{t+1} has QC0-Safe exactly T_{t+1} . (CI2) Exhaustive carrier-position/successor coverage. CI1 is proved for every finite $\text{Core.KOriginCarrier}$ reachable from the CI0 base and every reachable RTLCycleStep appendable to its final node; hence every nonfinal position of every represented carrier is covered, not merely one simulation path. A deterministic implementation theorem may collapse that successor set only if, for each reachable carrier-final node, the reachable appendable Core.RTLCycleStep set is a singleton or all of its members agree on every field consumed by CI1; B1 alone does not license dropping unobserved internal successors. CI0/CI1 may be supplied by an inductive simulation/refinement invariant or a verified certificate-extraction procedure, but they may not cite K.CertCov , the successor $\text{K.CertifiedCutpoint}$ fact, or J.CertExist-QSync itself.

Lemma J.QSyncCertInd-Trace (K-facing induction projects to the cycle-safety induction).

If J.QSyncCertInd_I holds, then J.QSyncTraceInd_I holds. Proof. Each QSyncCertState C carries its safety state literally as QC0-Safe, namely $\text{TraceOf}(C) = (\tau_post; \tau_ref^{\{G, E, w, H\}})$. Project CI0 to the same QC0-Safe base tuple, obtaining TI0. For every CI1 carrier edge, project the explicitly constructed safety-stage object S_t satisfying $\text{J.QSyncTraceStep_I}(\text{TraceOf}(C_t), R_t, T_{t+1}; S_t)$; Lemma $\text{J.QSyncTraceStep-Sound}$ is exactly the TI1 successor argument. No proof term from the later K-facing stage is available in the QSyncTraceStep schema, because ST6 excludes $v_post, p_post, v_ref, \sigma_ref, O$, Core.KObsNormRef , K.NormStable , $\text{OfferReach}/\text{OfferConf}/\text{KObsConf}$, DrainReach , and K.RLAdmReach . CI2 and its exhaustive

carrier-position/appendable-successor quantification project unchanged to TI2. Thus J.QSyncCertInd-Trace is a literal structural projection of stored safety states and per-edge safety-step certificates, not a proof that reconstructs safety after discarding K-facing fields. $\square$

Theorem J.TraceCertCov-QSyncCycle (QSync induction gives cycle-aligned safety package coverage).

Assume QualifiedSynchronousModel_RTL_B(I), QualifiedSynchronousModel_Sys_ref(I), the standing/static normal-compositional Theorem J.1 premises for the fixed instance other than per-prefix LocalGWR/TraceCertCov package totality, and J.QSyncTraceInd_I. Then J.TraceCertCov_I(CycleReach_post(I)) holds. Proof. Fix arbitrary $\tau_{\text{post}} \in \text{CycleReach_post}(I)$ with designated frame (s_n, o_n, e_n) . Lemma Core.KOriginCarrier-AtOrigin supplies a finite carrier C with final node $(\tau_{\text{post}}, s_n, o_n, e_n)$. Induct over the finite positions of C. TI0 supplies the route-complete QSyncTraceState at the explicit base node. At each carrier edge, TI1 constructs the exact J.QSyncTraceStep/J.GWR-CompExt safety certificate and J.QSyncTraceStep-Sound yields the successor state, while TI2 guarantees that the proof applies to every reachable appendable edge/carrier position rather than one simulation path. Hence the final node τ_{post} carries a QSyncTraceState tuple. TS2 is exactly route (a) of Definition J.TraceCertCov for that same τ_{post} , and TS3 retains the J.HCanon equality used by the safety transfer. Because τ_{post} was arbitrary, TraceCertCov_I(CycleReach_post(I)) holds. No J.OfferReach, J.OfferConf, J.KObsConf, K.DrainReach, K.RLAdmReach, or K.CertCov premise is used. $\square$

Corollary J.1-Safe-QSyncCycle (cycle-aligned source-side safety transfer).

Assume the hypotheses of Theorem J.TraceCertCov-QSyncCycle and the applicable source-side clause of Corollary J.1-Safe for an extensional predicate Φ on the finite canonical Σ_{DUT} histories of Definition J.OuterCanonHist satisfying J.IdealPrefixClosed_I, $\Sigma_{\text{DUT}}(\Phi)$. Then the corresponding conclusion of Corollary J.1-Safe holds for every $\tau_{\text{post}} \in \text{CycleReach_post}(I)$. Proof. The preceding theorem supplies J.TraceCertCov_I(D) with $D := \text{CycleReach_post}(I)$; instantiate Corollary J.1-Safe with that D. Lemma J.OuterCanon-HCanon supplies the required source/RTL outer-history equality from the retained J.HCanon field. $\square$

Lemma J.CycleIdealCover (cycle-aligned histories cover finite observable ideals).

Assume QualifiedSynchronousModel_RTL_B(I). Let $\tau_{\text{post}} \in \text{Reach_post}(I)$ be any reachable finite RTL_B prefix in the standard QSync scope, and let $H := \text{CanonHist}_{\Sigma_{\text{DUT}}}(\tau_{\text{post}})$ under Definition J.OuterCanonHist. Apply Lemma Core.KOriginCarrier-Decomp to the exact reachability witness for τ_{post} . If KD0 holds, $\tau_{\text{post}} < \kappa_0$ for the explicit initial/reset-established carrier node. Lemma J.OuterCanon-Ideal applied to $\tau_{\text{post}} \leq \kappa_0$ directly gives H as a finite ideal of $\text{CanonHist}_{\Sigma_{\text{DUT}}}(\kappa_0)$, so choose $\tau_{\text{post}}^{\text{cyc}} := \kappa_0 \in \text{CycleReach_post}(I)$. Otherwise KD1 supplies a finite Core.KOriginCarrier with final node $N_t = (\kappa_t, s_t, o_t, e_t)$, $\kappa_t \leq \tau_{\text{post}}$, and no later designated K-origin at or before τ_{post} . Thus κ_t is the current cycle-aligned prefix as a theorem-derived carrier projection, not an assumed “greatest K-origin” state. There are then two exhaustive outer-history cases. (i) No complete selected outer unit has been added to $\text{CanonHist}_{\Sigma_{\text{DUT}}}$ since κ_t . Then the extension from κ_t to τ_{post} adds only ϵ /non-unit activity and complete units erased by Π_{DUT}, E ; the closing clause of Lemma J.OuterCanon-Ideal gives $\text{CanonHist}_{\Sigma_{\text{DUT}}}(\tau_{\text{post}}) = \text{CanonHist}_{\Sigma_{\text{DUT}}}(\kappa_t)$, so choose $\tau_{\text{post}}^{\text{cyc}} := \kappa_t$. This includes a prefix anywhere after the K-origin but before the next retained outer unit: qualified settling, boundary processing, a post-boundary observation point, registered/result/update activity, a Core.SilentCycle segment, and any RESET or other complete proof/elaboration unit that Π_{DUT}, E erases. (ii) At least one complete selected outer unit has been added since κ_t . Then τ_{post} contains a complete selected unit of the already-entered real cycle whose Π_{DUT}, E image is retained. Applying Lemma

Core.RTLCycleCarrier to the carrier-projected designated origin/frame supplies a typed Core.RTLCycleStep to $\kappa_{\{t+1\}}$; Lemma Core.KOriginCarrier-Extend appends it to the carrier, and its endpoint $\tau_{\text{post}}^{\text{cyc}} = \kappa_{\{t+1\}}$ lies in CycleReach_post(I). A DeclaredTerminal successor may serve as that final K-origin without a later cycle. Since $\tau_{\text{post}} \leq \tau_{\text{post}}^{\text{cyc}}$ on the same legal carrier, Lemma J.OuterCanon-Ideal gives that H is a finite ideal of CanonHist_ Σ _DUT($\tau_{\text{post}}^{\text{cyc}}$), including dependencies whose intermediate RESET/elaboration/proof units are projected away. Whole-unit projection prevents rendezvous, paired synchronization, grouped E2 anchors, emitted R2C fixed-point units, RESET units, or declared atomic elaboration groups from being split by the ideal argument. KD0/KD1 and cases (i)/(ii) are exhaustive, including prefixes before the first designated origin and real cycles that contribute zero retained outer units. $\square$

Corollary J.1-Safe-QSync (standard QSync all-finite-prefix safety transfer).

Assume the hypotheses of Corollary J.1-Safe-QSyncCycle for an extensional predicate Φ on the finite canonical Σ _DUT histories of Definition J.OuterCanonHist satisfying J.IdealPrefixClosed_I, Σ _DUT(Φ). Then the corresponding source-side safety conclusion holds for every $\tau_{\text{post}} \in \text{Reach_post}(I)$. Proof. Fix arbitrary $\tau_{\text{post}} \in \text{Reach_post}(I)$. Lemma J.CycleIdealCover supplies $\tau_{\text{post}}^{\text{cyc}} \in \text{CycleReach_post}(I)$ such that CanonHist_ Σ _DUT(τ_{post}) is a finite ideal of CanonHist_ Σ _DUT($\tau_{\text{post}}^{\text{cyc}}$). Corollary J.1-Safe-QSyncCycle establishes Φ for the covering cycle-aligned history. J.IdealPrefixClosed_I, Σ _DUT(Φ) therefore gives Φ for CanonHist_ Σ _DUT(τ_{post}). Lemma J.OuterCanon-HCanon makes the conclusion independent of the enriched source/RTL canonical representative by construction. Since τ_{post} was arbitrary, the result holds for every reachable finite RTL_B prefix. $\square$

Scope note (ideal closure does not create per-prefix correspondence packages).

Lemma J.CycleIdealCover and Corollary J.1-Safe-QSync close the residual non-cycle-aligned safety case only for properties of finite canonical Σ _DUT histories satisfying Definition J.IdealPrefixClosed. They do not assert J.TraceCertCov for every non-cycle-aligned τ_{post} , do not construct a separate matching Sys_ref witness or J.GWR package for each such prefix, and do not extend raw latency/performance, properties that fail ideal-prefix closure, or non-QSync claims. In particular, a cross-interface scoreboard whose correctness relation depends on an event that may be omitted by a $\prec_{\Sigma}$ _DUT ideal is not covered by the ideal-closure shortcut merely because it is informally called a safety check. Applications that need direct correspondence evidence for an arbitrary prefix continue to discharge Definition J.TraceCertCov on the selected broader domain.

Definition J.QSyncCertCov (J-side total canonical-cutpoint package coverage).

J.QSyncCertCov(I) holds when for every $p_{\text{post}} \in \text{KCutReach_post}(I)$ there exists at least one tuple C satisfying QSyncCertState_I(C) whose post endpoint is p_{post} . This predicate records semantic package existence only; K.CertHdr binding and the downstream K.CertifiedCutpoint/K.CertCov packaging are performed in Appendix K.

Definition J.CertExtract_QS (carrier-indexed constructive QSync certificate extractor).

Assume the fixed standard normal-compositional hypotheses and J.QSyncCertInd_I. For a finite Core.KOriginCarrier_I object C with nodes $N_i = (\kappa_i, s_i, o_i, e_i)$, define CertExtract_QS(I,C,i) for a carrier of length n and $0 \leq i \leq n$ by finite recursion over the carrier position: at $i=0$ use CIO on the exact base node $(\kappa_0, s_0, o_0, e_0)$; at $i+1$ use the actual Core.RTLCycleStep stored between nodes i and $i+1$ and the C11 proof-producing step to obtain the successor QSyncCertState. Core.QSync-Post supplies the canonical $v_{\text{post}}^i / p_{\text{post}}^i$ endpoint derived from o_i ; p_{post}^i is not stored in the carrier. The returned tuple has QC1 post prefix exactly κ_i and the exact designated frame/epoch e_i threaded by KC3/RS5. The extractor's domain is therefore a carrier position together with its reachability/step evidence, not a bare KCut state. A caller starting from p_{post} first uses Corollary Core.KOriginCarrier-KCut to obtain the

represented carrier/origin position and canonical QSync endpoint. Canonicity of p_post does not imply uniqueness of the reachability/carrier witness used by the extractor or of the returned certificate package; $K.CertCov$ requires at least one sound package for that endpoint, not a canonical certificate witness.

Lemma J.CertExtract-QS-Sound (carrier extraction returns the represented certificate state).

Under the hypotheses of Definition J.CertExtract_QS, for every carrier C and position i , $CertExtract_QS(I, C, i)$ satisfies $QSyncCertState_I$ and its $QC1$ fields are exactly the node prefix/frame $(\kappa_i, s_i, o_i, e_i)$ together with the canonical $Core.QSync-Post$ endpoint p_post^i . Proof. Finite induction on i . The base is $CI0$. The successor case uses the stored $RTLCycleStep$, Lemma $Core.KOriginCarrier-Extend/KC3$ exact node sharing, and $CI1$; $CI2$ guarantees that the same step proof covers every reachable carrier edge represented by the theorem instance. $\square$

Theorem J.CertExist-QSync (canonical-cycle induction gives total J-side package existence).

Assume $QualifiedSynchronousModel_RTL_B(I)$, $QualifiedSynchronousModel_Sys_ref(I)$, the standing/static normal-compositional Theorem J.1 premises for the fixed instance other than per-prefix $LocalGWR/TraceCertCov$ package totality, and $J.QSyncCertInd_I$. Then $J.QSyncCertCov(I)$ holds. Proof. Fix arbitrary $p_post \in KCutReach_post(I)$. Corollary $Core.KOriginCarrier-KCut$, applied to its reachability witness, supplies the unique designated origin subprefix κ_n , frame (s_n, o_n, e_n) , a finite $KOriginCarrier$ C ending at that node, and the canonical $QSync-Post$ endpoint p_post derived from o_n . Definition $J.CertExtract_QS$ and Lemma J.CertExtract-QS-Sound then return a $QSyncCertState$ tuple for exactly that represented endpoint. Hence every reachable canonical $KCut$ has a semantic package and $J.QSyncCertCov(I)$ holds. This proof hides the existential carrier from the theorem's output interface but does not pretend that a bare $KCut$ state is enough input for constructive extraction. No $J.TraceCertCov$ premise, arbitrary normalization choice, independent per-cutpoint package-existence axiom, or generic fair-extension premise is used. $\square$

J.3 Causal Dependency and What System-Level Compatibility Guarantees (Informative)

The system-level result of Appendix J establishes compatibility of the aggregate observable behavior, i.e. $\tau_ref, system \approx \tau_post, system$ over the alphabet Σ of channel operations, synchronization events, and signal Read/Write actions.

This compatibility is intentionally *observational*: it constrains what an external environment can distinguish at Σ , not internal RTL micro-timing.

As a consequence, Appendix J should be read as preserving explicitly required observable ordering, atomicity, and FIFO/synchronization semantics, not incidental latency. The causal-dependency relation ($\rightarrow$) defined in the “Causal Dependency Between Processes” section remains useful for explaining information flow and for documenting which interactions a designer intends to be functionally significant. However, $\rightarrow$ is not itself the strict induction order for Theorem J.1, and the theorem does not require the unquotiented dependency graph $\rightarrow$ to be acyclic. The proof order is the required-order relation $\leq_J$ over observable units.

Accordingly, the system-level theorem implies the following required-order preservation principle:

- If two observable units γ_1 and γ_2 are ordered by the required-order relation $\gamma_1 \leq_J \gamma_2$, then the matched Sys_ref and RTL_B executions preserve that required order. This follows because each elementary $\leq_J$ generator is preserved by the corresponding rule: synchronization source order by E1; signal-anchor timestamping by E2; issue-side and request-attribution obligations by E3; channel FIFO

legality and rendezvous atomicity by E4; immediate synchronization-boundary constraints by E5; and paired-barrier semantics by the explicit synchronization well-formedness conditions. In particular, for each explicit abstract channel c , the matched executions preserve the committed $\text{Push}_c/\text{Pop}_c$ history: for $B(c)=0$, a committed transfer is one explicitly atomic rendezvous $\text{Push}_c(v)/\text{Pop}_c(v)$ unit; for $B(c)>0$, the k -th committed Pop_c observes the k -th earlier unmatched Push_c according to reset-empty, FIFO-order, no-underflow/no-overflow, and non-fall-through E4 semantics.

- If two observable units are not ordered by $<_J$ and are not members of the same explicitly atomic unit, then their relative cycle placement or interleaving is not constrained by the methodology, and differences are treated as incidental latency variation. A designer who needs a particular cross-process ordering to be preserved must express it through an explicit Σ -visible mechanism such as message-passing FIFO order, a rendezvous/handshake atomic unit, explicit synchronization, or signal-anchor semantics.

This perspective is what makes the “single testbench / no surprises” implication precise: for the fixed-stimulus execution sets covered by Theorem J.1, any testbench that observes only Σ and encodes its expectations through causal dependencies (channels, barriers, or explicit handshakes) cannot distinguish the matched Sys_ref / RTL_B execution pairs provided by the theorem.

Equivalently, RTL_B introduces no new Σ -observable behavior relative to Sys_ref under that fixed stimulus, and source-to-RTL transfer is asserted only for the annotated RTL-consistent source prefixes covered by Theorem J.1. (Here Sys_ref is as fixed in Remark 2. The rendezvous specialization Sys with $B(c)=0$ is covered only under the explicit conditions of Remark 4.)

Appendix K – Preservation of Internal Wait-Cycle Freedom (Reachable-Cutpoint Deadlock-Freedom) for the Prescribed Source Reference Model

K.1 Scope, Notation, and Definition of Internal Message-Passing Deadlock

Terminology alignment

Throughout Appendix K, when evaluating K-facing predicates at a cutpoint σ in $\text{KCutReach_ref}(I)$ or the corresponding certified RTL K-facing view, let t_σ denote the cycle of that cutpoint and use the cutpoint shorthand $\text{occ}_c(\sigma) \triangleq \text{occ}_c(t_\sigma)$. For any frontier item x on an explicit abstract channel $c = \text{BoundaryOf_P}(x)$ of Sys_ref , with owning dynamic source-level message instance $q = \text{MsgOf_P}(x)$, use $\text{BoundaryReq}(x, \sigma)$, $\text{ChannelEnabled}(x, \sigma)$, and $\text{ChannelDisabled}(x, \sigma)$ for the endpoint-owned request, enablement, and disabledness facts at that specific Core.KCut evaluated boundary. In the ordinary unelaborated case, where q has a unique evaluated boundary and a unique frontier item, $\text{BoundaryReq}(q, \sigma)$, $\text{ChannelEnabled}(q, \sigma)$, and $\text{ChannelDisabled}(q, \sigma)$ may be used as shorthands. In $\text{Sys_B}^{\text{elab}}$, however, the frontier-item form is authoritative because one source-level operation may own multiple proof-internal frontier items at different explicit boundaries. Thus “boundary request is asserted” means $\text{BoundaryReq}(x, \sigma)$, “channel-enabled” means $\text{ChannelEnabled}(x, \sigma)$, and “channel-disabled” means $\text{ChannelDisabled}(x, \sigma)$, all evaluated at the Core.KCut boundary fixed point of the selected cutpoint cycle and read through $q = \text{MsgOf_P}(x)$ and $\text{BoundaryOf_P}(x)$. These predicates are not applied to already-committed Σ -event labels except through the operation/frontier item that contributes $m(q)$ when it commits.

We work with a fixed design:

System Sys_ref: The collection of pre-HLS processes obeying the R-rules and B-rules, interpreted under the prescribed source reference model of Definition Core.1 and Appendix J, Remark 2. In this appendix, capacities/occupancies/enableness predicates are interpreted at the chosen abstract boundaries of Sys_ref; in elaborated cases, this means the explicit abstract staged/bundle/join channels of that model. Each explicit abstract channel c of Sys_ref has finite capacity $B(c) \geq 0$, chosen to match the corresponding RTL realization at that abstract boundary.

Post-HLS implementation RTL_B: The hardware implementation of the same processes in which each chosen abstract channel c is realized with finite capacity $B(c) \geq 0$. In this appendix, we compare Sys_ref (the prescribed source-side proof target) to RTL_B (the micro-architectural realization). The proof does not require mapping buffered RTL states to a distinct raw rendezvous ($B(c)=0$) state.

Witness-selection and post-view convention. Corollary J.1 supplies a selected reachable Sys_ref cutpoint σ_{ref} and a common record K with $K\text{Obs_E},w(\sigma_{\text{ref}}) = K\text{ObsPost_J}(p_{\text{post}};E,w,H,O)$ for the selected RTL K -facing cutpoint p_{post} . Appendix K does not compare arbitrary RTL microstate with source state. Define the K -facing post abstract view by applying the reconstruction functions of Definitions Core.18d–Core.18e to $K\text{ObsPost_J}$. Unless a raw RTL signal is expressly named as part of $J.\text{OfferConf}$, every post-side occurrence of NextFront, Pending, BoundaryReq, channel state, ChannelEnabled/ChannelDisabled, Front, WFG, DrainClosedNow, or Dead_K in Appendix K denotes that reconstructed view. If a reproducible source witness is desired for debugging, the Appendix L snooping wrapper may select one aligned to the RTL latency-sensitive events; uniqueness is not required by the theorem.

Stage-2 generalized in-flight drain applicability.

The generalized DrainClosedNow/InFlightEscape definitions are normative in this revision for two explicitly versioned profiles. (G1) Non-lowered profile: $\text{Sys_K} = \text{Sys_ref}$, the literal $K.\text{Dead}$ blocked-frontier carrier remains the existing blocking message-operation carrier typed by $\text{MsgOf}/\text{BoundaryReq}$, and the selected Q.2 action universe contains ordinary already-issued message actions plus qualified proof-internal staging/bundle/join actions represented by $E/K\text{Obs}$. (G2) Lowered profile: $\text{Sys_K} = \Lambda_W(\text{Sys_ref})$ may additionally contain guard/epoch-gate realization, $\text{Core.SyncCarrierWF}$ and the Stage-2B SyncFire/EpochGateAdvance schema are bound where synchronization lowering is used, $K.\text{InFlightQual}$ is discharged separately for Sys_K and Sys_ref , and $K.\text{Low.ModalQual}/K.\text{Low.InFlightPathCorr}/K.\text{Low.InFlightEscapeCorr}/K.\text{Low.DrainClosedNowCorr}$ are discharged for the exact lowering. Pure synchronization remains synchronization-typed and is not widened into MsgOf or message BoundaryReq; literal WFG/Dead_K remains message-rooted in each proof model. A lowered instance that does not discharge the Stage-2 modal lowering package is outside the current generalized theorem rather than silently reverting to the legacy snapshot predicate. **Certificate/version rule:** a lowered certificate issued under an earlier document/theorem version remains evidence only for that pinned earlier version. The current theorem version defines no legacy-fallback drain theorem; a lowered generalized claim requires a new or independently revalidated certificate bound to the Stage-2 modal schema.

A global state σ of either system consists of:

1. For each process P : a control location (program point) and local state.
2. For each explicit abstract channel c of Sys_ref: its abstract FIFO state at the chosen Sys_ref boundary—its current occupancy $\text{occ_c}(t)$ and (for buffered channels) its ordered contents. Here $\text{occ_c}(t)$ counts all items that have been committed by Push_c but have not yet been committed

by Pop_c at that chosen abstract boundary. In RTL_B, this count includes items residing anywhere in the concrete sub-realization that contributes to that particular abstract channel's back-annotated capacity B(c). Thus internal movement within the concrete realization of one abstract channel c is ϵ and does not create additional committed Push_c/Pop_c Σ -events beyond the boundary events counted by occ_c; however, transfers between distinct explicit abstract channels introduced in Sys_B^elab are represented in Sys_ref by those channels themselves and are not collapsed back into one outer Sys_B boundary.

3. Any additional architectural state not representing buffered messages on any channel (e.g., computation pipeline registers, arbitration state, bookkeeping, etc.).

Appendix K uses the normative wait-for graph WFG(σ) of Definition Core.WFG, restricted there to internal message-passing channels. We write WFG_ref for the graph reconstructed at a Sys_ref cutpoint and WFG_post for the corresponding K-facing graph at the selected RTL_B cutpoint. All vertices, coarse edge conditions, the edge-versus-blocked and multi-frontier clarifications, frontier-item attribution, ReleaseOwnerSets/WaitOwners projection, buffered/rendezvous treatment, and exclusion of environment-facing channels are exactly those of the Core/Q definitions; this appendix adds the release-aware deadlock, waiver, and preservation predicates but does not redefine the coarse WFG.

(No deassert-before-commit assumption.) Blocking Push_c/Pop_c requests are non-withdrawable within one reset epoch: once the request corresponding to a frontier item $x \in \Omega_{\text{msg},P}$ is asserted through its owning message instance $q = \text{MsgOf}_P(x)$, it is not deasserted before q commits, and Push_c payload remains stable while asserted. If an affecting RESET occurs first, RW2 ends that dynamic instance at the epoch boundary. No other nonterminal transition may withdraw or reattribute it. This rules out “try-and-withdraw” behavior for blocking transfers within the WFG abstraction.

K-facing cutpoint convention. All WFG and Dead_K reasoning is over Core.KCut states: the source state lies in KCutReach_ref(l), and the selected/certified RTL state lies in KCutReach_post(l). Core.4 quiescence means no internal ϵ -step is enabled in the corresponding model at its current cycle/phase; it does not mean that no Σ action or environment action is enabled. Purely combinational ready/valid propagation is additionally required to be at its fixed point by Core.KCut. Every source-side L2 SettlePoint is a KCut by Lemma Core.SettleIsKCut, although a Dead_K evaluation may occur later after Core.16-permitted drain has reached a drain-closed KCut. Core.SettleKBridge/K.DrainReach supplies that continuation relation. On RTL_B, Core.KCutClosure_post is the separate non-vacuity premise ensuring that each reachable K-facing cycle-evaluation origin reaches a finite KCut; with $K\epsilon$, Lemma Core.KCutObserve prevents a persistent relevant blocker from being hidden solely in pre-normalization RTL microstate.

Definition $K\epsilon$ (WFG-inert ϵ / no hidden micro-timing control decisions)

For the selected theorem instance l and proof model M used by Appendix K, $K\epsilon_{\{l,M\}}$ holds iff, after the WFG-relevant boundary-request / attribution cutpoint has been fixed, every ϵ -path $\sigma \xrightarrow{\epsilon^*} \sigma'$ in M that contains no Σ -visible committed Push_c/Pop_c event, no synchronization event, no signal Read/Write event, and no source-level issue/attribution transition that creates, removes, or reassigns a pending blocking boundary request is WFG-inert: it does not change (i) the potentially-blocking frontier $\text{Front}_P(\sigma)$ for any process P (i.e., which guarded blocking Push/Pop frontier items are minimal and pending), nor (ii) the truth or attribution of any boundary requests relevant to those frontier items, nor (iii) the channel-side enabling status (as defined in K.1) of any $x \in \text{Front}_P(\sigma)$, read through $q = \text{MsgOf}_P(x)$. Equivalently, these WFG-inert ϵ -steps do not change the channel state relevant to ChannelEnabled/ChannelDisabled (e.g., $\text{occ}_c(t)$ for $B(c) > 0$ buffered channels), and they do not create/remove rendezvous enablement for a frontier transfer except via Σ -visible committed

Push_c/Pop_c events. The model index M fixes the state/frontier universe, explicit boundaries, and ε -normalization relation on which these facts are interpreted.

This definition does not say that every non- Σ microstep is WFG-inert. In particular, an ε -normalization segment may include source-level issue/attribution steps that establish a persistent blocking request, such as moving from a cutpoint before a Pop has issued to a cutpoint where the Pop request is pending and BoundaryReq is true. Those steps may change Front_P, BoundaryReq, or request attribution, and are therefore not the ε -steps quantified over by the WFG-inert premise. Appendix K's WFG construction is applied only at the resulting Core.KCut cutpoints, after such request/attribution state has been fixed and the boundary network has settled.

In particular (pipelined loops / overlapped iterations): when HLS introduces overlap that may initiate later-iteration message reads “early,” we assume the design obeys the Automatic flush (drain-only blocking discipline) defined in Appendix B. Under that discipline, once the process has reached a stable blocked frontier, overlapped execution does not introduce any later-iteration blocking Push/Pop frontier item into Front_P(σ), so transient internal pipeline activity that merely moves already-accepted work inside the concrete realization remains WFG-inert ε and does not contribute wait-for edges. Overlapped execution in pipelined loops is modeled only as a potential retiming of ordinary Σ -visible boundary commits (Push_c/Pop_c) relative to the unpipelined source, and is accounted for by the back-annotated semantics at the chosen abstract boundary of Sys_ref (E4). In particular, the channel facts relevant to ChannelEnabled/ChannelDisabled and WFG edges—e.g., $\text{occ}_c(t)$ for $B(c) > 0$, and rendezvous enablement for $B(c) = 0$ —change only on Σ -visible committed Push_c/Pop_c events at that boundary, or on explicit source-level issue/attribution steps before the WFG cutpoint is formed; internal movement within the concrete realization of a single abstract channel after that cutpoint is WFG-inert ε and does not affect ChannelEnabled/ChannelDisabled or introduce/remove WFG edges.

Scope. All Appendix K results that cite $K\varepsilon$ require $K\varepsilon_{\{I,M\}}$ on the ε -normalization segments of the particular proof model M whose Core.KCut/WFG facts are being used. The legacy shorthand $K\varepsilon_I$ or $K\varepsilon$ may be used only when M is fixed unambiguously by the surrounding statement. Thus

K.DrainReach/Theorem K.1A consume the Sys_ref interpretation, the serial-reduction route consumes the Sys_K interpretation, and any use of Lemma Core.KCutObserve on RTL_B consumes the RTL_B interpretation. If an implementation intentionally uses internal micro-timing on such a segment to change Front_P membership, boundary-request attribution, or ChannelEnabled/ChannelDisabled facts without a corresponding Σ -visible event or explicit source-level issue/attribution transition, then $K\varepsilon_{\{I,M\}}$ is violated unless that behavior is made Σ -observable (e.g., via explicit modeling/snooping) or is witness-selected under the Appendix I.2 framework.

Model-indexing convention for Appendix K operational premises. Whenever Appendix K consumes a transition-system predicate whose truth may differ between Sys_ref, the lowered/normalized Sys_K, and RTL_B, the proof model is part of the predicate's meaning even where legacy prose omits the model subscript. In particular, $K\varepsilon$, nonwithdrawal, Core.Drain and its K.DrainEvidence discharge, Core.CycleClosure, Core.IssueFair, B2/B3/B5, and the source-model portions of A6–A11 are interpreted on the model whose execution or continuation is being proved. K.DrainReach and Theorem K.1A bind the source-side versions to Sys_ref; K.SerialRoutePremises and Lemma K.2b bind the serial-route versions to Sys_K; RTL-side KCut-observability uses the RTL_B $K\varepsilon$ instance. When Sys_K = Sys_ref, the corresponding source obligations coincide. When lowering is used, Lemma K.Low.C transports executions/cutpoints and Lemma K.Low.A5 transports A5-L, but neither lemma silently transports these other semantic premises or their evidence records. The qualification flow shall therefore discharge each required model instance separately or cite a named per-predicate preservation theorem whose hypotheses are checked for the selected lowering. Generalized drain adds

K.InFlightQual/Core.InFlightCycleExt/J.InFlightPathSound to this model-indexed class. In the non-lowered profile Sys_K=Sys_ref one source-model discharge may serve both roles. In the Stage-2 lowered

profile the Sys_K and Sys_ref qualifications are separate: Sys_K Core.CycleClosure carried by the serial route does not silently establish Sys_ref continuation existence. K.Low.ModalQual may cite an explicit preservation theorem for a named component, but otherwise each model instance is discharged independently. K.Low.C now transports the modal drain predicate only through the separately proved K.Low.InFlightPathCorr/InFlightEscapeCorr/DrainClosedNowCorr stack; it still does not transport Core.Drain, Core.CycleClosure, K ϵ , fairness, or the other model-sensitive premises named above.

A common instance is branching on the success/failure of a non-blocking poll in a way that changes the next Σ -visible candidate frontier **NextFront_P(σ)** (Definition Core.6; see Lemma J.KObs-Proc and Corollary J.1); in that case, either NB-CF must hold (Appendix I.2), or the poll outcomes must be witness-selected as stated there.

Non-blocking polling attempts and other purely internal actions do not contribute edges to the WFG; they are modeled as internal ϵ -steps. When a non-blocking call succeeds, that success is already represented at the Σ level as the corresponding committed Push_c/Pop_c event, but it does not itself add a wait-for edge because it is not a blocking operation.

Mapped-memory boundary rule for Appendix K. Core.MemFaithful does not broaden K ϵ into a general multi-cycle memory-progress theorem. A mapped memory may remain below the selected K boundary only when its hidden realization does not introduce, remove, or redirect a process-facing blocking cause or any frontier/request-attribution/enablement/release-support/WFG fact except through structure already represented in Sys_ref/Sys_B^{elab}; any ϵ -normalization activity used by K additionally satisfies K ϵ . A shared or arbitrated memory whose finite request/credit structure can fill and persistently backpressure a requester will generally require explicit K representation. If transaction-changing caching/merge/split would make the raw command/response stream unsuitable as an E4/Core.17 boundary, a Core.MemProfile shall instead place a transformation-invariant, K-faithful logical boundary above that stream. If no such boundary exists, the transformation is not covered by Appendix K for that instance.

SyncChannels and barrier well-formedness.

SyncChannel (barrier) operations are treated as pure synchronization events and are omitted from the Wait-For Graph, which tracks only blocking Push/Pop dependencies. We therefore interpret “deadlock” in this appendix as internal message-passing deadlock (i.e., a deadlocked set in which all processes are blocked on some internal Push/Pop dependency captured by the WFG). Any global stall at a barrier caused by mismatched or conditional participation (e.g., a process can permanently bypass the barrier or reach it a different number of times) is a specification-level error already present in the pre-HLS model; such stalls are not created by the HLS transformation and are excluded by an explicit barrier well-formedness assumption (A8 below), not by the internal message-passing deadlock premise. Under this assumption, omitting SyncChannels from the WFG is sound: a post-HLS deadlock implies the existence of a Push/Pop wait-cycle as characterized below. Stage-2 typing note. Definitions Core.7b–Core.7e and Lemmas J.KObs-SyncCarrier/J.KObs-SyncInFlight give a pending synchronization a typed carrier and local in-flight action/effect semantics for lowering/path correspondence, but they do not make that carrier a literal WFG source or broaden Dead_K to barrier deadlock. The A8/specification-level barrier-stall treatment remains unchanged. Synchronization carriers/actions are consumed only by the explicitly named in-flight/lowering correspondence interfaces.

Definition (Blocked on message passing). Using Front_P(σ) as defined above, we say that P is blocked on message passing in state σ iff Front_P(σ) is non-empty and, for every frontier item $x \in \text{Front_P}(\sigma)$, BoundaryReq(x, σ) holds and ChannelDisabled(x, σ) holds (equivalently, BoundaryReq(x, σ) $\wedge$ $\neg$ ChannelEnabled(x, σ)), read operation-attributively through the owning message instance $q = \text{MsgOf_P}(x)$. Note (ALL disabled $\Rightarrow$ stall). If any frontier item $x \in \text{Front_P}(\sigma)$ is channel-enabled at the

chosen Σ -visible boundary, then P is not “blocked on message passing” for WFG purposes. Consequently, an implementation may not be physically stalled at the chosen Σ boundary by a cross-channel “join” predicate while some frontier item $x \in \text{Front_P}(\sigma)$ at that boundary remains channel-enabled; any such conjunctive/join dependency must be represented as an explicit staged/bundle/join boundary channel so that, when the process is physically stalled, the relevant frontier items on that explicit boundary are all channel-disabled.

Legacy fast-path selector. For a non-empty D and $K = \langle E, w, H, O \rangle$, $\text{DrainCandidates_D}(K)$ is the finite set/multiset of attributed residual blocking offers o owned by $P \in D$ whose item lies outside $\text{FrontRec_P}(K)$ and is ChannelEnabledRec at K . This selector is retained for diagnostics, ordinary primitive fast-path checking, and backwards-readable certificates; it is not the general definition of drain closure once Q.2 proof-internal in-flight actions are admitted.

Ordinary embedding. Every member of $\text{DrainCandidates_D}(K)$ corresponds to a legal declared ordinary in-flight action whenever its current boundary commit is within the selected action/opportunity profile. Therefore a non-empty legacy DrainCandidates set witnesses generalized InFlightEscape whenever the corresponding action changes a D root or can participate in a finite declared path that does. The converse requires the primitive-only qualification below and does not hold for staging/join actions in general.

Definition K.WaitRoots (process-indexed blocked-frontier root family).

For a $\text{Core.FlightStateRec}$ state r and process P , define $\text{WaitRoot_P}(r)$ to contain: $\text{FrontRec_P}(r)$; the reconstructed $\text{ChannelEnabled}/\text{ChannelDisabled}$ values of its frontier members; and, for every disabled internal frontier member, the exact $\text{ReleaseOwnerSetsRec_E}$ family. For a non-empty process set D , $\text{WaitRoots_D}(r)$ is the family $P \mapsto \text{WaitRoot_P}(r)$ indexed by $P \in D$. This is deliberately a process-indexed family rather than an unordered flattened bag, so restriction to a subset is definitional.

Definition K.InFlightEscape and generalized DrainClosedNowRec.

Fix an applicable Appendix-K source proof model $M \in \{\text{Sys_ref}, \text{Sys_K}\}$, elaboration/lowering package E , and the in-flight action schema bound by the selected generalized profile. $\text{InFlightEscape_}\{M, E\}(D, K)$ holds iff there exist $n \geq 1$, reconstructed states $r_0, \dots, r_n$ and action identities $u_0, \dots, u_{n-1}$ such that $r_0 = \text{Core.FlightStateRec_}\{M, E\}(K)$, $\text{Core.InFlightStepRec_}\{M, E\}(r_i, u_i, r_{i+1})$ for every $i < n$, and $\text{WaitRoots_D}(r_j) \neq \text{WaitRoots_D}(r_0)$ for some $1 \leq j \leq n$. The edge relation has no D argument and no $\text{DrainScope}/\text{ownership}$ filter; D appears only in the root comparison. Define $\text{DrainClosedNowRec_}\{M, E\}(D, K) := \neg \text{InFlightEscape_}\{M, E\}(D, K)$. At an operational K -facing cutpoint σ of M in a qualified generalized profile, “ σ is drain-closed for D ” means $\text{DrainClosedNowRec_}\{M, E\}(D, \text{KObs_E}, w(\sigma))$. This remains a current-cutpoint predicate computed from a finite reconstructed transition system; it is not a claim that σ itself persists forever.

Continuation-soundness qualification. The predicate above is pure once M, E and K are fixed, but Appendix K may use it as the intended operational drain abstraction only under $\text{K.InFlightQual}/\text{J.InFlightPathSound}$. That level-3 theorem proves that the reconstructed finite paths are realizable; it is not stored in KObs . This explicit obligation surfaces a continuation-opportunity assumption that the legacy snapshot interpretation left implicit whenever an enabled drain object was observed after the current L3 selection opportunity had already closed.

Lemma K.WaitRootsProject, K.InFlightEscapeMono, and K.DrainClosedHeredity.

If $T \subseteq D$ then $\text{WaitRoots_T}(r) = \text{restrict_T}(\text{WaitRoots_D}(r))$ by definition. Hence any finite path that changes a T root also changes the corresponding D family, so $\text{InFlightEscape}(T, K) \Rightarrow \text{InFlightEscape}(D, K)$. By contraposition, $\text{DrainClosedNowRec}(D, K) \Rightarrow \text{DrainClosedNowRec}(T, K)$. These are pure reconstructed-

state lemmas and require no candidate monotonicity, SCC preservation, fairness, or per-action causal relevance.

Lemma K.PrimitiveDrainExact (ordinary source-offer fast path).

Under the restricted ordinary primitive profile in which the declared in-flight action universe contains only currently enabled already-issued non-frontier blocking Push/Pop source offers and every such action changes the corresponding owner's D-root immediately or is excluded by the primitive boundary semantics, generalized DrainClosedNowRec agrees with emptiness of DrainCandidates_D. Outside that profile, DrainCandidates_D is only a diagnostic/fast-path selector: proof-internal staging may require a path of length greater than one and may originate outside D.

Finite in-flight note. Finiteness is not derived from capacity alone and is no longer phrased as “the number of D-owned downstream drain commits.” Core.16(6), Core.InFlightDrainWF, and the Q.2 finite-choice theorem bound the full activation-local already-admitted action domain that may be explored, including proof-internal actions originating outside D when they can change D's roots. The well-founded rank/multiset decreases on every selected in-flight action and Core.16(2) prevents replenishment by new source launch/Issue. Core.SettleKBridge/K.DrainReach continue to account for actual reached drain commits and WFG-inert normalization on the exact source prefix; J.InFlightPathSound separately justifies the finite future path semantics used by generalized drain closure.

Definition K.Dead (ordinary internal message-passing deadlock predicate).

For a K-facing cutpoint $\sigma \in \text{KCutReach_M}(I)$ of an applicable Appendix-K proof model M, $\text{Dead_K}(\sigma)$ holds iff there exists a non-empty set of processes D such that σ is drain-closed for D and all of the following hold. The blocked-frontier carrier remains the literal blocking message-operation carrier of M. In profile G1, $M = \text{Sys_ref} = \text{Sys_K}$. In profile G2, M may be Sys_ref or the explicitly lowered Sys_K, but generalized drain is usable on the lowered model only under the Stage-2 modal qualification and the cross-model transport theorem below; synchronization itself is never inserted into the literal message carrier.

7. Every process P in D is blocked on message passing, and in particular there exists at least one frontier item $x \in \text{Front_P}(\sigma)$ on an internal message-passing channel c, with owning message instance $q = \text{MsgOf_P}(x)$, such that $\text{BoundaryReq}(x, \sigma)$ and $\text{ChannelDisabled}(x, \sigma)$ hold.
8. Connectivity: the coarse WFG induced by D is strongly connected and contains an internal process-dependency cycle; for $|D|=1$ this requires a genuine $P \rightarrow P$ dependency rather than relying on the vacuous one-vertex SCC convention.
9. Release closure: for every $P \in D$ and every internal channel-disabled $x \in \text{Front_P}(\sigma)$, every minimal internal process release support $S \in \text{ReleaseOwnerSets_E}(x, \sigma)$ intersects D.
Equivalently, no genuine internal modeled-process release alternative for any blocked frontier item is supported entirely by processes outside D.

When every relevant release family is singleton, $\text{ReleaseOwnerSets_E}(x, \sigma) = \{\{Q\}\}$, this release-closure clause is exactly the former no-outgoing-edge/closed-SCC test. For a conjunctive join, however, a support such as $\{Q, R\}$ remains trapped by D whenever at least one required member lies in D, even if another required member lies outside D; the coarse WaitOwners/WFG union is therefore not itself the exact closure predicate. Different frontier items remain disjunctive under the existing ALL-disabled process-blocking rule: because P becomes unblocked when any frontier member becomes enabled, the release-closure test ranges over every disabled member of Front_P .

In every qualified generalized profile, “drain-closed for D” in proof model M means

$\text{DrainClosedNowRec_}\{M, E\}(D, \text{KObs_M}(\sigma)) = \neg \text{InFlightEscape_}\{M, E\}(D, \text{KObs_M}(\sigma))$: the qualified finite reconstructed in-flight relation of that model has no path that changes D's literal blocked-frontier root family. This remains a current K-facing-state predicate computed from the current observation plus the

fixed qualified action schema; it is not a claim that σ persists forever. Continuation existence is consumed only by the model-indexed $K.InFlightQual/J.InFlightPathSound$ evidence. For G2 cross-model lowering, $K.Low.DrainClosedNowCorr$ below relates the two model-specific predicates at administratively normalized corresponding cutpoints; it does not identify their raw intermediate gate/guard states.

Members of D may additionally hold environment-facing frontier items, which remain outside ordinary WFG/ReleaseOwnerSets closure and are handled by the existing fixed-stimulus/environment-waiver conventions. On the RTL side Appendix K evaluates the corresponding reconstructed predicate $DeadKRec/DeadKRec^W$ on the certified $KObs$ view; it does not apply this operational source definition to an arbitrary RTL microstate. A process blocked solely on environment/testbench channels does not witness an internal message-passing deadlock in this appendix.

Lemma $K.ReleaseSCCReduction$ (connectivity is without loss of generality for internally supported release-closed sets). Fix a K -facing cutpoint $\sigma \in KCutReach_M(I)$. Suppose there exists a non-empty process set D_0 such that σ is drain-closed for D_0 , every $P \in D_0$ is blocked on message passing, D_0 satisfies Definition $K.Dead$'s release-closure clause, and every $P \in D_0$ has at least one internal channel-disabled frontier item $x \in Front_P(\sigma)$ with $ReleaseOwnerSets_E(x, \sigma)$ non-empty. Do not assume connectivity. Then there exists a non-empty $T \subseteq D_0$ satisfying all clauses of Definition $K.Dead$; hence $Dead_K(\sigma)$. Proof. Consider the coarse WFG induced by D_0 . For each $P \in D_0$, choose such an internal x and any $S \in ReleaseOwnerSets_E(x, \sigma)$. Release closure gives $S \cap D_0 \neq \emptyset$; choose $Q \in S \cap D_0$. Because $WaitOwners_E$ is the union of the release supports, $P \rightarrow Q$ is an induced-WFG edge, so every vertex has at least one outgoing edge within D_0 . Choose a terminal strongly connected component T of this finite induced graph. For $P \in T$ and any internal disabled $x \in Front_P(\sigma)$, let $S \in ReleaseOwnerSets_E(x, \sigma)$. Release closure in D_0 gives some $Q \in S \cap D_0$; then $Q \in WaitOwners_E(x, \sigma)$, so $P \rightarrow Q$ is an induced-WFG edge. Terminality forces $Q \in T$, and therefore $S \cap T \neq \emptyset$. Thus T is release-closed. Blockedness inherits processwise, and generalized drain closure inherits to T by Lemma $K.DrainClosedHeredity$ (via $K.WaitRootsProject/K.InFlightEscapeMono$). T is strongly connected; if $|T|=1$, the within- D_0 outgoing-edge property together with terminality forces the genuine self-loop required by Definition $K.Dead$. Hence T is a $Dead_K$ witness. $\square$ The non-empty-release-family premise is essential: an internal wait whose peer has terminated or otherwise has no genuine internal modeled-process release possibility may have an empty $ReleaseOwnerSets$ family, and Definition $K.Dead$ intentionally does not classify such a root as an internal wait cycle merely by release-closure vacuity.

Lemma $K.InFlightRelocate$ (activation-local witness relocation and candidate redispatch).

Fix a $Core.LAdm$ -admissible reachable qualified K -facing cutpoint σ_0 , a non-empty blocked candidate D at $K_0=KObs(\sigma_0)$, and the bound $K.InFlightQual/J.InFlightPathSound$ profile for the applicable $Core.16$ activation family. If $DrainClosedNowRec(D, K_0)$ is false, choose a finite reconstructed escape and, using $J.InFlightPathSound$, a matching finite legal no-new-launch/Issue operational extension. Stop at the first resulting K -facing cutpoint σ_1 whose reconstructed $WaitRoots_D$ differs from K_0 . This is a witness relocation, not an assertion that D or its SCC survives unchanged.

Redispatch rule. At σ_1 , stop only the relocation instance for the original D and immediately re-evaluate every non-empty candidate set permitted by Definition $K.Dead$, including proper subsets, supersets, and incomparable sets. A residual candidate that is already drain-closed is evaluated normally; a residual candidate with another in-flight escape starts a fresh relocation instance using the still-live activation evidence applicable to its processes/frontiers. Ending one process's activation does not retire independent still-live activations for other processes. If later source progress or reset creates a genuinely fresh activation, the old activation-local rank/evidence is not reused.

Termination/endpoint rule. Repeated relocation within one live activation context terminates by $Core.InFlightDrainWF$: each realized in-flight action strictly decreases the finite admitted-work measure,

and no new source work replenishes it. The endpoint need not preserve connectivity. For any drain-closed endpoint candidate D_0 that is non-empty, blocked, release-closed, and has non-empty internal release support for every member, apply $K.\text{ReleaseSCCReduction}$ to recover a literal Dead_K witness subset. Thus the proof needs neither monotonic D nor SCC preservation nor a cross-activation global measure. A5-L covers each resulting finite admissible K -facing endpoint; later independent source progress is handled as a fresh candidate/activation, not as continuation of the old normalization.

Terminated-process convention. For purposes of Appendix K, termination is interpreted per reset epoch. After a process P emits FINISH_P for a given reset epoch, the infinite execution is padded only by process-local idle stutter for P for the remainder of that epoch. This padding creates no new P -owned Σ -events, no new BoundaryReq , no new Front_P or NextFront_P item, and no new outgoing WFG edge from P . This epoch-local padding does not preclude a later reset boundary affecting P . If such a reset occurs and P later exits reset, the subsequent START_P is a new dynamic occurrence that begins a fresh reset epoch, and the earlier FINISH_P has no ordering, anchoring, or WFG force across that reset boundary. Any later termination of P is represented by a distinct dynamic FINISH_P occurrence. Thus a process blocked waiting for a complementary Push/Pop from a peer that has terminated in the current reset epoch is not, by itself, an internal message-passing deadlock under the Dead_K predicate. It is instead outside the Dead_K predicate unless the design includes an explicit modeled termination protocol that makes such waiting part of the internal channel protocol being analyzed.

Terminology (adopted reading: reachable release-closed internal wait-cycle cutpoint). The predicate just defined is, precisely, a reachable, drain-closed, release-closed internal wait-cycle cutpoint. It is deliberately weaker than a conventional permanent deadlock: a member of D may hold environment-facing frontier items excluded from ordinary $\text{ReleaseOwnerSets}/\text{WFG}$ closure, and—in a declared timed or reactive environment outside the B1-avail convention—a fair continuation in which the environment later enables such an item can leave the state. (Under the default B1-avail pull-based stimulus convention, an environment-facing input item disabled at a cutpoint is stimulus-exhausted and remains disabled forever, per Lemma $K.\text{EnvX-Auto}$ in K.4b; the later-enablement possibility, and with it the input-side $W.\text{Env}$ rescue class, arises only for declared timed/reactive environments.) Three consequences are adopted normatively throughout this appendix. (1) The property preserved by Theorem K.1—absence of reachable states satisfying this predicate—is an invariance (safety-style) reachability property over admissible executions, not a temporal liveness property in the eventually/always sense; the historical word “liveness” is retained in titles for continuity but shall be read with this meaning. (2) Informal prose such as “processes in D can only wait on each other” shall be read as Definition $K.\text{Dead}$ ’s exact release-closure condition: every internal modeled-process release alternative for every disabled frontier item intersects D . It does not mean that every owner in the coarse WaitOwners union lies in D , and it does not exclude coexisting environment-facing waits. (3) The A5 premise is correspondingly conservative: it can be violated by a transient release-closed internal wait-cycle cutpoint even in a design whose fixed environment is guaranteed to release an environment-facing frontier item so that the system always progresses; such a report is a true positive for the defined predicate and a deliberate conservatism with respect to permanent deadlock. Designs for which this conservatism is unacceptable may instead be assessed against a stronger persistent-deadlock predicate requiring the relevant blocking frontier to remain a trap under all admissible continuations; the theorems of this appendix neither require nor establish results for that stronger predicate, and adopting it would require re-proving Lemmas $K.\text{DeadChar}$ – $K.2b$ and Theorem K.1 with an added permanence argument.

Scope warning. This exclusion is only a scope convention for Theorem K.1. It does not imply that waiting forever for a terminated peer is benign or acceptable design behavior. If termination, EOF, done, cancellation, or end-of-stream behavior is part of the intended protocol, that behavior must be modeled explicitly in Sys_ref and RTL_B , or verified separately outside the Appendix K internal message-passing-deadlock theorem.

Our notion of the preserved property (historically labeled “liveness”) in this appendix is: no reachable internal message-passing-deadlock state (reachable closed internal wait-cycle cutpoint) as defined above, under the fixed stimulus and fair scheduling. Formally this is an invariance (safety-style) reachability property over admissible executions — see the Terminology note above — and it does not assert that a reached deadlock state is permanent when environment-facing frontier items are present. (Starvation of a single process in an otherwise live system is ruled out separately by weak fairness.) Implication note (relation to persistent deadlock). Within the release-closed internal wait-cycle class captured by `Dead_K`, the reachable-cutpoint property is conservative with respect to permanence: it excludes even a transient reachable `Dead_K` cutpoint, and therefore also excludes a persistent internal hang whose blocked condition contains such a drain-closed release-closed WFG/release-family component. This implication is intentionally limited to the `Dead_K` structural class. The theorem does not establish the broader operational guarantee that every form of permanent internal-communication non-progress is impossible. In particular, starvation or perpetual under-supply by a complementary owner that remains live on unrelated work can leave downstream operations permanently unanswered without producing a release-closed `Dead_K` component; Example K.CE-5/H2 is the canonical case. Waiting for a terminated peer and the other stalls excluded by the scope warnings above likewise require explicit protocol modeling or a separate argument. The stronger reachability form is retained as the normative `Dead_K` predicate because, within its stated class, it is safety-shaped—an invariance over reachable states—which makes it checkable by inductive invariants and preservable through finite-prefix correspondence (Definition J.Reach, Corollary J.1). A theorem covering permanent non-progress outside that release-closed wait-cycle class would instead require additional continuation/fairness and permanence reasoning and is not claimed here. Designs with intentional environment-rescued `Dead_K` cutpoints are handled by the Waiver `W.Env` mechanism (K.2).

Example K.InFlight-CE1 (offer-only drain can over-fire; two-stage rescue).

Take $D = \{C, R\}$. At $K0$, C has a persistent blocking `Pop_d` with d empty and `ReleaseOwnerSets = {{R}}`; R is blocked on an internal frontier whose minimal release support is $\{C\}$, so D is blocked, release-closed, and strongly connected. Separately, E contains already-admitted proof-internal token z in a positive-capacity chain $a \rightarrow b \rightarrow d$ with $a = [z]$, $b = []$, $d = []$ and no enabled non-frontier residual source offer owned by C or R . The legacy offer-only `DrainCandidates` test is empty and would therefore classify the cutpoint as drain-closed. Under Q.2, however, `MoveAB` followed in a later real cycle by `MoveBD` is a legal reconstructed finite path: the first action changes no D root, while the second makes d nonempty and changes C 's frontier enablement. Thus `InFlightEscape(D, K0)` holds and generalized `DrainClosedNowRec` is false. The example simultaneously shows why proof-internal actions outside D may matter and why immediate root change is insufficient. Its cross-cycle operational use requires `J.InFlightPathSound/Core.InFlightCycleExt`; on the no-lowering serial route that opportunity evidence is supplied from the already-bound `CycleClosure` plus the template/profile proof.

K.2 Assumptions and Imported Results

Standing Assumptions. We assume throughout Appendix K:

A1. R-rules and B-rules. The prescribed source reference model `Sys_ref` and the post-HLS implementation `RTL_B` satisfy the process- and channel-level semantic rules defined earlier (R-rules and B-rules).

A2. Finite Pipeline Depth (FPD) / B4 finite ϵ -stutter.

For each process P there is a finite bound D_P such that, along any consecutive epsilon-only evolution in which P is internally advancing toward its next observable Σ -action or its next Σ -action is locally enabled,

and P is not externally stalled by a peer/environment/back-pressure condition, P executes at most D_P consecutive epsilon-steps before either (i) executing a Σ -action, or (ii) reaching a stable waiting state in which no further epsilon-step is available until some external/peer/environment condition changes the relevant enabling predicates. This is not a bound on clock cycles spent waiting for external stimulus, peer action, or back-pressure relief; such waiting is governed by WF/B2 and B3. The system-level all-disabled quiescence bound is the separate B5 cycle-valued bound C_{sys} . A theorem instance may discharge B5 by supplying C_{sys} directly or by proving per-process cycle bounds C_P from D_P through the bucket-semantic charging lemma and taking $C_{sys} = \max_P C_P$. The identification $C_{sys} = \max_P D_P$ is permitted only when that bridge is proved.

A3. Weak Fairness (WF) (B2). If an action remains continuously enabled (its guard remains true), the scheduler eventually selects it.

A4. Channel Progress (B3 (Additional Semantic Assumptions (B-rules)); Appendix N). For each channel c with capacity $B(c)$, assume the progress invariants of Appendix N hold. This is an obligation on the scheduler/environment (e.g., fairness of arbiters/back-pressure) and is not implied by the local R-rules alone.

A5. Primary selected-instance Policy-S bounded-response premise (A5-S). For the selected Sys_K instance, let p_S^{det} be the deterministic Policy-S execution selected by $SDet$ after fixing capacities, elaboration, initial/reset state, stimulus, witness choices, environment behavior, arbitration, concrete pre-HLS scheduling semantics, Policy-S issue semantics, ϵ -normalization, and $K.Low$ lowering. A5-S requires: (i) $K.CompleteS$ evidence that p_S^{det} is maximal, fair, and complete; (ii) $K.RespCov$ evidence defining a checkable proof disposition and finite bound for every static internal process-facing blocking site and relevant dynamic class and proving that every dynamically reachable instance belongs to a monitored or statically bounded class, with coverage-preserving waivers and the separate environment-site rule; and (iii) $K.RespClean$ evidence that no finite prefix of the complete run witnesses a non-waived $K.RespFail$. $K.RespFail$ is a finite-prefix predicate covering certified-bound expiration without process-facing commit/return under the deadline convention of Definition $K.RespFail$, a declared maximal finite terminal run with the monitored frontier still pending, or an affecting RW2 RESET that clears the monitored dynamic instance before response without an explicit waiver. A5-S is used by Corollary $K.2c$ to derive $A5L_Sys_K$ and then, through $K.Low.A5$, A5-L only when $K.EnvMF$ and the shared $K.InterruptionGate$ applicability hold for the candidate components, with $K.ResetEpoch$ evaluated first and $K.TerminalDisposition$ required only on $K.ResetEpoch.Continue$.

For A5, Policy S is the conservative serial-blocking interpretation of Appendix G. Within each source interval, a process does not issue a source-later blocking operation until every source-earlier blocking operation required by the serial discipline has committed in a strictly earlier cycle. The condition is process-facing: proof-internal staged/bundle/join/epoch-gate transfers do not satisfy it unless the elaboration identifies them as the commit that releases the source-level call. Channel-side E4 movement, FIFO drainage, independent-process execution, and explicit elaborated transfers remain concurrent.

Maximality, completeness, and fairness of p_S^{det} are part of A5-S/ $K.CompleteS$, not $SDet$. $K.RespCov$ and $K.RespClean$ are additional obligations checked over that complete artifact, not definitions of completeness itself. A finite simulation discharges A5-S only when it reaches a declared maximal terminal state and has no pending covered frontier, or when a complete proof establishes the required monitor result. A finite prefix of a still-running system is insufficient. For a nonterminating design, the selected run requires an invariant, sound model checking, recurrence/lasso evidence, or another complete argument that every covered response occurs within its finite bound and no reset-pending failure occurs.

Instance dependence. A5-S and $SDet$ are not properties of the raw SystemC text and are not discharged by an arbitrary simulation. They are re-established whenever capacities, boundaries, elaboration,

lowering, stimulus, witness assignment, environment/reset contract, arbitration rule, concrete simulator scheduling semantics, Policy-S implementation, or normalization changes. Such a change defines a different selected theorem instance.

Practical discharge route (Corollary K.2c). Under SDet there is one selected Policy-S execution p_S^{det} for the selected instance, with a committed history unique modulo finite WFG-inert ε -stutter. $K.EnvMF$ restricts the supported environment/frontier cases. $K.ResetEpoch$ and $K.TerminalDisposition$ instantiate the shared $K.InterruptionGate$ schema in the normative nested order: reset first, then terminal only on $K.ResetEpoch.Continue$. A $DirectFail$ branch yields the corresponding finite $K.RespFail$ without invoking dominance; the two nested $Continue$ branches select $K.2b.E \rightarrow K.2b.P0/K.2b.R/K.2b.P \rightarrow K.2b.P-Frozen/K.2b.SF \rightarrow K.2b.Q$. A5-S/CF-S separately supply $K.CompleteS$, $K.RespCov$, and $K.RespClean$ evidence. Lemma K.2b shows that any internal deadlock reached by a finite Policy-L prefix admissible under Definition Core.LAdm forces that same selected execution to produce a finite non-waived $K.RespFail$ witness. Therefore a complete clean bounded-response certificate for p_S^{det} is sufficient on that supported fragment. No alternate Policy-S schedulers or interleavings exist within the selected instance, and no confluence theorem is relevant.

A source region whose liveness depends on favorable same-cycle or overlapped issue of multiple blocking operations is outside the scheduler-robust A5-L framework whenever an admissible serial-shaped Policy-L prefix reaches $Dead_K^W$. In that case A5-L is false, not merely difficult to discharge: direct Policy-L verification, structural conditions, invariants, exploration, and tool certificates cannot prove it for the unchanged instance. The design must be rewritten or buffered, placed under explicit protocol/synchronization structure that removes the serial deadlock prefix, or verified under a different scheduler-specific theorem not supplied by this document.

A5 and Appendix K do not assert freedom from specification-level barrier errors, reset stalls, or external-environment stalls that do not contain the internal wait structure represented by the operative predicate.

A6. Point-to-point source ownership and elaborated dependency projection. For every source-level logical message channel c , exactly one modeled process owns all $Push_c$ operations and exactly one modeled process owns all Pop_c operations; shared source resources are remodeled through explicit arbitration and point-to-point process-facing channels. At an ordinary primitive boundary this gives the unique complementary process owner and the singleton release family $\{Q\}$ used by WFG and the ordinary K.2b cases. In Sys_B^{elab} , an internal explicit channel endpoint may instead be driven by a stateless coupler and need not be assigned a fictitious process. The well-formed elaboration E shall provide the Q.1 ReleaseOwnerSets family and its WaitOwners union required by Core.WFG/Definition K.Dead, including family-level support soundness/completeness and the no-spurious-self-dependency rule. A coupling-sensitive disabled frontier may therefore have one conjunctive multi-owner support, several alternative supports, or both.

A7 (RTL determinism / witness selection / equal-KObs correspondence). Assume B1 for RTL_B . J.GWR-Comp/Theorem J.1 and Corollary J.1 supply one selected J.GWR source prefix τ_ref^G , its recorded K-facing Core.KObsNormRef suffix v_ref , and equal KObs at σ_ref for $\tau_ref^K := \tau_ref^G \cdot v_ref$. For Appendix K, J.RegPhaseReach is derived from the revised registered, deferred-hole, phase-normal construction (or directly supplied) on τ_ref^G ; K.NormStable validates the post-L3 observation-only suffix; and K.DrainReach validates Sys_ref nonwithdrawal, $K_{\varepsilon}\{I, Sys_ref\}$, and every activated Sys_ref Core.Drain/K.Drain obligation on τ_ref^K . Together these three components are $K.RLAdmReach$ and establish $\tau_ref^K \in Pref_L^{adm}(I)$. J.OfferReach, K.NormStable, J.OfferConf, J.KObsConf, RegPhaseReach, DrainReach, and K.RLAdmReach must bind the same RTL/source package, H,O,E,w, instance, and stimulus. Core.KCutClosure_{post} and K.CertCov are additionally required only for the universal reachable-cutpoint conclusion. In the standard QSync normal-compositional scope, Core.QSync-Post derives the closure premise and Lemma Core.QSync-KCutCanonical identifies the

complete RTL KCut domain with canonical endpoints; if the flow proves $J.QSyncCertInd$, Theorem $J.CertExist-QSync$ and Corollary $K.CertCov-QSync$ derive $K.CertCov$ by induction over those canonical cycle cutpoints. Otherwise $K.CertCov$ remains an independently discharged coverage obligation. When a deterministic, reproducible Sys_ref witness is desired for simulation/debug, apply the Appendix L snooping wrapper to select a particular witness aligned with the latency-sensitive ϵ -choices. B4/B5 remain available for the generic bounded-normalization uses stated elsewhere, but they do not by themselves establish the phase placement or no-feedback conditions of the source post-L3 $Core.KObsNormRef$ used by Corollary J.1; the standard $QSync$ route uses $Core.KObsSettle-Ref$ and $K.NormStable$ for that source suffix. Every Appendix K transfer of a reconstructed frontier, request, channel, enablement, WFG, drain, or deadlock predicate uses the same equal-KObs package. A discharge for an unrelated source cutpoint, different witness, different O, or noncanonical unmatched history is insufficient.

Premise $SDet$ (deterministic selected Policy-S simulator). Fix the selected Sys_K , initial/reset state, capacities/elaboration, stimulus and reactive environment, witness-selected ϵ outcomes, deterministic arbitration, concrete SystemC/pre-HLS scheduler, Policy-S issue rule, and ϵ -normalization. These define a functional Policy-S next-state semantics and select one run artifact p_S^{det} . Every finite or infinite artifact generated from the same initial state by those exact semantics has a committed history prefix-comparable with that of p_S^{det} modulo finite WFG-inert ϵ -stutter; hence no two incompatible committed histories arise from the selected instance. WC and $I.DR/I.DR'$ determine replay state/actions relative to the selected history; $SDet$ excludes multiplicity caused by an unfixed scheduler, arbiter, environment, witness, issue, or normalization choice. $SDet$ does not by itself assert maximality, fairness, completeness, response coverage, or response cleanliness, and a shorter finite artifact may be a proper prefix of the selected history. $K.CompleteS$ and $A5-S/CF-S$ separately establish maximality/fairness/completeness and $K.RespCov/K.RespClean$ for p_S^{det} . Changing any listed choice defines a different selected instance.

A8. Barrier well-formedness (SyncChannels). For each $SyncChannel$ instance, the set of designated participant processes reaches the barrier the same number of times under the fixed stimulus/initial state; equivalently, no participant can permanently bypass a barrier call while another participant is waiting at that barrier. (Barrier stalls are therefore outside the behaviors considered in Appendix K's internal message-passing deadlock analysis.)

A9. Single-transfer-per-cycle endpoint constraint (scalar channel interface). The selected instance satisfies the authoritative $Core$ single-transfer-per-cycle endpoint constraint; multi-lane/burst resources are modeled explicitly as specified there.

A10. Reset-empty FIFO assumption. The selected instance satisfies the authoritative $Core$ reset-empty FIFO convention at every reset boundary in scope.

A11. WFG-inert ϵ ($K\epsilon$). Every proof model M on which the selected Appendix K route consumes $K\epsilon$ satisfies $K\epsilon_{\{I,M\}}$. In particular, $K.DrainReach/Theorem K.1A$ require the Sys_ref instance, the serial-reduction route requires the Sys_K instance, and RTL_B KCut-observability requires the RTL_B instance when that consequence is invoked. If $Sys_K = Sys_ref$, one source-model discharge suffices.

No further numbered A-assumption is introduced beyond A1–A11; $SDet$ is a separately named selected-simulator premise. $A5-S$ is part of the selected-instance liveness obligation, not a conclusion derived from the local R-rules, WC alone, or the no-reverse rule.

Definition K.DrainEvidence (model-indexed Core.Drain discharge record).

For a fixed theorem instance I , proof model M , and exact source path or prefix p on which Appendix K consumes $Core.Drain_I(p)$, $K.DrainEvidence_{\{I,M\}}(p)$ is the model-indexed application evidence record whose completed fields discharge that semantic path property; it is not a second drain predicate. Legacy

prose using “K.Drain” denotes the obligation to supply this record for the applicable model/path, and legacy prose of the form “Core.Drain/K.Drain” denotes Core.Drain together with its required K.DrainEvidence discharge. The record binds the same theorem-instance, witness, environment, arbitration, reset, elaboration, and boundary-attribution choices as p and covers every Core.16 activation a reached on p for as long as the same fully disabled frontier persists. On the standard normal route, K.DrainReach-Comp factors the record into DR1–DR4. For each activation a, the DR1 activation field records one of two routes for the Core.16 blocked-point/frontier, no-new-later-work, frontier-minimality/non-frontier exclusion, and drain-only downstream-progress content: (DR1a) RTL-backed activation — bind an Appendix-B automatic-flush fact for the RTL region/cycle corresponding to a and for the same dynamic blocked frontier/request attribution, then use the bound J.LocalGWR/J.GWR-Comp construction-action package with I.A0/E3 and the PCert/ConstructionAction identity to lift that restriction to a, including any L1 A1 preparation that would count as a Core.16(2) source-later launch; or (DR1b) direct-source activation — supply a source structural proof of the same applicable Core.16 clauses at a. Different activations in one mixed design may use different routes; there is no instance-wide route choice. If the standard J/I package does not establish that a DR1a automatic-flush fact refers to the same dynamic source frontier rather than merely the same cycle, the application shall supply a named activation-correspondence theorem or certificate. The remaining record fields may cite the named common evidence used by DR2–DR4, including Core.IC/Appendix C request nonwithdrawal, E4/B-faithfulness, finite explicit storage/credit, the Q.2 finite action inventory, and Core.InFlightDrainWF; K ϵ and Core.SettleKBridge remain the separately named normalization/activation-to-KCut qualifications consumed by K.DrainReach. A uniform instance-level K.DrainEvidence proof schema may be instantiated afresh for each exact constructed prefix. No K.DrainEvidence prefix-extension rule is implied.

Definition K.InFlightQual (generalized drain action/path qualification record).

For any Appendix-K source proof model M selected by the theorem instance and fixed I,E, let K.InFlightQual__{I,M,E, Φ } package the independently checkable evidence that makes generalized DrainClosedNow in M an operationally sound abstraction: (IQ1) the exact Q.2 action/effect schema identity for M and Q.InFlightActionEffectSound, including Q.SyncInFlightActionEffectSound when the Stage-2 synchronization schema is selected; (IQ2) Q.InFlightChoiceFinite and the Core.InFlightDrainWF/DR4 finite admitted-population and rank proof; (IQ3) the selected Core.InFlightCycleExt or full Core.CycleClosure/equivalent route establishing required same-/cross-cycle action opportunity on profile Φ , with Core.SyncInFlightOpportunityQual when applicable; (IQ4) J.InFlightPathSound__{I,M,E, Φ } with prefix-compositional profile closure and exact reconstructed successors; and (IQ5) the declared precision/completeness class of the reconstructed relation and checker search. This record is model-indexed operational/certificate evidence, not a KObs field and not a second definition of DrainClosedNow.

Route-specific discharge. For the primary serial route, K.SerialRoutePremises binds Core.CycleClosure on Sys_K. If Sys_K=Sys_ref, that C2/C3-class evidence may discharge IQ3 for the source theorem through Core.CycleChoiceExt once Q.2 action legality and Φ -closure are supplied. If lowering is used, the same Sys_K evidence discharges only the Sys_K IQ3 obligation; the Sys_ref K.InFlightQual required by K.Low.ModalQual shall be discharged separately or by an explicit named Sys_K $\rightarrow$ Sys_ref continuation-preservation theorem. Structural, invariant, exploration, direct Policy-L, and tool-certificate A5-L routes similarly bind the K.InFlightQual required by the model on which their Dead_K claim is interpreted. Proving A5-L alone does not manufacture continuation evidence.

Definition K.A5L (model-indexed scheduler-robust bad-cutpoint exclusion). For a source proof model M used by Appendix K, write A5L_M(I,W) for the A5-L formula interpreted in M: every finite Policy-L prefix

admissible under M 's Core.LAdm/Pref_L^{adm} domain, with the fixed theorem-instance stimulus, capacities, elaboration/lowering, environment, arbitration, reset interpretation, and witness-selected outcomes, excludes every reached $\sigma \in \text{KCutReach}_M(I)$ satisfying the corresponding Dead_K^W predicate. The model index fixes the operation/frontier universe and explicit boundaries on which the predicate is evaluated; it is not an additional execution choice.

Premise A5-L (scheduler-robust liberal source liveness). A5-L is the abbreviation A5L_Sys_ref(I,W). Thus, for every $p_L \in \text{Pref}_L^{\text{adm}}(I)$ of the selected Sys_ref instance, under the fixed stimulus, capacities B(c), elaboration, and fixed witness-selected outcomes, no source K-facing cutpoint $\sigma \in \text{KCutReach}_{\text{ref}}(I)$ reached by p_L satisfies Dead_K^W. Theorem K.1A consumes this Sys_ref form for one source prefix whose Pref_L^{adm}(I) membership is supplied by K.RLAdmReach. Where synchronization waits or witness-selected non-NB-CF sites can participate in a potential wait cycle, the serial route first proves the corresponding A5L_Sys_K(I,W_K) result on the normalized blocking instance Sys_K and then transports it to A5-L by Lemma K.Low.A5.

Policy-S-primary architecture. SDet fixes one concrete deterministic Policy-S execution $p_{S^{\text{det}}}$. K.EnvMF routes candidates by environment and release structure: under StandardEnv/B1-avail, candidates satisfying K.JointFreeze-Ord's ordinary/single-release-support premises use that lemma and every other admitted candidate supplies K.JointFreeze; under timed/reactive or otherwise nondefault environments, only the certified single-internal-frontier, K.JointFreeze-Ord-eligible case is admitted. K.InterruptionGate supplies the common applicability branch discipline; K.ResetEpoch is evaluated first and K.TerminalDisposition only on K.ResetEpoch.Continue. Lemma K.2b compares a liberal counterexample with the selected execution and proves that a Policy-L internal deadlock forces a finite K.RespFail witness on $p_{S^{\text{det}}}$. Corollary K.2c therefore reduces A5-L to the single complete A5-S check consisting of K.CompleteS, total K.RespCov, and K.RespClean. There is no family of Policy-S scheduler interleavings inside one selected instance and no confluence question. Dead_KS, EnvStop, DoneStop, K.EnvOrd, and K.Done are optional diagnostic classifications only.

Normative organization of Appendix K. The spine is: the Dead_K predicate; K.0 frontier reconstruction; K.2 WFG extensionality; K.2a DrainClosedNow extensionality; the K.RLAdmReach qualification and K.CertifiedCutpoint interface; Theorem K.1A from A5-L for one certified cutpoint; Core.QSync-Post/Core.QSync-KCutCanonical plus J.QSyncCertInd/J.CertExist-QSync and K.CertCov-QSync for the standard universal package-totality route; Core.KCutClosure_post, K.CertCov, and Corollary K.1A-Total for the universal reachable-cutpoint conclusion; Sys_K and the SDet-selected $p_{S^{\text{det}}}$; K.EnvMF; the shared K.InterruptionGate schema with reset-first K.ResetEpoch and nested K.TerminalDisposition instantiations; deterministic serial-dominance K.2b; Corollary K.2c; A5-S on $p_{S^{\text{det}}}$; and Theorem K.1. Appendix K-P contains the operational companion proof. Structural and direct certificates remain alternate A5-L routes only when A5-L is true for the selected instance.

Waiver W.Env (declared environment-rescued cutpoints). The Appendix M.7a side-condition record may exclude a precisely identified cutpoint class when a declared environment-facing co-frontier is intended and proved to rescue it. The record shall identify the component class, rescuing item, environment assumption, and design justification. A waiver does not alter the default predicates and does not excuse an Example K.CE-style bypass without an accompanying no-bypass or equivalent argument.

Definition K.DeadW and response waivers. Let W be the union of declared KObs-facing waived cutpoint classes. Dead_K^W(σ) means Dead_K(σ) and $\sigma \notin W$; optional diagnostic Dead_KS^W is defined analogously. KObs-facing waiver closure requires KObsClosed_I,E,w(W) whenever W is non-empty. Separately, let W_resp be the declared dynamic K.Resp monitor-waiver classes. RespFail^W means a K.RespFail whose monitor instance is not in W_resp. K.RespCov requires coverage-preserving waiver closure by the authoritative pre-K.2b Environment and waiver rule below: for every non-waived liberal Dead_K candidate admitted to the K.2b premise, the candidate-to-serial-frontier coverage relation guarantees at least one mapped covered Policy-S response class outside W_resp. This condition is

established before invoking K.2b and is not defined by a monitor later “extracted by K.2b.Q.” Because the release-family correction can recognize coupling-sensitive Dead_K components that the former flat no-outgoing-edge test missed, K.RespCov/waiver coverage and any direct A5-L certificate created under the superseded flat-edge schema shall be revalidated against the current K.CertHdr schema/version; ordinary singleton point-to-point instances are unchanged. A response waiver depending on hidden issue timing, reset disposition, or future environment behavior must carry the corresponding operational proof. Neither W nor W_resp may be used as a broad label to hide an unclassified deadlock or end-of-test hang.

Scope of the preservation theorem. Theorem K.1A transfers a deadlock at one K.CertifiedCutpoint RTL cutpoint to a Core.LAdm-admissible Policy-L Sys_ref deadlock through equal KObs and contradicts A5-L; it consumes neither SDet nor the serial-route premises and makes no package-totality claim. Corollary K.1A-Total adds Core.KCutClosure_post and K.CertCov and obtains the universal reachable-cutpoint conclusion. Theorem K.1 adds SDet, K.CV, NB-Align/K.Low where applicable, K.Resp, and the remaining serial conditions, obtains A5-L from A5-S for the SDet-selected execution p_S^{det} via Corollary K.2c, and uses Core.KCutClosure_post together with K.CertCov for the user-facing universal RTL conclusion. Within Lemma K.2b, K.InterruptionGate is instantiated first by K.ResetEpoch and, only on K.ResetEpoch.Continue, by K.TerminalDisposition. Either DirectFail branch bypasses the dominance construction; the two nested Continue branches select $K.2b.E \rightarrow K.2b.P0/K.2b.R/K.2b.P \rightarrow K.2b.P\text{-Frozen}/K.2b.SF \rightarrow K.2b.Q$. K.CompleteS supplies maximality/fairness/completeness, K.RespCov supplies total finite-bound coverage, and K.RespClean is the contradiction used by K.2c. Generalized-drain qualification: every K.1A/K.1A-Total application under this document version binds K.InFlightQual for Sys_ref under the selected source action/profile schema. K.1A is deliberately agnostic to how A5-L was discharged: in G1, Sys_K=Sys_ref and the primary serial route may reuse its already-bound CycleClosure for the source continuation component; in G2, K.Low.ModalQual/generalized K.Low.A5 may be used upstream to transport A5L_Sys_K to the source A5-L consumed here, while the separate Sys_ref K.InFlightQual required by K.Low.ModalQual supplies the source modal/path qualification. Structural, invariant, exploration, direct Policy-L, and tool-certificate A5-L routes likewise bind the applicable Sys_ref K.InFlightQual or equivalent route-specific discharge. The per-cutpoint theorem itself consumes source DeadKRec/A5-L and source K.InFlightQual, not a no-lowering premise.

The no-reverse rule is not by itself a liveness theorem. Its practical force is captured by the deterministic Policy-S reduction: the selected simulator withholds source-later requests and serves as the conservative deadlock-stress execution. K.2b does not claim that p_S^{det} reproduces the Policy-L trace or the same component; it states that any Policy-L internal deadlock forces a permanent covered process-facing serial frontier and therefore a finite K.RespFail witness.

Example: blocking rendezvous operations whose liveness depends on same-cycle/eager issue. Consider two rendezvous channels a and b with $B(a)=B(b)=0$ and two processes; assume v and w are independent of any data/status returned by the preceding Pops:

```
P1: b.Pop(); a.Push(v)
P2: a.Pop(); b.Push(w)
```

Under Policy S’s conservative serial-blocking issue discipline, each process issues only its first source-order blocking operation: P1 waits on Pop_b and P2 waits on Pop_a; neither process reaches the complementary Push. That two-Pop execution is not merely a Policy-S artifact—it is itself a Core.LAdm-admissible Policy-L prefix. After the L2 settled snapshot exposes both Pops disabled, Core.Drain/K.Drain forbids newly issuing the source-later Pushes, and the resulting drain-closed source K-facing cutpoint contains the closed internal rendezvous deadlock. Thus A5-L is false for the unchanged design, even though, when the selected instance admits the straight-line Core.ReachPrep case, a different eager

Policy-L schedule may assert both distinct-interface requests and complete both rendezvous transfers in the same cycle; the RTL may likewise exhibit that behavior when its validated scheduling permits it. No direct Policy-L proof, structural proof, invariant, exploration, or alternate A5-L certificate can establish a false premise. The Appendix K scheduler-robust framework therefore cannot certify a design whose deadlock freedom depends on this favorable eager/overlapped issue; the design must be rewritten, buffered, explicitly synchronized, or verified under a different scheduler-specific theorem. Completeness boundary (normative): if some $p \in \text{Pref_L}^{\text{adm}}(I)$ reaches Dead_K^W , then $\neg \text{A5-L}(I)$, regardless of whether another admissible schedule is live.

The Front/NextFront bridge used later in Appendix K is the source-local reconstruction result of Lemma K.0. Cross-model frontier, raw rendezvous-request, enablement, and WFG agreement are obtained by applying the same reconstruction functions to the equal KObs record of Corollary J.1. No asserted rendezvous request is identified with frontier membership merely because it is present in O.

Remark K.0-pre (Derived bridge, not an extra premise).

The asserted blocking-message slice of NextFront is not an additional theorem premise. Core.7a defines the candidate blocking slice, Core.10 and Core.1C supply operation-attributive request persistence, and Lemma K.0 derives that the asserted slice is exactly the pending blocking frontier. BoundaryReq remains undefined for non-message frontier items, and successful non-blocking frontier items do not create pending blocking WFG members.

Definition K.StimX. Under convention B1-avail, the next fixed input item is available as soon as earlier items on that environment channel have been consumed. An environment-facing Pop is stimulus-exhausted exactly when no item remains; an environment-facing Push is exhausted only under a declared permanent-stop contract, and never under the default always-ready output environment. Diagnostic Definition K.WFG+ and K.DeadKS. For debugging and failure triage only, extend the ordinary process-level WFG with terminal nodes. For an environment-facing dependency whose next required interaction is permanently unavailable, create EnvStop_c and add the corresponding blocked-process diagnostic edge. For an ordinary point-to-point dependency, or an elaborated release support consisting of the singleton set {Q}, whose required process owner Q has normally completed before the demanded ordinal, create the corresponding DoneStop node. A non-singleton/multi-alternative ReleaseOwnerSets dependency may be classified as DoneStop only by a release-family-aware diagnostic certificate; otherwise it is left unclassified rather than flattened owner-by-owner. Ordinary and elaborated internal dependencies otherwise retain the coarse $P \rightarrow Q$ edges produced by Core.WFG. At a reachable drain-closed Policy-S K-facing cutpoint, Dead_KS is an optional diagnostic extension of the current release-aware Dead_K predicate with EnvStop/DoneStop terminals. These diagnostics remain optional and are not used to prove Corollary K.2c.

K.2.1 KObs Deadlock Factoring and Extensionality

These results factor the Appendix K current-cutpoint predicates through the common Core.18c KObs record. Generalized DrainClosedNow is modal only in the sense that it evaluates finite reachability in the KObs-pure reconstructed in-flight transition system. The operational realization theorem for that relation is a separate K.InFlightQual/J.InFlightPathSound premise. Core.Drain/K.Drain, future issue legality, fairness, and response-monitor obligations remain operational premises and are not reconstructed from KObs.

Definition K.EnvTerminalAdequate (environment-terminal reconstruction premise).

EnvTerminalAdequate(I) holds when the selected instance records either StandardEnv or an EnvReplayAdequate certificate. Under StandardEnv, Definition K.StimX and the default always-ready output convention determine permanent next-interaction availability from the committed consumption history; Lemma K.EnvX-Auto supplies the corresponding environment-co-frontier fact used by the serial route. Under EnvReplayAdequate, the certificate supplies a typed reconstructed environment state EnvStateRec_I(H,w) and proves that the permanent-availability predicates used by K.WFG+ are determined by H, w, and that state. This is an extensionality premise for optional diagnostics, not a claim that every well-typed KObs record is reachable.

Definition K.KObsDerived (K-specific reconstruction functions).

For a Core.18c record $K = \langle E, w, H, O \rangle$, define DeadKRec(K,I) by evaluating Definition K.Dead on the Core-reconstructed FrontRec, ReleaseOwnerSetsRec, WFGRec, process-completion, and DrainClosedNowRec_{Sys_ref,E} facts under the fixed qualified Sys_ref action schema. Define DeadKRec^W(K,I,W) by applying Definition K.DeadW to DeadKRec and the KObs-facing waiver set W. When EnvTerminalAdequate(I) holds, define EnvStopRec, DoneStopRec, WFGPlusRec, DeadKRec, and their KObs-closed waived variants by applying Diagnostic Definition K.WFG+ and K.DeadKS to the same Core-reconstructed process, frontier, request, channel, release-support, and committed-history facts together with the selected environment-reconstruction mode. These are Appendix K functions over a source-realizable Core record. No K.RespFail reconstruction is defined from K alone because monitor activation, age, bound class, and reset/terminal disposition state are operational data outside KObs.

Lemma K.KObs-Dead (DrainClosedNow and K-predicate factoring).

For every non-empty process set D in a qualified Sys_ref generalized-drain profile, DrainClosedNowRec_{Sys_ref,E}(D,K) is exactly the pure generalized predicate $\neg \text{InFlightEscape}_{\{\text{Sys_ref}, E\}}(D, K)$ computed from Core.FlightStateRec/Core.InFlightStepRec. When $K = \text{KObs_E}, w(\sigma)$ for a reachable source cutpoint, this is Definition K.Dead's operational cutpoint drain clause by definition. ReleaseOwnerSetsRec_E reconstructs the exact family-level internal release alternatives used by Definition K.Dead, while WaitOwnersRec/WFGRec retain their coarse union. The unwaived ordinary Dead_K predicate is therefore determined by K and the fixed E/action schema. If KObsClosed holds, Dead_K^W is determined by DeadKRec^W. Optional diagnostics factor as before. Proof. Lemmas J.KObs-Proc-J.KObs-WFG and J.KObs-InFlight reconstruct every current state/root/action-schema input used by WaitRoots and InFlightStepRec. Because InFlightEscape is finite reachability in that pure reconstructed relation, its truth is a function of K under fixed E; no operational continuation witness is stored in K. K.InFlightQual/J.InFlightPathSound is separately required by theorem applicability to justify that the reconstructed paths are operationally sound. KObsClosed supplies waiver extensionality. No K.RespClean/K.RespFail or continuation-level Core.Drain conclusion follows from KObs.

Corollary K.KObs-Ext (KObs extensionality).

Let $\sigma_1, \sigma_2 \in \text{KCutReach_ref}(I)$ be source K-facing cutpoints of the same fixed instance with $\text{KObs_E}, w(\sigma_1) = \text{KObs_E}, w(\sigma_2)$. Then their reconstructed process slices, explicit abstract channel states, NextFront and Front sets, operation-attributive BoundaryReq values, channel-enable predicates, ReleaseOwnerSets families, WaitOwners unions, ordinary coarse WFGs, Core.FlightStateRec/InFlightStepRec transition systems, generalized DrainClosedNow predicates, and unwaived Dead_K truth values are equal. If KObsClosed_I,E,w(W) holds, their Dead_K^W truth values are equal. If EnvTerminalAdequate(I) holds, their optional diagnostic EnvStop/DoneStop classifications, WFGPlusRec structures, and unwaived Dead_KS truth values are equal; with KObsClosed, their diagnostic Dead_KS^W values are equal. The

conclusion is independent of legal historical Issue timestamps not retained in O , and it makes no claim about K .Resp timer activation or age. Without waiver closure or environment-terminal adequacy, the extensionality conclusion is correspondingly limited. Proof. Functional extensionality of the named reconstruction definitions and Lemmas J.KObs-Proc–J.KObs-WFG, J.KObs-InFlight, and K.KObs-Dead. $\square$ Cross-model-use note. Corollary K.KObs-Ext is the source-side extensionality theorem for reachable records. Appendix J supplies J.OfferReach and J.OfferConf and proves through J.KObsConf that the selected post certificate and source cutpoint share one record. Lemma K.0, Lemma K.2, Lemma K.2a, and Theorem K.1A state the required cross-model consequences directly from that equality. Corollary K.0a is a derived convenience consequence for raw rendezvous endpoint-request equality and is not a separate downstream premise.

K.2.2 Lemma K.0 — Frontier Reconstruction from KObs

Purpose. Appendix K forms WFG edges from the minimal pending blocking frontier. The KObs layer already separates the candidate source frontier reconstructed from H from the currently asserted attributed offers reconstructed from O . This lemma identifies the operational source frontier with the corresponding pure reconstruction and states the cross-model consequence by equal-KObs extensionality.

Statement. Fix I , E , w and $\sigma \in \text{KCutReach_ref}(I)$, and let $K = \text{KObs_E},w(\sigma)$. For process P define $\text{BlockingMsgNextFrontRec_P}(K) := \{ x \in \text{NextFrontRec_P}(K) \mid x \in \Omega_msg,P \text{ and } \text{MsgOf_P}(x) \text{ is a blocking Push or Pop instance} \}$.

Then

$\text{Front_P}(\sigma) = \text{FrontRec_P}(K) = \{ x \in \text{BlockingMsgNextFrontRec_P}(K) \mid \text{BoundaryReqRec}(x,K) \}$.

Here NextFrontRec is witness-relative over the fixed Ω_P and $\leq_P$ supplied by E ; it is a candidate frontier, while BoundaryReqRec selects the currently outstanding blocking offers.

Cross-model consequence. Let $K_post = \text{KObsPost_J}(\rho_post;E,w,H,O)$ and $K_ref = \text{KObs_E},w(\sigma_ref)$ be the exact records selected by Corollary J.1. If $K_post = K_ref$, then for every process P , $\text{FrontRec_P}(K_post) = \text{FrontRec_P}(K_ref)$, and $\text{BoundaryReqRec}(x,K_post) = \text{BoundaryReqRec}(x,K_ref)$ for every typed blocking message-operation frontier item x . Thus the reconstructed post frontier is exactly the operational source frontier at σ_ref . This consequence uses only the common record; it does not compare historical Issue times or raw RTL control state.

Mixed blocking/non-blocking clarification. Successful non-blocking transfers may contribute committed Push/Pop events to H , and failed polls may contribute WC-selected ϵ outcomes, but neither creates a pending blocking frontier member. A blocking endpoint may nevertheless wait on a complementary process whose source endpoint uses a non-blocking call; the non-blocking side can change occupancy or complete the complementary transfer, but does not itself become a WFG source unless it also owns a separate pending blocking frontier.

Proof. For a reachable source cutpoint, Lemma J.KObs-Offers identifies Pending and BoundaryReq with PendingRec and BoundaryReqRec and therefore identifies Front_P with FrontRec_P . The displayed $\text{BlockingMsgNextFrontRec}$ characterization is the Core.7a and Definitions Core.14–Core.15 classification of the $\leq_P$ -minimal pending blocking items; source-prefix issue order, synchronization anchoring, and the exclusion of failed polls ensure that no earlier unrealized nonblocking, signal, or synchronization item invalidates the candidate-frontier characterization. The cross-model conclusion follows by substituting $K_post = K_ref$ into the pure reconstruction functions. $\square$

Corollary K.0a (derived convenience: rendezvous endpoint-request extensionality). For a point-to-point rendezvous channel c with $B(c)=0$, define $\text{ReqProdRec_c}(K)$ and $\text{ReqConsRec_c}(K)$ as the producer-direction and consumer-direction projections of O . If $K_post = K_ref$ as above, then $\text{ReqProdRec_c}(K_post) = \text{ReqProdRec_c}(K_ref)$ and $\text{ReqConsRec_c}(K_post) = \text{ReqConsRec_c}(K_ref)$. This

raw endpoint-request equality is independent of frontier membership: an already-asserted non-frontier blocking offer remains represented in O and can participate in rendezvous enablement.

K.2.3 Example K.0b — Front_P vs. $\text{BlockingMsgNextFront_P}$ Example

The following example illustrates the distinction between Front_P and $\text{BlockingMsgNextFront_P}$. In this example all message-operation items under discussion are blocking Push/Pop operations, so $\text{BlockingMsgNextFront_P}$ is the relevant WFG-facing slice of the source-level NextFront_P frontier over Ω_P . The broader notation MsgNextFront_P , when used elsewhere, denotes the general message-operation-item slice and may also include successful non-blocking transfer items; it is not the object used by the Front/NextFront bridge for WFG construction.

```
void P() {
  while (1) {
    wait();           // sync point
    int x = in.Pop(); // blocking Pop
    out.Push(x + 1);  // blocking Push
  }
}
```

Consider two source K-facing cutpoints for this same process:

Cutpoint σ_0 : just after $\text{wait}()$ has committed, before $\text{in.Pop}()$ has issued

- $\text{BlockingMsgNextFront_P}(\sigma_0) = \{ \text{Pop_in} \}$
- $\text{BoundaryReq}(\text{Pop_in}, \sigma_0) = \text{false}$
- $\text{Front_P}(\sigma_0) = \emptyset$

Cutpoint σ_1 : $\text{in.Pop}()$ has issued, rdy is asserted, but no data has arrived yet

- $\text{BlockingMsgNextFront_P}(\sigma_1) = \{ \text{Pop_in} \}$
- $\text{BoundaryReq}(\text{Pop_in}, \sigma_1) = \text{true}$
- $\text{Front_P}(\sigma_1) = \{ \text{Pop_in} \}$

So the difference is:

- $\text{BlockingMsgNextFront_P}$ says which blocking message-operation items are next candidates in the source-level frontier over Ω_P for WFG/frontier classification.
- Front_P says which of those next blocking message-operation items are actually outstanding pending blockers right now.

K.3 Lemma K.DeadChar — Characterization of Internal Message-Passing Deadlock (Buffered and Rendezvous Cases)

Elaboration prerequisite. The boundary used in this lemma is the selected explicit handshake boundary of the source-side or lowered proof model M after any required elaboration. For an ordinary primitive Sys_B boundary, Core.12/Core.13 give the complete full/empty/rendezvous characterization. For a coupling-sensitive $\text{Sys_B}^{\text{elab}}$ boundary, E4 supplies storage/commit legality while the actual $\text{ChannelEnabled/ChannelDisabled}$ result is the settled ready/valid value declared by E ; sync/epoch gating and join coupling are treated the same way and are not hidden exceptions. This lemma characterizes blockedness only at the supplied $\text{KCut } \sigma$. When σ is a post-L3 KObsEvalPoint , it does not assert that $\text{ChannelEnabled/ChannelDisabled}$ agrees with an earlier L2 SettlePoint ; any persistence from a Core.16 L2 activation to σ is supplied separately by $\text{Core.SettleKBridge/K.DrainReach}$.

Statement: Let M be a source-side or lowered proof model used by Appendix K and let $\sigma \in \text{KCutReach_M}(I)$. Suppose a nonempty process set D witnesses $\text{Dead_K}(\sigma)$. Then:

Every process P in D is blocked on a blocking channel operation (Push or Pop), not on an enabled internal ε -step.

Ordinary primitive buffered case ($B(c)>0$): if a blocked frontier item x is evaluated at an uncoupled primitive buffered boundary, a blocked Push implies $\text{occ}_c(\sigma)=B(c)$ and a blocked Pop implies $\text{occ}_c(\sigma)=0$.

Ordinary primitive rendezvous case ($B(c)=0$): if x is evaluated at an uncoupled primitive rendezvous boundary, blockedness means the matching complementary endpoint request is absent at σ .

Elaborated coupling-sensitive case: if x is evaluated at a $\text{Sys_B}^{\text{elab}}$ boundary whose MirrorAvail_E value is produced by declared coupler logic, ChannelDisabled means $\text{BoundaryReq}(x,\sigma)$ is true and that settled mirror value is false. The blocking structure is read from the E-declared dependency semantics rather than inferred from occupancy alone; $\text{ReleaseOwnerSets_E}(x,\sigma)$ preserves the minimal conjunctive/alternative internal process supports and WaitOwners_E is only their union, with local coupler/FIFO realization nodes contracted away.

In all cases, a Dead_K witness D satisfies Definition K.Dead's strong-connectivity and release-closure clauses; for ordinary singleton families this reduces to the familiar closed WFG component.

Proof. Definition K.Dead supplies, for every $P \in D$, a channel-disabled blocking message-passing frontier item, the family-level release-closure condition, and strong connectivity of the coarse WFG induced by D . Because $\sigma \in \text{KCutReach_M}(I)$, Definition Core.KCut together with Core.4 gives $\text{Quiescent_M}(\sigma)$, so no enabled internal ε -step remains at σ .

For ordinary primitive $B(c)>0$ and $B(c)=0$ boundaries, the stated local characterizations are exactly Core.12 and Core.13. For a coupling-sensitive $\text{Sys_B}^{\text{elab}}$ boundary, Core.11 plus Q.1's settled-handshake reconstruction identifies the false MirrorAvail_E value, and Q.1(k) identifies the corresponding ReleaseOwnerSets family and WaitOwners union without turning local couplers into processes or losing AND/OR release structure. Definition K.Dead then applies release closure to the family and strong connectivity to the induced coarse WFG. $\square$

K.4 Lemma K.2 — WFG Extensionality under Equal KObs

Statement. Let $\rho_{\text{post}} \in \text{KCutReach_post}(I)$ and $\sigma_{\text{ref}} \in \text{KCutReach_ref}(I)$ be the selected K-facing cutpoints of one Corollary J.1 application, and write

$K_{\text{post}} := \text{KObsPost_J}(\rho_{\text{post}};E,w,H,O)$,

$K_{\text{ref}} := \text{KObs_E},w(\sigma_{\text{ref}})$.

If $K_{\text{post}} = K_{\text{ref}}$, then for every reconstructed frontier item x ,

$\text{ReleaseOwnerSetsRec_E}(x,K_{\text{post}})=\text{ReleaseOwnerSetsRec_E}(x,K_{\text{ref}})$, and therefore

$\text{WaitOwnersRec_E}(x,K_{\text{post}})=\text{WaitOwnersRec_E}(x,K_{\text{ref}})$ and $\text{WFGRec}(K_{\text{post}})=\text{WFGRec}(K_{\text{ref}})$.

Equivalently, the K-facing post and source views have the same family-level release alternatives and the same coarse process WFG. The statement applies uniformly to buffered and rendezvous boundaries and to $\text{Sys_B}^{\text{elab}}$ explicit staged, bundle, join, and epoch-gate boundaries supplied by E.

Proof. $\text{ReleaseOwnerSetsRec}$ and WFGRec are typed pure functions of K and the qualified E template.

FrontRec and the source-attributed BoundaryReq roots come from O via Lemma K.O/J.KObs-Offers; explicit stored channel state comes from H/E via J.KObs-Chan; HandshakeRec_E reconstructs the settled ready/valid network; and J.KObs-WFG reconstructs $\text{ChannelEnabled}/\text{ChannelDisabled}$ together with $\text{ReleaseOwnerSetsRec}$, its WaitOwnersRec union, and the process-level WFG edge relation. Substitution of $K_{\text{post}}=K_{\text{ref}}$ therefore gives family and graph equality by functional congruence. No separate full/empty/rendezvous/coupler case split is required for the transfer theorem. Lemma K.DeadChar remains the local semantic characterization of why a frontier is disabled at a concrete boundary. $\square$

K.4a Lemma K.2a — DrainClosedNow Extensionality under Equal KObs

Statement. Let K_post and K_ref be the equal records of Lemma K.2 under the same fixed qualified $Sys_ref/E/action$ -schema identity, and let D be non-empty. Then $DrainClosedNowRec_ \{Sys_ref,E\}(D,K_post)=DrainClosedNowRec_ \{Sys_ref,E\}(D,K_ref)$. Consequently the reconstructed post view and selected source cutpoint have the same generalized in-flight escape/drain-closure truth value. This is equality of the $KObs$ -pure cutpoint predicate, not transport of $Core.Drain$ or future operational evidence.

Proof. Equal $KObs$ gives equal $Core.FlightStateRec$ values and, under the same $E/Q.2$ schema, equal $Core.InFlightStepRec$ relations by $J.KObs-InFlight$. $WaitRoots_D$ is a pure process-indexed function of the same reconstructed frontier/enablement/ReleaseOwnerSets facts. Therefore the finite reachability predicate $InFlightEscape$ and its negation $DrainClosedNowRec$ are equal by functional congruence. $K.InFlightQual/J.InFlightPathSound$ remains an operational theorem-applicability qualification and is not inferred from $KObs$ equality. $\square$

K.4b Deterministic Serial-Source Dominance and the Primary Policy-S Route

Organization. The serial route normalizes the selected instance, applies $SDet$ to select the Policy-S run artifact p_S^{det} and to exclude any incompatible committed history under the same selected functional semantics, and uses separately supplied $K.CompleteS$ evidence to establish that this artifact is maximal, fair, and complete. $K.RespCov$ assigns every static internal blocking site and relevant dynamic class a proof disposition and proves that every actual persistent process-facing serial blocker is covered by a monitor or static bounded-response proof; $K.RespClean$ proves that the complete selected run violates none of those obligations. Lemma K.2b proves that a Policy-L internal deadlock would force a finite $K.RespFail$ witness on p_S^{det} . The $Dead_KS/K.EnvOrd/K.Done$ owner-walk remains diagnostic machinery and is not needed for Corollary K.2c.

Definition K.BF (blocking-only fragment). Every wait relevant to the WFG is a persistent blocking Push/Pop at an explicit point-to-point boundary with finite B-faithful capacity; control, payloads, and later operation identities are deterministic functions of certified value/identity provenance, fixed stimulus, and fixed witness choices under $K.CV$, including Definition K.CVPred and $K.CV-WF$ below; shared arbitration is remodeled as explicit point-to-point subchannels through an arbiter whose choices are deterministic or witness-fixed. Bounded local progress: computation between message operations terminates, so at every point each process either normally completes, reaches its next message operation, or is blocked at one; a process that diverges in pure computation is outside the fragment. Persistence/interruption scope: because every WFG-relevant wait in K.BF is an ordinary blocking Push/Pop, the exhaustive $Core.IC$ lifecycle applies once its request has issued: the request remains the same asserted, attributed, pending dynamic instance until process-facing commit or an affecting RW2 RESET; a DeclaredTerminal disposition may end the execution while it remains pending. Independent nonterminal cancellation, withdrawal, retry, or reattribution of an outstanding blocking request is not part of the G-K theorem model and requires a separately defined protocol extension outside this theorem stack. Lemma K.2b.Q uses K.BF to identify a permanently non-advancing process with a persistent process-facing frontier that $K.RespCov$ must cover. No missing-supply owner trichotomy is assumed.

Lemma K.BF-Persist (frozen-frontier interruption completeness). Fix a K.BF instance, a candidate component D , and a selected WFG-relevant blocking frontier f_P whose attributed endpoint request is pending at a $Core.LAdm$ -admissible prefix. Along any finite continuation prefix before a DeclaredTerminal state, if f_P has not process-facing committed and no RESET affecting P or its explicit boundary has occurred, then the request for f_P remains asserted, pending, and attributed to the same dynamic operation. In particular, no independent nonterminal cancellation, withdrawal, retry, reattribution, retirement, or replacement can dissolve a K.2b frozen component. Proof. Definition K.BF establishes that f_P is an ordinary blocking Push/Pop request. $Core.IC(1)-(2)$ exhausts the ordinary

blocking-instance lifecycle and gives persistence plus stable attribution until process-facing commit or an affecting RW2 RESET. A DeclaredTerminal disposition is handled separately by K.TerminalDisposition and may end the run with the request still pending. Therefore, absent process-facing commit, affecting RESET, or DeclaredTerminal, the selected request remains the same pending dynamic instance. $\square$

Definition NB-Align (witness-selected non-blocking alignment). For the fixed theorem instance and witness choices, NB-Align holds iff every witness-selected non-NB-CF non-blocking site that can participate in a potential Appendix-K wait cycle is either (a) Appendix-L aligned, with its selected outcome dependency tied to an explicitly modeled message/synchronization obstruction and a modeled supplier that can be represented by the point-to-point guard/epoch-gate lowering of Definition K.Low, or (b) proved structurally unable to participate in any potential wait cycle of the selected instance. NB-CF sites require no such lowering. For every aligned site, the selected dynamic outcome/ordinal and modeled supplier used by K.Low must be the same ones justified by the Appendix-L WC/witness package and the fixed K.CV provenance; a relevant witness-selected site that is neither NB-CF, aligned, nor structurally excluded makes NB-Align false and places that witness schedule outside Lemma K.2b, Corollary K.2c, K.PSF, CF-S, and the Policy-S form of Theorem K.1. Theorem K.1A does not consume NB-Align.

Definition K.Low and normalized instance Sys_K. Let $\Lambda_W(\text{Sys_ref})$ be the witness-indexed lowering that represents synchronization by explicit epoch-gate channels and every aligned non-NB-CF non-blocking dependency by a fresh blocking guard channel. The default guard has $B(g)=1$ with ordinary non-fall-through E4 timing; after the supplier commits there may be a one-real-cycle guard skew. Such guard/epoch administrative residue is represented by explicit in-flight actions and is removed only by the Stage-2 lowering-administrative normalization below; it is not treated as ϵ merely because it is erased by projection. Define $\text{Sys_K} \triangleq \Lambda_W(\text{Sys_ref})$ where lowering is needed, and $\text{Sys_K} \triangleq \text{Sys_ref}$ otherwise. All serial-route predicates below are evaluated on Sys_K.

Definition K.Low.AdminStepRec and lowering-administrative normal form.

For a lowered model $\text{Sys_K} = \Lambda_W(\text{Sys_ref})$, $\text{LowAdminStepRec}_{\{I,E\}}(r_K, u, r_K')$ holds iff $\text{Core.InFlightStepRec}_{\{\text{Sys_K}, E\}}(r_K, u, r_K')$ and u is an action introduced solely by Λ_W 's guard/epoch realization whose K.Low projection is source stutter: it changes only lowering-created guard/epoch channel, token, credit, request/frontier, or deterministic handshake state; emits no source message/synchronization/signal unit; advances no source process control; and launches/issues no source-later work. This class includes EpochGateAdvance and the corresponding guard-channel retire/advance action (including a declared atomic variant) when that action is erased by π_K . It does not include SyncFire, an ordinary source-corresponding message commit, or an elaboration action whose projection changes source proof state.

Define $\text{LowStateCorr}_{\{I,E\}}(r_K, r_ref)$ iff r_ref is the lowering projection of r_K on every source-semantic field consumed by the reconstructed relation—source process/replay control, source message/synchronization history, source-attributed requests, source explicit-channel state after the declared projection, reset epoch, and fixed witness/environment facts—while allowing r_K to retain additional lowering-created guard/epoch state. Define $\text{LowAdminNF}_{\{I,E\}}(r_K)$ iff no LowAdminStepRec action is enabled from r_K . Define $\text{LowAdminNormalize}_{\{I,E\}}(r_K, r_K^*)$ iff r_K^* is reached from r_K by a finite zero-or-more sequence of LowAdminStepRec actions, $\text{LowAdminNF}(r_K^*)$ holds, every intermediate state has the same Sys_ref projection, and the sequence is legal under the same reset/environment/witness choices. Under K.Low.ModalQual, such normalization exists for every relevant reachable/lifted K-facing state and terminates by the lowering component of $\text{Core.InFlightDrainWF}$: the finite guard/epoch admitted population strictly decreases and no new source work replenishes it. A disabled guard/epoch object representing an unresolved source obstruction is

already administrative-normal and is not removed. LowAdminNormalize is not K_e : a real-cycle guard/gate action remains an explicit in-flight action even though π_K erases it.

Definition K.Low.LowWaitRoots (lowering-aware source root view).

For a Sys_ref reconstructed state r_ref under a fixed lowering certificate, LowWaitRoot_P(r_ref) is the finite typed root view obtained by applying the lowering rules to P's current source-side wait cause without changing the literal source WFG: (LR1) the rule-selected current ordinary blocking message frontier contributes its ordinary K.WaitRoot_P data; (LR2) an NB-Align obstruction contributes only when, under the same source/elaboration order used by Λ_W , that obstruction is the current lowering-view frontier cause rather than a source-later residual cause; it contributes a guard-cause record keyed by the selected dynamic site/outcome/provenance, modeled supplier, and the exact internal release-support family certified by K.Low.GuardClose; (LR3) a synchronization obstruction contributes only when it is the corresponding current lowering-view frontier cause; it contributes the Core.7b/Core.7c synchronization carrier key, pending/ready state, participant identity, and SyncReleaseOwnerSets data used by K.Low.EpochClose. The active-case selection/order is fixed by the source replay state, NB-Align certificate, synchronization carrier, and Λ_W mapping; no fake MsgOf/BoundaryReq is synthesized on Sys_ref. For $D \neq \emptyset$ define LowWaitRoots_D(r_ref) := $P \mapsto \text{LowWaitRoot_P}(r_ref)$ for $P \in D$.

At an administratively normalized pair (r_K, r_ref) satisfying LowStateCorr(r_K, r_ref), K.Low.RootCorr_D means that applying the IdClose/EpochClose/GuardClose projection to the literal Sys_K WaitRoots_D(r_K) yields exactly LowWaitRoots_D(r_ref), including frontier/cause identity, the rule-specific progress status, and exact minimal internal release-support families. In the epoch case this does not equate a disabled gate with SyncBlocked: a disabled gate may project to SyncPending with SyncReady=true and an empty blocked-support family, exactly as K.Low.EpochClose specifies. LowWaitRoots is proof infrastructure for lowering correspondence only. It does not replace source FrontRec/WFGRec, does not add synchronization to Dead_K, and is not a second user-facing deadlock predicate.

Definition K.Low.LowInFlightEscape and K.Low.ModalQual.

LowInFlightEscape_ $\{I, E\}$ (D, K_ref) holds iff a finite nonempty Sys_ref Core.InFlightStepRec path from Core.FlightStateRec_ $\{Sys_ref, E\}$ (K_ref) changes LowWaitRoots_D at some reached reconstructed state. Define LowDrainClosedNow(D, K_ref) := $\neg \text{LowInFlightEscape}(D, K_ref)$. This extended modal predicate is used only inside K.Low; ordinary source DrainClosedNow remains the literal-message-root predicate of Definition K.InFlightEscape.

K.Low.ModalQual_ $\{I, E, \Phi_K, \Phi_ref\}$ is the Stage-2 lowering qualification record. It binds: (MQ1) the exact Λ_W rule set plus the active IdClose/EpochClose/GuardClose local typing/release facts and the Stage-2 Q.2 action/effect schemas; (MQ2) finite LowAdminNormalize existence/termination and its lowering rank; (MQ3) LowStateCorr on every relevant projected/lifted state and RootCorr_D at every administratively normalized paired state, uniformly for every candidate D; (MQ4) the action projection/lifting and finite-administrative-suffix root-reflection cases needed by K.Low.InFlightPathCorr, uniformly over every relevant paired state and candidate D: every lowering-only admin edge preserves LowStateCorr and projects to source stutter; every source-corresponding Sys_K edge projects to the exact Sys_ref action/effect while preserving LowStateCorr; every Sys_ref edge has a finite lift beginning at a normalized corresponding state and ending, after any inserted administrative suffix, at another normalized corresponding state; and, for every source-corresponding Sys_K edge followed before the next source-corresponding edge by any finite zero-or-more LowAdminStepRec suffix, if that source edge or any action/state in that administrative suffix first enables, removes,

advances, or otherwise changes a literal D-root object, then the projected Sys_ref source edge changes the corresponding LowWaitRoot_D cause. This administrative-suffix clause includes the case in which the first literal D-root change occurs only on the second or a later erased administrative edge; an admin-only root change with no preceding reflected source-cause change is forbidden; (MQ5) K.InFlightQual_{l,Sys_K,E,Φ_K} and K.InFlightQual_{l,Sys_ref,E,Φ_ref} separately, or explicit named preservation theorems that discharge particular model-indexed fields; and (MQ6) schema/version/checker binding in K.CertHdr. In particular, Sys_K Core.CycleClosure is not silently reused as Sys_ref continuation evidence.

Theorem K.Low.InFlightPathCorr (global stuttering path projection/lifting).

Assume K.Low.ModalQual and let r_K^0, r_{ref}^0 be corresponding reconstructed K-facing states with LowStateCorr(r_K^0, r_{ref}^0), LowAdminNF(r_K^0), and RootCorr_D(r_K^0, r_{ref}^0). Then finite in-flight paths correspond up to lowering stutter in both directions.

Downward projection. Take any finite raw Sys_K in-flight path from r_K^0 . Traverse that exact path without modifying or reordering it. A LowAdminStepRec edge projects to source stutter and preserves LowStateCorr. Every other qualified edge applies the local IdClose/EpochClose/GuardClose or ordinary Q.2 projection case and contributes the exact corresponding Sys_ref action/effect. Erasing only the administrative edges therefore yields a finite Sys_ref in-flight path, and LowStateCorr holds between each raw Sys_K position and the source state obtained after projecting its prefix. No administrative normalization is inserted into this downward traversal. For root changes, consider the first position at which literal Sys_K WaitRoots_D differs from its start value. Because the start is LowAdminNF, that first change cannot be caused solely by an administrative edge whose relevant guard/epoch object was already enabled initially. If an administrative edge is the first syntactic edge that removes or advances a D-root object, let e be the most recent preceding source-corresponding edge and let α be the intervening finite administrative suffix (possibly containing earlier administrative edges that do not yet change the literal root). MQ4's administrative-suffix root-reflection clause requires the matching LowWaitRoot_D cause to change at the projection of e ; this covers in particular a first literal root change that appears only on the second or a later administrative edge. A source-corresponding edge that directly changes a literal D-root is the zero-length-suffix case and likewise changes the matching LowWaitRoot_D cause. Thus every literal Sys_K escape projects to a source LowWaitRoots escape.

Upward lifting. Take a finite Sys_ref in-flight path from r_{ref}^0 . Induct over its actions while maintaining an administratively normalized corresponding Sys_K checkpoint. Each source action uses the applicable identity/guard/epoch/local Q.2 lift case. The lift may contain the source-corresponding Sys_K action plus zero or more lowering-created guard/epoch administrative actions; append the finite LowAdminNormalize suffix guaranteed by MQ2 and MQ4 before lifting the next source action. The resulting checkpoint again satisfies LowStateCorr, LowAdminNF, and RootCorr. No inserted action launches/issues source-later work, changes projected source history except where the source action does, or crosses reset epochs without the recorded rule. If the source LowWaitRoots_D family changes, RootCorr at the first changed normalized checkpoint yields a literal Sys_K root change on the lifted path.

Both directions compose by induction on finite path length. The theorem requires no commutation or confluence claim for arbitrary raw Sys_K paths: downward projection follows the path as given, whereas normalization is used only for designated paired checkpoints and the constructed upward lift.

Operational existence/soundness of the two reconstructed relations is supplied separately by MQ5's model-indexed K.InFlightQual/J.InFlightPathSound evidence.

Corollary K.Low.InFlightEscapeCorr (escape correspondence at normalized pairs).

Under the hypotheses of K.Low.InFlightPathCorr and for every nonempty D,
 $\text{InFlightEscape}_{\{\text{Sys}_K, E\}}(D, K_K) \Leftrightarrow \text{LowInFlightEscape}_{\{I, E\}}(D, K_{\text{ref}})$
at administratively normalized corresponding K-facing observations K_K, K_{ref} . Proof. Left-to-right, project the chosen raw Sys_K escape exactly as in K.Low.InFlightPathCorr and take its first literal D-root-changing position. A source-corresponding edge gives the matching LowWaitRoots change directly. If the first syntactic root-removal/advance edge is administrative, LowAdminNF at the start plus MQ4's finite-administrative-suffix clause identify the most recent preceding source-corresponding edge whose following erased administrative suffix contains that first root change; the projected source edge already changes the corresponding LowWaitRoots_D cause, even when the literal Sys_K root change occurs only on the second or a later administrative edge of the suffix. Right-to-left, lift the source extended-root escape action-by-action, administrative-normalizing only the constructed lifted checkpoints; RootCorr at the first changed normalized checkpoint yields a literal Sys_K root change. $\square$

Lemma K.Low.DeadRootBridge (extended roots collapse to ordinary roots for a literal source Dead candidate).

Let K_{ref} be a reachable qualified Sys_{ref} K-facing observation and $D \neq \emptyset$ satisfy Definition K.Dead's literal message-blocked clause at K_{ref} : every PED owns a current pending channel-disabled blocking message frontier. Assume the same Core.16/no-new-later-work in-flight profile used by K.InFlightQual. Then, until ordinary WaitRoots_D changes, LowWaitRoots_D cannot change. A source-later synchronization or aligned-NB lowering cause for P cannot become the active lowering-view root while P's current pending message frontier remains unchanged: source/replay order plus Core.16 prohibit advancing/launching past that blocker, while any source-earlier lowering cause would already have been resolved before the current literal message frontier could be reached and pending. In-flight actions may drain already-admitted work but introduce no new source-later Issue that bypasses this argument. Hence, from such a starting candidate,

$\text{LowInFlightEscape}_{\{I, E\}}(D, K_{\text{ref}}) \Leftrightarrow \text{InFlightEscape}_{\{\text{Sys}_{\text{ref}}, E\}}(D, K_{\text{ref}})$.

The reverse direction is immediate because the ordinary message root is the active LowWaitRoot at the starting candidate. This lemma is candidate-scoped; it does not claim that the extended and ordinary root views coincide at arbitrary source states.

Corollary K.Low.DrainClosedNowCorr (modal drain correspondence for literal Dead candidates).

Let K_K, K_{ref} be an administratively normalized corresponding pair satisfying K.Low.ModalQual, and let D satisfy the literal source message-blocked premise of K.Low.DeadRootBridge. Then

$\text{DrainClosedNowRec}_{\{\text{Sys}_K, E\}}(D, K_K) \Leftrightarrow \text{DrainClosedNowRec}_{\{\text{Sys}_{\text{ref}}, E\}}(D, K_{\text{ref}})$.

Proof. K.Low.InFlightEscapeCorr identifies Sys_K escape with LowInFlightEscape on Sys_{ref} ; K.Low.DeadRootBridge identifies that extended escape with ordinary Sys_{ref} InFlightEscape; negate both sides. For an arbitrary source extended wait set that is not a literal source Dead candidate, the general correspondence is instead $\text{DrainClosedNowRec}_{\{\text{Sys}_K, E\}} \Leftrightarrow \text{LowDrainClosedNow}$; no claim equating LowDrainClosedNow with ordinary source DrainClosedNow is made. $\square$

Lemma K.Low.C (normalization correspondence) — statement. Assume the fixed stimulus/witness lowering and, when generalized modal drain is used, K.Low.ModalQual. (i) Projection erasing lowering-created guard/epoch administrative transitions maps admissible executions of Sys_K to admissible executions of Sys_{ref} with the same issue-policy class and source-visible committed history. (ii) Every witness-consistent admissible Sys_{ref} execution lifts to Sys_K ; at any lifted K-facing endpoint, finite LowAdminNormalize may be appended without changing the Sys_{ref} projection, yielding an administratively normalized corresponding K-facing pair. (iii) At such normalized pairs the Revision-A local rules continue to supply blocked-frontier typing, non-empty minimal release-family adequacy, and

release closure, while generalized drain is supplied globally: $K.Low.InFlightPathCorr$ gives $K.Low.InFlightEscapeCorr$ and, for a literal source message-blocked candidate, $K.Low.DeadRootBridge$ gives $K.Low.DrainClosedNowCorr$. If a literal Sys_K $Dead_K$ candidate is first encountered at a non-normalized K -facing state, finite $LowAdminNormalize$ may be taken before applying the correspondence: any administrative normalization step that changed that candidate's D -root family would itself witness $InFlightEscape$ and contradict its drain-closed premise, so its D roots remain unchanged to the normalized endpoint. Equivalently, literal Sys_K closed internal waits correspond to source extended closed waits evaluated with $LowWaitRoots/LowDrainClosedNow$; a literal source $Dead_K$ candidate is the special case in which the extended drain predicate equals ordinary source $DrainClosedNow$. On the extended-wait-to-literal direction, the resulting blocked, drain-closed, release-closed nonempty process set may be reduced to a literal Sys_K $Dead_K$ witness by $K.ReleaseSCCReduction$. These clauses refactor rather than discard $Revision-A$ $LowerClosure$: its three non-drain local fact families remain local, while its former local $DrainClosedNow$ family is replaced by the global modal path theorem. $K.Low.C$ still does not transport $K\epsilon$, $K.Drain/Core.Drain$, $nonwithdrawal$, $Core.CycleClosure$, $Core.IssueFair/B2/B3/B5$, or other separately model-indexed semantic premises; those require their own discharge or named preservation theorem.

Definition $K.OpKey$ (common cross-policy dynamic-operation key and source order). For each selected $K.BF$ theorem instance I , let $OpKey_I$ be a theorem-instance universe of typed process-local dynamic-operation occurrence keys under the fixed source code, stimulus, witness, elaboration/lowering, and reset-epoch schema. A key κ records at least its owner P , dynamic source-position/iteration occurrence identifier, operation kind/site, typed boundary/direction schema, and logical reset-epoch slot; the key exists independently of whether a particular admissible execution reaches that occurrence. Each admissible execution p has a partial realization map $Realize_p : OpKey_I \rightarrow Q_exec, P(p)$, and every realized execution-local operation q has a unique $Key_p(q)=\kappa$. Realization preserves kind/boundary/direction and the process-local source order. Define $\kappa_1 \leq^K_src \kappa_2$ on common keys by that source-induced order. The execution-internal relation $\leq_src$ of $Core.5$ remains unchanged; whenever Appendix K compares an operation from Policy S with one from Policy L , notation such as $g_P \leq_src f_P$ is a readability shorthand for $Key(g_P) \leq^K_src Key(f_P)$, never a direct comparison of elements of two execution-local Q_exec universes. Likewise, cross-policy "same dynamic operation identity" means the same $OpKey$ together with the typed fields/value facts determined by $K.CV$.

Definition $K.CV$ (design-level causal/value determinism and provenance). For a fixed selected $K.BF$ theorem instance I , $K.CV_I$ holds iff all of the following are certified under the same fixed source code, stimulus, witness choices, elaboration/lowering, reset/environment contract, and comparison semantics. (CV1) Definition $K.OpKey$ is instantiated by one theorem-instance $OpKey_I$ universe with partial per-execution realization maps and a checked key-level source order. (CV2) A common $K.CVPred$ provenance certificate exists on a typed $CVFact_I$ universe, with finite predecessor sets, predecessor-occurrence closure, deterministic $Eval_b$ equations for every fact used by the comparison, a checked none/some(desc) evaluator for each $OpFact(\kappa)$ that determines control/branch occurrence rather than assuming occurrence, the serial-route comparison-availability closure for every blocking $OpFact$ consumed by $K.2b.P0/K.2b.R/K.2b.P$, including the finite certified predecessor-materialization segment/stack descriptor consumed by $Core.FairPick FP4a$, and $K.CV-WF$ ($WellFounded(\leq_CV)$ or an equivalent strictly decreasing checked $CVRank$). (CV3) Whenever later control, payload selection, or operation selection depends on a cross-process observation beyond an ordinary blocking Pop transfer value—including anchored signal reads, direct-input(-sync) reads, shared/external-memory reads through the mapping layer, emitted $R2C$ fixed-point observation units, or synchronization outcomes—the observation is carried by an explicitly modeled causal mechanism and is a deterministic function of the certified value/identity provenance rather than incidental scheduling or micro-timing; the applicable $RULE\ 1/RULE\ 2$, direct-input/handshake, memory-mapping, $R2C$, and Appendix D/L isolation or witness-

alignment obligations therefore also hold. K.CV does not assert that a keyed operation occurs in every execution; K.2b.PO derives occurrence/reachability from the certified provenance and comparison-availability clauses. Definition K.CVPred and K.CV-WF in Appendix K-P define the low-level certificate structure used by CV2; Appendix M.7a specifies acceptable discharge evidence and is not a second definition of K.CV.

Definition K.SerialDomOrder (selected Policy-S dominance action-unit order). Fix the selected SDet Policy-S artifact p_S^{det} and elaboration E before the K.2b dominance proof begins. Contract each primitive rendezvous and every E/J-declared same-cycle atomic boundary-action group into one indivisible action unit; every other committed explicit-boundary Push/Pop/Sync action is a singleton unit. For units that occur in the selected Policy-S dominance footprint, define $<_{\text{SD}}$ by committed cycle followed by one deterministic within-cycle unit rank bound to the fixed simulator/arbitration/elaboration semantics. The within-cycle rank shall respect every required same-process/source predecessor, explicit synchronization predecessor, and E-declared atomic-unit constraint; otherwise-independent same-cycle units may be tie-broken by any fixed deterministic order recorded in the selected artifact. The resulting order is total on the occurring dominance units and well-founded by the lexicographic natural-number rank. K.2b.P's 'least missing' object is the $<_{\text{SD}}$ -least Policy-S action unit not yet represented on the liberal side; a transfer T means either a singleton unit or the corresponding member of an indivisible unit, and the unit is never split by the induction.

Stored/history support cited by Q.1(I) is 'earlier' only when its producing action unit is $<_{\text{SD}}$ -earlier than T, unless that support and T belong to the same declared atomic unit. K.SerialDomOrder is distinct from the cross-policy key order $<^{\text{K_src}}$ and from Core.16's process-local item order: key inequality alone does not establish a multi-frontier item-order fact. The relation $<_{\text{SD}}$ is intentionally partial on the full operational action universe: a unit that occurs only on a Policy-L continuation outside the selected Policy-S dominance footprint has no $<_{\text{SD}}$ rank. Accordingly, later references to a permitted liberal action occurring "before T commits" are temporal statements along that Policy-L continuation and do not assert an $<_{\text{SD}}$ comparison with T. Write $\text{Units}_{\text{SD}}(p)$ for the complete action units occurring in an operational prefix or execution p that lie in the $<_{\text{SD}}$ domain.

Lemma K.SerialDomPredFinite (finite strict dominance predecessors). Fix an action unit T in the selected Policy-S K.SerialDomOrder footprint and let t_T be its committed cycle. Then $\text{Pred}_{\text{SD}}(T) := \{U \mid U <_{\text{SD}} T\}$ is finite. For this lemma the relevant per-cycle carrier is $\text{CycleResult}_{\{\text{Sys}_K, I\}}.\text{committedUnits}$ at the selected Sys_K proof-model boundary before $\Pi_{\text{elab}}/\Pi_{\text{DUT}}$ outer projection; hence it contains every complete explicit-boundary Push/Pop/Sync or declared atomic action unit eligible for K.SerialDomOrder, including proof-internal $\text{Sys}_B^{\text{elab}}$ channel commits later erased from Σ_{DUT} . Proof. If $U <_{\text{SD}} T$, either U belongs to a cycle $j < t_T$ or U belongs to cycle t_T with a smaller deterministic within-cycle unit rank. There are finitely many natural-number cycles $j < t_T$, and each corresponding $\text{CycleResult}_{\{\text{Sys}_K, I\}}.\text{committedUnits}$ is finite by Definition Core.CycleResult. The strict same-cycle predecessors of T form a subset of the finite cycle- t_T inventory. A finite union of those finite sets is finite; contraction of rendezvous or declared same-cycle atomic groups cannot increase cardinality. No finiteness of the complete Policy-S execution is assumed. $\square$

Lemma K.KeyItemOrderBridge (common-key to process-item order). Fix one process P and two realized operations q_1, q_2 in the same reset epoch and applicable source interval, with common keys $\kappa_1 := \text{Key}(q_1)$, $\kappa_2 := \text{Key}(q_2)$ and concrete source/elaboration items y_1, y_2 representing those operations. If $\kappa_1 <^{\text{K_src}} \kappa_2$, then in an ordinary non-elaborated interval, or in a certified singleton/total-order elaborated interval whose item map preserves that source order, $y_1 <_P y_2$ and hence $y_1 \leq_P y_2$. For a genuine $\text{Sys}_B^{\text{elab}}$ multi-frontier, no such item-order conclusion follows from key order alone: the application must supply the explicit E/order certificate mapping the named keys/action units to the required $\leq_P$ or joint-frontier relation, and this lemma returns only that certified relation. Proof. Definition K.OpKey makes realization preserve process-local source order;

Core.5a/Core.OrdFrontSingleton supplies the ordinary item embedding, and the certified-singleton case supplies the same embedding as an explicit hypothesis. The multi-frontier case is exactly the separately checked elaboration bridge. $\square$

Lemma K.Low.A5 (lowered-to-source A5-L transport, generalized modal form). Let π_K be the K.Low projection from Sys_K cutpoints to Sys_ref cutpoints, let W_K be the pullback of the selected source waiver class W along π_K , and assume K.Low.ModalQual together with Lemma K.Low.C. Any additional lowering-only waiver shall be separately proved not to waive the normalized lift of a non-waived Sys_ref cutpoint. Then $A5L_Sys_K(l, W_K)$ implies $A5L_Sys_ref(l, W)$, i.e. A5-L.

Proof by contrapositive. Let a finite Core.LAdm-admissible Sys_ref prefix reach a non-waived literal Dead_K cutpoint σ_ref with witness D . By K.Low.C(ii), lift that exact witness-consistent prefix to Sys_K and append only finite LowAdminNormalize actions, obtaining an administratively normalized K-facing σ_K with the same Sys_ref projection; those inserted actions launch/issue no source-later work and preserve admissibility under the bound lowering profile. The source witness D satisfies the literal message-blocked premise of K.Low.DeadRootBridge. Its ordinary source drain closure therefore lifts by K.Low.DrainClosedNowCorr. The retained Revision-A IdClose/EpochClose/GuardClose local facts lift the applicable blocked-frontier typing, non-empty internal release supports, and release closure to a non-empty process set $D0$ at σ_K . Apply K.ReleaseSCCReduction to $D0$ to obtain a non-empty $T \subseteq D0$ satisfying literal generalized Dead_K in Sys_K at the same normalized cutpoint. Because W_K is the cutpoint pullback of W and normalization does not change the source projection, σ_K remains non-waived. This contradicts $A5L_Sys_K(l, W_K)$. If no lowering is required, LowAdminNormalize and all correspondence facts reduce to identity. $\square$

The theorem transports the generalized bad-cutpoint property only because K.Low.ModalQual contains both model-indexed K.InFlightQual records and the global path/root correspondence. It does not transport Core.CycleClosure, Core.Drain, $K\epsilon$, or any other model-sensitive operational premise as a side effect.

Example K.CE (why ordinary Policy-S Dead_K avoidance is not a sufficient serial check). Let P execute Pop_e ; $Push_c(v0)$, where e is an exhausted environment input and $Push_c$ is independent of Pop_e . Let downstream processes use c and then form an internal wait cycle. Under Core.ReachPrep, the ordinary straight-line $Push_c$ may become reached and issue in the L1 window after Pop_e has issued but before Pop_e process-facing commits, because its reachability/payload is independent of the unresolved Pop_e result; source order and issue-prefix closure are still preserved. Policy L can therefore reach the downstream internal cycle. Policy S withholds $Push_c$ and instead leaves covered process-facing frontiers pending along a chain ending at exhausted e . Thus absence of a Policy-S internal cycle alone does not imply A5-L; K.Resp detects the unanswered serial frontier without requiring a closed Policy-S WFG component.

Definition K.RespMon (process-facing bounded-response monitor). For a dynamic blocking source-operation instance q of process P in reset epoch e , let x be its E-typed process-facing frontier item or certified joint-frontier representative. The monitor triggers at the first selected Policy-S normalized cycle t in which q is the active source-minimal pending blocking frontier, its persistent request attribution is established, and q has not process-facing committed. The response is the commit/return that releases q from P 's source execution; proof-internal channel movement does not count unless E identifies it as that process-facing commit. The monitor carries a finite bound $B_resp(P, q, e, \text{state-class})$ measured in selected Policy-S clock cycles. Its age advances with the selected clock even while unrelated processes commit. For a certified multi-frontier source operation, the monitor is attached to operation return and the K-O2 composition lemma proves that permanent joint blockage leaves the monitor pending.

Definition K.RespFail, K.RespCov, and K.RespClean. K.RespFail is a predicate on finite Policy-S prefixes. A non-waived witness exists for a covered monitor when: (i) timeout—the monitor triggers with age zero in normalized cycle t ; a response through the boundary-action phase of cycle $t + B_resp$ is timely, and otherwise timeout is witnessed at the first normalized cutpoint after that phase; (ii) maximal-pending—the selected execution reaches a declared maximal finite terminal state while the monitor remains pending; or (iii) cancelled-pending—an affecting RESET governed by RW2 clears q before response. A temporarily silent/noncommitting cycle is not maximal: Core.CompleteCycle continues the clock and therefore continues K.Resp aging. An affecting reset is a failure unless the exact dynamic class is explicitly waived with the K.Resp coverage-preservation proof. Under the present G–K blocking-request model, case (iii) is the reset-epoch cancellation case; there is no independent nonterminal cancellation/withdrawal disposition for an outstanding blocking request. A modeled abort or end-of-test that ends the selected execution is represented by DeclaredTerminal and is therefore covered by maximal-pending when the monitor remains pending. K.RespCov is default-total over a declared coverage universe of static internal process-facing blocking sites and relevant dynamic classes. Its response-waiver closure is evaluated over the pre-K.2b admitted candidate domain defined in the following environment/waiver rule; it is not defined by quantifying over components only after K.2b succeeds. Each class has one disposition: a literal monitor, a validated static bounded-response proof, or an unreachability proof. Completeness requires every dynamically reachable instance to belong to a declared class; once reached, an unreachability disposition cannot apply. The user or checker need not first decide whether a site can become the persistent serial blocker; that fact motivates coverage but is not its quantification domain. A production-count argument alone is not an exemption. K.RespClean states that no finite prefix of the complete $p_S^{\wedge det}$ witnesses a non-waived K.RespFail.

Environment and waiver rule for K.RespCov. Every internal process-facing blocking site is covered by default. For StandardEnv/B1-avail, a source-earlier environment operation that committed in the frozen liberal prefix has the same consumption ordinal available to Policy S and, under persistent request, B2/B3, and Core.IssueFair, cannot be the permanent serial blocker; that theorem may discharge coverage for the corresponding dynamic instance. For a timed/reactive or otherwise nondefault environment, K.EnvMF first restricts every relevant liberal Dead_K component to the single-internal-frontier, K.JointFreeze-Ord-eligible release structure. Environment-facing operations that may block the selected Policy-S execution must then be monitored, proved available, or placed outside the serial route. A per-site monitor, availability proof, no-bypass argument, or waiver does not admit a timed/reactive liberal component with an extra frontier or non-singleton/multi-alternative release support; that combination is outside K.2b/K.2c. A response waiver is valid only if, for every non-waived liberal Dead_K candidate admitted to the K.2b premise before invoking K.2b.E/P/Q, the candidate-to-serial-frontier coverage relation guarantees at least one mapped covered Policy-S response class outside W_resp . This waiver-closure relation is a precondition of K.2b; it is defined over potential admitted candidates and does not quantify over components only after K.2b has extracted or refuted them.

Complete-run and bound rule. K.CompleteS remains the semantic proof that $p_S^{\wedge det}$ is the complete selected artifact. K.RespClean is a separate property checked over it. For a terminating artifact, the checker validates every monitor through the declared maximal terminal state; absence of current commits in an otherwise nonterminal cycle does not make the artifact complete, because Core.CompleteCycle continues the selected clock and monitor ages. For an infinite artifact, a finite trace is insufficient: an invariant, exhaustive model check, recurrence/lasso proof, or equivalent certificate must establish that monitor ages never exceed their bounds. For theorem soundness, the bound need only be finite and compliance must be proved over the complete run; a separately justified worst-case-latency proof is required only to classify a timeout as a design-specification violation rather than as a failed certificate attempt. No K.Resp bound is transferred to RTL by Theorem K.1.

Diagnostic Definition K.Cone, Condition K.EnvOrd, and Condition K.Done. K.Cone and its may-follow closure may be used to explain environment-rooted or completion-rooted response failures and to prove that particular eager bypasses are impossible. K.EnvOrd retains the transitive, loop-aware dynamic no-bypass meaning described below. K.Done remains only a normal-completion audit: it can prove DoneStop unreachable by showing that every normally terminating endpoint produces or consumes enough operations before FINISH, or by proving that no process normally completes. The latter does not establish continuing service on every endpoint and does not address the H2 perpetual-under-supply case. Consequently K.EnvOrd/K.Done may simplify diagnostic Dead_KS reports but never permit omission of K.Resp coverage and never discharge A5-L by themselves.

Example K.CE-2 (transitive starvation—diagnostic value and uniform K.Resp detection). Let e be an environment input channel with empty stimulus, and let c_E, c_V, a, b be internal point-to-point channels. E : Pop $_e$; Push $_cE$. W : Pop $_cE$; Push $_cV(v0)$, with $v0$ independent of the Pop $_cE$ payload. V : Pop $_cV$; Push $_a$; Pop $_b$. $P2$: Push $_b$; Pop $_a$. The straight-line dependency-independent successors Push $_cE$ and Push $_cV$ use the same Core.ReachPrep permission as Example K.CE: they may become reached/issued in L1 before the earlier blockers commit while source order remains intact. An admissible Policy-L schedule can therefore feed the downstream internal cycle $\{V, P2\}$, so A5-L fails. Policy S withholds the feeding operations and leaves the relevant downstream/internal serial frontiers pending. Total K.RespCov therefore produces a finite timeout or maximal-pending failure without needing to walk the chain to EnvStop. K.Cone/K.EnvOrd may still diagnose the transitive environment root and explain which bypass admitted the liberal cycle, but that diagnosis is not part of the soundness proof.

Example K.CE-3 (loop-carried bypass—diagnostic value and uniform K.Resp detection). Let e and c_E be as in Example K.CE-2, with E : Pop $_e$; Push $_cE(f(e))$ fully dependent, and let W be the pipelined loop W : loop { Push $_cV(v0)$; Pop $_cE$ }, with $v0$ independent of the Pop $_cE$ payload. Policy L can issue iteration $i+1$'s Push $_cV$ while iteration i 's Pop $_cE$ remains pending and feed the downstream cycle; Policy S withholds the crossing and leaves a covered process-facing serial frontier unanswered. K.Resp therefore rejects the selected run by a finite response failure. The may-follow form of K.EnvOrd remains useful diagnostically because it identifies the loop-carried crossing that a merely textual source-later analysis misses.

Example K.CE-4 (completed-producer starvation—diagnostic value and uniform K.Resp detection). Let Z : Push $_c(w)$; end produce exactly one item on internal channel c and normally complete, and let Y : Pop $_c$; Pop $_c$; Push $_d(v0)$ demand two, with $v0$ independent; V : Pop $_d$; Push $_a$; Pop $_b$ and $P2$: Push $_b$; Pop $_a$. Policy L may issue Push $_d$ before Y 's second Pop $_c$ settles and thereby reach the downstream internal cycle. Under Policy S, Push $_d$ is withheld and Y 's second Pop $_c$ remains unanswered after Z completes. K.RespCov monitors that process-facing Pop and therefore yields a finite timeout or maximal-pending failure. The diagnostic WFG+ may classify the cause as DoneStop $_c$, but K.2b no longer depends on completion being an exhaustive owner-walk case.

Example K.CE-5 / H2 (perpetual under-supplying owner). Let Z : Push $_c(1)$; while (1) { Push $_d1(0)$; Pop $_d2$; }, W : while (1) { Pop $_d1$; Push $_d2(0)$; }, X : Pop $_c$; Pop $_c$; Push $_g(v0)$ with $v0$ independent of the Pops, V : Pop $_g$; Push $_a$; Pop $_b$, and $P2$: Push $_b$; Pop $_a$. Policy L may issue Push $_g$ before Pop $_c\#2$ settles and then reach the internal cycle $\{V, P2\}$. Under Policy S, Push $_g$ is withheld; Z and W may continue their $d1/d2$ ping-pong forever while X and the downstream chain remain permanently unanswered. No closed Dead_KS component need exist because the owner Z stays live. K.RespCov nevertheless monitors the process-facing serial blockers, and their finite bounds are violated on the complete run. This example is the reason the owner trichotomy and terminal taxonomy are not soundness-critical in the revised route.

Definition K.JointFreeze (non-ordinary release/freeze condition). Fix a candidate Dead_K component D at the $K.2b.E$ activation cutpoint σ_L , the selected internal frozen operation f_P for each $P \in D$, and the fixed $K.\text{DomResetScope}$ DomEpoch . On the $\text{StandardEnv/B1-avail}$ branch, this certificate is required for every admitted candidate that does not satisfy the premises of Lemma K.JointFreeze-Ord; this includes a single-frontier item with a non-singleton or multi-alternative ReleaseOwnerSets family as well as traditional multi-frontier $\text{Sys_B}^{\text{elab}}$ cases. $K.\text{JointFreeze_I}(D, \sigma_L, \{f_P\})$ holds when the certificate proves the following over every finite Core.LAdm -legal continuation in the same DomEpoch while the selected f_P remain pending. (JF1) Cross-process launch/issue closure. For every $Q \in D$ and every not-yet-issued blocking operation q of Q whose source-evaluation launch, new issue, attributed request, or elaboration-descendant request could enable, process-facing commit, remove, or otherwise release any selected frozen f_P , there exists a still-pending frontier item $x \in \text{Front_Q}$ in the frozen component with $x \leq_Q q$ and $x \neq q$. Hence, while Front_Q remains nonempty and fully channel-disabled so that Core.16 remains active, q may neither be newly launched past the blocked point nor become IssueEnabled . The certificate shall treat incomparable $\text{Sys_B}^{\text{elab}}$ frontier items and non-singleton/multiple release supports explicitly; neither incomparability nor owner-set membership alone is equivalent to a releasing action. (JF2) Continuation preservation beyond the initial in-flight closure. At σ_L , the Dead_K premise already includes the model-specific generalized $\text{DrainClosedNowRec}_{\{ \text{Sys_K}, E \}}(D, K_L)$; under the bound $K.\text{InFlightQual}/J.\text{InFlightPathSound}$ this excludes every finite $\text{Core.InFlightStepRec}_{\{ \text{Sys_K}, E \}}$ path over the activation-local already-admitted population that changes D 's WaitRoots . JF2 does not duplicate that initial modal-drain proof. Its independent burden is to prove that every other phase-legal continuation action still permitted while D remains frozen—including work outside D , activity that launches/issues or otherwise creates work not present in the initial reconstructed in-flight closure, and any in-flight work subsequently admitted because of such continuation activity—preserves the selected f_P and cannot satisfy a release alternative whose required process support is disjoint from D or otherwise dissolve Definition K.Dead's release closure, except through a process-facing commit whose required release path is already excluded by JF1 or the initial $\text{Dead_K}/\text{DrainClosedNowRec}$ premise. JF2 may cite $K.\text{InFlightEscape}/\text{DrainClosedNowRec}$ and the bound $K.\text{InFlightQual}$ for the initial closure, $\text{Core.Drain}/K.\text{Drain}$ for the continuing no-replenishment discipline where applicable, B -faithfulness, $Q.1$ release-support semantics, the selected $Q.2$ action schema, and the concrete elaboration template. For a candidate satisfying Lemma K.JointFreeze-Ord's ordinary/single-release-support premises, no separate $K.\text{JointFreeze}$ certificate is required. $K.\text{JointFreeze}$ is a serial-route applicability condition preserving the selected frozen component; it is separate from the Core.16 comparison-advance condition consumed by $K.2b.P0/K.2b.R$.

Uniform $K.\text{JointFreeze}$ discharge (optional sufficient form). The $K.\text{JointFreeze}$ obligation family need not be materialized as one independent certificate per candidate. A flow may supply one versioned instance-level certificate, bound to the same selected instance, elaboration, environment model, and Policy-S semantics, that quantifies over every reachable/admitted StandardEnv candidate not covered by $K.\text{JointFreeze-Ord}$ and proves JF1/JF2 for each such candidate. This is a packaging convenience only: after instantiation, the uniform certificate must establish the complete candidate-relative $K.\text{JointFreeze}$ predicate above and does not admit any candidate for which that predicate fails.

Lemma K.JointFreeze-Ord (ordinary/single-release-support no-release). Let D be a candidate Dead_K component at the $K.2b.E$ activation cutpoint σ_L . Suppose every process $Q \in D$ has an active message frontier consisting solely of one selected frontier item x_Q with frozen source operation $f_Q := \text{MsgOf_Q}(x_Q)$, and either (a) that frontier is ordinary/non-elaborated, or (b) its elaboration certificate supplies the same source-order comparability and $\text{ReleaseOwnerSets_E}(x_Q, \sigma_L)$ consists of exactly one singleton support $\{\{R_Q\}\}$ controlled by one projected process owner R_Q . Assume D is release-closed and drain-closed under Definition K.Dead, $A6$ holds, Core.16 is active, and $\text{Core.Drain}/K.\text{Drain}$ holds. Then the selected frontiers cannot be released by newly issued work of D , by a phase-legal action of a

process $R \notin D$, or by any finite already-admitted $\text{Core.InFlightStepRec}_{\{\text{Sys}_K, E\}}$ path from σ_L , so the JF1/JF2 no-release conclusion used by K.2b.E holds without a separate K.JointFreeze certificate. Proof. Source-order/Core.16 excludes a new source-later releasing operation owned by Q. For an ordinary boundary, A6 gives the unique complementary owner and singleton release support; the certified elaborated case supplies the same structure directly. If an $R \notin D$ action could release x_Q , then $\{R\}$ would be a genuine singleton release alternative disjoint from D, contradicting Definition K.Dead's release-closure clause. Local stateless coupler activity cannot be an independent releasing action: after settling it only reflects its declared process/channel roots. The assumed generalized $\text{DrainClosedNowRec}_{\{\text{Sys}_K, E\}}(D, K_L)$ excludes every finite initial $\text{Core.InFlightStepRec}$ escape that changes D's roots, including qualified proof-internal drain/elaboration actions, while $\text{Core.Drain}/\text{K.Drain}$ supplies the continuing non-replenishment discipline. Hence initially admitted in-flight work cannot release the selected root; later outside-D actions are covered by the singleton release-closure argument above. $\square$

Definition K.CompareAdvanceLegal (dominance-comparison Core.16 advance compatibility). Fix a candidate component D at σ_L , its selected $\{f_P\}$, the certificate-bound $\text{K.DomResetScope}/\text{DomEpoch}$, and the K.2b.P least-missing/first-overtake comparison. $\text{K.CompareAdvanceLegal}_I(D, \sigma_L, \{f_P\})$ holds when the following condition is valid for every process P in K.DomResetScope and every common operation key κ consumed by Lemma K.2b.P0 or Lemma K.2b.R on a finite comparison tuple (π_S, π_L) while the selected f_P remain pending and DomEpoch is fixed. Suppose the strengthened comparison invariant has already matched the source-earlier keys required by Policy S, the endpoint transfers earlier than κ in the fixed dominance order, and the K.CVPred comparison-availability step has supplied every required predecessor fact while independently preserving the frozen component. Let y_κ denote the concrete source/elaboration message item or certified launch target associated with the κ operation whose preparation/Issue is being advanced. Then either Core.16 is inactive for P at the applicable settled frontier, or the comparison certificate proves the Core.16(2) item-order fact directly: no active $x \in \text{Front}_P$ at that comparison prefix satisfies $x \leq_P y_\kappa$ with $x \neq y_\kappa$. Consequently Core.16 forbids neither (a) the finite dependency-independent $\text{Core.ReachPrep}/\text{source-evaluation}$ launch needed by Lemma K.2b.P0 to reach $q_L := \text{Realize}(\kappa)$, nor (b), once q_L is reached and unissued, its subsequent endpoint-request Issue/assertion used by Lemma K.2b.R. This condition is applied before the P0 Core.ReachPrep extension and is retained through the later R request assertion. The same condition therefore discharges both distinct Core.16(2) obligations; issue legality is not inferred merely from successful reach, and reach legality is not inferred from later IssueEnabled. This condition applies whether $P \in D$ or $P \notin D$. For an ordinary non-elaborated/singleton or otherwise total-order/comparability-qualified frontier, Lemma K.CompareAdvanceLegal-Ord derives the required item-order fact from the convenient no-key-earlier condition $\text{Key}(\text{MsgOf}_P(x)) <^{K_src} \kappa$ through Lemma K.KeyItemOrderBridge. If an explicit $\text{Sys}_B^{\text{elab}}$ package can give a K.2b.P0/K.2b.R-consumed owner a genuine multi-frontier on a covered comparison prefix, the flow shall instead provide checked elaboration-specific evidence for the item-order/joint-frontier relation required by Core.16(2), or an equivalent direct proof that the named launch and Issue are Core.16-legal. A bare key-order inequality or incomparability is not sufficient unless the certificate also proves the bridge from that key relation to the applicable $\leq_P$ relation. Consistency with K.JointFreeze is mandatory: no same prefix/key may simultaneously be classified by JF1 as a releasing launch/issue that Core.16 must suppress and by K.CompareAdvanceLegal as a comparison advance that must remain legal. If both classifications apply, no joint serial-route certificate exists for that candidate and the candidate requires another A5-L discharge.

Lemma K.CompareAdvanceLegal-Ord (ordinary/singleton comparison advance legality). On a K.2b.P least-missing/first-overtake comparison tuple satisfying the strengthened transfer/provenance invariant, let κ be a blocking operation key owned by process P for which the selected Policy-S prefix

requires the same source-earlier blocking-transfer prefix before κ may issue, let y_κ be its concrete source/elaboration message item, and suppose P 's applicable active message frontier is ordinary non-elaborated or otherwise certified singleton with the same relevant total-order/source-comparability property. Then the direct item-order conclusion used by $K.CompareAdvanceLegal$ holds for y_κ , regardless of whether P belongs to D . In particular, the lemma discharges every ordinary/singleton $K.2b.P0/K.2b.R$ comparison key and may be cited pointwise by the $K.CVPred$ recursion for a lower-ranked blocking $OpFact$ predecessor once that predecessor's own lower-ranked facts are available. Proof. If $Core.16$ is inactive, the claim is immediate. Otherwise suppose $Core.16(2)$ would suppress the proposed launch/Issue for y_κ . Then some active $x \in Front_P$ satisfies $x \leq_P y_\kappa$ and $x \neq y_\kappa$. Lemma $K.KeyItemOrderBridge$ packages the ordinary $Core.5a/Core.OrdFrontSingleton$ embedding and the certified-singleton total-order/source-comparability embedding: the active item $x \leq_P y_\kappa$ with $x \neq y_\kappa$ therefore corresponds to an owning blocking operation whose key is source-earlier than κ . Policy S therefore requires that operation's process-facing transfer before κ may issue. By the least-missing/first-overtake comparison invariant, that earlier transfer is already present on the liberal side; $E3/source$ -order prefix closure together with $Core.IC$ then prevents the same earlier operation from remaining an uncommitted active frontier item. Contradiction. Hence no active item is an applicable $\leq_P$ predecessor of y_κ , and $Core.16(2)$ permits both the finite dependency-independent preparation/evaluation that reaches $Realize(\kappa)$ and, after reach, the request assertion. $\square$

Definition $K.SegmentAdm$ (construction-side admittedness of certified comparison segments). Fix an admitted candidate D at σ_L , its selected $\{f_P\}$, $K.DomResetScope/DomEpoch$, the fixed $SDet$ Policy- S artifact, and the $K.CVPred/P0$ certified segment descriptors. At a finite $Core.LAdm$ prefix η satisfying the base $K.2b.E$ frozen-comparison invariant Inv (the semantic conditions $I1$ – $I5$ defined in the proof of Lemma $K.2b.E$, excluding segment-bookkeeping condition $I6$), an active certified segment stack S is $K.SegmentAdm_I(D, \{f_P\}, \eta, S)$ when: (SA1) the next whole unit of the top segment is operationally/phase legal at η , is assigned to the phase mechanism that actually owns that unit under $Core.FairPick/Core.CycleChoiceExt$, and is compatible with the fixed witness, scheduler, arbitration, environment, reset, and normalization choices; (SA2) every $Core.16$ -sensitive launch/reach in that unit is covered by the recorded $K.CVPred$ disposition or the applicable $K.CompareAdvanceLegal$ result, and an observation-only unit certified $Core.16$ -inert remains so; (SA3) the unit neither releases the selected frozen component, changes $DomEpoch$, launches prohibited replenishing work, nor introduces an unmatched $K.SerialDomOrder$ dominance transfer; (SA4) if the unit requires a lower-CVRank predecessor segment whose required comparison fact is not already materialized at the current prefix, that segment is pushed as one lower-ranked obligation; an already-satisfied predecessor subtree is skipped by the deterministic prefix-indexed $K.SegNorm$ rule below, and suspension/resumption occurs only between whole declared atomic units; and (SA5) after one whole unit plus the mandatory $Core.CycleChoiceExt$ completion permitted by the fixed semantics, either the top segment completes or its remaining suffix and every suspended caller again satisfy SA1-SA4. $K.SegmentAdm$ is a predicate on prefixes during the continuation construction, not a redefinition of the upstream $K.CVPred$ existence certificate. The $K.2b.E$ proof must establish it relative to the base Inv for every segment stack emitted by the fixed $K.SegmentPlan$ below, and then carry the separate active-stack bookkeeping in $Inv+ := Inv \wedge I6$. If a fixed scheduler/environment choice, phase-owner mismatch, or recursive predecessor disposition makes SA1-SA5 unprovable, that candidate is outside the $K.CV/K.2b$ serial route; the proof may not silently drop the segment obligation.

Definition $K.SegmentPlan$ (prefix-local serial-comparison segment designation). For one admitted candidate $D/\{f_P\}/K.DomResetScope/DomEpoch$ and the fixed $SDet$ Policy- S artifact, $K.SegmentPlan$ is the deterministic $SegReq$ transition rule supplied to the $K.2b.E$ $Core.FairPick$ witness before dependent choice. It is evaluated on proof-construction state (η, S_active) , consisting of the finite frozen Policy- L comparison prefix η and the active certified segment stack, if any. If S_active is nonempty, $SegReq$

retains that same designated top-level obligation until it completes or the candidate is rejected; it may push only a lower-CVRank nested predecessor segment required by the fixed descriptor, and it does not recompute or replace the caller merely because another dominance unit later becomes eligible. Only when no top-level segment is active does the plan inspect the $K.SerialDomOrder$ dominance domain against the fixed Policy-S artifact. If no Policy-S action unit is missing from η , or if the next required endpoint roots are already reached/asserted, emit no segment obligation. Otherwise let T be the $<_{SD}$ -least missing action unit whose $<_{SD}$ -earlier dominance units required by the comparison are already matched in η . The plan emits the finite certified $K.CVPred/PO$ stack(s) $SegPred(\kappa)$ then $SegReach(\kappa)$, as needed, for the endpoint operation key or finite endpoint-key family whose source-attributed request roots are required by T ; for a primitive rendezvous or declared atomic action unit the required endpoint obligations are treated as one indivisible T -level plan and the boundary unit itself is never split. The emitted descriptors and their CVRank nesting come entirely from the fixed $K.CVPred/PO$ certificate; $K.SegmentPlan$ chooses no new witness outcome, scheduler action, environment value, or reset behavior. Its no-active-stack designation is prefix-local and fixed independently of the later proof invocation that cites $PO/R/P$; once designated, the top-level obligation is sticky until completion/rejection under the construction-state rule above. $K.SegmentAdm$ must prove that every emitted stack remains continuously admitted until completed; failure of that proof rejects the candidate. $K.SegmentPlan$ itself makes no theorem-level claim that every action unit later selected by a least-missing reductio has already been emitted. That completed-continuation consequence is the separate Lemma $K.SegmentPlan-EventuallyTarget$ in Appendix K-P; downstream $PO/R/P$ proofs shall cite that lemma rather than infer eventual designation from this definition.

Lemma $K.CompareFact-Mono$ (comparison-prefix fact persistence). Fix the same $K.2b.E$ constructed Policy-L continuation, candidate, $K.DomResetScope$, and $DomEpoch$, and let $\eta \leq \eta'$ be two finite prefixes of that continuation satisfying the base frozen invariant Inv . Then: (CM1) every $K.SerialDomOrder$ action unit already matched in η remains matched in η' ; (CM2) every keyed $K.CV$ predecessor fact already materialized and carried by the comparison invariant at η remains available at η' with the same keyed identity/value; and (CM3) therefore any certified predecessor subtree that the prefix-indexed segment normalization classifies as already satisfied at η remains satisfied at η' . This lemma concerns historical matched units and keyed provenance facts only; it does not assert that transient $ChannelEnabled$, current occupancy inequalities, or other live handshake predicates are monotone. Proof. Execution-prefix extension never deletes a completed unit or historical keyed occurrence. $K.CV/K.CVPred$ fixes the fact keys and deterministic values, while the fixed $DomEpoch$ prevents reset-driven replacement of the compared dynamic identity. Hence the historical comparison record only grows on the stated fields. $\square$

Definition $K.SegWork$ (certificate-tree remaining work) and deterministic normalization. For every finite $SegPred/SegReach$ descriptor emitted by $K.SegmentPlan$, form the transitive certified expansion tree $SegTree$ from the fixed $K.CVPred/PO$ certificate before dependent choice begins. Each node is one whole phase-bearing operational unit; its ordered child subtrees are exactly the lower-CVRank predecessor descriptors that the fixed certificate may require before that unit can proceed. $SegTree$ is finite: each $Pred_CV(b)$ is finite and a recursive child has strictly smaller natural CVRank, so strong induction on CVRank proves the complete predecessor expansion below each fact finite; the top-level $SegPred/SegReach$ unit sequence is itself finite. For a finite operational prefix η and a cursor C into this fixed tree/stack, define $SegNorm(\eta, C)$ to be the unique deterministic administrative normalization obtained by scanning children in the fixed certificate order, skipping an entire child subtree when its required historical comparison fact is already satisfied at η , descending into the first unresolved required child, and popping completed child/caller frames until the cursor names the next unresolved whole operational unit or the tree is finished. The skip predicate is prefix-indexed, but the tree and child order are certificate-determined and continuation-independent. Define $K.SegWork(\eta, C) \in \mathbb{N}$ as the number of unresolved nodes represented by $SegNorm(\eta, C)$. $K.CompareFact-Mono$ makes skipping monotone under

prefix extension, so a subtree skipped at η cannot later reappear. Administrative normalization terminates structurally on the finite tree; it is proof bookkeeping, not a source transition and not a new Core phase.

Lemma K.SegWork-Complete (finite completion of one sticky emitted stack). K.SegWork is structural and has no K.SegmentAdm hypothesis. At a K.2b.E use site, however, Inv+/I6 supplies K.SegmentAdm for the active emitted stack and FP4a supplies eventual realization of each next continuously admitted whole unit. Each such realized unit strictly decreases the unresolved K.SegWork budget after deterministic normalization, while prefix extension can only preserve or remove skipped work by K.CompareFact-Mono. Because the budget is natural-valued, a sticky top-level emitted stack cannot remain active forever: it completes after finitely many realized whole units unless the candidate is rejected because admittedness fails. Push/pop/skip are administrative cursor normalization and need not be modeled as operational transitions. $\square$

Definition K.EnvMF (Policy-S environment/release-structure applicability). For every application of Lemma K.2b.E to a component D at cutpoint σ_L , one of the following shall hold. (1) StandardEnv/B1-avail with the default always-ready output environment: if the selected candidate satisfies all premises of Lemma K.JointFreeze-Ord, that lemma supplies the no-release fact; every other admitted candidate—including a single frontier with non-singleton/multiple ReleaseOwnerSets support or any explicit multi-frontier—shall supply K.JointFreeze_I(D, σ_L , {f_P}). (2) Timed/reactive or otherwise nondefault environment: every $P \in D$ shall have exactly one frontier item at σ_L , namely the selected internal blocker f_P, and the candidate shall satisfy the ordinary/single-release-support premises of Lemma K.JointFreeze-Ord. Thus no environment-facing co-frontier, second internal frontier item, or non-singleton/multi-alternative release support is admitted on clause (2). Lemma K.EnvMF-Uniform below gives an instance-level sufficient discharge of clause (2), avoiding candidate-by-candidate K.EnvMF certification when its structural hypotheses hold. A timed/reactive/nondefault candidate outside those premises is outside Lemma K.2b.E, Lemma K.2b, Corollary K.2c, K.PSF, and the Policy-S form of Theorem K.1 unless a different A5-L route is used. K.EnvMF governs preservation of D; K.2b.P0/K.2b.R comparison-advance legality for all consumed owners is governed separately by K.CompareAdvanceLegal.

Lemma K.EnvMF-Uniform (uniform timed/nondefault-environment sufficient discharge). Fix a timed/reactive or otherwise nondefault environment contract. Suppose that, at every reachable K-facing cutpoint under that fixed environment contract, every process P that could belong to a candidate Dead_K component has at most one message frontier item; any such item is internal; its qualified ReleaseOwnerSets_E consists of exactly one singleton support; and the remaining premises of Lemma K.JointFreeze-Ord hold for the corresponding candidate. Because membership in a Dead_K component requires P to be blocked on message passing and to have an internal disabled frontier item, the relevant frontier is nonempty; the at-most-one condition therefore makes that item exactly the selected internal frontier required by Definition K.EnvMF clause (2). Hence every candidate D, σ_L satisfies K.EnvMF(2) without candidate-specific certification. Common ordinary case. For an ordinary non-elaborated instance at primitive point-to-point boundaries, the extra structural burden of this lemma is normally discharged by existing results: Lemma Core.OrdFrontSingleton gives $|\text{Front_P}(\sigma)| \leq 1$, candidate Dead_K membership supplies the required nonempty internal disabled item and therefore makes that unique item internal, and Definition Core.WFG gives $\text{ReleaseOwnerSets_E}(x, \sigma) = \{\{Q\}\}$ at an ordinary primitive point-to-point boundary. Thus, once the remaining K.JointFreeze-Ord premises are already qualified, a timed/reactive/nondefault ordinary instance may cite Core.OrdFrontSingleton plus the ordinary Core.WFG release-support fact to obtain the instance-wide K.EnvMF clause-(2) discharge; no separate per-candidate K.EnvMF certificate is required. This is a sufficient instance-level structural discharge only: it does not admit any timed/reactive/nondefault candidate excluded by Definition K.EnvMF(2).

Lemma K.EnvX-Auto (environment co-frontiers). Under B1-avail and the default always-ready output environment, every environment-facing frontier item held by a blocked member of a Dead_K component is stimulus-exhausted. Proof. Blockedness requires every frontier item to be channel-disabled. A remaining input item or ready output environment would enable the environment-facing item. Hence no separate environment-freeze premise is needed under StandardEnv. Scope emphasis (normative): B1-avail is a genuine environment-model restriction, not a bookkeeping convention—it asserts that stimulus availability is consumption-ordered and untimed. A testbench that delivers stimulus at declared clock times, or a nondefault output environment that may release back-pressure later, is outside B1-avail. For such an environment the Policy-S route is supported only through K.EnvMF’s certified single-internal-frontier clause together with K.JointFreeze-Ord’s ordinary/single-release-support premises. Any timed/reactive/nondefault candidate with an additional frontier member or non-singleton/multi-alternative release support requires a non-serial A5-L proof or another theorem and is not admitted by a substitute exhaustion/no-rescue certificate in this revision.

Definition K.InterruptionGate (shared nested comparison-interruption schema). Fix a theorem instance I , a candidate liberal component D , its Core.LAdm-admissible counterexample prefix p_L ending at σ_L , and the selected frozen frontier operation f_P for each $P \in D$. A gate specification G supplies two gate-specific predicates. First, $\text{Interrupt_G}(I, D, \{f_P\}, \eta)$ is a prefix/state-local interruption predicate evaluated only on finite Core.LAdm-admissible extension prefixes η of p_L that are consistent with the fixed instance choices and throughout which every selected f_P remains pending; it may inspect the fixed contract, η 's final state, and the legality of the next proof-relevant transition/unit (with any executed real-cycle occurrence recorded in the enclosing Core.CycleResult), but it shall not depend on a continuation already constructed by Lemma K.2b.E. Second, DirectFail_G supplies an independently certified mapping $P \in D$, f_P , a process-facing Policy-S serial frontier x_{P^S} , and $g_P := \text{Op}(x_{P^S})$ with $g_P \leq_{\text{src}} f_P$, together with a finite selected Policy-S prefix on which a non-waived K.Resp obligation for x_{P^S} has already triggered and remains pending, plus gate-specific evidence that the corresponding interruption/disposition occurs and therefore witnesses K.RespFail. The direct mapping and disposition shall be established without K.2b.R, K.2b.P, or K.2b.Q Step 1. A K.InterruptionGate_G discharge uses one of two branch forms: Continue, when every such frozen prefix η satisfies $\neg \text{Interrupt_G}(I, D, \{f_P\}, \eta)$; and DirectFail, when DirectFail_G is established. If neither branch is established, the candidate is outside the Policy-S reduction and requires another A5-L discharge. Multiple gate instances are not flattened into one unordered enumeration: the theorem states their evaluation order explicitly. For Lemma K.2b the required order is K.ResetEpoch first and, only on K.ResetEpoch.Continue, K.TerminalDisposition second. Exhaustiveness for K.BF. For the K.BF serial-reduction fragment, Core.IC and Lemma K.BF-Persist establish that, apart from process-facing commit, the only modeled interruptions that can terminate or invalidate persistence of a selected frozen blocking instance are an affecting RW2 RESET or a DeclaredTerminal disposition ending the execution while the request remains pending. Accordingly, Reset and Terminal are the complete interruption-gate instantiations consumed by Lemma K.2b. Adding another gate is a theorem-model extension and requires an explicit operational definition plus re-proof of K.BF-Persist and the affected K.2b obligations; it is not a routine additional instantiation of this schema.

Definition K.DomResetScope (dominance reset scope). For a candidate D , p_L , $\{f_P\}$, fixed Sys_K elaboration, fixed source/dynamic operation universe, and selected Policy-S comparison artifact, first define $\text{K.CompareScopeFootprint_I}(D, \{f_P\})$ to be the owner/boundary footprint determined from that fixed artifact and the certified K.CV comparison/provenance closure: it contains every modeled process and explicit abstract boundary that can own an endpoint operation or explicit-boundary transfer consumed by K.2b.P0/K.2b.R/K.2b.P/K.2b.P-Frozen or by the derived K.2b.SF serial-frontier extraction for this candidate. The footprint is computed before, and independently of, the later liberal continuation constructed by Lemma K.2b.E; it is a function of the fixed elaboration, source/dynamic operation

universe, selected Policy-S artifact, candidate keys, and checked comparison/provenance schema, not of which liberal comparison steps later succeed. A $K.DomResetScope$ certificate supplies finite sets S_p^{dom} of processes and S_c^{dom} of explicit abstract boundaries. The certificate is valid iff: (D1) $D \subseteq S_p^{dom}$; (D2) for every $c \in S_c^{dom}$, every modeled process that owns a source-level process-facing endpoint of c or is named by the selected elaboration E 's qualified process-dependency/release-support interface for c belongs to S_p^{dom} ; elaboration-local stateless couplers are not process identities and are not added to S_p^{dom} ; (D3) $K.CompareScopeFootprint_I(D, \{f_P\})$ is covered by S_p^{dom}/S_c^{dom} : every process and every boundary in that precomputed footprint is contained in the corresponding scope set; and (D4) S_p^{dom} and S_c^{dom} are determined from the fixed elaboration, source/dynamic operation universe, selected Policy-S comparison artifact, and $K.CompareScopeFootprint$ certificate and do not depend on the later liberal continuation constructed by Lemma K.2b.E. Define $K.DomResetScope_I(D, \{f_P\}) := (S_p^{dom}, S_c^{dom})$ for any validated certificate. Minimality is not required; a larger conservative scope is permitted. For any comparison prefix η , $DomEpoch_I(\eta)$ is the vector of reset-epoch identifiers of the processes and explicit boundaries in this validated scope. A RESET is dominance-affecting exactly when its RW scope intersects this dominance scope. RW5 permits RESET occurrences disjoint from $K.DomResetScope$; such resets may change epochs outside the dominance scope but cannot change any dynamic identity, boundary state, or request fact consumed by K.2b.P0/R/P/P-Frozen/SF. $K.ResetEpoch.Continue$ therefore keeps $DomEpoch$ fixed without requiring a global no-reset execution.

Definition K.ResetEpoch (reset instantiation of K.InterruptionGate). For the candidate D , p_L , and $\{f_P\}$, let $Interrupt_Reset(I, D, \{f_P\}, \eta)$ hold exactly when the fixed reset contract permits, as the next proof-relevant transition/unit from η 's final state, a RESET whose scope intersects $K.DomResetScope_I(D, \{f_P\})$. When that RESET executes within a real cycle, the occurrence is recorded in the enclosing legal $Core.CycleResult.resetUnits$. Let $DirectFail_Reset$ be the independently mapped reset witness: $P \in D$, selected liberal f_P , process-facing Policy-S serial frontier x_P^S in the same pre-reset endpoint epoch, $g_P := Op(x_P^S)$ with $g_P \leq_{src} f_P$, an already-triggered non-waived $K.Resp$ obligation that remains pending, and the first matched RESET affecting that monitored frontier whose emitted RESET unit causes RW2 to clear that exact dynamic instance, yielding cancelled-pending $K.RespFail$. Define $K.ResetEpoch$ from the reset gate with the following strengthened Continue branch. (R-L) The generic $K.InterruptionGate.Continue$ condition holds on every finite frozen $Core.LAdm$ Policy-L extension prefix of p_L . (R-S) Independently of K.2b.P0/K.2b.R/K.2b.P/K.2b.P-Frozen/K.2b.SF/K.2b.Q Step 1, the fixed reset contract and selected Policy-S artifact provide a checked reset-exclusion domain covering every Policy-S prefix that K.2b.P0/K.2b.R/K.2b.P/K.2b.P-Frozen/K.2b.SF/K.2b.Q may consume for this candidate; no RESET whose RW scope intersects $K.DomResetScope$ is permitted/occurs on that covered Policy-S comparison domain. A simple sufficient R-S discharge is that the fixed reset contract permits no such RESET anywhere on p_S^{det} ; the uniform U2 condition below is stronger still. A more precise prefix-domain certificate is allowed only when its domain is defined from the fixed selected Policy-S artifact, $K.CompareScopeFootprint$, and reset contract before the dominance proof and is proved to cover every prefix later consumed by P0/R/P/P-Frozen/SF/Q. $K.ResetEpoch.DirectFail$ remains the matched-reset direct-failure branch. Thus the Continue branch is dominance-scope-reset-free on both the frozen liberal comparison and the selected Policy-S comparison used by K.2b.Q. A reset intersecting $K.DomResetScope$ that is not excluded by R-L/R-S and is not covered by $DirectFail_Reset$ makes the candidate ineligible for the Policy-S reduction and requires another A5-L discharge; a reset disjoint from $K.DomResetScope$ is permitted and framed by RW5. All request-identity, endpoint-ordinal, nonwithdrawal, serial-frontier, and persistence arguments on the Continue branch are scoped by the fixed $DomEpoch$ rather than by a nonexistent single global reset epoch.

Definition K.TerminalDisposition (terminal instantiation of K.InterruptionGate). This gate is instantiated only after $K.ResetEpoch.Continue$ has been established for the candidate D . For the same p_L and $\{f_P\}$,

let $\text{Interrupt_Terminal}(I, D, \{f_P\}, \eta)$ hold exactly when η 's final state satisfies $\text{DeclaredTerminal_Sys_K}, I$. Ordinary Core.SilentCycle progression does not satisfy this predicate. Let $\text{DirectFail_Terminal}$ be the independently mapped terminal witness: $P \in D$, selected liberal f_P , process-facing Policy-S serial frontier x_P^S , $g_P := \text{Op}(x_P^S)$ with $g_P \leq_{\text{src}} f_P$, a finite selected Policy-S prefix on which a non-waived $K.\text{Resp}$ obligation for x_P^S has already triggered and remains pending, and independently checked evidence that ρ_S^{det} reaches the specified declared maximal terminal state while that same monitor remains pending, yielding maximal-pending $K.\text{RespFail}$. Define $K.\text{TerminalDisposition} := K.\text{InterruptionGate_Terminal}$. Thus $K.\text{TerminalDisposition.Continue}$ is the terminal-free comparison branch and $K.\text{TerminalDisposition.DirectFail}$ is the declared-terminal direct-failure branch. Modeled abort, watchdog, cycle-count, or end-of-test dispositions are handled here only when they satisfy DeclaredTerminal ; independent nonterminal cancellation/withdrawal is outside the present G–K model. The nesting is normative: $K.\text{TerminalDisposition}$ is not evaluated on $K.\text{ResetEpoch.DirectFail}$. Uniform instance-level gate discharge (normative sufficient condition). The candidate-relative $K.\text{DomResetScope}/K.\text{ResetEpoch}/K.\text{TerminalDisposition}$ obligations above need not be enumerated one Dead_K component at a time when the flow proves the following stronger facts for the fixed selected instance and selected Policy-S comparison artifact. Supply one finite conservative dominance scope $U = (U_p, U_c)$, fixed from the selected elaboration and source/dynamic operation universe before any liberal counterexample is chosen, such that: (U1) U contains every process that can participate in an Appendix-K blocking component and every process/boundary identity that $K.2b.PO/K.2b.R/K.2b.P/K.2b.P\text{-Frozen}$ or the derived $K.2b.SF$ extraction can consume; for every boundary in U_c , every modeled process that owns a source-level process-facing endpoint of that boundary or is named by E 's qualified process-dependency/release-support interface for that boundary lies in U_p ; stateless couplers themselves are not process identities and are not added to U_p ; and U is independent of the later $K.2b.E$ continuation. Thus U satisfies $K.\text{DomResetScope}$ conditions D1-D4 whenever used for a candidate. The scope may conservatively contain all Sys_K processes and all relevant explicit abstract message-passing boundaries; minimality is not required. (U2) No RESET permitted by the fixed reset contract has RW scope intersecting U . (U3) Either DeclaredTerminal is unreachable during the applicable comparison phase, or every reachable DeclaredTerminal source state has no pending $K.BF$ WFG-relevant blocking frontier owned by U_p or represented at U_c . Under U1-U3, use U as $K.\text{DomResetScope}$ for every candidate: U2 proves both the R-L and R-S parts of $K.\text{ResetEpoch.Continue}$ uniformly, and because each frozen candidate retains a selected $K.BF$ blocking frontier in U , U3 proves $K.\text{TerminalDisposition.Continue}$ uniformly on the reset-Continue branch. A clean $\text{CF-S}/K.\text{PSF}$ discharge may therefore cite this one instance-level record instead of enumerating candidate-specific gate records or DirectFail mappings. This is a sufficient, not necessary, discharge; if any clause cannot be proved, the existing candidate-specific nested-gate schema remains applicable without change.

Lemma $K.2b.E$ (fair complete frozen extension). Let $\rho_L \in \text{Sys_K.Pref_L}^{\text{adm}}(I)$ end at a source K -facing cutpoint $\sigma_L \in K\text{CutReach_Sys_K}(I)$ satisfying Dead_K^W , let D be a selected Dead_K component witnessing Dead_K^W at σ_L , and let f_P be the component frontier operation selected for each $P \in D$. By Definition $K.\text{Dead}$, D is drain-closed at σ_L ; this fact is extracted from the deadlock hypothesis and is not an additional cutpoint premise. Assume $K.\text{EnvMF}(I, D, \sigma_L)$, with Lemma $K.\text{JointFreeze-Ord}$ discharging every candidate satisfying its ordinary/single-release-support premises and $K.\text{JointFreeze}$ required on every other admitted StandardEnv release/frontier structure, $K.\text{ResetEpoch.Continue}$, and—on that reset-free branch— $K.\text{TerminalDisposition.Continue}$ for this application. Fix the $K.\text{SegmentPlan}$ SegReq rule determined by this candidate, $K.\text{SerialDomOrder}$, the fixed SDet Policy-S artifact, and its $K.\text{CVPred}/PO$ descriptors before dependent choice. Under the Sys_K interpretations of Definition Core.LAdm , Core.CycleClosure , $\text{Core.Drain}/K.\text{Drain}$, B5, and a Core.FairPick_I , Policy L witness that constructively discharges the existing $\text{Core.IssueFair}/B2/B3$ behavioral obligations for the particular continuation and binds that $K.\text{SegmentPlan}$ through FP4a, together with the $K.\text{SegmentAdm}$

construction-side discharge relative to the base frozen invariant Inv for every segment stack emitted by the plan and the structural $\text{K.SegWork}/\text{K.SegWork-Complete}$ discharge for that fixed certificate expansion, with K.2b.E carrying the additional sticky-stack bookkeeping in $\text{Inv}^+ := \text{Inv} \wedge \text{I6}, \text{Sys_K}$ nonwithdrawal, $\text{K}\epsilon_{\{I, \text{Sys_K}\}}$, K.BF , A6 source/process ownership and the elaborated $\text{ReleaseOwnerSets}/\text{WaitOwners}$ specification, and the same fixed stimulus, witness, environment, reset, arbitration, termination contract, and normalization choices, there exists an infinite fair complete continuation $\bar{p}_L \in \text{Exec_L}^{\text{adm}}(I)$ extending p_L in which every f_P remains pending while the $\text{K.DomResetScope DomEpoch}$ remains fixed and every continuously admitted K.SegmentPlan obligation is realized by FP4a . Under StandardEnv , Lemma K.EnvX-Auto prevents later environment rescue of any co-frontier; for a StandardEnv candidate outside K.JointFreeze-Ord 's premises, K.JointFreeze additionally prevents a newly issued incomparable/internal operation or permitted elaboration action from releasing the selected frozen frontiers. Under a timed/reactive or otherwise nondefault environment, K.EnvMF excludes all co-frontiers and non-ordinary release alternatives by requiring each $P \in D$ to have only the selected internal frontier and the candidate to satisfy K.JointFreeze-Ord 's release-support premises. The continuation may schedule all enabled work outside the frozen component and may drain already-issued downstream work, but Lemma K.JointFreeze-Ord or K.JointFreeze , as applicable, together with $\text{Core.Drain}/\text{K.Drain}$ prevent any newly introduced work from releasing the frozen frontiers. A disabled frozen request is not ChannelEnabled and therefore creates no B2 obligation to commit; Lemma K.BF-Persist applies the exhaustive Core.IC blocking lifecycle to that WFG-relevant blocker; and an operation excluded by the applicable $\text{K.JointFreeze-Ord}/\text{K.JointFreeze}$ discharge is not IssueEnabled and therefore creates no Core.IssueFair obligation to issue. This lemma supplies only the reset-free, terminal-free continuation used by the dominance comparison. $\text{K.ResetEpoch.DirectFail}$ and $\text{K.TerminalDisposition.DirectFail}$ are direct K.RespFail branches of Lemma K.2b and do not invoke this extension.

Definition $\text{K.SerialRoutePremises}$ (authoritative deterministic serial-reduction applicability record). Fix the same Sys_K theorem instance, stimulus, capacities, elaboration, environment, witness, concrete Policy-S simulator/scheduler and arbitration semantics, Policy-S issue semantics, ϵ -normalization, reset and termination/end-of-test contracts, $\text{K.InterruptionGate}$ schema, and K.Resp configuration used by Lemma K.2b . $\text{K.SerialRoutePremises}$ holds exactly when: (SR1) SDet , the Sys_K interpretations of the source-side $\text{A1}–\text{A4}$ content and B5 , and the Sys_K interpretations of the $\text{Policy-L Core.IssueFair}/\text{B2}/\text{B3}$ obligations are available in the constructive Core.FairPick form required to build the K.2b.E continuation, including FP4a certified-segment progress with the candidate-specific K.SegmentPlan fixed before continuation construction; (SR2) the applicable Sys_K source-model content of $\text{A6}–\text{A11}$, K.BF , A6 source/process ownership, and the elaborated $\text{ReleaseOwnerSets}/\text{WaitOwners}$ specification hold; whenever K.2b consumes a coupling-sensitive boundary, the strengthened Q.1(I) serial-route handshake-support/persistence template fact also holds; (SR3) K.CV , Sys_K request nonwithdrawal, $\text{K}\epsilon_{\{I, \text{Sys_K}\}}$, the Sys_K interpretations of K.Drain and Core.CycleClosure hold; (SR4) K.EnvMF holds for every candidate used by K.2b.E , and $\text{K.CompareAdvanceLegal}$ holds for every candidate comparison consumed by $\text{K.2b.P0}/\text{K.2b.R}/\text{K.2b.P}$, with the ordinary/singleton or explicit $\text{Sys_B}^{\text{elab}}$ discharge routes stated by those definitions, and K.SegmentPlan is defined from the fixed Policy-S artifact/ K.SerialDomOrder and K.SegmentAdm is established construction-side for every $\text{K.CVPred}/\text{P0}$ certified segment stack it emits on an admitted candidate; (SR5) the shared $\text{K.InterruptionGate}$ schema is instantiated by K.ResetEpoch for every candidate component, with $\text{K.ResetEpoch.Continue}$ carrying both its R-L liberal-prefix and R-S selected- Policy-S reset exclusions, and only on $\text{K.ResetEpoch.Continue}$ by $\text{K.TerminalDisposition}$, either candidate-by-candidate or through the uniform $\text{U1}–\text{U3}$ sufficient discharge; (SR6) the liberal prefix and its applicable continuation use the same fixed theorem-instance choices; and (SR7) NB-Align and the applicable Definition K.Low correspondence obligations hold whenever synchronization waits or witness-selected non- NB-CF sites require lowering, including K.Low.C

for the normalized comparison; when the current generalized-drain theorem is claimed on a lowered instance, SR7 additionally binds $K.Low.ModalQual$, including separate Sys_K/Sys_ref $K.InFlightQual$ and the $K.Low.InFlightPathCorr/InFlightEscapeCorr/DrainClosedNowCorr$ stack. The separate $K.Low.A5$ transport back to Sys_ref is required by Corollary K.2c/Theorem K.1 when lowering is used, but is not a premise of Lemma K.2b. $K.Low.C$ is not a transport theorem for SR1–SR3: when Sys_K differs from Sys_ref the Sys_K premises above must be discharged on Sys_K or obtained from explicit named preservation lemmas. This record packages K.2b applicability only: $K.RespCov$ and $K.CompleteS$ are supplied separately when Lemma K.2b is invoked directly, while A5-S supplies $K.RespCov$, $K.CompleteS$, and $K.RespClean$ when Corollary K.2c or Theorem K.1 is used. The fields above also provide the structural/provenance/reset inputs consumed by the derived Lemma K.2b.SF; SF is not an additional SR field or independent premise, and its separate $K.CompleteS$ hypothesis is supplied at theorem use. A5-L is not a field of $K.SerialRoutePremises$ because the serial route derives it.

Lemma K.2b (deterministic serial-source dominance with bounded-response extraction). Fix Sys_K , stimulus, capacities, elaboration, environment, witness, concrete Policy-S simulator/scheduler and arbitration semantics, Policy-S issue semantics, ε -normalization, the reset and termination/end-of-test contracts, the $K.InterruptionGate$ schema, and $K.Resp$. Assume $K.SerialRoutePremises$ for these fixed choices and assume $K.RespCov$. Let p_S^{det} be the Policy-S execution selected by the SDet field of $K.SerialRoutePremises$, with separate $K.CompleteS$ evidence establishing maximality, fairness, and completeness. If some Policy-L prefix p_L in Sys_K 's $Pref_L^{adm}(I)$ domain reaches a source K-facing cutpoint whose K_L satisfies $DeadKRec^W$, then p_S^{det} has a finite prefix containing a non-waived $K.RespFail$ witness. The proof follows the gate order, not a flat case split. First evaluate $K.ResetEpoch$. Its DirectFail branch supplies the independently mapped pending Policy-S monitor and matched affecting RESET, so RW2 yields cancelled-pending directly. Only on $K.ResetEpoch.Continue$ evaluate $K.TerminalDisposition$. Its DirectFail branch independently supplies the mapped pending monitor and DeclaredTerminal occurrence, yielding maximal-pending directly. Only when both nested gates take Continue does Lemma K.2b.E construct the infinite fair frozen continuation; Core.FairPick/Core.FairCycleDovetail prove that the constructed continuation satisfies Core.IssueFair and B2/B3, K.2b.R and K.2b.P establish deterministic serial dominance on it, Corollary K.2b.P-Frozen and Lemma K.2b.SF establish the frozen serial-frontier/control-path bridge, and K.2b.Q extracts the finite timeout/maximal-pending witness from the complete selected Policy-S artifact. If either required gate instance satisfies neither shared-schema branch, the candidate is outside the Policy-S reduction. The direct-failure evidence is independent of K.2b.P0/K.2b.R/K.2b.P/K.2b.P-Frozen/K.2b.SF/K.2b.Q Step 1, so no gate branch is proved by the dominance continuation it is used to select.

Proof status. Revision C completes the planned editorial/type/interface closeout after Revision B. ReachProgress, CouplerPersistence, and LowerClosure remain paper-closed at the stated abstraction level and applicability premises. Revision C adds no new proof premise or scheduling restriction: it makes the explicit-boundary-action versus process-facing-commit typing explicit; makes the K.2b.P action-unit case coverage exhaustive through $K.Low$ for synchronization/guard/epoch-gate units; rewrites K.2b.P-Frozen as one typed composition using the $E/K.CV$ -identified $T_{\{f_P\}}$; makes $K.Ser$ terminal-offer admissibility exactly equivalent to $DrainBag_Proc(I)=\emptyset$ for both singleton and qualified multi-frontier cases; adds uniform K.2b.R citations in the buffered dominance cases; names the loose Core.16(7) and Q.1(j)–(l) references; topologically reorganizes Appendix K-P with internal subsections and places the K.2b.E proof before its dominance consumers; and normalizes the active P0/R/P/P-Frozen/SF/Q consumer narration plus Appendix R/S references. A targeted Appendix-J check confirms that the Revision-B model-relative Core.CycleResult.committedUnits clarification is consistent with J.OuterCanonHist: J already carries complete observable/proof units before applying Π_DUT,E to form the outer Σ_DUT quotient, including cases where complete proof/elaboration units are erased. No identified paper-level serial-reduction proof obligation remains at this abstraction level. Outstanding

work is mechanization, tool/flow qualification, per-instance certificate discharge, and ordinary implementation validation. This is a paper-audit status statement, not mechanized certification.

K.4c Corollary K.2c — The Deterministic Policy-S Execution Suffices

Premise A5-S (primary Policy-S check). Let p_S^{det} be the Policy-S execution selected by SDet. A5-S requires K.CompleteS evidence establishing maximality, fairness, and completeness of that run, K.RespCov establishing total finite-bound monitor coverage, and K.RespClean establishing that no finite prefix of the complete run witnesses a non-waived K.RespFail. K.RespCov and K.RespClean are separate response-side obligations for the same complete artifact; this complete-run response evidence is not a KObs cutpoint predicate and not an RTL latency statement. CF-S records these separate layers together with the shared K.InterruptionGate schema and either the candidate-specific K.ResetEpoch/K.TerminalDisposition evidence required by K.2b or one uniform instance-level U1-U3 record that proves all required Continue branches. A5-S alone is not an A5-L discharge outside those applicability conditions.

Corollary K.2c (deterministic Policy S is a sufficient bounded-response check). Under K.SerialRoutePremises for the fixed selected instance and the applicable K.Low.A5 transport when lowering is used, A5-S implies A5-L. The universal reset/terminal gate premises may be discharged either candidate-by-candidate or by the uniform instance-level U1-U3 sufficient condition above. A5-S supplies the K.RespCov and K.CompleteS evidence required by Lemma K.2b and the K.RespClean contradiction used below. More precisely, Lemma K.2b first proves $A5L_Sys_K(I, W_K)$ for the selected normalized blocking instance. If that lowered property were false, its finite Core.LAdm-admissible counterexample prefix would satisfy K.EnvMF and one shared-schema branch of K.ResetEpoch. K.ResetEpoch.DirectFail independently supplies a finite non-waived cancelled-pending K.RespFail at the matched affecting RESET. On K.ResetEpoch.Continue, evaluate K.TerminalDisposition: its DirectFail branch independently supplies the mapped pending Policy-S response obligation and actual DeclaredTerminal disposition, yielding maximal-pending directly. Only under K.ResetEpoch.Continue followed by K.TerminalDisposition.Continue does Lemma K.2b.E supply the infinite fair frozen continuation required by K.2b, after which K.2b.P0/R/P, Corollary K.2b.P-Frozen, K.2b.SF, and K.2b.Q force a finite non-waived K.RespFail on the complete selected p_S^{det} . Every branch contradicts K.RespClean, so $A5L_Sys_K(I, W_K)$ holds. Lemma K.Low.A5 then transports that result to $A5L_Sys_ref(I, W)$, which is A5-L. The corollary makes no claim for candidates that fail K.EnvMF or for which a required gate instantiation satisfies neither Continue nor DirectFail. □ Generalized-drain scope. If $Sys_K = Sys_ref$, the transport is identity and one source-model K.InFlightQual suffices. If lowering is used, the corollary may discharge the current generalized A5-L premise only when SR7 binds K.Low.ModalQual and the new generalized K.Low.A5 theorem applies; in particular the Sys_ref continuation/path qualification is not inherited merely from the Sys_K serial continuation.

Definition K.CompleteS and practical completeness rule. The SDet-selected p_S^{det} is complete when it is maximal and fair for the selected deterministic simulator semantics under the Core.CompleteCycle projection of the selected Core.CycleResult semantics. A terminating run must reach a declared maximal terminal state; a cycle with no selected Σ commit is not terminal merely for that reason. A nonterminating run requires invariant, model-checking, recurrence/lasso, or equivalent completeness evidence. K.CompleteS identifies the entire artifact over which K.RespClean is checked; it does not itself imply any response bound. Changing capacities, elaboration, witness, stimulus, environment/reset contract, arbitration, simulator scheduler, Policy-S implementation, response-monitor configuration, waiver set, or normalization creates a different selected instance.

Determinism and bound note. Corollary K.2c concerns the selected artifact p_S^{det} generated by the concrete Policy-S simulator semantics and certified complete by K.CompleteS. It neither assumes nor proves confluence among multiple Policy-S histories. A successful proof may use any finite response

bounds if it proves complete-run compliance; increasing an unproved bound after a timeout changes the certificate attempt but not the functional design instance. A separately justified bound is needed to call the timeout a functional performance violation. WC and deterministic replay support reproducibility relative to the selected history; SDet fixes the remaining global choices that could change that history or the monitor outcomes.

K.4d Structural Sufficient Conditions for A5-L

The theorems of this subsection are alternate direct discharges of A5-L. They do not use Policy S, Lemma K.2b, or Corollary K.2c. They remain useful for nonterminating systems, designs outside the serial fragment, timed/reactive multi-frontier instances, or flows that prefer a compact structural certificate—but only when A5-L is actually true. They cannot certify the eager-dependent crossing example above, because that design has an admissible serial-shaped prefix that already falsifies A5-L.

Definition K.Dep (full static dependency graph). G_dep is the directed graph over modeled processes that over-approximates every possible coarse Core.WFG edge in the selected instance. For ordinary primitive point-to-point channels it contains the familiar data edge from consumer to producer, capacity/back-pressure edge from producer to consumer, and both directions for rendezvous; synchronization/epoch-gate and explicit arbiter-process edges are included as before. For $Sys_B^{\wedge}elab$, G_dep additionally contains every modeled-process pair that can occur in $WaitOwners_E = U ReleaseOwnerSets_E$ for any reachable disabled frontier under the selected elaboration template. Stateless couplers, FIFO stages, and local elaboration nodes are not graph vertices; their possible paths are contracted to the process pairs declared by E. The union is intentionally conservative: exact $Dead_K$ closure uses $ReleaseOwnerSets$, but every release-closed $Dead_K$ component still contains a directed cycle in this coarse union graph (including a genuine self-loop for a one-process component). Therefore the existing acyclicity/rank/cycle-breaking structural routes remain sound without hypergraph analysis. Caution (normative): G_dep shall be computed on this full union dependency relation; the data-flow topology alone is insufficient, and omitting an elaboration-projected dependency is unsound.

Theorem K.Str-1 (acyclic dependency). If G_dep is acyclic, including absence of genuine self-loops, then A5-L holds for the selected instance. Proof. At every reachable source K-facing cutpoint under every admissible issue policy, each coarse WFG edge is an instance of a G_dep edge by Core.11–Core.13 together with the Q.1 $ReleaseOwnerSets/WaitOwners$ reconstruction. Every Definition K.Dead witness contains a directed cycle in the WFG induced by D because it is strongly connected and release-closed; hence it would yield a G_dep cycle. ■

Theorem K.Str-2 (ranked dependency). If there is a function r from processes (or from process–boundary pairs) to a well-ordered set such that every G_dep edge realizable as a WFG edge at a reachable cutpoint strictly decreases r , then A5-L holds, since a WFG cycle would yield an infinite descent in r . Edges provably unrealizable at reachable cutpoints may be excluded from the obligation, with each exclusion proved as an invariant of the selected instance and recorded in the side-condition discharge record. ■

Theorem K.Str-3 (credit / initial-token invariant). If every cycle of G_dep contains at least one edge that is unrealizable as a coarse WFG edge at any reachable cutpoint, then A5-L holds. For an ordinary uncoupled primitive buffered boundary, a proved invariant such as $0 < occ_c < B(c)$ whenever both endpoints are pending can establish such an exclusion. At a coupling-sensitive boundary, occupancy alone is insufficient: the invariant/certificate must establish unrealizability using the applicable settled $MirrorAvail_E$ and $ReleaseOwnerSets/WaitOwners$ semantics. ■ Each invariant is itself a proof obligation over the selected instance, including its capacities and elaboration, and shall be recorded in the side-condition discharge record.

K.4e Certificate and Discharge Formats (Normative)

A concrete theorem application uses one versioned certificate package. It may contain CF-0 cutpoint certificates and one A5-L discharge payload. A K-facing CF-0 carries or references the theorem-derived/direct J.RegPhaseReach provenance for the selected J.GWR prefix τ_{ref}^G , the Core.KObsNormRef/K.NormStable evidence for the exact observation suffix v_{ref} , and the direct or K.DrainReach-Comp-derived K.DrainReach proof on τ_{ref}^K ; their three-part composition is K.RLAdmReach for the source prefix ending at σ_{ref} . A universal K.1A-Total or K.1 claim records the selected K.CertCov route as well. The package may reference shared proof artifacts rather than duplicating them, but every referenced item must bind through the same K.CertHdr instance identity. Definition K.CertHdr (common certificate header). The header binds every artifact to one theorem instance and records schema/model versions; source, Sys_ref/Sys_K, and RTL identities; elaboration, capacities, projections, ownership, ReleaseOwnerSets/WaitOwners schema, the applicable Q.1(I) serial-route handshake-support/CouplerPersistence schema/version, and lowering; the Q.2 in-flight action/effect schema identity/version, declared precision class, and model/profile identity when generalized drain is used, including Core.SyncCarrierWF and SyncFire/EpochGateAdvance schema identities when synchronization lowering is selected, and for profile G2 the K.Low.ModalQual/LowAdminNormalize/LowWaitRoots/InFlightPathCorr checker schema and both model-indexed K.InFlightQual identities; fixed stimulus, environment, reset, witness/WC provenance; the Core.CycleResult/Core.CycleClosure semantics identity where a complete source or Policy-S run is consumed, including full nextState environment/auxiliary typing and reset-unit emission, and the Core.FairPick constructive-witness schema/version when a proof-built continuation consumes FP4a; and, for the Policy-S route, the concrete deterministic simulator/scheduler, deterministic arbitration, Policy-S issue semantics, ε -normalization identity, SDet evidence, K.Resp monitor schema, finite-bound function, reset/end-of-test disposition, the K.InterruptionGate schema/version, each candidate K.DomResetScope/DomEpoch definition and provenance, K.ResetEpoch evidence for candidate components, K.TerminalDisposition evidence on K.ResetEpoch.Continue candidates, and response-waiver policy. It also records cutpoint waivers, route, side-condition references, the K.DrainEvidence schema/version and, whenever K.DrainReach-Comp is used, the per-activation DR1a/DR1b route-map provenance, Core.QSync structural-evidence provenance and theorem references, and, when the standard QSync induction/extraction route is used, the Core.RTLDesignatedOriginFrame/Core.KOriginCarrier schema identity plus initial-epoch/base-node provenance and the J.CertExtract_QS theorem/checker reference; it records the K.CertCov coverage method when applicable, generator/checker versions, and the trusted computing base. Any QS/settling premise accepted solely from an unverified HLS/synthesis-tool report shall be named in the declared trusted base; independently checked concrete evidence records the checker/certificate identity instead. Different header identities denote different selected instances. Revision-A schema rule. A package that relied on the superseded pointwise-only Q.1(I) handshake-support wording or on a pre-FP4a Core.FairPick witness is not silently grandfathered: when the Policy-S route consumes those interfaces, the package shall be revalidated and rebound under the revised K.CertHdr schema identity. This follows the existing superseded-schema revalidation rule used for release-family corrections; it does not change the functional design instance, but it changes the proof/certificate identity accepted for the theorem claim.

Definition K.CertifiedCutpoint (K-ready package for one reachable RTL cutpoint). Fix an RTL K-facing cutpoint $p_{\text{post}} \in \text{KCutReach_post}(I)$, a finite reachable RTL_B prefix τ_{post} , and an explicitly recorded finite normalization witness v_{post} satisfying NormPost_I($\tau_{\text{post}}, v_{\text{post}}, p_{\text{post}}$).

CertifiedCutpoint_I($\tau_{\text{post}}, v_{\text{post}}, p_{\text{post}}$) holds when one K.CertHdr-bound package records that triple and establishes: (1) J.LocalGWR/J.GWR-Comp or J.GWR-Direct for the selected prefix; (2) the exact

Corollary J.1 source package $(\tau_ref^G, v_ref, \sigma_ref, E, w, H, O)$, with $Core.KObsNormRef_I(\tau_ref^G, v_ref, \sigma_ref)$, $\tau_ref^K := \tau_ref^G \cdot v_ref$, and $\sigma_ref \in KCutReach_ref(I)$; (3) $J.HCanon$ on τ_ref^G , $K.NormStable$ and Lemma $K.NormStable-H$ across v_ref , plus $J.OfferReach$, $J.OfferConf$, and $J.KObsConf$ at σ_ref ; (4) $J.RegPhaseReach$ for the exact τ_ref^G construction prefix and its endpoint s_ref , theorem-derived on the normal route or directly validated; (5) $K.DrainReach$, including $Core.SettleKBridge$, for the exact K -facing prefix τ_ref^K/σ_ref , derived by Lemma $K.DrainReach-Comp$ on the standard normal route when its hypotheses are bound or directly validated otherwise; and hence (6) the three-part $K.RLAdmReach$ and membership of τ_ref^K in $Pref_L^{adm}(I)$. The package proves one RTL prefix/normalization/ K -cutpoint triple and its one exact source construction/observation-normalization counterpart only and does not imply $K.CertCov$. When the generalized drain theorem version is selected, the same $K.CertHdr$ -bound application also references the uniform $K.InFlightQual$ model/ E /profile evidence used by $K.1A$; it need not duplicate that instance-level proof in every cutpoint package.

Definition $K.CertCov$ (K -ready package coverage). $K.CertCov(I)$ holds when every reachable RTL K -facing cutpoint $p_post \in KCutReach_post(I)$ has at least one theorem-constructible or explicitly recorded reachable-prefix/normalization pair (τ_post, v_post) satisfying $NormPost_I(\tau_post, v_post, p_post)$ and a corresponding $CertifiedCutpoint_I(\tau_post, v_post, p_post)$ package. A total proof-producing extractor or inductive existence theorem may represent this family without eagerly materializing one file record per cutpoint, provided the checker can derive or validate the package for any queried reachable cutpoint under the bound $K.CertHdr$ instance. On a constructive route, the query shall be accompanied by or recover an independently checked reachability/carrier representation; a bare state value with no proof that it is the represented reachable $KCut$ is not an extractor input. Under the standard $Core.QSync$ route, the canonical same-cycle normalization/endpoint supplied by Theorem $Core.QSync-Post$ may be used for this K -facing $NormPost$ selection. This canonicalization does not replace Appendix I/J $ReplayNormalizationConvention$ or its selected replay-normalization/congruence witnesses. $K.CertCov$ does not require every arbitrary reachable RTL prefix to admit a normalization before the next Σ action; coverage is over the complete reachable K -facing domain on which $DeadKRec^W$ is defined.

Equivalently, the package-existence relation is total on $KCutReach_post(I)$. For Corollary $K.1A$ -Total and Theorem $K.1$, $K.CertCov$ is paired with $Core.KCutClosure_post(I)$. In the standard $QSync$ scope, $Core.QSync-Post$ proves the closure/non-vacuity side, while Theorem $J.CertExist-QSync$ plus Corollary $K.CertCov-QSync$ below provide a generic inductive route to the package-totality side from $J.QSyncCertInd$. Thus $K.CertCov$ remains the downstream semantic interface, but it need not be supplied as an unrelated per-cutpoint axiom or pre-enumerated package set when the $QSync$ certificate-induction hypotheses are discharged. In that standard $QSync$ route, the $QC5$ component of $J.QSyncCertInd$ may be obtained uniformly through Lemma $K.DrainReach-Comp$ from its bound instance-level evidence; $J.CertExist-QSync$ does not itself create or assume those underlying drain facts.

Corollary $K.CertCov-QSync$ ($QSync$ certificate existence discharges $K.CertCov$).

Assume the standard $QSync$ normal-compositional scope and the hypotheses of Theorem $J.CertExist-QSync$. Then $K.CertCov(I)$ holds. Proof. $J.CertExist-QSync$ gives $J.QSyncCertCov(I)$ by recovering, for each queried $p_post \in KCutReach_post(I)$, the represented $Core.KOriginCarrier$ /origin position and applying the carrier-indexed $J.CertExtract_QS$. Fix such a p_post and its returned $QSyncCertState$ tuple C . $QC1$ supplies the canonical $NormPost$ pair and designated origin/frame; $QC2$ supplies $J.LocalGWR/J.GWR-Comp$ and the exact $J.GWR$ source prefix τ_ref^G ; $QC3$ supplies $J.HCanon$, $Core.KObsNormRef/K.NormStable/K.NormStable-H$, and $J.OfferReach/J.OfferConf/J.KObsConf$ for the exact K -facing extension τ_ref^K ; $QC4$ supplies $J.RegPhaseReach$ on τ_ref^G ; and $QC5$ supplies $K.DrainReach$ on τ_ref^K and therefore the three-part $K.RLAdmReach$ with $\tau_ref^K \in Pref_L^{adm}(I)$. On the standard normal route $QC5$ is ordinarily discharged by Lemma $K.DrainReach-Comp$ from the already-

bound independent DR1–DR4 evidence together with the theorem-derived J.RegPhaseReach, attribution/history, QSync-Ref L2 settling, and K.NormStable observation-suffix facts of the same package. Bind those already-proved semantic facts and the carrier/extractor theorem references to the K.CertHdr for the fixed theorem instance. The resulting record satisfies Definition K.CertifiedCutpoint for p_post . Because p_post was arbitrary, Definition K.CertCov holds. The theorem/extractor reference may represent the family and materialize individual packages on demand; K.CertHdr adds version/provenance/trust accounting, not an additional semantic existence assumption. $\square$ Generalized-drain note: this certificate-existence theorem does not manufacture K.InFlightQual. K.1A-Total binds a separate uniform Sys_ref generalized-drain qualification in addition to the per-cutpoint packages. In G1 the serial route may discharge its continuation component from the already-bound Sys_ref=Sys_K Core.CycleClosure; in G2 the Sys_ref K.InFlightQual is discharged separately as required by K.Low.ModalQual. K.CertCov itself is neutral to which valid A5-L route supplied the source premise.

CF-0 — KObs cutpoint correspondence certificate. This certificate supports Corollary J.1 and, when augmented with item (8), Theorem K.1A for one cutpoint; it neither discharges A5-L nor proves K.CertCov. It records or commits to: (1) K.CertHdr; (2) the selected RTL prefix/NormPost pair and the selected source $\tau_ref^G/Core.KObsNormRef v_ref/\tau_ref^K/\sigma_ref$ objects; (3) canonical H; (4) typed residual offers O; (5) J.HCanon on τ_ref^G , the primitive K.NormStable checks and derived K.NormStable-H transport, and J.OfferReach; (6) complete RelevantReqSet/ReqProjection equality, attribution, cardinality, prefix-closure, and no-hidden-request evidence for J.OfferConf; (7) B-faithfulness/E4, reset, and ϵ -opacity for J.KObsConf; and (8), for Appendix K, the J.RegPhaseReach provenance on τ_ref^G plus K.DrainReach evidence on τ_ref^K whose checked three-part composition with K.NormStable is K.RLAdmReach for the exact source package; when K.DrainReach-Comp supplies that field, CF-0 also carries or references the bound K.DrainEvidence schema identity and its activation-indexed DR1a/DR1b route map. The phase provenance includes Core.RegSeq, enriched deferred-hole/NoDependentUse records, field-sensitive footprints, J.RegPhaseNF, Horizon-batched J.UnitExt, and the no-unresolved-hole construction-endpoint invariant, or a direct equivalent. The normalization provenance includes Core.KObsSettle/Core.KObsNormRef and the no-feedback-to-L3 restriction. Derived Front/WFG/deadlock records are checker-reconstructed. Historical Issue/ReturnedAtCycle logs and K.Resp timers need not be duplicated, but referenced artifacts must validate their consumed obligations. Under the generalized drain schema, item (1) also binds the Q.2 action/effect schema and K.InFlightQual profile identity used to recompute InFlightEscape/DrainClosedNow; the per-cutpoint package need not duplicate a uniform instance-level path-soundness proof.

CF-S — Primary deterministic Policy-S bounded-response certificate. The payload records the header; the functional Policy-S simulator semantics, initial state, fixed witness/stimulus/environment, deterministic scheduler and arbitration, capacities, elaboration, lowering, normalization, response-waiver policy, K.CVPred/K.CV-WF provenance, including the K.CVPred comparison-availability closure used by K.2b.P0/R/P and the explicit Core.16 reach-legality disposition for every recursively materialized non-transfer predecessor, and the shared K.InterruptionGate schema with either the per-candidate certificate-bound K.DomResetScope/DomEpoch and ordered K.ResetEpoch/K.TerminalDisposition instantiations or one uniform instance-level gate record satisfying U1-U3 above; the selected run artifact p_S^{det} as a sequence of Core.CycleResult records or an equivalent symbolic representation whose CompleteCycle projection is checkable; SDet evidence; fairness/maximality and Sys_K B2/B3/B5/K.Drain/K ϵ /nonwithdrawal evidence, including the exhaustive Core.IC lifecycle and Lemma K.BF-Persist; and K.CompleteS. For a successful clean discharge, K.ResetEpoch.Continue is the normal prospective reset evidence, and only on that branch K.TerminalDisposition.Continue is the normal prospective terminal evidence; the reset Continue discharge is proved contract-wise over both the R-L frozen Policy-L prefix class and the independently defined R-S selected-Policy-S comparison-prefix domain covering P0/R/P/P-Frozen/SF/Q, while terminal Continue is proved over its applicable frozen-

prefix class; neither is inferred from absence of observed failures. Either DirectFail branch may be proved conditionally for a hypothetical candidate or instantiated to classify a direct failure; a concrete DirectFail instantiation is itself a K.RespFail witness and cannot coexist with K.RespClean on a successful clean run. The generic direct-failure payload shape binds P , f_P , x_P^A , g_P with $g_P \leq_{\text{src}} f_P$ and an already-triggered pending non-waived monitor independently of K.2b.P0/R/P/P-Frozen/SF/Q Step 1. The ResetEpoch.DirectFail specialization additionally binds the common pre-reset epoch, matched affecting RESET in CycleResult.resetUnits, and RW2 clearing of that exact monitor, yielding cancelled-pending. The TerminalDisposition.DirectFail specialization instead binds the independently validated declared maximal terminal state while that monitor remains pending, yielding maximal-pending. CF-S also records timer progression through CycleResult.nextState, complete-run results, K.EnvMF applicability, including K.JointFreeze for every admitted StandardEnv candidate not covered by K.JointFreeze-Ord; K.CompareAdvanceLegal coverage for every K.2b.P0/K.2b.R-consumed owner/key, including the P0 preparation/launch and R Issue/assertion halves and any required explicit Sys_B^elab multi-frontier certificate and the JointFreeze/CompareAdvanceLegal consistency check; the K.CVPred recursive dispositions may cite Lemma K.CompareAdvanceLegal-Ord pointwise for qualifying ordinary/singleton blocking OpFact predecessors and must identify inert-observation or equivalent launch-legality evidence for the remaining predecessor reaches; and waiver preservation, and references the instance-bound Core.CycleClosure/Core.FairPick/B5 discharge used to construct the Policy-L continuation in Lemma K.2b.E. Core.FairPick does not strengthen the behavioral Core.IssueFair/B2/B3 property, but it is an additional constructive evidence form required because K.2b.E builds a Policy-L continuation rather than consuming one already supplied. Checker obligations include validating the K.CVPred provenance equations, comparison-availability closure, every recursive predecessor Core.16 reach-legality disposition, and K.CV-WF/rank certificate used by K.2b.P0/P1/R/P; checking the theorem-side K.SerialDomPredFinite, K.CompareFact-Mono, K.SegWork/K.SegWork-Complete, K.KeyItemOrderBridge, and K.SegmentPlan-EventuallyTarget interfaces used to close same-continuation target readiness; validating the common gate schema and its K.BF exhaustiveness once and then either (i) validating each certificate-supplied K.DomResetScope against D1–D4, including K.CompareScopeFootprint provenance, and its associated DomEpoch, each instantiation's Interrupt predicate, the reset R-S Policy-S prefix-domain coverage, and each DirectFail specialization, or (ii) validating the single uniform instance-level gate record against U1-U3 and its claimed universal Continue discharge; in either route the required reset-first nesting and rejection of any clean candidate for which the applicable gate discharge fails remain enforced. For a candidate satisfying K.JointFreeze-Ord's ordinary/single-release-support premises, the K.JointFreeze field is a theorem reference rather than a separate certificate; ordinary comparison cases may likewise cite K.CompareAdvanceLegal-Ord when its separate premises hold. Definition K.A5Cert (A5-L discharge package). The package consists of K.CertHdr plus exactly one declared route payload: CF-S; a proof reference for K.Str-1; CF-1; CF-2; CF-3; CF-4; CF-5; or a direct Policy-L invariant/exploration/proof artifact. The package records the route and target property and shall expose enough data for an independent checker to validate the route without trusting a tool-emitted conclusion string. Policy-S route packages target K.RespClean and include dominance, coverage, finite-bound, waiver-preservation, and complete-run evidence; direct Policy-L packages target Dead_K^W. Optional diagnostic environment-terminal records do not substitute for the response payload.

CF-1 — Rank certificate. A function r from processes (or process–boundary pairs) of the selected instance to a declared well-ordered set, together with the Definition K.Dep edge enumeration, such that every realizable edge strictly decreases r (Theorem K.Str-2). Checker obligations: the edge enumeration covers data, capacity back-pressure, rendezvous, epoch-gate, and arbiter edges of the selected instance including its elaboration and lowering; every non-excluded edge strictly decreases r ; every excluded edge carries a recorded invariant proof of unrealizability.

CF-2 — Credit / cycle-breaking certificate. For each cycle of the full union dependency graph G_{dep} , provide an invariant—such as token presence, occupancy bounds at an ordinary primitive boundary, a circulating-credit invariant, or a coupling-aware settled-handshake invariant—establishing that at least one edge of the cycle is unrealizable as a coarse WFG edge at any reachable cutpoint (Theorem K.Str-3). Checker obligations: the dependency-cycle enumeration is complete; each invariant is inductive over the selected instance; and the proved edge exclusion applies at the explicit capacities, MirrorAvail/release-support semantics, and elaborated boundaries. Petri-net or siphon analyses may be used by a tool as one way to generate this certificate, but no Petri-net equivalence theorem is part of normative Appendix K.

CF-3 — Inductive-invariant certificate. Supply an abstraction map α from the selected operational transition system into a finite abstraction A . For direct Policy-L verification, the observation component shall be KObs, or an explicitly proved quotient sufficient to reconstruct $DeadKRec^W$. For the Policy-S response route, A shall additionally carry the monitor activation, dynamic attribution, age, bound-class, reset/terminal-disposition, and maximality state needed to reconstruct $K.RespFail$; KObs alone is intentionally insufficient. Supply an invariant Inv such that $\alpha(Init)$ is contained in Inv , $Post_A(Inv)$ is contained in Inv , and Inv excludes the route's bad event. Checker obligations: α and $Post_A$ soundly overapproximate the selected operational relation; every cutpoint or monitor fact is reconstructed from the appropriate observation/monitor components and ambient instance data; reset/environment/waiver distinctions are represented; witness-selected non-blocking outcomes are source-describable through NB-CF or $K.Low$; and all invariant inclusions are mechanically validated. Equality of KObs records alone shall not be treated as transition equivalence or as proof of future Issue legality, fairness, DrainDiscipline, or response timing.

CF-4 — Exploration certificate. Record an exhaustive or soundly bounded model-checking result over the selected Policy-L relation establishing unreachability of $DeadKRec^W$, or over the selected Policy-S relation establishing absence of $K.RespFail$, with $K.CertHdr$, the explored transition-system identity, abstraction map, cutpoint normalization rule, monitor semantics, tool/version, bound or completeness justification, fairness interpretation, and coverage argument. The checker shall recompute $Dead_K$ from KObs for the direct route and shall recompute monitor outcomes from the augmented Policy-S state for the response route. A finite exploration depth or simulation prefix without a proof of completeness is supporting evidence only. For generalized drain, a positive InFlightEscape result records a concrete finite action path with enough successor-state derivation/hashes for independent recomputation. A negative result is exact only after exhaustive search of the declared finite reconstructed relation; a truncated no-escape search is UNKNOWN/INCOMPLETE unless a separately proved conservative precision mode is explicitly declared. Search precision/completeness is distinct from operational completeness of the action schema and both statuses are bound by $K.CertHdr$.

CF-5 — Tool-emitted certificate. Any CF-S or CF-1 through CF-4 emitted by the HLS tool or a companion tool over the selected instance may exploit the tool's knowledge of the scheduled control graph, stage allocation, actual FIFO and skid-buffer capacities, request ownership, join and arbitration structure, automatic-flush implementation, and hidden storage or credits. It shall use $K.CertHdr$ and the applicable CF-0/A5 payload schema and be validated by an independent checker whose trusted base is recorded. The checker shall validate structure, evidence references, and reconstructed K-facing predicates, not merely the certificate generator's final claim. Labeling rule (normative): a certificate stated over the generated RTL itself, rather than over the selected Sys_ref instance, is admissible evidence of deadlock-freedom for that design but bypasses the Appendix K preservation architecture — it is output verification, not a discharge of A5-L — and shall be labeled as such in the record. A tool may exploit hidden scheduled/stage state when generating an in-flight certificate only if the independently checked Q.2 abstraction/refinement evidence exposes that state in the selected proof model or proves it

irrelevant to action identity/effect. Tool knowledge alone is not a new KObs field or a waiver of Q.InFlightActionEffectSound/J.InFlightPathSound.

Soundness restriction (normative). An arbitrary eventuality claim—such as ‘the arbiter eventually grants’—does not by itself discharge A5-L, because A5-L prohibits reaching the closed wait-cycle cutpoint. The K.Resp route is different: Lemma K.2b proves that any such liberal cutpoint forces a permanent covered serial frontier and therefore a finite timeout/maximal-pending or matched-reset cancelled-pending event under a certified finite bound. Only response properties satisfying K.RespCov, the K.2b trigger mapping, complete-run checking, and waiver preservation may be used in Corollary K.2c. An unproved timeout threshold or a finite clean simulation is not a certificate.

K.4f The Policy-S Standard Fragment (K.PSF)

Definition K.PSF (Policy-S standard applicability fragment). A selected instance is in K.PSF when it is in Sys_K form; channels are point-to-point with finite B-faithful capacities; SDet holds for the concrete Policy-S simulator; K.CV holds, including K.CVPred/K.CV-WF and its certified predecessor-segment descriptors, and arbiters are deterministic and part of that simulator semantics; K.EnvMF holds, with timed/reactive or otherwise nondefault instances permitted to cite Lemma K.EnvMF-Uniform once its instance-level hypotheses are proved, while StandardEnv candidates not covered by K.JointFreeze-Ord retain the K.JointFreeze requirement; for an ordinary non-elaborated primitive point-to-point timed/nondefault instance already satisfying the remaining K.JointFreeze-Ord premises, the K.EnvMF-Uniform structural part reduces to Core.OrdFrontSingleton plus the ordinary Core.WFG singleton release-support fact; K.CompareAdvanceLegal holds for every candidate comparison, with ordinary/singleton main comparison cases theorem-derived and explicit Sys_B^elab multi-frontier owners checked regardless of membership in D; K.SegmentPlan is fixed from K.SerialDomOrder/the selected Policy-S artifact and K.SegmentAdm is discharged for every certified K.CVPred/P0 segment stack it emits on an admitted candidate; K.CVPred recursively records the explicit Core.16 reach-legality disposition for every non-transfer predecessor materialization, using Lemma K.CompareAdvanceLegal-Ord pointwise for qualifying ordinary/singleton blocking OpFact predecessors and explicit reach evidence otherwise; the shared K.InterruptionGate schema is fixed and its universal candidate obligations are discharged either by the existing per-candidate K.DomResetScope/DomEpoch, K.ResetEpoch (including both R-L and R-S Continue coverage), and nested K.TerminalDisposition records or by the uniform instance-level U1-U3 sufficient condition above; requests are nonwithdrawable under the exhaustive Core.IC lifecycle, with Lemma K.BF-Persist supplying the frozen-frontier specialization for every WFG-relevant K.BF blocking request; $\text{K}\epsilon_{\{I, \text{Sys}_K\}}$, the Sys_K K.Drain and nonwithdrawal obligations hold; the Sys_K interpretations of Core.CycleClosure and B5 hold for the selected Policy-L semantics, and its Sys_K Core.IssueFair/B2/B3 discharge is available in the constructive Core.FairPick form required by Lemma K.2b.E, including FP4a bound to that K.SegmentPlan; whenever a coupling-sensitive boundary is consumed, the qualified E package supplies the strengthened Q.1(l) CouplerPersistence proof for that template; Q.1(d)–(k) well-formedness alone does not establish CouplerPersistence for mutual-stall, shared-capacity, or arbitrated/non-monotone handshake topologies, and if the selected template cannot supply that persistence proof then the candidate is outside K.PSF/CF-S and requires another A5-L discharge route; and a finite-bound K.Resp configuration with total K.RespCov is supplied. The structural/provenance/reset fields above are the K.PSF-side inputs consumed by derived K.2b.SF; SF itself is not another K.PSF premise, and its K.CompleteS input is supplied by A5-S/CF-S. Completeness, fairness, and K.RespClean of p_S^{det} belong to A5-S/CF-S. Generalized-drain publication scope: a K.PSF instance may invoke the generalized K.1 theorem either in G1 with Sys_K=Sys_ref or in G2 with lowering. In G2, the K.PSF/K.SerialRoutePremises package shall additionally bind K.Low.ModalQual, including separate Sys_K/Sys_ref K.InFlightQual and the completed

modal path/escape/drain correspondence; without that package the lowered instance remains outside the current generalized theorem.

Theorem K.PSF-1. For a K.PSF instance, the SDet-selected Policy-S execution p_S^{det} establishes $A5L_Sys_K(I, W_K)$, and hence A5-L by Lemma K.Low.A5, when CF-S supplies K.CompleteS and K.RespClean for the totally covered finite-bound monitor set. Proof: Lemma K.2b, Corollary K.2c, and K.Low.A5. $\square$ Generalized-drain specialization: if $Sys_K = Sys_ref$, $A5L_Sys_K$ is already A5-L. If lowering is used, the result applies to the current generalized theorem only when the K.PSF/SR7 package includes K.Low.ModalQual and the generalized K.Low.A5 transport above; otherwise the lowered instance remains outside this theorem version.

Intent. K.PSF is the recommended standardized applicability target for new latency-insensitive designs. The general user-facing evidence is the conservative Policy-S execution checked by complete process-facing bounded-response obligations. For finite transactions and bounded hardware protocols, these are ordinary timeout/service assertions and are expected to be highly automatable. A correct design with fair but genuinely unbounded response gaps lies outside this serial fragment and must use a structural, direct Policy-L, invariant, exploration, or tool-certificate route. Dead_KS/K.EnvOrd/K.Done remain optional diagnostic aids rather than primary proof premises. Ordinary-frontier simplification: for an ordinary non-elaborated instance whose relevant message frontiers are singleton, neither a separate K.JointFreeze certificate nor a separate K.CompareAdvanceLegal certificate is an additional per-candidate obligation; Lemmas K.JointFreeze-Ord and K.CompareAdvanceLegal-Ord derive those frontier facts from the standard source-order, Core.16, ownership, comparison-invariant, and drain premises. Explicit certificates are needed where the corresponding ordinary lemma does not apply: for K.JointFreeze this includes a single Sys_B^{elab} frontier item with a non-singleton or multi-alternative release family as well as explicit multi-frontier structure, while K.CompareAdvanceLegal retains its separate ordinary/singleton theorem-derived cases. A stronger uniform instance-level K.JointFreeze certificate may discharge all of its admitted non-ordinary candidates at once. In the common no-relevant-reset/clean-terminal case, the uniform instance-level gate discharge above replaces candidate-by-candidate reset/terminal gate enumeration without weakening the underlying universal premise.

K.5 Theorems K.1A, K.1A-Total, and K.1 — Reachable-Cutpoint Deadlock-Freedom Preservation (“Liveness Preservation”)

Theorem K.1A (certified-cutpoint preservation from the scheduler-robust premise). Statement: Assume A1–A4 and A6–A11, WC/Core.RegSeq, the Appendix I/J correspondence prerequisites, with A6 interpreted at every explicit boundary on which KObs/WFG facts are evaluated through its source-ownership/elaborated-dependency-projection clauses. Assume also the current generalized-drain source-side scope and a bound $K.InFlightQual_I, Sys_ref, E, \Phi_ref$ for the selected Sys_ref action/path profile. This per-cutpoint theorem is route-neutral with respect to the A5-L discharge: G1 obtains A5-L directly on $Sys_ref = Sys_K$, while G2 may obtain it upstream through K.Low.ModalQual and generalized K.Low.A5; K.1A itself always evaluates the common KObs-derived DeadKRec on Sys_ref and consumes the source-side K.InFlightQual rather than assuming no lowering. Fix one $p_post \in KCutReach_post(I)$ and one recorded reachable-prefix/normalization pair (τ_post, v_post) with $NormPost_I(\tau_post, v_post, p_post)$ whose package satisfies CertifiedCutpoint_I($\tau_post, v_post, p_post$). Thus Corollary J.1 yields equal KObs for p_post and σ_ref from the selected J.GWR construction prefix $\tau_ref^{\wedge}G$ plus its recorded Core.KObsNormRef suffix v_ref ; Corollary J.RegPhaseReach-Comp supplies J.RegPhaseReach on $\tau_ref^{\wedge}G$; K.NormStable qualifies the observation-only extension $\tau_ref^{\wedge}K := \tau_ref^{\wedge}G \cdot v_ref$; and K.DrainReach proves Sys_ref nonwithdrawal, $K \in \{I, Sys_ref\}$, and Core.Drain/K.Drain on $\tau_ref^{\wedge}K$. Their three-part composition K.RLAdmReach places $\tau_ref^{\wedge}K$ in $Pref_L^{\text{adm}}(I)$. If W is non-empty, also assume KObsClosed_I($E, w(W)$). If A5-L holds for the selected Sys_ref instance, then p_post does not satisfy $DeadKRec^{\wedge}W$, equivalently the $Dead_K^{\wedge}W$ predicate of

the selected theorem instance. With $W=\emptyset$, the fixed certified cutpoint is not an internal message-passing-deadlock cutpoint. The theorem consumes neither $K.CV$, $NB\text{-}Align$, nor Lemma $K.2b$, and it makes no $K.CertCov$ or package-totality claim.

Proof (by contradiction).

1. Assume, for contradiction, that the fixed certified RTL K -facing cutpoint $p_post \in KCutReach_post(I)$ has K -facing reconstructed observation K_post satisfying $DeadKRec^W(K_post, I, W)$. By Definitions $K.Dead$ and $K.DeadW$ and the reconstruction interface, this includes a non-empty blocked release-closed strongly connected component, its reconstructed $ReleaseOwnerSets$ family, and $DrainClosedNow$ for that component. The asserted RTL requests and channel facts used in this observation are covered by $J.OfferConf$ and $B\text{-}faithfulness/E4$; no unrecorded raw RTL request may be omitted.
2. By $CertifiedCutpoint_I(\tau_post, v_post, p_post)$, apply the recorded Corollary $J.1$ instance to this same finite prefix and package. It supplies the reachable source K -facing cutpoint $\sigma_ref \in KCutReach_ref(I)$, reached by $\tau_ref^K := \tau_ref^G \cdot v_ref$, and $K_ref = KObs_E, w(\sigma_ref)$ such that $K_post = K_ref$. $J.OfferReach$ ensures that σ_ref is the endpoint of the recorded K -facing observation-normalized extension of the selected $J.GWR$ source witness rather than an arbitrary or unrealizable record.
3. By Lemma $K.0$, Lemma $K.2$, and Lemma $K.2a$ — equivalently, by functional extensionality of the reconstruction functions — $DeadKRec^W(K_ref, I, W)$ has the same truth value as $DeadKRec^W(K_post, I, W)$. When W is non-empty, $KObsClosed_I, E, w(W)$ makes the waiver predicate well-defined on the common observation, and Lemma $K.KObs\text{-}Dead$ identifies the reconstructed source predicate with $Dead_K^W(\sigma_ref)$. Hence $Dead_K^W(\sigma_ref)$ holds.
4. The same $K.CertifiedCutpoint$ package supplies theorem-derived/direct $J.RegPhaseReach$ for τ_ref^G , $K.NormStable$ for the recorded v_ref observation suffix, and direct or $K.DrainReach\text{-}Comp$ -derived $K.DrainReach$ for the exact K -facing prefix τ_ref^K . $RegPhaseReach$ proves $Policy\text{-}L$ legality and $L0\text{--}L4$ phase ordering of the $J.GWR$ -constructed part; $K.NormStable$ proves that the post- $L3$ observation suffix adds no scheduling, request, NB , reset, or replay-significant action and preserves the construction invariants; and $K.DrainReach$ proves Sys_ref nonwithdrawal, $K \in \{I, Sys_ref\}$, and every activated Sys_ref $Core.Drain/K.Drain$ clause through σ_ref . Therefore the three-part $K.RLAdmReach$ establishes $\tau_ref^K \in Pref_L^{adm}(I)$. Since τ_ref^K ends at σ_ref , σ_ref is the reachable $Core.LAdm$ -admissible K -facing cutpoint prohibited by $A5\text{-}L$. No phase inference is made from equal $KObs$, and no drain inference is made from $J.RegPhaseNF$ or $K.NormStable$ alone.
5. This contradicts $A5\text{-}L$. Therefore the fixed certified cutpoint p_post cannot satisfy $Dead_K^W$.

□

Mechanization note. The proof core is Corollary $J.1$, equality of typed $KObs$, pure reconstruction of $Dead_K^W$, theorem-derived/direct $J.RegPhaseReach$ on τ_ref^G , $K.NormStable/Core.KObsNormRef$ on v_ref , and direct or $K.DrainReach\text{-}Comp$ -derived $K.DrainReach$ on τ_ref^K . Corollary $J.RegPhaseReach\text{-}Comp$ proves $J.RegPhaseReach$ upstream from $Core.RegSeq$, enriched holes, field-sensitive footprints, $J.RegPhaseNF$, and bucketed $J.UnitExt$; $Core.KObsSettle\text{-}Ref$ plus the primitive $K.NormStable$ checks form the post-boundary observation interface; and Theorem $K.1A$ consumes only the composed $K.RLAdmReach$ fact, which now concludes admissibility of the prefix actually ending at σ_ref . $K.Drain$ continuation evidence is not derived from $KObs$, phase normality, or normalization stability. Package totality remains outside this theorem. Theorem $J.GWR\text{-}CompExt$ extends τ_ref^G under $J.LocalGWRExtCert$; the K -facing v_ref/τ_ref^K pair is selected after that extension and is not itself the object of the extension theorem. $K.DrainReach\text{-}Comp$ is then re-invoked on the enlarged package rather than treating the old $DrainReach$ fact as incrementally prefix-extensible.

Corollary K.1A-Total (universal reachable-cutpoint preservation under $K.CertCov$ and $KCut$ closure). Assume the premises of Theorem K.1A hold uniformly for the selected instance, assume $Core.KCutClosure_post(I)$, and assume $K.CertCov(I)$. Then no $\rho_post \in KCutReach_post(I)$ satisfies $Dead_K^W$. With $W = \emptyset$, RTL_B is internally message-passing-deadlock-free over the complete reachable K -facing cutpoint domain, and by Lemma $Core.KCutObserve$ a persistent K -relevant blocked configuration cannot evade that domain solely through endless pre-normalization ϵ activity. Standard $QSync$ instantiation. In the standard theorem scope, $Core.QSync$ -Post supplies $Core.KCutClosure_post$; if $J.QSyncCertInd$ is discharged, Theorem $J.CertExist$ - $QSync$ and Corollary $K.CertCov$ - $QSync$ additionally supply $K.CertCov$. Thus both universal-domain obligations are theorem-derived from checked structural plus base/step semantic-compliance evidence, with the K -facing package induction additionally consuming the bound Sys_ref drain/nonwithdrawal/ $K\epsilon_{\{I, Sys_ref\}}/channel/attribution$ evidence used by Lemma $K.DrainReach$ -Comp, rather than from arbitrary normalization or sampled cutpoints. Proof. $Core.KCutClosure_post$ supplies a finite $KCut$ for every reachable RTL K -facing cycle-evaluation origin; $K\epsilon_{\{I, RTL_B\}}$ preserves any blocker that persists through that RTL normalization. Any alleged prohibited $KCut$ has a $K.CertifiedCutpoint$ package by $K.CertCov$ and is therefore excluded by Theorem K.1A. $\square$

Theorem K.1 (user-facing deterministic Policy-S preservation under cutpoint closure and certificate coverage). Assume Theorem K.1A's per-cutpoint premises uniformly, $Core.KCutClosure_post(I)$, $K.CertCov$, $K.SerialRoutePremises$ for the same selected instance, and the applicable $K.Low.A5$ transport when lowering is used. If $A5$ -S holds for the $SDet$ -selected execution ρ_S^{det} —supplying $K.RespCov$, $K.CompleteS$, and $K.RespClean$ —then no reachable RTL $KCut$ satisfies $Dead_K^W$. Consequently, RTL_B cannot reach a persistent K -relevant internal message-passing blocking configuration whose normalized K -facing observation satisfies $Dead_K^W$. $Core.KCutObserve$ additionally prevents a K -relevant blocked configuration that persists through the selected same-cycle normalization from evading the $KCut$ observation domain solely through pre-normalization ϵ activity; this observation fact does not by itself imply that the observed $KCut$ is drain-closed or satisfies $Dead_K^W$. Proof: K.2c gives $A5$ -L. In the standard $QSync$ normal-compositional scope, $Core.QSync$ -Post derives $Core.KCutClosure_post$ and, when $J.QSyncCertInd$ is discharged, with $QC5$ supplied by $K.DrainReach$ -Comp or an equivalent direct discharge, $J.CertExist$ - $QSync$ plus $K.CertCov$ - $QSync$ derive $K.CertCov$; Corollary K.1A-Total then gives the universal RTL conclusion. A flow that does not use that induction route may instead establish $K.CertCov$ by one of the alternate methods in M.7a. Without $K.CertCov$, the same $A5$ -L discharge supports only the certified-cutpoint conclusion of Theorem K.1A; without $KCutClosure_post$, no universal claim is made that persistent pre-normalization RTL blocking states become observable as $KCuts$. The Policy-S form makes no claim for timed/reactive/nondefault candidates outside $K.EnvMF$'s single-internal-frontier, $K.JointFreeze$ -Ord-eligible release structure or for candidates for which a required nested gate instantiation satisfies neither Continue nor DirectFail. $\square$

Stage-2 theorem-version qualification: the generalized in-flight $DrainClosedNow$ form applies either in profile G1 ($Sys_K=Sys_ref$) or in profile G2 when the exact lowering carries $K.Low.ModalQual$ and generalized $K.Low.A5$. A lowered instance lacking that modal package is outside the current generalized theorem; it does not fall back to the legacy snapshot drain predicate.

Implementation status. The operational Policy-S check is specified in terms of conventional pre-HLS simulation and bounded-response assertions and is expected to be highly automatable for finite transactions and bounded protocols. Treating a clean run as a verified discharge of $A5$ -L retains proposed-reduction status until the K -P proof obligations and required certificate interfaces are mechanized and qualified. For the universal RTL conclusion, the paper-level standard $QSync$ route is now complete at the theorem-interface level: $Core.QSync$ -Post derives $Core.KCutClosure_post$, and $J.CertExist$ - $QSync$ / $K.CertCov$ - $QSync$ derive $K.CertCov$ from an independently checked $J.QSyncCertInd$ base-and-step invariant covering every reachable RTL successor. This removes the former generic package-existence gap but does not make coverage automatic: a concrete flow must still prove or

independently check QSync and the QSyncCertInd base/step obligations (or use an alternate K.CertCov route), and no verified generic certificate generator is supplied here. In particular, the standard QSync package-totality claim presumes that a uniform instance-level DR1-DR4 proof schema/evidence has been established and can be instantiated afresh for each exact constructed prefix so K.DrainReach-Comp can combine it with the J/QSync construction's theorem-derived J.RegPhaseReach, attribution/history, and source-settling facts to discharge QC5, or that an equivalent direct QC5 proof is supplied; the package-existence induction does not manufacture the independent liveness evidence.

Discharge routes for A5-L (normative summary). Primary: an SDet-qualified CF-S certificate and K.2c from A5-S for the SDet-selected p_S^{det} , including K.CompleteS, K.RespCov, and K.RespClean.

Alternatives are structural, invariant, exploration, CF-5, or direct Policy-L artifacts. The header records the selected normalized instance and, when Policy S is used, the deterministic simulator/scheduler, arbitration, Policy-S issue semantics, normalization identity, SDet, response-monitor schema and finite bounds, completeness/fairness, waivers, and trust boundary. Generalized-drain qualification: in profile G1 the primary CF-S/K.2c route uses $Sys_K = Sys_ref$ and binds K.InFlightQual once for that model. In profile G2 the primary route proves A5L_Sys_K and invokes generalized K.Low.A5 under K.Low.ModalQual, which includes separate Sys_K/Sys_ref K.InFlightQual and global modal lowering correspondence. Alternative direct A5-L routes bind the K.InFlightQual appropriate to the model on which they prove the bad-cutpoint exclusion. No route may invoke the retired legacy snapshot DrainClosedNow correspondence.

Appendix K-P – Proofs Companion for the Serial-Reduction Route

Status. This companion collects the deterministic serial-reduction proof. SDet fixes the selected functional Policy-S semantics and one run artifact ρ_S^{det} at the outset; K.CompleteS separately establishes that this artifact is maximal, fair, and complete. Below write $\rho_S := \rho_S^{\text{det}}$. The soundness-critical proof retains K.Low.C, identity/dataflow lemmas, serial-prefix reachability, progress dominance, joint frontier freeze, and finite bounded-response extraction. The former missing-supply owner trichotomy, stuck-owner walk, EnvStop/DoneStop closure, and K.EnvOrd/K.Done collapse are retired from the trusted route and remain at most diagnostic. Optional serial replay is unchanged. Generalized-drain use of this companion is profile-dependent: $\text{Sys_K} = \text{Sys_ref}$ uses the identity route; a lowered Sys_K uses the Stage-2 modal lowering package and generalized K.Low.A5. The K.Low material below now contains the global path/escape/drain proof that replaces Revision-A's local snapshot-drain family. K-P KObs interface convention. For every source K-facing cutpoint $\sigma \in \text{KCutReach_Sys_K}(I)$ used below, write $K_ \sigma := \text{KObs_E}, w(\sigma) = \langle E, w, H_ \sigma, O_ \sigma \rangle$, where $O_ \sigma = \text{ResOffers_E}, w(\sigma)$. For $K = \langle E, w, H, O \rangle$, use $\text{owner}(o)$, $\text{item}(o)$, $\text{op}(o)$, $\text{boundary}(o)$, $\text{dir}(o)$, and $\text{epoch}(o)$ for the projections of the attributed residual offer tuple $o = \langle P, x, q, c, \text{dir}, e \rangle$ of Definition Core.18b. Write $\text{Proc}(I)$ for the finite process set of the selected instance. For a process set D define $\text{Offers_D}(K) := \{o \in O \mid \text{owner}(o) \in D\}$; $\text{FrontOffers_D}(K) := \{o \in \text{Offers_D}(K) \mid \text{item}(o) \in \text{FrontRec_owner}(o)(K)\}$; and $\text{DrainBag_D}(K)$ as the finite multiset of offers $o \in \text{Offers_D}(K)$ whose item lies outside $\text{FrontRec_owner}(o)(K)$. $\text{DrainCandidates_D}(K)$ is the legacy/fast-path submultiset of $\text{DrainBag_D}(K)$ whose item is ChannelEnabledRec at K . $\text{CmtRec_P}(K)$, $\text{PushOrdRec_c}(K)$, and $\text{PopOrdRec_c}(K)$ denote the committed-identity and per-endpoint ordinal projections of H . DrainCandidates_D is not the general definition of drain closure; generalized DrainClosedNow uses K.InFlightEscape. These names are pure snapshot selectors; at an operational cutpoint $K_ \sigma$ they agree with the corresponding Sys_K source facts by Lemmas J.KObs-Proc–J.KObs-WFG and K.KObs-Dead.

Operational and elaboration boundary. The graph and residual-offer arguments in this appendix range over the full E-typed frontier items and tuples, including explicit $\text{Sys_B}^{\text{elab}}$ staged, bundle, join, guard, and epoch-gate boundaries. When the proof uses an operation symbol f_P , g_P , or q , it abbreviates $\text{MsgOf_P}(x)$ only after the corresponding typed item x and boundary have been fixed; the item identity, boundary, direction, epoch, and owner remain part of the formal object. A committed explicit-boundary action unit is not automatically a process-facing source-operation commit: proof-internal $\text{Sys_B}^{\text{elab}}$ channel movement counts as the process-facing commit/return of q only when E and the K.CV/OpKey attribution identify that action unit as the boundary event that completes q . K.2b.P proves inclusion of explicit-boundary dominance action units; K.2b.P-Frozen, K.2b.SF, K.2b.Q, and K.Resp may infer a process-facing response only through that typed identification. Ordinary non-elaborated point-to-point boundaries are the identity special case. KObs deliberately omits historical Issue times and does not determine future issue legality. Core.Drain/K.Drain remains the continuation-level DrainDiscipline used to forbid replenishment past a blocked frontier. DrainCandidates remains a snapshot fast-path selector. Generalized DrainClosedNowRec is the KObs-pure finite-reachability predicate of Appendix K and requires the separately bound K.InFlightQual/J.InFlightPathSound operational justification. State-changing E4 movement inside the selected elaboration is represented by the applicable complete Q.2 proof-model in-flight action unit and is not duplicated as a source residual offer; only genuinely WFG-inert same-cycle normalization is handled through $K\epsilon/B4/B5$ -style ϵ reasoning.

K-P.1 Lowering and comparison foundations

Lemma K.Low.IdClose (unchanged ordinary blocking boundary; local simulation case). At any ordinary blocking boundary untouched by Λ_W , projection/lift is identity on the owning frontier item, BoundaryReq/ChannelEnabled, explicit channel state, ReleaseOwnerSets/WaitOwners, and every Q.2 in-

flight action/effect at that boundary. Hence blockedness, non-empty release support, release closure, and the corresponding root/action transition facts are preserved in both directions. This lemma is a local case for $K.Low.RootCorr$ and $K.Low.InFlightPathCorr$; it no longer states an independent modal $DrainClosedNow$ correspondence.

Lemma $K.Low.EpochClose$ (synchronization-to-epoch-gate local typing/action case). For each synchronization wait lowered to an E-declared explicit epoch-gate boundary, assume $Core.SyncCarrierWF_I$ and let the lowering certificate bind the Sys_K epoch-gate frontier item g_P to the corresponding Sys_ref synchronization carrier $sc=SyncCarrier_P(x,\sigma_ref)$, with the same owner, dynamic endpoint identity, $SyncTxn$ key/reset epoch, and declared participants. At corresponding pre-commit states, a disabled epoch-gate frontier projects to $SyncPending$ and conversely; when the synchronization is not ready, the gate's exact minimal internal $ReleaseOwnerSets$ family equals $SyncReleaseOwnerSets$, while a ready carrier has empty blocked-support family and exposes $SyncFire$. $SyncFire$ is the source-corresponding transaction; any residual $EpochGateAdvance$ is lowering-administrative and projects to stutter. The local certificate proves exact action/effect ordering for this rule: a source $SyncFire$ lifts to the declared Sys_K transaction segment followed by zero or more finite epoch-gate administrative actions, while every epoch-gate administrative action erases and frames source state. It also proves the local $RootCorr$ case at $LowAdminNF$ checkpoints. It does not assert step-for-step correspondence or an independent $DrainClosedNow$ theorem; those are derived only by the global Stage-2C path theorem.

Lemma $K.Low.GuardClose$ (aligned-NB guard local typing/action case). For each witness-selected aligned non-NB-CF dependency lowered to a fresh guard boundary g_s , the certificate fixes the selected source outcome/provenance, modeled supplier, guarded consumer, and exact guard realization. While the selected source obstruction remains infeasible, a disabled guard frontier projects to that active source guard-cause and has the non-empty singleton $\{\{supplier\}\}$ release family (or the declared equivalent atomic support); conversely the source obstruction lifts to that disabled guard frontier. When source-side supplier progress resolves the obstruction, the corresponding source action lifts first, after which any enabled guard transfer/retire is a finite lowering-administrative action erased by projection; in the default $B(g_s)=1$ form this may require a later real cycle and is not $K\epsilon$. The certificate proves exact local action/effect and $RootCorr$ facts at $LowAdminNF$ checkpoints. It does not independently prove modal drain closure; that conclusion is global because one escape path may traverse identity, guard, epoch, and ordinary staging cases before any candidate root changes.

Proof of Theorem $K.Low.InFlightPathCorr$ and its corollaries.

Fix an administratively normalized corresponding start pair. For downward projection, traverse the given finite Sys_K path exactly once. Maintain $LowStateCorr$ between each raw K position and the source state obtained by projecting the path prefix. $IdClose$ and every source-corresponding epoch/guard/ordinary Q.2 edge apply their exact local projection theorem and advance the source path; $LowAdminStepRec$ edges preserve the source state and are erased. No normalization or reordering is inserted into this downward proof. For escape reflection, choose the first literal D-root change. Since the start is $LowAdminNF$, an administrative action cannot have an initially enabled D-root effect. If the first syntactic root-removal/advance action is administrative, apply MQ4 to the most recent preceding source-corresponding action together with the entire intervening finite administrative suffix: whether the literal root change occurs on the first, second, or a later erased administrative edge, the projected source action must already change the matching $LowWaitRoot$ cause. Otherwise the source-corresponding root-changing action maps directly as the zero-length-suffix case. Thus a raw lowered escape always yields an extended source escape.

For upward lifting, induct over the finite Sys_ref path. $IdClose$ lifts directly. A source synchronization step uses $EpochClose$: lift $SyncFire$ and then append the finite epoch administrative suffix selected by

the lowering certificate; an aligned-NB supplier/outcome step uses GuardClose and appends its finite guard suffix. Other qualified source-corresponding Q.2 actions use their declared lift case.

LowAdminNormalize is applied only to this constructed lifted path, restoring LowStateCorr/LowAdminNF/RootCorr before the next source action. Exact source-visible histories and reset epochs agree by K.Low.C(i)–(ii) and the local effect theorems. This proves K.Low.InFlightPathCorr without a general admin/action commutation theorem.

For K.Low.InFlightEscapeCorr, use the same first-change argument downward and RootCorr at normalized lifted checkpoints upward. For K.Low.DeadRootBridge, fix PED with a current literal pending disabled message root. Until that ordinary root changes, Core.16/no-new-later-work, frontier minimality, and source/replay order prevent a source-later synchronization/NB lowering cause from becoming the current lowering-view root, while a source-earlier cause cannot remain unresolved after the current literal message frontier has been reached. Therefore LowWaitRoots_D cannot change first. This proves extended-escape iff ordinary-escape for a literal source Dead candidate. Negating the two escape equivalences proves K.Low.DrainClosedNowCorr. $\square$

Lemma K.Low.C (normalization correspondence) — restated with proof. Parts (i) and (ii) retain the Revision-A execution projection/lifting proof, strengthened only to append the finite LowAdminNormalize suffix selected by K.Low.ModalQual; every such suffix is source stutter, preserves the fixed witness/reset/environment interpretation, and terminates by the lowering rank. For part (iii), do not prove modal drain closure independently in IdClose/EpochClose/GuardClose. Those lemmas supply the three retained local fact families—blocked/root typing, non-empty release-support adequacy, and release closure—and the local action cases consumed by K.Low.RootCorr/K.Low.InFlightPathCorr. The global path theorem then yields K.Low.InFlightEscapeCorr. On a literal source Dead_K candidate, K.Low.DeadRootBridge converts the source extended-root escape predicate back to ordinary message-root InFlightEscape, and K.Low.DrainClosedNowCorr supplies the required bidirectional generalized drain fact.

Left to right, let D witness literal generalized Dead_K at a Sys_K K-facing cutpoint. First apply finite LowAdminNormalize. If any step of that administrative suffix changed D's literal root family, the suffix itself would witness InFlightEscape, contradicting the assumed drain closure; hence D's blocked root family is unchanged at the normalized endpoint. Now project each active literal lowering/root case to the source LowWaitRoots view; InFlightEscapeCorr projects drain closure to LowDrainClosedNow, and the local release facts project the exact support families/release closure. Hence the source contains a closed extended internal wait in the lowering sense. Right to left, begin with a non-empty closed extended source wait set and lift/normalize it. RootCorr and InFlightEscapeCorr give literal Sys_K blocked/drain facts at the normalized endpoint; the local support lemmas give non-empty release families and release closure. Apply K.ReleaseSCCReduction to obtain a literal Sys_K Dead_K witness subset. When the source premise is itself a literal Dead_K witness, K.Low.DeadRootBridge identifies its ordinary DrainClosedNow with the extended lowering view, yielding the exact fact used by K.Low.A5. Waiver pullback remains cutpoint-based and is unaffected by SCC subset selection. This refactors rather than discards Revision-A LowerClosure: only the drain family moved to the global theorem. $\square$

Definition K.Id (committed-identity state and serial frontier). For an execution prefix p and process P , $\text{Cmt}_P(p)$ is the set of dynamic source-operation identities of P that have process-facing committed in p , and $\text{Push}_c(p)$, $\text{Pop}_c(p)$ are the per-endpoint commit ordinals at channel c . At a terminal cutpoint σ with $K_\sigma = \langle E, w, H_\sigma, O_\sigma \rangle$, write $\text{CmtRec}_P(K_\sigma)$, $\text{PushOrdRec}_c(K_\sigma)$, and $\text{PopOrdRec}_c(K_\sigma)$ for the same facts reconstructed from H_σ . Because operations on a single message-passing interface issue and commit in source order (R4/E3 with FIFO transfer order and point-to-point ownership), the per-channel per-endpoint commit ordinal of any fixed dynamic operation is the same in every admissible execution

in which it commits; per-channel counts therefore identify committed transfers by identity through Lemma K.2b.P1. Across distinct interfaces, however, Policy L may commit a source-later operation while a source-earlier operation remains uncommitted, so $\text{Cmt_P}(p_L)$ need not be a source-order prefix, and a scalar commit count identifies neither the committed set nor the frontier under Policy L. Under Policy S, the process-facing commits of blocking Push/Pop operations are prefix-ordered in the serial blocking subsequence: within one source interval, a source-later blocking Push/Pop may not issue until every Policy-S-required source-earlier blocking Push/Pop has committed, while Core.SyncFence prevents interval advance with an applicable earlier blocking operation still uncommitted. Accordingly, in the K.BF serial fragment $\text{SerFront_P}(p_S)$ — P's serial frontier — is its source-minimal uncommitted blocking operation, with every Policy-S-required source-earlier blocking operation committed. Successful non-blocking operations may nevertheless commit while an earlier blocking operation remains pending when permitted by Core.ReachPrep , $R4/\text{Core.RegSeq}$, and the selected witness; failed non-blocking polls do not commit and are not members of this blocking serial frontier. All cross-policy progress comparisons identify dynamic operations through Definition K.OpKey and remain statements over operational prefixes, keyed committed-identity sets, and per-channel ordinals; the reconstructed KObs projections are used only when a terminal cutpoint is named. Scalar counts are consumed only as inequalities (Lemma K.2b.P). The earlier identification of a process's liberal frontier as operation $\text{prog_P} + 1$ was unsound and remains retired.

Definition K.CVPred and K.CV-WF (well-founded value/identity provenance). Fix a K.BF theorem instance I and its common OpKey_I universe of Definition K.OpKey. A K.CV provenance certificate supplies the common semantic fact-key universe CVFact_I used below. This definition supplies the CV2 certificate field of Definition K.CV; it does not assume K.CV as a prior premise. It contains at least $\text{OpFact}(\kappa)$ for $\kappa \in \text{OpKey_I}$, whose deterministic value is an optional typed operation descriptor: none when the fixed predecessor/control facts exclude that keyed occurrence, and $\text{some}(\text{desc})$ when κ is selected, with desc recording operation kind/site identity, typed boundary/direction, and any Push payload; $\text{ObsFact}(u)$, recording the value or status of each explicitly modeled observation that K.CV permits to influence a later operation; and $\text{XferFact}(c,n)$, the distinguished ObsFact for the payload of the n-th committed transfer at explicit boundary c. Fixed constants, fixed stimulus values, and witness-fixed choices are roots rather than recursive facts. For an admissible execution p, $\text{Occurs_p}(\kappa)$ means $\text{Realize_p}(\kappa)$ is defined; whenever $\text{Occurs_p}(\kappa)$, $\text{OpFact}(\kappa)=\text{some}(\text{desc})$ for that realization. The K.CV certificate supplies, for every non-root fact b, a finite predecessor set $\text{Pred_CV}(b)$ and a deterministic evaluator Eval_b . Occurrence closure is explicit: whenever a fact b occurs in an admissible execution, every $a \in \text{Pred_CV}(b)$ also occurs in that execution; and whenever the predecessor facts are available, Eval_b determines the same fact value from the fixed roots and those keyed predecessor values. For $\text{OpFact}(\kappa)$, this evaluator includes the control/branch/iteration selection needed to determine none versus $\text{some}(\text{desc})$, not merely the fields of an already-assumed occurrence. The certificate shall validate this provenance against $\text{Core.RegSeq}/\text{Core.ReachPrep}$ and the applicable explicit causal mechanisms; for $\text{XferFact}(c,n)$, it also validates the producer OpKey attribution and ordinal relation using point-to-point ownership, $R4/E3$, $E4/\text{FIFO}$ or rendezvous semantics, and the fixed elaboration. Define $a \prec_{\text{CV}} b$ iff $a \in \text{Pred_CV}(b)$. The same OpKey/CVFact keys, predecessor sets, and evaluators are bound to the fixed source, elaboration, stimulus, witness, and reset schema across the admissible executions compared by Lemmas K.2b.P0/P1; they are not reconstructed from incidental commit timing. The K.CV discharge shall additionally supply or derive K.CV-WF: $\text{WellFounded}(\prec_{\text{CV}})$. A checked CVRank into the natural numbers or another well-founded lexicographic order, with $a \prec_{\text{CV}} b$ implying $\text{CVRank}(a) < \text{CVRank}(b)$, is a sufficient certificate form. No $\prec_{\text{CV}}$ edge is introduced merely because two requests mutually enable a rendezvous, share a cycle, or are related by the broad Appendix-G causal-dependency graph $\rightarrow$. Thus a legal cycle in $\rightarrow$ does not imply a cycle in $\prec_{\text{CV}}$. Conversely, an actual circular value/control provenance

for which no K.CV-WF certificate exists is outside K.CV and therefore outside the K.2b/K.2c serial-reduction route.

Comparison-availability closure (normative K.CVPred serial-route clause). For every blocking OpFact(κ) consumed by Lemma K.2b.P0, K.2b.R, or K.2b.P, the K.CV certificate shall additionally establish cross-policy predecessor availability for the least-missing/first-overtake comparison. Fix Policy-S and Policy-L comparison prefixes in one K.DomResetScope DomEpoch, with κ reached on the Policy-S side. Once the Policy-L side contains, with the common keyed values, every source-earlier blocking operation that Policy S requires to have committed before κ may issue and every dominance-domain endpoint-transfer fact that must precede κ in the selected serial prefix, and no still-frozen Core.16 frontier of κ 's owning process that is key-earlier than κ prohibits the owner-local preparation, there exists a finite phase-legal Policy-L extension that preserves the frozen component and fixed DomEpoch, introduces no unmatched dominance-domain endpoint transfer, and contains every $a \in \text{Pred_CV}(\text{OpFact}(\kappa))$ that occurs in the Policy-S prefix with the same keyed value. The certificate records that extension as a finite FP4a-certified predecessor-materialization segment stack SegPred(κ): each non-transfer predecessor reach is represented by whole phase-bearing operational units, recursive predecessor segments have strictly smaller CVRank and are pushed before their caller resumes, and no primitive rendezvous or E-declared atomic boundary-action unit is split. SegPred(κ) is fixed from the K.CVPred certificate, the selected Policy-S comparison artifact, and the current finite comparison tuple; its existence and step-local legality do not assert that an arbitrary Policy-L continuation performs it. An XferFact predecessor may be discharged only by showing that it is already one of the matched earlier serial transfers. A non-transfer OpFact/ObsFact predecessor is discharged recursively along the checked CVRank using its certified Core.ReachPrep/observation reach rule; after occurrence is established, Lemma K.2b.P1 supplies value/identity equality. Recursive Core.16 reach-legality disposition (normative). Before any recursive non-transfer predecessor-reach extension is accepted, the K.CVPred certificate shall record one of the following dispositions for that extension. (a) If the predecessor is a blocking OpFact(κ_a) whose materialization performs source preparation/evaluation or another new launch, the comparison shall establish the same no-active-key-earlier condition used by K.CompareAdvanceLegal after the lower-ranked predecessor facts of κ_a are available. For an ordinary/non-elaborated or otherwise comparability-qualified singleton frontier, Lemma K.CompareAdvanceLegal-Ord may be cited pointwise only when, at the recursive comparison prefix, every blocking transfer that the selected Policy-S semantics requires to precede κ_a 's Issue is already present in the matched liberal transfer prefix with the required keyed identity/ordinal; none of those prerequisite transfers may be created by the predecessor-materialization extension itself. If that qualification test is not established, the pointwise lemma citation is unavailable and the certificate must instead supply independent owner/frontier Core.16 legality for the predecessor reach or reject the candidate. An explicit Sys_B^{elab} multi-frontier requires checked elaboration/order evidence. Only the reach/launch half is consumed when κ_a is materialized solely as a provenance predecessor. (b) If an ObsFact(u) reach rule consists only of observation, settling, or other actions that launch no new source-later work and issue no new message request, the reach rule shall certify that extension Core.16-inert. (c) If an ObsFact or other non-transfer predecessor reach rule requires source-control advance or another launch not represented by a blocking OpFact key covered by clause (a), its reach certificate shall supply the equivalent owner/frontier proof that no active Core.16 frontier is source/key-earlier than each launched operation. The adjective "phase-legal" is not a substitute for one of these explicit dispositions. If no applicable disposition can be proved, comparison availability is false for κ and the candidate is outside the K.CV/K.2b serial route. The recursion is well-founded because CVRank strictly decreases on Pred_CV edges. This clause supplies occurrence/reachability only; K.2b.P1 remains conditional on occurrence and does not by itself establish predecessor availability. The extension-preservation requirement is global over the frozen component, not merely owner-local: if a required non-transfer predecessor is produced by another process in D and

its certified reach/observation rule would have to advance or release that process's frozen frontier, issue work prohibited by Core.16/K.JointFreeze, change DomEpoch, or otherwise violate preservation of D, then the comparison-availability clause is false for that κ and the candidate is outside the K.CV/K.2b serial route. The owner-local Core.16 caveat therefore does not authorize materialization through another frozen process. K.CVPred owns this recursive predecessor-availability and SegPred certificate obligation; K.CompareAdvanceLegal remains the named coverage condition for the main K.2b.P0/K.2b.R comparison key, while Lemma K.CompareAdvanceLegal-Ord or equivalent elaboration-specific no-key-earlier evidence may be reused pointwise inside the K.CVPred recursion as specified above. The dependency direction is deliberate: K.CVPred certifies that SegPred exists and that each unit has the required local legality disposition independently of the later K.2b.E continuation; K.SegmentAdm is the construction-side proof that the designated stack remains admitted on the continuation being built, and Core.FairPick FP4a is what realizes it. If K.SegmentAdm cannot be established for the fixed choices, comparison availability is unusable for that candidate and the candidate is rejected rather than treating the existential segment as executed.

K-P.2 Frozen-continuation construction

Lemma K.2b.E (fair complete frozen extension)—Proof. At every finite prefix constructed below while the selected f_P remain pending, K.ResetEpoch.Continue forbids any RESET whose scope intersects K.DomResetScope as the next transition; RW2 therefore never clears the selected requests on this branch. K.TerminalDisposition.Continue makes every such frozen-prefix final state nonterminal, so the construction has no finite declared-terminal escape. Lemma K.BF-Persist supplies the remaining interruption-completeness fact by applying the exhaustive Core.IC blocking lifecycle to the selected WFG-relevant K.BF blockers. Thus every persistence and dynamic-identity claim below remains within the fixed DomEpoch for K.DomResetScope and the same selected frozen request instances; RW5-framed resets disjoint from that scope remain permitted. Start from the finite Core.LAdm-admissible prefix p_L and its drain-closed component D, retaining the persistent requests f_P already present at σ_L . First discharge K.EnvMF. Under StandardEnv/B1-avail, Lemma K.EnvX-Auto makes every disabled environment-facing co-frontier permanently unavailable. Under a timed/reactive or otherwise nondefault environment, K.EnvMF certifies that each $P \in D$ has exactly one internal frontier item and that the candidate satisfies Lemma K.JointFreeze-Ord's single-release-support premises, so no environment-facing co-frontier or non-ordinary release alternative can later deactivate Core.16 and release the component. Under StandardEnv/B1-avail, invoke Lemma K.JointFreeze-Ord when its premises hold; every other admitted release/frontier structure uses K.JointFreeze, whose JF1/JF2 clauses cover releasing issues and permitted drain/elaboration actions without flattening the ReleaseOwnerSets family. Hence no releasing operation is an FP1 issue candidate and no permitted completion action can dissolve D merely through the multi-frontier incomparability that Core.16 alone does not order. Already-issued source-later work is finite at the cutpoint, may drain under the selected progress semantics, and cannot replenish the frozen set. Each frozen f_P itself remains ChannelDisabled, so it is never an FP2/B2 commit candidate. Define the base frozen-comparison invariant Inv for a finite extension prefix to mean: (I1) the same D and selected f_P remain pending and attributed to the same dynamic instances; (I2) the K.DomResetScope DomEpoch is unchanged; (I3) Core.16 remains active wherever required; (I4) no prohibited replenishing source-later work has been launched; and (I5) the prefix remains Core.LAdm-legal with the fixed witness/environment/arbitration choices. Define $Inv+ := Inv \wedge I6$, where (I6) is the construction-side segment-bookkeeping condition: if a top-level certified segment stack is active, it is the sticky stack designated by the most recent no-active-stack evaluation of K.SegmentPlan, possibly extended only by lower-CVRank nested pushes required by its fixed certificate; its current top satisfies K.SegmentAdm relative to Inv, and every suspended caller retains its certified remaining suffix. K.SegmentPlan is not re-evaluated to replace an active top-level obligation; only after

that obligation completes may the plan inspect the successor prefix for the next least eligible missing T. K.SegmentPlan, the segment-capable Core.FairPick witness, and the certificate-determined SegTree/descriptor/rank data are fixed before dependent choice. This construction proves completion for every stack actually emitted by SegReq; it deliberately makes no forward claim that an arbitrary target selected only later by the K.2b.P reductio was emitted. That consequence is proved separately by Lemma K.SegmentPlan-EventuallyTarget over the completed continuation. The initial prefix satisfies Inv and has no active segment, so I6 is vacuous and Inv+ holds. Assume a constructed prefix satisfies Inv+. If no FP4a segment is active, Core.FairPick selects phase-legal issue and boundary work permitted by the operational semantics and not excluded by the applicable K.JointFreeze-Ord/K.JointFreeze clauses; membership of an owner in D does not by itself freeze every operation of that owner. If no top-level FP4a segment is active and K.SegmentPlan emits a certified segment stack, deterministic K.SegNorm evaluates the fixed tree at the current prefix; K.CompareFact-Mono permits already-satisfied historical predecessor subtrees to be skipped permanently, while K.SegmentAdm relative to Inv and FP4a make the next unresolved whole unit the relevant constructive obligation in its owning phase. An unresolved lower-CVRank predecessor is pushed and discharged first, and suspension occurs only between whole K.SerialDomOrder/declared atomic units. K.JointFreeze-Ord or K.JointFreeze exclude any releasing new launch/issue, K.EnvMF excludes an environment co-frontier escape, and Lemma K.BF-Persist plus nonwithdrawal retain each frozen request. The recorded K.CVPred dispositions and K.CompareAdvanceLegal establish the next segment unit's Core.16 legality; SA3-SA5 ensure that realizing that unit and any mandatory typed cycle completion preserve the frozen component, fixed DomEpoch, and admitted remaining stack. If those facts fail for a fixed scheduler/environment choice, K.SegmentAdm is false and the candidate is outside K.2b rather than an execution branch of this proof. K.ResetEpoch.Continue excludes an affecting RESET at every intermediate frozen prefix, and K.TerminalDisposition.Continue makes the resulting frozen-prefix state nonterminal. The two-stage construction of Lemma Core.FairCycleDovetail therefore applies: its first Core.CycleChoiceExt call obtains an actual finite L2 settle path after the selected L1 segment; BoundaryPick is evaluated only at that settled state; its second Core.CycleChoiceExt call produces a typed CycleResult containing the selected fair/segment work. Every additional action in the completion is phase-legal and, by Core.Drain/K.Drain together with Lemma K.JointFreeze-Ord on its eligible branch or K.JointFreeze on the other admitted StandardEnv release-structure branch, cannot replenish or release D; by K.SegmentAdm it also preserves any active certified segment suffix. Hence the successor prefix again satisfies Inv+. Iterate these typed Inv+-preserving CycleResult successors. Because every Inv+ state is nonterminal on the nested Continue branch, dependent choice yields an infinite CompleteCycle continuation $\bar{p}_L$. FP4 of Core.FairPick proves Core.IssueFair and B2 on $\bar{p}_L$; for every stack actually emitted by the fixed SegReq rule, I6/K.SegmentAdm plus FP4a and Lemma K.SegWork-Complete prove finite completion; and FP5 supplies the B3 channel-progress cases; the separately assumed B5 closure property applies to this operational continuation. If a selected cycle has no Σ commit, its typed completion is a Core.SilentCycle rather than an ϵ /identity step, and the EnvStep component of CycleResult advances the time-aware environment. Thus $\bar{p}_L \in \text{Exec}_L^{\text{adm}}(I)$, extends p_L , preserves every f_P pending with the same dominance-scope DomEpoch, and carries the construction-side segment-admittedness/history interface later consumed by K.SegmentPlan-EventuallyTarget and K.2b.P0/R/P. K.ResetEpoch.DirectFail and K.TerminalDisposition.DirectFail do not invoke this construction; each is discharged by its independently certified direct K.RespFail branch. $\square$

K-P.3 Deterministic dominance and ReachProgress

Lemma K.2b.P1 (keyed dataflow determinacy) — restated with proof. In the blocking-only fragment, for the fixed instance, stimulus, witness choices, and OpKey_I universe: for every $\kappa \in \text{OpKey}_I$, any two admissible executions—of either policy—in which Occurs $_{\rho}(\kappa)$ holds realize κ with the same operation

kind/site identity, typed boundary/direction, and any K.CV-determined Push payload; and for every channel c and ordinal n , the payload of the n -th committed transfer at c is the same in every admissible execution in which that transfer fact occurs. Proof. Strengthen the induction claim to all facts in CVFact_I and perform well-founded induction on $\prec_{\text{CV}}$ using K.CV-WF ; this induction does not use the broad causal-dependency graph $\rightarrow$. Fix a fact b and assume the execution-independent equality claim for every $a \prec_{\text{CV}} b$. By provenance adequacy in Definition K.CVPred, every admissible execution in which b occurs evaluates b with the same deterministic Eval_b from the same fixed roots and the same keyed predecessor facts $\text{Pred}_{\text{CV}}(b)$. Occurrence closure supplies an occurrence of every $a \in \text{Pred}_{\text{CV}}(b)$ in each such execution; each predecessor has smaller $\prec_{\text{CV}}$ rank and therefore has the same value/identity by the induction hypothesis. The fixed roots are identical by the theorem-instance stimulus/witness binding. Hence b has the same value/identity in every execution in which it occurs. For $\text{OpFact}(\kappa)$, this yields the same typed realization fields and any Push payload for the common operation key κ ; for $\text{XferFact}(c, n)$, the certified producer-key attribution plus point-to-point ownership, R4/E3, and E4/FIFO or rendezvous ordinal semantics make the n -th transfer payload the deterministic value of its lower-ranked producer/provenance facts. Other K.CV observation kinds are handled by their explicitly modeled deterministic Eval_b equations. Same-cycle rendezvous request coincidence is an enablement condition only and is not a $\prec_{\text{CV}}$ value predecessor, so it creates no induction cycle. $\square$ This lemma is deliberately conditional on occurrence; existence/reachability of the corresponding operation in the other execution is the separate content of Lemma K.2b.P0. At any named terminal cutpoint, the determined committed keys, payload ordinals, and process-local replay facts are exactly the corresponding projections of H in its KObs record; this observation does not replace the operational induction.

Lemma K.2b.P0 (keyed next-operation occurrence and reachability) — Statement. Fix a common key $\kappa \in \text{OpKey}_I$ owned by process P that is reached in the selected Policy-S execution; fix the candidate component D , its K.2b.E activation cutpoint σ_L and selected $\{f_P\}$; and fix Policy-S/Policy-L comparison prefixes π_S and π_L within the same $\text{K.DomResetScope DomEpoch}$ for that candidate. The K.CVPred comparison-availability clause first supplies the certified predecessor-materialization stack $\text{SegPred}(\kappa)$, whose completion preserves the fixed DomEpoch and frozen component, adds no unmatched K.SerialDomOrder dominance transfer, and yields a comparison prefix in which every K.CV predecessor fact required by $\text{OpFact}(\kappa)$ is present with the same keyed value as in π_S . Using that completed predecessor prefix, assume every source-earlier key that Policy S requires to complete before κ may issue has committed as required. Before taking any further $\text{Core.ReachPrep/source-evaluation}$ step to reach κ , assume the applicable $\text{K.CompareAdvanceLegal}_I(D, \sigma_L, \{f_P\})$ coverage for κ on this comparison tuple; for ordinary/singleton frontiers Lemma K.CompareAdvanceLegal-Ord supplies it, while any explicit $\text{Sys_B}^{\text{elab}}$ multi-frontier owner consumed by K.2b.P0 requires checked elaboration-specific evidence regardless of whether $P \in D$. Then either κ is already realized/reached or there is a second finite FP4a-certified segment $\text{SegReach}(\kappa)$, within the same DomEpoch , that reaches $q_L := \text{Realize}(\kappa)$ and preserves the frozen component and K.SerialDomOrder matched-prefix invariant. The realized q_L has the operation kind/site, typed boundary/direction, and payload determined by $\text{OpFact}(\kappa)$. This lemma certifies the finite $\text{SegPred}(\kappa)/\text{SegReach}(\kappa)$ descriptors and their semantic effect; it does not by itself assert that the fixed K.SegmentPlan eventually emits them on an arbitrary already-constructed continuation. For the K.2b.P least-missing use, actual same-continuation target readiness is supplied by Lemma K.SegmentPlan-EventuallyTarget below: its ET1 branch emits/completes these P0 descriptors on $\bar{p}_L$, while ET2 observes that the required target roots are already reached/asserted and no segment realization is needed. This lemma establishes occurrence/reachability only; request assertion and persistence are supplied by K.2b.R.

Proof. Lemma K.2b.P1 alone cannot prove this conclusion because P1 is conditional on occurrence. The K.CVPred comparison-availability certificate provides $\text{SegPred}(\kappa)$, and well-founded CVRank recursion

plus the recorded dispositions (a)/(b)/(c) establish the step-local legality and preservation facts for every unit of that stack. Completion of SegPred makes every input of Eval_OpFact(κ) available in the liberal comparison prefix. Because the selected Policy-S execution reaches κ , its evaluator value is some(desc); determinism of the common K.CV evaluator therefore gives the same some(desc) value from the equal keyed predecessors on the liberal side, establishing that the same keyed occurrence is control-selected rather than an untaken-branch artifact. Definition K.OpKey supplies that potential process-local occurrence independently of execution, and realization preserves source order. Before advancing the liberal preparation cursor for κ itself, apply K.CompareAdvanceLegal. If Core.16 is active, its conclusion gives the direct Core.16(2) item-order/no-predecessor fact for the κ launch target; therefore the finite dependency-independent source-evaluation/preparation needed to reach κ is not prohibited as work launched past a blocked point. Core.ReachPrep and K.BF bounded local progress establish the finiteness and operational shape of SegReach(κ), while Core.CycleChoiceExt supplies a typed completion whenever the segment crosses a source-cycle boundary. This proves existence of the two certified finite segments. This proves existence, typing, and local legality of the finite certified descriptors. P0 itself stops there: it does not infer that SegReq eventually selects this κ after the continuation has been constructed. In a K.2b.P least-missing application, Lemma K.SegmentPlan-EventuallyTarget combines K.SerialDomPredFinite, K.CompareFact-Mono, K.SegWork-Complete, the sticky SegReq rule, and the completed K.2b.E continuation to obtain the required ET1/ET2 target-readiness branch. If an emitted segment cannot satisfy K.SegmentAdm under the fixed scheduler/environment/freeze choices, the candidate is outside the serial route; existential reachability is never substituted for execution on $\bar{p}_L$. The subsequent request Issue is not part of this lemma; K.2b.R consumes the same K.CompareAdvanceLegal fact for that second Core.16(2) obligation. $\square$

Lemma K.SegmentPlan-EventuallyTarget (least-missing target readiness on the completed continuation). Assume the K.2b.E Continue branch has produced the fixed infinite continuation $\bar{p}_L$ with Inv+, the fixed K.SegmentPlan, K.SegmentAdm, K.CompareFact-Mono, and K.SegWork/K.SegWork-Complete. Let T be an action unit occurring in the selected Policy-S K.SerialDomOrder dominance footprint and assume explicitly that T does not occur in $\bar{p}_L$ (equivalently $T \notin \text{Units_SD}(\bar{p}_L)$). Assume every $U <_{\text{SD}} T$ does occur in $\bar{p}_L$, as in the K.2b.P least-missing reductio. Then there exists a finite prefix $\eta \leq \bar{p}_L$, after any previously active sticky top-level segment has completed, at which one of exactly two target-readiness cases holds. (ET1) The T-related endpoint/root preparation is still missing and the no-active-stack SegReq evaluation emits the fixed T-level SegPred/SegReach descriptor (or indivisible finite endpoint-family descriptor); I6/K.SegmentAdm, K.SegWork-Complete, and FP4a complete that emitted stack on the same $\bar{p}_L$ and leave the required liberal realization/root reached for K.2b.R. (ET2) The required T-related endpoint roots are already reached or asserted at η , so SegmentPlan correctly emits no segment; the proof passes directly to K.2b.R, which supplies any still-missing Issue/assertion under its fairness/legality hypotheses and then uses nonwithdrawal/K.BF-Persist for persistence. Proof. By Lemma K.SerialDomPredFinite, $\text{Pred_SD}(T)$ is finite. By the least-missing hypothesis every member occurs somewhere in $\bar{p}_L$, so choose one finite prefix η_0 containing all of them. Lemma K.CompareFact-Mono makes those matched units and any already-materialized lower-ranked K.CV predecessor facts permanent on later prefixes. If a sticky top-level segment is active at η_0 , I6 supplies its admittedness and Lemma K.SegWork-Complete plus FP4a finishes it after finitely many whole-unit realizations. At the first subsequent no-active-stack prefix, T is still absent by hypothesis, no $<_{\text{SD}}$ -earlier unit has become missing, and no newly eligible earlier target can displace T. The deterministic SegReq rule therefore either emits T's fixed descriptor because target roots remain missing, giving ET1, or observes that those roots are already reached/asserted, giving ET2. This is a theorem about the completed $\bar{p}_L$ and is not an Inv/Inv+ conjunct or a premise used to construct that continuation. $\square$

Lemma K.2b.R (serial-prefix request reachability) — Statement. Under K.ResetEpoch.Continue and K.TerminalDisposition.Continue, let D be the frozen component associated with the infinite fair

continuation $\bar{p}_L$ of Lemma K.2b.E, and let q_S be a blocking endpoint operation of process P reached by the selected Policy-S execution, with common key $\kappa := \text{Key}(q_S)$. By the definition of $K.\text{DomResetScope}$, P and q_S 's explicit boundary belong to the dominance scope whenever κ is consumed by this comparison; let $e_{\{P,\kappa\}}$ denote the corresponding endpoint/boundary reset epoch within the fixed DomEpoch . Fix Policy-S and Policy-L comparison prefixes π_S and π_L immediately before a first alleged Policy-S overtake, with π_L a prefix of $\bar{p}_L$. Assume every source-earlier key that Policy S requires before κ , every $K.\text{SerialDomOrder}$ action unit earlier than κ in the $K.2b.P$ dominance domain, and every $K.CV$ predecessor fact required by $\text{OpFact}(\kappa)$ are present in π_L with the keyed identities/values supplied by the strengthened $K.2b.P$ transfer/provenance invariant; its $K.CV$ predecessor portion is obtained from the $K.CVPred$ comparison-availability closure. For a least-missing $K.2b.P$ application, assume additionally the $K.\text{SegmentPlan-EventuallyTarget}$ target-readiness conclusion for the κ/T endpoint family: ET1 has completed the required $\text{SegPred}/\text{SegReach}$ stack on $\bar{p}_L$ or ET2 establishes that the required liberal root is already reached/asserted. Assume also the applicable $K.\text{CompareAdvanceLegal}_I(D, \sigma_L, \{f_P\})$ coverage for this comparison tuple; for ordinary/singleton frontiers Lemma $K.\text{CompareAdvanceLegal-Ord}$ supplies it, while any explicit $\text{Sys}_B^{\text{elab}}$ multi-frontier owner consumed by $K.2b.P0/K.2b.R$ requires checked elaboration-specific evidence regardless of whether $P \in D$. If κ 's liberal realization has not committed in π_L , then $\bar{p}_L$ has a finite extension of π_L , with the same DomEpoch and endpoint epoch, in which the attributed endpoint request for $q_L := \text{Realize}(\kappa)$ is asserted. On this $K.BF$ reset-free, terminal-free comparison branch, once asserted, Lemma $K.BF\text{-Persist}$ retains that request as the same pending keyed dynamic request until q_L commits. No reachability or nonwithdrawal claim crosses a dominance-affecting RESET or a declared-terminal branch; $K.\text{ResetEpoch.DirectFail}$ and $K.\text{TerminalDisposition.DirectFail}$ are handled directly through $K.\text{RespFail}$.

Proof. Under Policy S, every source-earlier blocking operation of P required by the serial discipline has committed before q_S 's request may issue. By the comparison hypothesis, those committed operation keys, returned values, earlier endpoint-transfer facts, and the remaining $K.CV$ predecessors of $\text{OpFact}(\kappa)$ also occur with the same keyed values in the liberal prefix. For the least-missing use, apply Lemma $K.\text{SegmentPlan-EventuallyTarget}$ rather than inferring plan emission from $P0$. In ET1, the fixed $P0$ $\text{SegPred}/\text{SegReach}$ descriptors have been emitted and completed on the same $\bar{p}_L$ by $I6/K.\text{SegmentAdm}$, $K.\text{SegWork-Complete}$, and $FP4a$, so $q_L := \text{Realize}(\kappa)$ is reached. In ET2, q_L or its source-attributed request root is already reached/asserted and no segment realization is required. Lemma $K.2b.P1/K.CV$ supplies its common typed identity fields and payload. Because P and q_L 's explicit boundary belong to $K.\text{DomResetScope}$ and this comparison remains within the fixed DomEpoch , q_L is in the corresponding endpoint/boundary reset epoch $e_{\{P,\kappa\}}$. If π_L has already issued q_L , nonwithdrawal gives the conclusion. Otherwise reuse $K.\text{CompareAdvanceLegal}$, whose reach/launch half was already consumed by $K.2b.P0$ and whose Issue/assertion half applies now. If P 's applicable active message frontier is ordinary/non-elaborated or otherwise certified singleton, Lemma $K.\text{CompareAdvanceLegal-Ord}$ derives the required Core.16 compatibility from the least-missing/first-overtake invariant, $E3/\text{source-order}$ prefix closure, and the fact that any genuinely source-earlier required blocking transfer is already matched on the liberal side, with Lemma $K.\text{KeyItemOrderBridge}$ providing the ordinary/certified-singleton key-to-item-order step. This argument does not depend on $P \in D$. If an explicit $\text{Sys}_B^{\text{elab}}$ package gives P a multi-frontier on the comparison prefix, the explicit $K.\text{CompareAdvanceLegal}$ certificate supplies the corresponding result, again regardless of membership in D . Thus, whenever Core.16 is active for P , the direct no-item-predecessor/ Core.16 legality conclusion holds for q_L , and the specific no-new-Issue clause of $\text{Core.16}(2)$ does not suppress q_L . This is the second half of the same comparison-advance fact that justified $P0$'s preparation launch. The former shortcut " $P \notin D$, so no freeze restriction applies" is not used: a process outside D may itself acquire a fully disabled frontier and is covered by the same comparison-advance interface. Hence q_L is either already asserted or remains legally IssueEnabled under Core.16 at every applicable frozen comparison

prefix. Policy L admits the issue because it admits every Policy-S issue and assertion of a persistent blocking request does not require the channel transfer itself already to be enabled. At every finite comparison prefix on which the relevant frozen requests remain pending, P and q_L 's explicit boundary are members of $K.DomResetScope$ by the comparison-domain definition, so $K.ResetEpoch.Continue$ forbids a RESET affecting either as the next transition and preserves the same DomEpoch and $e_{\{P,\kappa\}}$; $K.TerminalDisposition.Continue$ makes the corresponding frozen-prefix final state nonterminal. Because q_L remains legally IssueEnabled at every applicable L1 opportunity, Core.IssueFair therefore schedules the assertion after finitely many such opportunities; the infinite fair continuation cannot postpone it forever. Lemma K.BF-Persist together with nonwithdrawal then retains the attributed request as the same pending keyed dynamic instance until q_L commits: the exhaustive Core.IC lifecycle admits no independent nonterminal withdrawal/cancellation transition, $K.ResetEpoch.Continue$ excludes an affecting RESET on the comparison prefix, and $K.TerminalDisposition.Continue$ excludes a declared-terminal escape. RW2 would end the realized dynamic instance at an affecting RESET, which is why the conclusion is expressly DomEpoch- and endpoint-epoch-local and $K.ResetEpoch.DirectFail$ is handled separately. $\square$

Lemma K.2b.P (deterministic serial progress dominance) — Statement. Under $K.ResetEpoch.Continue$ and $K.TerminalDisposition.Continue$, fix a Policy-L prefix and an infinite fair reset-free, terminal-free continuation $\bar{p}_L$. Assume explicitly the K.2b.E frozen-continuation invariant for this same candidate D, selected $\{f_P\}$, and $K.DomResetScope$: $\bar{p}_L \in Exec_L^{adm(I)}$ extends p_L ; throughout every finite prefix of $\bar{p}_L$ each selected f_P remains the same asserted, attributed, pending dynamic instance; and the $K.DomResetScope$ DomEpoch remains fixed. Lemma K.2b.E is the standard producer of exactly this invariant, but provenance from that lemma is not a substitute for the hypothesis in this statement. Let $p_{S^{det}}$ be the Policy-S execution selected by SDet, with separate $K.CompleteS$ evidence establishing maximality, fairness, and completeness. Every $K.SerialDomOrder$ explicit-boundary action-unit identity and payload committed in $p_{S^{det}}$ whose owning endpoints/boundary belong to $K.DomResetScope$ for this candidate, while the comparison remains in the fixed DomEpoch, also occurs in $\bar{p}_L$. For an ordinary process-facing boundary this action unit is the source-operation commit itself; for Sys_B^{elab} proof-internal movement the conclusion remains an internal explicit-boundary inclusion unless $E/K.CV$ separately identifies that unit as the process-facing commit/return of the source operation. The conclusion is transfer inclusion over the dominance comparison footprint; it does not assert identical cycle timing, the same global interleaving, or cross-epoch inclusion for reset activity framed outside that scope.

Proof. Use Definition $K.SerialDomOrder$ and choose the $<_{SD}$ -least Policy-S dominance-domain action unit T that is missing from $\bar{p}_L$. Let c be the explicit E-typed boundary named by T. The case split below is exhaustive for the $K.SerialDomOrder$ units consumed by K.2b.P: ordinary Push/Pop units use the buffered or rendezvous cases; coupling-sensitive explicit boundaries use the qualified elaborated case; synchronization waits and aligned non-blocking dependencies consumed by the serial route have already been represented in Sys_K by $K.Low$ as explicit epoch-gate/guard channel action units and therefore enter the same explicit-boundary cases. Declared same-cycle atomic groups remain indivisible throughout. Every $<_{SD}$ -earlier action unit occurs in $\bar{p}_L$ by minimality. Apply Lemma $K.SegmentPlan-EventuallyTarget$ using the explicit hypothesis $T \notin Units_SD(\bar{p}_L)$. In ET1 the target-related certified SegPred/SegReach stack is emitted and completed on this same continuation; in ET2 the required roots are already reached/asserted. In either branch K.2b.R performs the explicit handoff from reached roots to Issue/assertion when necessary and then supplies persistent source-attributed request roots through nonwithdrawal/ $K.BF-Persist$. $K.CVPred/K.2b.P0/P1$ supply the same required operation identities and values. A declared atomic action unit is handled as one indivisible T and is never split by this induction. One specialization cuts across the channel-kind cases below: if T contains the transfer corresponding to a selected frozen frontier item f_P , minimality still supplies the required earlier transfers and support

facts, but any forced occurrence of that indivisible unit closes directly against the K.2b.E freeze invariant. If c is coupling-sensitive under $\text{Sys_B}^{\text{elab}}$, use the strengthened Q.1(l) serial-route handshake-support/persistence property: E declares the finite channel-state/history and request roots on which T 's settled MirrorAvail depends; those stored/history supports are $<_{\text{SD}}$ -earlier than T or belong to the same declared atomic unit; and the same persistent roots are present on the liberal side. Deterministic settling gives $\text{ChannelEnabled}(T)$, while $\text{CouplerPersistence}$ proves that every subsequently permitted action before T commits on any fixed-DomEpoch continuation satisfying the base frozen invariant Inv preserves the declared support facts or directly preserves $\text{ChannelEnabled}(T)$. Therefore the B2 continuous-enablement antecedent, rather than merely pointwise enablement, holds. If the boundary is ordinary primitive, use the explicit buffered/rendezvous persistence cases below. This separates the hardware-specific persistence proof from the generic least-missing induction and does not treat nonfull/nonempty occupancy as sufficient at a coupled boundary.

Primitive buffered Pop case. Assume $B(c) > 0$, c is an uncoupled primitive buffered boundary, and T is the n -th Pop_c transfer. Non-fall-through E4 implies that at least n Push_c transfers precede the Policy-S Pop. By $<_{\text{SD}}$ minimality those Pushes occur in $\bar{p}_L$. If the n -th Pop is still absent, $\text{PopOrd}_c < n$ while $\text{PushOrd}_c \geq n$, hence $\text{occ}_c > 0$. By K.2b.R, the target Pop's source-attributed request has been asserted and K.BF-Persist/nonwithdrawal retain that same pending request on the fixed DomEpoch branch; Core.12 therefore makes that persistent Pop request ChannelEnabled . This enablement is persistent, not merely pointwise: before the n -th Pop commits, later Pushes cannot remove an already-present token and any additional Pop that could reduce the available count would itself be the n -th-or-later same-endpoint Pop and is excluded by endpoint order/least-missing choice. Equivalently, with the target n -th Pop absent, the inequality $\text{PushOrd}_c - \text{PopOrd}_c > 0$ remains true at every applicable settled boundary opportunity until T commits. Thus the B2 continuous-enablement antecedent holds, and B2/B3 force T , contradicting minimality.

Primitive buffered Push case. Assume $B(c) > 0$, c is an uncoupled primitive buffered boundary, and T is the n -th Push_c transfer. The Policy-S commit implies the ordinary E4 nonfull inequality immediately before T . Because T is the $<_{\text{SD}}$ -least missing dominance-domain unit and Push_c transfers commit in same-interface/source order, the liberal side contains exactly the first $n-1$ Push_c transfers and cannot contain the n -th or any later Push_c transfer; hence $\text{PushOrd}_c(\bar{p}_L) = n-1$ at the comparison point. The least-missing argument also places in $\bar{p}_L$ every Pop_c transfer that is $<_{\text{SD}}$ -earlier than T , so $\text{PopOrd}_c(\bar{p}_L)$ is at least the corresponding Policy-S pre- T Pop count. Therefore $\text{occ}_c = (n-1) - \text{PopOrd}_c$ is no greater than the Policy-S occupancy immediately before T and is strictly less than $B(c)$. By K.2b.R, the target Push's source-attributed request has been asserted and K.BF-Persist/nonwithdrawal retain that same pending request on the fixed DomEpoch branch; Core.12 therefore makes that persistent Push request ChannelEnabled . The nonfull condition then remains true until T commits: with the n -th Push absent, no later same-endpoint Push can occur, while any Pop only decreases occupancy. Hence the B2 continuous-enablement antecedent holds, and B2/B3 force T , again contradicting minimality.

Primitive rendezvous case. Assume $B(c)=0$ and c is an uncoupled primitive rendezvous boundary. Apply K.2b.R to both endpoint operations. After the later assertion point both attributed requests remain present by nonwithdrawal until the atomic rendezvous commits; no independent endpoint transfer exists that can consume only one side. Core.13 therefore makes the rendezvous ChannelEnabled at every applicable settled opportunity with payload fixed by K.2b.P1/K.CV. This request-pair persistence supplies the B2 continuous-enablement antecedent, so B2/B3 force the indivisible rendezvous unit.

Coupling-sensitive elaborated case. For a $\text{Sys_B}^{\text{elab}}$ boundary whose complementary ready/valid value is produced by declared coupler logic, the strengthened Q.1(I) property replaces the primitive occupancy/peer-request inference. After the $\leq_{\text{SD}}$ least-missing induction has matched the required earlier support actions and K.2b.R has supplied the persistent request roots, deterministic settling makes T ChannelEnabled exactly as in the Policy-S support state. Q.1(I) CouplerPersistence then proves that on every Core.LAdm-admissible fixed-DomEpoch continuation satisfying the base K.2b.E frozen invariant Inv, every permitted action before T commits preserves the required support facts or directly preserves ChannelEnabled(T). Thus T remains continuously B2-enabled; the ordinary B2/B3 discharge for the primitive storage/rendezvous actions composing the qualified stateless template forces T. Hence T also occurs in $\bar{p}_L$, a contradiction.

Conclusion. Every possible type of least-missing K.SerialDomOrder action unit in the comparison footprint yields a contradiction. Therefore every explicit-boundary dominance action unit committed by p_S^{det} in the K.DomResetScope dominance domain while DomEpoch is fixed occurs, with the same dynamic identity, channel ordinal where applicable, and payload/value facts, in $\bar{p}_L$.

Synchronization/epoch-gate/guard cases are included through the K.Low explicit-boundary representation described above. This is one-way action-unit inclusion only; it does not by itself reclassify a proof-internal elaborated transfer as a process-facing commit, and $\bar{p}_L$ may contain additional liberal transfers/actions and may order independent units differently. $\square$

Corollary K.2b.P-Frozen (frozen-transfer no-overtake consequence). Under the hypotheses of Lemmas K.2b.E and K.2b.P for the same candidate D, selected $\{f_P\}$, K.DomResetScope/DomEpoch, and the same frozen continuation $\bar{p}_L$, with the explicit K.2b.E frozen-continuation invariant passed as the K.2b.P hypothesis, fix $P \in D$. Let $T_{\{f_P\}}$ denote the K.SerialDomOrder explicit-boundary action unit that the E/K.CV attribution identifies as the process-facing commit/return of the selected blocking frontier operation f_P . Such an identification is required here; an unrelated proof-internal $\text{Sys_B}^{\text{elab}}$ transfer is not sufficient. Then p_S^{det} cannot commit $T_{\{f_P\}}$ while the comparison remains in the fixed DomEpoch. Within the same source interval, Policy S therefore cannot issue a source-later blocking Push/Pop before f_P process-facing commits; Core.SyncFence likewise forbids advancing through a succeeding selected synchronization while an applicable f_P remains in that synchronization's blocking prefix. The corollary makes no claim that every source-later non-blocking operation is withheld. Proof. Suppose p_S^{det} commits $T_{\{f_P\}}$. Lemma K.2b.P, applied to the same $\bar{p}_L$ supplied by K.2b.E, forces that same typed explicit-boundary action unit to occur in $\bar{p}_L$. By the defining E/K.CV identification of $T_{\{f_P\}}$, that occurrence is exactly the process-facing commit/return of f_P . But K.2b.E keeps f_P as the same asserted, attributed, pending dynamic instance throughout $\bar{p}_L$, so its process-facing commit/return does not occur there. Contradiction. The same-interval blocking-issue consequence follows from Core.PolicyS, and the cross-interval consequence from Core.SyncFence. $\square$

K-P.4 Serial-frontier response extraction

Lemma K.2b.SF (Policy-S control-path agreement and serial-frontier extraction). Under K.SerialRoutePremises, separate K.CompleteS evidence for p_S^{det} , the same fixed candidate $D/\{f_P\}/K.\text{DomResetScope}/\text{DomEpoch}$, the K.2b.E frozen-continuation invariant, Corollary K.2b.P-Frozen, and the R-S Policy-S reset-exclusion part of K.ResetEpoch.Continue, each $P \in D$ eventually

reaches in p_S^{det} a process-facing Policy-S serial frontier x_P^{AS} with operation g_P satisfying $\text{Key}(g_P) \leq^{\text{K_src}} \text{Key}(f_P)$, and from some finite prefix onward that frontier remains non-advancing and does not process-facing commit unless a K.RespFail disposition is produced. This is the control-path bridge used by K.2b.Q Step 1; it is not a consequence of bounded local progress alone.

Proof. Fix $P \in D$ and $\kappa_f := \text{Key}(f_P)$. Follow P 's Policy-S execution through the finite dynamic source/control prefix up to κ_f . K.BF bounded local progress excludes infinite pure computation between message operations. At each potential control divergence before κ_f , choose the first differing control/operation fact. K.CV/K.CVPred makes that fact a deterministic function of fixed roots, already matched earlier transfer facts, and lower-ranked certified predecessor observations; K.2b.P0/P1 and the strengthened comparison-availability invariant supply the same keyed predecessor identities and values on the liberal side. Therefore a first branch/operation-selection divergence before κ_f is impossible. Policy S consequently either reaches the operation keyed by κ_f or is stopped first at a source-earlier blocking operation g_P . If an earlier blocking frontier responds, Policy-S serial discipline and bounded local progress advance through it; because the dynamic prefix before κ_f is finite, this cannot repeat indefinitely without either reaching f_P or encountering a first permanently non-advancing earlier blocker. Corollary K.2b.P-Frozen excludes process-facing commit of f_P in the fixed DomEpoch . The R-S reset exclusion prevents an affecting RESET from changing the compared Policy-S epoch during this extraction, while K.CompleteS supplies the complete artifact on which the argument is evaluated. Hence in all cases there is a finite point after which P is non-advancing at a process-facing serial frontier x_P^{AS} whose operation key is at or before κ_f . Persistence of an issued K.BF request follows from $\text{Core.IC/K.BF-Persist}$; if the frontier has not yet issued, K.CV/K.CVPred and the no-first-divergence argument establish the reached Policy-S control point, $\text{Core.PolicyS/Core.16}$ determine the actual Policy-S issue legality, and the fairness/completeness evidence carried by K.CompleteS for the selected deterministic Policy-S scheduler supplies the finite issue step. $\text{K.CompareAdvanceLegal}$ is a Policy-L comparison-advance certificate and is not invoked on this Policy-S control-path step. Any later declared terminal or permitted reset disposition is handled by $\text{K.RespFail/K.TerminalDisposition/K.ResetEpoch}$ rather than by treating bounded local progress as a control-path theorem. $\square$

Lemma K.2b.Q (deterministic bounded-response extraction)—Statement. Assume $\text{K.SerialRoutePremises}$, separate K.CompleteS evidence for p_S^{det} , K.RespCov , $\text{K.ResetEpoch.Continue}$ including its R-S selected-Policy-S reset exclusion, $\text{K.TerminalDisposition.Continue}$, Lemma K.2b.E, and Lemma K.2b.SF. Let a Policy-L prefix reach σ_L with $\text{DeadKRec}^{\text{W}}(\text{K}_L)$. Then the SDet -selected execution p_S^{det} has a finite prefix containing a non-waived K.RespFail witness. No conclusion about Dead_KS , an owner terminal, or a closed serial WFG component is required.

Proof, Step 1 (joint frontier freeze and permanent serial blockage). Let D be the non-waived Dead_K^{W} component reconstructed at K_L . For each $P \in D$ select the internal frozen item/operation f_P supplied by the candidate and K.2b.E interfaces. Exact residual offers and source-prefix closure show that every process-facing blocking Push/Pop required by the Sys_K/K.BF serial prefix and source-earlier than f_P has committed by σ_L ; failed non-blocking polls are handled through their WC-selected return outcomes and are not claimed to have committed. K.EnvMF supplies either $\text{StandardEnv/B1-avail}$, using Lemma K.JointFreeze-Ord for candidates satisfying its ordinary/single-release-support premises and explicit K.JointFreeze for every other admitted release/frontier structure, or a timed/reactive/nondefault environment restricted to the K.JointFreeze-Ord -eligible single-internal-frontier case. Invoke Lemma K.2b.E to obtain the infinite fair complete Core.LAdm continuation in which all f_P remain pending. The R-L part of $\text{K.ResetEpoch.Continue}$ applies at each finite frozen liberal prefix and prevents a relevant RW2 reset there; its independent R-S part excludes dominance-affecting resets on every selected Policy-S comparison prefix consumed below, so the liberal and Policy-S comparisons remain in the fixed DomEpoch . Lemma K.2b.E has already established the required no-release result for this same D through K.JointFreeze-Ord or K.JointFreeze as applicable; K.2b.Q does not repeat the closure proof.

Corollary K.2b.P-Frozen, together with K.2b.R and the Policy-S blocking-prefix discipline, prevents p_S from committing the selected frozen blocking operation or issuing past it in the serial blocking subsequence. Now apply Lemma K.2b.SF: the K.CV/K.CVPred control-path agreement induction, not bounded local progress by itself, establishes that each PED eventually becomes permanently non-advancing at a process-facing Policy-S serial frontier x_{P^S} with operation g_P satisfying $\text{Key}(g_P) \leq^{\text{K_src}} \text{Key}(f_P)$ (the Appendix-K shorthand $g_P \leq_{\text{src}} f_P$), and that frontier never process-facing commits on the reset-Continue comparison branch. For an admitted non-ordinary StandardEnv candidate, the paper-level K.JointFreeze certificate is the soundness premise and K-O2 mechanizes its JF1/JF2 clauses; timed/reactive/nondefault candidates outside K.JointFreeze-Ord's release-structure premises are outside the lemma.

Proof, Step 2 (coverage and non-waiver). By the default-total definition of K.RespCov, every dynamically reachable internal x_{P^S} produced by Step 1 already has a checkable proof disposition; no modal selection argument about whether its static site can become a serial blocker is permitted. A validated unreachability disposition cannot apply to the reached dynamic instance, and a validated static bounded-response disposition supplies the same finite-response fact as a literal monitor. If x_{P^S} is environment-facing and the instance uses StandardEnv/B1-avail, then $g_P <_{\text{src}} f_P$ committed in the liberal prefix; its fixed consumption ordinal is therefore available in Policy S, and persistent request plus Core.IssueFair and B2/B3 would force its commit, contradicting permanent blockage. Thus such an environment frontier cannot be the Step 1 blocker. For a timed/reactive or otherwise nondefault environment, K.EnvMF has already excluded every liberal candidate outside the single-internal-frontier, K.JointFreeze-Ord-eligible release structure; any environment-facing serial blocker arising on $p_{S^{\text{det}}}$ requires an explicit K.Resp monitor or availability proof. Coverage-preserving waiver closure ensures that the original non-waived component D cannot map solely to waived dispositions. Therefore at least one non-waived covered response obligation triggers at a finite Policy-S cycle t and remains unsatisfied thereafter unless the process-facing response occurs; any reset or run-ending disposition is classified separately by Definition K.RespFail and the applicable reset/terminal branch.

Proof, Step 3 (finite response-failure witness). Let B be the finite bound of the monitor selected in Step 2. If p_S continues through the boundary-action phase of cycle $t+B$ without response, the first normalized cutpoint after that phase witnesses timeout. If p_S reaches a declared maximal finite terminal state before that post-phase cutpoint while the monitor remains pending, the terminal prefix witnesses maximal-pending. The R-S part of K.ResetEpoch.Continue has already excluded every dominance-affecting RESET on the Policy-S comparison prefixes consumed by Steps 1–3; therefore these cases exhaust the selected K.CompleteS artifact on the Continue branch: a nonterminating complete run eventually reaches the finite timeout bound, while a terminating complete run ends only at DeclaredTerminal. When an affecting RESET is permitted instead, the candidate may use K.ResetEpoch.DirectFail only with its independently mapped already-triggered non-waived monitor, in which case RW2 yields cancelled-pending before this Continue-branch Step 3 is entered; an affecting reset covered by neither R-S nor DirectFail makes the candidate ineligible for the serial route. Therefore p_S contains a finite non-waived K.RespFail. $\square$

Diagnostic note. The retired owner walk may still be run after K.RespFail to classify a likely root as an internal cycle, EnvStop, DoneStop, or another owner-chain condition. Example K.CE-5/H2 shows why that taxonomy cannot be used as an exhaustive proof premise: a complementary owner can remain live forever on unrelated channels while permanently under-supplying the demanded endpoint. Failure to classify a response failure affects diagnostics only, not Corollary K.2c.

Proof of Lemma K.2b. Apply K.Low.C if lowering is needed, then instantiate the shared K.InterruptionGate in the normative order. First evaluate K.ResetEpoch. On K.ResetEpoch.DirectFail, use the independently certified $P \in D$, selected liberal frontier f_P , Policy-S serial frontier x_{P^S} , and $g_P := \text{Op}(x_{P^S})$ with $g_P \leq_{\text{src}} f_P$ in the same pre-reset epoch. The associated non-waived K.Resp monitor

has already triggered and remains pending before the first matched affecting RESET; RW2 clears that exact monitored dynamic instance at the RESET unit, so K.RespFail supplies cancelled-pending immediately. This branch does not invoke K.2b.Q Step 1. Only on K.ResetEpoch.Continue evaluate K.TerminalDisposition. On K.TerminalDisposition.DirectFail, the independently certified mapping supplies the same D-to-serial-frontier data, a finite prefix of p_S^{det} with the mapped non-waived monitor already triggered and pending, and independently checked evidence that p_S^{det} reaches the specified declared maximal terminal state while that monitor remains pending; K.RespFail therefore supplies maximal-pending directly. Only on K.TerminalDisposition.Continue invoke Lemma K.2b.E to obtain the infinite fair frozen continuation; K.ResetEpoch.Continue already supplies both R-L and R-S reset exclusion, after which K.2b.R/P, Lemma K.2b.SF, and K.2b.Q derive the response failure. Thus each DirectFail branch is discharged independently, and the dominance continuation is constructed only after both nested Continue checks succeed. $\square$

K-P.5 Optional serial replay and mechanization obligations

Lemma K.Ser (conditional serial trace replay). Statement. Fix the selected instance Sys_K, the fixed stimulus, and the fixed witness choices, and let p_L be a finite Policy-L prefix admissible under Definition Core.LAdm ending at a source K-facing cutpoint $\sigma_L \in \text{KCutReach_Sys_K}(I)$ with $K_L = \langle E, w, H_L, O_L \rangle$. Suppose: (a) commit-prefix condition — for every process P, CmtRec_P(K_L) is a source-order prefix of P's dynamic operation sequence; (b) serial-precedence acyclicity — the union of the causal dependencies of p_L (per-channel transfer order and capacity dependencies, rendezvous atomic pairing, epoch-gate and guard dependencies, and process-local source order) with the Policy-S precedence constraints, requiring each source-earlier blocking operation of a process to commit strictly before a source-later one issues, is acyclic; and (c) terminal offer admissibility — every $o \in O_L$ lies on its owner's reconstructed minimal frontier, $\text{item}(o) \in \text{FrontRec_owner}(o)(K_L)$, equivalently $\text{DrainBag_Proc}(I)(K_L)$ is empty; in the ordinary Core.OrdFrontSingleton case this is the unique source-minimal uncommitted blocking item, while a qualified Sys_B^{elab} frontier may contain several minimal members. Thus σ_L contains no asserted residual offer outside the minimal frontier that Policy S would have withheld; and (d) replay-environment stability — either the instance uses StandardEnv/B1-avail, or the application supplies an explicit environment-stutter-equivalence proof showing that every idle cycle inserted solely to satisfy Policy-S strictly-earlier-cycle precedence preserves the fixed environment inputs/reactions and all replay-relevant request, enablement, and observation facts until the next recorded source action. Then there exists a finite admissible Policy-S execution of the same instance, stimulus, and witness with the same per-channel committed transfer sequences and payloads ordinal for ordinal, the same committed-identity sets and canonical process-local replay history H_L , and therefore the same ProcRec, ChanRec, NextFrontRec, and source-relevant terminal values as K_L . If a separate terminal-offer replay check also establishes the same residual-offer set O_L , the two terminal cutpoints have equal KObjs; equal O is not inferred from the committed history alone. Proof sketch. Linearize the finite acyclic combined precedence relation and replay commits in that order. Insert an idle cycle solely when condition (d) permits it: this is automatic under StandardEnv/B1-avail, while a timed/reactive or otherwise nondefault environment requires the explicit stutter-equivalence proof and may not be delayed by proof fiat. The induction invariant — equal per-channel counts and payloads, equal process-local committed histories, equal channel state — is preserved at each step: source-earlier blockers have committed by the added precedence constraints, so each issue is Policy-S-legal; channel enabling conditions transfer by the equal-channel-state invariant; Lemma K.2b.P1 supplies operation-identity and payload agreement; and I.DR supplies per-process state agreement from equal replay histories. The optional equal-O conclusion requires one additional finite check that the terminal serial requests are asserted at the same typed boundaries and epochs. Mechanization note: a finite structural induction with no fairness or limit reasoning; mechanization is required before the lemma is cited as certified.

Engineering note (serial replay of recorded traces; scope of Lemma K.Ser). Lemma K.Ser characterizes when a recorded liberal or RTL-matched execution can be replayed under the conservative serial scheduler with identical committed histories — and, with the additional terminal-offer replay check, identical KObs. KObs alone is not a sufficient replay certificate. The recorded artifact must additionally contain the historical dynamic operation identities, Issue events, persistent-request intervals and attribution, failed non-blocking outcomes as witness selections, elaborated-boundary transfers, and arbiter selections needed to justify the acyclic operational replay; a committed-event history determines neither those issuance obligations nor the intermediate enablement facts. O_L records only requests still outstanding at the terminal cutpoint. Conditions (a) and (c) are the commit-prefix/minimal-frontier tests that reject recorded runs whose terminal history/offers rely on eager issue beyond what Policy S can replay; Examples K.CE and K.CE-2 are minimal failures, and their liberal histories are reproducible by no Policy-S execution. The optional replay lemma is also environment-scoped: its inserted idle cycles are justified directly under StandardEnv/B1-avail and require an explicit environment-stutter-equivalence proof otherwise; a time-aware/reactive environment may not be silently held fixed. Lemma K.Ser is therefore deliberately not a substitute for Lemma K.2b: the non-serializable case is where the dominance content of the serial-reduction route resides.

Mechanization obligations. K-O0: mechanize Definition K.OpKey, its partial per-execution realization maps and key-level $\leq^{\text{K_src}}$ order, and Definition K.SerialDomOrder over indivisible Policy-S action units; mechanize Definition K.CVPred on the common CVFact_I key universe, validate predecessor occurrence closure, the per-fact deterministic provenance equations, the serial-route comparison-availability closure with SegPred/SegReach certified segment descriptors, and the explicit recursive Core.16 reach-legality disposition for each non-transfer predecessor (pointwise ordinary/singleton blocking-OpFact proof only after the prerequisite-transfer qualification test, Core.16-inert observation reach, or equivalent owner/frontier launch proof), establish K.CV-WF (or an equivalent checked CVRank whose strict decrease proves it), and mechanize Lemma K.2b.P1 by well-founded induction on $\leq_{\text{CV}}$; do not use the possibly cyclic Appendix-G causal-dependency graph $\rightarrow$ as that induction order. K-O1: mechanize Definition Core.IssueFair, Core.FairPick FP4a certified finite-segment progress, the requirement that SegReq be a fixed construction-state transition rule before dependent choice, with no active-stack designation prefix-local and an active top-level obligation sticky until completion/rejection, phase-owner routing through IssuePick/L2-CycleChoiceExt/BoundaryPick, the certificate-tree K.SegWork measure with deterministic prefix-indexed K.SegNorm/skip behavior, the atomic-unit boundary rule, and the Core.16 compatibility fact. K-O2: mechanize Lemma K.JointFreeze-Ord and Definition K.JointFreeze, including JF1 coverage of every releasing launch/issue across incomparable Sys_B^elab frontier members, JF2 preservation under permitted drain/elaboration actions, and process-facing monitor-trigger composition. K-O3: prove Lemma K.2b.P over the K.DomResetScope dominance domain with fixed DomEpoch on the reset-free, terminal-free infinite-fair branch and derive Corollary K.2b.P-Frozen by composition with the same K.2b.E-constructed continuation $\bar{p}_L$, with $\bar{p}_L$ an explicit common parameter of the Lean statements for K.2b.E and K.2b.P and with the K.2b.E frozen-continuation invariant itself an explicit K.2b.P hypothesis, so the corollary composes by direct application rather than by provenance prose or section-variable capture, using the strengthened induction invariant that carries both all earlier serial endpoint transfers and every lower-ranked K.CV predecessor fact required to invoke K.2b.P0/K.2b.R; prove that each comparison-availability extension is represented by the certified segment stack, preserves the frozen component, introduces no unmatched K.SerialDomOrder dominance transfer, and carries one of the required recursive Core.16 reach-legality dispositions rather than relying on “phase-legal” as an unanalyzed assertion; mechanize the base Inv, $\text{Inv}^+ := \text{Inv} \wedge \text{I6}$, sticky K.SegmentPlan construction-state rule, K.SegmentAdm relative to Inv, K.SerialDomPredFinite from model-relative CycleResult_{Sys_K}.committedUnits, K.CompareFact-Mono, deterministic prefix-indexed K.SegNorm and K.SegWork/K.SegWork-Complete, and Lemma K.SegmentPlan-EventuallyTarget as a

theorem over the completed $\bar{p}_L$ with the explicit $T \notin \text{Units_SD}(\bar{p}_L)$ hypothesis and ET1/ET2 handoff; P0/R/P must cite that lemma rather than infer eventual plan emission from Inv+. K-O4: mechanize Definition K.CompareAdvanceLegal and Lemma K.CompareAdvanceLegal-Ord; prove coverage for every K.2b.P0/K.2b.R-consumed owner/key in K.DomResetScope, including $P \notin D$ and explicit $\text{Sys_B}^{\text{elab}}$ multi-frontier owners, and expose K.CompareAdvanceLegal-Ord as the reusable pointwise ordinary/singleton proof for qualifying blocking OpFact predecessor reaches required by K-O0, where qualification requires every Policy-S-required source-earlier blocking transfer for the predecessor key to be already present in the matched liberal transfer prefix before predecessor materialization and not created by that materialization extension; prove the direct Core.16(2) item-order/no-predecessor conclusion for the concrete launch target, using key order only through Lemma K.KeyItemOrderBridge or the explicit elaboration-specific bridge, and prove that this condition discharges both preparation/launch and Issue/assertion legality; enforce the JointFreeze/CompareAdvanceLegal consistency rejection rule; then validate K.CompareScopeFootprint and the certificate-supplied finite $S_p^{\text{dom}}/S_c^{\text{dom}}$ against K.DomResetScope conditions D1–D4, derive DomEpoch, mechanize Lemma K.2b.P0's keyed occurrence/reachability result, and then prove Lemma K.2b.R using the shared K.InterruptionGate.Continue rule: instantiate Interrupt_Reset as prefix-local legality of a RESET intersecting K.DomResetScope as the next proof-relevant transition/unit, prove the R-L liberal-prefix Continue clause and the independent R-S coverage/exclusion theorem for every selected Policy-S prefix later consumed by P0/R/P/P-Frozen/SF/Q, with any executed occurrence recorded in the enclosing CycleResult.resetUnits and Interrupt_Terminal as DeclaredTerminal of the frozen-prefix final state; prove both predicates prefix-locally, show every K.2b.P0/R/P/P-Frozen/SF-consumed process/boundary identity lies in K.DomResetScope, and preserve DomEpoch-local request attribution through Lemma K.BF-Persist. K-O5: mechanize Lemma K.2b.SF by the first-control-divergence/K.CVPred induction using Core.PolicyS/Core.16, Corollary K.2b.P-Frozen, the R-S reset-exclusion domain, the selected deterministic Policy-S scheduler and K.CompleteS fairness/completeness, and Core.IC/K.BF-Persist after issue; do not invoke K.CompareAdvanceLegal on the Policy-S side. Also mechanize default-total K.RespCov plus the authoritative pre-K.2b candidate waiver closure. K-O6: mechanize the common K.InterruptionGate.DirectFail payload shape independently of K.2b.P0/R/P/P-Frozen/SF/Q Step 1, then prove the reset specialization yields cancelled-pending through matched RESET/RW2 and the terminal specialization yields maximal-pending through DeclaredTerminal. K-O7: mechanize the schema's K.BF exhaustiveness fact, anti-circularity rule, and the normative nesting K.ResetEpoch.Continue $\rightarrow$ K.TerminalDisposition; prove the Core.CycleClosure C2 partial-cycle extension interface and Lemma Core.CycleChoiceExt for the selected source semantics; mechanize Definition Core.FairPick, including local obligation finiteness/keying, the actual fixed-scheduler/arbitration starvation proof, FP4a certified-segment selection with nested lower-rank stack/atomic-unit discipline, SyncFence prerequisite segments, and B3 exposure; mechanize Lemma Core.FairCycleDovetail with its two C2/CycleChoiceExt applications (L1 selection $\rightarrow$ provisional typed settle prefix $\rightarrow$ L3 selection $\rightarrow$ final typed CycleResult) and the Inv+-preservation argument used by K.2b.E, including K.SegmentAdm for every active certified segment; then use dependent choice only over those already-typed successors and derive Core.IssueFair/B2/B3 from FP4/FP5 while using FP4a only for the proof-built certified segments. Mechanize the strengthened Q.1(I) CouplerPersistence property and at least one qualified non-vacuous template discharge, plus the primitive buffered Pop/Push and rendezvous continuous-enablement arguments. Mechanize K.Low.IdClose/K.Low.EpochClose/K.Low.GuardClose bidirectionally and route the right-to-left closure through K.ReleaseSCCReduction before K.Low.A5 waiver pullback. Either DirectFail branch exits without invoking that construction. The supported environment branches remain those of K.EnvMF. K-O8 is needed only for the optional diagnostic drain-normalization/Dead_KS construction. These operational results shall be proved from Core.ReachPrep, Core.CycleResult/Core.CompleteCycle/Core.CycleClosure/Core.CycleChoiceExt, Core.16, Appendix C

Issue/Commit linkage, RW2, the correctly model-indexed nonwithdrawal and $K\epsilon$ instances, $K.OpKey$, $K.SerialDomOrder/K.SerialDomPredFinite/K.KeyItemOrderBridge$, $K.CV$ including $K.CVPred/K.CV-WF$, $K.CompareFact-Mono/K.SegWork/K.SegmentPlan-EventuallyTarget$, B1-B5, $Core.IssueFair$, $K.BF$, point-to-point E4 ownership, $K.EnvMF$, $K.CompareAdvanceLegal$, $K.InterruptionGate$ with $K.DomResetScope$ and its two exhaustive $K.BF$ instantiations, $K.Low.C/K.Low.A5$, and $K.Resp$ —not from finite state-space reasoning. $KObs$ selector facts do not replace operational monitor premises.

Appendix L – Snooping Introduction, Implementation and Concerns

Snooping Introduction

If designs are completely latency insensitive, verification of the pre-HLS versus post-HLS models is straightforward. However, if the design contains some components such as arbiters which have latency-sensitive behavior, and if that latency-sensitive behavior is in some cases externally visible to the DUT, then verification becomes somewhat more complex. Techniques can be applied to simplify verification.

Consider a design which has an arbiter like Matchlib toolkit example 09*. The arbiter uses non-blocking PopNB() on its inputs, and blocking Push() on its output to emit the winner of the arbitration. Since the latency between the pre-HLS and post-HLS designs will differ, the order of transactions presented to the arbiter in the two scenarios will differ, and thus the order of the winners will differ. This may result in verification mismatches between the two models if the order of the winners is externally visible to the DUT.

To force the pre-HLS and post-HLS simulations to match, we can run the two designs side-by-side. We can snoop the inputs to the arbiter in the post-HLS model, and only allow the inputs to the pre-HLS arbiter to proceed when the corresponding inputs are seen in the post-HLS model. This will force the order of the inputs to be equivalent between the pre-HLS and post-HLS models, and thus both arbiters will pick the same winners. When this technique is used, the pre-HLS simulation will be throttled by the post-HLS simulation, but the overall verification will still work properly.

Another related example involves interrupt request signals feeding into an interrupt controller within a CPU model. Typically, each interrupt request signal is a single bit `sc_signal<>`, indicating that an interrupt request is pending. If the requests originate from accelerator blocks that are being synthesized through HLS, then because of the latency differences between the pre-HLS and post-HLS models, the order of interrupt requests arriving at the interrupt controller will differ between the two models, and this will likely result in verification mismatches if the differences are externally visible to the DUT. To force the order of the requests to match, we snoop the requests arriving at the controller in the post-HLS model, and only then allow the requests to be seen in the pre-HLS model.

Simplified Snooping Approaches

In the snooping example directly above in which the arbiter is snooped, the entire post-HLS RTL DUT is simulated alongside the pre-HLS model to force the arbiter requests to be aligned in time. This is the most general case and assumes the input delays to the arbiter are difficult to determine via analysis.

Sometimes there are simpler cases where the input delays to the arbiter in the RTL DUT are easier to determine. For example, each input to the arbiter might arrive a fixed number of clock cycles after one of the primary inputs to the RTL DUT is pushed by the testbench. In this case, a much simpler model can be used to align the pre-HLS model with the post-HLS model. We can simply monitor the primary inputs to the pre-HLS model and then apply the fixed cycle delays to the pre-HLS arbiter inputs.

The two cases above illustrate two ends of a spectrum of possible approaches for extracting the delays from the RTL DUT. Between these two points there exist other possible approaches.

Snooping Implementation

To enable perfect matching between the pre-HLS and post-HLS system simulations, latency-sensitive global signals and latency-sensitive non-blocking message-passing operations need to be synchronized between the two simulations if their latency differences are externally visible to the DUT.

As explained earlier in this appendix, in the general case the entire post-HLS DUT RTL must be run alongside the pre-HLS system to achieve alignment. Examples of signals that need to be synchronized include global interrupt request signals (as discussed earlier in this appendix) and rdy/vld signal pairs for non-blocking operations on message-passing channels. All such synchronization signals and their corresponding handshake signals must be level-stable:

- A latency-sensitive signal s is **level-stable** if, once s becomes 1 in cycle t , it must remain 1 until the corresponding handshake signal h is 1 in some cycle $t' \geq t$.

Once the set of signals that need to be aligned between the two simulations is identified, the general rule to implement snooping is simple:

- For each polarity-normalized Boolean commit-enabling snoop signal (for example, the vld/rdy handshake signals of snooped non-blocking operations, or level-stable request lines), make all readers in the pre-HLS simulation see the logical AND of the value driven in the pre-HLS simulation and the recorded post-HLS observation of the same dynamic instance ordinal — captured at the post-HLS-side event and retired at the pre-HLS-side handshake of that ordinal, per Wrapper Realization Requirements W1–W4 below. Payload/data values are never ANDed or substituted; they are checked for equality at the matched ordinal (Requirement W4). The post-HLS simulation runs free: its readers see only its own driven values, and the wrapper observes but never gates, delays, or modifies any post-HLS-side signal.

Often the post-HLS side will not run ahead of the pre-HLS side at the snoop points, because HLS typically only adds (rather than subtracts) latency within each process. This tendency is a heuristic, not an invariant: HLS can also subtract latency, for example by removing conservative waits present in the source, by improving loop initiation intervals, or by realizing channels with fall-through timing. The one-sided rule above is therefore normative, and a symmetric variant — driving the logical AND of the two simulations' nets into the readers on both sides, so that the pre-HLS side can gate the post-HLS side — must not be used, even as a fallback when the post-HLS side runs ahead. Under B1 the post-HLS Σ -trace under fixed stimulus is unique, so there is no post-HLS nondeterminism for a wrapper to select among: gating a post-HLS-side reader (in particular an internal net such as an arbiter input or an interrupt request line) does not select an admissible RTL_B behavior; it produces a different circuit RTL_B' with different internal latencies, and any result obtained about RTL_B' does not transfer to the RTL_B that ships. Gating the pre-HLS side, by contrast, is witness selection over an intentionally under-specified specification (see Snooping Concerns below) and is always permitted.

Two situations must be distinguished when the post-HLS side does run ahead at a snoop point. (a) Benign cycle skew: the RTL asserts a snooped level-stable signal some cycles before the pre-HLS side does, but the order of Σ -visible events at the snoop points is unchanged. Because the compatibility

results of Appendices G–K are stated over Σ -event order and values, with finite ϵ -stutter tolerated (B4/B5), a matching pre-HLS witness still exists. The wrapper handles this case by recording the RTL assertion (level-stability makes the capture clean; the recorded assertion is held until the pre-HLS-side handshake of the same ordinal, per Requirement W1 below) and letting the pre-HLS side catch up, arbitrarily behind in host time. Cycle skew is not an error, and a cycle-locked per-cycle “perfect match” comparison must not be used as a validity gate, since it would misclassify a mere HLS speedup as a failure. (b) Divergence: no admissible pre-HLS execution can consume or produce the RTL’s event sequence — its ordinal order, kinds, and payloads — at the chosen boundary; the witness required by WC does not exist. This is the genuine failure mode; it violates Condition L.Dir below, must cause the wrapper to abort loudly, and is triaged as described under Snooping Concerns. Comparison of the pre-HLS-side AND against the value driven on the post-HLS side remains a mandatory monitor (Requirement W4 below), but its role is diagnostic: a per-cycle mismatch reports skew, while the L.Dir violation itself is defined at the event level — ordinal order, kind, and payload — and is detected by an order/kind/payload check together with a watchdog/timeout that converts witness non-existence, or an inconclusive failure to realize the witness within an unproved bound (Requirement W4), into a reported failure rather than a silent gate or an indefinite hang.

Wrapper Realization Requirements (Recorded-Observation AND)

The AND rule above is stated over a recorded observation, not over the live post-HLS net. A naive realization in which the pre-HLS reader sees the AND of the two currently driven values is a cycle-window mechanism: it delivers a snooped event to the pre-HLS side only if the post-HLS assertion is still asserted when the pre-HLS side is ready to commit. Because the post-HLS side runs free, its assertion window ends at its own handshake, which is unrelated to whether the pre-HLS side has caught up; level-stability guarantees that the signal remains asserted until the post-HLS-side handshake, not until the pre-HLS-side handshake. The live-AND realization is therefore faithful exactly when, for every dynamic instance ordinal k at every snoop point, the pre-HLS-side commit of instance k falls inside the post-HLS-side assertion window of instance k . This per-ordinal window-overlap condition is strictly stronger than Condition L.Dir, and it is an empirical per-run property: it shall be checked, not assumed.

Where the window-overlap condition fails — including on runs that are L.Dir-clean, i.e., runs exhibiting only benign cycle skew — the live AND fails in three ways. First, it drops events: the post-HLS assertion retires before the pre-HLS side arrives, the AND never rises for the pre-HLS side, and the event is silently lost (for example, a lost interrupt downstream, or a pre-HLS process waiting forever on a request that never becomes visible), converting an L.Dir-clean run into an apparent failure or value divergence — precisely the misclassification this appendix forbids. Second, it aliases ordinals: if the post-HLS side runs two or more transactions ahead on the same signal, the pre-HLS side’s k -th commit can be enabled by the post-HLS side’s $(k+1)$ -th assertion, and pulse-type signals held across two post-HLS instances can merge into one pre-HLS observation, or drop. Dropped and aliased events violate the no-add/remove/reorder premise of Lemma L.2 and break the per-instance μ_Q matching supplied by WC: the harness is then sampling a wire, not replaying the witness. Third, it detects nothing: genuine order divergence (a real L.Dir violation) manifests as a hang indistinguishable from a dropped-event hang, so the triage of Snooping Concerns cannot begin; moreover, the harness-induced infinite wait appears as a wait-for dependency in any Appendix K analysis — a phantom deadlock manufactured by the harness rather than by the design, and outside the finite-delay tolerance that B2 fairness extends to the wrapper on L.Dir-clean runs.

W1 (Recorded per-ordinal observation). For each snoop point, the wrapper shall capture each post-HLS snooped event — the committed transfer for message-passing snoop points, the assertion edge for level-stable global signals such as interrupt request lines — indexed by its dynamic instance ordinal, and shall hold the captured observation for the pre-HLS side until the pre-HLS-side handshake of the same ordinal, at which point the observation is retired. For a single-outstanding interface this reduces to one sticky flop, set by the post-HLS event and cleared by the pre-HLS-side handshake; in the general case it is a small per-snoop-point event FIFO. The AND shall consume the captured observation, never the raw net; level-stability is what makes the capture clean, and pulse-like signals shall be queued as distinct edge events so that dynamic instances are neither merged nor dropped.

W2 (Gate only the commit-enabling mirror signal). Per operation kind, the wrapper shall gate only the mirror signal that enables the pre-HLS-side commit: the observed vld for a pre-HLS-side Pop, and the observed rdy for a pre-HLS-side Push. The endpoint-owned request driven by the pre-HLS side itself is never modified, consistent with the Appendix C non-withdrawal discipline.

W3 (Registered sampling point). In a shared-clock co-simulation, the post-HLS observation shall be registered by at least one cycle before it enters the AND, so that delta-cycle ordering races between the two simulations cannot affect the observed value. This is safe by construction: it adds only pre-HLS-side latency, which is witness selection.

W4 (Monitor and watchdog; Lemma L.5(iv)). At each pre-HLS-side commit at a snoop point, the wrapper shall compare the committed event against the recorded post-HLS event of the same ordinal, in order, kind, and payload. Payloads must match (K.CV); the wrapper shall never substitute a post-HLS payload into the pre-HLS side, since substitution relabels Σ -events and masks value divergence. A pre-HLS-side snoop commit with no recorded post-HLS counterpart, or an order/kind/payload mismatch at a matched ordinal, shall abort loudly as an L.Dir violation. A recorded event outstanding beyond a configured watchdog bound, or a pre-HLS-side request that remains unmatched beyond that bound, shall likewise abort loudly, but as an L.Dir/WC discharge failure: if the bound has been proved sufficient for the selected instance, the abort may be classified as an L.Dir violation; otherwise it shall be triaged as an inconclusive witness-realization failure — the compared run remains outside the discharged theorem scope, but the abort shall not be converted into a claimed design deadlock or a claimed proven violation. In all cases the abort converts witness non-existence, or failure to realize the witness within the bound, into a reported failure rather than a silent gate or an indefinite hang. Per-cycle skew shall be logged separately as diagnostics and shall not be treated as a failure.

W5 (Coverage remains a separate obligation). Requirements W1–W4 concern the fidelity of the replay mechanism at each snoop point; they do not discharge the first premise of Lemma L.2. If the snooped set misses an ϵ -modeled choice that can affect Σ -visible control flow, frontier membership, request attribution, guard values, or payload values, the replayed witness is underdetermined regardless of the recording discipline, and the resulting failure shall be triaged as harness incompleteness under Snooping Concerns.

Remark (relation to the naive live AND). The live-AND and recorded-observation realizations coincide on single-outstanding snoop points at which the pre-HLS consumer commits within every post-HLS assertion window — for example, a non-blocking polling consumer that polls every cycle while the pre-HLS side remains at-or-ahead in cycle time at that snoop point throughout the run. Because W4 requires the W1 recording in any case as its comparison reference, gating with the live net offers no economy;

the recorded observation is therefore the normative gating term, and the live net may be used only as an additional diagnostic tap.

Shared reactive testbenches. Once the pre-HLS side is permitted to lag by more than a cycle, a single shared reactive testbench instance is problematic: its reactions are driven by whichever model it is wired to, so the two models no longer receive the same stimulus, contrary to the fixed-stimulus setting of the formal results. The flow shall therefore either replicate the stimulus/testbench per model, or structure the shared testbench as a function of the DUT-visible trace so that the wrapper’s recorder can replay its reactions to the lagging side under the same recorded-observation discipline of W1–W4.

Definition L.1 (Witness selector / Snooping).

Fix an initial state, fixed external stimulus, and an RTL_B execution prefix τ_{post} . A witness selector W for a source process P is a partial strategy that, at each ϵ -quiescent source cutpoint σ_{ref} and each source-level choice point whose outcome is ϵ -modeled but may affect later Σ -visible behavior, returns the outcome selected by the matched RTL_B execution.

Such choice points include failed/successful outcomes of non-blocking PushNB/PopNB calls when the outcome affects later control flow, frontier membership, request attribution, or payload/guard values; and any other ϵ -only implementation choice that can affect the next Σ -visible frontier.

Cadence-sensitive non-blocking polling. If a source process computes externally visible behavior from the number or cadence of failed PushNB/PopNB polls — for example, retry counters, timeout counters, or arbitration policies whose winner depends on how long an input remained empty/full — then that polling logic is latency-sensitive in the sense of Appendix D. Such a process is not covered by the ordinary Appendix I/J latency-insensitive source-core argument merely by treating the failed polls as hidden ϵ -steps. To bring such a design into scope, the cadence-sensitive polling logic shall be isolated as a latency-sensitive local protocol block, or shall be aligned by an Appendix L snooping/witness construction that makes the selected source execution an admissible execution of the same isolated protocol behavior observed in RTL_B. The observable interface to the rest of the system shall then be the chosen boundary of the isolated block, not the internal failed-poll cadence itself.

The selected outcome must be causally available from the matched RTL_B prefix and the fixed global Σ -causal history. W is not allowed to depend on future RTL_B events beyond the matched prefix needed to identify the current source choice. For cadence-sensitive polling logic, the witness must select an admissible execution of the isolated latency-sensitive block; it may not manufacture a retry/timeout outcome that is inconsistent with the RTL_B behavior being used as the witness, the fixed stimulus, or the chosen observable boundary of that block.

Lemma L.2 (Snooping establishes WC).

Let τ_{post} be an RTL_B execution under fixed stimulus, and let W be the witness selector induced by the Appendix L snooping wrapper for the corresponding source execution. Assume:

1. the snooped signals / outcomes include every ϵ -modeled source choice whose result can affect the next Σ -visible control flow, frontier, request-attribution, guard value, or payload value;
2. the snooping wrapper does not add, remove, reorder, or relabel any Σ -visible event except by selecting ϵ -only source outcomes at the chosen observable boundary;
3. each selected ϵ -only outcome is consistent with the matched RTL_B prefix and the fixed global Σ -causal history;
4. failed non-blocking polls remain ϵ -events and successful non-blocking transfers produce exactly the corresponding committed Push_c / Pop_c Σ -event; and

5. if the source contains cadence-sensitive non-blocking polling whose externally visible behavior depends on the number or cadence of failed polls, that polling logic has been isolated or snoop-aligned as a latency-sensitive local protocol block as described in Appendix D and Definition L.1. The WC witness is then applied to the chosen observable boundary of that block, not to an unconstrained hidden failed-poll cadence inside the ordinary latency-insensitive source core.

Then the source execution selected by W satisfies the witness-compatibility premise WC used in Appendix I.2 and Appendix J.

Corollary L.3 (NB-CF identity witness).

If NB-CF holds for process P , then failed non-blocking poll outcomes do not affect the next Σ -visible control flow, frontier membership, request attribution, guard values, or payload values. Therefore the witness selector W may be taken to be the identity / arbitrary selector on failed non-blocking outcomes, and WC holds without additional snooping for those polls.

Corollary L.4 (Cadence-sensitive polling is handled by isolation).

If NB-CF does not hold because the source observes the number or cadence of failed PushNB/PopNB polls — for example, retry counters, timeout counters, or arbitration policies whose later Σ -visible behavior depends on how long a channel remained empty/full — then the design is latency-sensitive at that polling site. Such a site must be isolated or snoop-aligned under Appendix D / Appendix L before applying the ordinary Appendix I/J source-core proof. If no such isolation, snooping, or equivalent explicit timing model is supplied, the process is outside the scope of the Appendix I/J trace-compatibility theorem for the chosen observable boundary.

Condition L.Dir (One-directional alignment / RTL-side non-interference).

Fix the free-running RTL_B execution τ_{post} under the fixed stimulus (B1) and the Appendix L wrapper for a chosen set of snoop points. L.Dir holds for a compared run iff the source-side snoop sequence selected by the witness selector W of Definition L.1 is exactly an ordinal-preserving consumption of the free-running RTL_B snoop sequence at the chosen boundary: (i) each pre-HLS-side snoop commit consumes the next recorded RTL_B event of the same snoop point, with matching kind and payload; (ii) no pre-HLS-side snoop commit occurs without such a next recorded RTL_B event; and (iii) every recorded RTL_B snoop event within the compared run is eventually either consumed by the source witness or reported outstanding according to the proved or declared comparison bound of Requirement W4 above. Throughout, the alignment never gates, delays, or modifies any RTL_B-side signal or event (RTL-side non-interference): the wrapper never drives a post-HLS-side reader with a value different from the value the free-running RTL_B drives. Cycle-level lag of the pre-HLS side (benign skew) is permitted under L.Dir provided the ordinal-preserving consumption above is maintained via recorded/buffered replay of the level-stable assertions; only order, kind, or payload divergence — non-existence of the WC witness at the chosen boundary — violates L.Dir. L.Dir is a named side condition with its own entry in the side-condition discharge record (Normative Formal Core). A run that violates L.Dir provides no theorem-backed guarantee. The harness shall abort loudly on any detected divergence and on any exceeded Requirement W4 watchdog bound, with the classification of bound-exceeded aborts governed by Requirement W4, and every abort shall be triaged as described under Snooping Concerns below.

Lemma L.5 (Online realization of W / harness soundness).

The equivalence and liveness results of Appendices I–K are stated over the free-running RTL_B execution τ_{post} under fixed stimulus (B1), together with the source-side witness selector W of Definition L.1. This statement is one-directional by construction: no premise of those results requires RTL_B to be stallable by the source, and τ_{post} is a fixed object rather than a participant that can be gated. Separately, suppose the co-simulation wrapper (i) observes but never gates, delays, or modifies any post-HLS-side

signal or event; (ii) records snooped level-stable assertions and replays them to the pre-HLS side in RTL order, permitting arbitrary finite host-time skew (Wrapper Realization Requirements W1–W3); (iii) satisfies the premises of Lemma L.2 (see Requirement W5); and (iv) monitors Condition L.Dir and aborts loudly upon violation (Requirement W4). Then on every run on which the wrapper does not abort, the wrapper computes W online, the selected source execution satisfies the WC premise used by Appendices I–J (by Lemma L.2), and the results proved over free-running $RTL_B + W$ apply to that run. Bidirectional stalling therefore never enters any theorem whose conclusion is transferred to the shipping RTL: the post-stalls-pre direction is witness selection over the intentionally under-specified source, and the pre-stalls-post direction is excluded by the monitored Condition L.Dir rather than assumed, endorsed, or repaired by gating the RTL.

Snooping Concerns

The snooping technique is used to align pre-HLS and post-HLS latency-sensitive global signals and latency-sensitive non-blocking message-passing operations in cases where their latency differences are externally visible at the DUT boundary. This includes cadence-sensitive polling blocks, such as retry/timeout or polling-arbiter logic, when their externally visible behavior depends on the number or cadence of failed non-blocking polls. When such a wrapper is applied, the pre-HLS model is throttled to follow the latency choices made by the post-HLS RTL, and the resulting traces at the observable interfaces are forced to agree. Snooping does not make failed polls into committed message-transfer events; failed polls remain ϵ -steps. Rather, snooping selects an admissible execution of the latency-sensitive local protocol block, while the rest of the system is compared at the block’s chosen observable boundary.

Readers with a formal background may be concerned by this technique, because it appears to use the post-HLS RTL implementation to modify the behavior of the pre-HLS specification. From a formal point of view, the important observation is that the pre-HLS model is intentionally under-specified with respect to micro-timing/latency: subject to the R-rules, B-rules, weak fairness, and the explicit Σ -visible ordering, FIFO, signal-anchor, synchronization, and atomicity constraints used by $\approx$, it can admit multiple ϵ -/latency-different executions that respect the same required-order constraints over the chosen observable units. This is compatible with B1, which asserts that the post-HLS Σ -projected trace is unique under a fixed stimulus; the pre-HLS side may still have multiple admissible realizations that (when projected to Σ) match that unique RTL Σ -trace.

Snooping does not change what executions are allowed by the pre-HLS specification. Instead, it selects—purely for verification purposes—one admissible pre-HLS execution whose latency-sensitive events align with those observed in the RTL. In other words, the implementation is used to pick a witness execution of the specification, not to redefine the specification.

Triage of L.Dir violations. A violation of Condition L.Dir (order, kind, or payload divergence at a snoop point, including a pre-HLS-side snoop commit with no recorded RTL_B counterpart) is a relational failure: the compared run is outside the discharged theorem scope, and no theorem-backed claim attaches to it. It shall not be defined a priori as a bug in the pre-HLS model, because witness non-existence has three possible root causes, and defining the event as a model bug by fiat forecloses detection of the third: (1) an ill-formed pre-HLS model — for example, reliance on the ordering of causally independent events, cadence-sensitive polling that was not isolated per Corollary L.4, or a snooped signal that is not level-stable — in which case the failure is of the same character as a standalone pre-HLS test failure and the

model is outside the well-formedness envelope of the stress-testing section below; (2) an incomplete snooping harness — the snooped set misses an ε -modeled choice covered by the first premise of Lemma L.2, so the replayed witness is underdetermined; or (3) a non-compliant tool/RTL — the RTL exhibits behavior outside the admissible envelope (for example, an R/B-rule or B-faithfulness violation), in which case the Post $\rightarrow$ Ref result itself fails and the divergence is evidence of a synthesis defect. In a flow in which the harness has been certified against Lemma L.2 and the tool is trusted, the pre-HLS model is the default suspect and the failure should be filed against it first; the formal status of the run, however, is “L.Dir/WC undischarged — outside theorem scope,” not “pre-HLS model incorrect,” until triage assigns the root cause. A watchdog abort under a bound that has not been proved sufficient for the selected instance is triaged in the same way but is recorded as an inconclusive witness-realization failure per Requirement W4, not as a proven L.Dir violation. Benign cycle skew (unchanged Σ -event order with the RTL merely earlier in cycle time) is not an L.Dir violation and shall not be classified as a model defect. Experienced SoC architects tend to view this slightly differently, in terms of design tradeoffs and verification cost.

First, it is usually possible at the architectural level to avoid latency-sensitive behaviors entirely, for example by insisting that all communication be latency-insensitive and fully synchronized. However, the quality-of-results (QOR) costs of such designs may be unacceptable. Introducing latency-sensitive behavior—non-blocking arbiters, latency-sensitive global interrupt signals, and so on—is a deliberate choice made by the designer to meet performance, power, or area goals.

Second, in many practical systems, latency-sensitive behavior is not functionally observable at the DUT boundary under the trace-compatibility predicate $\approx$ with Σ instantiated to include only the DUT-boundary observables (Appendix G; Normative Formal Core above). As an example, consider a DUT that contains a single-port RAM used to exchange data between two internal processes. An arbiter is required to control access to that RAM port, and the arbiter uses latency-sensitive non-blocking Pop operations on its request channels. During HLS, internal latencies will change, so the order in which the arbiter sees pending requests and chooses winners may differ between the pre-HLS and post-HLS models.

In this example:

- The individual RAM read/write operations and the internal arbitration decisions are not directly visible at the DUT interface.
- Only the higher-level IO of the two processes (for example, their message-passing interfaces to the rest of the system) is externally visible.

One might initially conclude that it is necessary to snoop the arbiter’s inputs to make the pre-HLS and post-HLS traces match. However, the modeling rules impose two additional constraints:

1. **Weak fairness for the arbiter.** The arbiter must satisfy the weak fairness assumptions, so that any request that remains pending is eventually granted in both the pre-HLS and post-HLS models.
2. **Causal-dependency discipline on shared memory.** The system must make any functionally significant shared-memory dependence explicit, either through synchronization that orders the relevant accesses or through declared array-access mapping / memory-serialization semantics. This discipline ensures that sufficient synchronization or declared serialization exists between the processes so that there are no read/write races through the RAM in either model.

Under these conditions, the different arbitration orders merely change the relative timing of internal operations and of externally visible actions that are not ordered by the explicit required Σ -constraints. They do not change the FIFO, synchronization, signal-anchor, atomicity, or other required ordering constraints at the DUT boundary. In the terminology of Appendices G and J, the differences are incidental variations in latency rather than changes to the required-order relation $\leq_J$ over the chosen observable units, and therefore they are not considered observable differences in behavior at the DUT boundary. **In such cases, snooping is not required.** (See also Note 1 below).

(More precisely: the trace-compatibility predicate $\approx$ is parameterized by the chosen observable alphabet Σ . If internal functionality (such as the above RAM arbiter) uses channels/signals that are not designated observable in the chosen instantiation of Σ (see Normative Formal Core), then mismatches on those internal actions are outside $\approx$ by construction: they are not in Σ . If those internal channels/signals are designated observable (e.g., by snooping for debug, or by expanding Σ for Appendix K deadlock analysis), then they are Σ -events and must match under $\approx$. In addition, even when B1 holds (unique post-HLS Σ -trace under fixed stimulus), the pre-HLS model may still admit multiple ε -/latency-different executions consistent with the same required Σ -ordering, FIFO, signal-anchor, synchronization, and atomicity constraints; Theorem J.1 captures the intended quantification (“existence of a matching witness execution prefix”) rather than requiring a single pre-chosen trace.)

Experienced SoC architects therefore try to avoid designs in which latency-sensitive internal behaviors leak directly into the observable behavior of the system under test, since this both complicates verification and makes RTL behavior less predictable. When they do introduce such behaviors—examples include non-blocking arbiters whose winners affect externally visible ordering, global interrupt request signals at the DUT boundary—they typically know exactly where those latency-sensitive interfaces are and can isolate them.

A useful analogy is a subsystem whose clock frequency is adjusted dynamically based on on-die temperature. As temperature changes, the externally visible behavior of the SoC can change (for example, in terms of throughput or timing of events), and designers may choose such a temperature-dependent scheme to meet overall system requirements. To verify such a system, we may wish to compare a concrete RTL trace captured at a specific temperature profile with a higher-level reference model. To do so, the reference model must be driven with the **same** temperature evolution as the RTL saw. If that temperature profile is not otherwise under direct control, the simplest way to obtain it is to “snoop” the temperature as observed in the RTL trace and replay it into the reference model. In this analogy, temperature is an external parameter of the environment rather than a fundamental part of the functional specification, but it still influences observable behavior. Snooping simply provides a mechanism to recover that parameter from the implementation when it cannot be easily prescribed a priori. Similarly, snooping of latency-sensitive IO provides a mechanism to supply implementation-specific latency choices — subject to the fairness assumptions and the explicit required Σ -ordering, FIFO, signal-anchor, synchronization, and atomicity constraints — back to the pre-HLS specification so that the two models can be compared under a common environment.

Note 1 (Unobservable non-blocking micro-timing).

Safety / functional equivalence. The trace-compatibility predicate $\approx$ is parameterized by the chosen observable alphabet Σ (Appendix G). If, for safety-only checking, we instantiate Σ to include only DUT-boundary observables (Σ_{DUT}) and we assume that any latency-sensitive internal behavior is not visible at that boundary under $\approx$ —i.e., it may change only internal micro-timing, but it does not change the values or presence of Σ_{DUT} events, and does not change the ordering of causally *dependent* Σ_{DUT}

events—then snooping of those internal latency-sensitive interfaces is not required for proving Σ_{DUT} -safety. Any mismatches on actions outside Σ_{DUT} are outside $\approx$ by construction. When source-reference safety is proved universally, or when one complete pre-HLS run satisfies Definition J.RefProjCover, Corollary J.1-Safe combines that result with Theorem J.1 Post $\rightarrow$ Ref to obtain RTL_B safety at Σ_{DUT} . This common-case transfer does not use the restricted Ref $\rightarrow$ Post clause and does not require snooping.

Liveness / deadlock analysis. For message-passing deadlock analysis (Appendix K WFG), any latency-sensitive interface whose ε -level choices can affect whether a blocking Push/Pop is enabled, or can create/remove a wait-for dependency without an intervening committed Σ -event, must be accounted for. Practically, the default is to snoop and replay every such latency-sensitive (typically non-blocking) arbitration/probe interface in the DUT. Snooping can be avoided only if the design is certified “WFG-inert” with respect to that interface: it cannot change WFG edges except via committed Push_c/Pop_c (or explicit synchronization) events.

Stress Testing Pre-HLS Models for Latency Robustness

The formal trace-compatibility results in Appendices G–K establish that, under the scheduling rules and fairness assumptions, the post-HLS RTL is trace-compatible with the pre-HLS/source-reference model at all observable interfaces.

These results implicitly assume that the pre-HLS specification is *well-formed*: it must not rely on incidental, implementation-specific timing alignments for functional correctness. Designs that do rely on such incidental alignments fall outside the formal guarantees.

Designs that use non-blocking message-passing operations (e.g., PushNB/PopNB, ac_channel nb_read/nb_write) or latency-sensitive global signals are particularly vulnerable to this kind of fragility. For such designs, it is strongly recommended to subject the pre-HLS model to aggressive *latency stress tests* in which message and handshake latencies are varied across executions. The intent is to validate that the design behaves correctly across the range of weakly fair schedules permitted by the modeling rules, not just under one convenient execution order.

More concretely, verification should check that:

- **Causal dependencies are explicit.** All functionally significant causal-dependency relationships are enforced via message-passing, SyncChannel/ac_sync operations, explicit signal handshakes/anchors, declared memory-serialization constraints, or Appendix L snooping/witness-selection boundaries, rather than by relying on the incidental execution order of concurrent processes. In the terminology of Appendix G, the design shall not depend on the ordering of causally independent events for correctness.
- **Weak fairness does not hide latent deadlocks or starvation.** The design continues to make progress under adversarial but *weakly fair* scheduling—i.e., enabled actions may be delayed arbitrarily long but not forever—consistent with the B2/B3 obligations on the scheduler and environment summarized in Appendix N.

If the pre-HLS model fails under such randomized-latency stress scenarios, then the specification itself is outside the formal model of this document: it violates the design rule that well-formed systems must not rely on the relative ordering of causally independent events. In that situation, the trace-compatibility theorems still apply to well-formed traces, but they no longer guarantee that the “golden” specification is robust under the full range of latency variations allowed by the environment.

The Matchlib library provides practical mechanisms to automate these stress tests in pre-HLS simulation, including:

- **Random stall injection on message-passing channels**, to simulate variable communication delays while preserving rendezvous semantics.
- **Latency and capacity back-annotation**, to model specific per-channel buffer depths and delays, including the capacities $B(c)$ chosen during HLS.

Worked examples of this methodology are provided in Catapult Matchlib examples 60* and 72*, which illustrate how to configure randomized stalls and back-annotated latencies when validating that a design remains well-formed and latency-robust.

Appendix M – Formal Results Overview

M.1 Purpose and status

This appendix gives a readable overview of the formal results and the relationship among their principal components. It is informative: the definitions, premises, theorem statements, proof obligations, applicability restrictions, and coverage requirements in Appendices G-L, Appendix N, and the Normative Formal Core remain authoritative. The overview intentionally uses formal names where they help a reviewer locate the controlling result, but it does not reproduce the complete proof API.

The document proves two related but distinct kinds of result. The safety layer establishes ordered observable trace compatibility between the post-HLS implementation model and a prescribed source reference model. The liveness layer excludes a specified prohibited internal wait structure under stronger and more global conditions. Neither layer independently certifies that a particular source design, HLS invocation, generated RTL instance, environment, or back-annotation satisfies the premises required to instantiate the result.

M.2 Models and observation boundary

The implementation-side model is `RTL_B`. The source-side theorem target is the selected `Sys_ref` under the fixed stimulus and the recorded environment, reset, arbitration, witness, and abstraction choices. In the ordinary bounded-channel case, `Sys_ref` is `Sys_B` with capacities `B(c)` faithful to the selected explicit abstract channel boundaries. The rendezvous case is the special case `B(c)=0` at the relevant ordinary boundaries. When the implementation contains staged, bundled, joined, coupled, or synchronization-gated realization structure that cannot be represented faithfully by one ordinary bounded FIFO, `Sys_ref` is `Sys_B^elab` for a well-formed elaboration package as described in Appendix Q.

The safety relation is defined over a selected observable alphabet and an ordered trace-compatibility relation rather than over equality of internal states or equality of cycle-by-cycle execution. Observable units include the committed channel transfers, synchronization events, and anchored signal reads and writes selected by the formal model. Independent units may commute or be grouped into cycles differently when the required ordering, value, atomicity, and observation-boundary conditions are preserved. Proof-internal elaboration events and epsilon steps are hidden or projected as specified by the applicable abstraction map.

Shared-memory communication between processes is covered by the proved safety result only under the J.Mem mapping-layer lowering condition stated in the Normative Formal Core. That condition makes memory-carried dependencies visible through operations already represented in the proof alphabet. A direct global memory-state extension is discussed as a possible future extension but is not part of the present theorem.

M.3 Principal safety result: Post -> Ref

The ordinary safety direction is Post -> Ref. For a covered reachable `RTL_B` prefix, the flow supplies Definition J.LocalGWR or, alternatively, a directly validated J.GWR package with its required provenance and checks. J.LocalGWR organizes design-specific evidence into process-local, explicit-channel, named atomic-interaction, environment/reset, construction-action, footprint, request-snapshot, generated-dependency, local-agreement, and coverage components, together with the applicable channel-capacity faithfulness obligations.

Theorem J.GWR-Comp is the generic local-to-global construction. It generates the dependency structure, proves the required acyclicity and canonical construction order, combines the local preparation scripts, settles shared non-blocking decisions under the registered phase discipline, and

produces one reachable globally witness-realizable Sys_ref package. Proposition J.0 then composes the selected source/RTL pair to establish the ordered observable compatibility required by Theorem J.1. The design-specific premises are primarily local even though the conclusion is system-wide; the global gluing argument is proved once rather than supplied as an unexplained replay premise for each design. Theorem J.GWR-CompExt is the corresponding cross-prefix theorem: under J.LocalGWRExtCert it reuses the already selected source chain and appends only the genuinely new later-Horizon suffix. This locality characterization concerns the finite-prefix J.1 safety construction. In the standard QSync normal-compositional scope, J.2b now exposes a smaller instance-wide safety projection: J.QSyncTraceInd carries explicit J.QSyncTraceStep records containing only the exact cycle-aligned J.LocalGWR/J.LocalGWRExtCert/J.GWR-CompExt and J.HCanon facts and Theorem J.TraceCertCov-QSyncCycle derives J.TraceCertCov on J.CycleReach_post. The richer J.QSyncCertInd used by the universal Appendix K route implies J.QSyncTraceInd but additionally carries the Offer/KObs/drain facts required downstream by K.

The practical safety interpretation is that a covered RTL_B behavior cannot introduce a new canonical observable safety behavior that is absent from the selected source reference model. For source-side safety signoff, Corollary J.1-Safe transfers an extensional safety predicate on CanonHist_ Σ _DUT from the covered Sys_ref prefixes to the covered RTL_B prefixes; Lemma J.OuterCanon-HCanon makes enriched-representative invariance automatic for such a predicate. When one complete pre-HLS execution is used as the source-side verification artifact, Definition J.RefProjCover is the additional coverage fact showing that its observable prefixes represent all covered source-reference observable prefixes relevant to the claimed safety property. In the standard QSync route, J.1-Safe-QSyncCycle removes the need to posit an unrelated route-complete package separately for every completed-cycle RTL prefix once J.QSyncTraceInd has been discharged. Lemma J.CycleIdealCover then shows that every reachable finite canonical Σ _DUT history is a finite ideal of one of those covered cycle-aligned histories, and Corollary J.1-Safe-QSync extends any safety predicate satisfying Definition J.IdealPrefixClosed to all reachable finite RTL_B prefixes. This ideal-closure step does not assert per-prefix J.TraceCertCov; general J.TraceCertCov remains the interface when a claim requires direct correspondence packages for non-cycle-aligned prefixes or uses a broader non-QSync coverage route.

M.4 Restricted Ref -> Post use and witness selection

The converse direction is intentionally not an unrestricted theorem saying that every raw source simulation execution is reproduced by RTL_B. HLS may change latency, cycle grouping, arbitration timing, non-blocking outcomes, and other witness-sensitive choices while remaining within the scheduling contract. Accordingly, direct Ref -> Post transfer is limited to annotated RTL-consistent source prefixes or to source executions selected by the Appendix L snooping and witness-selection mechanism.

This asymmetry is important for interpreting the methodology. The primary safety argument does not require a claim that one arbitrary pre-HLS run is cycle-for-cycle or choice-for-choice identical to the implementation. Instead, it shows that each covered implementation behavior has a compatible source-reference explanation. A particular source run can be used directly in the reverse direction only when the additional annotation or witness-selection conditions establish that it is the run corresponding to the implementation behavior under review.

M.5 Cutpoint correspondence used by liveness

The liveness proof begins from a stronger cutpoint interface than ordinary trace compatibility. Corollary J.1 and CF-0/J.KObsConf connect one selected reachable RTL_B cutpoint to one reachable Sys_ref cutpoint through an equal KObs record. KObs contains the canonical replay history and the complete

attributed residual-offer information needed to reconstruct the relevant frontier and wait-for structure. It is an observation record for the bad-state predicate, not by itself a complete operational state for future Policy-L or Policy-S execution.

For the exact source witness selected by that correspondence, $J.RegPhaseReach$ supplies theorem-derived L0–L4 phase-correct reachability for the $J.GWR$ construction prefix τ_{ref}^G .

$Core.KObsNormRef/K.NormStable$ separately qualify the post-L3 observation-only suffix v_{ref} and produce the K-facing prefix τ_{ref}^K ending at σ_{ref} . $K.DrainReach$ then supplies nonwithdrawal, $K\epsilon$, and activated $Core.Drain/K.Drain$ evidence on τ_{ref}^K . The checked three-part composition is $K.RLAdmReach$, which places the prefix that actually reaches the K-facing cutpoint in the admissibility domain consumed by Appendix K. Keeping these components separate prevents trace correspondence or generic ϵ -normalization from silently assuming the phase, observation-stability, or drain properties needed by the liveness argument.

M.6 Deadlock-preservation result

Primary user workflow. The primary user-facing discharge route is deterministic Policy-S serial reduction. For one fixed selected Sys_K instance, including capacities, elaboration, initial/reset state, stimulus, witness choices, environment behavior, arbitration, concrete pre-HLS scheduler, Policy-S issue semantics, lowering, and normalization, $SDet$ selects one Policy-S execution artifact p_S^{det} . Policy S is conservative: within each source interval, a process does not issue a source-later blocking operation until every required source-earlier blocking operation has committed in a strictly earlier cycle. The flow supplies total finite-bound $K.RespCov$ for every relevant internal process-facing blocking site or dynamic class. For a terminating selected instance, the design user runs this instrumented model to a declared maximal terminal state; the complete artifact must contain no covered timeout, no covered frontier pending at termination, and no non-waived reset-pending failure.

Why the run suffices. $A5-S/CF-S$ combines three separately checked layers for the selected execution: $K.CompleteS$ establishes maximality, fairness, and completeness; $K.RespCov$ establishes total finite-bound response coverage; and $K.RespClean$ establishes that the complete run contains no non-waived $K.RespFail$ witness. Lemma K.2b shows that a prohibited deadlock reached by an admissible Policy-L prefix would force that same selected Policy-S execution to produce a finite response failure. Corollary K.2c therefore derives $A5-L$ when $SDet$, $K.EnvMF$, $K.ResetEpoch$, $K.TerminalDisposition$, and the remaining reduction premises hold. This replaces exploration of alternative Policy-S scheduler interleavings with one selected complete artifact; no confluence theorem is required. For a terminating selected instance, the complete-run response check itself does not require model checking or a hand-written liveness invariant, although the flow must still qualify the selected semantics, finite bounds, applicability, coverage, fairness, and other stated premises. Separately, the proof of Lemma K.2b.E requires an instance-bound $Core.FairPick$ constructive witness for the fixed Policy-L semantics used to build the hypothetical frozen continuation. That witness is not supplied by the observed Policy-S run or by $K.CompleteS$; it is a flow/checker obligation that refines the same behavioral $Core.IssueFair/B2/B3$ requirements into prefix-local selectors and starvation proofs.

What transfers to RTL. Theorem K.1A is first a certified-cutpoint theorem. At a reachable K-relevant RTL_B cutpoint carrying a complete $K.CertifiedCutpoint$ package, the equal- $KObs$ source witness, $K.RLAdmReach$, $A5-L$, and the applicable point-to-point, channel, structural, waiver, environment, progress, and fairness premises exclude the prohibited closed internal wait structure. A universal statement additionally requires $Core.KCutClosure_post$ and $K.CertCov$. In the standard $QSync$ normal-compositional route, $Core.QSync-Post$ derives closure and canonical normalization; Lemma $Core.QSync-KCutCanonical$ shows that every reachable $KCut$ is one of those canonical endpoints; and a checked $J.QSyncCertInd$ base/step invariant feeds Theorem $J.CertExist-QSync$ and Corollary $K.CertCov-QSync$ to

derive total certificate coverage over that domain. For the standard normal compositional discharge, the QC5 drain field of that invariant is supplied by Lemma K.DrainReach-Comp from independently qualified K.Drain/Core.IC/K ϵ and finite-drain/channel evidence combined with J.RegPhaseReach on τ_ref^G , Core.KObsNormRef/K.NormStable on v_ref , and the J/QSync construction's attribution/history and QSync-Ref L2-settling facts; the package-existence theorem does not manufacture the independent liveness evidence. Corollary K.1A-Total then combines both with the per-cutpoint theorem. Thus one clean Policy-S execution discharges the source-side A5-L premise through K.2c; it does not by itself prove the implementation-side QSync or QSyncCertInd obligations, and its finite response bounds are not transferred as RTL latency bounds. Alternative K.CertCov routes remain available where the QSync induction route is not used.

Deadlock observation scope. Appendix K uses a wider proof-internal observation boundary than a DUT-boundary-only safety check. For Appendix K, Σ includes every explicit abstract message-passing channel whose endpoint transfers can contribute a WFG edge, including proof-internal Sys_B[^]elab boundaries where applicable. Each such boundary therefore carries the applicable E4/B-faithfulness and J.OfferReach/J.OfferConf/J.KObsConf evidence used by the KObs bridge. Internal RTL staging already represented within B(c) remains ϵ -level behavior and need not be exposed as a separate channel. Scope, alternatives, and status. Timed, reactive, or otherwise nondefault environment behavior is inside the serial route only when every relevant candidate has exactly the selected internal frontier and satisfies K.JointFreeze-Ord's ordinary/single-release-support premises; an additional frontier or non-singleton/multi-alternative ReleaseOwnerSets support is outside that route. This timed/nondefault K.EnvMF condition may be discharged candidate-by-candidate or once for the fixed instance through Lemma K.EnvMF-Uniform. In the common ordinary non-elaborated primitive point-to-point case, Core.OrdFrontSingleton supplies the at-most-one frontier fact, Dead_K candidate membership makes that unique frontier item internal, and the ordinary Core.WFG rule supplies ReleaseOwnerSets_E(x, σ)={{Q}}; once the remaining K.JointFreeze-Ord premises are qualified, no separate candidate-specific K.EnvMF certificate is needed. Under StandardEnv/B1-avail, K.JointFreeze-Ord-eligible candidates use the theorem-derived lemma, while every other admitted release/frontier structure requires K.JointFreeze_I(D, σ_L ,{f_P}); failure of that certificate likewise puts the candidate outside the serial-reduction route and requires another A5-L discharge. The required K.JointFreeze family may be established candidate-by-candidate or by one stronger instance-level certificate that proves JF1/JF2 uniformly for every admitted non-ordinary candidate. For an ordinary non-elaborated instance whose relevant message frontiers have singleton release supports, this combines with Lemma K.CompareAdvanceLegal-Ord to eliminate both special per-candidate certificates when the comparison-advance premises also hold. The reduction remains conditional: structural, invariant, direct Policy-L, or other A5-L routes remain available only if A5-L is actually true for the selected instance.

Common-case interruption-gate discharge. The reset/terminal applicability conditions do not require enumeration of hypothetical liberal deadlock components when the stronger uniform instance-level condition of Appendix K is proved. One conservative scope may be validated once; if no permitted reset intersects it and declared termination cannot occur while a relevant blocking frontier in that scope remains pending, all required K.ResetEpoch.Continue and nested K.TerminalDisposition.Continue facts follow uniformly. If those stronger instance-level facts are unavailable, the existing candidate-specific nested-gate route remains required.

M.7 Implementation qualification and evidence

The formal proofs specify a contract between source modeling, implementation behavior, and validation evidence. They do not prove that ordinary synthesis automatically discharges the named conditions. A concrete theorem application must identify the selected Sys_ref and abstraction boundaries; establish

source well-formedness and witness conditions; provide the local process, channel, atomic-interaction, environment, reset, footprint, request-snapshot, agreement, and coverage evidence required by the safety construction; and establish faithful realization of every selected channel capacity or elaborated boundary.

For liveness, the application must additionally establish the exact cutpoint correspondence, phase and drain bridge, A5-L and its chosen discharge route, fairness and progress assumptions, nonwithdrawal, applicable waiver rules, environment/frontier scope, point-to-point ownership or explicit arbiter modeling, and universal certificate coverage when the total conclusion is claimed. Tool documentation, generated certificates, static checks, formal proofs, simulation monitors, randomized-latency stress tests, and targeted RTL/interface checks may all contribute evidence, but the evidence must be sufficient for the named premise it is used to discharge.

At the architecture level, the proof separates operational semantics, safety/replay construction, KObs cutpoint correspondence, and complete-run Policy-S evidence. This separation prevents trace compatibility, cutpoint reconstruction, and source-side liveness evidence from being treated as interchangeable assumptions. M.7b gives a standard QSync packaging view over this separation so that a tool or user-facing flow may expose a small number of top-level certificate bundles while M.7a remains the definitive ledger of the individual obligations.

Division of verification responsibilities. The following table separates the principal dynamic design-user activity from per-instance flow evidence, reusable tool qualification, and framework work that remains to be completed. Ownership may be shared in a particular implementation; the table identifies the expected primary source of evidence.

Primary source of evidence	Principal items	Current interpretation
Design user	Run the selected instrumented Policy-S model to its declared maximal terminal state for a terminating instance; review response failures, declared waivers, and design/environment contracts.	This complete execution is the principal dynamic user artifact. It supplies K.RespClean only when bound to the qualified selected instance and accompanied by the remaining evidence.
Per-instance tool / verification flow	Select and bind Sys_ref/Sys_K, capacities, elaboration, stimulus, witness, reset and environment contracts, and discharge the applicable per-instance side conditions recorded in M.7a, including K.PSF/K.EnvMF/K.ResetEpoch applicability, K.JointFreeze where an admitted StandardEnv candidate is not covered by K.JointFreeze-Ord, K.CompleteS, total K.RespCov, B-faithfulness, K.Drain/K _e , Core.FairPick when K.2b.E constructs a continuation, J.TraceCertCov for safety coverage, and J.QSyncCertInd/K.CertCov when the	These obligations are not discharged merely by a clean observed run. They may be generated or checked automatically, but remain theorem premises for the selected design instance.

Primary source of evidence	Principal items	Current interpretation
	corresponding universal K-facing claim is made.	
Reusable tool qualification	Qualify the concrete Policy-S simulator semantics and deterministic scheduling/arbitration used by SDet; validate standard lowering/elaboration schemes, certificate schemas, generators, and independent checkers.	Much of this qualification can be reusable across designs. SDet and the selected certificate binding must nevertheless be re-established whenever capacities, elaboration, lowering, arbitration, normalization, environment/reset behavior, or another history-affecting choice changes.
Framework / mechanization work not yet completed	Mechanize the ReleaseOwnerSets reconstruction/closure obligations and the Appendix K-P obligations, including K.JointFreeze/K-O2 and K-O3/K-O4 for freeze preservation, progress dominance, and request reachability; mechanize and qualify Core.FairPick/Core.FairCycleDovetail for proof-built continuations; qualify the certificate interfaces; and mechanize/qualify the J.QSyncCertInd -> J.CertExist-QSync -> K.CertCov-QSync theorem route (or an alternate verified package-existence route) for concrete universal-coverage flows; define/qualify reusable concrete Core.MemProfile library patterns (with ping_pong_mem/native_ram_fifo as intended first candidates) and independent checks for their transformation-invariant J/K boundaries. The first arbitrated-memory qualification should explicitly determine whether a transformation-invariant, K-faithful boundary exists for the declared finite queue/credit realization; if it does not, the qualified covered profile shall disable the incompatible transaction-changing transformations rather than treat profile construction as a drafting exercise.	The paper now supplies the generic QSync package-existence theorem route and the corresponding paper-level soundness arguments for the current formal interfaces. The remaining work is mechanization, independent checking/tool qualification, and concrete discharge of the design-wide refinement/fairness/freeze obligations; the reduction and certificate stack therefore retain proposed, not fully mechanized and flow-qualified, status. Concrete qualified memory-profile instances are likewise reusable qualification work; until bound and checked, they do not enlarge the theorem-covered design class, and a first qualification may validly conclude that particular transaction-changing transformations must be disabled for the K-covered instance.

Interpretation of the discharge ledger. M.7a is an explicit accounting of the conditions required for the selected formal claim; it is not a list of separate manual tasks imposed on the design user. Many entries formalize engineering conditions or verification assumptions that were largely already present in either an HLS-generated or manually written RTL flow but might otherwise remain unnamed or implicit. The ledger makes those dependencies visible, assigns responsibility for establishing them, and permits them to be checked, reused, packaged, or qualified systematically. Depending on the selected theorem and instance, an entry may be inapplicable, satisfied by a simple design restriction, theorem-derived, discharged through reusable tool or library qualification, or established automatically by the per-instance flow. The methodology may require more explicit evidence and accounting than an informal flow, but that additional explicitness should not be confused with newly creating the underlying engineering burden.

M.7a Side-condition discharge record

Before applying Theorem J.1, Corollary J.1 cutpoint transfer, or the Appendix K liveness-preservation theorem to a concrete design, the verification flow shall maintain an explicit side-condition discharge record. The record need not use one physical file, but every referenced artifact shall be versioned and bound to one selected theorem instance through the common certificate header of Definition K.CertHdr. For each theorem premise or separately named qualification obligation identified below, the record shall state whether the item is inapplicable to the selected theorem or flow-qualification scope, satisfied by a simple syntactic/design restriction, discharged by tool/RTL evidence, or discharged by a source/RTL witness construction. A premise or qualification condition required by the selected theorem or flow-qualification scope may not be marked inapplicable. Remark Core.Det uses this same evidence axis; derived results such as I.DR or theorem-derived Core.KCutClosure_post are not converted into independent assumptions merely because their proof artifacts are recorded. The named audit entries are Core.RegSeq registered sequential semantics; Core.ReachPrep overlap-aware source reach/preparation; Core.SelectLatitude finite-prefix source-selection/no-maximal-progress conformance; ClockScope single-global-clock theorem scope; Core.QSync qualified synchronous same-cycle settling; Core.KObsSettleQualified post-L3 observation-only source K-facing settling when triggered; Core.CycleClosure full-state source cycle qualification/closure; Core.FairPick constructive fairness witness for proof-built source continuations; Core.IssueFair source-issue weak fairness; B2/B3 scheduler/channel progress; B5/B6-Terminal quiescence-closure/terminal-idle adequacy; B1 observable RTL projection determinism; B4/FPD finite internal stutter/bounded local source progress; A6 point-to-point ownership/explicit arbitration modeling; A8/A9/A10 barrier/cardinality/reset-empty well-formedness; R/E scheduling-contract conformance (R1–R5/E1–E5, including R2C where applicable); Core.IC request-lifecycle/Issue–Commit conformance; Core.SyncFence synchronization-completion conformance; K.CertPkg certificate-package integrity; WC witness-compatible ϵ choices/non-blocking outcomes; J.LocalGWR local witness-construction certificates and optional J.GWR-Direct; J.LocalGWRExtCert/J.GWR-CompExt exact cross-prefix source-witness extension when incremental cycle induction is used; Sys_B^elab; B-faithfulness; J.QSyncTraceInd cycle-aligned safety package induction; J.TraceCertCov Post $\rightarrow$ Ref safety certificate coverage; J.OfferReach/K.NormStable/J.OfferConf/J.KObsConf KObs cutpoint correspondence; K.RLAdmReach admissibility of the exact selected source K-facing prefix; Core.KCutClosure_post theorem-derived K-facing cutpoint existence/non-vacuity for universal RTL conclusions; J.QSyncCertInd canonical-cycle correspondence/package induction for the standard QSync coverage route; K.CertCov total K-ready package coverage for universal reachable-cutpoint conclusions; K.Drain; K.BF; K ϵ ; SDet deterministic selected Policy-S simulator; A5-S selected-instance serial liveness; A5-L scheduler-robust liberal liveness; K.CV value determinism/provenance; NB-Align witness-selected non-blocking alignment; K.Low witness-indexed blocking lowering/correspondence; K.EnvMF together with K.JointFreeze-Ord/K.JointFreeze for

frozen-component preservation; K.CompareAdvanceLegal dominance-comparison Core.16 launch/issue compatibility for every K.2b.P0/K.2b.R-consumed owner/key; K.InterruptionGate with K.ResetEpoch/K.TerminalDisposition; K.Resp validated bounded-response coverage; optional K.EnvOrd/K.Done diagnostics; W.Env waivers; L.Dir one-directional alignment/RTL-side non-interference; Core.MemFaithful mapped-memory boundary faithfulness; Core.MemSyncOrder array/synchronization commit-order conformance; J.Mem shared-memory lowering; Core.MemProfile qualified memory mapping profile when used; and RW reset well-formedness (RW1–RW5), as applicable to the selected theorem scope. M.7b below provides a standard logical packaging of these entries for QSync flows; it does not alter this discharge record or change the trigger, discharge, or evidence requirements of any entry.

R/E scheduling-contract conformance — R1–R5/E1–E5, including R2C where applicable.

Trigger: Required for every theorem or flow-qualification scope that invokes the Appendix G–K scheduling contract. Apply the applicable R1–R5 source rules to the selected Sys_ref behavior and the applicable E1–E5 implementation/compatibility rules to the compared RTL_B behavior at the selected abstraction and observation boundaries. The R2C fixed-point semantics and matching obligations are required whenever combinational processes or explicitly modeled combinational networks are in scope. Discharge: Bind the concrete source/tool/RTL instance to the applicable R/E rules. Establish the required source dynamic-operation/source-order and synchronization/signal-anchor legality; post-HLS synchronization-order preservation; sequential signal visibility/anchoring or R2C fixed-point observation and matching; prefix-closed message issue order; E4 explicit-boundary/channel-capacity legality; and E5 synchronization-boundary issue/completion ordering. For ordinary scalar blocking operations on one endpoint, E3(i) and the same-interface portion of E3(iii) may cite the separately discharged Core.IC/BoundaryReq qualification from which that Issue order is derived; E3(ii) and the cross-interface prefix-closure content remain independent implementation scheduling checks. The record may cite constitutive Sys_ref semantics where a source rule is built into the normative operational model, but corresponding RTL_B uses shall be backed by tool/RTL/checker evidence or a proved refinement/conformance result. This roll-up does not replace separately named entries such as Core.RegSeq, Core.SelectLatitude, Core.IC, Core.SyncFence, B-faithfulness, Core.MemFaithful, Core.MemSyncOrder, J.Mem/Core.MemProfile, or RW; those remain separately discharged when triggered.

Evidence examples: source well-formedness/lint and simulator-semantics qualification; HLS schedule and issue-order certificates; RTL/interface assertions or formal checks for synchronization and signal anchoring; an R2C combinational fixed-point/unique-settling certificate; and explicit-boundary E4/capacity conformance checks.

Core.RegSeq — registered sequential-interface scope and conformance.

Trigger: Required for every Appendix I/J/K theorem instance containing a sequential process whose message requests, Push payloads, or continuation can depend on a message result.

Discharge: Classify each process as R2C combinational or Core.RegSeq sequential. For each sequential process, prove or validate that process-driven request and Push-data outputs for Horizon t are determined from registered state and operands stable before the L2 settle point; that direct inputs satisfy their E2/direct-input stability contract; and that every use of a cycle-t returned data/status result has strictly later Horizon. Tool-side Cat# or equivalent schedule/RTL evidence and source-side simulation, lint, wrapper, or Appendix L evidence shall identify and reject, isolate, or explicitly remodel zero-time NB-result cascades and retry loops. Evidence examples include schedule reports, control-step allocation, generated RTL cone/register checks, source lint, and cycle-accurate transactor or snooping artifacts.

Core.ReachPrep — overlap-aware source reach/preparation.

Trigger: Required whenever the selected Sys_ref/Policy-L theorem instance permits a source-

later dynamic operation to become reached before an earlier same-interval blocking operation process-facing commits, including the straight-line eager cases used by Examples K.CE/K.CE-2, the Appendix K A5-L blocking-rendezvous example, and pipelined/overlapped cases.

Discharge: Identify the source/tool rule or certificate that validates each permitted finite dependency-independent L1 preparation segment under Definition Core.ReachPrep, including dynamic source order, no synchronization crossing, Core.RegSeq/NoDependentUse dependencies, and the Core.16 post-settle prohibition. For every RTL_B prefix included in a Post → Ref coverage claim, the record shall verify that the selected Sys_ref ReachPrep semantics admit every dependency-independent preparation/reach segment required by the accepted J.LocalGWR or J.GWR construction for that prefix; the source semantics may be more permissive than the observed RTL schedule but shall not exclude preparation needed to realize a covered RTL witness. The record shall state explicitly whether ordinary straight-line q0; q1 eager reach is admitted. If it is not admitted for an instance, examples or proof arguments that rely on that straight-line liberal behavior may not be used unchanged.

Evidence examples: a schedule/preparation certificate, source dependency/footprint check, a qualified eager-source simulator semantics, or a tool-generated mapping from RTL issue stages to legal Core.ReachPrep segments.

Core.SelectLatitude — finite-prefix source-selection latitude / no-maximal-progress conformance.

Trigger: Required as a source-model qualification for finite-prefix Post → Ref construction or any other covered use that may postpone an otherwise legal L1 issue or L3 boundary action. The normative abstract Sys_ref discharges this condition definitionally through Core.SelectLatitude; a concrete executable source/reference model used as the theorem carrier shall establish conformance rather than importing Core.IssueFair or B2 into the finite-prefix theorem. The rule applies under both Policy L and Policy S.

Discharge: Establish that the selected source operational semantics has no forced maximal-progress rule at L1 issue selection or L3 boundary-action selection: an otherwise legal action may remain unselected at a finite opportunity while Core.16, Core.SyncFence, R/E/E4/E5, reset, arbitration, atomicity, endpoint-cardinality, and fixed witness/environment choices remain authoritative. Show that such nonselection is ordinary Core.LLegal/Policy-S cycle legality, preserves issued-request nonwithdrawal/attribution, and can be represented by the applicable legal CycleResult/CompleteCycle, including unit-empty SilentCycle cases when the fixed semantics admits them. Absence of issue shall not be used as evidence of Core.16 activation. Core.IssueFair and B2 remain separate complete-execution eventual-selection obligations and are not required merely to construct a finite J witness. On the deterministic Policy-S route, SDet separately fixes the selected serial execution; SDet is not part of Core.SelectLatitude and does not turn finite-prefix J into a fairness theorem. For an executable reference simulator, evidence shall also show that no hidden eager/maximal-progress implementation rule removes the finite deferral behavior required by an accepted J.LocalGWR/J.GWR construction.

J.RefProjCover interaction: Core.SelectLatitude enlarges the admissible source latency realizations. A one-run J.RefProjCover discharge therefore needs separate coverage/latency-invariance evidence when different legal selection delays can change the canonical Σ_{DUT} history; pure unit-empty finite delay whose canonical history is unchanged is already absorbed by the J.HCanon/stutter treatment.

Evidence examples: a theorem for the abstract Sys_ref transition semantics; source-simulator operational-semantics qualification; a checker showing that L1/L3 selection may be non-maximal while all legality constraints are preserved; or a tool-generated finite-deferral/construction witness bound to the selected theorem instance.

ClockScope — single-global-clock theorem scope.

Trigger: Required for every theorem instance using the formal results of Appendices G–K.

Discharge: Confirm that the selected theorem instance is interpreted under one global clock/cycle relation for every process and explicit abstract channel to which the G–K rules are applied. A larger multi-clock design may be partitioned into synchronous islands, but any clock-domain crossing used by the selected theorem instance must be excluded from the single-clock comparison or represented by a separately specified and verified CDC/refinement boundary under an explicit multi-clock semantic layer. In the absence of that layer, the G–K theorems shall not be applied across clock domains. Evidence examples include a clock-domain/elaboration manifest, structural CDC partition report, or an equivalent checked instance-scope declaration.

Core.QSync — qualified synchronous same-cycle settling scope and conformance.

Trigger: Source-side Core.QSync is required on the normal J.LocalGWR $\rightarrow$ J.GWR-Comp $\rightarrow$ finite-prefix Theorem J.1 route, because J.RegPhaseNF/J.UnitExt form a finite L2-settled source snapshot for every constructed Horizon. A directly validated J.GWR package may instead carry independently checked finite L2-settled segments for its Horizons and need not separately discharge global source-side Core.QSync solely for Theorem J.1. Post-side Core.QSync is required whenever the application invokes post KCut/NormPost settling, including Corollary J.1 cutpoint transfer and every Appendix K application. Any R2C combinational-process fixed-point obligation remains independently applicable.

Discharge: Establish the hypotheses of Definition Core.QSync for the concrete instance. Use the ClockScope entry as the concrete audit evidence for QS1. For the source side, show the Core.Settle L2 frame, stable sequential drives under Core.RegSeq/direct-input contracts, inclusion of R2C combinational components in the settle network, QS3 relation-to-network realization/adequacy for every phase-legal same-cycle settling step, finite acyclic deterministic cycle-local settling, and explicit-boundary completeness. For Sys_B^{elab}(E), cite the Appendix Q qualified-settling/template evidence required by Q.1(i) for the elaboration-owned subnetwork and additionally establish template/design settling composition for the complete source network. Every dependency edge crossing the design/template partition shall be included in the whole-network analysis, the combined transition relation shall satisfy the QS3 realization/adequacy condition, and the combined network shall satisfy QS2–QS5; no combinational settling cycle may be introduced across the partition. For RTL_B, establish the analogous K-origin frame, stable registered drives/cycle-t inputs, QS2a origin recoverability for every reachable post KCut in the standard K-facing domain, QS3 relation-to-network realization/adequacy for the complete K-facing same-cycle settling relation, finite acyclic deterministic combinational/handshake settling, and complete selected-boundary mapping/B-faithfulness. The QS2a evidence shall bind the designated same-cycle state/origin/epoch triple through Core.RTLDesignatedOriginFrame_I(s_t,o_t,e_t) and show that the reachable suffix from o_t to the KCut contains only the qualified settling relation before any later registered update; quiescence alone is not accepted as origin recovery. When J.QSyncTraceInd or J.QSyncCertInd is used, the record shall also bind every admitted exact initial carrier node (κ_0, s_0, o_0, e_0) needed by the covered reachability carriers, show that all such nodes share the theorem-instance prescribed initial/reset state and e_0 post-initialization/reset epoch valuation, and expose the corresponding Core.RTLCycleStep records (or an equivalent representation from which all RS1–RS6 fields are mechanically reconstructible). The checker shall verify that each endpoint is the designated frame boundary, that RS6 means no intervening designated origin rather than no arbitrary OF1–OF4-compatible microstate, and that RS5/KC3 thread the exact epoch valuation into the next node. Lemmas Core.KOriginCarrier-AtOrigin/-Decomp then derive the finite historical carrier used by the induction; this is typing/provenance, not a new cross-

cycle progress premise. The checker shall validate enough of these structural facts to invoke Theorems Core.QSync-Ref/Core.QSync-Post.

Evidence examples: a syntactic/design restriction plus source/elaboration template proof; an independently checked combinational-dependency/clock/register-frame certificate; a combined dependency-rank or template/design interface-compatibility certificate covering cross-partition edges; structural RTL/netlist analysis; a qualified template-library theorem; or equivalent tool/RTL evidence. For QS2a specifically, sufficient evidence may be a checked RTL cycle/frame-decomposition theorem or certificate proving that every reachable post KCut in the selected K-facing domain recovers exactly one designated Core.RTLDesignatedOriginFrame_l(s_t,o_t,e_t) in the same cycle, that the registered-state/cycle-input roots fixed by QS2 remain unchanged on the recovered suffix, and that every boundary-relevant change on that suffix is confined to the qualified QS3 settling relation before any later registered update. QS2a-specific evidence examples include a structural RTL control/register-frame checker, an inductive invariant over the concrete RTL transition system establishing that designated-frame decomposition, exhaustive finite-state checking, or a separately verified tool-generated frame-origin certificate. Trust accounting follows K.CertHdr: an unverified tool report used as an axiom enlarges the declared trusted base, while an independent check of the concrete artifact does not require trusting the producing HLS tool for that fact.

Scope boundary: Core.QSync is same-cycle L2/RTL settling evidence only. It does not by itself discharge the separate Core.KObsSettleQualified source observation obligation or Core.CycleClosure full-state/source-cycle/environment definedness, B2/B3 cross-cycle progress, or the terminal Core.EnvTerminal premise.

Core.KObsSettleQualified — post-L3 observation-only source K-facing settling qualification.

Trigger: Required only when a Corollary J.1 / Appendix K source K-facing package obtains σ_{ref} by Core.KObsNormRef from a J.GWR post-L3 construction endpoint. It is not a premise of Theorem J.1, J.QSyncTraceState/J.QSyncTraceInd, J.TraceCertCov-QSyncCycle, J.CycleIdealCover, or the standard safety-only M.7b profile.

Discharge: Bind the exact source model/elaboration and verify KSQ1–KSQ4 of Definition Core.KObsSettle: the qualified stateless boundary equations/topology remain valid under each selected post-L3 architectural/request root; the L3Obs observation subphase disables all non-observation ϵ /actions; the observation network is finite, acyclic, deterministic, and complete for the K-facing boundary values; and its fixed-point outputs cannot feed back into the already-closed L3 selection, J.BufSameCycle, or E4 fall-through decisions. On the standard normal-compositional K-facing route, this record together with source Core.QSync permits Lemma Core.KObsSettle-Ref to construct the recorded Core.KObsNormRef suffix. The K.NormStable audit remains separate and checks replay/request/NB/reset invariance of the exact v_{ref} path. Evidence examples: a source/elaboration phase-semantics theorem; a checked post-L3 combinational dependency graph/topological order over the same declared coupler/handshake equations; and a checker proof that the post-boundary L3Obs subphase has no transition back into L3 selection.

Core.CycleClosure — full-state source cycle qualification and closure. Trigger: Required when a theorem constructs a complete/fair source continuation through Core.LAdm, notably Lemma K.2b.E/K-O7. It is not required by finite-prefix Theorem J.1 or by RTL K-facing cutpoint closure. Discharge: Bind Definition Core.CycleResult and its Core.CompleteCycle projection to the selected source/environment semantics; prove the Core.CycleClosure record: full-state CycleResult typing, nextState clock/environment/auxiliary advancement, exact scoped RESET emission under RW1-RW5, nonterminal source-cycle existence, finite L1 preparation, finite L2 settling (Core.QSync-Ref or an allowed equivalent route), compatible EnvStep definedness, and

the C2 partial-cycle extension property needed by Lemma Core.CycleChoiceExt: every finite phase-legal selected issue/boundary segment used by a continuation proof must admit completion to a typed CycleResult. Verify that source-side SilentCycle is a CycleResult with no selected Σ commit rather than an ε /identity step, and that any proof-internal RESET remains present in resetUnits. For Lemma K.2b.E, C2 is used only through the two-stage Core.FairCycleDovetail construction; CycleClosure itself does not discharge fairness or legalize an L3 action chosen before the actual L2 settled state is known. Evidence examples: mechanized source cycle semantics with a cycle-completion lemma; source FPD/preparation plus QSync certificate and checked partial-cycle extension; environment state-machine totality proof; or a qualified source-model theorem whose concrete artifact is independently checked. Generalized-drain route note: full Core.CycleClosure is a sufficient but not universally mandatory discharge of Core.InFlightCycleExt. The profile-G1 no-lowering serial route already carries it; an alternate A5-L route may instead prove the restricted in-flight profile directly.

Core.FairPick — constructive fairness witness for proof-built source continuations.

Trigger: Required when Lemma K.2b.E or another theorem constructs a complete/fair Policy-L execution from a finite prefix rather than consuming an already supplied execution. It is not separately required merely to state Core.IssueFair/B2/B3 of an existing run.

Discharge: Provide the FP1–FP5 IssuePick/BoundaryPick witness, local obligation finiteness/keying, two-stage settle-then-boundary applicability, and starvation proof for the particular fixed theorem instance. Distinguish proof-selectable Policy-L issue latitude from choices fixed by Core.LLegal/I. A round-robin/oldest-enabled selector may be introduced only for genuinely proof-selectable issue choices; any fixed scheduler component, arbitration/grant, environment, witness, reset, or other bound choice must be followed and proved fair enough for FP4/FP5. This is stronger constructive evidence than an abstract fairness assertion, but it does not strengthen the behavioral fairness property being established.

Evidence examples: mechanized dovetailing over proof-selectable Policy-L issue choices plus a rank proof for fixed arbiters; a qualified source operational-semantics theorem returning prefix-local fair selectors; or an independently checked scheduler/arbiter certificate exposing the required selection and starvation ranks.

Core.IssueFair — weak fairness of source issue.

Trigger: Required for complete Core.LAdm executions consumed by Lemma K.2b.E/K.2b/K.2b.R and wherever the Policy-S route invokes issue fairness.

Discharge: For each relevant reached operation, prove that if its issue transition remains legal at every applicable L1 opportunity within the reset epoch, it is eventually issued. The legality predicate must include Core.16; an operation blocked as source-later is not IssueEnabled and therefore creates no IssueFair obligation. Intermittently legal operations do not trigger this weak-fairness antecedent. When the theorem constructs the complete run itself, notably K.2b.E, the separate Core.FairPick entry supplies the required prefix-constructive issue-selection witness; abstract fairness of some already-given execution is insufficient.

Evidence examples: simulator scheduler proof, source operational-semantics theorem, fairness assertion over issue-enable state, or mechanized dovetailing argument.

B2/B3 — boundary-action and channel progress.

Trigger: Required whenever a complete/fair continuation is used by Lemma K.2b.E, the K.2b dominance/response arguments, K.CompleteS, or another theorem that explicitly requires such a run. Appendix I.5 and the finite-prefix J.1 correspondence do not require B2/B3 merely to construct finite replay prefixes.

Discharge: Record the concrete scheduler/arbiter and channel evidence establishing B2 weak fairness for continuously ChannelEnabled Push/Pop/Sync actions and the B3 P_push/P_pop

channel-progress obligations on every relevant explicit abstract channel. This entry is distinct from `Core.IssueFair`, which concerns request assertion before a boundary action is enabled. This obligation includes arbitration introduced by HLS itself, such as resource-sharing, memory-port, or interface/channel-selection arbitration, whenever that arbitration controls selection among B2-covered boundary actions. HLS-generated arbitration need not be reproved manually by each design user: when applicable, the discharge may cite reusable tool/vendor or qualified-library evidence for the generated arbitration class, bound to the concrete theorem instance; otherwise the per-instance flow must supply equivalent RTL/checker evidence. When K.2b.E constructs the continuation, the separate `Core.FairPick` entry supplies the boundary-selection/B3-exposure witness and binds it to the actual fixed scheduler/arbiter; a proof may not replace a fixed unfair arbitration choice with an invented fair one.

Evidence examples: fair-arbiter RTL proof, ready/valid liveness assertions, channel-library qualification, or a mechanized scheduler invariant.

B5/B6-Terminal — quiescence closure and terminal idle adequacy.

Trigger: B5 is required where its bounded quiescence/normalization role is invoked. The B6 terminal-idle branch is required only when an infinite execution is extended by idle ticks after a permanently idle state.

Discharge: Prove the B5 closure premise for the selected bucket semantics and, for any B6 suffix, prove `Core.TerminalIdle`: no future registered/internal machine transition and no permitted future environment evolution can change non-clock state or re-enable work.

`Core.EnvTerminal` supplies only the environment component; `Core.CycleClosure`'s environment-definedness clause or `Core.EnvTerminal` alone is not sufficient. `StandardEnv` stimulus exhaustion plus a machine-side stable-terminal/permanent-idle proof is one sufficient route. A currently idle but nonterminal state must instead continue through `Core.SilentCycle`.

Evidence examples: bounded quiescence proof plus stimulus-exhaustion and stable-machine certificate, explicit terminal state with no registered successor effects, or a model-checked permanent-idle invariant.

B1 — observable RTL projection determinism. Trigger: Required wherever B1 is a theorem premise, including the finite-prefix J.1 correspondence and Appendix K routes that cite RTL observable determinism. Discharge: Under the fixed initial state and stimulus, prove uniqueness of the RTL_B observable Σ projection required by Definition B1. Internal ϵ paths, issue annotations, request attribution, and next-cycle totality need not be unique; those stronger properties are discharged separately. Evidence examples: deterministic RTL semantics/refinement proof, qualified simulation theorem, or formal trace-projection uniqueness check.

B4/FPD — finite internal stutter / bounded local source progress. Trigger: Required wherever Lemma I.1, finite replay/preparation normalization, B5 charging, or the generic source-cycle finiteness argument relies on FPD/B4. Discharge: For each relevant process P, provide the finite D_P bound (or an equivalent well-founded measure) showing that consecutive internally advancing ϵ -work reaches the next Σ action or a stable external/peer wait in finitely many steps. A certified Lemma I.3 preparation segment may cite its local certificate; generic `Core.CycleClosure` use requires the corresponding uncapped per-cycle source-control finiteness evidence for its source-cycle-existence clause. Evidence examples: finite pipeline/FSM-depth proof, source/control graph ranking, simulator semantics theorem, or tool-generated progress certificate.

A6 — point-to-point source ownership / elaborated dependency projection. Trigger: Required wherever Appendix K constructs process-level WFG edges or consumes ordinary complementary-owner reasoning. Discharge: show one producer process and one consumer

process for every source-level logical message channel, remodel shared source resources through explicit arbitration, and, for $\text{Sys_B}^{\text{elab}}$, provide the Q.1 ReleaseOwnerSets family together with its WaitOwners union, including the family-level support-soundness/no-spurious-rescue and support-completeness/no-missed-rescue checks under the declared template semantics. Internal stateless coupler endpoints are not required to be modeled processes. Evidence examples: source/elaboration ownership manifest, structural connectivity check, arbiter-lowering certificate, and the qualified elaboration-template release-support/WFG contraction proof.

A8/A9/A10 — barrier, endpoint-cardinality, and reset-empty well-formedness. Trigger: A8 is required whenever SyncChannels/barriers are omitted from the internal Push/Pop WFG; A9 is required for scalar per-cycle endpoint semantics; A10 is required for buffered channels at initial/reset boundaries. Discharge: For A8 prove matched barrier participation/no permanent bypass under the fixed stimulus; for A9 prove at most one Push and at most one Pop commit per channel per cycle unless an explicit multi-lane model is used; for A10 prove logical buffered occupancy is zero at each selected reset boundary or explicitly model any preload. Evidence examples: source/elaboration checks, barrier-count/participation proof, interface cardinality assertions, and reset-state/channel initialization checks.

Core.IC — request lifecycle / Issue—Commit conformance.

Trigger: Required on RTL_B whenever the selected theorem or certificate uses implementation-side Issue/BoundaryReq/Commit attribution or blocking-request persistence, including the applicable E3/E5 trace obligations, J.OfferConf/J.KObsConf request abstraction, K.Drain, or K.BF. On Sys_ref , Core.IC is constitutive operational legality and is not a separate premise to be assumed or independently discharged.

Discharge: For every admitted explicit boundary and endpoint direction in the selected scope, establish the applicable Core.IC lifecycle and request-identification rules. First, the selected boundary mapping shall satisfy Core.10's fixed, bidirectionally complete request-identification rule and exhibit the implementation request condition/owner decode; an arbitrary physical ready/valid level caused by complementary channel/storage availability or declared coupler logic is not a source request. For blocking operations, also discharge Core.IC(1)'s general live-owner invariant; once issued that process-owned request remains asserted with stable Push payload until its process-facing commit or an affecting RW2 RESET; no hidden nonterminal cancellation, withdrawal, retry, replacement, or reattribution is permitted. After blocking q_0 commits, if the same process-owned request remains asserted in the next cycle and q_1 is the next same-direction blocking source operation in that source interval, q_1 owns the continuing request and $\text{Issue}(q_1)$ is that next cycle; ownerless blocking-request latency is not an attribution option. The scalar same-endpoint serialization rule and synchronization-boundary attribution/Core.SyncFence rules must also hold. For ordinary scalar blocking E3(i), this Core.IC/BoundaryReq discharge is the implementation evidence from which the same-interface Issue order is derived; it does not discharge the distinct-interface no-reverse or cross-interface prefix-closure checks. A selected commit must also satisfy Core.8/Core.11/E4; Core.17/ ΣMatch additionally catches any resulting committed ordinal/value mismatch, while ungranted cycles rely on the fixed request-identification evidence itself. Evidence examples include an explicit-boundary/interface mapping, RTL/interface assertion or monitor, structural analysis, a formal refinement check, or a checked request-attribution/ordinal certificate tied to the selected boundary and dynamic per-endpoint source sequence. A protocol requiring an additional nonterminal request disposition is outside the present Core.IC model unless that operational extension and its correspondence obligations are separately defined and proved.

Core.SyncFence — synchronization-completion conformance.

Trigger: Required whenever a synchronization call s selected into S can have a blocking Push/Pop q in $\text{pref}_S(s)$ on a side of the comparison. The obligation is vacuous for a synchronization boundary with no such blocking predecessor.

Discharge: Establish the applicable R3/E5/Core.SyncFence condition that every such q process-facing commits no later than s and that no issued-but-uncommitted q survives the synchronization into the succeeding source interval. For source operational semantics, a same-cycle q/s completion is represented by the selected L3 boundary-action batch with q accounted for before the interval update. For RTL_B, no physical sub-cycle ordering is required: the implementation evidence must establish the same commit cycle and the certificate/annotation convention must attribute the predecessor completion before advancing the formal source-interval bookkeeping. In $\text{Sys}_B^{\text{elab}}$, bind q to the designated process-facing completion event rather than an arbitrary internal staging transfer. Evidence examples include schedule/control-state evidence, an RTL/interface monitor or assertion, a structural/formal refinement check, or a checked process-facing completion map in the elaboration certificate.

K.CertPkg — certificate identity, schema, and trust boundary. Trigger: Required whenever a generated or manually assembled certificate package is used to discharge Corollary J.1, Theorem K.1A, Corollary K.1A-Total, Theorem K.1, A5-S/A5-L, or CF-1 through CF-5. Discharge: Provide the common K.CertHdr fields: certificate-schema and document/theorem version; a digest or unambiguous identifier for the selected source, $\text{Sys_ref}/\text{Sys_K}$, and RTL artifact; E, B_E , explicit boundaries, projections, ownership, and K.Low lowering where applicable; fixed stimulus, environment, reset policy, witness w , and WC provenance; KObs-facing waiver set W and closure evidence; the selected A5-L discharge route; the Core.QSync structural-evidence provenance, the separately triggered Core.KObsSettleQualified source observation record, and the Core.QSync-Ref/Core.KObsSettle-Ref/Core.QSync-Post theorem references used to derive standard L2 settling, source post-L3 observation normalization, and RTL K-facing closure, together with the K.CertCov coverage route when a universal reachable-cutpoint conclusion is claimed; and, for the Policy-S route, the concrete deterministic simulator/scheduler, deterministic arbitration, Policy-S issue semantics, ϵ -normalization identity, including the recorded Core.KObsNormRef/K.NormStable source suffix when CF-0 is K-facing, SDet evidence, K.Resp monitor schema, finite bounds, reset/end-of-test disposition, response waivers, and coverage-preservation evidence; plus the theorem scope, clock-scope, Core.IC, and Core.SyncFence side-condition references where applicable, other side-condition references, generator/checker versions, and declared trusted base. Every subordinate KObs or A5 certificate shall cite the same header identity. Derived K-facing predicates and K.Resp outcomes shall be recomputed by the checker from their respective artifacts rather than accepted as authoritative fields. A QSync property accepted only from an unverified tool report is part of the declared trusted base; a property independently checked from the bound artifact is recorded as checked evidence instead. Evidence examples: A signed or hashed manifest binding all artifacts, an independently checked structural QSync certificate, qualified checker/tool versions, and reproducible source/RTL identities. Generalized-drain addition: bind Q.2 action/effect schema/version, K.InFlightQual model/profile identity, finite-choice/rank proof, Core.InFlightCycleExt or equivalent route, J.InFlightPathSound, declared operational precision/completeness, and any exact/partial checker-search status. A certificate created under the former offer-only DrainClosedNow schema remains interpretable only against the earlier document/theorem version recorded in its K.CertHdr; it does not establish the generalized-drain theorem. This revision defines no current-version legacy-fallback theorem for lowered instances. A lowered certificate can support a current generalized-drain claim only after the Stage-2

lowered qualification exists and the certificate is newly issued or independently upgraded/rechecked against that then-current schema and its modal K.Low obligations.

- **WC** — witness-compatible ϵ choices / non-blocking outcomes.
 Trigger: Required when failed non-blocking polls or other ϵ -modeled choices can affect later Σ -visible control flow, frontier membership, request attribution, guard values, payload values, or other later Σ -visible choices.
 Discharge: Provide evidence that proves the actual WC premise used by Appendices I–J. The audit entry shall identify the provenance route by which WC was established:
 - NB-CF route: show that failed PushNB/PopNB outcomes do not affect later Σ -visible control flow, frontier membership, request attribution, guard values, payload values, or other later Σ -visible choices. Under this route, the failed-poll witness may be the identity or arbitrary selector, subject to the applicable latency-sensitive local-protocol conditions.
 - Appendix L snooping route: provide a witness selector W satisfying Lemma L.2. Show that the snooped outcomes cover every ϵ -modeled source choice that can affect later Σ -visible behavior, that the wrapper changes only ϵ -choice selection, and that it does not add, remove, reorder, or relabel Σ -visible events.
 - Direct witness-construction route: otherwise construct and prove the μ_Q / WC witness used by Appendix I/J.
 - Mixed per-site route: partition the relevant non-blocking or ϵ -choice sites among the applicable NB-CF, snooping, or direct-construction routes, and show that the partition covers every relevant site.
 NB-CF and Appendix L snooping are therefore provenance routes for proving WC, not separate theorem premises that replace WC. When useful for review or automation, the WC audit entry may include optional per-site metadata identifying each relevant call site or ϵ -choice site, its selected provenance route, and the corresponding source restriction, tool/RTL evidence, wrapper evidence, or witness construction.
 Evidence examples: Source inspection or a static control/dataflow check establishing NB-CF; a generated WC witness or certificate; a wrapper proof or monitor specification satisfying Lemma L.2; a co-simulation/snooping harness, subject to the L.Dir entry below; or a complete per-site coverage report for a mixed discharge.
- **Sys_B^{elab}** — elaborated source reference model.
 Trigger: Required when isolated outer Sys_B capacities are not sufficient to represent the RTL storage/handshake realization; examples include multi-input joins, staged/bundle channels, stateless couplers (including process-boundary couplers), shared-capacity effects, and sync/epoch gating.
 Discharge: provide a well-formed Q.1 elaboration package E containing the explicit finite storage/rendezvous channels, channel projection, outer trace projection, finite multi-effect elaboration-token projection, signed outer state-accounting map, complete stateless ready/valid/data equations, boundary/frontier attribution, and the process-level ReleaseOwnerSets family with its WaitOwners union. The package shall satisfy all applicable Q.1(a)–(l) obligations and, when generalized drain is used, the applicable Q.2 finite in-flight action/effect obligations; in particular Q.1(a)–(c) supply the outer trace/token/accounting preservation and state-vs-history agreement, Q.1(i) supplies elaboration-side same-cycle settling, Q.1(j) supplies handshake/release-family reconstruction and congruence, Q.1(k) supplies family-level support soundness/completeness and WFG-union contraction, and Q.1(l) supplies the template-level serial-route handshake support when K.2b reaches a coupling-sensitive boundary. The signed A_E range at a legal outer cycle boundary is derived from the Q.1(b) agreement plus outer E_4 legality and is not a separate M.7a discharge item. For mutual-stall, shared-capacity, or

arbitrated/non-monotone handshake topologies consumed by K.2b, Q.1(d)–(k) qualification does not by itself discharge Q.1(l): the selected template shall provide the qualified CouplerPersistence proof or the candidate is outside K.PSF/CF-S and requires another A5-L route. This is an explicit applicability warning, not an additional premise beyond Q.1(l). These template facts may be proved once per qualified FIFO/coupler elaboration pattern and bound to the concrete E instance. Q.2 supports the no-lowering G1 profile and the Stage-2 G2 lowering profile. Stage 2A/2B supply the separately typed SyncCarrier plus SyncFire/EpochGateAdvance local action schema; pure synchronization is still not a fake message action. A G2 application additionally binds K.Low.ModalQual and the completed Stage-2C path/escape/drain correspondence for the exact lowering. Whole-source Core.QSync still combines template evidence with the design-owned Core.RegSeq/direct-input/R2C/non-template facts. Evidence examples: tool-generated elaboration plus independently checkable FIFO/coupler template certificate, structural back-annotation report, Appendix-Q construction, qualified elaboration-library theorem, or manual elaboration justified against RTL boundary connectivity.

- B-faithfulness — faithfulness of each back-annotated capacity B(c) and chosen explicit abstract channel boundary.

Trigger: Required for every back-annotated explicit storage/handshake boundary whose B(c), occupancy, BoundaryReq, settled ChannelEnabled/ChannelDisabled, or Appendix K dependency/WFG behavior is used by the selected theorem instance. For Sys_B^elab this includes the declared combinational boundary mapping that determines the complementary ready/valid value; the stateless coupler itself carries no capacity.

Discharge: Show all of the following for the chosen abstract boundary of c:

Boundary completeness — every Σ -visible transfer of c in RTL_B crosses the chosen abstract boundary as the corresponding committed Push_c or Pop_c, with no hidden side entrance or exit.

Exact boundary behavior — committed transfers obey the corresponding E4 capacity/order/rendezvous legality, and the mapped RTL ready/valid/data signals at the selected BoundaryEvalPoint agree with the ChannelEnabled/ChannelDisabled behavior of the selected Sys_ref. For ordinary Sys_B this means the primitive E4 full/empty/rendezvous equivalence. For Sys_B^elab it means the same storage legality plus the E-declared settled coupler network; nonfull/nonempty status alone is not required to assert a complementary signal at a coupling-sensitive boundary. Realized stored occupancy remains bounded by the declared explicit capacities.

No hidden enablement, disablement, or storage — every grant, throttle, mutual-stall, join, epoch-gate, or backpressure condition that changes the selected boundary handshake is either part of the declared stateless E network or part of the fixed scheduler/arbitrator semantics; every stateful capacity/credit effect is represented by B(c) or an explicit finite-capacity Sys_B^elab channel. No unmodeled condition may change the reconstructed settled handshake.

Internal-movement opacity and explicit-action accounting — movement below the selected proof boundary that changes no Q.2 explicit abstract state/action carrier remains internal ε -behavior and creates no additional Σ -visible Push/Pop event or Appendix K WFG blocking cause. By contrast, any state-changing transfer across an E-declared explicit Sys_B^elab staging/bundle/join boundary used by generalized drain shall appear as the corresponding complete Q.2 proof-model in-flight action unit with exact E4/history/token effects, even when Π_{elab} later erases it from the outer Σ projection. Such an action is not duplicated as a source residual offer.

Evidence examples: A trusted back-annotation algorithm; a per-channel capacity and boundary certificate; a structural RTL check; an RTL/interface monitor; a formal refinement check

comparing the realized boundary behavior with E4; or equivalent tool/RTL-flow evidence establishing the complete B-faithfulness obligation.

J.LocalGWRExtCert / J.GWR-CompExt — exact incremental source-witness extension for cycle induction.

Trigger: Required whenever a proof carries one selected J.GWR source witness across a larger RTL prefix and needs exact source-prefix nesting, including every TI1/CI1 carrier edge in the standard QSync induction route. It is not required for an isolated one-prefix Theorem J.1 application that does not relate its source witness to an earlier selected prefix.

Discharge: For each old/new cycle-aligned RTL package pair, bind the exact LocalGWR packages and prove J.LocalPrefixAgree/J.LocalGWRExtCert: old local records, ConstructionActions, footprints, dynamic identities, snapshots, channel/reset/environment histories, Dep_J edges, Horizon/stage metadata, and stable keys restrict exactly; no new predecessor enters the old action ideal; old CanonicalRank/ValidatedConstructionOrder is the restriction of the new one; the old final Horizon is closed against retroactive insertion; bridge choices before the old endpoint are stable; and the new suffix has complete cross-prefix join dependencies. Then apply J.GWR-CompExt to the already selected old $W/\tau_ref^A G$ and retain its J.GWRPrefixExt result. On a QSync carrier edge, the RTLCycleStep RS4/RS5 unit/epoch delta shall agree with the package extension. The extension object is $\tau_ref^A G$ only; any K-facing $v_ref/\tau_ref^A K$ is reselected after extension.

Evidence examples: a mechanized restriction theorem for generated LocalGWR records; an incremental certificate generator whose checker compares old/new package hashes and dependency restrictions; a proof-producing carrier-step refinement pass; or an equivalent verified source-witness extension relation.

- J.QSyncTraceInd — cycle-aligned safety package induction for the standard QSync route.
Trigger: Required when J.TraceCertCov_I(CycleReach_post(I)) is discharged through Theorem J.TraceCertCov-QSyncCycle for either Corollary J.1-Safe-QSyncCycle or the all-finite-prefix Corollary J.1-Safe-QSync. It is not required for a one-prefix Theorem J.1 application, for a direct J.GWR route, or for a broader safety domain discharged independently through J.TraceCertCov. If J.QSyncCertInd is already discharged for the standard Appendix K route, Lemma J.QSyncCertInd-Trace supplies J.QSyncTraceInd and no second induction proof is required.
Discharge: Bind TI0 to every admitted exact Core.KOriginCarrier base node (κ_0, s_0, o_0, e_0) in the covered reachability domain, including the common prescribed initial/post-reset state and epoch valuation. For every appendable RTLCycleStep, construct the explicit J.QSyncTraceStep safety certificate. Its ST2/ST3 fields shall carry the J.LocalGWRExtCert/J.GWR-CompExt result for the exact old/new LocalGWR packages and extend the already selected $\tau_ref^A G$; ST4 carries source Core.QSync-Ref settling, I.A0/LocalAgree/channel/reset handling, and ST5 carries the J.HCanon update. ST6 is normative: the safety-step checker shall not consume KObs, Offer, K.NormStable, K.DrainReach, or K.RLAdmReach fields. Lemma J.QSyncTraceStep-Sound establishes the successor QSyncTraceState. TI2 requires exhaustive carrier-position and appendable-successor coverage. A single simulation path is insufficient unless an independent determinism theorem collapses the reachable successor set.
Evidence examples: a mechanized safety-step refinement proof; a checked incremental LocalGWR/GWR-CompExt certificate generator plus J.HCanon update; bounded/exhaustive carrier-step proof; or literal projection of the safety stage stored inside an already qualified J.QSyncCertInd proof.
- J.TraceCertCov — Post $\rightarrow$ Ref certificate coverage for safety claims.
Trigger: Required whenever Corollary J.1-Safe is applied directly over a coverage domain D, and as the cycle-aligned package interface discharged by Theorem J.TraceCertCov-QSyncCycle in the

standard QSync route.

Discharge: Identify D and prove that every $\tau_post \in D$ carries one route-complete $Post \rightarrow Ref$ package. On the normal compositional route, provide the applicable $J.LocalGWR$ evidence, source-side $QualifiedSynchronousModel_Sys_ref(I)/Core.QSync-Ref$ settling evidence required by $J.GWR-Comp$, and all remaining applicable $Post \rightarrow Ref$ premises of Theorem J.1. On the direct route, provide a directly validated $J.GWR$ package whose checked reachable phase-normal chain contains a finite L2-settled segment for every represented Horizon, together with all remaining applicable $Post \rightarrow Ref$ premises. Bind all evidence to the same theorem instance, stimulus, observation boundary, and selected RTL_B prefix. For a direct universal RTL safety application of Corollary J.1-Safe, D shall contain every reachable RTL_B prefix relevant to Φ . This safety-side coverage fact does not imply $K.CertCov$. In the standard normal-compositional QSync route, it is sufficient for prefix-closed finite-history safety to take $D = CycleReach_post(I)$, discharge $J.TraceCertCov_I(D)$ from $J.QSyncTraceInd$ by Theorem $J.TraceCertCov-QSyncCycle$, and then invoke Lemma $J.CycleIdealCover$ / Corollary J.1-Safe-QSync. No separate $J.TraceCertCov$ package is thereby asserted or required for each non-cycle-aligned ideal. Lemma $J.QSyncCertInd-Trace$ permits reuse of an already discharged K -facing $J.QSyncCertInd$ proof. When a safety property Φ is evaluated by a checker using a chosen linearization or other noncanonical encoding of $CanonHist_Σ_DUT$, the discharge record shall also establish that the checker's evaluation of Φ is extensional on the quotient of Definition $J.OuterCanonHist$ and, for an ideal-closure route, that prefix closure is checked against the induced outer relation $<_{Σ_DUT}$ rather than against the chosen linearization or full replay/source order.

- $J.OfferReach$ / $J.OfferConf$ / $J.KObsConf$ — $KObs$ cutpoint correspondence. Trigger: Required whenever Corollary J.1 or Appendix K transfers

Pending/BoundaryReq/ChannelEnabled/Front/WFG/deadlock facts. $J.OfferReach$ binds the exact $J.GWR$ source prefix τ_ref^G , recorded $Core.KObsNormRef$ suffix v_ref , and K -facing endpoint σ_ref and realizes H/O at $\tau_ref^K := \tau_ref^G \cdot v_ref$. $K.NormStable$ is separately checked so Lemma $K.NormStable-H$ transports the $J.HCanon$ history from τ_ref^G to the K -facing prefix without silently treating all ϵ behavior as replay-inert. $J.OfferConf$ proves the complete bidirectional inventory of $KObs$ -relevant source-attributed own-side requests; stateless coupler mirror signals are not extra offers. $J.KObsConf$ additionally binds the selected E/B -faithful RTL boundary mapping so the same explicit channel state/request roots evaluate through the declared finite ready/valid network to the same $HandshakeRec/WaitOwners$ result. The evidence shall be packaged as or mapped onto CF-0 for the exact cutpoint package. A well-typed record, an offer-only match without handshake-network conformance, a generic unphased $NormRef$ suffix, or an unrelated source witness is insufficient.

$K.RLAdmReach$ — composed $Core.LAdm$ qualification of the exact selected source K -facing prefix.

Trigger: Required whenever Theorem K.1A, Corollary K.1A-Total, or Theorem K.1 uses the source cutpoint selected by Corollary J.1. It is not required for an ordinary safety-only Corollary J.1 application.

Discharge: Bind the exact package $(\rho_post, \tau_ref^G, v_ref, \sigma_ref, E, w, H, O)$ used by $Core.KObsNormRef/K.NormStable/J.OfferReach/J.OfferConf/J.KObsConf$ and define $\tau_ref^K := \tau_ref^G \cdot v_ref$. On the normal compositional route, reference Corollary $J.RegPhaseReach-Comp$ and its theorem-derived $J.RegPhaseReach$ fact for τ_ref^G , obtained from $Core.RegSeq$, the enriched Lemma I.3 records, $J.RegPhaseNF$, and Horizon-batched $J.UnitExt$. Validate the primitive $K.NormStable$ conditions on v_ref , including its post-L3 $Core.KObsSettle$ phase placement, no feedback to L3, no request/NB/replay-significant/reset change, and $K\epsilon$ -inertness after request roots are fixed; Lemma $K.NormStable-H$ then transports

H to τ_{ref}^K . Bind Lemma K.DrainReach-Comp's independent DR1–DR4 evidence to the exact package: the K.Drain record including the checked implementation-to-source Core.16/no-new-launch lift on τ_{ref}^G , Core.IC Sys_ref nonwithdrawal, $K\epsilon_{\{I, \text{Sys_ref}\}}$, and the applicable finite-drain/E4/Sys_B^{elab} channel facts. The lemma explicitly uses K.NormStable to cover the v_{ref} suffix that J.GWR-Comp did not construct and supplies K.DrainReach/Core.SettleKBridge on τ_{ref}^K . A direct/alternate route may instead validate the three components directly. The checker combines J.RegPhaseReach(τ_{ref}^G), K.NormStable(v_{ref}), and K.DrainReach(τ_{ref}^K) to establish $\tau_{\text{ref}}^K \in \text{Pref_L}^{\text{adm}}(I)$. KObs equality, OfferReach alone, a bare “admissible=true” flag, or ambient premise names do not discharge this obligation.

Core.KCutClosure_post — K-facing cutpoint existence / non-vacuity (derived in the standard QSync scope).

Trigger: Required by Corollary K.1A-Total and the universal user-facing Theorem K.1 conclusion. It is not required to apply Theorem K.1A to one already selected certified KCut.

Standard discharge: Cite the accepted Core.QSync audit entry for RTL_B and Theorem Core.QSync-Post. No independent per-instance semantic normalization argument is required beyond the structural QSync evidence: the theorem proves that every reachable RTL K-facing cycle-evaluation origin admits one finite same-cycle normalization to a unique $p_{\text{post}} \in \text{KCutReach_post}(I)$ before a later cycle-advancing registered update. With $K\epsilon$, Lemma Core.KCutObserve supplies the persistent-blocker observation consequence. Record the theorem/certificate identity in K.CertHdr.

Scope note (intentional standard-scope narrowing): Core.KCutClosure remains the downstream semantic interface consumed by Appendix K. The standard G–K scope now requires the finite-acyclic QSync settling discipline and therefore withdraws the former independent standard-scope Core.KCutClosure_post discharge alternatives based solely on an arbitrary inductive RTL invariant or checked normalization certificate. A non-QSync design is not asserted incorrect, but it is outside the standard user-facing theorem scope unless an explicitly defined and separately qualified extension proves the same Core.KCutClosure interface under its alternative settling semantics.

J.QSyncCertInd — canonical-cycle correspondence/package induction for the standard QSync coverage route.

Trigger: Required only when K.CertCov is discharged through Theorem J.CertExist-QSync / Corollary K.CertCov-QSync. It is not required for one-cutpoint Theorem K.1A or when an alternate independently checked K.CertCov route is used. When it is discharged, Lemma J.QSyncCertInd-Trace supplies J.QSyncTraceInd by literal projection of the stored safety state/step objects, so the same proof supports the standard QSync safety route.

Discharge: Bind CI0 to every admitted exact Core.KOriginCarrier base node (κ_0, s_0, o_0, e_0), including the common prescribed initial/post-reset state/epoch valuation, its canonical Core.QSync-Post endpoint, and QC0-Safe=TI0. For each appendable carrier edge, CI1 must first construct a complete J.QSyncTraceStep using only the safety-side evidence and J.LocalGWRExtCert/J.GWR-CompExt, producing the exact extended successor τ_{ref}^G and J.HCanon state. Only after that safety stage is complete may the K-facing stage select the canonical RTL endpoint, a fresh post-L3 Core.KObsNormRef suffix v_{ref} and σ_{ref} , validate K.NormStable and OfferReach/OfferConf/KObsConf, and re-derive K.DrainReach/Core.SettleKBridge on the fresh τ_{ref}^K . The old $v_{\text{ref}}/\tau_{\text{ref}}^K$ is never incrementally extended, and DrainReach is re-established for the enlarged prefix. The successor QSyncCertState shall store QC0-Safe exactly as the safety-stage output. CI2 requires exhaustive carrier-position and appendable-successor coverage; the proof may not assume K.CertCov or a successor K.CertifiedCutpoint.

Evidence examples: an inductive simulation whose edge invariant has a separately typed safety field and K-facing field; a verified two-stage certificate generator/checker; or a proof object whose first projection is `J.QSyncTraceStep` and whose second projection discharges the K-facing fields.

`K.CertCov` — total K-ready package coverage for universal reachable-cutpoint conclusions.

Trigger: Required whenever Corollary `K.1A-Total` or the universal user-facing Theorem `K.1` conclusion is claimed, together with `Core.KCutClosure_post`. It is not required to apply Theorem `K.1A` to one already certified cutpoint.

Standard QSync discharge: Establish the `Core.QSync` audit entries for `RTL_B` and `Sys_ref`, establish a uniform instance-level `K.DrainEvidence/DR1–DR4` proof schema whose `DR1` field is activation-indexed and permits per-activation `DR1a/DR1b` routing, instantiated afresh for each exact constructed prefix consumed by Lemma `K.DrainReach-Comp` (or supply an equivalent direct `QC5` discharge), and prove `J.QSyncCertInd`. The normal `J/QSync` construction itself supplies `J.RegPhaseReach` on τ_ref^G , attribution/history, `QSync-Ref L2` settling, and the `Core.KObsNormRef/K.NormStable` observation-suffix facts used by the lemma; they are not an additional per-cutpoint `DR5` premise. Theorem `J.CertExist-QSync` then proves `J.QSyncCertCov` over the complete canonical `KCut` domain by using `Core.KOriginCarrier-KCut` plus the carrier-indexed `J.CertExtract_QS`; Corollary `K.CertCov-QSync` binds those semantic tuples to `K.CertHdr` records and derives `K.CertCov`. This is the preferred generic package-existence route in the standard scope.

Alternate discharge: A proved total certificate-extraction function on `KCutReach_post(l)` whose input includes an independently checked reachability/carrier representation (rather than a bare unproven state), another inductive simulation/refinement relation whose invariant contains the complete package data, exhaustive finite-state exploration, a verified certificate generator, or a separately qualified tool-specific package-existence theorem may establish the same `K.CertCov` interface.

Scope and anti-shortcut rule: Coverage is required exactly on the reachable `Core.KCut RTL` domain on which the K-facing deadlock predicate is defined; arbitrary transient RTL microstates and arbitrary pre-normalization prefixes need not carry `CF-0` packages. Merely supplying a package for one observed run, a finite set of sampled K-facing cutpoints, or a clean source simulation does not establish `K.CertCov`. `Core.KCutClosure_post` separately proves that a persistent K-relevant blocking configuration cannot remain hidden indefinitely in excluded transient states.

- `K.DrainEvidence` (legacy `K.Drain`) — model-indexed `Core.Drain` discharge and settle-to-`KCut` bridge. Trigger: Required for Appendix K whenever the selected proof model has source-later same-interval blocking work already issued while an earlier minimal blocking frontier remains uncommitted. This includes pipelined-loop overlap, other explicitly modeled eager/overlapped issue under Policy L, and ordinary non-overlapped control when Lemma `K.2b` or drain normalization relies on the absence of newly launched work past the blocked point. Discharge: For each exact model/path consumed by Appendix K, supply Definition `K.DrainEvidence` and cover every `Core.16` activation reached on that path. The record shall establish the full seven-clause Definition `Core.16` discipline and shall record the `DR1` route separately for each activation: `DR1a` binds an Appendix-B automatic-flush fact for the corresponding RTL region/cycle and the same dynamic blocked frontier/request attribution, then uses `I.A0/E3` plus `LocalGWR/PCert/ConstructionAction` identity to perform the checked implementation-to-source no-new-launch lift; `DR1b` supplies the direct source structural proof for an activation not discharged through automatic flush. A mixed design may use `DR1a` for some activations and `DR1b` for others; one instance-wide route tag is insufficient. If the standard `J/I` package cannot

prove that a DR1a flush fact refers to the same dynamic frontier rather than merely the same cycle, bind a named activation-correspondence theorem/certificate. Common fields may cite Core.IC/Appendix C for nonwithdrawal, E4/B-faithfulness and finite explicit storage/credit for finite non-replenishing drain, and the applicable Sys_B^{elab} sync-boundary/epoch-gate evidence. Separately discharge Core.SettleKBridge from every persisting L2 activation to the selected later drain-closed KCut, with the $K\epsilon_{\{I,M\}}$ instance for that proof model, WFG-inert normalization, and every intervening drain commit represented explicitly. Core.SettleKBridge is conditional on persistence; it does not prove that the blocked frontier persists. On the standard J.GWR-Comp route, automatic flush remains implementation-side evidence only and K.DrainReach-Comp performs the checked lift to the exact J-constructed source witness. By K.NormStable NS1/NS5, the final v_ref observation suffix contains no drain commit; the exact drain-commit history/attribution bookkeeping is supplied theorem-wise by Core.18a/J.OfferReach/J.HCanon plus J.GWR-Comp/I.A0 and QSync-Ref. Ordinary source operations on distinct interfaces remain source-ordered; any multi-frontier behavior used by Appendix K must be represented by explicit source/elaboration structure rather than by implicit incomparability. Under Definition Core.LAdm this discipline and bridge are part of Appendix K admissibility once the settled-frontier trigger occurs; they are not part of the basic Policy-L issue rule before that trigger. Normal-route lifting: a uniform instance-level K.DrainEvidence/DR1–DR4 schema may be instantiated afresh for each exact J-constructed witness; no K.DrainEvidence or DrainReach prefix-extension rule is available. Model binding: the exact J/K source witness uses K.DrainEvidence on Sys_ref; the serial-route continuation uses K.DrainEvidence on Sys_K. A lowering correspondence certificate does not by itself discharge either model instance.

K.BF — blocking-only serial fragment. Trigger: Required by Lemma K.2b, Corollary K.2c, K.PSF, and the Policy-S route whenever the bounded-response reduction is used. Discharge: Show that every WFG-relevant wait is a persistent blocking Push/Pop on a finite B-faithful point-to-point boundary; control, payload, and later operation identity are deterministic under K.CV/fixed witness choices; shared arbitration is explicit and deterministic or witness-fixed; and local computation between message operations is bounded so pure-computation divergence cannot replace the process-facing serial frontier used by K.2b.Q. Because those waits are ordinary blocking requests, Core.IC supplies the exhaustive lifecycle: once issued, the request remains the same pending dynamic instance until process-facing commit or an affecting RW2 RESET, while DeclaredTerminal may end the execution with it still pending. Lemma K.BF-Persist packages that fact for the frozen-frontier proof. A protocol with independent nonterminal cancellation, withdrawal, retry, or reattribution of an outstanding blocking request is outside the G–K theorem model and requires a separately defined operational extension before any A5-L route in this theorem stack can be used. Evidence examples: source/control-flow qualification, explicit arbiter lowering, finite local-progress proof, per-boundary B-faithfulness certificate, and conformance of every admitted blocking interface to Core.IC/RW2.

- $K\epsilon$ — WFG-inert ϵ / no hidden micro-timing control decisions.

Trigger: Required for Appendix K whenever ϵ -steps occur between the Σ -visible/cutpoint states used by the selected proof model.

Discharge: Bind the proof model explicitly and establish $K\epsilon_{\{I,M\}}$. The exact Corollary-J.1/K.DrainReach source witness requires $K\epsilon_{\{I, Sys_ref\}}$; the deterministic serial-reduction route requires $K\epsilon_{\{I, Sys_K\}}$; and a use of Core.KCutObserve to rule out a blocker hidden in RTL pre-normalization requires $K\epsilon_{\{I, RTL_B\}}$. If $Sys_K = Sys_ref$, the two source-model obligations coincide. When lowering is nontrivial, a Sys_ref discharge shall not be reused for Sys_K merely because K.Low.C holds: either discharge $K\epsilon$ on both models or cite a checked lowering-specific

K ϵ preservation theorem. Show that qualifying ϵ -only steps do not create, remove, or redirect the relevant Appendix K WFG blocking causes except through explicit Σ -visible events or modeled state changes allowed by the selected model.

Evidence examples: source/RTL structural argument bound to M, scheduler or normalization proof, bounded pipeline/drain argument, per-lowering-rule K ϵ preservation lemma, formal check over the WFG abstraction, or tool certificate.

- **SDet** — deterministic selected Policy-S simulator. Trigger: Required for K.2b, K.2c, K.1, K.PSF-1, or CF-S. After fixing Sys_K, capacities/elaboration, initial/reset state, stimulus, environment function, witness-selected ϵ outcomes, deterministic arbitration, concrete SystemC/pre-HLS scheduling, Policy-S issue semantics, and ϵ -normalization, SDet defines a functional Policy-S next-state semantics and selects one run artifact p_S^{det} . For every finite or infinite run artifact p generated from the same initial state under exactly those selected semantics, the committed histories of p and p_S^{det} are prefix-comparable modulo finite WFG-inert ϵ -stutter; equivalently, the selected semantics generate no second incompatible committed history. A finite artifact may therefore be a proper prefix of p_S^{det} without contradicting SDet. WC and deterministic replay give reproducibility relative to the selected history; SDet is the separate global premise fixing every remaining history-affecting choice. Changing any such choice creates a different theorem instance. Discharge by a simulator-semantics proof, deterministic qualification, schedule/arbitration certificate, or mechanized functional-determinism and history-comparability theorem. SDet does not assert that p_S^{det} is maximal, fair, complete, or bounded-response clean. Those facts are separate K.CompleteS and A5-S/CF-S obligations.
- **A5** — primary selected-instance Policy-S bounded-response discharge (A5-S). Trigger: Required whenever Theorem K.1 is invoked; Theorem K.1A and Corollary K.1A-Total instead consume A5-L. A5-S is a property of the execution p_S^{det} selected by SDet for the selected normalized instance Sys_K, including capacities, elaboration, fixed stimulus, witness choices, deterministic simulator/scheduler and arbitration semantics, environment/reset contracts, K.Low lowering, and the K.Resp configuration. It combines three obligations for that exact run: (1) K.CompleteS evidence establishing maximality, fairness, and completeness over the CompleteCycle projection of the selected CycleResult semantics; (2) K.RespCov evidence defining a checkable disposition for every static internal process-facing blocking site and relevant dynamic class and proving that every dynamically reachable instance belongs to a monitored or statically bounded class, together with finite bounds and waiver-preservation, while environment-facing sites are handled by the separate StandardEnv/monitor/exclusion rule; and (3) K.RespClean evidence proving that the complete run contains no non-waived K.RespFail witness. Discharge: Supply CF-S identifying p_S^{det} and carrying all three obligations. A finite prefix of a nonterminating execution is supporting evidence only unless accompanied by a complete invariant, model-checking, recurrence/lasso, or equivalent argument that all monitor ages remain bounded. K.Resp evidence is source-side A5-L-discharge evidence; it does not establish an RTL response-time bound.

K.InFlightQual — generalized in-flight drain action/path qualification. Trigger: Required whenever Definition K.Dead/DeadKRec uses the current generalized DrainClosedNow := $\neg$ InFlightEscape. It is not required for an isolated functional-safety Theorem J.1 claim that does not consume Appendix K deadlock preservation. Discharge: bind the action/profile schema applicable to the selected theorem version; for profile G1 this is the no-lowering Sys_ref=Sys_K schema; for profile G2 bind separate Sys_K and Sys_ref qualifications plus the Stage-2 synchronization/guard/epoch constructors required by the lowering; prove Q.InFlightActionEffectSound and Q.InFlightChoiceFinite; prove Core.InFlightDrainWF/DR4, including SyncTxn/gate ranking when those constructors are selected; bind

Core.InFlightCycleExt, full Core.CycleClosure, or an equivalent route-specific continuation-existence theorem, and when SyncFire/EpochGateAdvance is present discharge Core.SyncInFlightOpportunityQual; and prove J.InFlightPathSound with prefix-compositional profile closure and exact reconstructed successors. On the primary no-lowering serial route, the already-required Sys_K Core.CycleClosure supplies the C2/C3-class continuation component because Sys_K=Sys_ref. Direct/structural/invariant/exploration/tool A5-L routes must supply the path qualification separately. Evidence examples: a qualified elaboration-library Q.2 theorem plus source-cycle extension proof; an independently checked finite transition/profile certificate; a model-checker abstraction with proved successor/path refinement; or a mechanized theorem deriving the restricted profile from Core.CycleClosure.

- A5-L — scheduler-robust liberal source liveness. Statement: A5-L abbreviates Definition K.A5L on Sys_ref: no finite Policy-L prefix admissible under Definition Core.LAdm for the selected Sys_ref instance reaches a source K-facing cutpoint $\sigma \in \text{KCutReach_ref}(I)$ satisfying Dead_K^W . A5-L is consumed by Theorem K.1A for one certified cutpoint, by Corollary K.1A-Total together with Core.KCutClosure_post and K.CertCov for the universal RTL conclusion, and by Theorem K.1 after the selected A5-S route discharges it. Discharge in the generalized-drain scope: primarily by an SDet-qualified CF-S certificate and Corollary K.2c. In profile G1 Sys_K=Sys_ref and no lowering transport is required; in profile G2 the CF-S result is A5L_Sys_K and generalized K.Low.A5 transports it under K.Low.ModalQual; alternatively by K.Str-1/K.Str-2/K.Str-3, direct Policy-L verification, an inductive invariant, exploration, or CF-5 directly on Sys_ref. These alternatives do not weaken A5-L: if any admissible Sys_ref Policy-L prefix—including a Policy-S-shaped prefix—reaches Dead_K^W , A5-L is false and no alternate discharge can certify the unchanged instance. Certificate/profile binding: K.A5Cert/K.CertHdr records the selected route and theorem version. In G1, a CF-S serial package is bound to Sys_ref=Sys_K and the applicable source K.InFlightQual. In G2, the CF-S result establishes A5L_Sys_K and the package additionally binds K.Low.ModalQual, W_K, the generalized K.Low.A5 transport, and both model-indexed K.InFlightQual records; the transported result is the same source A5-L consumed by K.1A. Direct source-side alternatives bind the applicable Sys_ref K.InFlightQual without a lowering transport. Theorem K.1A is therefore route-neutral once source A5-L and source K.InFlightQual are established. No current theorem version silently falls back to the legacy snapshot drain predicate.
- K.CV — design-level value determinism.
Serial-route provenance trigger (K.OpKey/K.CVPred/K.CV-WF): Required whenever Lemma K.2b.P0 or K.2b.P1 is consumed, including every use of Lemma K.2b, Corollary K.2c, K.PSF, CF-S, or the Policy-S form of Theorem K.1, regardless of whether the design has cross-process observations beyond blocking Pop values.
Additional K.CV observation-check trigger: Required when a process can branch on, compute payloads from, or select later operations based on cross-process observations beyond ordinary blocking Pop transfer values — anchored signal reads, direct-input(-sync) reads, shared/external-memory reads through the array-access mapping layer, emitted R2C fixed-point observation units, or synchronization outcomes.
Discharge: Satisfy Definition K.CV. In particular, for its serial-route provenance field, supply or derive Definition K.OpKey plus the Definition K.CVPred certificate: one theorem-instance OpKey_I universe with partial per-execution realization maps and a checked key-level source order, one common CVFact_I key universe, finite Pred_CV(b) sets, explicit predecessor occurrence closure, deterministic Eval_b equations adequate for every admissible execution in the comparison scope, and for each OpFact(κ) a checked none/some(desc) evaluator that includes control/branch occurrence selection rather than assuming occurrence, plus K.CV-WF:

WellFounded($\leq_{CV}$), or an equivalent checked CVRank whose strict decrease proves it. The same K.CVPred certificate shall also discharge the serial-route comparison-availability closure: for every blocking OpFact(κ) consumed by K.2b.P0/R/P, once the liberal comparison has matched the source-earlier blocking operations and earlier serial-transfer facts required before κ may issue, all Pred_CV(OpFact(κ)) facts must be available with the same keyed values after a finite phase-legal extension that preserves the frozen component/DomEpoch and introduces no unmatched dominance-domain endpoint transfer; transfer predecessors must already be among the matched earlier transfers, while lower-ranked non-transfer predecessors are discharged recursively by CVRank and their certified reach rules. Each recursive non-transfer predecessor reach shall also carry an explicit Core.16 disposition: a qualifying ordinary/singleton blocking OpFact predecessor may cite Lemma K.CompareAdvanceLegal-Ord pointwise after its lower-ranked facts are available only if every Policy-S-required source-earlier blocking transfer for that predecessor is already in the matched liberal transfer prefix with the required keyed identity/ordinal before materialization, and the materialization extension itself supplies none of those prerequisites; a pure observation/settling reach shall be certified Core.16-inert; and any other reach that launches source work shall provide equivalent owner/frontier no-key-earlier evidence. “Phase-legal” alone is not an accepted discharge. The $\leq_{CV}$ relation shall contain actual value/control provenance only; rendezvous mutual enablement, same-cycle coincidence, and the broad Appendix-G causal-dependency graph $\rightarrow$ do not become predecessor edges merely by being causal/temporal neighbors. A design with circular value/control provenance that cannot satisfy K.CV-WF is outside the K.2b/K.2c serial route. For the additional observation-check trigger, also show that every such dependency is carried by an explicitly modeled causal mechanism and that the observed value is a deterministic function of the certified value/identity provenance feeding that mechanism, not of incidental schedule timing. These additional design checks include RULE 1 / RULE 2 anchored signal IO, proper handshakes or direct-input contracts on cross-process signals, mapping-layer discipline on shared memories, and R2C well-formedness; latency-sensitive sites are isolated or witness-aligned per Appendices D and L. Evidence examples: a source/tool-generated value-provenance graph with a checked ranking function; a mechanized well-foundedness proof over typed dynamic operation/observation keys; source inspection or lint establishing RULE 1 / RULE 2 and handshake conformance where the additional observation-check trigger applies; a coverage-justified Appendix L latency-robustness stress-testing result with provenance/rank evidence; a refinement argument; or an Appendix D isolation argument. Randomized stress testing by itself is supporting evidence, not a complete K.CV discharge, unless accompanied by a coverage/refinement argument and the required K.OpKey/K.CVPred/K.CV-WF evidence. This K.CV evidence is necessary but not sufficient for the separate NB-Align condition (next entry), which is discharged independently.

- NB-Align — witness-selected non-blocking alignment.

Statement: Definition NB-Align in Appendix K is authoritative. It requires every witness-selected non-NB-CF non-blocking site that can participate in a potential wait cycle either to satisfy the Appendix L alignment needed to tie its selected outcome dependency to an explicitly modeled message/synchronization obstruction and the supplier used by Definition K.Low, or to be proved structurally unable to participate in such a wait cycle. An aligned site is lowered to a blocking guard on an explicit point-to-point supplier channel; a relevant site that is neither NB-CF, aligned, nor structurally excluded makes NB-Align false and places the selected witness schedule outside Lemma K.2b / Corollary K.2c / Theorem K.1. Theorem K.1A does not consume NB-Align. Trigger: Required whenever Lemma K.2b or Corollary K.2c is invoked (including through Theorem K.1) and the design contains witness-selected non-blocking outcomes that are not NB-CF.

Discharge: Show that each witness-selected non-blocking site is NB-CF, is Appendix L-aligned so Definition K.Low applies to it, or is structurally excluded from every potential wait cycle of the selected instance.

Evidence examples: An NB-CF classification per Appendix L; a per-site Appendix L alignment argument identifying the modeled supplier used by the Definition K.Low guard channel; or a structural argument that no witness-selected site can participate in a wait cycle of the selected instance. NB-Align is distinct from, and not implied by, K.CV.

- K.Low — witness-indexed blocking lowering, modal path correspondence, and A5 transport. Trigger: Required whenever the serial-reduction route (Lemma K.2b / Corollary K.2c / Theorem K.1) is applied to a design containing explicit synchronization waits or NB-Aligned witness-selected non-blocking sites in potential wait cycles. Discharge: Exhibit $\Lambda_W(\text{Sys_ref})$, the exact IdClose/EpochClose/GuardClose local certificates, and—when the current generalized drain theorem is claimed—K.Low.ModalQual. The modal package binds finite LowAdminNormalize, lowering-aware source LowWaitRoots, local root/action correspondence, separate Sys_K/Sys_ref K.InFlightQual, and the global K.Low.InFlightPathCorr theorem; its corollaries establish InFlightEscape and DrainClosedNow correspondence at normalized paired cutpoints. Define waiver pullback W_K and discharge the generalized Lemma K.Low.A5 so $A5L_Sys_K(I, W_K)$ transports to $A5L_Sys_ref(I, W)$. Evidence examples: the elaboration/lowering record; per-site guard and epoch-gate mappings; Core.SyncCarrierWF/SyncFire/EpochGateAdvance evidence where applicable; finite lowering administrative rank; local action/effect certificates; source and lowered path-soundness certificates or explicit preservation lemmas; K.Low.InFlightPathCorr checker proof; and schema/version binding in K.CertHdr. The Revision-A blocked-frontier, non-empty release-family, and release-closure local facts remain valid; their former local snapshot DrainClosedNow family is retired in favor of the global modal theorem. K.Low still does not transport Core.Drain, Core.CycleClosure, $K\epsilon$, fairness, or other model-indexed premises unless a named preservation theorem is separately supplied.

K.Low.ModalQual / MQ1–MQ6 — lowered generalized-drain modal qualification.

Trigger: Required for every profile-G2 generalized-drain claim with $\text{Sys_K} = \Lambda_W(\text{Sys_ref})$. It is not implied merely by choosing K.Low, proving the Revision-A local closure lemmas, or carrying Sys_K Core.CycleClosure on the serial route.

Discharge: (MQ1) bind the exact Λ_W rule set, active IdClose/EpochClose/GuardClose local typing/release facts, and Stage-2 Q.2 action/effect schemas; (MQ2) prove finite LowAdminNormalize existence/termination with a checked lowering rank; (MQ3) prove LowStateCorr on every relevant projected/lifted state and RootCorr_D at every administratively normalized paired state uniformly for every candidate D; (MQ4) prove the bidirectional local action simulation/lifting interface consumed by K.Low.InFlightPathCorr, including admin-edge source stutter, exact projection of every source-corresponding Sys_K edge, a finite normalized Sys_K lift for every Sys_ref edge, and finite-administrative-suffix root reflection: after any source-corresponding edge, every finite zero-or-more erased guard/epoch suffix up to the next source-corresponding edge must attribute any first literal D-root enable/remove/advance/change back to a corresponding LowWaitRoot_D change on that projected source edge, including changes that occur only on the second or a later admin edge; (MQ5) discharge K.InFlightQual separately for Sys_K and Sys_ref, or cite an explicitly checked named preservation theorem for a particular model-indexed field; and (MQ6) bind the exact schema, checker, generator, template/library qualification, and theorem/document versions in K.CertHdr.

Checker status: MQ4 is a universal qualification premise, not a consequence manufactured by

K.Low.InFlightPathCorr. The global path theorem composes the checked MQ2–MQ5 cases; it does not replace them. A reusable lowering-library/template proof may discharge MQ1–MQ4 once for a parameterized family only when an independent checker verifies that the concrete instance satisfies that proof's structural and parameter side conditions. Failure or omission of any required MQ4 direction places the lowered instance outside G2 rather than weakening the theorem or falling back to the legacy snapshot drain predicate.

- K.EnvMF — Policy-S environment/release-structure applicability. Trigger: Required for Lemma K.2b.E, Lemma K.2b, Corollary K.2c, Theorem K.PSF-1, Theorem K.1, and CF-S. Discharge: Establish either (a) StandardEnv/B1-avail with the default always-ready output environment; candidates satisfying Lemma K.JointFreeze-Ord's ordinary/single-release-support premises cite that lemma, while every other admitted candidate—including a single-frontier item with non-singleton/multiple ReleaseOwnerSets support and every admitted multi-frontier—supplies Definition K.JointFreeze, proving JF1 cross-process launch/issue closure and JF2 no-release under permitted drain/elaboration actions; or (b) a timed/reactive or otherwise nondefault environment together with exactly one selected internal frontier item per process and the full K.JointFreeze-Ord single-release-support premises for that candidate. For clause (b), Lemma K.EnvMF-Uniform is the standard candidate-independent sufficient discharge: prove its at-most-one/internal/singleton-release conditions once over every reachable K-facing cutpoint and the lemma establishes K.EnvMF(2) for every candidate. In an ordinary non-elaborated primitive point-to-point instance, that structural part normally reduces to citations of Lemma Core.OrdFrontSingleton and Definition Core.WFG's ReleaseOwnerSets_E(x, σ)= $\{Q\}$ fact; Dead_K candidate membership supplies nonemptiness/internality, so once the remaining K.JointFreeze-Ord premises are qualified no separate per-candidate K.EnvMF certificate is needed. A timed/reactive/nondefault candidate with an environment-facing co-frontier, additional frontier member, or non-singleton/multi-alternative release support is outside the Policy-S serial-reduction route. It may use Theorem K.1A only with a non-serial proof of A5-L, or a different theorem outside this document. The K.JointFreeze family may be discharged either candidate-by-candidate or through one stronger instance-level certificate proving JF1/JF2 uniformly for every admitted StandardEnv candidate not covered by K.JointFreeze-Ord. A StandardEnv candidate for which the required K.JointFreeze cannot be discharged is likewise outside K.2b/K.2c. K.EnvMF together with K.JointFreeze-Ord/K.JointFreeze preserves the frozen component; K.CompareAdvanceLegal is the separate audit entry for K.2b.P0/K.2b.R launch-and-issue compatibility.
- K.CompareAdvanceLegal — dominance-comparison Core.16 launch-and-issue compatibility. Trigger: Required by Lemma K.2b.P0/K.2b.R/K.2b and therefore by K.2c/K.PSF/CF-S for every candidate comparison. Discharge: Cover every process/key in K.DomResetScope that K.2b.P0 or K.2b.R may consume, whether or not the owner belongs to D. The certificate's normative conclusion is the Core.16(2) item-order/no-predecessor fact for the concrete launch/Issue target y_k : when Core.16 is active, no $x \in \text{Front}_P$ satisfies $x \leq_P y_k$ with $x \neq y_k$. For an ordinary non-elaborated active frontier, or a certified singleton frontier carrying the same relevant total-order/source-comparability evidence, cite Lemma K.CompareAdvanceLegal-Ord and the checked least-missing/first-overtake transfer/provenance invariant; that lemma derives the item-order conclusion from the no-key-earlier argument through the named Lemma K.KeyItemOrderBridge. If a Sys_B^elab package can present a multi-frontier for a consumed owner on a covered comparison prefix, supply an explicit elaboration/order certificate proving the required $\leq_P$ /joint-frontier fact or equivalent direct Core.16 legality before the P0 preparation/reach extension and through the later R request assertion; key incomparability or a key-order inequality alone is insufficient unless the bridge to the applicable item order is

proved. The record shall state that this one condition discharges both distinct Core.16(2) obligations: the broader no-new-launch rule for P0's Core.ReachPrep/source-evaluation advance and the specific no-new-Issue rule for R's request assertion. The checker shall also enforce the consistency rule with K.JointFreeze: the same operation/prefix may not be both JF1-suppressed as a releasing launch/issue and certified here as a required legal comparison advance. A conflict or an uncovered elaborated multi-frontier owner puts that candidate outside the Policy-S reduction and requires another A5-L discharge. Reuse inside K.CVPred: when a lower-ranked predecessor is itself a blocking OpFact in the same ordinary/singleton comparison class, the predecessor-reach disposition may cite Lemma K.CompareAdvanceLegal-Ord pointwise only after the recursive qualification test confirms that every Policy-S-required source-earlier blocking transfer for that predecessor is already in the matched liberal prefix before materialization and is not created by the materialization extension; non-operation observations and other launch forms remain governed by the explicit K.CVPred recursive-disposition rules rather than being silently folded into this entry.

- K.InterruptionGate / K.ResetEpoch / K.TerminalDisposition — nested comparison-interruption applicability. Trigger: Required whenever Lemma K.2b, Corollary K.2c, K.PSF, CF-S, or the Policy-S form of Theorem K.1 uses the serial reduction. Discharge the shared schema once, including its K.BF exhaustiveness rule: every gate instance has a prefix-local Continue predicate and an independently mapped DirectFail payload; DirectFail must bind P, f_P, x_P^A, g_P with $g_P \leq_{src} f_P$ and an already-triggered pending covered monitor independently of K.2b.P0/K.2b.R/K.2b.P/K.2b.P-Frozen/K.2b.SF/K.2b.Q Step 1, and a candidate satisfying neither branch is outside the route. For each candidate, first construct K.CompareScopeFootprint from the fixed selected Policy-S artifact and certified comparison/provenance closure, then bind finite $S_P^A \text{ dom } S_C^A \text{ dom}$ and validate K.DomResetScope conditions D1–D4—including D membership, both-endpoint boundary closure, coverage of the precomputed comparison owner/boundary footprint, and independence from the later K.2b.E continuation—and derive DomEpoch. Instantiate reset first. K.ResetEpoch.Continue has two checked parts: R-L proves that no RESET intersecting K.DomResetScope is legal as the next proof-relevant transition/unit on any applicable frozen Policy-L extension prefix; R-S independently proves the same dominance-affecting reset exclusion over every selected Policy-S prefix that K.2b.P0/K.2b.R/K.2b.P/K.2b.P-Frozen/K.2b.SF/K.2b.Q may consume. The R-S domain must be fixed from the selected Policy-S artifact, K.CompareScopeFootprint, and reset contract before the dominance proof; whole-artifact reset exclusion is a sufficient implementation. Reset accounting is cycle-level: any RESET occurrence that executes within a relevant real cycle, whether RW5-disjoint or the affecting RESET matched by DirectFail, appears in CycleResult.resetUnits. K.ResetEpoch.DirectFail binds the matched affecting RESET and exact RW2 clearing, yielding cancelled-pending. An affecting reset covered by neither R-S nor DirectFail rejects the candidate. Only on K.ResetEpoch.Continue instantiate terminal. K.TerminalDisposition.Continue proves every applicable frozen-prefix final state is non-DeclaredTerminal; K.TerminalDisposition.DirectFail independently binds the declared maximal terminal state while the same mapped monitor remains pending, yielding maximal-pending. For a successful clean CF-S discharge the two nested Continue branches are the normal prospective evidence; concrete DirectFail witnesses are themselves K.RespFail classifications and are not required on every clean run. Under the per-candidate route, the audit record shall name the K.InterruptionGate schema version, the K.BF exhaustiveness citation, K.CompareScopeFootprint, each K.DomResetScope certificate, D1–D4 validation/provenance and DomEpoch artifact, the reset-first nesting, R-L and R-S reset evidence, each Interrupt predicate, and each invoked DirectFail specialization. Uniform instance-level alternative: a clean CF-S/K.PSF application may instead supply one fixed conservative scope

U satisfying U1–U3; U2 provides both R-L and R-S reset exclusion universally. This alternative may not be used if any U1–U3 clause is unproved.

- **K.Resp** — validated process-facing bounded-response coverage. Trigger: Required for Lemma K.2b, Corollary K.2c, Theorem K.1, K.PSF-1, and CF-S. K.RespCov is default-total over a declared coverage universe of static internal process-facing blocking sites and relevant dynamic classes. Each class has one checkable disposition: (a) a monitor with an unambiguous trigger at the first normalized cycle in which the source operation is the active minimal pending frontier and its persistent boundary request is established, the process-facing commit/return as response, a finite Policy-S-cycle bound, and reset/end-of-test handling; (b) a validated static bounded-response proof; or (c) a proof that the class is unreachable in the selected instance. Completeness requires every dynamically reachable instance to belong to a declared class; once an instance is reached, disposition (c) cannot apply. K.RespCov response-waiver closure is checked over every non-waived liberal Dead_K candidate admitted to K.2b before K.2b.E/P/Q is invoked; the candidate-to-serial coverage relation must leave at least one mapped covered response class non-waived, so the waiver definition does not depend on a component first being “extracted by K.2b”. A production-count argument alone is not an exemption, because the complementary owner may itself be blocked. Under StandardEnv, a source-earlier environment interaction that committed in the liberal witness may be proved incapable of remaining the serial blocker by replay availability and fairness. Under a timed/reactive or otherwise nondefault environment, K.EnvMF first requires the relevant liberal Dead_K components to be single-internal-frontier; environment-facing sites that may block the selected Policy-S execution then require monitoring, a specific availability proof, or exclusion from the serial route. No monitor or no-bypass proof admits a timed/reactive multi-frontier liberal component. K.RespFail is a finite-prefix predicate/witness: certified-bound expiration without response under Definition K.RespFail’s deadline convention, a declared-terminal maximal finite selected run with the frontier still pending, or an affecting RW2 RESET that clears the monitored dynamic instance before response unless explicitly waived. A terminating abort/end-of-test disposition is represented by DeclaredTerminal and is covered by the maximal-pending case. K.RespClean proves absence of non-waived K.RespFail witnesses over the complete p_S^{det} . A successful complete-run proof may use any finite bound; a bound violation under an unproved design-latency bound makes that certificate attempt inconclusive rather than proving a design bug.
- **K.EnvOrd / K.Done** — optional diagnostic starvation classification. These conditions are no longer part of the soundness-critical Policy-S discharge and cannot substitute for K.RespCov/K.RespClean. They may be used to classify a reported K.RespFail through the historical WFG+/EnvStop/DoneStop owner-walk machinery or to simplify diagnostic reporting from Dead_KS to Dead_K. K.EnvOrd builds the finite may-follow starvation cone and proves the corresponding dynamic no-bypass property. K.Done audits normal completion sufficiently to eliminate DoneStop terminals; the alternative that every process is perpetual eliminates normal completion only and says nothing about live endpoint under-supply. In particular, neither K.Done nor process perpetuity permits omission of a K.Resp monitor. The K.CE-2/K.CE-3/K.CE-4 and H2 examples remain useful diagnostics, but no Corollary K.2c conclusion depends on completeness of this terminal taxonomy.
- **W.Env** — environment-rescue and response-monitor waiver. Trigger: Optional; used only for a specifically declared cutpoint or dynamic monitor class whose environment/reset/end-of-test behavior is intentional and proved compatible with the theorem scope. For KObs predicates, record the component class, rescuing item, environment contract, and KObs-closure justification. For K.Resp, record the exact dynamic frontier class, the reason timeout/reset interruption is permitted, and a coverage-preservation proof that a non-waived Policy-L

deadlock component cannot map only to waived serial monitors. A broad end-of-test or reset waiver is insufficient. Waivers are removed explicitly from Dead_K^W and optional diagnostic Dead_KS^W , and from K.RespFail^W , and propagate to Theorems K.1A/K.1. A waiver depending on hidden issue timing or unconstrained future behavior must carry the corresponding operational proof; a label alone does not discharge the obligation.

- L.Dir — one-directional alignment / RTL-side non-interference.

Statement: L.Dir is the scope restriction stated normatively in Condition L.Dir (Appendix L), which is its authoritative wording: the source-side snoop sequence selected by W is exactly an ordinal-preserving consumption of the free-running RTL_B snoop sequence at the chosen boundary — each pre-HLS-side snoop commit consumes the next recorded RTL_B event of the same snoop point with matching kind and payload, no pre-HLS-side snoop commit occurs without such a next recorded RTL_B event, every recorded RTL_B snoop event within the compared run is eventually consumed or reported outstanding per the Requirement W4 bound, and no RTL_B-side signal or event is gated, delayed, or modified. Benign cycle skew is permitted; only order, kind, or payload divergence — non-existence of the WC witness at the chosen boundary — violates the condition. A run violating L.Dir is outside the Appendix I–K theorem scope for that comparison and carries no theorem-backed guarantee.

Trigger: Required whenever WC is discharged through an Appendix L snooping / witness-selector co-simulation harness (Lemma L.2 / Lemma L.5).

Discharge: Provide a per-run monitor per Lemma L.5(iv) — realized by a wrapper conforming to the Appendix L Wrapper Realization Requirements W1–W5 — that aborts loudly on violation, together with the run’s clean-monitor result and a statement of whether the Requirement W4 watchdog bound is proved sufficient for the selected instance (bound-exceeded aborts under an unproved bound are inconclusive witness-realization failures, not proven violations); or a static/structural argument — for example, latency back-annotation or delay padding of the pre-HLS model, or latency-interval analysis at each snoop point in the spirit of the simplified snooping approaches — showing that the post-HLS side cannot run ahead in event order; or a model-checking proof that order divergence is unreachable for the selected instance.

Evidence examples: A harness monitor log showing no L.Dir abort, with cycle-skew statistics reported separately as diagnostics; a static latency analysis at the simplified-snooping end of the Appendix L spectrum; or a wrapper certification against Lemma L.2 together with a watchdog configuration that converts witness non-existence into a reported failure rather than a hang.

L.Dir violations are triaged per the Snooping Concerns section of Appendix L: the pre-HLS model is the default suspect in a certified-harness, trusted-tool flow, but harness incompleteness and tool non-compliance must not be excluded a priori.

- J.LocalGWR — local certificates for the compositional J.GWR construction.

Trigger: Required whenever Theorem J.1’s normal $\text{Post} \rightarrow \text{Ref}$ route, Corollary J.1, or any result consuming their witnesses is applied, unless the optional J.GWR-Direct route below is used.

Discharge: For each compared RTL_B prefix, provide the package of Definition J.LocalGWR. The package shall cover every process with J.PCert; every explicit abstract channel with J.CCert and B-faithfulness; every rendezvous, paired synchronization, R2C, anchored synchronization/signal, RESET, and other explicitly atomic unit with J.ACert; and the applicable J.EnvCert and J.ResetCert obligations. It shall also provide the finite typed construction-action inventory, per-action ReadSet/WriteSet, ownership facts, and locally derived epoch/horizon/stage/ordinal metadata for J.FoundationRank, request-snapshot keys for every witness-significant non-blocking call, the locally generated semantic D1–D4 edges and D5 construction-stratification edges required by Definition J.DepJ, including J.HorizonOrderSafe commutation evidence for every D5-only pair, and J.LocalAgree for every duplicated endpoint, identity, payload, reset-epoch, request-

attribution, boundary, footprint, and snapshot field. The package shall prove complete coverage and the premises needed for the case-by-case J.DepRank proof: every relevant process, explicit boundary, construction action, non-blocking decision, and atomic unit in the selected prefix is represented exactly once or by a uniquely identified shared record. J.CanonicalRank is reconstructed from the checked dependency graph and is not a separately supplied certificate field.

The discharge may not contain a renamed global realizability premise. In particular, it may not assume that the local certificates admit a realizable merge, that an arbitrary phase/rank already gives a legal replay order, that the next complete unit has a global extension, or that a matching Sys_ref execution already exists. Dependency edges must arise only from the enumerated local rules of Definition J.DepJ, and non-commuting action pairs must be covered by Lemma J.DepComplete or by a component-local footprint/commutation certificate. Global well-foundedness, coexistence, and extension are conclusions of Lemma J.DepRank, Lemma J.DepAcyclic, J.PrepareCommute, J.UnitExt, and Theorem J.GWR-Comp.

Evidence examples: per-process schedule/replay scripts with typed action footprints; per-channel boundary, occupancy, same-cycle transfer, and request-snapshot certificates; static or generated atomic-unit identity records; locally justified Dep_J edge records; source/RTL structural checks; component-local footprint/commutation certificates for custom Sys_B^elab elements; a translation-validation package whose individual records map to J.PCert/J.CCert/J.ACert; or equivalent independently checkable evidence.

J.GWR-Direct — optional direct global witness-realizability route.

Trigger: Optional alternate route when a flow does not use J.LocalGWR/J.GWR-Comp for a selected prefix.

Discharge: Supply a directly validated package satisfying Definition J.GWR, including the complete reachable Sys_ref witness chain, all operational annotations, and a direct-provenance J.ValidatedConstructionOrder with an independently checked well-founded predecessor relation, complete unit enumeration, required-order refinement, operational-prerequisite coverage, and ideal-prefix proofs. This route may be implemented by translation validation, bounded symbolic replay, a validated global trace certificate, or a previously proved tool-specific theorem. It is not required when J.LocalGWR and Theorem J.GWR-Comp are used, and it may not cite the theorem-derived J.CanonicalRank without also supplying the corresponding LocalGWR package.

Historical Issue timestamps may remain inside either route's operational witness evidence, but they are not fields of KObS and are not independently compared across cutpoints.

- Core.MemSyncOrder — array/synchronization commit-order conformance.

Trigger: Required whenever the selected design/flow instance contains array or memory accesses in a process that are source-ordered with synchronization calls selected into S, whether the storage is process-internal or externally mapped.

Discharge: Check the main-text "Scheduling of Array Accesses" commit-time fence at the schedule/mapping-layer level: mapped storage commits for accesses before s occur no later than s's commit, and mapped storage commits for accesses after s occur strictly later. Any allowed cache/merge/split/reorder transformation shall carry mapping-layer evidence that preserves the fence. A single merged storage commit may not represent accesses lying on opposite sides of the same synchronization boundary; such a merge is forbidden. This is a scheduling-contract qualification check recorded by M.7a, not an Appendix I replay or Theorem J.1 semantic premise, and it does not add memory events to Σ . For a J.1 application with cross-process shared-memory value flow, additionally discharge J.Mem; when transformed physical memory activity is hidden below the selected proof boundary, additionally discharge

Core.MemFaithful.

Evidence examples: HLS schedule/control-step reports; memory mapping metadata; generated-RTL memory-enable/address/commit assertions; or a mapping-layer proof/checker.

- Core.MemFaithful — mapped-memory boundary faithfulness.

Trigger: Required whenever mapped-memory channels/signals or transformed physical memory activity are excluded from the selected Σ boundary while their logical memory behavior remains in a J or K theorem instance, including any cache/merge/split/reorder transformation hidden below the selected boundary.

Discharge: Bind the declared untransformed logical mapping-layer semantics as the reference; identify the selected logical boundary/alphabet and allowed transformation set; and show that the transformed realization induces the same Core.17/E1–E5 selected-boundary behavior as that reference, without creating, deleting, duplicating, or relabeling selected events and while preserving required endpoint occurrence order, values, ordering, and atomicity. A message channel or anchored signal selected at the boundary remains governed by E4 or R2/E2 and cannot be selectively hidden. For Appendix K, additionally show that hidden realization does not introduce/remove/redirect a process-facing blocking cause or selected frontier/request/enablement/release-support/WFG fact except through represented Sys_ref/Sys_B^{elab} structure, with applicable ε -normalization satisfying $K\varepsilon$. A persistently backpressuring hidden resource therefore requires a K-faithful logical boundary above the transaction-changing realization or the transformation is outside K scope.

Evidence examples: trusted mapping/back-annotation algorithm, generated certificate, structural RTL check, interface monitor, formal refinement/equivalence check, or other qualified mapping-layer validator. Core.MemFaithful is not derived by E4, Core.17, Proposition J.0, or Theorem J.1.

- J.Mem — shared-memory lowering.

Trigger: Required whenever the compared design contains a cross-process shared-memory dependency through which one process's memory write can affect another process's control flow, payload/value expressions, or the identity or order of its later operations.

Discharge: Show every such dependency is lowered by the array-access mapping layer onto covered proof-visible operations (explicit point-to-point E4 channels, including a qualified logical memory boundary, and/or Σ -visible anchored signal IO), or supply the memory-state invariant extension described in the informative note in Normative Formal Core. The audit is dependency-based rather than per physical RAM access, but it shall be complete: show that all dependencies satisfying the trigger are represented and that no other unlowered cross-process memory-carried path satisfying the trigger exists. If a transaction-changing transformation would alter the event stream otherwise used for this lowering, additionally use a Core.MemProfile boundary satisfying Core.MemFaithful or the separate memory-state extension.

Evidence examples: use of a qualified shared-memory library/profile presenting the logical boundary; mapping-layer configuration review; data/control-flow analysis establishing the trigger set; and a structural completeness check that no unlowered cross-process array dependency remains.

- Core.MemProfile — qualified memory mapping profile (when used).

Trigger: Required when the selected J/K instance relies on a reusable logical memory boundary above transaction-changing cache/merge/split/reorder behavior, including the standard route for transformed cross-process shared memory and for a persistently backpressuring memory whose raw command/response stream cannot itself serve as the E4/Core.17 boundary.

Discharge: Bind a qualified profile satisfying Core.MemProfile MP1–MP7: declared untransformed reference/boundary/alphabet/transformation set; Core.MemFaithful; complete

J.Mem lowering; Core.MemSyncOrder; the Appendix-K WFG-inert or K-faithful-boundary disposition; A6-compatible explicit arbitration and point-to-point process-facing ownership for shared/multi-client resources, with applicable Appendix Q.1 obligations when the exposed logical boundary is Sys_B^elab; and RW1–RW5/reset disposition. Bind the profile identity/version and qualification evidence to the concrete instance through K.CertHdr. If the K disposition cannot provide a transformation-invariant K-faithful boundary for a persistently backpressuring realization, disable the transaction-changing transformation for the covered instance or mark that memory outside the Appendix-K theorem scope. Evidence examples: qualified library-template theorem/certificate, independent checker for the declared logical boundary, reusable arbitration/reset qualification, and instance binding metadata. The profile interface is normative; qualification of particular library patterns is separate reusable flow work.

- RW — reset well-formedness (RW1–RW5).

Trigger: Required whenever dynamic reset can occur in the compared executions.

Discharge: Show every dynamic reset occurrence satisfies RW1–RW5 so that Lemma J.RS (MatchedResetEpochSplice) applies at each RESET boundary unit.

Evidence examples: reset-tree/scope review; RTL check that both endpoints of each live channel reset together or that an explicit modeled flush/abort protocol is used; confirmation that outstanding requests clear and affected channels re-initialize reset-empty.

This record is not itself an additional semantic assumption beyond the named premises used in Appendices I–K. It also records separately named qualification obligations, notably Core.MemFaithful and Core.MemSyncOrder, without converting them into Appendix I or Theorem J.1 premises. It makes the required single-global-clock scope check and the discharge of Core.IC, Core.SyncFence, Core.ReachPrep, Core.SelectLatitude, Core.CycleClosure, Core.IssueFair, B1, B2/B3, B4/FPD, B5/B6-Terminal, A6, A8/A9/A10, R/E scheduling-contract conformance, WC, J.LocalGWR (or J.GWR-Direct), elaboration, B-faithfulness, Core.MemFaithful, J.OfferReach, J.OfferConf, J.KObsConf, Core.MemSyncOrder, J.Mem, Core.MemProfile when used, RW, the model-indexed K.DrainEvidence/K ϵ /nonwithdrawal/fairness records required by each selected proof model, K.BF, SDet, A5-L, and the selected A5-L discharge route—including K.CompleteS, K.RespCov, K.RespClean, and the shared K.InterruptionGate schema with K.ResetEpoch/nested K.TerminalDisposition when the deterministic Policy-S route is used—and K.CV explicit for the selected theorem or flow-qualification scope.

M.7b Standard QSync certificate profile

Normative packaging status. For the standard normal-compositional QSync flow, the detailed M.7a discharges may be organized and exposed through the four logical certificate bundles below. These bundles are packaging interfaces only: they do not introduce a new semantic assumption and do not replace, merge, weaken, or make optional any theorem premise, qualification obligation, coverage condition, or evidence requirement defined elsewhere in this document. M.7a remains the definitive per-instance discharge ledger. If an item is required by the selected theorem or flow-qualification scope, it shall still be discharged as specified in M.7a; if an item is theorem-derived, the bundle may reference the applicable theorem and its checked inputs rather than duplicate the derived fact as an independent premise.

Common binding and physical representation. The four bundle names in this profile are logical interface names, not new theorem predicates. A conforming flow may represent a bundle in one physical file, several files, a proof object, a database record, or a set of versioned references. Every artifact used in one theorem claim shall map unambiguously to the M.7a obligations it discharges and shall be bound to the same selected theorem instance through K.CertHdr/K.CertPkg, including the

source, Sys_ref/Sys_K and RTL identities, capacities/elaboration, observation boundaries, stimulus, environment, reset, witness/waiver provenance, and applicable tool/checker versions. A stronger checked bundle may cite theorem-derived projections instead of requiring a second independently generated certificate.

1. Instance and QSync qualification bundle. This bundle fixes the theorem instance and carries or references the primarily structural, scheduling-contract, and implementation-mapping qualifications used across J and K. In the standard QSync profile it includes the applicable ClockScope and source/RTL Core.QSync QS1–QS5 evidence, including RTL QS2a KCut-origin recoverability, the designated Core.RTLDesignatedOriginFrame boundary mapping, the explicit initial carrier node/epoch valuation, and RS6 consecutive-designated-origin checking when the cycle-induction route is used; Core.RegSeq, Core.ReachPrep, and Core.SelectLatitude where triggered; Core.IC and Core.SyncFence where triggered; the applicable M.7a R/E scheduling-contract conformance (R1–R5/E1–E5, including R2C where applicable); selected abstract boundaries, capacities, B-faithfulness and Core.ElabProj; and the applicable Core.MemFaithful, Core.MemSyncOrder, J.Mem, Core.MemProfile, RW, direct-input, and elaboration/template-composition qualifications recorded by M.7a. For the richer K-facing source-cutpoint profile, but not for the safety-only profile, this bundle additionally carries or references Core.KObsSettleQualified_Sys_ref(l); it may then cite Core.KObsSettle-Ref for the post-L3 Core.KObsNormRef endpoint. It may cite Core.QSync-Ref/Core.QSync-Post and Core.QSync-KCutCanonical when their hypotheses have been checked. This bundle establishes that the selected source/RTL instance and its cycle/boundary abstraction are qualified; it does not by itself prove J trace correspondence, KObs equality, K.CertCov, or A5-L.

2. J correspondence/refinement bundle. This bundle carries the source/RTL witness-construction and correspondence evidence for the selected coverage profile. Its local/global core is J.LocalGWR with J.GWR-Comp, or a permitted J.GWR-Direct package, together with the applicable WC discharge, the exact selected J.GWR witness prefix τ_ref^G , and the J.HCanon history relation. Whenever the profile is incremental across carrier edges, the bundle additionally carries the old/new LocalGWR restriction proof J.LocalGWRExtCert and the theorem-derived J.GWRPrefixExt result from J.GWR-CompExt; independent successful per-prefix J.GWR constructions are not a substitute for this exact extension evidence. For the standard QSync safety profile, it carries or references the carrier-indexed J.QSyncTraceInd and, for each edge, the explicitly safety-only J.QSyncTraceStep together with the theorem-derived Core.KOriginCarrier decomposition. Theorem J.TraceCertCov-QSyncCycle then supplies J.TraceCertCov on CycleReach_post, Corollary J.1-Safe-QSyncCycle supplies cycle-aligned safety, and Lemma J.CycleIdealCover / Corollary J.1-Safe-QSync extend ideal-prefix-closed safety to all reachable finite outer observable prefixes without adding per-prefix certificate packages. The 6' K-facing endpoint repair does not add v_ref , σ_ref , K.NormStable, or DrainReach to that safety form. For the richer K-facing profile, the bundle additionally carries the carrier-indexed J.QSyncCertInd/QSyncCertState and may expose J.CertExtract_QS as the constructive on-demand extractor. Each CI1 record is structurally two-stage: its first field is the same J.QSyncTraceStep safety certificate, while later fields record the canonical post endpoint, fresh Core.KObsNormRef suffix v_ref and endpoint σ_ref , primitive K.NormStable checks plus K.NormStable-H, exact J.OfferReach/J.OfferConf/J.KObsConf correspondence, theorem-derived J.RegPhaseReach provenance on the extended τ_ref^G , and the freshly re-derived K.DrainReach/Core.SettleKBridge on τ_ref^K required by QC5. Carrying K.DrainReach here is a package-location choice and does not reclassify the underlying Sys_ref K.DrainEvidence/K ϵ /nonwithdrawal obligations as Appendix-J safety premises. Lemma J.QSyncCertInd-Trace projects the richer induction by retaining QC0-Safe and the per-edge J.QSyncTraceStep objects literally; it does not reconstruct safety from K-facing evidence. On the standard normal route, this bundle may reference the instance bundle's Core.KObsSettleQualified record together with Core.KObsSettle-Ref/K.NormStable and the K.DrainReach result produced by Lemma K.DrainReach-Comp rather than duplicate those cutpoint-specific proofs.

3. **K applicability and provenance bundle.** This bundle carries the route-applicability and proof-construction qualifications that make Appendix K sound beyond J correspondence. For $\text{Sys_B}^{\text{elab}}$ it includes the qualified FIFO/coupler handshake reconstruction and $\text{ReleaseOwnerSets}/\text{WaitOwners}$ process-dependency specification, and, when the primary Policy-S route consumes a coupling-sensitive boundary, the strengthened Q.1(l) serial-route handshake-support/CouplerPersistence template fact, including its K.SerialDomOrder support classification and a non-vacuous qualified persistence proof. For mutual-stall, shared-capacity, or arbitrated/non-monotone handshake topologies, Q.1(d)–(k) qualification alone is insufficient; absence of the qualified CouplerPersistence proof places the candidate outside $\text{K.PSF}/\text{CF-S}$ and requires another A5-L route. It otherwise includes the applicable $\text{K.BF}/\text{Core.IC}$ persistence and A6 source ownership facts; $\text{K.OpKey}/\text{K.SerialDomPredFinite}/\text{K.CVPred}/\text{K.CV-WF}$ provenance together with the theorem-side $\text{K.CompareFact-Mono}$, $\text{K.SegWork}/\text{K.SegWork-Complete}$, $\text{K.KeyItemOrderBridge}$, and $\text{K.SegmentPlan-EventuallyTarget}$ interfaces; the explicitly model-bound $\text{K.DrainEvidence}/\text{K}\epsilon/\text{nonwithdrawal}$ records (Sys_ref for the J/K witness path and Sys_K for the serial route); K.EnvMF and K.JointFreeze where required; $\text{K.CompareAdvanceLegal}$; $\text{K.SegmentPlan}/\text{K.SegmentAdm}$ and the FP4a-capable Core.FairPick construction; $\text{Core.CycleClosure}/\text{Core.IssueFair}/\text{B2}/\text{B3}/\text{B5}$; $\text{K.InterruptionGate}/\text{ResetEpoch}/\text{TerminalDisposition}$, including $\text{K.CompareScopeFootprint}$ and reset R-L/R-S coverage; NB-Align/K.Low/waiver conditions; and the remaining M.7a premises. K.JointFreeze-Ord -eligible ordinary/single-release-support cases may cite K.JointFreeze-Ord ; qualifying comparison-advance cases separately cite $\text{K.CompareAdvanceLegal-Ord}$ when its own premises hold. Collectively, these individual audit entries discharge the fields of Definition $\text{K.SerialRoutePremises}$; the record is packaging, not an additional independent premise. This bundle establishes applicability of the K reduction; it does not substitute for the selected complete Policy-S response result. The independent drain/conformance items in this bundle are the $\text{K.DrainEvidence}/\text{DR1-DR4}$ fields consumed by Lemma K.DrainReach-Comp , including the activation-indexed $\text{DR1a}/\text{DR1b}$ route map for mixed designs. The correspondence bundle separately supplies theorem-derived J.RegPhaseReach on τ_ref^G , J/I attribution/history, Core.QSync-Ref L2 settling, and the $\text{Core.KObsNormRef}/\text{K.NormStable}$ observation-suffix evidence; the lemma uses K.NormStable explicitly to extend DR1 across v_ref . Those theorem-derived/normalization facts need not be duplicated as an additional DR5 certificate when QC5 is discharged uniformly on the normal route.

4. Policy-S response bundle. For the primary bounded-response discharge of A5-L, this bundle binds the SDet -selected Policy-S simulator/run artifact and the response-monitor semantics to the same theorem instance, and carries or references K.CompleteS , total finite-bound K.RespCov , K.RespClean , the applicable response-waiver configuration, and the $\text{CF-S}/\text{A5-S}$ result. It references rather than duplicates the provenance, lowering, fairness, environment, reset/terminal, and other route-applicability evidence owned by the preceding bundles. With the K applicability/provenance bundle, Corollary K.2c uses this evidence to establish the required A5-L result on the supported fragment. If A5-L is instead discharged by K.Str-1 , CF-1 through CF-5 , a direct Policy-L invariant/exploration proof, or another permitted route, the corresponding K.A5Cert route payload replaces this Policy-S response bundle for that theorem application; the other J/K correspondence and implementation-qualification obligations remain governed by their own triggers.

Standard profile composition. For a standard QSync safety claim satisfying $\text{J.IdealPrefixClosed}$, the profile consists of the instance/QSync qualification bundle plus the safety form of the J correspondence/refinement bundle; the K applicability/provenance and Policy-S response bundles are not safety premises. $\text{Core.KOriginCarrier-AtOrigin}$ supplies the finite designated-origin chain and J.QSyncTraceInd establishes the cycle-aligned packages along it through per-edge $\text{J.QSyncTraceStep}/\text{J.GWR-CompExt}$ certificates, and $\text{J.CycleIdealCover}/\text{J.1-Safe-QSync}$ supply all-finite-prefix ideal closure without adding a new certificate obligation. If the richer J.QSyncCertInd K-facing bundle is already available, $\text{J.QSyncCertInd-Trace}$ supplies the safety induction by projecting those same

stored safety-step objects. For one-cutpoint generalized Theorem K.1A, the flow uses the applicable Sys_ref K.InFlightQual; when A5-L is obtained through a lowered serial route, the upstream A5 certificate additionally carries K.Low.ModalQual/generalized K.Low.A5, while the K.1A cutpoint package itself remains a Sys_ref package. The flow uses the exact K-facing J/cutpoint package, the triggered K applicability evidence including K.InFlightQual, and any valid A5-L discharge; universal QSync package induction and K.CertCov are not required merely for that one cutpoint. For the universal user-facing Theorem K.1 through the primary Policy-S route, all four bundles are used. In G1 with Sys_K=Sys_ref, checked QSync structure gives the standard KCut closure/canonical-domain results; J.QSyncCertInd together with the carrier-indexed J.CertExtract_QS/J.CertExist-QSync and K.CertCov-QSync derives K.CertCov; the K applicability bundle supplies the uniform Q.2/K.InFlightQual action/path qualification; and the Policy-S response bundle discharges A5-L through K.2b/K.2c before K.1A-Total is applied. In G2, the same universal QSync/cutpoint-coverage stack remains on the Sys_ref/RTL correspondence side, while the K applicability/provenance and A5 certificate additionally bind K.Low.ModalQual, both model-indexed K.InFlightQual records, W_K, and generalized K.Low.A5 to transport A5L_Sys_K to the source A5-L consumed by K.1A-Total. No G2 claim may reuse G1 continuation evidence across models without the separately named qualification or preservation evidence.

Independent-checker and API boundary. This profile intentionally does not prescribe a concrete file format, serialization, command-line interface, database schema, or certificate transport. Those are tool-flow API choices. Whatever representation is chosen, an independent checker shall validate the referenced subordinate evidence and reconstruct theorem-derived or KObs-derived facts where the existing certificate definitions require that behavior; a producer's final pass/fail Boolean is not a substitute for the M.7a discharges. A user-facing implementation may therefore report four top-level bundle statuses and provide drill-down to the named M.7a obligations without exposing every formal predicate as a separate top-level user action.

M.8 Practical interpretation

The architecture provides two different forms of scalability. Safety scales by keeping most design-specific evidence local to processes, explicit channel boundaries, and named atomic interactions, while using one generic theorem to construct the global replay witness. Liveness scales by separating the system-wide source premise A5-L from the available methods for proving it and by making the remaining progress, environment, drain, and cutpoint-coverage obligations explicit.

For a qualified HLS flow, the intended engineering result is that the pre-HLS SystemC model remains the primary place for system-level functional reasoning, coverage, and debug, while RTL work is concentrated on showing that the implementation conforms to the scheduling contract and its selected abstraction. The approach does not eliminate implementation verification; it changes its purpose from repeating the full functional test campaign at RTL to establishing the evidence needed to justify transfer from the source-reference verification result.

The same proof architecture is not inherently limited to HLS-generated RTL. Hand-written RTL, modified generated RTL, or RTL produced by another generation flow can use the same results when the same process, channel, observation-boundary, reset, environment, progress, and implementation-side obligations are established. The user-visible and tool-neutral form of the contract is also what makes the document suitable as a starting point for standardization discussions.

Appendix N – Scheduler and Environment Fairness Assumptions

The formal results in Appendices I–K rely on several scheduler/environment assumptions. The B-rule table in the Core gives B1–B6. This appendix restates boundary-action fairness/progress in ready/valid form. For ordinary primitive channels, B3 is exactly the P_push/P_pop discharge interface below. For Sys_B^elab, those clauses apply to the primitive finite-capacity/rendezvous elements, while Q.1 supplies the stateless coupler composition fact that maps changes in those explicit causes and process request roots to the next settled ChannelEnabled/ChannelDisabled valuation. A coupler itself has no separate fairness or progress state.

- **B2 (Weak fairness of the scheduler).** If the enabling predicate for an action (Push, Pop, Sync) remains continuously true from some cycle onward, the scheduler must eventually select that action within a finite, but unspecified, number of cycles.

Core.IssueFair (source issue; separate from B2). If a reached source operation remains legally IssueEnabled at every applicable L1 opportunity from some cycle onward within the same reset epoch, it is eventually issued. This is weak fairness over request assertion, not over commit. Core.16 is part of issue legality, so a source-later operation prohibited by an active blocked-frontier discipline is not IssueEnabled and imposes no IssueFair obligation. B2 begins after an applicable Push/Pop/Sync boundary action is ChannelEnabled. Core.LAdm/K.2b consume Core.IssueFair as a premise of complete admissible continuations; finite-prefix Appendix J does not require a generic issue-fair extension theorem. Constructive use in Lemma K.2b.E is through Definition Core.FairPick and Lemma Core.FairCycleDovetail. This does not strengthen the behavioral IssueFair/B2/B3 property, but it does require a stronger evidence form: prefix-local selector/starvation proofs for the particular constructed execution. Proof-selectable Policy-L issue latitude may be dovetailed; instance-fixed scheduler/arbitrator/environment/witness choices must be followed and justified, and the selected L3 action is chosen only after an actual typed L2 settle prefix has been obtained.

- **B3 Channel Progress Invariants (ready/valid form).** For a primitive channel element c , blocking Push/Pop endpoint requests are non-withdrawable during the outstanding attempt within one reset epoch. If a primitive buffered Push is blocked because the represented storage is full and the complementary Pop request persists, some Pop eventually commits and discharges the full condition. If a primitive buffered Pop is blocked because the represented storage is empty and the complementary Push request persists, some Push eventually commits and discharges the empty condition. For a primitive $B(c)=0$ rendezvous, persistent matching endpoint requests eventually complete the atomic rendezvous. Push payload remains stable while its request is outstanding; an affecting RESET ends the dynamic instance under RW2.

For Sys_B^elab, these primitive obligations are composed through the declared finite stateless ready/valid network. A boundary may remain ChannelDisabled while its local storage is nonfull/nonempty because another explicit handshake cause is missing. Q.1 requires that every such cause be exposed through explicit stored channel state or modeled process-facing request roots and that the next settled valuation respond deterministically when those roots change. Therefore B3 does not assert that every coupler-disabled boundary eventually enables; a genuine closed wait cycle may persist and is exactly what Appendix K analyzes. B2 applies only once the actual settled ChannelEnabled predicate remains continuously true.

Clarification (scope of the B3 conclusion). For ordinary primitive $B(c)>0$ channels, P_push concludes a complementary Pop discharge and P_pop concludes a complementary Push discharge; neither directly commits the original blocked endpoint. After the discharge, Core.11–Core.13 determine the actual settled ChannelEnabled value. In Sys_B^elab that value includes the declared coupler network, so occupancy becoming nonfull/nonempty does not by itself create a B2 obligation unless the complementary ready/valid value is also asserted. For primitive $B(c)=0$ rendezvous the matching discharge is the atomic rendezvous itself.

- **B5 (System quiescence closure).** If at some cycle no observable actions are enabled in any process, and no external or environmental change during the closure interval changes the relevant enabling predicates, then within a bounded number of cycles the system reaches an ϵ -fixed point. Thereafter, while no external or environmental change re-enables observable or internal work, the system executes no additional ϵ -steps. This prevents a dead, all-disabled state from being hidden behind an infinite tail of internal activity, without requiring the system to remain quiescent after a later reset, input change, externally supplied message, or other environmental event re-enables work.

B6 (terminal idle only). B6 may provide an indefinite idle-clock suffix only when `Core.TerminalIdle` holds: the system is at a genuine B5/global fixed point, future registered/internal machine evolution cannot re-enable work, and `Core.EnvTerminal` establishes the same no-future-enabling fact for the environment. A time-aware environment can be perfectly total while delivering an input at a later cycle, so `Core.CycleClosure`'s environment-definedness clause does not imply `Core.EnvTerminal`; likewise environment terminality alone does not prove that an internal timer/FSM cannot advance. Temporary cycles with no committed Σ action are ordinary `Core.SilentCycle` transitions, not B6 terminal stutter. These B-rules are **theorem-applicability assumptions**, not guarantees automatically provided by the HLS tool, and their discharge is not exclusively an environment responsibility. Depending on the selected theorem instance, they may constrain source/scheduler semantics, HLS-generated or hand-written RTL arbitration and back-pressure, channel-library behavior, or external masters. In particular, arbitration introduced by HLS for resource sharing, memory-port scheduling, channel selection, or similar implementation functions must satisfy B2 whenever it controls selection among continuously B2-enabled actions in the theorem scope. The corresponding implementation-side evidence is recorded under the Appendix M.7a B2/B3 discharge; reusable tool or library qualification may supply that evidence when applicable.

Priority arbiter example: fair vs unfair priority

Consider a simple two-input priority arbiter inside the post-HLS RTL. Each input is driven by a message-passing channel; the arbiter issues Pop operations to select which request to serve in each cycle.

- Input 0: high-priority channel `c_high`
- Input 1: low-priority channel `c_low`

Both channels may have pending requests at the same time.

1. Fair priority arbiter (satisfies B2 / B3).

A fair implementation might still give `c_high` strict priority in *individual* decisions, but it ensures that a continuously pending `c_low` request cannot starve. In RTL terms, this can be realized by, for example:

- A **round-robin or rotating-priority** arbiter that periodically moves the “highest” priority to the next requester, or
- A **bounded-burst priority** scheme in which `c_high` may win at most N consecutive cycles while `c_low` is requesting; after that, the next grant is forced to `c_low`.

Under these implementations:

- If `Pop_low`'s `ChannelEnabled` predicate remains true indefinitely (the low-priority channel has a pending request and the relevant abstract boundary availability predicate holds), the scheduler/arbiter must eventually select `Pop_low`. The selected `Pop_low` then commits in the selected cycle. This satisfies B2 for that action.
- If the low-priority FIFO is full and keeps requesting service, the system ensures that some complementary `Pop_low` eventually commits, discharging data and freeing space.

This is consistent with P_push/P_pop in B3; it is a channel-progress obligation that may be needed before a blocked producer's Push becomes ChannelEnabled.

Intuitively, the arbiter is still “priority-based,” but its micro-architecture ensures that every continuously enabled requester is eventually selected, and that full/empty channel states covered by B3 cannot persist forever under the stated persistent-request premises.

2. Unfair priority arbiter (violates B2 / B3).

A more naive design might implement:

```
if (req_high) grant_high();  
else if (req_low) grant_low();
```

combined with an environment in which req_high can remain asserted indefinitely. In this case, even if req_low is also continuously asserted:

- Pop_low's ChannelEnabled predicate may be true forever, but Pop_low is never selected by the scheduler/arbiter.
- Low-priority requests can starve indefinitely, even though they are continuously eligible for selection.

This behavior violates B2, because B2 requires eventual selection of any continuously enabled action. If c_low's FIFO is full and the only way to make space is to service it, permanent starvation can also violate B3's progress expectations for that channel, because the full condition may never be discharged under the stated persistent-request premises.

In such a design, the safety properties E1–E5 may still hold (trace-compatibility on actions that do occur), but the liveness/results that depend on B2/B3—especially the starvation-vs-deadlock arguments in Appendix K—no longer apply.

Intentional priority does not waive these liveness premises. If a fixed source or implementation arbitration policy can leave a continuously ChannelEnabled lower-priority action unselected indefinitely, B2 fails and the B2-dependent Appendix K liveness results are unavailable even when the starvation is intentional and source and RTL agree. If instead a reached source operation remains continuously IssueEnabled at the applicable L1 opportunities but is never issued, the relevant failure is Core.IssueFair, not B2. If source control never reaches the operation, that is source-control starvation. If a reached operation is IssueEnabled only intermittently, or an issued boundary action is ChannelEnabled only intermittently, the corresponding Core.IssueFair or B2 weak-fairness antecedent does not hold; starvation under such intermittent enablement is therefore outside that weak-fairness guarantee by design, not a violation of it. Unless separately covered, these cases are outside the applicable internal-message-deadlock guarantee.

Patterns that typically satisfy the fairness assumptions

The following design patterns normally satisfy B2 and are compatible with B3/B5, provided the rest of the system is well-behaved:

- **Round-robin or rotating-priority arbiters.** Any requester that remains enabled will eventually be at the front of the rotation and will be granted.
- **Age-based or credit-based arbiters.** Requesters accumulate age or credits while waiting; the arbiter prefers older or more-starved requests, guaranteeing that long-pending actions eventually win.
- **Bounded-burst fixed priority.** Fixed priority is combined with a counter that limits how long a higher-priority requester can dominate while lower-priority requesters are pending. After the burst limit is reached, lower-priority requests are forced to win until the system is “caught up.”

- **Back-pressure with bounded stall.** When ready/valid or similar handshake signals are used, the design (or its environment) must enforce that ready cannot remain low forever while valid remains high, and vice versa. For example:
 - Downstream consumers are required (by specification) to service queues at least once every N cycles when data is present.
 - External bus masters are configured such that they cannot indefinitely defer reading from full DUT FIFOs that are logically part of the interface contract.
- **Clock-gating and power-management schemes that respect B5.** Once no observable actions are enabled, the design either:
 - Quietly stabilizes (no further internal toggling), or
 - Explicitly enters a low-power state from which it only wakes when some observable action again becomes enabled.

In each case, the intent is the same: continuous ChannelEnabled status implies eventual scheduler/arbitrer selection and commit under B2; persistent full/empty/rendezvous-blocked request premises imply eventual discharge under B3; and once nothing is enabled, the system converges rather than oscillating internally.

Patterns that tend to violate the fairness assumptions

Conversely, the following patterns are likely to **break B2/B3/B5** and therefore fall outside the scope of the liveness arguments in this document:

- **Pure fixed-priority arbitration with unbounded high-priority traffic.** A static priority tree with no rotation or aging, combined with workloads in which a high-priority master can keep its request asserted indefinitely, can starve lower-priority channels forever.
- **“Best-effort” external masters with no progress guarantee.** For example:
 - A software driver that reads from a DUT output FIFO only when it happens to poll, and that may be pre-empted indefinitely by higher-priority threads.
 - An external bus or DMA engine that is architecturally allowed to ignore some requesters forever if higher-priority traffic remains heavy.
 Such environments can violate **P_push/P_pop** by allowing FIFOs to remain full or empty indefinitely while the DUT is continuously requesting progress.
- **Circular back-pressure loops with no escape.** A closed wait cycle alone is the Appendix K case described above and need not violate **B3**. B3 is violated only if a stated progress antecedent remains continuously true but its required discharge never occurs.
- **Internal oscillation without quiescence.** A scheduler or control FSM that can toggle internal ϵ -level state forever, even after all observable actions are disabled, violates **B5**. While such designs are uncommon in practice, patterns involving mis-configured clock gating, asynchronous feedback, or “keep-alive” timers that never expire can have this effect.

For cases above that actually violate B2/B3/B5, the trace-compatibility theorems **still describe what happens when actions do occur**, but the liveness claims that depend on B2/B3/B5 (for example, “a continuously enabled channel will not starve”) are no longer guaranteed. Designers and verification engineers should therefore either:

- Architect their arbiters and environments to satisfy these B-rules, or
- Treat the liveness guarantees in Appendix K as *out of scope* for those particular interfaces, and rely only on the safety-oriented parts of the model.

Appendix O – Support for Multiple Clock Domains (Informative Sketch)

This appendix is an informative sketch of how the single-clock formalism of Appendices G–K could be organized in a multi-clock design. The formal results of Appendices G–K are stated for a single global clock unless an explicit multi-clock semantic layer is added.

Within a single clock domain d , one may treat that domain as a synchronous island with a local cycle counter clk_d and apply the existing R- and E-rules to processes wholly inside that domain, interpreting $\text{clk}(\cdot)$ as $\text{clk}_d(\cdot)$ for local synchronization, signal I/O, and message-passing events. Inter-domain communication should be represented by explicit clock-domain-crossing components, such as CDC FIFOs, whose source-side and destination-side commit semantics are specified at their respective clock-domain boundaries.

A full formal lift requires additional definitions not supplied by Appendices G–K themselves: a global event order or partial order relating local-domain cycles, precise CDC commit semantics, occupancy sampling rules across source and destination clocks, synchronizer fairness/metastability abstraction assumptions, and progress obligations for CDC components. Once those additional definitions are supplied, the intended proof strategy is to treat each synchronous island using the existing single-domain rules and to treat each CDC component as an explicit channel/refinement boundary with its own stated E4-style legality and progress obligations. Without those additional definitions, Appendix O should not be read as claiming that the single-clock proofs apply unchanged to arbitrary multi-clock systems.

Appendix P – AI Discussion of Alternative Formal Frameworks

The following links provide an AI discussion of alternative formal frameworks compared to the one presented in this document:

https://drive.google.com/file/d/1ccU-eRW7H2ggh-AZyKnCOzO-KLPejob_/view?usp=sharing

<https://chatgpt.com/share/69457119-ec30-8006-8684-9a2a8b19f96f>

<https://drive.google.com/file/d/1R3-46DtZLsBn1hdTbLEXyC0FVB50Gymz/view?usp=sharing>

Appendix Q – Multiple Input Pipelines Back-Annotation

Appendix J Remark 2 discusses back-annotation of pipelined loops with a single message passing input. In more complex cases such as HW pipelines with multiple message-passing inputs, the back-annotation mechanism may elaborate the Sys_B (source-level) simulation by introducing additional internal realization structure. In this document, we adopt the following simplified modeling discipline:

1. Storage/accounting. Every stateful buffering, skid/elastic stage, slot, or credit that can hold or backpressure a message is represented by an explicit finite-capacity channel with $B(\cdot)$ and $occ_{\cdot}(t)$.
2. Handshake logic. Every non-storage join, mutual-stall, bundle, gate, or glue function is represented as stateless deterministic combinational ready/valid/data logic. It may connect internal elaborated boundaries or sit directly at a process-facing outer boundary.
3. Settled boundary meaning. $BoundaryReq$ remains the source-operation-attributed own-side request. $ChannelEnabled/ChannelDisabled$ is the resulting settled handshake at that boundary after the complete declared combinational network has evaluated. E4 governs storage/rendezvous legality and committed-transfer accounting; at a coupling-sensitive boundary the complementary ready/valid value may be stricter than the local not-full/not-empty condition.

Observation-boundary convention. $\Pi_{elab}/\pi_{\Sigma,E}$ selects the outer Σ_{DUT} history. Proof-internal storage channels may be projected away from Σ_{DUT} while still contributing explicit channel state to Appendix K. Stateless coupler nodes are not channels, do not contribute capacity, and are not WFG process vertices.

Concretely, a combinational coupler reads upstream valid/data and downstream ready values and deterministically computes downstream valid/data and upstream ready values. It contains no state and no credit/capacity. Any stateful feedback passes through an explicit finite-capacity channel. The same definition covers a coupler placed immediately at a process boundary: it remains part of the elaboration's same-cycle handshake network and does not become a source process or a residual-offer owner.

Example (two inputs, staged Pops, $II=1$, latency=4):

Consider a 4-stage RTL pipeline that (i) performs Pop_{c1} in stage 1, (ii) performs Pop_{c2} in stage 2, and (iii) performs $Push_{cout}$ in stage 4, with $II=1$. To model the pipeline's internal staging and the stage-2 input coupling in the source-level Sys_B^{elab} simulation while keeping the same outer projected Σ boundary, the back-annotation elaboration may introduce the following internal realization structure:

- An explicit bounded **channel F1 of capacity 1** on the $c1$ path to represent stage-1 in-flight storage of values that have been accepted from $c1$ but have not yet reached the stage-2 join point.
- An explicit internal rendezvous channel F2 on the $c2$ path at the stage-2 join boundary, with $B(F2)=0$. F2 represents the $c2$ -side handoff into the join and contributes no independent storage. Its source-attributed request is one root of the settled join handshake; whether the evaluated F2 boundary is $ChannelEnabled$ is determined after the coupler also considers the F1 token and downstream F34 space.
- A purely combinational coupler/join at the stage-2 boundary that permits the joined action only when (a) a token is available from F1, (b) the $c2$ -side F2 request/handshake is present, and (c) downstream F34 has space. The coupler therefore computes the complementary ready/valid values seen at the affected boundaries; it does not change their stored occupancies and owns no operation. If any of the three conditions is missing, the affected source-attributed $BoundaryReq$ may remain true while $ChannelEnabled$ is false after settling. Appendix K sees the resulting dependency through the E-declared $ReleaseOwnerSets$ family; $WaitOwners$ is only the union used for the coarse graph, and the coupler itself is not turned into a process.
- An explicit bounded **channel F34 of capacity 2** after the coupler to represent the in-flight capacity of stages 3–4 for the joined/paired work.

This example illustrates why multi-input pipelines cannot always be captured by a single per-channel capacity number in isolation: $c1$ and $c2$ do not have independent “slack” because acceptance across the

stage-2 join boundary is coupled. In this particular example, the c1 side has one stored stage-1 token available through F1, the c2 side has no independent pre-join storage and is therefore modeled by the rendezvous channel F2 with $B(F2)=0$, and the joined/paired work has two downstream storage slots through F34.

All storage-bearing buffering/capacity resides in the explicit bounded channels F1 and F34, and in any additional positive-capacity stages the realization uses. The explicit rendezvous channel F2 has $B(F2)=0$ and contributes no storage; it is an explicit proof-internal join boundary used to type the mutual-stall condition. The coupler itself remains stateless combinational handshake logic.

This elaboration refines the abstract bounded channels at the chosen outer observation boundary without changing the outer event labels or E4 capacity/order meaning. It may introduce proof-internal storage-channel commits, and it may place stateless coupling logic at internal or outer process-facing boundaries. Any elaboration must satisfy:

- No new outer observables: $\pi_{\Sigma,E}$ neither creates nor relabels outer Push/Pop events.
- E4 storage legality: every explicit finite-capacity/rendezvous channel has its own $B(\cdot)$, $\text{occ}_{\cdot}(t)$, FIFO/rendezvous legality, and committed-transfer accounting. Outer projected commits remain E4-legal under Π_{elab} .
- Declared handshake completeness: every ready/valid/data value used by ChannelEnabled/ChannelDisabled is produced by the E-declared finite same-cycle network from fixed process request roots, explicit channel state, and permitted fixed inputs. A coupling-sensitive boundary may therefore be disabled despite local capacity, but never because of an undeclared predicate.
- No double counting: movement within one represented storage realization is ϵ unless it crosses an explicit abstract channel boundary; a committed proof-internal channel transfer may be projected away from Σ_{DUT} but remains part of the proof-internal history.
- Conservation/accounting: projected-away internal movements preserve outer occupancy/accounting except when E maps a transfer to the corresponding outer commit. Bundle/join accounting is declared by E.

• **Multi-input pipelines / combinational couplers (mutual stall + explicit capacity):**

Any downstream storage reachable only through a join is modeled as explicit finite-capacity channel state rather than as hidden coupler capacity. The coupler may enforce mutual stall by computing ready/valid from several explicit channel/request conditions, but it remains stateless. Accordingly, the coupler does not redefine E4's storage capacity/order legality; it determines the actual settled handshake eligibility at the boundary. This same rule applies if the coupler is placed at the process-facing outer boundary.

Projection/accounting and handshake invariant for $\text{Sys}_B^{\text{elab}}$ (normative implementation of Core.ElabProj). Π_{elab} preserves the selected outer event history and occupancy accounting. Separately, the E-declared combinational network preserves the actual handshake structure: a source-attributed BoundaryReq is never withdrawn by the coupler, every complementary ready/valid value is a deterministic function of the declared current roots, and a boundary action can commit only when both its BoundaryReq and settled complementary signal are asserted and E4 legality is satisfied. Cross-channel/join dependence may therefore appear directly in ChannelEnabled/ChannelDisabled at any boundary designated by E, including a process-facing outer boundary; what is forbidden is hidden or stateful coupling outside E.

For Appendix K, E also declares the template-qualified ReleaseOwnerSets_E family for each false complementary signal after contracting local stateless logic and explicit channel-internal paths. Each

family member is one minimal conjunctive set of modeled process owners sufficient to release that blocked condition; different members are alternatives. WaitOwners_E is the union of those sets and is used only for the coarse directed WFG/structural graph. Local coupler nodes are never WFG vertices. A purely local combinational path does not create $P \rightarrow P$ merely because it lies inside P ; a $P \rightarrow P$ dependency exists only when a genuine release support includes source/process behavior owned by P . Conversely, a genuine internal process-mediated release alternative may not disappear during contraction.

- WFG/release-family compatibility (Appendix K): if a process cannot advance a pending blocking frontier, the disabled boundary status must be visible after settling and the qualified elaboration template must provide the complete minimal internal process release-support family for that frontier. The certificate obligations are family-level: (1) no spurious rescue alternative—every declared support S is genuinely sufficient, as a joint owner support under the template semantics, to release that particular blocked condition; and (2) no missed rescue—every genuine internal modeled-process release possibility under the declared template semantics contains at least one declared minimal support S . Minimal supports are kept as an antichain. Taking the union gives WaitOwners_E and the conservative coarse WFG, but exact $K.\text{Dead}$ closure is evaluated on $\text{ReleaseOwnerSets_E}$. External/terminal alternatives remain outside ordinary WFG and use the existing waiver/diagnostic treatment. Couplers have no independent transition or owner; changes in their outputs arise only from changes in explicit channel state or process-facing roots.

Pure combinational reevaluation that changes no BoundaryReq , explicit channel state, settled $\text{ChannelEnabled/ChannelDisabled}$, ReleaseOwnerSets , or WaitOwners fact is $\epsilon/\text{WFG-inert}$. A stateful staging element that can block progress must be an explicit finite-capacity channel; a stateless coupler that participates in blocking remains represented only through the settled handshake value and the template-qualified release-support/dependency projection, even when located at the process boundary. Proof-internal channel commits may be projected away from Σ_{DUT} by Π_{elab} .

- Combinational well-formedness: the complete coupler/ready-valid network is finite, deterministic, and acyclic after cutting at explicit registered/storage boundaries. Any feedback or remembered condition required for correctness is represented by explicit finite-capacity/registered state. This condition is exactly what permits one settled handshake valuation to be reconstructed from E plus the current state/request roots.

Accordingly, the formal proof need not depend on one particular back-annotation algorithm. It depends only on the supplied $\text{Sys_B}^{\text{elab}}(E)$ satisfying the storage/accounting rules, the complete stateless handshake equations, the same-cycle settling qualification, and the process-dependency projection below. Appendix J reconstructs the settled handshake from $E+H+O$ (plus the replay roots explicitly permitted by E); Appendix K uses the same reconstructed handshake and contracts its blocking dependencies to the original modeled processes. The outer Σ_{DUT} history and aggregate storage accounting remain governed by $\Pi_{\text{elab}}/\pi_{\Sigma}, E$.

Lean-oriented clarification / proof-parameter convention

The formal theorems do not require a unique algorithm that constructs $\text{Sys_B}^{\text{elab}}$ from RTL_B . Instead, the chosen comparison boundary Sys_ref is part of the theorem instance. For ordinary bounded-FIFO cases, Sys_ref may be Sys_B . For cases involving proof-relevant internal staging, multi-input joins, combinational couplers, sync-boundary epoch gates, or shared-capacity effects, Sys_ref shall be supplied as $\text{Sys_B}^{\text{elab}}(E)$ for some well-formed elaboration package E .

Thus elaboration is not an extra semantic freedom inside the proof. It is an explicit input to the proof instance, and all Appendix I–K definitions are interpreted over that chosen Sys_ref .

Definition Q.1 (Elaboration package for $\text{Sys_B}^{\text{elab}}$).

For a chosen outer source reference boundary Sys_B and outer observable alphabet Σ_{DUT} , an elaboration package E consists of:

1. an elaborated source reference model $\text{Sys_B}^{\text{elab}}(E)$;
2. a finite set of explicit abstract channels $\text{Chan_elab}(E)$, each with finite capacity $B_E(d) \geq 0$ and nonnegative cycle-boundary occupancy $\text{occ}_d(t)$;
3. a distinguished subset of outer-boundary channels Chan_outer and a channel projection/accounting-fiber map

$$\Pi_{\text{elab}} : \text{Chan_elab}(E) \rightarrow \text{Chan_outer} \cup \{\perp\};$$

4. a trace projection map $\pi_{\Sigma,E}$ from proof-internal Σ_{elab} traces to outer Σ_{DUT} traces;
5. a state-accounting map A_E with signed integer codomain, where $A_E(c, \sigma)$ computes the outer accounting value for c from the occupancies and token/accounting state of the elaborated channels in $\Pi_{\text{elab}}^{-1}(c)$; and
6. a finite proof-internal elaboration-token universe $\text{ElabToken}(E)$, a token projection

$$\Pi_{\text{token},E} : \text{ElabToken}(E) \rightarrow \text{Set}(\text{Chan_outer}),$$

with $\Pi_{\text{token},E}(u)$ finite for every token u , together with an E -declared attribution that is functional when defined: each relevant staged/bundle/join transfer occurrence is assigned to at most one token. The attribution may be undefined for an occurrence with no outer effect. $\Pi_{\text{token},E}$ identifies the finite set of outer channels whose $\pi_{\Sigma,E}$ effects are accounted for by that token; $\pi_{\Sigma,E}$ determines the corresponding outer Push/Pop labels, payloads, and ordering.

The package is well formed only if the following obligations hold:

(a) Outer-trace preservation. $\pi_{\Sigma,E}$ neither creates, drops, duplicates, nor relabels any outer-boundary $\text{Push}_c(v) / \text{Pop}_c(v)$ event.

(b) Occupancy/accounting preservation. For every outer channel c and every ε -quiescent cycle-boundary state σ at time t , let π_t be the c -projection of the outer history $\pi_{\Sigma,E}$ strictly before cycle t . The signed state-accounting value computed from the elaborated state shall agree with that independently reconstructed outer history:

$$A_E(c, \sigma) = \text{push}(\pi_t) - \text{pop}(\pi_t).$$

Here, because $c \in \text{Chan_outer}$, $E4$, occ_c , and $B(c)$ mean the corresponding outer Sys_B channel semantics, not the $B_E(d)/\text{occ}_d$ semantics of the explicit elaborated channels in item 2. By that outer $E4$ legality, this same signed balance equals the legal nonnegative outer occupancy $\text{occ}_c(\sigma)$ and satisfies $0 \leq \text{push}(\pi_t) - \text{pop}(\pi_t) \leq B(c)$. Thus, viewing the legal occupancy as an integer, $A_E(c, \sigma) = \text{occ}_c(\sigma)$ and $0 \leq A_E(c, \sigma) \leq B(c)$. This range is a derived consequence of the Q.1(b) agreement plus outer $E4$ legality, not a consequence of A_E 's signed codomain. Projected $\text{Push}_c / \text{Pop}_c$ commits update A_E exactly as

required by outer E4 for channel c.

(c) Internal refinement and multi-effect projection. An internal staged/bundle/join transfer with no outer effect may be projected away by $\pi_{\Sigma,E}$ and may have no attributed token. Any internal staged/bundle/join transfer that contributes one or more outer effects shall have the item-6 attribution defined and is therefore attributed by E to exactly one token $u \in \text{ElabToken}(E)$; $\Pi_{\text{token},E}(u)$ is the uniquely determined finite outer-channel projection for that token, and $\pi_{\Sigma,E}$ determines the corresponding one-or-more outer Push/Pop commit effects consistent with it. A single token may therefore account for multiple outer channels/effects. No outer effect may be invented, omitted, duplicated, or multiply charged by the token projection. Projected-away internal transfers preserve every outer signed accounting value $A_E(c,\sigma)$.

(d) Complete stateless handshake network. Every same-cycle join, mutual-stall, bundle, sync/epoch gate, shared-capacity handshake, and process-boundary coupler that can affect a selected boundary ready/valid/data value is included in E as deterministic combinational logic. Such logic introduces no stored state and may be attached to an internal or outer process-facing boundary.

(e) Settled boundary visibility. For every evaluated frontier boundary, E identifies the own-side source-attributed BoundaryReq root and the complementary mirror value used by Core.11. Primitive channel storage legality is E4; the actual ChannelEnabled/ChannelDisabled value is the settled handshake produced by the complete E network. No undeclared grant, throttle, backpressure, or guard may affect it.

(f) Combinational well-formedness. After cutting at explicit registered/finite-capacity state, the complete elaboration-owned ready/valid/data network is finite, deterministic, and acyclic; any feedback/storage required for correctness is represented by explicit state. For the standard ReleaseOwnerSets route, the availability-dependency portion used to attribute a false ready/valid value shall additionally be monotone in its designated process-facing release roots once explicit registered/arbitration state is fixed. Non-monotone priority or mutual-exclusion logic whose release meaning depends on both assertion and deassertion shall be represented through an explicit arbiter/state boundary or carry a separately qualified release-support certificate rather than being inferred by root polarity alone.

(g) Endpoint roles and process ownership. Every source-level logical message channel retains exactly one producer process and one consumer process, and every source-attributed frontier item records its owning process/operation/direction. An explicit $\text{Sys_B}^{\text{elab}}$ storage boundary may have a stateless coupler on one side; the coupler is an endpoint realization component, not a modeled process. Shared source resources that truly have multiple process owners are remodeled through explicit arbitration and point-to-point process-facing channels. For each coupling-sensitive disabled frontier, E carries a qualified symbolic ReleaseOwnerSets schema whose state-instantiated value is the family of minimal internal process-owner release supports; WaitOwners_E is defined as its union. For an ordinary primitive boundary the family is exactly $\{\{Q\}\}$ for the unique complementary process owner Q.

Clarification (source-level logical channel versus elaborated explicit boundary). The one-producer/one-consumer statement above applies to a source-level logical message channel. In $\text{Sys_B}^{\text{elab}}(E)$, E may refine that channel into explicit proof-internal channels or boundaries whose individual complementary endpoints are driven or observed by stateless combinational couplers rather than directly by a modeled source process. Such a coupler may be implemented as a combinational process or module in SystemC/RTL terminology, but it is not a modeled process for source ownership, WFG vertices, Dead_K, or process-local source order. Source-operation ownership remains $\text{MsgOf_P}(x)$, while handshake

evaluation occurs at $\text{BoundaryOf_P}(x)$. The ordinary $\{\{Q\}\}$ reduction applies only when the complementary endpoint of the evaluated boundary is itself the unique modeled complementary process owner Q . If that endpoint is a stateless elaboration coupler, the boundary is coupling-sensitive for ReleaseOwnerSets purposes and E shall use the qualified $Q.1(k)$ dependency contraction; the resulting support may be conjunctive, alternative, or both.

(h) Frontier-item declaration and boundary attribution. E declares every proof-internal frontier item used by $\text{Sys_B}^{\text{elab}}$, including its owning process, $\Omega_P/\Omega_msg, P$ membership, owning source-level message instance $\text{MsgOf_P}(x)$, evaluated handshake boundary $\text{BoundaryOf_P}(x)$, and $\leq_P$ relation. $\text{BoundaryReq}(x, \sigma)$ remains the persistent own-side request attributed to that operation/frontier item. A stateless coupler may transform the complementary mirror signal at the same boundary but does not acquire MsgOf ownership and does not become a new residual offer. For elaborated calls with several frontier items, E states how those items and any proof-internal commits project through $\pi_Σ, E/\Pi_elab/\Pi_token, E$ and A_E back to the outer operation/event/accounting view.

Clarification (typing correction versus topology change). Reclassifying an existing E -declared boundary from an outer/source logical channel to a proof-internal elaborated boundary is a typing correction only when the declared storage, transfer semantics, handshake equations, and projection/accounting structure are otherwise unchanged. By contrast, introducing or removing a positive-capacity explicit channel, staging buffer, registered boundary, or other state-bearing element changes $\text{Sys_B}^{\text{elab}}(E)$, its explicit channel state, and its settled handshake equations, and generally requires a fresh $Q.1(j)/(k)$ reconstruction and release-support derivation. A proof may not carry over equations or ReleaseOwnerSets from a no-storage topology merely by renaming its boundaries.

(i) Qualified same-cycle settling. The elaborated source model as a whole shall satisfy the Core.Settle L2 frame and the source-side hypotheses of Definition Core.QSync . The elaboration package is responsible for the settling facts contributed by the staged/bundle/join/coupler/epoch-gate storage and handshake structure. For that elaboration-owned cycle-local subnetwork, every applicable $\rightarrow_{\epsilon, L2}$ step shall respect the $L2$ frame—preserving the cycle index, registered process/FSM/pipeline state, registered explicit-channel storage/occupancy/credit state, and cycle- t exogenous inputs, with no explicit-boundary Σ commit or next-cycle registered update—and, after cutting at the state-bearing boundaries, the subnetwork shall be finite, deterministic, combinational acyclic, and complete for every ready/valid/data and boundary predicate it owns. The package shall provide or cite a compositional/template proof of these template-side facts; that proof may be qualified once for each legal elaboration template and then bound to the concrete E instance.

Whole-source composition requirement. The template-side facts alone do not imply $\text{QualifiedSynchronousModel_M}(I)$ for $M = \text{Sys_B}^{\text{elab}}(E)$. The selected theorem instance shall additionally establish the applicable design-owned source-settling facts, including stable sequential drives under $\text{Core.RegSeq/direct-input}$ contracts, inclusion and deterministic evaluation of $R2C$ combinational components, the applicable non-template $L2$ -frame facts, and completeness at non-template selected boundaries. It shall also establish a checked template/design settling-composition condition: every dependency edge crossing the design/template partition is included in the complete same-cycle dependency analysis, and connecting the two subnetworks introduces no violation of $QS2-QS5$. In particular, the dependency graph of the complete combined same-cycle settling network shall satisfy $QS4$. A sufficient discharge is a checked combined dependency rank or topological order in which every design-local, template-local, and cross-partition settling edge advances the order. The template-side, design-side, and composition evidence together establish the corresponding $\text{QualifiedSynchronousModel_M}(I)$ fact for $M = \text{Sys_B}^{\text{elab}}(E)$, so Theorem Core.QSync-Ref supplies existence and uniqueness of the source SettlePoint/KCut . This obligation makes the former implicit

source-settling termination premise explicit; Definition Core.Settle by itself does not assume a fixed point exists.

(j) Handshake and release-support reconstructibility. At every source K-facing cutpoint, the complete settled boundary valuation and the state-instantiated ReleaseOwnerSets family are pure functions of E, reconstructed explicit channel state, the source/replay roots explicitly permitted by E, and the attributed outstanding own-side requests. Equivalently, evaluating the same finite acyclic equations and qualified release-support schema on equal Core.18c KObs records yields equal MirrorAvail/ChannelEnabled values and equal ReleaseOwnerSets; WaitOwners equality follows by union. No coupler state, per-cutpoint prime-implicant extraction, or historical issue timestamp is required. The RTL boundary mapping used by J.KObsConf shall validate the same equations/signals and release-support schema.

(k) Release-support declaration, correctness, and WFG contraction. The qualified elaboration template declares a symbolic release-support schema once for each supported coupling topology; the theorem instance evaluates that schema at the reconstructed settled state to obtain ReleaseOwnerSets_E(x,σ). It is not specified as a per-cutpoint Boolean-minimization algorithm. For each disabled frontier x, the resulting finite antichain of minimal non-empty process-owner sets shall satisfy two family-level obligations. Support soundness/no spurious rescue: every declared S is a genuine jointly sufficient internal modeled-process support for releasing x under the declared template semantics. Support completeness/no missed rescue: every genuine internal modeled-process release possibility for x under the declared template semantics contains at least one declared minimal support S. Define WaitOwners_E(x,σ):=UReleaseOwnerSets_E(x,σ); Core.WFG uses that union as a conservative directed graph, while Definition K.Dead applies the exact family-level release-closure test. Purely local combinational realization does not create a process self-dependency; a self-dependency is retained only when a genuine release support includes another source/process action owned by that same process. External/terminal alternatives are handled outside ordinary WFG. A qualified template may prove these obligations once for all legal instances of its topology. Satisfaction of this clause establishes correctness of the release-support schema relative to the qualified E template semantics. Faithfulness of those template equations, signals, and the release-support schema to the implementation is a separate boundary-mapping obligation under Q.1(j) and J.KObsConf; passing clause (k) alone does not establish RTL correspondence.

(l) Serial-route handshake support and CouplerPersistence (required only when K.2b consumes a coupling-sensitive boundary). For every such boundary action unit T, the qualified elaboration template identifies a finite support consisting of explicit channel/history facts and currently asserted source-attributed request roots sufficient to determine T's settled enablement. The stored/history support must be produced by K.SerialDomOrder action units <_SD-earlier than T, or be represented with T as one declared atomic boundary-action unit when same-cycle indivisibility is required. Therefore, after the least-missing induction has matched those earlier support facts and K.2b.R has supplied the same persistent request roots, deterministic settling gives the same ChannelEnabled result needed by K.2b.P. In addition the template shall prove CouplerPersistence_E(T): for every Core.LAdm-admissible continuation in the fixed K.DomResetScope DomEpoch that satisfies the base K.2b.E frozen-comparison invariant Inv (I1–I5, not the construction-bookkeeping Inv+), once those declared earlier support facts and persistent request roots are present, every subsequently permitted whole action unit before T commits preserves either (a) the declared support facts sufficient to recompute ChannelEnabled(T), or (b) directly the truth of ChannelEnabled(T) at the next applicable settled boundary opportunity. Here “before T commits” is temporal order along the considered Policy-L continuation: a permitted liberal-only unit may be quantified over even though it lies outside the domain of <_SD. The property is scoped only to that fixed-DomEpoch base-Inv continuation class; it is not a universal monotonicity claim for arbitrary design behavior. A reset/terminal interruption is handled only by the existing

K.ResetEpoch/K.TerminalDisposition DirectFail/Continue structure. CouplerPersistence converts the one-time deterministic-settling result into the continuous B2 antecedent consumed by K.2b.P. This remains a template/proof-interface obligation, not an extra runtime guard or hidden state; the template proof is fixed independently of the later K.2b.E continuation and may be qualified once per legal elaboration template, then bound to the concrete E instance.

Worked Q.1(l) persistence discharge (non-vacuity witness). Consider a qualified stateless two-input join whose output is the sole producer of an explicit point-to-point buffered channel d. Let T be the join output transfer. The template declares as support the two source-attributed input request roots, their already-matched payload/history facts, and the output-space fact $\text{occ_d} < B(d)$; the two requests and any stored/history producers are $<_{SD}$ -earlier than T unless the join and T are declared one atomic unit. Once both input requests are asserted, nonwithdrawal keeps them present until the join/T unit consumes them. By A6 point-to-point ownership together with the template's sole-producer binding, no other Push_d can fill the available slot before T; any Pop_d only decreases occupancy and therefore preserves output space. Nonwithdrawal keeps the input roots present; the base K.2b.E invariant Inv preserves the frozen component and fixed operational choices, while A6—not Inv—excludes a competing producer, and the fixed-DomEpoch condition excludes an affecting reset. Hence every permitted pre-T action preserves the declared support and ChannelEnabled(T), establishing CouplerPersistence_E(T) and the continuous B2 antecedent. This worked template demonstrates that clause (l) is not satisfied merely by excluding all coupling-sensitive boundaries; more complex joins/bundles may use an equivalent qualified invariant over their declared support equations.

Definition Q.2 (profile-indexed finite in-flight action/effect schema; Stage-1 base and Stage-2 synchronization extension).

Profile scope. Q.2 supports both current generalized profiles. G1 uses $\text{Sys_K} = \text{Sys_ref}$ with ordinary already-issued message actions plus qualified explicit staging/bundle/join actions. G2 adds separately typed synchronization/lowering constructors and lowering-created guard/epoch administrative actions, while literal Definition K.Dead remains rooted in blocking message-operation frontiers in each proof model. Pure synchronization is not widened into MsgOf or message BoundaryReq. A lowered theorem application is accepted only when K.Low.ModalQual and the global Stage-2C path/escape/drain correspondence are bound to the exact Q.2 schemas; local Stage-2B action typing alone is insufficient. For each qualified elaboration E and selected profile, InFlightActionSchema_E declares a finite set of action templates and, for every instantiated action identity u, the following current-state semantics: (Q2a) typed identity and participating explicit boundaries/tokens/transaction keys; (Q2b) state-based admission carrier showing that the work is currently present in O, explicit ChanRec/token/storage state, or a Stage-2 SyncCarrierRec; (Q2c) exact enablement/applicability equation from the reconstructed current roots; (Q2d) indivisible atomic grouping where same-cycle effects may not be split; (Q2e) exact successor effects on explicit channel contents/occupancy/credit, attributed requests/process-facing commit state when applicable, synchronization transaction/process-interval state when applicable, reset/epoch/token/accounting state, and H-retained proof units; (Q2f) frame conditions for unrelated state and an explicit prohibition on embedding new source ReachPrep/Issue inside an in-flight action; (Q2g) deterministic post-action re-settling through Q.1(i)/(j) and the applicable synchronization-root equations; and (Q2h) the $\pi_{\Sigma, E/\Pi_{\text{elab}}/\Pi_{\text{token}}, E/A_E}$ /lowering projection-accounting effect. The schema may not consult historical Issue time or undeclared hidden state.

Stage-2B synchronization action constructors.

A synchronization-qualified action schema may instantiate two distinct constructors. (1) SyncFire_I(κ), only from the Core.7b/Core.18d-1 carrier of an already-pending ready transaction κ , with Core.SyncFence-valid atomic synchronization/E2-anchor effects exactly as in Core.7d. A peer that has

not reached the transaction is not an action carrier; its required progress remains represented in SyncReleaseOwnerSets. (2) EpochGateAdvance_{{l,E}(g,κ)}, only on Sys_K for explicit lowering-created gate state bound to κ, with the exact gate-state effect of Core.7e and with lowering projection equal to stutter. A schema may contain both constructors for the same logical synchronization occurrence, but it shall not identify them or make one constructor's event/effect stand in for the other. No constructor issues or reaches a post-sync source operation.

Theorem Q.InFlightActionEffectSound (dangerous-direction local fidelity).

For every qualified Q.2 template and reachable operational state represented by reconstructed state r , each declared action identity u denotes a genuine action kind of the selected operational proof model, its reconstructed carrier/applicability equation agrees with that action's local state precondition at an applicable settled opportunity, and, if u is selected/occurs, its declared successor effect is exact: applying Q2(e)–(h) and deterministic re-settling yields the reconstructed successor r' corresponding to the operational post-action state. Thus the schema may neither invent a nonexistent action kind/local enablement condition nor assign a real action a fictitious successor. Either error is a dangerous-direction soundness failure because it can fabricate a WaitRoots change and therefore fabricate InFlightEscape, hiding a real Dead_K. This theorem deliberately does not prove that an applicable opportunity for u exists from every K-facing phase/state, or that such opportunities compose across cycles. Those continuation-existence claims belong to Core.InFlightCycleExt/J.InFlightPathSound.

Lemma Q.SyncInFlightActionEffectSound (synchronization/epoch local fidelity).

Assume Core.SyncCarrierWF and a Stage-2B synchronization-qualified Q.2 schema. For SyncFire(κ), reconstructed applicability is exact iff the same currently represented transaction is pending/ready, participant/reset identities match, and Core.SyncFence is satisfied; occurrence has exactly the Core.7d atomic synchronization, replay/interval, carrier-retirement, framing, and re-settling effect. For EpochGateAdvance(g,κ), reconstructed applicability is exact iff the declared lowering gate state is currently present and E4/handshake-enabled; occurrence has exactly the Core.7e gate/token/accounting effect and projects to source stutter. Neither constructor may fabricate the other's observable unit. Proof obligations are local state/action correspondence only; Core.SyncInFlightOpportunityQual supplies operational opportunity and Stage-2C K.Low.InFlightPathCorr supplies cross-model segment correspondence.

Theorem Q.InFlightChoiceFinite and optional completeness qualification.

At every Core.FlightStateRec state, the instantiated enabled action inventory is finite because E declares a finite action-template universe over finite explicit channel/token state and the current process population is finite. Under Stage 2B, synchronization constructors are instantiated only from the finite current set of reached/pending SyncTxn keys (deduplicated per atomic transaction, not once per endpoint) and the finite set of explicit lowering-created gate objects bound to those keys. This proof shall not be inferred merely from the finite committedUnits of a completed CycleResult. A template may additionally prove Q.InFlightActionEffectComplete for a declared operational domain. Completeness is a precision theorem, not the dangerous-direction soundness theorem: omitting a genuine legal action can over-report Dead_K, whereas adding a fictitious action or effect can hide Dead_K.

Worked-artifact status. Artifacts A–C below are illustrative schema-validation instances used to exercise the generic Q.2 architecture; they do not by themselves qualify a concrete design or reusable template library. Every concrete template consumed by a theorem application must separately bind its exact Q.2 action/effect schema and discharge the applicable Q.InFlightActionEffectSound/Q.InFlightChoiceFinite obligations together with the required Core.InFlightCycleExt/J.InFlightPathSound opportunity/path

qualification; a G2 lowering application additionally binds the applicable synchronization and K.Low.ModalQual evidence.

Worked Q.2 artifact A — atomic two-input join.

Let J be a qualified stateless two-input join with current source-attributed blocking requests q_1 and q_2 , reconstructed stable payloads v_1 and v_2 , deterministic function F , and an explicit sole-producer output FIFO d with $\text{occ}_d < B(d)$. Declare $\text{JoinFire}_J(q_1, q_2, e)$ as one atomic action whose carrier is exactly those two requests/payloads, the common epoch, d state, F , and the fixed E projection/token metadata. When enabled, the action retires exactly the process-facing request instances that E identifies as completed by the atomic join, appends $z = F(v_1, v_2)$ once to d , emits exactly the E -declared proof/outer effects, frames unrelated state, and re-settles the stateless network. If the eventual blocked candidate D contains a consumer C of d while the two input owners lie outside D , JoinFire remains in $\text{Core.InFlightStepRec}$ because the action relation has no D /ownership filter; after re-settling it may change C 's WaitRoots_D enablement. This is the concrete regression against a $\text{DrainScope}_E(u) \cap D$ filter.

Worked Q.2 artifact B — two-stage non-fall-through chain.

Let a, b, d be explicit positive-capacity $B=1$ channels with state $a=[z]$, $b=[]$, $d=[]$, and let C have a persistent blocking Pop_d . E declares $\text{MoveAB}(z)$, atomically $\text{Pop}_a/\text{Push}_b$, and $\text{MoveBD}(z)$, atomically $\text{Pop}_b/\text{Push}_d$. By E4 beginning-of-cycle non-fall-through semantics, the token pushed into empty b by MoveAB cannot be consumed from b in the same cycle, so MoveBD requires a later real cycle. MoveAB leaves $\text{WaitRoots}_{\{D\}}(K)$ unchanged for D containing C ; MoveBD later makes d nonempty and changes C 's reconstructed frontier enablement. Therefore the required escape witness has length two. This artifact proves that immediate-root-change is insufficient and that $J.\text{InFlightPathSound}$ must be compositional across real cycle boundaries. The admitted-population rank may assign $a > b > d > \text{retired}$ and use the multiset extension; IF2/Core.16(2) prevents replenishment by new source Issue.

Worked Q.2 artifact C — ready synchronization plus residual epoch-gate skew.

Let a two-party transaction κ have both modeled endpoint carriers already pending and SyncReady at a K -facing state. On Sys_ref , $\text{SyncFire}(\kappa)$ is one atomic synchronization unit and advances both participants only to the immediate post-sync interval entry. In a qualified non-atomic lowered epoch-gate template, the corresponding Sys_K realization may require one or more explicit gate-realization steps after the projected synchronization occurrence before all lowering-created gate state is retired. The subsequent $\text{EpochGateAdvance}(g, \kappa)$ changes only that declared lowering gate/token/accounting state and is erased by the $K.\text{Low}$ projection. Thus a lowered K -facing state after the projected synchronization but before the residual gate advance may contain a temporary literal gate blocker even though its Sys_ref projection is already past κ . Q.2 must expose every such certified residual EpochGateAdvance as an in-flight action so that the temporary obstruction is not classified as drain-closed. An atomic epoch-gate template with no residual step is simply the zero-stutter special case. The artifact demonstrates why Stage 2 requires stuttering path correspondence rather than a step-for-step action bijection.

Qualification boundary. A real implementation whose join/stage/synchronization/gate semantics depend on an internal register, token-valid bit, credit counter, arbitration state, participant state, or other state not represented by the qualified proof model cannot use a Q.2 instance merely because a tool knows that hidden state. Expose the state at a qualified abstract boundary, reconstruct it through the typed $\text{SyncCarrier}/\text{participant}$ interface, or prove it irrelevant to action identity/effect. Pure synchronization remains separately typed and is never forced into $\text{MsgOf}/\text{message BoundaryReq}$. Except for the explicitly outer projection/accounting semantics of Q.1(a)–(c), when $\text{Sys_ref} = \text{Sys}_B^{\text{elab}}(E)$, unqualified $B(\cdot)/\text{occ}_\cdot$ and E4 used for Sys_ref operational channel semantics refer to the explicit storage/rendezvous channels declared by E . Within Q.1(a)–(c), $c \in \text{Chan_outer}$

instead denotes the corresponding outer Sys_B channel, with its outer B(c), occ_c, and E4 legality used as the refinement target. BoundaryReq refers to the source-attributed own-side request at the selected frontier boundary; ChannelEnabled/ChannelDisabled refers to the settled handshake produced by E; ReleaseOwnerSets carries the exact internal process release alternatives used by K.Dead; and Appendix K's coarse WFG uses the WaitOwners union. Outer observations/accounting are recovered only through the coordinated $\pi_{\Sigma,E}$, Π_{elab} , $\Pi_{token,E}$, and A_E interface.

Notation convention for A_E. This document writes the signed state-accounting function in channel-first form $A_E(c, \sigma)$. A mechanization may equivalently use the state-first form $A_E(\sigma, c)$ so that state-indexed maps compose naturally with elaborated-state projections. The two spellings denote the same signed integer-valued accounting function with the argument order reversed; no semantic distinction is intended. At the ε -quiescent cycle-boundary states covered by Q.1(b), its legal nonnegative range is theorem-derived from agreement with the E4 outer-history balance, not encoded by the function codomain.

Locality of elaboration

The elaboration package E is normally constructed from local process/channel realization templates: finite storage elements and stateless ready/valid couplers are selected from the process's pipelining, input/join, synchronization, and boundary structure, and independently qualified templates compose through their declared interfaces. This locality is an engineering/compositionality property, not a prohibition on proof-internal auxiliary handshake nodes. Such nodes remain realization detail; the template contracts them into ReleaseOwnerSets and then the WaitOwners union before Core.WFG is formed over modeled processes. Cross-process obligations arise only through the process-facing channel/request dependencies that already connect the source processes; $\Pi_{elab}/\pi_{\Sigma,E}$ continue to project storage/event accounting per selected outer channel.

Appendix R – Compact Derived Formal Core and Mechanization View

R.1 Purpose and normative status

This appendix is a derived companion to the normative theorem stack. It introduces no definition, premise, execution domain, reconstruction function, or theorem interface consumed by an earlier section. The normative dependency order is: the Normative Formal Core; Appendix I process-local replay and preparation; Appendix J system construction, KObs reconstruction, and equal-KObs correspondence; and Appendix K wait-for/deadlock factoring and preservation. This appendix may be used as a compact review or mechanization map, but citations needed to instantiate a theorem shall point to those normative locations.

Deletion-safety rule. Removing this appendix shall not make any earlier formula, proof obligation, certificate field, or theorem statement undefined. A future edit that introduces an upstream citation to an Appendix R item is therefore an architectural regression.

R.2 One-way dependency structure

Core $\rightarrow$ I $\rightarrow$ J $\rightarrow$ K $\rightarrow$ R.

The Core fixes Sys_ref, explicit abstract channels and elaboration, Σ/ε , global Core.4 quiescence, L2 Core.Settle points and Core.QSync, the Core.KCut comparison domain, Core.ReachPrep, dynamic frontier items and source order, Commit/Issue/BoundaryReq, Core.IC/Core.SyncFence, Core.MemFaithful/Core.MemSyncOrder/Core.MemProfile/J.Mem, Core.IssueFair,

ChannelEnabled/ChannelDisabled, Pending and Front, Core.7a BlockingMsgNextFront, Core.ElabProj, Core.WFG, Core.RegSeq, Core.LLegal, Core.SelectLatitude, Core.Phase, Core.PolicyL/Core.PolicyS, Core.16/Core.Drain discipline, Core.InFlightStep/Core.InFlightDrainWF/Core.InFlightCycleExt, Core.CycleState/Core.RTLKOriginFrame/Core.RTLDesignatedOriginFrame/Core.CycleResult/Core.RTLCycleStep/Core.KOriginCarrier/Core.CompleteCycle/Core.SilentCycle, DeclaredTerminal/Core.TerminalIdle, Core.EnvStep/Core.EnvTerminal, Core.CycleClosure, Core.FairPick/Core.FairCycleDovetail including construction-side FP4a certified-segment progress, Core.LAdm, and the Core.18c KObs datatype plus Core.FlightStateRec/Core.InFlightStepRec pure reconstruction functions. Appendix I proves process-local deterministic replay and finite preparation. Appendix J composes the local evidence, proves the J.KObs reconstruction lemmas, and establishes Corollary J.1 equal-KObs correspondence. Appendix K factors the deadlock predicates through that record and adds the operational liveness, drain, fairness, response, terminal/reset applicability, and certificate-coverage obligations. Appendix R only summarizes the completed chain.

R.2a K-observation architecture

The formal development has three explicit layers. First, the operational semantics define Core.RegSeq, Core.ReachPrep, Core.SelectLatitude, ReturnedAtCycle, Policy L/S, channel commits, Core.IssueFair, Core.Settle/Core.QSync, the separately triggered Core.KObsSettle/Core.KObsSettleQualified K-facing source observation interface, Core.CycleState/Core.RTLKOriginFrame/Core.RTLDesignatedOriginFrame/Core.CycleResult/Core.RTLCycleStep/Core.KOriginCarrier/Core.CompleteCycle/Core.SilentCycle, DeclaredTerminal/Core.TerminalIdle, environment/cycle closure, the constructive Core.FairPick/Core.FairCycleDovetail fair-continuation interface, Core.Drain/K.Drain, and the generalized Core.InFlightStep/Core.InFlightDrainWF/Core.InFlightCycleExt interfaces, including the Stage-2 synchronization/gate constructors when selected. Second, Appendix I records enriched deferred NB holes and field-sensitive local preparation; Appendix J derives the dependency order, proves J.RegPhaseNF, performs Horizon-batched J.UnitExt, constructs the J.GWR prefix $\tau_ref^G/J.RegPhaseReach$, proves exact cross-prefix restriction/extension through J.LocalGWRExtCert/J.GWR-CompExt when carrier induction is used, and proves Theorem J.1/J.HCanon. The QSync safety induction packages each carrier edge as a J.QSyncTraceStep whose schema contains only the incremental J/I/QSync-Ref/reset/J.HCanon facts; the richer certificate induction stores that safety step as its first stage and appends K-facing work afterward. Third, the K cutpoint layer appends the recorded Core.KObsNormRef suffix v_ref to form τ_ref^K , checks K.NormStable, and combines H and O in KObs; Core.FlightStateRec/Core.InFlightStepRec then reconstruct the finite in-flight transition system without adding KObs fields. J.KObsConf transfers one selected cutpoint; K.DrainReach supplies the remaining nonwithdrawal/ $K\epsilon$ /Core.16 bridge on τ_ref^K ; K.InFlightQual/J.InFlightPathSound supplies the separately qualified finite-path operational realization used by generalized DrainClosedNow, with Lemma K.DrainReach-Comp explicitly using K.NormStable to cover the suffix outside J.GWR-Comp; and K.RLAdmReach is the three-part composition of J.RegPhaseReach, K.NormStable, and K.DrainReach. K.CertifiedCutpoint and Theorem K.1A operate per cutpoint; Core.QSync-Post derives Core.KCutClosure_post from the qualified synchronous structural evidence; Core.KOriginCarrier-AtOrigin/-Decomp make finite cycle ancestry explicit; J.CertExtract_QS folds the QSync certificate induction over a represented carrier position; and K.CertCov/K.1A-Total supply the separate certificate-totality/universal-preservation layer. The Policy-S reduction then adds K.EnvMF, K.DomResetScope, the shared K.InterruptionGate schema with K.ResetEpoch and nested K.TerminalDisposition instantiations, K.BF, K.Resp, and the derived operational chain $K.2b.E \rightarrow K.2b.P0/K.2b.R/K.2b.P \rightarrow K.2b.P-Frozen/K.2b.SF \rightarrow K.2b.Q \rightarrow K.2b/K.2c$. The ReachProgress construction in that chain uses K.SerialDomPredFinite, K.CompareFact-Mono, K.SegWork, K.KeyItemOrderBridge, and K.SegmentPlan-

EventuallyTarget; these names summarize normative Appendix-K/K-P results and add no Appendix-R premise.

(1) Operational Issue is retained. Issue transitions and the Issue/Commit linkage remain part of the operational semantics and continue to govern R4/E3/E5, Policy L, Policy S, Core.IC, Core.Drain/K.Drain, the applicable progress and fairness premises including K-O1, Lemmas K.2b.R/K.2b.P/K.2b.SF/K.2b.Q, and the K.Resp monitor trigger. KObs removes historical issue timestamps only from the cross-model cutpoint-observation and certificate interface; it does not redefine Issue(q), identify Issue(q) with Commit(q), or permit an operational proof to ignore whether work was already issued before a block.

(2) KObs is an observation-layer quotient, not a normalized execution. The factoring theorems prove that K-relevant cutpoint predicates depend only on replay history plus the current attributed residual offers. They do not construct a replacement execution in which committed operations issue in their commit cycles, and no issue-erasure normal form is assumed.

(3) Residual offers are typed over frontier items and explicit boundaries. A residual offer identifies the owning process P, frontier item $x \in \Omega_msg, P$, owning dynamic message-call instance $q = MsgOf_P(x)$, evaluated explicit abstract boundary $c = BoundaryOf_P(x)$, endpoint direction, and reset epoch. Its operation kind is the explicitly derived field $kind(o) := kind(q)$, with well-typedness requiring producer-valid exactly for a Push instance and consumer-ready exactly for a Pop instance. A bare set of source calls q is insufficient in Sys_B^elab, where one source-level call may own several proof-internal frontier items at different explicit boundaries.

(4) The full elaboration package E is a parameter of KObs. E includes the selected Sys_ref form, explicit storage/rendezvous channels and capacities, $\Pi_elab/\pi_Σ, E$, frontier ownership/boundary maps, the complete stateless ready/valid/data equations, the qualified ReleaseOwnerSets schema and derived WaitOwners union, and any qualified serial-route handshake-support template facts. KObs is not parameterized by B alone; stateless coupler values and release families are reconstructed from E rather than stored as new KObs state.

(5) Cross-model residual-offer conformance is J.OfferConf: every KObs-relevant source-attributed own-side request is represented bidirectionally and uniquely in O. Stateless coupler-derived mirror signals are not residual offers and do not need fictitious operation owners; J.KObsConf plus the E/B-faithful boundary mapping validates their deterministic reconstruction. This preserves operation attribution while still making every ChannelEnabled/WFG fact reconstructible.

(6) Residual-offer reachability remains separate from boundary conformance and liveness admissibility. J.OfferReach establishes that a proposed KObs record is in the image of the exact reachable Sys_ref K-facing cutpoint reached by the recorded $\tau_ref^G/Core.KObsNormRef$ extension, with the same annotated replay history and witness w; K.NormStable supplies the separate observation-suffix invariance needed to transport the J.HCanon history. J.OfferConf establishes agreement between the record's residual offers and the RTL requests at the selected cutpoint. J.KObsConf requires these correspondence components, while K.RLAdmReach is the separate three-part proof that the source K-facing prefix is Core.LAdm-admissible. OfferConf alone cannot manufacture an unreachable source offer set, and I.ReplayObs/I.DR replay matching alone cannot determine I.CutpointOfferObs or which uncommitted requests are presently asserted.

(7) Generalized cutpoint drain closure is distinct from the operational drain/continuation qualifications. In any qualified generalized profile, DrainClosedNowRec_{M,E}(D,KObs) means that the finite D-independent Core.InFlightStepRec_{M,E} relation reconstructed from the current KObs contains no finite path that changes WaitRoots_D for the selected proof model M. In G1, M=Sys_ref=Sys_K; in G2, the Sys_ref and Sys_K relations are separately qualified and K.Low.ModalQual supplies the cross-model modal correspondence at administratively normalized paired cutpoints. Legacy DrainCandidates_D remains an ordinary/diagnostic enabled-offer selector. Core.Drain/K.Drain continues to govern activation-local no-new-later-work/nonwithdrawal/finite non-replenishment on actual source paths,

while $K.InFlightQual/Core.InFlightCycleExt/J.InFlightPathSound$ separately justifies that the reconstructed modal paths are operationally realizable. None of those evidence records is a $KObs$ field.

(8) The minimal K -specific residual-offer abstraction carries no pending Push payload. Payload stability while a blocking Push request is asserted remains required by $Core.IC$ and Appendix C, and the committed payload remains checked by the value-labeled replay/trace results. A proof or checker needing mid-flight data-wire correspondence may add that separate implementation obligation without enlarging the deadlock observation itself.

(9) Elaboration and issue policy remain separate. Pipeline storage, bundle/join structure, and internal stage movement are represented by E and the explicit Sys_B^{elab} channels. Policy L/S governs issue of the actual source-level operation instances and their elaboration-provided frontier items. $KObs$ records current source-attributed offers but does not turn intra-realization movement into additional source calls or double-count capacity already represented by $E4$ occupancy.

(10) $KObs$ -facing waiver classes are observationally closed. For any theorem, certificate, or checker that claims a waiver-parameterized cutpoint predicate is determined by $KObs$, write $KObsClosed_I,E,w(W)$ for the requirement that, for all $\sigma_1, \sigma_2 \in KCutReach_ref(I)$ of the same fixed instance, $KObs_E,w(\sigma_1) = KObs_E,w(\sigma_2)$ implies $(\sigma_1 \in W \text{ iff } \sigma_2 \in W)$. $K.Resp$ monitor waivers are a separate complete-run certificate interface: they are keyed by the dynamic process-facing frontier instance and must satisfy $K.RespCov$'s coverage-preservation rule. A waiver class that depends on historical Issue timing, implementation microstate, monitor age, or any other fact not reconstructible from $KObs$ cannot be used in $KObs$ extensionality unless a separate proof establishes $KObsClosed_I,E,w(W)$.

(11) The primary serial route and diagnostic terminal reconstruction are separate instance data. The ambient theorem instance records the $K.Resp$ monitor schema, finite-bound function, reset/end-of-test disposition, and response-waiver policy used by $CF-S$. Those monitor fields are not part of $KObs$ and are not transferred to RTL as latency claims. Separately, an instance may request diagnostic reconstruction of $WFG+$, $EnvStop$, $DoneStop$, $Dead_KS$, $K.EnvOrd$, or $K.Done$. For $EnvStop$ diagnostics, the instance records either the standard $B1$ -avail, pull-based, consumption-ordered input model with default always-ready outputs, or an explicit $EnvReplayAdequate$ certificate showing that permanent next-interaction availability is a function of H and w . Without one of these environment conditions, $KObs$ still factors the ordinary internal $Dead_K$ predicate but makes no extensionality claim for $EnvStop$ -dependent diagnostic predicates.

(12) The certificate boundary is layered and explicit. A proof-backed application records: (a) $K.CertHdr$, including $Core.RegSeq/Cat\#$ and source-simulation conformance plus the $Core.QSync$ structural-evidence provenance; (b) $CF-O$ with $H,O,J.HCanon/Core.KObsNormRef/K.NormStable/J.OfferReach/J.OfferConf/J.KObsConf$ and, for Appendix K, $J.RegPhaseReach$ provenance on τ_ref^G plus $K.DrainReach$ on τ_ref^K ; (c) the selected $A5$ payload; and (d) the $Core.QSync$ -Post theorem reference establishing $Core.KCutClosure_post$ plus $K.CertCov$ for a universal claim. $J.RegPhaseReach$ is theorem-derived on the compositional route from enriched holes, field-sensitive footprints, $J.RegPhaseNF$, and Horizon-batched $J.UnitExt$; $Core.KObsSettle-Ref$ plus $K.NormStable$ form the distinct source observation-normalization bridge; $DrainReach$ remains separately justified and is not derivable from $KObs$, phase normality, or normalization stability, but on the normal compositional route its exact- K -facing-prefix form is theorem-derived by $K.DrainReach-Comp$ from separately checked instance-level drain/conformance facts plus those upstream outputs. Deferred-hole records are proof-internal and empty at the $J.GWR$ construction endpoint and remain empty across $K.NormStable$; they do not enlarge $KObs$. Derived $Front/WFG/deadlock$ values remain checker-reconstructed.

R.3 Compact theorem-instance package

A theorem instance fixes the design, source and RTL identities, Sys_ref form, elaboration package E and Π_{elab} , explicit abstract boundaries and capacities, initial/reset state, stimulus and environment model, witness choices, arbitration and ε -normalization conventions, observation boundary, applicable memory mapping profile/reference semantics, waiver classes, and—when the Policy-S route is used—the concrete deterministic simulator and $K.\text{Resp}$ monitor configuration. These bindings are supplied by the Core and the certificate interfaces in J and K rather than by this appendix.

At a selected source K -facing cutpoint $\sigma_{\text{ref}} \in K\text{CutReach_ref}(I)$, Core.18c records $K\text{Obs_E}, w(\sigma_{\text{ref}}) = \langle E, w, H, O \rangle$. $J.\text{OfferReach}$ proves source realizability and $J.\text{OfferConf}$ proves the attributed own-side request abstraction; $J.K\text{ObsConf}$ binds the same E -declared handshake/release-support template to the RTL cutpoint. $J.K\text{Obs-Proc}/J.K\text{Obs-Chan}/J.K\text{Obs-Offers}$ reconstruct replay, stored channel state, and request/frontier roots; $J.K\text{Obs-WFG}$ then evaluates the stateless ready/valid network and ReleaseOwnerSets schema, derives WaitOwners by union, and reconstructs ChannelEnabled plus the coarse process WFG. $K.K\text{Obs-Dead}/K.K\text{Obs-Ext}$ factor the downstream release-aware deadlock predicates.

R.4 Compact safety and cutpoint bridge

Derived summary R-C1. Under the hypotheses of Theorem J.1 and Corollary J.1, every covered reachable RTL_B prefix has a compatible reachable Sys_ref witness under E1–E5, and the selected ε -normalized RTL/source cutpoints share one Core.18c $K\text{Obs}$ record. Equal reconstructed K -facing facts are functional consequences of that typed equality; they are not independent state-equality assumptions and do not transfer response-monitor age or continuation-level drain discipline. In the standard cycle-induction route, $J.\text{LocalGWRExtCert}/J.\text{GWR-CompExt}$ preserve exact nesting of the already selected $J.\text{GWR}$ construction prefixes across carrier edges, and $J.\text{QSyncTraceStep}$ is the independently typed safety edge consumed by $J.\text{QSyncTraceInd}$. The richer $J.\text{QSyncCertInd}$ reuses that same safety edge before adding any K -facing normalization/offer/drain fields.

R.5 Compact liveness bridge

Derived summary R-C2. For the exact source package selected by Corollary J.1, $J.\text{RegPhaseReach}$ supplies the Core.Phase L0–L4 phase-correct reachability component on the $J.\text{GWR}$ construction prefix $\tau_{\text{ref}}^{\wedge G}$. $\text{Core.KObsNormRef}/K.\text{NormStable}$ then form the observation-only suffix v_{ref} and K -facing prefix $\tau_{\text{ref}}^{\wedge K}$ ending at σ_{ref} without reopening L3 selection, and Lemma $K.\text{NormStable-H}$ preserves the enriched history. $K.\text{DrainReach}$ separately supplies Sys_ref nonwithdrawal, $K\varepsilon_{\{I, \text{Sys_ref}\}}$, and every activated Sys_ref $\text{Core.Drain}/K.\text{Drain}$ obligation on $\tau_{\text{ref}}^{\wedge K}$. Their three-part composition $K.\text{RLAdmReach}$ places $\tau_{\text{ref}}^{\wedge K}$ in $\text{Pref_L}^{\wedge \text{adm}}(I)$. If A5-L excludes $\text{Dead_K}^{\wedge W}$ throughout that admissible domain, Theorem K.1A excludes the corresponding reconstructed deadlock at the certified RTL cutpoint; $\text{Core.KCutClosure_post}$ and $K.\text{CertCov}$ are additionally required for the universal reachable-cutpoint conclusion. On the standard normal-compositional route, $\text{Core.KObsSettle-Ref}$ and Lemma $K.\text{DrainReach-Comp}$ supply the K -facing normalization/stability/drain facts from the already-qualified instance evidence; direct validation remains the fallback.

For the principal serial-reduction route, SDet selects one concrete Policy-S execution and CF-S supplies complete bounded-response coverage and cleanliness under $K.\text{EnvMF}$, the shared $K.\text{InterruptionGate}$ schema, and the existing K.2b constructive continuation machinery. In profile G1 ($\text{Sys_K} = \text{Sys_ref}$), the Core.CycleClosure already carried by $K.\text{SerialRoutePremises}$ supplies the C2/C3-class continuation component used by $\text{Core.InFlightCycleExt}/J.\text{InFlightPathSound}$, while Q.2 and $K.\text{InFlightQual}$ supply action/effect, finite-rank, and path-profile evidence. In profile G2, the same serial evidence qualifies the Sys_K relation only; the Sys_ref $K.\text{InFlightQual}$ required by $K.\text{Low.ModalQual}$ is separately discharged or

explicitly preserved. `K.Low.InFlightPathCorr` then relates the two reconstructed modal systems through finite guard/epoch administrative normalization, and generalized `K.Low.A5` transports `A5L_Sys_K` to the `Sys_ref A5-L` consumed by `K.1A`. `K.ResetEpoch/K.TerminalDisposition` and the `K.2b.E → K.2b.P0/R/P → K.2b.P-Frozen/SF → K.2b.Q` chain remain otherwise unchanged. Proof-internal action occurrence is not automatically process-facing source-operation completion; `E/K.CV` attribution remains authoritative.

R.6 Mechanization decomposition

A faithful mechanization should preserve the same module boundaries: (1) Core types and operational semantics; (2) Appendix I local replay/preparation; (3) Appendix J finite-ideal composition and `KObs` reconstruction; (4) Appendix K current-`KObs` factoring plus reconstructed in-flight transition/path soundness, admissible-prefix qualification, serial reduction, and certificate coverage; and (5) this appendix as generated documentation or a compact theorem-export layer. In particular, the mechanization should not define a second `Sys_ref`, a second admissibility relation, or a second `KObs` correspondence inside the companion layer. The Appendix-J module should expose `J.GWR-CompExt` as an exact-prefix extension theorem over τ_ref^G and represent `QSyncTraceStep` as a safety-only datatype/relation with no constructors or fields importing the later K-facing package; `QSyncCertInd` should extend that safety object rather than re-prove it from richer fields.

R.7 Audit checklist

A structural audit passes when: every shared definition is located in the Core before first use; Appendix I depends only on the Core and its explicit premises; Appendix J imports I and the Core; Appendix K imports the Core, I/J results, and its named liveness premises; no earlier section cites Appendix R for a required definition or lemma; and Appendix R's compact statements are traceable to named normative results without adding stronger claims.

Appendix S – Lean Proof Strategy Document

Scope of this inventory. The list below gives selected foundational import anchors for the mechanization spine; it is not an exhaustive inventory of every theorem premise or qualification obligation. Appendix M.7a remains the definitive consolidated side-condition discharge ledger.

A separate Lean proof strategy document is under development. It should import, rather than duplicate, the applicable normative interfaces fixed here. Foundational anchors include: `Core.1/Core.ElabProj`; `Core.2–Core.4` observable/ ϵ /quiescence semantics; Appendix G `R1–R5/E1–E5`, including the `R2C` fixed-point semantics/matching used by `R2/E2`;

`Core.Settle/Core.KObsSettle/Core.KObsNormRef/Core.KCut/Core.SettleIsKCUT/Core.KObsSettle-Ref`; `Core.QSync/Core.QSync-Ref/Core.QSync-Post/Core.QSync-KCutCanonical`; `Core.8/Core.9` Commit/Issue, `Core.IssueFair`, `Core.10` endpoint-owned `BoundaryReq`, and `ReturnedAtCycle`;

`Core.RegSeq/Core.SelectLatitude/Core.IC/Core.SyncFence`; `Core.7b–Core.7e` synchronization-carrier/`SyncFire/EpochGateAdvance` typing, `Core.SyncCarrierWF/Core.SyncInFlightOpportunityQual`; `Core.MemFaithful/Core.MemSyncOrder/Core.MemProfile/J.Mem`; `Core.11–Core.13`

`MirrorAvail_E/ChannelEnabled` semantics; `Core.14/Core.15` Pending/Front;

`Core.WFG/ReleaseOwnerSets_E/WaitOwners_E`; `Core.Phase/Core.LLegal/Core.PolicyL/Core.PolicyS`;

`Core.16/Core.Drain/Core.SettleKBridge/Core.InFlightStep/Core.InFlightDrainWF/Core.InFlightCycleExt`;

`Core.CycleState/Core.RTLKOriginFrame/Core.RTLDesignatedOriginFrame/Core.CycleResult/Core.RTLCycleStep/Core.KOriginCarrier/Core.KOriginCarrier-Extend/Core.KOriginCarrier-`

`AtOrigin/Core.KOriginCarrier-Decomp/Core.KOriginCarrier-`

`KCut/Core.CompleteCycle/Core.SilentCycle/Core.RTLCycleCarrier/Core.CycleClosure/Core.CycleChoiceExt/Core.FairPick` (including `FP4a` certified-segment progress)/`Core.FairCycleDovetail`;

Core.KCutClosure/Core.KCutObserve;
 Core.EnvStep/Core.EnvTerminal/DeclaredTerminal/Core.TerminalIdle/Core.LAdm; Core.17/ Σ Match
 system-level observable compatibility; and Core.18/Core.18a–Core.18e, including the Core.18c KObs
 datatype, Core.18d-1 synchronization-carrier/action reconstruction,
 Core.FlightStateRec/Core.InFlightStepRec, and Core-owned reconstruction functions. Appendix-I/J
 anchors include I.DeferredNBHole/NoDependentUse; J.ConstructionAction and the field-sensitive
 location vocabulary; J.ReqSnapshot/J.NBDecision/J.BufSameCycle; Dep_J semantic-versus-D5
 stratification/J.HorizonOrderSafe/FR/acyclicity/canonical order; J.RegPhaseNF/J.RegPhaseReach;
 PCert/CCert/LocalGWR plus J.LocalPrefixAgree/J.LocalGWRExtCert; frame/local-
 lift/commutation/enablement/Horizon-batched UnitExt; Proposition J.0; J.GWR-Direct/J.GWR-
 Comp/J.GWR/J.GWRPrefixExt/J.GWR-CompExt;
 J.CycleReach_post/J.QSyncTraceState/J.QSyncTraceStep/J.QSyncTraceStep-
 Sound/J.QSyncTraceInd/J.QSyncCertState/J.QSyncCertInd/J.QSyncCertInd-Trace/J.TraceCertCov-
 QSyncCycle/J.1-Safe-QSyncCycle/J.CycleIdealCover/J.1-Safe-QSync;
 J.TraceCertCov/J.OuterCanonHist/J.OuterCanon-
 HCanon/J.RequiredPredCycleOrder/J.RequiredPredPrefix/J.OuterCanon-
 Ideal/J.RefProjCover/J.IdealPrefixClosed/J.1-Safe; J.HCanon/J.OfferReach/J.OfferConf/J.KObsConf;
 J.KObs-Proc/J.KObs-Chan/J.KObs-Offers/J.KObs-WFG/J.KObs-SyncCarrier/J.KObs-SyncInFlight/J.KObs-
 InFlight/J.InFlightPathSound; J.CertExtract_QS/J.CertExtract-QS-Sound; and J.CertExist-QSync. Appendix-
 Q anchors include Q.1(a)–(c) trace/token/accounting projection; Q.1(d)–(h) stateless handshake-
 network completeness, settled-boundary visibility, combinational well-formedness, endpoint/process
 ownership, and frontier/boundary attribution; Q.1(i) qualified settling, Q.1(j) handshake/release-family
 reconstructibility, Q.1(k) release-support correctness/WFG-union contraction, and Q.1(l) serial-route
 handshake support plus CouplerPersistence where applicable; Q.2 InFlightActionSchema,
 Q.InFlightActionEffectSound, Q.SyncInFlightActionEffectSound, and Q.InFlightChoiceFinite. Appendix-K
 anchors include the Appendix K-P operational/elaboration typing convention distinguishing explicit-
 boundary action-unit occurrence from process-facing source-operation commit/return;
 K.EnvTerminalAdequate/K.KObsDerived;
 K.WaitRoots/K.InFlightEscape/K.WaitRootsProject/K.InFlightEscapeMono/K.DrainClosedHeredity/K.Prim
 itiveDrainExact; K.Dead/K.KObs-Dead/K.KObs-Ext/K.2a; K.ReleaseSCCReduction/K.InFlightRelocate;
 K.InFlightQual; K.DrainReach/K.DrainReach-Comp/K.RLAdmReach;
 K.OpKey/K.SerialDomOrder/K.SerialDomPredFinite/K.KeyItemOrderBridge/K.CVPred/K.CV-
 WF/K.CompareFact-Mono/K.SegmentPlan/K.SegmentAdm/K.SegWork/K.SegWork-
 Complete/K.SegmentPlan-EventuallyTarget/K.2b.P0/K.2b.P1/K.2b.R/K.2b.P/K.2b.P-
 Frozen/K.2b.SF/K.2b.Q;
 K.Low.IdClose/K.Low.EpochClose/K.Low.GuardClose/K.Low.AdminStepRec/LowAdminNF/LowAdminNor
 malize/LowWaitRoots/K.Low.ModalQual/K.Low.InFlightPathCorr/K.Low.InFlightEscapeCorr/K.Low.Dead
 RootBridge/K.Low.DrainClosedNowCorr/K.Low.C/K.Low.A5; K.DomResetScope/DomEpoch;
 K.JointFreeze-Ord/K.JointFreeze/K.EnvMF/K.EnvMF-Uniform;
 K.CompareAdvanceLegal/K.CompareAdvanceLegal-Ord; K.InterruptionGate and its exhaustive K.BF
 reset/terminal instantiations with Continue/DirectFail branch typing and reset-first nesting;
 K.Resp/K.RespCov; K.CertHdr/K.CertifiedCutpoint/K.CertCov; K.1A/K.1A-Total; Corollary K.2c; and K.PSF.
 Appendix R remains a deletion-safe, nonnormative derived companion. The exported interface must
 preserve the distinction between supplied local facts, theorem-derived order and phase normality, the
 exact-prefix drain bridge, the reconstructed in-flight path-soundness layer, the KObs observation
 boundary, and universal package coverage. The separate Lean document should be updated in the same
 named revision before mechanization relies on superseded interfaces, including the old stop-at-hole or
 unit-at-a-time forms.

Mechanization status rule. Encode Core.IC and Core.SyncFence as legality/invariant content of the Sys_ref operational semantics, not as unconstrained global axioms. Proofs about RTL_B shall consume separately named conformance evidence (directly or through the corresponding R/E and Appendix M.7a discharge records) before using the analogous lifecycle or fence facts on the implementation side. Encode Core.SelectLatitude likewise as constitutive source operational legality/non-maximal selection, not as an axiom of eventual progress and not as a use of Core.IssueFair or B2 in finite-prefix Appendix J.

Appendix T – Scheduling-Specific Obligations Beyond Standard Simulation Theory

T.1 Purpose and Nature of the Formal Result

Status. Appendix T is informative and non-normative. It highlights selected scheduling-specific library gaps but does not add, remove, redefine, or exhaustively enumerate formal obligations. Normative authority remains with the Normative Formal Core and the applicable theorem interfaces; Appendix M.7a supplies the definitive consolidated side-condition discharge record, and Appendix S records the Lean mechanization decomposition and selected import anchors.

Standard formal-methods libraries provide reusable foundations for labeled transition systems, weak or epsilon-abstracted transitions, forward simulation, finite and infinite executions, trace inclusion, well-founded induction, fixed-point reasoning, fairness, and related proof techniques. Those foundations can substantially reduce the generic infrastructure needed for a Lean mechanization of this document. They do not, however, establish the HLS scheduling contract or its safety and liveness guarantees by themselves.

The top-level safety result is not ordinary trace inclusion or trace equivalence. It is the document's witness-relative ordered trace-compatibility relation over the selected observable alphabet. Matching is defined over complete observable units and bucketed executions, preserves the required scheduling-specific ordering constraints, permits independent observable units to commute or to appear in different cycle groupings in Sys_ref and RTL_B, and treats the Post → Ref and restricted Ref → Post uses differently. A generic theorem such as "forward simulation implies trace inclusion" therefore does not directly yield the result proved here.

The liveness result is likewise not ordinary per-process progress. It is a conditional global preservation statement about complete admissible executions, explicit channel boundaries, persistent blocking requests, a finite reconstructed already-admitted in-flight action relation with separately proved path realization, wait-for components, fairness, environment behavior, and a conservative relationship between overlapped and serial source execution. Generic liveness and graph libraries can express parts of this reasoning, but the scheduling-specific bridge must still be supplied.

Accordingly, this appendix selectively highlights important semantic and proof obligations that remain specific to the scheduling contract after generic formal-methods infrastructure is imported. It is illustrative rather than exhaustive: omission of an obligation here does not make that obligation generic or remove it from a faithful mechanization. It intentionally does not restate complete theorem-premise inventories, certificate field layouts, or step-by-step proof chains. Those authoritative details remain in the Normative Formal Core and Appendices I, J, K, and K-P; Appendix M.7 and M.7b describe discharge and certificate packaging; Appendix R provides the compact derived architecture map; and Appendix S records the Lean proof strategy, mechanization decomposition, and selected import anchors; it is not the definitive side-condition ledger.

T.2 Safety-Specific Obligations

T.2.1 Initial-State and Reset-Epoch Correspondence

A generic simulation theorem can use an initial-state relation once one is supplied, but it does not establish the particular source/RTL correspondence required here. The post-reset RTL_B state must correspond to the prescribed Sys_ref state at the selected abstraction boundary, including the process, channel, request, synchronization, and source-visible state that the theorem instance treats as relevant. Dynamic reset makes this more than a one-time startup obligation. Each affected reset epoch must re-establish the applicable source/RTL relationship, including the reset well-formedness, channel and request accounting, and any direct-input anchoring contract. The RESET observable unit, RW1-RW5, and the Appendix-J reset-splice construction supply the scheduling-specific semantics; generic libraries supply only the surrounding induction and transition-system machinery.

T.2.2 Dynamic Source Enablement and Operation Attribution

When RTL_B exhibits an observable unit, the proof must establish that the corresponding specific dynamic source operation can legally occur from the currently constructed Sys_ref state. Equality of labels is not enough: the proof may need the process control position, branch and loop-instance identity, source-order predecessors, synchronization anchors, channel state, witness-selected outcomes, and dynamic-operation attribution.

This obligation is specific to HLS because the implementation may contain added FSM states, overlapped loop iterations, eager issue, or internal staging that do not appear as source control steps. Appendix I supplies process-local preparation/replay machinery and Appendix J combines it with channel, atomic-interaction, and global extension evidence. A generic simulation library can host that construction, but it cannot infer the HLS-specific enablement relation or the correct dynamic occurrence.

One especially important example is the R4/E3 safe message-issue discipline. The obligation is prefix-closed: at each applicable comparison cutpoint, if a source-later message operation has issued, every required source-earlier operation in the dynamic source order must already have issued. This is stronger than comparing timestamps of events that are already present, and Issue is proof-semantic state rather than merely a Σ -visible committed-trace event. A generic trace-inclusion or forward-simulation library can represent the resulting relation once it is defined, but it does not supply the dynamic source order, operation attribution, or downward-closed issued-set invariant required by this scheduling contract.

T.2.3 Same-Cycle Independence, Atomic Units, and Buffered Sequentialization

A hardware cycle may contain multiple observable events for which no meaningful order exists, while other collections of events must be treated as one indivisible observable unit. The formalization must therefore preserve scheduling-specific independence and required ordering rather than impose an arbitrary total order. Rendezvous transfers, synchronization handshakes, and the selected combinational fixed-point observation are examples of atomic units whose internal pieces may not be split across matching steps.

Positive-capacity channels add another scheduling-specific case. Under E4, same-cycle Push and Pop decisions are made from the common pre-cycle occupancy rather than by fall-through. In the interior-occupancy case the corresponding source effects can be sequentialized without changing the channel result; at empty or full occupancy the same-cycle complementary operation cannot create availability that was absent at the boundary. Generic concurrency or trace libraries can represent independence once specified, but they do not establish these E4-specific commutation and atomicity facts.

T.2.4 Witness Compatibility, Common Snapshots, and Deterministic Replay

The source reference model may admit internal choices that are not independently visible in the observable trace, including non-blocking outcomes, arbitration choices, epsilon-state choices, and replay paths. The proof witness must therefore be tied to the behavior actually represented by RTL_B rather than chosen merely because it makes a source execution possible. WC is the scheduling-specific obligation that supplies this connection.

Witness-significant non-blocking decisions also require a common request/channel snapshot. A process-local replay path cannot decide such a call in isolation if its outcome depends on requests or availability

produced elsewhere in the same construction horizon. Appendix J therefore reconstructs the relevant shared snapshot and orders request writes against the decision that reads them. The generic library contribution is ordinary state and dependency reasoning; the snapshot contents and legal non-blocking outcome are consequences of the scheduling semantics.

Once the fixed stimulus, WC-compliant witness, dynamic-operation attribution, and complete process-local observable history are fixed, the logical replay checkpoint must be determined well enough to justify later source behavior. The document deliberately distinguishes that logical replay state from raw operational preparation state and from currently asserted uncommitted offers. A generic determinism theorem does not choose this quotient or prove that it is the correct replay boundary for this scheduling model.

T.2.5 Explicit Boundary Semantics, Capacity, and Boundary Faithfulness

The safety proof depends on the behavior of selected explicit abstract communication boundaries, not merely on abstract message values. E4 supplies the legal Push/Pop behavior for rendezvous and positive-capacity channels, including availability and occupancy updates. When staged, bundled, joined, coupled, or sync-gated implementation structure cannot be faithfully represented by one ordinary E4 channel, Appendix Q supplies explicit elaborated boundary structure instead.

Back-annotated capacity is therefore not just a numeric upper bound. B-faithfulness requires the selected RTL boundary and its represented internal storage or credit to exhibit the same boundary-visible acceptance, retention, backpressure, and transfer behavior as the declared abstract channel model. Generic FIFO or simulation libraries do not identify the correct physical boundary, prove that hidden storage is completely represented, or establish the required source/RTL availability correspondence. Mapped-memory abstraction has the parallel scheduling-specific obligation: Core.MemFaithful fixes the untransformed logical mapping-layer semantics as the reference and requires any hidden transformed realization to preserve the selected memory boundary, while Core.MemProfile packages the reusable qualified case. Generic abstraction or refinement libraries do not choose that boundary, establish the memory-specific completeness/J.Mem conditions, or prove the K-facing blocking/dependency faithfulness.

T.2.6 Proved Local-to-Global Witness Construction

Process-local replay and preparation facts do not by conjunction produce one globally reachable source execution. The global proof must reconcile shared channel state, non-blocking snapshots, atomic interactions, reset/environment effects, field-sensitive footprints, and cross-process dependencies while allowing genuinely independent work to commute.

Appendix J supplies the scheduling-specific construction that derives a validated proof order and a phase-correct reachable source witness from those local facts. Generic graph algorithms, topological ordering, well-founded induction, and commutation lemmas are reusable once the dependency relation has been defined and proved sound. They do not establish dependency completeness, prove that omitted edges correspond to commuting actions, or show that the resulting order respects the document's registered-cycle semantics.

This is also why the formal development distinguishes the constructive J.GWR route from a direct route. A direct package may supply an independently validated order and extension proof, but it must satisfy the same scheduling-specific interface; an unexplained time order or an arbitrary legal source trace is not sufficient.

T.2.7 Qualified Synchronous Settling, Canonical Cutpoints, and KObs Correspondence

Generic epsilon-closure or fixed-point machinery does not by itself establish the cycle structure required by the standard qualified-synchronous scope. The proof must identify the registered K-origin, hold the required registered and exogenous state fixed during same-cycle settling, establish finite source and RTL settling to the selected endpoints, and on the RTL side recover the correct cycle origin for each canonical

K-facing cutpoint. Core.QSync and the associated source/post closure results supply those scheduling-specific hypotheses and theorems.

Trace compatibility alone is also insufficient for Appendix K. A process that has not yet reached a blocking call can have the same committed observable history as one that currently has an attributed uncommitted boundary request. The KObs interface therefore carries the additional residual-offer information needed to reconstruct process frontier, enablement, wait-for structure, and the selected deadlock predicate at the comparison cutpoint.

The corresponding implementation obligation is completeness at the selected request-abstraction boundary: every relevant asserted RTL request must be represented and uniquely attributed, not merely the requests for which a convenient certificate record was produced. The details of KObs, OfferReach/OfferConf, K.CertifiedCutpoint, and the checker/certificate split are defined in Appendices J and K and summarized in Appendix M.7; Appendix T needs only the point that this observation boundary is scheduling-specific and is not supplied by ordinary trace inclusion.

T.2.8 Coverage Beyond One Prefix: Cycle Induction and Ideal Closure

A theorem that constructs a matching source witness for one supplied RTL prefix does not by itself prove that suitable packages exist for every reachable implementation history. The standard QSync route therefore uses Definition Core.RTLDesignatedOriginFrame and Definition Core.RTLCycleStep for one-cycle typing, then Definition Core.KOriginCarrier and Lemmas Core.KOriginCarrier-AtOrigin/-Decomp to expose the finite exact chain from the initial/reset-established node through every completed real cycle to the represented cycle position. Across each such edge, J.LocalGWRExtCert/J.GWR-CompExt prove that the already selected source construction prefix extends exactly rather than being replaced by another observationally matching witness, and J.QSyncTraceStep packages the safety-only part of that incremental refinement. The carrier stores the designated origin, full cycle state, exact prefix and reset-epoch valuation at each node and the exact CycleResult between nodes; it deliberately does not store the same-cycle settled KCut p_i , because Core.QSync-Post derives that endpoint uniquely from o_i . The standard base-and-step correspondence argument is therefore indexed by carrier nodes/appendable steps, with exhaustive successor coverage rather than evidence from one observed simulation path. In the richer K-facing induction, that safety step is completed first; v_{ref}/τ_{ref}^K and DrainReach are then freshly selected/re-derived, so K-facing evidence cannot be used to prove the safety edge. Generic induction can consume this finite chain, but it cannot manufacture the designated boundary, exact unit/reset accounting, or initial/reset epoch valuation.

For finite-history safety properties, cycle-aligned package coverage must then be related to finite observable prefixes that may end inside a cycle. The Appendix-J ideal-cover result supplies that scheduling-specific history relation for properties satisfying the stated ideal-prefix-closure condition. Generic induction and prefix reasoning are reusable, but they cannot supply the implementation carrier, QSync-silent settling facts, atomic-unit boundaries, or the document-specific notion of an admissible observable ideal.

The full definitions and proof obligations are in Appendix J, with the safety and K-facing certificate triggers summarized in Appendix M.7. Appendix T therefore does not duplicate the J.GWR-CompExt/QSyncTraceStep/QSyncTraceInd/QSyncCertInd field structure or their individual sub-obligations.

T.2.9 Finite Local Traces Inside Infinite Global Executions

A process may stop producing local observable units while the complete system continues indefinitely. A correct infinite witness cannot be created by simply appending local idle cycles as though the whole design had terminated. The proof must instead select or construct an admissible global continuation and then project that execution onto the locally inactive process.

Only when the entire system satisfies the document's true terminal fixed-point conditions may indefinite global idle stutter be justified directly. This distinction between local trace finiteness,

temporary silent cycles, and genuine global terminal idleness is part of the scheduling model; generic infinite-trace libraries provide the execution machinery but not the hardware-specific terminal criterion.

T.3 Liveness-Specific Global Obligations

The safety obligations above are largely local or compositional. The liveness and deadlock-preservation obligations are global: they depend on the interaction of processes, channels, persistent requests, fairness assumptions, environment behavior, drainable work, reset/termination behavior, and the wait-for structure. They cannot generally be discharged by proving each process locally live.

T.3.1 Admissible Global Executions and Fair Progress

Generic temporal-logic and fairness libraries can state that continuously enabled actions eventually occur, but the document must first define what counts as a legal and admissible Policy-L execution. Core.LLegal, Core.Phase, and Core.LAdm bind the scheduling policy, fixed instance choices, cycle phases, request nonwithdrawal, settling and drain discipline, and the distinction between finite admissible prefixes used for reachability and complete executions used for fairness and limit arguments. B2, B3, and the applicable environment assumptions then express the required scheduler/channel progress over complete admissible executions. B5 supplies the stronger global fixed-point condition where genuine terminal quiescence is required. Standard libraries can support fairness proofs for already-supplied executions once these predicates are supplied, but they do not define the HLS-specific enabledness, persistence, channel-progress, or terminal semantics.

Some proof sites require more than checking fairness of an execution that is already given. In particular, Lemma K.2b.E constructs a complete fair continuation from a finite prefix. Core.FairPick packages the prefix-local issue and boundary selections and starvation-freedom evidence used for that construction, while Core.FairCycleDovetail proves that the selected fairness actions can be embedded in typed complete cycles before dependent choice extends them indefinitely. Generic fairness and dependent-choice libraries can host this construction, but they do not supply the HLS-specific selection functions, phase legality, cycle-completion compatibility, or fairness exposure needed to build the continuation.

T.3.2 Drain Normalization and Persistent Blocking Structure

Epsilon-quiescence is not the same as generalized drain closure. A K-facing state can have no enabled internal epsilon step while already-admitted proof-model work can still advance through explicit staged/join/channel actions and eventually change the blocked-frontier root family. The generalized predicate therefore evaluates finite reachability in a KObs-reconstructed, D-independent in-flight transition system rather than only checking enabled non-frontier residual source offers. The two-stage E4 non-fall-through example in Appendix Q demonstrates why a first internal transfer may leave the candidate roots unchanged and only a later-cycle transfer expose the rescue.

The scheduling-specific Core.16/Core.Drain discipline still prevents new source-later launch/Issue and bounds the finite non-replenishing already-admitted population. Q.2 supplies exact action/effect semantics and finite choice; Core.InFlightDrainWF supplies the well-founded rank;

K.WaitRoots/InFlightEscape defines the cutpoint abstraction; and model-indexed

K.InFlightQual/Core.InFlightCycleExt/J.InFlightPathSound proves reconstructed paths operationally realizable. Continuation existence lives in that proof layer, not in KObs and not as a claim that Dead_K is permanent. For lowered profile G2, finite lowering-only guard/epoch administrative skew is normalized explicitly rather than called ϵ , and K.Low.InFlightPathCorr plus K.Low.DeadRootBridge establish the modal drain correspondence needed by generalized K.Low.A5.

T.3.3 Conservative Serial-Source Deadlock Reduction

The central non-generic liveness step is the relationship between overlapped/liberal source execution and the selected conservative serial source execution. The document does not assume that ordinary trace inclusion or local progress makes these schedules interchangeable. It proves, on a stated applicability fragment, that a relevant liberal internal deadlock would force a permanent process-facing

obstruction in the selected deterministic Policy-S semantics and therefore a source-side bounded-response failure.

That reduction requires scheduling-specific facts about dynamic operation identity and provenance across the two policies, source-order comparability, persistent blocking-request lifecycle, point-to-point boundary ownership, legal comparison/reach steps, preservation of frozen blocking structure, and explicit handling of reset or declared-terminal interruptions. Lean can reuse finite maps, graph theory, well-founded induction, and rank arguments, but the relation being ranked and the proof that it faithfully represents these HLS operations are part of this document, not a standard library result.

At the conceptual level the argument is: expose a persistent liberal blocking component at a valid drain-closed cutpoint; relate the frozen liberal blocking component to the selected serial execution through the K.2b.P/K.2b.R dominance comparison and K.2b.Q frontier extraction, without releasing the obstruction; and use complete finite-bound response monitoring to contradict a clean Policy-S run. The exact K.2b sublemmas, interruption-gate structure, provenance interfaces, environment/frontier applicability limits, and response-certificate fields are specified in Appendix K/K-P and Appendix M.7. Repeating them here would not add a distinct library-gap argument.

The bounded-response certificate is source-side proof machinery for this reduction. It does not assert that the RTL completes within the same bound. This distinction is scheduling-specific and remains important even when generic temporal or response-property libraries are used to encode the monitor.

T.3.4 Universal Reachable-Cutpoint Coverage

A per-cutpoint preservation theorem and a proof that one serial source execution is clean still do not establish a universal RTL deadlock-freedom claim. The proof must also establish that every reachable K-facing RTL cutpoint belongs to the intended canonical domain and carries the complete source/RTL correspondence package consumed by Appendix K.

In the standard QSync normal-compositional scope, the document derives the canonical cutpoint domain from the qualified cycle structure and derives package totality from an implementation-wide base-and-step correspondence invariant. Alternate independently checked coverage routes remain possible. Generic invariant induction can support this proof, but it cannot supply the QSync origin/cutpoint relation, KObs correspondence, or the exact package whose existence is being propagated.

Certificate records and checker modules are evidence interfaces for these obligations, not substitutes for the mathematical obligations themselves. Their detailed field ownership, trigger conditions, and packaging are therefore maintained in Appendix M.7/M.7b and the normative J/K definitions rather than repeated in Appendix T.

T.4 Summary

Reusable Lean and formal-methods libraries can provide much of the generic proof infrastructure: transition systems, finite and infinite executions, epsilon closure, simulation, trace lifting, partial-order and graph utilities, well-founded induction, fixed-point reasoning, fairness, and temporal properties. The document should reuse those foundations wherever they match the required abstraction.

Among the scheduling-specific obligations highlighted here are the semantics and bridges that connect generic formal foundations to HLS behavior: dynamic source-operation attribution and enablement; prefix-closed dynamic message-issue ordering; same-cycle independence and atomic observable units; witness-compatible replay and common non-blocking snapshots; explicit channel-boundary and back-annotated-capacity semantics; local-to-global source-witness construction; qualified cycle settling and canonical K-facing cutpoints; KObs request correspondence; cycle-wide package coverage and ideal closure; proof-built fair continuation from typed cycle choices; drain-aware persistent-blocking reasoning; and the conservative reduction from liberal overlapped deadlock to a bounded-response failure in the selected serial source execution.

These examples explain why importing a standard simulation or temporal-logic library is necessary but not sufficient. The normative definitions and theorem interfaces remain in the Core and Appendices I-K/K-P, the evidence and qualification interfaces remain in Appendix M.7/M.7b, and the mechanization structure remains in Appendices R and S. Appendix T is therefore a selective map of important scheduling-specific proof burdens, not an exhaustive mechanization checklist or a second copy of the detailed proof architecture.

Appendix U – Revision Record

31 August 2026 — Stage-2C modal lowering correspondence and lowered generalized-drain re-enable.

The lowering repair now distinguishes raw lowering states from administratively normalized paired cutpoints. New `K.Low.AdminStepRec/LowAdminNF/LowAdminNormalize` makes enabled guard/epoch realization explicit finite in-flight work rather than ϵ ; disabled objects that still represent an unresolved source cause remain in normal form. New `K.Low.LowWaitRoots` supplies a lowering-only extended source root view for ordinary message, NB-Align guard, and typed synchronization causes without changing source `FrontRec/WFG/Dead_K`. `K.Low.ModalQual` binds both model-indexed `K.InFlightQual` records, finite administrative normalization, root/action correspondence, local `IdClose/EpochClose/GuardClose` cases, and schema versions. New Theorem `K.Low.InFlightPathCorr` proves bidirectional finite stuttering projection/lifting: lowering-only actions erase downward and are inserted upward, with exact normalized checkpoints. `K.Low.InFlightEscapeCorr` derives escape equivalence between literal `Sys_K` roots and the extended source lowering view. New `K.Low.DeadRootBridge` proves that when the source starting candidate is a literal message-blocked `Dead_K` candidate, the extended root view cannot change before ordinary `WaitRoots` changes, because `Core.16/no-new-later-work` and source order prevent reaching a later sync/NB lowering cause past the unchanged blocker. Therefore `K.Low.DrainClosedNowCorr` recovers exact generalized drain correspondence for the candidates used by `K.Low.A5`. Revision-A `IdClose/EpochClose/GuardClose` retain their blocked/root typing, non-empty release-family, and release-closure facts and become local simulation cases; their old local `DrainClosedNow` family is retired. `K.Low.C/K.Low.A5`, `K.2c/K.PSF/K.1`, `Q.2/M.7`, and certificate versioning are updated accordingly. Lowered profile G2 is now permitted only when `K.Low.ModalQual` is discharged, including separate `Sys_K` and `Sys_ref` continuation/path qualifications; `Sys_K` `Core.CycleClosure` is not silently transported to `Sys_ref`. Pure synchronization remains outside `MsgOf/message` `BoundaryReq` and literal `Dead_K` remains message-rooted. Closeout clarification: `MQ4` now states finite-administrative-suffix root reflection explicitly, including root changes first realized on a second or later erased administrative edge, and Appendix M.7a records `K.Low.ModalQual/MQ1–MQ6` as a separate per-instance evidence item; `K.Low.InFlightPathCorr` composes that checked premise rather than deriving it. Final $q \rightarrow K$ scope closeout: Theorem K.1A and its summary/M.7a text are route-neutralized so G2 lowering may discharge A5-L upstream through `K.Low.ModalQual/K.Low.A5` while K.1A continues to consume the separate `Sys_ref` `K.InFlightQual`; the stale Stage-1 G1-only premise is removed.

31 August 2026 — Stage-2B synchronization/epoch in-flight action semantics (no theorem-scope expansion). Definitions `Core.7d/Core.7e` add two deliberately distinct constructors: `SyncFire( $\kappa$ )`, the real already-pending/ready source synchronization transaction, and `EpochGateAdvance( $g, \kappa$ )`, a lowering-only explicit gate action erased by `K.Low` projection. `SyncFire` obeys `Core.SyncFence`, commits the existing atomic synchronization/E2-anchor unit, advances participants only to the immediate post-sync control/interval entry, and performs no post-sync `ReachPrep/Issue`; a peer that has not reached the synchronization remains a `SyncReleaseOwnerSets` release owner rather than hidden drain work. `EpochGateAdvance` changes only declared lowering gate/token/accounting state and may discharge a temporary post-sync gate skew without inventing another synchronization event.

Core.InFlightStep/state-based admission/Core.InFlightDrainWF are extended to these typed carriers, with a SyncTxn/gate-aware well-founded rank; new Core.SyncInFlightOpportunityQual specializes the existing InFlightCycleExt continuation obligation without using B2/B3 as an existence constructor. Core.18d-1 and new J.KObs-SyncInFlight reconstruct the source local action/effect semantics from existing KObs; Q.2 is generalized to profile-indexed Stage-1 plus Stage-2 constructors, adds Q.SyncInFlightActionEffectSound and a ready-sync/residual-gate worked artifact, and retains finite-choice requirements. K.Low.EpochClose now binds both constructors but explicitly leaves their stuttering sequencing/lifting to Stage-2C K.Low.InFlightPathCorr. No synchronization is widened into MsgOf/message BoundaryReq, literal WFG/Dead_K remains message-rooted, and at that revision point lowered generalized-deadlock claims remained unavailable pending Stage 2C. The Stage-2C entry above records completion of that modal correspondence and its qualification boundary.

31 August 2026 — Stage-2A synchronization-carrier typing (lowering prerequisite; no theorem-scope expansion). The source synchronization object deferred by Stage 1 is now typed without widening the message machinery. New Core.7b/Core.7c define $\Omega_{\text{sync},P}$, SyncOf_P, dynamic SyncTxn identity, finite participant sets, SyncPending/SyncReady/SyncBlocked, exact minimal internal SyncReleaseOwnerSets, and Core.SyncCarrierWF; MsgOf_P and message BoundaryReq/ChannelEnabled/ChannelDisabled remain defined only for message-operation frontier items. Core.18d-1 reconstructs the typed carrier from existing KObs through ProcRec plus committed synchronization/reset history, so no new KObs field or fake residual offer is added, and Lemma J.KObs-SyncCarrier proves reachable-state reconstruction. K.Low.EpochClose is refactored to retain Revision-A blocked/frontier, owner/epoch, non-empty release-family, release-closure, and projection facts as a typed epoch local case while deferring operational synchronization/gate action semantics to Stage 2B, now completed in the immediately preceding revision entry. The typed carrier remains outside literal WFG/Dead_K and does not change A8 barrier-stall scope. Lowered generalized-drain claims remain unavailable under the current theorem version at that revision point until the global K.Low.InFlightPathCorr $\rightarrow$ InFlightEscapeCorr $\rightarrow$ DrainClosedNowCorr proof was completed. Stage 2B supplied the local action semantics; Stage 2C now supplies the global modal lowering theorem and qualified re-enable.

31 August 2026 — Stage-1 generalized in-flight drain integration (non-lowered message-rooted profile). The legacy offer-only DrainClosedNow selector is replaced, within the explicitly qualified Sys_K=Sys_ref/no-K.Low theorem scope, by a KObs-pure finite in-flight reachability abstraction. Core adds InFlightStep, InFlightDrainWF, InFlightCycleExt, FlightStateRec, and InFlightStepRec; Q.2 adds finite action/effect schemas, dangerous-direction effect soundness, finite choice, and worked Join2/two-stage E4 artifacts; J adds KObs in-flight reconstruction and compositional InFlightPathSound; K adds process-indexed WaitRoots, D-independent InFlightEscape, generalized DrainClosedNow, root-projection/heredity, primitive fast-path equivalence, activation-local relocation/candidate redispatch, K.InFlightQual, and the literal two-stage premature-Dead_K regression. K.2a/K.KObs factoring and certificate/M.7 interfaces are aligned to the new schema. The primary no-lowering serial route reuses its already-bound Core.CycleClosure for the continuation-existence component; alternative A5-L routes must supply Core.InFlightCycleExt or equivalent path evidence. Lowered instances are explicitly outside this Stage-1 generalized-drain theorem version: Revision-A K.Low blockedness/release-family/release-closure work is retained, but modal drain correspondence is deferred to a global K.Low.InFlightPathCorr $\rightarrow$ InFlightEscapeCorr $\rightarrow$ DrainClosedNowCorr proof after the source synchronization carrier/action semantics are separately typed. No synchronization is widened into MsgOf/message BoundaryReq, and no two silent DrainClosedNow semantics are permitted under one theorem version. Certificate compatibility is version-pinned: a lowered certificate issued under an earlier offer-only theorem/document version remains evidence only for that earlier version and does not establish this Stage-1 generalized-drain theorem. No current-version legacy-fallback theorem is introduced; current-

version lowered generalized claims await Stage 2 and require new or independently revalidated certificate evidence against the completed lowered modal schema.

30 August 2026 — Appendix-Q clarification-only revision. Q.1(g) now distinguishes source-level logical channels from proof-internal $\text{Sys_B}^{\text{elab}}$ boundaries, states explicitly that a stateless SystemC/RTL combinational process or module is not a modeled process for source ownership or WFG purposes, and makes the negative case for the ordinary $\{\{Q\}\}$ release-family reduction explicit at coupler-terminated boundaries. A clarification after Q.1(h) distinguishes a boundary-typing correction from a storage/topology change that requires fresh Q.1(j)/(k) reconstruction and release-support derivation. Q.1(k) now states explicitly that its soundness/completeness obligations establish release-family correctness relative to the qualified E template, while implementation faithfulness belongs separately to the Q.1(j)/J.KObsConf boundary-mapping obligation. No drain, K.Low, KObs carrier, theorem-scope, or scheduling-semantics change is made in this revision.

29 August 2026 — drain-evidence interface and KObs-WFG audit hardening. The former run-in K.Drain obligation is promoted to Definition K.DrainEvidence, preserving Core.Drain as the semantic path predicate while giving the application evidence a model/path-indexed certificate type. The normal K.DrainReach-Comp route now requires activation-indexed DR1 coverage: each Core.16 activation chooses DR1a RTL-backed automatic-flush lift or DR1b direct source structural proof, and mixed designs may use both routes in one instance. DR1a binds not only the corresponding RTL region/cycle but the same dynamic blocked frontier/request attribution; if the existing J/I construction package cannot establish that identity, a named activation-correspondence theorem/certificate is required. DR2–DR4 remain the common nonwithdrawal, $K\epsilon$, finite-drain, and sync-boundary fields; K.NormStable NS1/NS5 is cited explicitly for the no-drain-commit v_ref suffix. Core.SettleKBridge now states that persistence is a hypothesis rather than a result and exposes the finite ϵ -segment/drain-commit induction shape for mechanization. J.KObs-WFG cites Q.1(j) for reconstruction exhaustiveness and Q.1(k) separately for release-support family correctness. K.CertHdr, CF-0, M.7a, and M.7b are aligned to the K.DrainEvidence schema and its per-activation route-map provenance. No scheduling capability, theorem scope, or deadlock predicate is changed.

29 August 2026 — Revision C serial-route closeout and type/cross-reference normalization. No new serial-reduction proof premise or scheduling restriction is introduced. Appendix K-P now states explicitly that K.2b.P proves inclusion of E-typed explicit-boundary action units, while K.2b.P-Frozen/SF/Q and K.Resp infer process-facing commit/return only when E/K.CV identifies the relevant action unit as completion of the source operation; the frozen corollary uses a typed $T_{\{f_P\}}$ rather than the former $T=f_P$ shorthand. The buffered Pop/Push cases cite K.2b.R uniformly, and the K.2b.P case list states how synchronization, epoch-gate, guard, rendezvous, and coupling-sensitive units are covered. K.Ser condition (c) is corrected to minimal-frontier membership $\text{item}(o) \in \text{FrontRec_owner}(o)(K_L)$, exactly equivalent to $\text{DrainBag_Proc}(l)(K_L) = \emptyset$, with Core.OrdFrontSingleton only the singleton special case. Loose citations are normalized to Core.16(7) and Q.1(j)–(l). Appendix K-P is divided into five internal subsections and the K.2b.E proof is moved before the dominance results that consume its continuation. Active reset/certificate/derived narration is normalized to the P0/R/P/P-Frozen/SF/Q chain; Appendix R and Appendix S record the same typed interface. A targeted Appendix-J audit confirms that Revision B's model-relative Core.CycleResult.committedUnits clarification does not widen J's outer-history semantics: J.OuterCanonHist already begins from complete observable/proof units and applies Π_DUT, E afterward, explicitly permitting erased proof/elaboration units. The serial-reduction paper proof is therefore closed at its stated abstraction level; remaining work is mechanization, tool/flow qualification, per-instance certificate discharge, and implementation validation.

29 August 2026 — Revision B serial-route proof/interface consolidation. ReachProgress is completed at paper level without changing Core.IssueFair or narrowing Core.LAdm. Definition Core.CycleResult is clarified model-relatively so $\text{CycleResult}_{\{\text{Sys_K}\}}.\text{committedUnits}$ retains complete explicit proof-model

boundary units before Π_elab/Π_DUT outer projection. New Lemma K.SerialDomPredFinite derives finiteness of the strict $<_SD$ predecessor set of any fixed Policy-S target from the finite per-cycle committedUnits inventories, without assuming the complete selected run is finite. New Lemma K.CompareFact-Mono states persistence of matched dominance units and carried historical K.CV predecessor facts under prefix extension; it is used both by the sticky-segment work measure and by the least-missing proof. New K.SegWork/K.SegWork-Complete build a certificate-determined finite expansion tree by strong induction on CVRank, use deterministic prefix-indexed normalization to skip already-satisfied subtrees monotonically, and prove finite completion of each actually emitted sticky stack when I6/K.SegmentAdm and FP4a supply admitted operational progress. New Lemma K.KeyItemOrderBridge names the previously implicit ordinary/certified-singleton key-to-item-order step and keeps genuine Sys_B^elab multi-frontier order certificate-driven. New Lemma K.SegmentPlan-EventuallyTarget is deliberately a consequence theorem over the completed K.2b.E continuation, not an Inv/Inv+ conjunct: for an explicitly absent $<_SD$ -minimal target it proves either ET1 segment emission/completion on the same continuation or ET2 already-reached/asserted roots, with K.2b.R as the explicit Issue/persistence handoff. K.2b.P0/R/P are rewritten to cite that theorem rather than claiming prior plan emission. The response-waiver text now points to the authoritative pre-K.2b K.RespCov Environment/waiver rule; K.2b.SF no longer invokes the Policy-L K.CompareAdvanceLegal certificate on the Policy-S issue step; optional K.Ser replay may insert idle cycles only under StandardEnv/B1-avail or an explicit environment-stutter-equivalence proof. K.2b.SF is propagated through reset-domain, consumer, proof-chain, mechanization, Appendix R, and Appendix S narration as a derived lemma, not as a new SR/K.PSF premise. Appendix M.7a now repeats the K.PSF exclusion for unqualified mutual-stall/shared-capacity/arbitrated non-monotone coupling-sensitive templates. Revision C remains limited to low-severity/editorial/global-normalization cleanup unless later review identifies a new substantive defect.

29 August 2026 — Revision A.1 invariant/segment-plan clarification. The Revision-A ReachProgress construction is tightened without changing the serial-route theorem scope. K.2b.E now names I1–I5 as the base frozen-comparison invariant Inv and defines $Inv+ := Inv \wedge I6$; K.SegmentAdm and Q.1(I) CouplerPersistence consume only Inv, removing the circular reading $Inv+ \rightarrow I6 \rightarrow K.SegmentAdm \rightarrow Inv+$. Core.FairPick FP4a and K.SegmentPlan now use a fixed construction-state rule: a top-level certified segment designated when no obligation is active is sticky until completion or candidate rejection, with only certificate-required lower-CVRank predecessor stacks pushed beneath it, so later eligibility changes cannot replace an in-progress obligation. K.SerialDomOrder is stated explicitly to be partial outside the selected Policy-S dominance footprint, while Q.1(I)'s “before T commits” quantifier is temporal along the Policy-L continuation and may include liberal-only units with no $<_SD$ rank. The worked join discharge now attributes the no-competing-producer fact to A6 point-to-point ownership/template binding rather than to the frozen invariant. K.PSF also states explicitly that Q.1(d)–(k) alone do not prove CouplerPersistence for mutual-stall, shared-capacity, or arbitrated/non-monotone handshake templates; an unqualified such template is outside K.PSF/CF-S and requires another A5-L route. The explicit active-stack termination measure and the planned K.2b.SF dependency-narration sweep remain Revision-B work.

29 August 2026 — Revision A serial-reduction soundness closure (ReachProgress, CouplerPersistence, LowerClosure). The audit's temporary H1/H2/H3 labels are not reused because Appendix U already uses H1/H2/H3 for older architecture revisions and Example K.CE-5/H2 is a separate perpetual-under-supply example. New Definition K.SerialDomOrder fixes the well-founded total order over indivisible Policy-S boundary-action units before the least-missing induction; primitive rendezvous and declared same-cycle atomic units cannot be split. Core.FairPick gains FP4a certified finite-segment progress without strengthening Core.IssueFair or Core.LAdm. Definition K.SegmentPlan fixes the prefix-local SegReq designation rule before dependent choice, and Definition K.SegmentAdm relative to the base K.2b.E Inv

plus the strengthened K.2b.E Inv+ carries admittedness of the resulting nested K.CVPred/P0 segment stack on the proof-built continuation. K.CVPred now separates certificate-side existence/step legality from construction-side admittedness: SegPred/SegReach are fixed from the comparison certificate, K.SegmentPlan emits the relevant least-missing obligation prefix-locally, K.SegmentAdm preserves it on the continuation, and FP4a realizes it through the phase mechanism that owns each unit; a candidate whose segment admittedness cannot be proved is rejected. This FairPick change has no Appendix-J consumer: Appendix J remains a finite-prefix construction layer and does not import Core.FairPick. Q.1(I) is strengthened from pointwise handshake reconstruction to CouplerPersistence over all Core.LAdm-admissible fixed-DomEpoch continuations satisfying the base K.2b.E frozen invariant Inv, and a worked two-input stateless-join template demonstrates a non-vacuous discharge. K.2b.P now states the continuous-enablement arguments explicitly for buffered Pop, buffered Push, primitive rendezvous, and coupling-sensitive templates. K.PSF, Appendix M.7a, K.CertHdr binding/versioning, Appendix R's Revision-A FairPick/liveness summary, and Appendix S import anchors are aligned; Policy-S packages relying on the superseded Q.1(I) or pre-FP4a constructive witness require revalidation under the revised K.CertHdr schema identity. K.Low.C is repaired bidirectionally through K.Low.IdClose, K.Low.EpochClose, and K.Low.GuardClose, each carrying blocked-frontier, drain, non-empty release-family, and release-closure facts. The extended-wait-to-literal direction invokes K.ReleaseSCCReduction and K.Low.A5 now observes that the returned $T \sqsubseteq D_0$ changes only the witness set, not the cutpoint on which waiver pullback is evaluated. The Proof-status paragraph no longer claims that all remaining work is mechanization-only: Revision A closes these high-severity paper obligations, while the separately planned Revision-B interface/citation items and Revision-C cleanup remain.

28 August 2026 — exact J.GWR prefix extension and structural safety-step closeout (items 2 and 3).

New Definition J.LocalPrefixAgree/J.LocalGWRExtCert makes old/new LocalGWR package restriction explicit, including ConstructionAction/Dep_J/Horizon/stable-rank restriction, no-new-predecessor/ideal preservation, bridge-choice stability, cross-prefix join coherence, and the closed-Horizon rule that prevents a larger package from retroactively inserting a unit into an already exported L3 bucket. New J.GWRPrefixExt and Theorem J.GWR-CompExt then reuse the previously selected $\tau_{\text{ref}}^{\text{G/W}}$ chain exactly and append only the genuinely new later-Horizon/unit-empty-bridge suffix; independent successful J.GWR-Comp applications no longer stand in for prefix extension. The extension target is $\tau_{\text{ref}}^{\text{G}}$ only. New Definition J.QSyncTraceStep and Lemma J.QSyncTraceStep-Sound package each carrier edge as an independently checkable safety object containing LocalGWR extension, J.GWR-CompExt, source Core.QSync-Ref settling, reset/I.A0/LocalAgree updates, and J.HCanon, with a normative ST6 schema exclusion of every K-facing KObs/Offer/NormStable/Drain field. QSyncTraceInd T11 is rewritten to construct that object. QSyncCertState now contains QC0-Safe:=QSyncTraceState(TraceOf(C)); CI1 is explicitly two-stage, first constructing the same QSyncTraceStep and only afterward selecting the canonical RTL KCut, a fresh $v_{\text{ref}}/\sigma_{\text{ref}}$, K.NormStable/Offer/KObs, and freshly re-deriving K.DrainReach on $\tau_{\text{ref}}^{\text{K}}$. J.QSyncCertInd-Trace is therefore a literal projection of stored safety states/steps rather than a proof reconstructed after discarding K-facing fields. M.7a/M.7b and Appendices R/S/T are aligned. This closes the separately tracked closeout items 2 and 3; the earlier carrier, outer-history, and 6' endpoint architectures are unchanged.

28 August 2026 — designated-carrier terminal-semantics closeout. Definition Core.DeclaredTerminal is tightened to make a modeled terminal/abort/end-of-test disposition absorbing for ordinary CycleResult/CompleteCycle semantics; a finite run may end there, while any indefinite post-terminal extension is only the separate B6 idle-tick convention and requires Core.TerminalIdle. Core.RTLCycleStep and J.CycleReach_post are aligned so a DeclaredTerminal successor is the final ordinary designated-origin node rather than merely a node from which another cycle is optional. Lemma Core.KOriginCarrier-AtOrigin now proves RS6 explicitly from the one-clock CycleResult plus uniqueness of the designated

frame boundary, and Lemma Core.KOriginCarrier-Decomp cites the designated-origin existence obligation that supplies its KCO base node. This is a carrier-closeout consistency repair only; closeout items 1, 7, and 8 remain closed, and the separately tracked J.GWR prefix-extension theorem (item 2) and explicit safety-step/projection stratification (item 3) remain open.

28 August 2026 — finite designated K-origin carrier and constructive certificate-extraction closure (closeout items 1 and 8). Core.RTLDesignatedOriginFrame now distinguishes the single declared standard-QSync pre-settle origin from other same-cycle states that may satisfy the broader RTLKOriginFrame compatibility relation; RTLCycleStep RS6 is correspondingly scoped to consecutive designated origins, and M.7a owns the boundary/consecutiveness check. New Core.KOriginCarrier records the explicit initial node (κ_0, s_0, o_0, e_0), exact shared one-cycle RTLCycleStep edges, reset-epoch threading, terminal handling, and derived-not-stored QSync KCut endpoints. Core.KOriginCarrier-AtOrigin/-Decomp close the finite ancestry/decomposition gap for cycle-aligned and arbitrary reachable RTL prefixes, so J.CycleIdealCover no longer assumes an unexplained greatest preceding K-origin. J.QSyncTraceInd/J.QSyncCertInd and their coverage theorems are reindexed over carrier nodes and appendable steps. New carrier-indexed J.CertExtract_QS plus its soundness lemma makes constructive extraction consume an explicit reachability/carrier position; J.CertExist-QSync hides that carrier existentially after Core.KOriginCarrier-KCut recovers it from a reachable KCut. The source half of J.RequiredPredPrefix is closed through new Lemma J.RequiredPredCycleOrder, and TI0/CI0 explicitly bind the initial epoch valuation. This pass closes closeout items 1 and 8 together with the three item-7 residuals; the separately tracked J.GWR prefix-extension theorem (item 2) and explicit safety-step/projection stratification (item 3) remain open and are not implemented here.

28 August 2026 — typed RTL cycle/K-origin bridge (closeout item 7) plus outer-ideal proof-explicitness cleanup. New Definition Core.RTLKOriginFrame makes the RTL K-origin an explicit same-cycle view of a full Core.CycleState and records the reset-epoch valuation used by QS2a. New Definition Core.RTLCycleStep packages one exact real cycle from a cycle-aligned prefix/current K-origin through the bound Core.CycleResult to the successor Core.CycleState/K-origin, with exact committedUnits/resetUnits accounting, RW1–RW5 epoch update, exact prefix nesting, one-cycle index advance, and DeclaredTerminal successor handling. Lemma Core.RTLCycleCarrier now returns that typed object only for an already-entered real cycle; it remains non-total and does not assert a successor from an arbitrary K-origin. J.CycleReach_post, TI1/TI2, CI1/CI2, J.CycleIdealCover, the Core.QSync QS2a/M.7a wording, and Appendices R/S/T are aligned to the typed interface. Separately, new Lemma J.RequiredPredPrefix makes the execution-prefix branch of J.OuterCanon-Ideal explicit by combining the existing strict-cycle ordering of ordinary $\prec_J$ generators with RW-backed proof-unit/reset ordering; the erased-only equality clause and Practical-scope RESET-contraction wording are aligned. Scope of this revision: it closes closeout item 7 and the two residual outer-history wording/proof-explicitness items. It does not yet build the multi-cycle KOriginCarrier/extractor machinery of items 1+8, prove J.GWR-CompExt (item 2), or add the QSyncTraceStep safety-stratification object (item 3).

28 August 2026 — outer required-order alignment follow-up. Definition J.OuterCanonHist now makes the relationship between $\prec_{req}$ and $\prec_J$ explicit: on ordinary selected observable units $\prec_{req}$ is exactly the Theorem-J.1 required-predecessor relation $\prec_J$, while selected proof-internal units add only their separately stated normative predecessor/reset-epoch edges. Full per-process replay or lexical source order retained by Core.18a is not an ideal-order generator merely by being replay metadata. J.OuterCanon-HCanon and J.OuterCanon-Ideal are aligned to that distinction; the practical/scope prose now names $\prec_{\Sigma_DUT}$; J.CycleIdealCover case (i) cites J.OuterCanon-Ideal for equality when only erased units are added; M.3 drops the now-redundant standalone J.HCanon-invariance phrasing; and M.7a records quotient-extensional checker behavior when a noncanonical representation is used. This follow-up does not implement item 7 or the remaining carrier/extension/induction closeout items.

28 August 2026 — outer canonical-history projection and cycle-ideal zero-unit closure (closeout items 4 and 5). Definition `J.OuterCanonHist` now gives the previously missing authoritative meaning of `CanonHist_Σ_DUT` intrinsically for every reachable finite source/RTL prefix, using its complete selected units, $\Pi_{\text{elab}}/\pi_{\Sigma,E}$ whole-unit projection, and an induced outer required-order relation that preserves dependencies through erased proof/elaboration units; no per-prefix `J.GWR/J.TraceCertCov` package is needed merely to form the outer history. New Lemmas `J.OuterCanon-HCanon` and `J.OuterCanon-Ideal` prove, respectively, descent of enriched canonical equality to that intrinsic outer history and same-carrier/required-prefix ideal preservation; `J.RefProjCover`, `J.IdealPrefixClosed`, `J.1-Safe`, and the `QSync` safety corollaries are aligned so an extensional predicate on the outer quotient needs no separate informal `J.HCanon-invariance` premise. Lemma `J.CycleIdealCover` is rewritten with an exhaustive outer-history case split: if a current real cycle has added no retained complete outer unit, the last reachable `K`-origin already has the same outer history regardless of whether the prefix is in settling, boundary/`L3Obs/L4/update` activity, a `SilentCycle`, or contains only projected-away `RESET/proof` units; if at least one retained outer unit is present, `Core.RTLCycleCarrier` completes the already-entered cycle and `J.OuterCanon-Ideal` supplies down-closure in the successor `K`-origin history. Appendix S imports the new outer-history interfaces. Scope of this revision: it closes only agreed closeout items 4 and 5. The typed RTL cycle/`K`-origin carrier refinement (item 7), carrier-chain/extractor repairs (items 1 and 8), `J.GWR` prefix-extension theorem (item 2), and `QSyncTraceStep` safety stratification (item 3) remain for later passes.

28 August 2026 — source `J.GWR/K`-facing endpoint separation and post-L3 observation normalization. The `K`-facing source path no longer identifies the `J.GWR` post-boundary construction endpoint with the later `Core.KCut`. Definition `Core.KObsSettle/Core.KObsNormRef` adds an observation-only post-L3, pre-L4 evaluation of the already-selected boundary state; its recomputed handshake values cannot feed back into L3 selection, `J.BufSameCycle`, or E4 fall-through decisions. A separate `Core.KObsSettleQualified` predicate makes the post-L3 network/phase/no-feedback checks `K`-facing-only so the safety `QSync` premise is not strengthened. `J.RegPhaseReach` is retyped over the `J.GWR` construction prefix τ_{ref}^G ending at s_{ref} ; Corollary J.1 records a distinct v_{ref} and $\tau_{\text{ref}}^K := \tau_{\text{ref}}^G \cdot v_{\text{ref}}$ ending at σ_{ref} . New `K.NormStable` gives checker-visible primitive invariance obligations for that suffix and Lemma `K.NormStable-H` derives canonical-history preservation rather than accepting it as an independent certificate field. `K.RLAdmReach` is rewritten as the three-part composition `RegPhaseReach( $\tau_{\text{ref}}^G$ )+K.NormStable( $v_{\text{ref}}$ )+K.DrainReach( $\tau_{\text{ref}}^K$ )`, so the prefix proved admissible is the one that actually reaches the `Dead_K` source cutpoint consumed by A5-L. `K.DrainReach-Comp` explicitly uses `K.NormStable` to close the suffix outside `J.GWR-Comp`. `J.OfferReach/J.KObsConf`, `QSyncCertState/CI1`, `K.CertifiedCutpoint`, CF-0, K.1A step 4, M.7a/M.7b, Appendix R/S anchors, and explanatory summaries are aligned. The safety-only `QSyncTraceState/TraceCertCov/CycleIdealCover` chain remains unchanged in semantic scope and projects away v_{ref} , σ_{ref} , `K.NormStable`, and `K.DrainReach`. Scope of this revision: this 28 August pass closes only issue 6', the `J.GWR` construction-endpoint versus source `K`-facing cutpoint separation and its immediate typing/wording cleanup. It intentionally does not implement the separately tracked `CanonHist_Σ_DUT/ideal-cover` repair, RTL cycle/`K`-origin carrier, `J.GWR` prefix-extension theorem, or `CI1` safety-step stratification; those remain separate closeout items.

27 August 2026 — Appendix K model-typing and serial-reduction proof-interface closure. `Kε` is made explicitly model-indexed, with a general Appendix-K convention requiring separate `Sys_ref`, `Sys_K`, and `RTL_B` discharges whenever those models are consumed; `K.SerialRoutePremises` now binds SR1–SR3 to `Sys_K` and `K.DrainReach/K.1A` bind their operational evidence to `Sys_ref`, while `K.Low.C` is explicitly not a transport theorem for those premises. `K.2b.P` now takes the `K.2b.E` frozen-continuation invariant as an explicit hypothesis. `K.ResetEpoch.Continue` gains an independent selected-Policy-S reset-exclusion clause (R-S), closing the K.2b.Q reset-domain gap while retaining `DirectFail` for matched reset failures.

New Lemma K.2b.SF supplies the K.CV/K.CVPred control-path induction needed to extract the permanent Policy-S serial frontier. K.CompareAdvanceLegal now concludes the Core.16 item-order legality fact directly, with key order only a sufficient bridge in ordinary/singleton cases. The primitive buffered-Push proof states the exact Push-count bookkeeping. K.Resp waiver closure is stratified over the pre-K.2b admitted candidate domain. The default $B(g)=1$ guard skew is identified as a real-cycle drain-candidate skew rather than an ε -delay. K.DomResetScope now uses an explicitly precomputed K.CompareScopeFootprint. M.7a/M.7b and the K-P mechanization obligations are aligned to these interfaces. No theorem claim is broadened and no new scheduling capability is introduced.

27 August 2026 — formal-interface closure and label/route cleanup. K.CV is promoted to one authoritative Appendix-K definition, and K.CVPred/K.CV-WF no longer presuppose an ambient K.CV discharge. $K\epsilon$ and NB-Align are given explicit normative definitions. Definition K.SerialRoutePremises replaces the $K.1 \rightarrow K.2c \rightarrow K.2b$ “remaining premises” pointer chain without changing the serial-route assumptions. Core.DeclaredTerminal and Core.TerminalIdle are split into separately citable definitions. The former near-collision label “K.1a” is renamed K.DeadChar, the stale J.GWR-Direct V1–V5 reference is corrected to V1–V6, and Theorem J.1’s restricted reverse direction is relabeled as selected-pair compatibility rather than constructive existence; its semantics are unchanged.

26 August 2026 — B2 ownership and qualification-routing clarification.

Appendix N no longer classifies all B-rules categorically as execution-environment assumptions. It now states that the fairness/progress rules are theorem-applicability assumptions whose discharge may fall on source/scheduler semantics, HLS-generated or hand-written RTL arbitration/back-pressure, channel-library behavior, or external masters, and it points directly to the Appendix M.7a B2/B3 discharge. The M.7a entry now states explicitly that HLS-generated resource, memory-port, or interface/channel-selection arbitration is included whenever it controls B2-covered boundary actions, and that reusable tool/vendor or qualified-library evidence may discharge a generated arbitration class when bound to the concrete theorem instance. The unfair-priority example now distinguishes continuously ChannelEnabled starvation (B2), continuously IssueEnabled-but-unissued source behavior (Core.IssueFair), and source-control cases in which neither fairness antecedent fires. No theorem premise or finite-prefix Appendix J result is changed.

26 August 2026 — blocking Issue/BoundaryReq attribution closure.

The ordinary scalar blocking request semantics are tightened without changing allowed HLS scheduling behavior. Core.9 and Appendix C now treat Issue as the cycle index of the operational request event rather than an independently selectable witness timestamp, and Core.10 distinguishes the process-owned BoundaryReq from raw physical ready/valid values produced by complementary channel/storage availability or declared coupler logic. Core.IC(3) now removes the former $u>t+1$ ownerless-attribution latitude: after blocking q_0 commits, a continuing process-owned request in $t+1$ belongs to the next same-direction blocking source operation in that source interval and gives it $\text{Issue}=t+1$; genuine scheduler delay remains available through Core.SelectLatitude only while that next BoundaryReq has not begun. E3 records that the ordinary scalar same-interface blocking order is derived from the separately qualified Core.IC/BoundaryReq lifecycle and endpoint serialization, while distinct-interface no-reverse and cross-interface prefix closure remain substantive implementation checks. Appendix M.7a is aligned to that evidence split. A same-day cleanup fixes/bidirectionally completes the BoundaryReq mapping, adds the general live-owner invariant, records commit/ Σ Match backstops, corrects I.4 identity provenance, and defines “operation-attributive” as Core.10 operation-specific rather than witness-selectable. Non-blocking and Sys_B^{elab} x-form handling are unchanged. This supersedes the 16 August revision-record statement that Core.IC back-to-back attribution itself permits inserted ownerless latency; Appendix H.3 remains unresolved; Core.17/R2 also select the final source Write at a shared E2 anchor, and Appendix N distinguishes closed deadlock from B3 progress failure.

25 August 2026 — main-body architect-facing terminology and memory-layering cleanup.

The main-body “Scheduling of Array Accesses” section, together with the pipelining, Direct Inputs, relaxed message scheduling, Issue/Commit, RULE 2, and Summary discussions, is restated in HW-architect-facing language while retaining formal cross-references. The substantive requirements are unchanged: every mapped memory retains a declared logical mapping and comparison boundary; cache/merge/split/reorder transformations remain permitted only where behavior at the declared boundary is preserved as a per-instance discharged obligation; mapped message and signal behavior cannot be hidden by a memory mapping; the synchronization commit fence and the no-cross-boundary-merge rule are unchanged; persistent memory backpressure remains represented or qualified for the deadlock result, whether or not the memory carries cross-process value flow; every cross-process memory-carried value dependency remains represented; and conflict-free pipelined memory reordering remains permitted only when explicitly authorized and consistent with those rules. The full formal conditions remain in the Normative Formal Core and Appendices J, K, and M.7a, including the Core.MemFaithful, Core.MemSyncOrder, J.Mem, and Core.MemProfile machinery and the direct memory-state-extension route; a “Formal treatment.” pointer paragraph is added at the end of the array-access section. Two Summary statements are restated rather than merely reworded: the pipelining bullet no longer asserts that latency-sensitive or reordering-sensitive designs change only in “expected” ways, and instead directs such designs to the explicit latency-sensitive and mapped-memory contracts; and its precondition is corrected from “external array accesses are not reordered” to the weaker, body-consistent requirement that mapped memory transformations preserve the declared logical behavior and the synchronization and blocking rules. No theorem, premise, applicability condition, evidence burden, or admitted transformation is changed.

24 August 2026 — elaboration accounting/projection and mechanization-interface alignment.

Appendix Q and the Core.ElabProj interface are aligned to the mechanization-facing elaboration model without changing the physical nonnegative occupancy semantics of explicit storage channels. Q.1 now types A_E as signed state accounting, declares a finite ElabToken universe plus a finite multi-valued Π_token, E outer-channel projection and event-to-token attribution, and permits one elaboration token to account for multiple outer Push/Pop effects while retaining unique declared projection and no-double-counting requirements. Q.1(b) preserves the distinction between state-computed A_E and independently reconstructed outer history: their agreement is the elaboration obligation, while $0 \leq A_E \leq B(c)$ is derived from that agreement plus outer E4 legality rather than added as a separate M.7a burden. The occupancy glossary, Q.1(h), the A_E notation note, Core.ElabProj, and the existing M.7a Sys_B^elab entry are aligned to that split. Core.CycleState is promoted to its own named declaration before Core.CycleResult. Appendix S now states locally that its list is a selected foundational mechanization-spine inventory rather than the exhaustive side-condition ledger, and adds the missing Core.Settle/QSync/Core.SettleKBridge, model-indexed cycle/KCut, KObs, and major J/K export anchors. At that revision, Theorem K.1 used a direct pointer through Corollary K.2c to the residual K.2b premise list; that pointer arrangement is superseded by Definition K.SerialRoutePremises in the 27 August 2026 revision, while the “K.2c gives A5-L” proof citation remains unchanged. The redundant K.RespSafe alias is removed and the A5-S prose now states K.RespCov and K.RespClean directly; Corollary K.0a is marked as a derived convenience result and its cross-model-use note is aligned; and the missing sentence break before the K.Low “Evidence examples.” label is repaired. No new mid-settle A_E -invariance obligation is introduced, and no new M.7a discharge category is added. Q.1(b) and the post-Q.1 locality note now distinguish explicitly between outer Sys_B E4/B(c)/occ_c semantics used as the Q.1(a)–(c) refinement target and the explicit B_E(d)/occ_d channel semantics used operationally inside Sys_B^elab(E). Q.1 item 6 and clause (c) also state the token attribution rule consistently: attribution is functional when defined, is required and unique for every internal staged/bundle/join transfer occurrence with outer effects, and may be absent for a transfer with no outer effect. The Appendix S foundational spine

additionally names Core.IssueFair, Core.10, Core.14/Core.15, and Core.17/ Σ Match, and restores the concrete stop-at-hole/unit-at-a-time legacy-interface pointer.

21 August 2026 — H3 finite-prefix source selection-latitude closure.

New Definition Core.SelectLatitude makes the source scheduler latitude used by Post $\rightarrow$ Ref witness construction constitutive rather than an implicit appeal to fairness: under both Policy L and Policy S, otherwise legal L1 issue and L3 boundary actions need not be selected maximally in a finite prefix, subject to Core.16 and all existing legality/fixed-instance constraints. Core.IssueFair and B2 remain separate complete-execution eventual-selection obligations. Core.Phase and Core.CycleClosure C2 are sharpened to make selected/non-maximal L1/L3 endpoints explicit; Core.LAdm confirms that SelectLatitude-permitted nonselection and resulting legal CompleteCycle/SilentCycle behavior remain Policy-L-legal. Definition J.GWR G3, Lemma J.UnitExt, and the expanded J.GWR-Comp proof admit finite evidence-determined invariant-preserving unit-empty cycle bridges between nonempty Horizon buckets when cycle-sensitive source/environment evidence requires them, without introducing empty rank buckets or changing Dep_J/FR/J.DepRank/J.DepAcyclic/J.CanonicalRank. Bridge length is not identified with raw RTL Horizon gaps unless the selected cycle-sensitive evidence requires that alignment. M.7a adds the source-model conformance audit and the J.RefProjCover latency-coverage warning; M.7b and Appendices R/S are aligned. Theorem J.1 is unchanged and no B2/B3 premise is added to its finite-prefix direction.

21 August 2026 — H1 mapped-memory observation-boundary and transformation-scope closure.

The array-access rules are recast around proof role rather than physical internal/external placement. Core.MemFaithful is added as an admissibility condition on the existing Σ -selection freedom, with the untransformed logical mapping-layer semantics as its reference object and explicit Core.17/E1–E5 plus Appendix-K blocking/WFG preservation requirements. J.Mem keeps its existing semantic cross-process dependency trigger and gains an explicit completeness requirement. New Core.MemProfile defines the reusable qualified-memory route for transaction-changing cache/merge/split/reorder behavior: the selected logical boundary lies above the transformation, preserves Core.MemSyncOrder, satisfies J.Mem, carries A6-compatible arbitration/point-to-point ownership and RW reset treatment, and for K either proves hidden WFG-inertness or exposes a transformation-invariant K-faithful boundary representing the relevant capacity/backpressure/release dependencies. If no such K boundary exists for a persistently backpressuring memory, transaction-changing transformations are outside the covered Appendix-K instance. Appendix F/G/H, K, M.7a/M.7b, Appendix R/S, and the framework-work list are aligned. Core.17/ Σ Match, E4, J.HCanon, and J.KObs-Chan are unchanged; concrete ping_pong_mem/native_ram_fifo profile qualification remains reusable follow-on flow work. Post-review cleanup hoists the mapping-layer declaration, transformation gate, selected-event rule, Core.MemSyncOrder fence, and Appendix-K disposition so they apply to all mapped memories rather than only the J.Mem cross-process case; the case split now controls only J.Mem/Core.MemProfile treatment. Appendix F restores consecutive deadlock-constraint numbering and adds an architect-facing memory warning, Appendix T names the Core.MemFaithful analogue, the framework-work row fixes its sentence break and records that initial arbitrated-memory profile qualification may conclude that transformations must be disabled, and MP6 now cross-references Appendix Q.1 when the qualified logical boundary is Sys_B^{elab}.

20 August 2026 — elaborated-boundary ownership/reset-scope consistency cleanup. Residual pre-coupler “both endpoint processes” wording is removed from Theorem K.1A, K.DomResetScope D2, the uniform U1 gate discharge, and RW1. A6 remains point-to-point at the source logical-channel level, while Sys_B^{elab} explicit boundaries may terminate in stateless couplers that are not modeled processes. K.1A now cites A6’s source-ownership/elaborated-dependency-projection conditions; D2/U1 close the dominance process scope over the modeled process identities declared by E’s process-facing/release-support interface; and RW1 interprets reset endpoint ownership through E’s boundary

attribution/projection rather than inventing coupler processes. No theorem scope or admitted design class is changed.

20 August 2026 — K.DrainReach source-lift completion and K.EnvMF uniform-discharge surfacing.

Lemma K.DrainReach-Comp is tightened so its substantive lifting step is explicit rather than treating Appendix-B automatic flush as if it were already path-indexed source Core.Drain. For pipelined/overlapped control, Appendix-B flush supplies implementation-side no-new-launch/drain evidence; the lemma uses the bound J.LocalGWR/J.GWR-Comp construction actions with I.A0/E3 issue-prefix closure to transfer the A2 request restriction to the exact constructed τ_{ref} , and the K.Drain record explicitly covers any A1 preparation that would itself count as the broader Core.16(2) source-later launch. J.RegPhaseReach is named as the theorem-derived Policy-L-legality premise. The former DR5 history/attribution item is no longer presented as independent evidence: J.GWR-Comp/I.A0, Core.18a/J.OfferReach/J.HCanon, and Core.QSync-Ref already supply the exact-operation, intervening-commit, and settled-endpoint bookkeeping. Appendix B, K.Drain, QC5/CI1, K.CertCov, M.6/M.7a, and the mechanization summaries are aligned to this split. Lemma K.EnvMF-Uniform is also surfaced from Definition K.EnvMF, K.PSF, M.6, and the definitive M.7a K.EnvMF audit entry. The common ordinary non-elaborated primitive point-to-point timed/nondefault case is stated explicitly: Core.OrdFrontSingleton supplies the at-most-one frontier fact, Dead_K membership supplies the nonempty internal item, and Core.WFG supplies the singleton ReleaseOwnerSets fact, leaving only the already-required K.JointFreeze-Ord premises and eliminating separate candidate-specific K.EnvMF certification when those instance-level conditions hold.

20 August 2026 — Sys_B^elab handshake/WFG simplification, release-family refinement, and coupler-model alignment.

The Core and Appendix Q state the elaborated hardware model directly as explicit finite-capacity/rendezvous storage plus stateless deterministic ready/valid/data couplers, including permitted process-boundary couplers. Core.11–Core.13 distinguish primitive E4 storage legality from the actual settled complementary handshake value at coupling-sensitive boundaries. The initial flat WaitOwners contraction is refined so the qualified template now preserves ReleaseOwnerSets_E: a family of minimal conjunctive owner supports with alternatives kept distinct; WaitOwners remains only their union for the conservative directed WFG and structural routes. Definition K.Dead uses family-level release closure plus induced-WFG connectivity, fixing the AND/OR loss while preserving the ordinary singleton case. Q.1(k) now requires family-level no-spurious-rescue and no-missed-rescue obligations, and the standard template route restricts availability attribution to monotone release-root composition unless non-monotone arbitration is made explicit or separately certified. Core.18e, J.KObsConf/J.KObs-WFG, K.KObs-Dead/K.KObs-Ext, and Lemma K.2 reconstruct and transfer the same release families from the existing E,w,H,O record, with no new KObs state field. K.JointFreeze/K.EnvMF are re-keyed on release structure rather than frontier cardinality, closing the single-frontier/multi-owner hole including the timed/reactive branch. Appendix G E4 wording, J.NBDecision/J.BufSameCycle/J.UnitEnable, K.Str-1/K.Str-3/CF-2, fall-through guidance, process-boundary coupler guidance, diagnostics, and the M.7a audit text are aligned. Because the corrected Dead_K may recognize additional coupling-sensitive components relative to the superseded flat-edge schema, affected A5-L/K.RespCov/waiver certificates must be revalidated under the current K.CertHdr version; ordinary primitive singleton dependencies are unchanged. A same-date cleanup adds Lemma K.ReleaseSCCReduction, with the explicit non-empty internal-release-family premise needed to exclude terminated/no-release roots; aligns Q.1(k) completeness with the declared template semantics; removes the stale multi-frontier-only K.PSF wording; permits one stronger instance-level K.JointFreeze certificate to package all admitted non-ordinary candidates; aligns the M.7a A6/Sys_B^elab audit rows with ReleaseOwnerSets; and surfaces the timed/reactive joined-release applicability limit in the Abstract and Formal Guarantees at a Glance.

19 August 2026 — Policy-S buffered-communication applicability clarification.

Explanatory material is added to the front deadlock summary, Appendix F, and the informative Policy-S simulation discussion to clarify the practical scope of the conservative serial-issue route. The front summary now states the limitation briefly: Policy S can be conservative when deadlock freedom depends on favorable overlap of blocking requests, especially in tightly coupled rendezvous-style communication, while noting that pipelined, buffered, and credit-based communication often removes that dependence in typical modern designs. Appendix F and the informative Policy-S note retain the detailed canonical crossed two-process/two-rendezvous-channel explanation, the $B(c)/Sys_B^{elab}$ treatment of real communication slack, and the caveat that positive finite capacity alone is not a proof of A5-L. No theorem statement, premise, certificate requirement, or Policy-S definition is changed.

19 August 2026 — Appendix T selective-scope clarification and residual guide/reference cleanup.

Appendix T is explicitly classified as informative, non-normative, selective, and non-exhaustive: it highlights important scheduling-specific obligations that standard Lean/formal-methods libraries do not supply, while the Normative Formal Core and applicable Appendices G–K remain authoritative for definitions and theorem interfaces, Appendix M.7a remains the consolidated discharge record, and Appendix S remains the mechanization-facing strategy. T.2.2 highlights the R4/E3 prefix-closed dynamic issue-order obligation, T.3.1 distinguishes proof-built fairness evidence through $Core.FairPick/Core.FairCycleDovetail$ from fairness of an already-supplied execution, and T.4 remains deliberately selective. The Guide safety reading path is aligned with the J.2b standard QSync all-finite-prefix safety route as well as its richer K-facing package-totality route. The duplicate K.2.1 Standing Assumptions label is removed without renumbering the KObs subsection sequence, the M.7 H1/H2/H3 historical shorthand is replaced by current-interface terminology, and T.3.3 states the serial comparison without implying a reversed K.2b.P/K.2b.R mapping direction. The former T.2.9a organization referenced by the 18 August revision record was consolidated during the Appendix T refocusing into the current T.2.9/T.3 structure. A follow-on scope repair adds a first-class M.7a R/E scheduling-contract conformance entry covering R1–R5/E1–E5 and R2C where applicable, aligns the M.7b instance/QSync bundle and Appendix S import anchors to that definitive ledger, and clarifies in the Guide/T.1 that Appendix S supplies mechanization decomposition/import anchors rather than the exhaustive side-condition inventory. No theorem premise, safety/deadlock scope, or certificate-coverage requirement is changed.

18 August 2026 — operational-invariant/conformance split and abstract Commit cleanup.

$Core.IC$ and $Core.SyncFence$ are reclassified from “Axiom” headings to normative operational definitions/invariants of Sys_ref , with an explicit status note preventing a mechanization from treating them as unconstrained trusted assumptions. Appendix M.7a adds a first-class $Core.IC$ request-lifecycle/Issue–Commit conformance entry alongside the existing $Core.SyncFence$ entry, and the M.7b qualification bundle, K.CertPkg accounting, Appendix R, and Appendix S are aligned; RTL use therefore remains evidence-backed while source-side use is constitutive transition legality. Definition $Core.8$ now defines a boundary commit by the selected proof model’s legal E4 channel-commit transition and treats simultaneous mapped valid/ready observation plus payload sampling as the RTL_B realization/conformance fact. The two introductory informal commit descriptions are narrowed correspondingly to the RTL-facing ready/valid reading. The existing process-facing $Commit(q)/CommitCycle(q)$ distinction and later L4/ $Core.RegSeq$ result-availability rule are unchanged. No theorem premise, safety/deadlock scope, or certificate-coverage requirement is weakened.

18 August 2026 — QSync finite-prefix ideal-cover closure and residual carrier/closure repairs.

Appendix J.2b adds Lemma J.CycleIdealCover and Corollary J.1-Safe-QSync. The existing J.QSyncTraceInd / J.TraceCertCov-QSyncCycle route remains the package-producing result on the cycle-aligned safety domain; the ideal-cover lemma extends the safety conclusion to all reachable finite outer observable prefixes without asserting a separate J.TraceCertCov/J.GWR package for each non-cycle-aligned ideal. A residual carrier assumption is made explicit by new Lemma $Core.RTLCycleCarrier$: a reachable RTL prefix

that has already entered a legal real cycle and contains a complete selected unit inherits the enclosing Core.CycleResult and may be carried through that same cycle to its successor K-origin, without adding generic cross-cycle progress, fairness, or totality. J.CycleIdealCover now cites that fact and its case split explicitly includes a nonempty current-cycle ideal that may equal the full observable bucket before the registered update. New Definition J.IdealPrefixClosed defines the safety closure condition as downward closure over finite required-order ideals, not merely suffix deletion from a linearized trace, and the safety corollaries and practical scope text distinguish naturally ideal-closed invariants from cross-interface scoreboards that may require direct J.TraceCertCov coverage. The one-off “cycle-aligned carriers” wording is replaced by “prefixes ending at reachable K-origins,” Appendix S imports Core.RTLCycleCarrier and J.IdealPrefixClosed, and the then-current T.2.9a bold-heading formatting is restored; the 19 August Appendix T refocusing later consolidates that material into the current T.2.9/T.3 structure. No K-facing certificate, deadlock premise, or per-prefix correspondence obligation is weakened.

17 August 2026 — Commit typing/process-facing completion, Appendix C alignment, and K.DeadChar operational-domain cleanup.

Definition Core.8 now makes Commit(q) the source operation’s process-facing commit occurrence and CommitCycle(q) its cycle index, while preserving separate boundary-local cycle indices for proof-internal E4 commit occurrences; it also states explicitly that commit/release is distinct from later result availability governed by Core.Phase L4/Core.RegSeq. Appendix C now gives only an informal operational reading of Commit and defers the formal Commit(q)/CommitCycle(q) meaning to Core.8; its blocking-request linkage uses CommitCycle(q) explicitly. Lemma K.DeadChar is retargeted from an operational RTL microstate statement to the source-side/lowered KCut domain on which Definition K.Dead is defined; its buffered and rendezvous conclusions now cite Core.12/Core.13 directly and its proof is correspondingly shortened. No theorem premise, coverage obligation, or user-facing theorem scope is changed.

17 August 2026 — Dead_K consolidation, Core A9/A10 authority cleanup, and introductory deadlock-scope clarification.

Definition K.Dead now has one authoritative statement: the duplicated bullets and explanatory paragraph are removed, while the environment-only exclusion is retained in the authoritative definition. Core and pre-K references now cite the Core scalar endpoint and reset-empty conventions directly; Appendix K A9/A10 are instance assumptions referring back to those Core definitions rather than restating them as competing definitions. Formal Guarantees at a Glance adds one short plain-language sentence noting the conservative applicability boundary for communication overlap and clock-timed/reactive testbench behavior. No theorem statement, theorem premise, or proof obligation is otherwise changed.

17 August 2026 — standard QSync certificate-profile packaging.

Appendix M adds M.7b, a normative packaging profile for the standard normal-compositional QSync flow. It groups the detailed M.7a discharge ledger into four logical interfaces: instance/QSync qualification, J correspondence/refinement, K applicability/provenance, and the primary Policy-S response bundle. The section states explicitly that these are packaging interfaces only: M.7a remains definitive, no theorem premise or qualification obligation is weakened or made optional, theorem-derived facts may be referenced rather than reasserted, and every artifact remains bound through K.CertHdr/K.CertPkg. The profile distinguishes the safety-only J.QSyncTraceInd use from the richer J.QSyncCertInd K-facing use, records the J.QSyncCertInd-Trace reuse rule, explains the standard universal K.CertCov-QSync composition, and permits alternate K.A5Cert routes in place of the Policy-S response bundle when A5-L is discharged another way. No theorem statement, proof obligation, or theorem scope is changed.

16 August 2026 — QSync cycle-aligned safety coverage projection.

Appendix J.2b now exposes the safety content already carried by the QSync base-and-step construction without importing the K-only correspondence/drain burden into safety. New Definition J.CycleReach_post names exact reachable RTL prefixes ending at K-origins; J.QSyncTraceState retains the exact-prefix J.LocalGWR/J.GWR-Comp evidence and the J.HCanon component needed by the safety transfer; and J.QSyncTraceInd gives base, one-cycle preservation, and exhaustive-successor obligations that explicitly do not consume J.OfferReach/J.OfferConf/J.KObsConf/K.DrainReach/K.RLAdmReach. Lemma J.QSyncCertInd-Trace proves the richer K-facing induction projects to this safety induction, with the C11 proof recording that the dropped facts are downstream outputs rather than inputs and with J.RegPhaseReach re-derivable from J.GWR-Comp. Theorem J.TraceCertCov-QSyncCycle derives J.TraceCertCov on CycleReach_post, and Corollary J.1-Safe-QSyncCycle instantiates the existing safety theorem on that domain. M.3, M.7a, Appendix S, Appendix T, the appendix map, and the front coverage summary are aligned. The revision deliberately does not claim arbitrary intra-cycle ideal coverage: QSync settling is Σ -silent and R2C fixed-point units are atomic, but the separate question whether every later-in-cycle complete-unit ideal is covered by a reachable succeeding cycle-aligned prefix is left for a possible cycle-completion/ideal-cover lemma.

16 August 2026 — R2C/RESET carrier, request attribution, memory-order audit, and strict same-endpoint issue cleanup.

A focused formal-consistency pass makes the following narrow repairs. Core.2 now describes combinational observations as emitted R2C fixed-point units matched by component identity and local emitted-occurrence ordinal rather than by global cycle bucket. The Core.17 observable-unit carrier is clarified to include explicitly selected proof-internal RESET units without adding RESET to Σ . Appendix C and Core.IC replace the ambiguous back-to-back Issue=t+1 wording with a continued-request attribution rule that permits inserted latency, and the strict same-endpoint issue rule is limited to distinct blocking instances on one scalar A9-governed explicit boundary/direction; R4/E3 cite that stricter special case while retaining their general non-strict inequalities. Core.MemSyncOrder and M.7a bind the main-text array/synchronization commit-time rule to a separate local schedule/mapping-layer qualification check without adding memory events to Σ or extending the J.1 replay state; it is not an Appendix I or Theorem J.1 replay premise, while J.Mem remains the additional cross-process shared-memory value-flow condition for J.1. Core Commit/Issue/BoundaryReq/IC terminology and the mirrored Appendix C wording now use “explicit abstract boundary” for the operational endpoint object, without redefining proof-internal Σ or outer Σ_{DUT} trace visibility. K.DomResetScope D3 now covers identities consumed by K.2b.P0 as well as K.2b.R/K.2b.P, matching the stronger uniform U1 scope. No additional Appendix F design rule is introduced.

16 August 2026 — deadlock observation-boundary disclosure and uniform interruption-gate discharge.

Two qualification/workflow clarifications are added without changing the existing safety theorem or H1 coverage architecture. First, the front summary and Appendix M.6 now state that Appendix K deadlock qualification uses every WFG-relevant explicit abstract message-passing boundary, while RTL micro-staging already absorbed into B(c) remains internal; the existing E4/B-faithfulness and KObs residual-offer obligations are unchanged. Second, Appendix K adds a normative uniform instance-level sufficient discharge for the candidate-relative reset/terminal interruption gates: one conservative scope satisfying U1-U3 proves all required K.ResetEpoch.Continue and nested K.TerminalDisposition.Continue branches at once. CF-S, K.PSF, Corollary K.2c, M.6, and M.7a accept that single record as an alternative to per-candidate gate enumeration; the original per-candidate schema remains the fallback and the theorem's universal quantification is not weakened.

16 August 2026 — certified-singleton proof restoration and recursive predecessor qualification test.

A small regression in Lemma K.CompareAdvanceLegal-Ord is repaired: the proof now covers both branches of its own singleton hypothesis, using Core.5a/Core.OrdFrontSingleton in the ordinary case

and the explicit total-order/source-comparability certificate in the certified-singleton extension. The K.CVPred recursive Core.16 disposition is also tightened so pointwise reuse of K.CompareAdvanceLegal-Ord for a lower-ranked blocking OpFact is permitted only when every Policy-S-required source-earlier blocking transfer for that predecessor is already present in the matched liberal transfer prefix with the required keyed identity/ordinal before materialization; the predecessor-materialization extension may not create any such prerequisite. K-O0/K-O4 and the M.7a K.CV/K.CompareAdvanceLegal audit entries are aligned. No other H1/H2/H3, QSync, FairPick, JointFreeze, or ordinary-instance burden rule is changed.

16 August 2026 — recursive predecessor Core.16 disposition and ordinary-certificate simplification.

Two residual Core.16 coverage details are tightened without changing the H1/H2/H3 architecture. Lemma K.JointFreeze-Ord now proves the full JF1/JF2 ordinary/singleton no-release result, including phase-legal work owned outside D via A6 point-to-point ownership plus closed-WFG component closure. The K.CVPred comparison-availability recursion now requires an explicit Core.16 reach-legality disposition for every recursively materialized non-transfer predecessor: qualifying ordinary/singleton blocking OpFact predecessors reuse K.CompareAdvanceLegal-Ord pointwise, pure observation/settling reaches are certified Core.16-inert, and other launch-carrying reaches supply equivalent owner/frontier evidence; “phase-legal” alone is insufficient. K-O0/K-O3/K-O4, CF-S, M.7a, and the mechanization-facing summaries are aligned. K.PSF and M.6 now state the instance-level burden reduction explicitly: an ordinary non-elaborated singleton-frontier instance needs neither a separate K.JointFreeze certificate nor a separate main K.CompareAdvanceLegal certificate, because both frontier facts are theorem-derived. No QSync/H3, FairPick/H2, or other Appendix-J interface is changed.

16 August 2026 — Core.16 launch/issue symmetry and ordinary-freeze lemma.

The serial-reduction interfaces are tightened in two small but load-bearing ways. New Lemma K.JointFreeze-Ord proves the ordinary/non-elaborated or otherwise certified singleton-frontier no-release case from Core.OrdFrontSingleton, E3/source-prefix closure, Core.16, component closure, and Core.Drain/K.Drain, leaving certificate Definition K.JointFreeze only for admitted StandardEnv multi-frontier components. The former K.CompareIssueLegal interface is superseded by Definition K.CompareAdvanceLegal and Lemma K.CompareAdvanceLegal-Ord, which establish the Core.16 comparison fact before K.2b.P0 and explicitly discharge both distinct Core.16(2) obligations: P0’s finite Core.ReachPrep/source-evaluation launch and K.2b.R’s later request Issue/assertion. Coverage ranges over every K.2b.P0/K.2b.R-consumed owner/key in K.DomResetScope, inside or outside D; explicit Sys_B^elab multi-frontiers require checked elaboration/order evidence, and the JointFreeze/CompareAdvanceLegal consistency rejection rule is retained. K.2b.P0/R/E/Q, K.2b/K.PSF/CF-S, K-O2/K-O4, M.6/M.7a, Appendix S, and Appendix T are aligned. No change is made to the QSync/H3 certificate-existence architecture or the H1 provenance repair.

16 August 2026 — K.CompareIssueLegal scope separation and outside-D repair (intermediate; superseded).

The serial-reduction comparison interface is tightened to remove the remaining $P \notin D$ Core.16 shortcut. Definition K.JointFreeze is returned to its no-release role (JF1/JF2 only). New Definition K.CompareIssueLegal separately covers Core.16 issue compatibility for every K.2b.R-consumed owner/key in K.DomResetScope, whether inside or outside D; Lemma K.CompareIssueLegal-Ord derives the ordinary/singleton case from the least-missing/first-overtake invariant, E3/source-order prefix closure, and matched earlier transfers, while explicit Sys_B^elab multi-frontiers require checked elaboration/order evidence. The interface includes an explicit consistency rule rejecting any prefix/key simultaneously classified as JF1-suppressed and comparison-required. K.2b.R no longer claims that $P \notin D$ escapes Core.16. K.2b/K.PSF/CF-S, K-O2/K-O4, M.6/M.7a, Appendix S, and Appendix T are aligned. The preceding JF3 revision entry is retained as historical record but is superseded by this cleaner separation;

no QSync/H3 or H1 provenance interface is changed. This intermediate interface is superseded by the later K.CompareAdvanceLegal/K.JointFreeze-Ord revision above.

16 August 2026 — K.JointFreeze propagation and comparison-legality cleanup.

Definition K.JointFreeze is tightened in three ways: JF1 now states the full-frontier condition under which Core.16 remains active; notation is normalized to $K.JointFreeze_I(D, \sigma_L, \{f_P\})$; and new JF3 makes the StandardEnv multi-frontier K.2b.R issue-legality fact an explicit paper-level certificate clause rather than content deferred to K-O2. K.2b.R now consumes JF3 directly. K.2b.Q Step 1 consumes the joint freeze already established by K.2b.E/K.EnvMF and no longer re-derives the obsolete unqualified source-later claim; K-O2 is explicitly only the mechanization obligation for JF1–JF3. Appendix M.6, the M.7a roll-up and K.EnvMF entry, the K-P mechanization obligations, and the mechanization-facing summary are aligned. No change is made to the QSync/H3 certificate-existence architecture or to the H1 provenance repair.

16 August 2026 — residual H1/H2/H3 formal and audit tightening.

The StandardEnv multi-frontier branch now consumes explicit Definition K.JointFreeze, whose JF1/JF2 obligations close the remaining K.2b.E freeze seam for incomparable $Sys_B^{\wedge}elab$ frontiers; ordinary/single-frontier cases continue to use Core.OrdFrontSingleton/Core.16/Core.Drain. Definition Core.QSync gains QS2a RTL-KCut origin recoverability, and Core.QSync-KCutCanonical now derives canonical-domain coverage from that structural cycle decomposition rather than assuming an origin for an arbitrary quiescent KCut. K.CVPred clarifies that cross-process predecessor materialization must preserve every frozen member of D. Core.FairPick is re-described as a stronger constructive evidence form for the existing behavioral fairness obligations, with proof-selectable issue latitude separated from instance-fixed scheduler/arbitration/environment choices. M.6/M.7/M.7a, the formal-appendix guide, Appendix N, and Appendices S/T are aligned; J.2b now states explicitly why its cycle-aligned LocalGWR induction does not by itself discharge arbitrary-prefix J.TraceCertCov.

16 August 2026 — QSync canonical-cycle certificate-existence repair (H3).

The former universal package-totality gap is closed at the paper-theorem level for the standard QSync normal-compositional scope. Definition Core.QSync now includes QS2a RTL-KCut origin recoverability, and Lemma Core.QSync-KCutCanonical uses that structural clause to identify every reachable standard-scope RTL KCut with the unique Core.QSync-Post endpoint of its recovered cycle K-origin; quiescence alone is no longer treated as supplying the origin. New Definition J.QSyncCertState packages the substantive per-cutpoint J/K correspondence data without assuming K.CertCov; Definition J.QSyncCertInd gives an anti-circular base-and-universal-one-cycle-step refinement obligation; and Theorem J.CertExist-QSync proves total J-side package existence by finite cycle induction. New Corollary K.CertCov-QSync binds those tuples through K.CertHdr and derives K.CertCov. Appendix K.1A-Total/K.1, A7, K.4e, M.6/M.7a, and Appendix T are aligned. A single observed run still cannot prove package totality, and a concrete flow must mechanize or independently check QSync plus the QSyncCertInd invariant (or use an alternate K.CertCov route). H1 and H2 repairs are unchanged.

16 August 2026 — K.2b.E constructive fair-continuation repair (H2).

The fair-extension step used by the Policy-S reduction is now constructive rather than schematic. Definition Core.FairPick packages the existing Core.IssueFair and B2/B3 discharges as prefix-local issue/boundary selectors with local finiteness, fixed-scheduler legality, EnvStep-compatible per-cycle selections, starvation-freedom, SyncFence-prerequisite, and B3-exposure evidence; it preserves the same behavioral Core.IssueFair/B2/B3 property but requires a stronger constructive evidence form than an abstract fairness assumption. Lemma Core.FairCycleDovetail proves phase legality with two applications of Core.CycleChoiceExt: first obtain a typed completion of the selected L1 segment through an actual finite L2 SettlePoint, then choose the fair L3 segment at that settled state and invoke C2 again for the final CycleResult. Lemma K.2b.E now states the Core.FairPick constructive discharge explicitly and proves an explicit frozen-prefix invariant, applies that construction cycle by cycle, and invokes

dependent choice only over already-typed invariant-preserving successors; FP4/FP5 then establish Core.IssueFair and B2/B3, while B5 remains a separate closure premise. Lemma K.2b now names B2/B3/B5 and the constructive fairness discharge explicitly. K.PSF, CF-S, K-O7, Appendix M.7a, Appendix N, and the mechanization-facing Appendices S/T are aligned; the later residual repair also promotes the StandardEnv multi-frontier freeze condition to explicit K.JointFreeze rather than leaving it only to K-O2. That H2 repair itself does not change the Appendix J finite-prefix theorem, H1 K.CV provenance repair, or user-facing scheduling rule; the separate H3 revision recorded immediately above adds the QSync package-totality theorem route.

16 August 2026 — K.2b.P cross-policy provenance-availability repair (H1).

Lemma K.2b.P now carries a joint least-missing-transfer/provenance invariant rather than inferring all K.CV predecessor occurrences from earlier endpoint transfers. Definition K.CVPred gains a normative serial-route comparison-availability closure: after the liberal comparison has matched the source-earlier blocking operations and serial-transfer facts required before a Policy-S endpoint may issue, every $\text{Pred_CV}(\text{OpFact}(\kappa))$ fact needed by K.2b.P0/K.2b.R must be available with the same keyed value after a finite phase-legal extension that preserves the frozen component/DomEpoch and introduces no unmatched dominance-domain endpoint transfer. Transfer predecessors are discharged by the earlier-transfer induction; lower-ranked non-transfer predecessors are discharged recursively by CVRank and their certified reach rules, with K.2b.P1 used only after occurrence. K.2b.P0, K.2b.R, CF-S, K-O0/K-O3, Appendix M.7a, and the mechanization-facing summaries are aligned. No Appendix J finite-prefix result, K.2b.E fairness argument, or user-facing scheduling rule is changed.

15 August 2026 — minimal mechanization-driven semantic clarification pass.

A narrow clarification pass incorporates only semantic distinctions exposed by pre-mechanization, without changing theorem architecture or introducing Lean-specific representation choices. Core.18 now defines cross-model KObs correspondence through the common typed record and leaves J.KObsConf as the downstream theorem establishing it. The historical-name lookup marks Core.15a retired in favor of Core.WFG. Appendix G states explicitly that function-like Issue/Commit notation abbreviates the Appendix-C relational existence/uniqueness facts. R2C now identifies the cycle-entry valuation and the intended unique-settled-result meaning of determinism while allowing internal ϵ /stutter re-evaluation; it also states that Sys_ref and RTL_B need not share an internal combinational model or absolute cycle buckets. The cycle-locked R2C clause now makes per-side participation/completeness, implementation/certificate justification, and the Core.17/E2 single-clause relationship explicit. Core.16 clarifies that “launch” is intentionally broader than message Issue. Existing reactive-environment replay wording and the conditional certification status of K.Ser are unchanged.

14 August 2026 — deliberate proof-audit repair: cross-policy keys, lowering transport, dependency stratification, and cycle-extension interface.

A focused pre-mechanization audit produced eight repairs without changing the user-facing scheduling contract. (1) Definition K.OpKey now supplies a theorem-instance common dynamic-operation key universe, partial per-execution realization maps, and a key-level cross-policy source order; execution-local Core.5 universes remain unchanged. (2) New Lemma K.2b.P0 separates cross-policy occurrence/reachability from K.2b.P1’s conditional-on-occurrence value/identity determinacy, and K.2b.R now consumes P0 explicitly. (3) Definition K.A5L model-indexes the scheduler-robust premise; the Policy-S route proves the lowered Sys_K form and new Lemma K.Low.A5 transports it to the Sys_ref A5-L consumed by K.1A, with waiver pullback made explicit. (4) Definition J.DepJ now distinguishes semantic D1–D4 prerequisites from D5 Horizon proof stratification; J.DepSound is limited to semantic edges and new Lemma J.HorizonOrderSafe validates D5-only commutation. (5) Core.CycleClosure C2 now includes finite partial-cycle extension, exported by new Lemma Core.CycleChoiceExt and consumed by K.2b.E’s fair continuation construction. (6) K.2b.R’s Policy-S premise is corrected from commitment before q is “reached” to commitment before q’s request may issue. (7) J.RefProjCover/J.1-Safe

consistently use finite canonical ideals/down-closed prefixes. (8) the Appendix K finite-drain note now cites Core.16(6)/K.Drain as the actual finiteness premise rather than appearing to derive finiteness from capacity alone. M.7a, K-P mechanization obligations, and Appendices S/T are aligned to these interfaces.

14 August 2026 — Core.17 Σ Match, persistent-deadlock scope, and QSync adequacy repairs.

Definition Core.17 now includes the canonical Σ -event/value occurrence matching as an explicit Σ Match conjunct of the normative observable-compatibility relation, and Appendix G is identified as an expansion rather than a redefinition of that convention. Appendix K's persistent-deadlock implication note is narrowed to the closed internal wait-cycle class actually captured by Dead_K and explicitly excludes starvation/perpetual under-supply cases such as K.CE-5/H2 from the broader "never hangs permanently" reading. Definition Core.QSync QS3 now includes a transition-relation-to-settling-network realization/adequacy condition, and the QSync-Ref proof sketch and Appendix M.7a evidence checklist are aligned to that condition. No M.7a premise-entry expansion is made because the current revision already contains B1, B2/B3, B4/FPD, B5/B6-Terminal, A6, A8/A9/A10, K.BF, and Core.IssueFair audit entries.

14 August 2026 — K.CV audit-trigger and provenance-closure cleanup.

Appendix M.7a now separates the unconditional serial-route K.CVPred/K.CV-WF trigger from the additional design-level checks for cross-process observations beyond blocking Pop transfer values, so a pure blocking Push/Pop Policy-S route may not mark the P1 provenance obligation inapplicable. Definition K.CVPred now states predecessor occurrence closure explicitly and K-O0 checks it. Lemma K.2b.R now attributes dynamic identity, payload, and typed boundary/direction to K.2b.P1/K.CV while deriving reset-epoch agreement from K.DomResetScope and fixed DomEpoch. No Appendix J finite-prefix or serial-dominance architecture is otherwise changed.

14 August 2026 — K.2b.P1 well-founded value/identity provenance repair (H2 closed).

Lemma K.2b.P1 no longer invokes an undefined "causal order of commits." New Definition K.CVPred/K.CV-WF gives the K.CV serial fragment a common semantic fact-key/provenance relation with deterministic per-fact evaluators and an explicit WellFounded certificate or checked rank. P1 now proves the strengthened all-facts determinacy claim by well-founded induction on $\prec_{CV}$. Same-cycle rendezvous enablement and the broader Appendix-G causal-dependency graph $\rightarrow$ are explicitly not predecessor edges merely by causal or temporal proximity, so legal cycles in $\rightarrow$ do not undermine the proof. K.BF, K-O0 and the K-P mechanization obligations, CF-S, K.PSF, M.7a K.CV, Appendix R, Appendix S, and Appendix T are aligned. No Appendix J construction order is imported into K.2b.P1, and finite-prefix Post $\rightarrow$ Ref is unchanged.

14 August 2026 — M.6 reset-scope disclosure and DomResetScope certificate typing cleanup.

M.6 now states the actual K.ResetEpoch applicability rule: Continue excludes every RESET whose RW scope intersects the candidate's K.DomResetScope, RW5 permits disjoint resets, and an intersecting RESET that does not satisfy the independently mapped ResetEpoch.DirectFail conditions lies outside the serial route and requires another A5-L discharge. Definition K.DomResetScope is retyped as a certificate-supplied finite process/boundary scope with explicit D1–D4 coverage, endpoint-closure, and anti-circularity conditions; larger conservative scopes are permitted and minimality is not required. K-O4, CF-S, M.7a, and Appendix T are aligned. No finite-prefix Appendix J result is changed.

14 August 2026 — ReachPrep coverage-direction and rendezvous-example clarification.

Core.ReachPrep now states the Post $\rightarrow$ Ref coverage direction explicitly: every covered RTL_B prefix must have selected Sys_ref ReachPrep semantics permissive enough for the dependency-independent preparation/reach segments required by its accepted J.LocalGWR or J.GWR construction, while more-permissive source preparation remains conservative. The Appendix K blocking-rendezvous A5-L example is named as a straight-line ReachPrep-dependent example, states that its Push operands are independent of the preceding Pop results, and makes the eager Policy-L alternative conditional on the

selected instance admitting that ReachPrep case. Appendix M.7a carries the same coverage and example-dependency obligations.

14 August 2026 — K.BF gate exhaustiveness and dominance-reset-scope cleanup.

K.InterruptionGate now states explicitly that, for the K.BF serial-reduction fragment, Core.IC and Lemma K.BF-Persist make Reset and Terminal the complete non-commit interruption-gate instantiations; any third gate is a theorem-model extension requiring re-proof. New Definition K.DomResetScope binds the processes and explicit boundaries whose dynamic transfer identities can participate in K.2b.R/P, with DomEpoch denoting their reset-epoch vector. K.ResetEpoch.Continue now excludes RESETs intersecting that dominance scope while RW5-framed disjoint resets remain permitted. K.2b.R and K.2b.P are correspondingly scoped to the fixed DomEpoch, eliminating the former use of K.ResetEpoch.Continue to forbid resets of $P \notin D$ that the old gate definition did not cover. K-O3/K-O4/K-O7, K.CertHdr, CF-S, K.PSF, M.7a, Appendices R/S/T, and Corollary K.2c are aligned. Core.CycleState is also added to the Appendix R and Appendix S Core-interface inventories. No finite-prefix Appendix J result or step-0–2 semantic interface is changed.

14 August 2026 — shared nested interruption-gate schema (step 3 complete).

Definition K.InterruptionGate now factors the common Policy-S comparison-interruption machinery into a prefix-local Continue branch and an independently mapped DirectFail branch with a single anti-circularity rule. K.ResetEpoch and K.TerminalDisposition are retained as gate-specific instantiations rather than parallel ad hoc definitions: reset is evaluated first; only K.ResetEpoch.Continue permits evaluation of the terminal gate; and only the two nested Continue branches enter K.2b.E/R/P/Q. ResetEpoch.DirectFail remains the matched-RESET/RW2 cancelled-pending route, while TerminalDisposition.DirectFail remains the DeclaredTerminal maximal-pending route. Lemma K.2b/K.2c, K-O4/K-O6/K-O7, K.CertHdr, CF-S, K.PSF, Theorem K.1, Appendix M.7a, Appendix R, Appendix S, and Appendix T are aligned. Steps 0–2 are unchanged, and no flat interruption enum or new third cancellation gate is introduced.

14 August 2026 — full-state CycleResult and cycle-closure consolidation (step 2 complete).

Definition Core.CycleResult is introduced as the primary typed real-cycle object. Its nextState is the full theorem-instance state, including registered/internal machine state, time-aware environment state, global cycle index, and cycle-valued auxiliary/monitor state; committedUnits records selected ordinary complete observable units / Σ commits and resetUnits records each dynamic scoped RESET occurrence. Core.CompleteCycle is retained as the nextState projection of CycleResult for compatibility with existing theorem statements, and Core.SilentCycle is the no-selected- Σ -commit case; a reset-only cycle may therefore be silent at the selected outer alphabet while still carrying the proof-internal RESET occurrence. DeclaredTerminal remains a predicate on states rather than a CycleResult outcome. RESET is explicitly an emitted in-cycle unit, not a sum-type branch: RW4/RW5 and Lemma J.RS are aligned so partial resets preserve unaffected-scope continuation and the J induction remains indexed by the RESET unit itself. The former standalone environment-totality and source-cycle-closure premises are folded into the normative Core.CycleClosure record together with full-state step typing, source-cycle existence/finiteness, environment-step definedness, auxiliary/monitor advancement, and reset-emission fidelity. K.2b.E/K.2b, K-O7, M.7a, K.CertHdr/CF-S/K.CompleteS, Appendix R, Appendix S, and Appendix T are aligned. Step 3 interruption-gate unification is not included.

14 August 2026 — finite-prefix Reach interface and generic Exec/J.FE removal (step 1 complete).

Appendices I/J now use Reach_post/Reach_ref as the sole generic finite-prefix execution domain. The restricted Ref $\rightarrow$ Post clause and Definition J.RTLConsistentRefPrefix are retyped over annotated reachable prefixes; because the supplied τ_{post}^* is already a reachable RTL witness, the reverse proof still takes $\tau_{\text{post}} := \tau_{\text{post}}^*$ without any fair-extension argument. The generic Exec_post/Exec_ref definitions and Lemma J.FE are removed. Core.PostCycleTotal is deleted after the dependency audit found no remaining semantic consumer; Core.CycleClosure is narrowed to the source complete-

cycle/environment closure used by Core.LAdm/K.2b.E. Appendix I.5 is explicitly local/finite and makes no global infinite-execution existence claim. Appendix G, M.7a, Appendix R, Appendix T, B6 explanatory text, Core.IssueFair/B2-B3 scope, and the J.1 proof prose are aligned. Complete/fair execution obligations remain only where later liveness results explicitly require them, including Exec_L^adm/K.2b.E and K.CompleteS. No CycleResult or interruption-gate unification refactor is included in this revision.

14 August 2026 — blocking-request lifecycle closure and frozen-frontier interruption-completeness repair (step 0 complete).

Core.IC now states an exhaustive ordinary blocking-request lifecycle for the G–K theorem model: after issue, an outstanding blocking dynamic instance remains asserted and stably attributed until its process-facing commit or an affecting RW2 RESET; no other nonterminal transition may deassert, retire, reattribute, cancel, or replace it, and DeclaredTerminal may instead end the execution with the request still pending. Core.SilentCycle, the Appendix C/G/K/N no-deassert-before-commit statements, and Core.IssueFair are aligned to that reset-epoch-scoped rule. Definition K.BF and Lemma K.BF-Persist now consume the Core lifecycle rather than introducing a separate cancellation exception; K.BF-Persist therefore closes the frozen-frontier interruption enumeration used by Lemma K.2b.E/K.2b.R. Definition K.RespFail retains cancelled-pending only for an affecting RW2 RESET; terminating abort/end-of-test behavior is represented by DeclaredTerminal and uses maximal-pending. K.TerminalDisposition clause (2), K.2b.Q Step 3, the K.2b proof, K-O4/K-O6/K-O7, CF-S, K.PSF, Appendix M.7a, Appendix R, and Appendix T are aligned. The K.2b.R heading-format regression is repaired. Protocols with independent nonterminal cancellation/withdrawal of outstanding blocking requests require a separately defined operational extension and are outside the present G–K theorem stack. No J.FE/CycleResult/gate-unification refactor is included in this revision.

14 August 2026 — K.ResetEpoch prefix-local applicability repair.

K.ResetEpoch clause (1) is revised from a universal property over already-frozen complete extensions to a non-vacuous prefix/state-local reset-contract condition: at every finite admissible comparison prefix while all selected frozen requests remain pending in the same epoch, no affecting RESET is permitted as the next transition, so RW2 cannot clear those requests before the gate applies. K.TerminalDisposition clause (1) is put in the parallel prefix/state-local form by requiring every such frozen-prefix final state to be non-DeclaredTerminal. Lemma K.2b.R, the proof of K.2b.E, K.2b.Q Step 1, K-O4/K-O7, and the Appendix M.7a audit entries are aligned. K.2b.E continues to construct the infinite fair continuation from machine-closure, drain, fairness, and issue-fairness premises; neither applicability gate assumes that continuation exists. All previously recorded H1, terminal-branch, watchdog-scope, and finite-prefix Post → Ref conclusions are unchanged.

14 August 2026 — K.2b declared-terminal applicability, termination-contract scope, and audit-consistency repair.

Definition K.TerminalDisposition now separates the reset-free serial route into a terminal-free dominance branch and an independently certified Policy-S terminal/cancellation failure branch, parallel to K.ResetEpoch. Lemma K.2b.E/K.2b.R/K.2b.P/K.2b.Q and their Appendix K-P proofs use only the infinite-fair continuation on the terminal-free branch; a declared terminal no longer substitutes for old finite-maximal reasoning. The direct terminal branch must independently bind the liberal frontier to a non-waived Policy-S response monitor and prove the actual Policy-S terminal/cancellation disposition while that monitor remains pending, avoiding circular dependence on K.2b.P/Q. The clause-(1) branches of both K.ResetEpoch and K.TerminalDisposition are stated as contract-level applicability properties rather than referring to a continuation constructed by K.2b.E; the later prefix-local repair below makes their mechanizable finite-prefix form precise. DeclaredTerminal scope is clarified: a modeled cycle-count/watchdog/end-of-test disposition inside the selected machine/test semantics counts, whereas an external harness watchdog that merely stops host simulation does not truncate Core.CompleteCycle or

defeat the terminal-free gate. CF-S and M.7a identify clause (1) as the normal prospective clean-run applicability evidence and clause (2) as an independently certified direct-failure branch that may be conditional or instantiated but is not required as an observed witness on every clean run. The remaining Policy-S 'maximal finite' paraphrases are aligned to the normative declared-maximal-terminal wording. Corollary K.2c, K.PSF, Theorem K.1, CF-S, K-O3–K-O7, Appendix M.6/M.7a, Appendix R, and Appendix T remain aligned. The legacy Core.PolicyL returned-result sentence is synchronized with Core.ReachPrep, the stale J.FE environment-receptiveness name is removed, Core.CycleClosure receives an explicit Core definition, and M.7a includes B1, B4/FPD, A6, A8/A9/A10, K.BF, and K.TerminalDisposition audit entries. Finite-prefix Post $\rightarrow$ Ref remains unchanged.

14 August 2026 — fair-extension, complete-cycle, and source-issue formalization repair.

The machine-closure path has been revised to separate same-cycle settling, ordinary silent clock cycles, and terminal B6 stutter. Definition Core.ReachPrep now makes the intended Policy-L eager-reach semantics explicit, including dependency-independent ordinary straight-line successors used by K.CE/K.CE-2; Definition Core.IssueFair gives the previously unnamed source-issue weak-fairness predicate; and new Core.CompleteCycle/Core.SilentCycle, EnvCycleTotal/EnvTerminal, SourceCycleClosure, and PostCycleTotal definitions type cross-cycle progression. Lemma J.FE is rewritten as a phase-structured source construction plus a distinct RTL totality/progress discharge, so a presently idle state is no longer inferred to be a B5 fixed point and B6 is reserved for Core.TerminalIdle, which requires both machine-side and environment-side permanent no-future-enabling evidence. Core.LAdm, K.2b.E, K-O1/K-O3/K-O6/K-O7, K.Resp maximality/aging, Appendix N, and Appendix M.7a are aligned. The generic J.FE route is explicitly separated from Appendix K's stronger Exec_L^adm continuation route. These changes do not alter the finite-prefix Post $\rightarrow$ Ref safety construction.

13 August 2026 — Appendix Q compositional source-QSync repair.

Appendix Q.1(i) now distinguishes reusable elaboration-template settling facts from the whole-source Core.QSync discharge. Template-side finite/deterministic/acyclic settling of staged/bundle/join/guard/coupler/epoch-gate logic no longer implies whole-source QS2–QS5 by itself. For Sys_B^elab(E), the concrete theorem instance must also supply the design-owned Core.RegSeq/direct-input/R2C/non-template-boundary settling facts and a checked template/design settling-composition condition covering every cross-partition dependency; QS4 is applied to the complete combined same-cycle settling graph. Appendix M.7a now names this composition obligation and lists a combined dependency-rank/topological-order certificate as a sufficient mechanically checkable discharge. The Core.QSync-Ref proof note and Sys_B^elab audit entry are aligned accordingly. This is a qualification/discharge-interface correction only: Definition Core.QSync, QS1–QS5, Theorems Core.QSync-Ref/Core.QSync-Post, J.1 scope, and H3 are unchanged.

13 August 2026 — J.TraceCertCov source-settling propagation.

Definition J.TraceCertCov and its Appendix M.7a discharge entry now state the route-sensitive source-settling evidence explicitly rather than leaving it only inside “remaining Post $\rightarrow$ Ref premises”: the normal J.LocalGWR/J.GWR-Comp route carries QualifiedSynchronousModel_Sys_ref(I)/Core.QSync-Ref, while a direct J.GWR package may instead carry checked finite L2-settled segments for its represented Horizons. Corollary J.1-Safe and the abbreviated front-matter safety-signoff prose now point to that same route-sensitive requirement. This is a propagation/qualification clarification only; it adds no unconditional post-side Core.QSync premise to plain finite-prefix J.1 and does not change H3.

13 August 2026 — qualified synchronous hardware interpretation.

An informative hardware-interpretation paragraph now precedes Definition Core.QSync. It states the intended single-clock cycle-level model directly: registered state and cycle inputs are fixed, sequential process outputs are derived from registered/fixed operands without same-cycle interface-result through-paths, the complete same-cycle combinational/handshake network settles deterministically with pure combinational feedback outside the standard scope, and boundary actions precede the next

registered update. It also clarifies that `SettlePoint`, `BoundaryEvalPoint`, and `Core.KCut` are proof views of the ordinary settled cycle snapshot rather than additional hardware states, while preserving explicit latency-sensitive/WC and Policy-L nondeterminism.

13 August 2026 — Stage A terminology, L2 frame repair, and qualified synchronous-settling Stage B.

Stage A adds an explicit determinism taxonomy and separates δ -settling, real silent τ -cycles, and B6 idle ticks without redefining the general ε relation. Definition `Core.Settle` now states the normative L2 frame: cycle index, registered process/FSM/pipeline and explicit-channel storage/occupancy/credit state, and selected cycle- t exogenous/environment values are frozen during $\rightarrow \varepsilon, L2$; no Σ commit or next-cycle registered update occurs. `SettlePoint` existence is no longer implicit in the definition. Stage B adds Definition `Core.QSync` and Theorems `Core.QSync-Ref/Core.QSync-Post`. The standard K-facing/cutpoint scope now requires finite acyclic deterministic same-cycle settling over explicit state boundaries; `QSync-Ref` proves `source/ Sys_B^elab SettlePoint/KCut` existence and uniqueness, and `QSync-Post` derives `Core.KCutClosure_post` plus a canonical K-facing `NormPost` endpoint. Appendix Q adds the matching elaboration obligation; Appendix M.7a adds a first-class `Core.QSync` audit entry and treats `Core.KCutClosure_post` as theorem-derived in the standard scope; `K.CertHdr` records the structural evidence, theorem provenance, and any trusted-base delta when a settling claim is accepted without independent artifact checking. The normal compositional finite-prefix J.1 route now explicitly consumes source-side `QSync-Ref` settling existence through `J.RegPhaseNF/J.UnitExt`, while a directly validated J.GWR package may carry equivalent per-Horizon settling evidence; plain J.1 does not thereby acquire post-side `QSync` unless a post `KCut/NormPost` interface is used. This revision intentionally narrows the standard `Core.KCutClosure_post` discharge scope: the former independent inductive-invariant and checked-normalization alternatives are withdrawn as standard-scope routes and require an explicitly qualified non-`QSync` theorem extension. `QS1` now states the single-clock semantic condition directly, with M.7a `ClockScope` serving as its evidence entry. Appendix I/J replay normalization remains separate. No change is made to Appendix H.3 in this revision.

13 August 2026 — Lean/scheduling alignment clarifications.

The post-side interpretation of `Core.SyncFence/E5` now states explicitly that same-cycle predecessor/synchronization accounting is a trace/certificate attribution convention and does not impose an observable physical sub-cycle ordering within one RTL clock edge. Appendix M.7a adds named `ClockScope` and `Core.SyncFence` discharge entries, and `K.CertPkg` records their references where applicable. Theorem J.1's restricted `Ref  $\rightarrow$  Post` clause now exhibits its existential witness explicitly as $\tau_post := \tau_post^*$. Lemma K.2b and K.2b.E now state that drain-closedness is obtained from the `Dead_K / DeadKRec^W` hypothesis itself rather than imposed as an additional narrowing of A5-L's source-`KCut` domain. Definition L.1 remains intentionally stated over generic ε -quiescent source witness-choice cutpoints rather than K-facing `Core.KCut` points.

12 August 2026 — synchronization-completion fence formalization (H4).

The previously prose-only rule that a synchronization call cannot complete while an earlier blocking `Push/Pop` in its source interval remains uncommitted is promoted into the formal contract. New `Core.SyncFence` gates source synchronization commits, including same-cycle message/synchronization completion and `Sys_B^elab` process-facing commit semantics; `Core.Phase L3` and the operational transition relation consume the fence. R3 and E5 now contain explicit blocking-commit-before-sync clauses rather than relying on a parenthetical appeal to blocking-call control flow. Appendix C, Core.14, Core.18e, the drain-closed definition, the compact E1–E5 table, the system-level E5 proof, and Lemma J.KObs-Offers are aligned. This makes the interval scoping of `Pending_P/Front_P/drain` candidates lossless and closes the residual-offer versus `PendingRec` reconstruction seam without changing the intended deadlock predicate.

12 August 2026 — L2 settle / global ε -quiescence, KCut closure, and K-cutpoint-domain correction.

Core.4 defines model-specific global ε -quiescence; Core.Settle defines the complete L2 phase-restricted ε relation and BoundaryEvalPoint; Remark Core.1a classifies phase advancement as meta-level rather than ε/Σ activity; and Lemma Core.SettleIsKCut proves that every phase-bearing source L2 SettlePoint is a source KCut. Core.Phase L2/Core.16 use SettlePoint for activation, while WFG, KObs, Dead_K, Corollary J.1, and K.DeadChar use the Core.KCut comparison domain. Core.SettleKBridge remains the named activation-to-later-drain-closed-KCut obligation and is consumed by K.DrainReach/Core.LAdm. RTL_B retains its own Quiescent/NormPost/KCut_post semantics without importing L0–L4, while Core.KCutClosure_post plus Core.KCutObserve supply the non-vacuity/observability obligation ensuring that every RTL K-facing cycle-evaluation origin reaches a finite KCut and that persistent relevant blockers cannot remain forever outside that domain. K.CertCov remains certificate totality over KCutReach_post and is paired with KCutClosure_post for universal conclusions. Stale Core-table, Appendix I.2, $K\varepsilon$, and K.Low.C cutpoint terminology is aligned; Appendix T, M.7a, the front guarantee summary, and the certificate header are updated accordingly. B5 remains the stronger global fixed-point condition, not a synonym for Core.4 quiescence.

1 August 2026 — logical replay-checkpoint clarification and B4/B5 unit correction.

Definition I.ReplayObs now denotes the logical replay checkpoint represented at an operational state, not the raw post-preparation cursor/store. Replay-inert L1 preparation is separated from witness-fixed epsilon transitions that may update replay fields; normalized I.ReplayObs-Congruence and I.DR now use paired congruence of exact selected normalization paths rather than per-path neutrality or global confluence. I.CutpointOfferObs, I.DR', and Appendix T are aligned. B4/FPD use the per-process epsilon-step bound D_P; B5 uses a separate cycle bound C_sys, optionally derived through per-process C_P charging lemmas. Lemma I.1 and A2 likewise keep the units distinct. Appendix Q records the equivalence of channel-first A_E(c, σ) and state-first A_E(σ , c) notation.

31 July 2026 — replay/offer separation, R2C cycle-lock completeness, and B3 conclusion clarification.

Appendix I now separates the replay-derived process slice I.ReplayObs from the issue-schedule-dependent current blocking-offer slice I.CutpointOfferObs. Lemmas I.ReplayObs-Congruence, I.DR, and I.DR' no longer infer Pending_P, Front_P, or BoundaryReq from committed observable-unit history; Appendix J continues to construct and audit those facts through O, J.OfferReach, J.OfferConf, and J.KObs-Offers. Core.18d, Appendix R, and Appendix T are aligned to the split. R2C now defines ComponentParticipatesInCycle and requires one emitted fixed-point unit for every participating cycle of a cycle-locked component, while forbidding units outside the participation domain. B3 now states explicitly that its buffered P_push/P_pop clauses conclude the complementary discharge; E4 plus B2 govern eventual commit of any original request that remains pending and continuously enabled afterward.

31 July 2026 — ordinary-source frontier-uniqueness lemma.

Lemma Core.OrdFrontSingleton now derives the at-most-one-member property of Front_P for ordinary non-elaborated source intervals from the dynamic-program-order convention of Clarification Core.5a and the one-call/one-frontier-item ordinary-source mapping. Appendix B, Definition Core.15, and Clarification Core.WFGb now cite the lemma explicitly.

26 July 2026 — reset-branch mapping and multi-frontier legality tightening.

K.ResetEpoch clause (2) now supplies an independent, certificate-bound mapping from a selected liberal component frontier to a process-facing Policy-S serial frontier and binds the exact pending monitor and RW2 clearing event, rather than relying on the reset-free K.2b.Q mapping. K.2b.R now states Core.16 issue legality against the full frozen frontier set and tracks the StandardEnv multi-frontier case through K-O2/K-O4. The Formal Guarantees at a Glance section, the dynamic-reset main-body guidance, Appendix M.6/M.7a, CF-S, and Appendix T now disclose and name the dynamic-reset applicability condition consistently.

26 July 2026 — serial-reduction seam and front-matter scope clarification.

K.2b.R now proves explicitly that Core.16 does not block the reached serial-prefix request, removes the redundant independent-work disjunct, and scopes request persistence to one reset epoch. K.2b.P now states its elaborated-boundary coverage, explains why the same-cycle tie-break is harmless under begin-of-cycle non-fall-through E4 semantics, and includes an explicit composition note with K.2b.E. Definition K.ResetEpoch separates the reset-free dominance branch from an already-triggered matched-reset cancelled-pending branch; pre-reset cutpoints that satisfy neither branch are explicitly outside the Policy-S reduction. K.4b, K-P, CF-S, K.PSF, Theorem K.1, Appendix M.7a, and Appendix T are aligned. The four-item Formal Guarantees at a Glance section now restores a compact scope exclusion, instance-qualification warning, and direct pointer to Appendix M.7 without reintroducing proof-interface labels or the detailed table.

26 July 2026 — K.2b proof completion and user-facing responsibility/status clarification.

Appendix K-P now contains explicit proofs of Lemma K.2b.R serial-prefix request reachability and Lemma K.2b.P deterministic serial progress dominance, including the least-missing-transfer argument and separate buffered-Pop, buffered-Push, and rendezvous cases. K-O3 now tracks K.2b.P mechanization and K-O4 tracks K.2b.R; K.4b and Appendix T are aligned. The Formal Guarantees at a Glance section has been reduced to four user-level statements and proof-interface labels have been removed from that section. The detailed responsibility/current-status table has been moved to Appendix M.7. Appendix M.6 continues to distinguish the design user's complete instrumented Policy-S execution from the full theorem discharge and to identify valid universal-coverage routes without claiming that a generic construction already exists.

25 July 2026 — WFG, elaboration-map, and drain-interface completion.

Definition Core.WFG now restores the edge-versus-blocked-process and explicit multi-frontier out-degree clarifications from the former Appendix K definition, while continuing to cover mixed buffered/rendezvous graphs. Definition Core.ElabProj promotes the four-role Π_{elab} interface—trace projection, occupancy projection, conservation/accounting, and proof-internal WFG visibility—into the Core; Appendix Q now normatively instantiates that interface. Definition Core.Drain removes the remaining Core-to-K definitional reference, and K.Drain is stated as the application evidence obligation that discharges it. Core.6, Core.7, Core.15, Clarification Core.5a, and Core.PolicyS now have navigable formal headings; Remark K.0-pre restores the derived-bridge framing. Definition Core.PolicyL now gives the liberal issue policy a citable heading parallel to Core.PolicyS. The K.Drain and Appendix Q normative run-in labels are restored to bold, and the A9–A11 assumption paragraph has normalized run boundaries so that only the assumption labels are emphasized.

25 July 2026 — formal-core split attribution and side-condition organization cleanup.

Remark Core.1a and Appendix T now attribute Policy-L step legality, cycle-phase semantics, and admissibility to Core.LLegal, Core.Phase, and Core.LAdm respectively. J.Mem and RESET/RW1–RW5 now appear under a dedicated Core subsection, and the RESET label is restored as a bold run-in. The normative wait-for-graph definition is renumbered Core.15a to preserve ascending definition order at its logical location after Core.15, and Definition Core.17 now refers to the Core rather than to an appendix. The active historical-name lookup maps old R.20 to Core.15a; older preserved revision text is otherwise unchanged.

25 July 2026 — formal-core dependency audit follow-up.

The follow-up audit removes the remaining $J \rightarrow K$ proof citation by limiting Corollary J.1 to equal-KObs and J-owned reconstruction consequences; Appendix K now performs Dead_K and waiver/diagnostic factoring. Appendix K imports Definition Core.15a for WFG instead of restating a second normative-looking graph definition. The Core KObs layer now exports only Core-owned reconstruction functions; Definition K.EnvTerminalAdequate and Definition K.KObsDerived, together with K.StimX and K.WFG+, now reside in Appendix K before the factoring lemmas. The KObs architecture explanation has moved to downstream Appendix R, the side-condition discharge record to Appendix M.7a, and Lemma J.RS to

Appendix J. Core section headings are now descriptive rather than reusing definition numbers, the WFG definition fills the former Core.15a gap, the reading paths follow document/dependency order, and the Core scope text distinguishes Appendix G’s raw Sys presentation from the Sys_ref comparison used by Appendices I–K.

Historical-name lookup for preserved earlier revision entries: unlabelled Policy L → Core.PolicyL; R.RegSeq → Core.RegSeq; R.LAdm → Core.LLegal/Core.Phase/Core.LAdm; R.16 → Core.16/Core.Drain; R.18e → Core.18e; R.20 and the interim Core.15a label → Core.WFG; former Definition R.1a → Core.ElabProj; R.21b–R.21e → J.KObs-Proc/J.KObs-Chan/J.KObs-Offers/J.KObs-WFG; and R.21f–R.21g → K.KObs-Dead/K.KObs-Ext. Core.15a is a retired/interim label: occurrences in preserved revision-history text describe the name in force at that time and shall not be used as current normative cross-references; the current primary name is Core.WFG.

25 July 2026 — normative formal-core dependency reorder.

The former “Common Formal Definitions for Appendices I–K” has been promoted and moved immediately before Appendix G as the “Normative Formal Core for Appendices G–K.” The operational Sys_ref/Policy L/Policy S material, Core.RegSeq, Core.LAdm, the exact Core.16 drain discipline, the shared frontier/request definitions including Core.7a BlockingMsgNextFront, the normative Core.15a WFG definition, and the Core.18c KObs datatype/reconstruction functions now occur before first use. Appendix J now contains the source-side KObs reconstruction lemmas J.KObs-Proc/J.KObs-Chan/J.KObs-Offers/J.KObs-WFG; Appendix K contains K.KObs-Dead and K.KObs-Ext. Appendix R has been retained as a downstream compact/mechanization companion with an explicit deletion-safety rule and no normative upstream dependencies. Active cross-references have been redirected from the former late R-numbered definitions and lemmas to their new Core/J/K locations; historical revision entries are preserved unchanged. The dependency audit now has no Core/J forward reference to Appendix K for WFG, and Appendix R remains deletion-safe.

24 July 2026 — Policy-S user workflow and status clarification.

The Abstract, Introduction, Formal Guarantees at a Glance, Guide to the Formal Appendices, and Appendix M.6 now present the deterministic Policy-S route in user-workflow order: one selected, complete instrumented pre-HLS execution is the primary design-user activity, while the flow separately establishes SDet, applicability, completeness, total finite-bound response coverage, and certificate validity. The text now distinguishes the operational practicality of simulation monitors from the outstanding mechanization and qualification of the K-P reduction and certificate interfaces, and it states separately that K.CertCov is required for a universal reachable-RTL-cutpoint conclusion. The underlying theorem premises and conclusions are unchanged.

24 July 2026 — formal cross-reference, deadlock naming, snooping, and K.2b.Q cleanup.

The stale clause-based Corollary J.1 signal-grouping reference has been redirected to the J.RegPhaseNF/J.UnitExt unit treatment. Appendix K now names the ordinary deadlock predicate Definition K.Dead and uses Dead_K/Dead_K^W consistently instead of overloading “K.1” as a predicate or component name; the phantom Definition K.Dead citation is thereby made valid. The J.Mem extension note no longer reuses invariant clause number I4. Appendix G now states snooping alignment conditionally on existence of a WC/L.Dir witness and points failed realization to Appendix L triage. K.2b.Q Step 1 now limits its source-prefix commitment claim to the process-facing blocking Sys_K/K.BF serial prefix and explicitly excludes failed non-blocking polls. Appendix C now contains the normative ReturnedAtCycle definition directly rather than a dangling main-body cross-reference, and Definition R.18e cites both K.Dead and K.DeadW for the ordinary and waiver-parameterized reconstructions.

24 July 2026 — main-body formal-detail cleanup.

The pre-appendix scheduling discussion has been simplified without changing the formal contract. The Basic Conceptual Model now explains overlapping issue without proof-internal multi-frontier or elaboration terminology; the R1/E1 and R3/E5 rationale has been removed from the pipelined-loop

section; and the synchronization-boundary subsection now gives only the operational drain/next-interval rule with an Appendix B reference. ReturnedAtCycle and the detailed registered sequential-interface semantics have been removed from the main body and replaced by one plain-language registered-cycle rule, with the normative ReturnedAtCycle definition relocated to Appendix C and the dependent-use rule retained in R.RegSeq. The two formal-architecture bullets at the end of the main Summary have been removed; their unique Policy-S coverage content and a concise statement of proof-layer separation are consolidated in Appendix M.

24 July 2026 — introductory restructuring and formal-results navigation revision.

The former theorem-by-theorem opening discussion has been replaced by a short front abstract, a practical Introduction, and a separate Formal Guarantees at a Glance section. Broad claims concerning RTL verification cost, customer experience, implementation coverage, and source-level signoff have been qualified to avoid implying unconditional tool or RTL certification. A new unlettered Guide to the Formal Appendices has been inserted immediately before Appendix G without changing any existing appendix identifier. Appendix M has been rewritten as Formal Results Overview, preserving the precise Post $\rightarrow$ Ref safety direction, restricted Ref $\rightarrow$ Post use, local-to-global J.GWR-Comp architecture, KObs/K.RLAdmReach cutpoint bridge, certified-cutpoint versus total-coverage distinction, Policy-S/A5-S serial-reduction route, applicability limits, and implementation-evidence obligations.

18 July 2026 — M1 safety-transfer direction and common-case pre-HLS signoff clarification.

The document now states explicitly that ordinary source-side safety transfer uses Theorem J.1 Post $\rightarrow$ Ref: RTL_B introduces no new canonical Σ_{DUT} safety behavior absent from Sys_ref. Definition J.RefProjCover names the optional observable-prefix coverage fact under which one complete pre-HLS execution represents all covered source-reference observations, and Corollary J.1-Safe packages both the universal-source and one-run transfer cases. The Appendix G “No Surprises” discussion and Appendix L Note 1 cross-reference this corollary. The restricted Ref $\rightarrow$ Post clause remains an annotation-dependent selected-behavior realization clause and is not presented as the common safety-signoff mechanism. The former “source-side pass only through the restricted reverse direction” wording in the Practical Implication section is replaced by this safety-versus-realization distinction.

18 July 2026 — H4 eager-issue completeness boundary and H5 environment/frontier scope correction.

H4 is made explicit: because Policy-S-shaped prefixes are Policy-L-admissible, the two-channel crossing example contains an admissible two-Pop deadlock prefix and therefore falsifies A5-L for the unchanged design. The misleading suggestion that another A5-L proof route could rescue that instance is removed; alternate proof routes remain available only when A5-L is true. H5 is closed by the named applicability condition K.EnvMF. Certified multi-frontier components are supported by K.2b.E/K.2b/K.2c only under StandardEnv/B1-avail; timed/reactive or otherwise nondefault environments are restricted to a certified single frontier consisting solely of the selected internal blocker. K.EnvX-Auto, A5-S/CF-S, K.PSF, Theorem K.1, K-O2/K-O7, the Appendix K-P proof, Appendix T, the side-condition record, and the summaries are aligned. No general future-environment no-rescue certificate is introduced.

18 July 2026 — K.1A statement and interface-seam correction.

Theorem K.1A’s statement now explicitly restores A5-L, conditional KObsClosed waiver closure, the point-to-point evaluated-boundary premise, and the certified-cutpoint conclusion that its proof already consumed. K.CertCov now quantifies over an explicit reachable prefix/cutpoint pair. ReturnedAtCycle has one normative main-text definition and an Appendix C pointer. Definition J.RegPhaseReach and Corollary J.RegPhaseReach-Comp are separate citable anchors. R.RegSeq requires checked identification before rejection or theorem-scope exclusion of zero-time dependent NB cascades. The historical 17 July sequential-result-availability entry is restored verbatim; supersession remains recorded only in the later 18 July H3 entry.

18 July 2026 — consolidated H1/H2/H3 registered-phase revision.

H1 remains split into Definition K.CertifiedCutpoint/Theorem K.1A for one cutpoint and K.CertCov/Corollary K.1A-Total for universal coverage. H3 is reclassified as a formalization-fidelity defect: the intended registered sequential-interface semantics has no same-Horizon decision-to-dependent-request edge, but the former stop-at-NBHole replay forced decision-before-independent-continuation. Definition R.RegSeq and ReturnedAtCycle now cover successful and failed NB results, request and Push payload outputs, stable direct inputs, Cat#/source conformance, and later-Horizon dependent use. Lemma I.3 promotes NBHole to an enriched carried ghost obligation with result destinations and NoDependentUse. J.ConstructionAction adopts a normative field-sensitive location vocabulary; A3 is narrowed to the shared decision/result; Dep_J/FR are revised; J.RegPhaseNF proves the R.LAdm L1/L2/L3 normal form modulo J.HCanon using J.BufSameCycle; and J.UnitExt/J.GWR-Comp use outer Horizon induction with an inner L3 pass. No generic micro-position or new action taxonomy is introduced. For H2, Corollary J.RegPhaseReach-Comp now derives J.RegPhaseReach on the compositional route and supplies L0–L4; K.DrainReach remains the explicit nonwithdrawal/ $K\epsilon$ /R.16/K.Drain bridge, and K.RLAdmReach is their conjunction. CF-0, K.CertifiedCutpoint, K.1A, the side-condition record, Appendix S/T, summaries, and the no-unresolved-hole cutpoint invariant are aligned. The separate Lean strategy must propagate the same interface change before mechanization.

18 July 2026 — H1 certificate coverage and H2 admissibility-interface revision.

The Appendix K preservation theorem was split into a per-cutpoint result and a coverage corollary. The consolidated H1/H2/H3 revision above retains that H1 architecture but refines H2: the L0–L4 portion is now J.RegPhaseReach, derived from the revised Appendix I/J construction, while K.DrainReach carries the remaining nonwithdrawal/ $K\epsilon$ /R.16 evidence. Their conjunction is the K.RLAdmReach consumed by K.1A.

17 July 2026 — H2 bounded-response serial-route revision.

The primary K.2b/K.2c/CF-S route now uses validated process-facing bounded-response monitoring rather than the incomplete missing-supply owner trichotomy and Dead_KS terminal taxonomy. Definitions K.RespMon/K.RespCov/K.RespFail/K.RespClean require default-total internal-site proof coverage, process-facing trigger/response attribution, maximal-finite and cancellation handling, complete-run evidence, and waiver preservation. Lemma K.2b.E and mechanization obligation K-O7 explicitly construct the fair complete frozen Policy-L continuation from a bare A5-L counterexample prefix; K.2b.Q then proves that the liberal Dead_K cutpoint forces a finite Policy-S response failure, and K.2c converts K.RespClean into A5-L. The perpetual under-supplying-owner example is added explicitly. Dead_KS, EnvStop, DoneStop, K.EnvOrd, and K.Done remain optional diagnostics only. CF-S, K.PSF, Theorem K.1, Appendix R’s KObs boundary, Appendix T, the side-condition record, and both the abstract and main-body Summary are aligned; no Policy-S response bound is transferred to RTL.

17 July 2026 — Policy-L admissibility and settle-window clarification.

Definition R.LAdm now supplies one normative execution domain for Appendix K. It distinguishes Policy-L step legality, finite-prefix admissibility for A5-L reachability, and complete admissible executions for fairness, stabilization, and limit arguments; binds the fixed stimulus/environment/witness/arbitration choices; and defines the L1 pre-settle issue window, the L2 ϵ -settle snapshot, the exact activation point of Definition R.16/K.Drain, and next-cycle availability of committed results. A5-L, K.1A, K.2b, K.Ser, Examples K.CE-2/K.CE-3/K.CE-4, Appendix R, and Appendix T now cite this single definition. The change resolves the ambiguity without changing which independent eager issues Policy L permits or making K.Drain part of the basic Policy-L issue rule.

17 July 2026 — sequential-result-availability clarification.

The Issue/Commit semantics now state explicitly that a result committed by a sequential process in cycle t may affect a dynamically succeeding dependent operation only in a strictly later cycle. The clarification is added to the main Issue definition, Appendix C linkage rules, the Sys_ref/Policy-L

operational interpretation, Lemma I.3 and Corollary I.4, J.DepSound, J.FoundationRank, and the D1 case of Lemma J.DepRank. Same-cycle rendezvous, buffered-channel updates, eager issue, and pipeline overlap remain permitted for requests whose reachability, control, and operands were established independently before the cycle boundary.

15 July 2026 — J.GWR formal-ordering and SDet clarification revision.

The J.DepAcyclic proof now uses one graph-independent semantic foundation rank and the new Lemma J.DepRank; the prior union-of-local-orders argument is removed, and Height/J.CanonicalRank are defined only after acyclicity. J.RequiredPrefixCandidate removes the circular reference from J.PCert to the not-yet-derived canonical order. J.ConstructionOrderProvenance/J.ValidatedConstructionOrder provide one interface for theorem-derived and directly validated J.GWR packages. SDet now states functional history determinism as prefix-comparability/no-incompatible-history for run artifacts, while K.CompleteS separately establishes maximality, fairness, and completeness. Remaining legacy time-chain terminology is replaced by the derived or directly validated construction-order ideal chain.

15 July 2026 — J.GWR snapshot, same-cycle, and dependency-completeness refinement.

The local-to-global construction now derives its proof order from a generated typed dependency graph rather than accepting J.UnitRank/phase metadata. Definition J.ConstructionAction introduces explicit ReadSet/WriteSet footprints; J.DepSound and J.DepComplete prove prerequisite soundness and non-commutation coverage; J.FoundationRank/J.DepRank prove every edge advances one graph-independent semantic rank; J.DepAcyclic follows; and J.CanonicalRank is reconstructed only afterward. Lemma J.BufSameCycle proves legal sequentialization of simultaneous positive-capacity Push/Pop commits without permitting empty/full fall-through. Definition J.ReqSnapshot and Lemma J.NBDecision move witness-significant non-blocking outcomes out of process-local replay paths and evaluate them from one common attributed boundary snapshot. Lemma I.3, J.PCert/J.CCert/J.LocalGWR, J.PrepareCommute/J.UnitEnable/J.UnitExt, the expanded J.GWR-Comp proof, the side-condition record, and Appendices S/T are updated accordingly.

15 July 2026 — J.GWR compositional-construction revision.

Definition J.GWR remains the stable semantic interface but is no longer a mandatory unexplained Post → Ref premise. J.LocalGWR supplies the local certificates, and Lemma I.3 supplies finite process-owned preparation scripts with shared-decision holes. J.Frame/J.LocalLift/J.PrepareCommute/J.UnitEnable/J.UnitExt provide generic gluing; J.GWR-Comp constructs a reachable Sys_ref witness and J.GWR package. J.O remains the separate pairwise composition lemma.